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Chapter 1: Orienting Yourself - The Use of Coordinates

This chapter teaches you how to describe the exact position of any point on a flat surface using nothing but two numbers. That sounds small, but it is the foundation of coordinate geometry, which you will use in every higher class. The uploaded PDF covers the 2-D Cartesian system (axes, origin, ordered pairs, quadrants, plotting), distances parallel to the axes, the distance formula built on the Baudhayana-Pythagoras Theorem, reflections in the axes, midpoint and trisection ideas, the collinearity test, and points on a circle centred at the origin. In this practice book we push every one of those ideas one level higher, with harder numbers, missing-value reasoning, and proof-style thinking.

1.1 Complete Concept Cheat Sheet (read this before any question)

1.1.1 The coordinate system and its parts

What is it? A coordinate system is a grid-based framework that locates every point in a plane using an ordered pair of numbers. Two perpendicular number lines are drawn: the horizontal one is the **x-axis** and the vertical one is the **y-axis**. They meet at the **origin O = (0, 0)**. Values to the right of O on the x-axis and above O on the y-axis are positive; values to the left and below are negative.

When do I use it? Whenever a question gives you points, positions on a map or a room layout, distances between locations, or asks you to 'plot', 'locate', 'identify the quadrant', or 'find the distance between points'.

How do I recognise it? Any mention of (x, y) pairs, axes, quadrants, graph paper, or a floor plan with walls along two directions. **What does each part mean?** In the pair (x, y), the first number x is the **perpendicular distance from the y-axis** (measured along the x-axis direction), and the second number y is the **perpendicular distance from the x-axis**. The pair is called an *ordered pair* because the order is fixed: (3, 5) and (5, 3) are different points. Logically, you can understand it as 'walk x units horizontally, then y units vertically from O'.

Tiny example. The point (2, 3) is 2 units right of O and 3 units up. The point (-2, 3) is 2 units LEFT and 3 up. Same distances, different signs, different points entirely.

1.1.2 Points on the axes and the quadrant sign rules

- A point on the x-axis always looks like (x, 0). Its y-coordinate is 0 because it has zero height above the x-axis. Positive x means right of O, negative x means left of O.
- A point on the y-axis always looks like (0, y). Its x-coordinate is 0. Positive y means above O, negative y means below O.
- The origin (0, 0) lies on BOTH axes; it belongs to no quadrant.
- Quadrant I: (+, +). Quadrant II: (-, +). Quadrant III: (-, -). Quadrant IV: (+, -). Memorise the cycle: starting from I and going anticlockwise the signs are ++, -+, --, +-. A quick memory trick: signs of y follow the C-shape of Roman numerals I, II, III, IV written anticlockwise; signs of x are + in I, - in II, - in III, + in IV.
- Any point with x = 0 and y = 0 must be the origin. Any point with both coordinates non-zero cannot lie on either axis.

> **Identify-the-method trick**
> See a question that says 'on the x-axis' or 'on the y-axis'? Instantly write the fixed coordinate: (something, 0) or (0, something). This single habit unlocks a whole family of missing-coordinate questions.

1.1.3 Distances parallel to the axes

What is it? If two points share the same y-coordinate, the segment joining them is horizontal, and the distance is simply the absolute gap between the x-values: distance = $|x_2 - x_1|$. If two points share the same x-coordinate, the segment is vertical and the distance is $|y_2 - y_1|$.

When do I use it? Whenever two given points have one identical coordinate - check this FIRST before reaching for the distance formula. **How do I recognise it?** Same second number means a horizontal move (like walking along a corridor east-west); same first number means a vertical move. **Each term:** the absolute value bars guarantee a positive (non-negative) length; order of subtraction does not matter.

Tiny example. Distance between (2, 5) and (9, 5) is $|9 - 2| = 7$. Distance between (4, -3) and (4, 6) is $|6 - (-3)| = 9$. Note the classic sign trap: $6 - (-3) = 9$, not 3.

1.1.4 The distance formula (the star of the chapter)

What is it? For any two points $A(x_1, y_1)$ and $D(x_2, y_2)$, the distance is $AD = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$. **Where does it come from?** Drop a right-angled helper triangle: the horizontal leg is $|x_2 - x_1|$ and the vertical leg is $|y_2 - y_1|$. The segment AD is the hypotenuse, so the Baudhayana-Pythagoras Theorem ($leg^2 + leg^2 = hypotenuse^2$) gives the formula. It is not a magic rule - it is Pythagoras wearing coordinates.

When do I use it? When the two points differ in BOTH coordinates. **How do I recognise it?** Words like 'find the length of AB', 'how far apart', 'show that the triangle is isosceles', 'find the perimeter' with slanted sides. **Each term:** subtract corresponding coordinates, square each difference (squares kill any minus sign, so the order of the points does not matter), add, then take the positive square root.

Can it be derived logically? Yes - see the helper-triangle picture in Figure 2. **Tiny example.** From (1, 2) to (4, 6): differences are 3 and 4, so distance = $\sqrt{(9 + 16)} = \sqrt{25} = 5$. Whenever the squared sum is a perfect square you get a clean integer - examiners love the triples 3-4-5, 5-12-13, 6-8-10, 8-15-17, 9-12-15, 7-24-25.

- **Method 1 (helper triangle):** sketch, mark legs |dx| and |dy|, apply Pythagoras. Best for learning and for verifying.
- **Method 2 (formula directly):** substitute into $\sqrt{(dx^2 + dy^2)}$. Best for speed once confident.
- **Method 3 (shift to origin):** subtract the first point from both - distance is unchanged - then compute from (0, 0). Great mental shortcut: distance of (x, y) from origin is $\sqrt{(x^2 + y^2)}$.

> **Common confusion**
> Students often compute $\sqrt{(dx)} + \sqrt{(dy)}$ or $dx + dy$ instead of $\sqrt{(dx^2 + dy^2)}$. Remember: square EACH difference first, THEN add, THEN take one single square root at the very end. Never take a root midway.

1.1.5 Reflections in the axes

What is it? Reflecting a point in the x-axis flips its height: (x, y) becomes (x, -y). Reflecting in the y-axis flips its horizontal position: (x, y) becomes (-x, y). Reflecting in BOTH axes (equivalent to a half-turn about O)

gives (-x, -y). **Why only one sign changes:** a mirror in the x-axis changes only how far up or down you are; your sideways position stays the same.

Key property: reflection preserves distances. If A reflects to A' and B reflects to B', then AB = A'B'. The PDF proves this by computing both distances - the squares of the coordinate differences come out identical. Use this to check your reflection answers. **Tiny example.** A(2, 3) reflects in the x-axis to A'(2, -3). Take another point B(6, 3); it reflects to B'(6, -3). Now AB = |6 - 2| = 4 and A'B' = |6 - 2| = 4: the distance is preserved because the reflection moves the WHOLE segment, not just one endpoint. (If you compare A' with the unreflected B you get √52 - but that was never a preserved pair; always compare each point with its own image.)

1.1.6 Midpoint and equal-distance ideas

What is it? The midpoint of a segment splits it into two equal halves. The PDF discovers the connection experimentally: if M(m_1, m_2) is the midpoint of S and T, then each coordinate of M is the average of the corresponding coordinates: $m_1 = (x_S + x_T)/2$ and $m_2 = (y_S + y_T)/2$. You can verify with the distance formula: distances from M to S and from M to T come out equal. **Reverse use:** if you know one endpoint A and the midpoint M, the other endpoint is B = ($2m_1 - x_A$, $2m_2 - y_A$) - just undo the averaging. **Trisection:** points P and Q that cut AB into three equal parts satisfy P = A + (B - A)/3 and Q = A + 2(B - A)/3; the midpoint of P and Q equals the midpoint of AB, which is a lovely self-check.

When do I use the average idea? Keywords: 'midpoint', 'bisects', 'equally spaced', 'middle point of the segment'. **Tiny example.** Midpoint of (-4, 1) and (6, 5) is ((-4 + 6)/2, (1 + 5)/2) = (1, 3).

1.1.7 Collinearity test without plotting

What is it? Three points are collinear (lie on one straight line) if one of the three distances equals the sum of the other two. Concretely, compute AB, BC, AC and check whether the largest equals the sum of the other two. **Why it works:** if B truly sits on the straight path from A to C, then walking A to B to C covers exactly the straight distance AC - no detour is possible on a straight line (triangle inequality with equality).

When do I use it? Questions like 'are these points on the same straight line?' or 'check without plotting'. **Trap:** the check must be done with EXACT values. If the distances involve roots, compare exact squared forms or keep radicals; do not round early. **Tiny example.** (0,0), (3,4), (6,8): AB = 5, BC = 5, AC = 10, and 5 + 5 = 10, so collinear.

1.1.8 Points on a circle centred at the origin

What is it? A circle with centre O(0, 0) and radius r is the set of points whose distance from O equals exactly r. So a point (x, y) lies ON the circle if $x^2 + y^2 = r^2$, INSIDE if $x^2 + y^2$ is less than r^2, and OUTSIDE if it is greater. This is just the distance-from-origin shortcut applied with an equality/inequality. **How do I recognise it?** 'Show that these points lie on a circle with centre the origin - find the radius' or 'is the point inside, on, or outside the circle?'. **Tiny example.** (1, -8), (-4, 7), (-7, -4): squares give 65, 65, 65 - all equal, so all three lie on the circle $x^2 + y^2 = 65$, radius √65.

1.1.9 Exam tips for this chapter

1. Before any distance computation, CHECK whether a coordinate is shared - the parallel-to-axes shortcut saves time and avoids errors.
2. Keep the ordered-pair order sacred. (x, y) and (y, x) coincide only when x = y.

3. Squaring removes minus signs, so $(x_2 - x_1)^2 = (x_1 - x_2)^2$. Do not waste time worrying about subtraction order.
4. Memorise Pythagorean triples; they turn ugly roots into integers.
5. For 'inside/on/outside circle' questions, compare SQUARES of distances - never square roots.
6. After finding a reflection, quickly verify one preserved distance.
7. In multi-part geometry questions, draw a rough sketch even if the question says 'without plotting' - your sketch guides the algebra; the algebra is what you present.

1.2 Question Types / Concept Map for Chapter 1

Almost every coordinate question in exams falls into one of the following types. Use this map to label what you are looking at before solving - half the battle is knowing WHICH type of fight it is.

Type	Name	How to recognise it in the question	Core method
T1	Direct identification	Given coordinates, find quadrant / axis / sign pattern	Sign rules of the four quadrants
T2	Plotting and reading	Mark points, read positions from a room/map figure	Ordered pair = walk x then y
T3	Axis-parallel distance	Two points share one coordinate	|dx| or |dy| shortcut
T4	Distance formula	Two points, both coordinates differ	$\sqrt{(dx^2 + dy^2)}$
T5	Reverse / missing value	Distance given, one coordinate unknown (find k)	Set up equation, solve, check BOTH roots
T6	Triangle / quadrilateral classification	Show isosceles / right / square / rhombus / parallelogram	Compute all sides (and diagonals), compare
T7	Reflection reasoning	Image of a point in x-axis / y-axis / both	Flip the relevant sign; verify preserved distance
T8	Midpoint / trisection	'M is the midpoint', 'points of trisection', 'bisects'	Average coordinates; reverse-average for missing endpoint
T9	Collinearity test	'On the same line?', 'without plotting'	Largest distance = sum of other two (exact values)
T10	Circle membership	'lie on circle with centre origin', inside/outside check	Compare $x^2 + y^2$ with r^2
T11	Applied grid problems	City streets, screen pixels, room maps	Model positions as coordinates, then use T3/T4
T12	Perimeter / area from coordinates	Vertices given, find perimeter or area	Distance formula on each side; 1/2 base x height for right triangles
T13	Conceptual trap questions	Order of pair, meaning of coordinates, statements to justify	Reason from definitions, not memory

Type	Name	How to recognise it in the question	Core method
T14	Multi-concept HOTS	Combines distance + midpoint + circle or reflection chains	Break into labelled mini-steps, verify at each junction

1.3 Speed Tricks & Trap Alarms

Ye table tumhara "dekho aur maaro" radar hai - question ka trigger phrase dikha, toh yahan ka trick fire karo. Inhe ratt lo, exam me sochna kam padega aur speed double ho jayegi.

Instant trick	Fire it when...
Same y? Distance = $\|x_2 - x_1\|$. Same x? Distance = $\|y_2 - y_1\|$	Points share a coordinate - skip the formula, just subtract.
Distance from origin = $\sqrt{(x^2 + y^2)}$. Memorise triples: (3,4,5), (6,8,10), (5,12,13), (8,15,17)	"How far from the origin?" - spot the triple, answer in 2 seconds.
Reflect in x-axis: (x, -y). In y-axis: (-x, y). In both: (-x, -y)	Any "mirror image / reflection" wording.
Midpoint = averages of the coordinates; reverse: B = 2M - A	"Midpoint", "bisects", or one endpoint + midpoint given.
Right angle test: (biggest side)^2 = sum of squares of other two	"Prove/verify right angle", triangle classification.
Point on circle centre O radius r $\iff x^2 + y^2 = r^2$	"On/inside/outside the circle" - compare x^2+y^2 with r^2.
"On the x-axis" $\rightarrow$ y = 0; "on the y-axis" $\rightarrow$ x = 0	Building coordinates of axis points.

1.4 The 120-Question Workout

Poora chapter ab ek hi ring mein: **120 questions**, chaar level zones mein banta hua. Warm-up se shuru karo, Boss Zone tak pahuncho. Har question ke turant baad boxed step-by-step solution hai - pehle khud try karo, atako toh solution dekho, aur "Why this step?" + Hinglish note se logic pakka karo. Agar in 120 ko solve kar liya toh is chapter ka koi bhi twisted question tumhe nahi rok payega.

Level 1 · Warm-up Zone: basics and confidence:

QUESTION 1 | Basic | Quadrant sign rules (T1)

Q1. Name the quadrant (or axis) in which each point lies, and state the sign pattern: A(7, -9), B(-6, 2), C(-8, -11), D(9, 4), E(0, -3), F(-12, 0).

SOLUTION — step by step

Step 1. Given: six points with their coordinates. Needed: location of each point.

Step 2. Recall the sign pattern: Quadrant I (+, +), Quadrant II (-, +), Quadrant III (-, -), Quadrant IV (+, -). Points with x = 0 lie on the y-axis; points with y = 0 lie on the x-axis.

Step 3. A(7, -9): x is positive, y is negative, so A is in Quadrant IV with signs (+, -).

Step 4. B(-6, 2): x negative, y positive, so Quadrant II with signs (-, +).

Step 5. C(-8, -11): both negative, so Quadrant III with signs (-, -).

Step 6. D(9, 4): both positive, so Quadrant I with signs (+, +).

Step 7. E(0, -3): x-coordinate is 0, so E lies on the y-axis, 3 units below the origin, in no quadrant.

Step 8. F(-12, 0): y-coordinate is 0, so F lies on the x-axis, 12 units left of the origin.

Answer: A in Q IV (+, -); B in Q II (-, +); C in Q III (-, -); D in Q I (+, +); E on the y-axis below O; F on the x-axis left of O.

Hinglish: *Beta, yahan sirf signs dekhne hain. Pehle number jo hai wo x ki duniya hai - positive matlab right side, negative matlab left. Doosra number y hai - positive matlab upar, negative matlab neeche. A mein x positive, y negative, matlab right aur neeche = Quadrant IV. Bas ye pattern yaad rakho: I mein dono plus, II mein x minus y plus, III mein dono minus, IV mein x plus y minus. Aur agar koi coordinate 0 hai toh point axis par baitha hai, quadrant mein nahi.*

QUESTION 2 | Basic | Points on the axes (T1/T2)

Q2. Write the coordinates of (i) the point on the x-axis that is 6 units to the left of the origin, (ii) the point on the y-axis that is 4.5 units below the origin, (iii) the point where the two axes meet.

SOLUTION — step by step

Step 1. Given: descriptions along the axes. Needed: coordinates.

Step 2. A point on the x-axis has the form (x, 0); 'left of the origin' means the x-value is negative.

Step 3. So 6 units left of O gives (-6, 0).

Step 4. A point on the y-axis has the form (0, y); 'below the origin' means the y-value is negative.

Step 5. So 4.5 units below O gives (0, -4.5).

Step 6. The axes meet at the origin, whose coordinates are (0, 0).

» Why this step? We fix the zero coordinate FIRST because 'on the axis' forces it; then only the sign and the size of the other coordinate remain.

Answer: (i) (-6, 0) (ii) (0, -4.5) (iii) O(0, 0)

Hinglish: *X-axis par rehne wale har point ka second number hamesha 0 hota hai - ye pakka rule hai. Left jaana hai matlab negative. Isliye (-6, 0). Y-axis par rehne wale ka pehla number 0 hota hai, aur neeche matlab negative y, isliye (0, -4.5). Origin dono axis ka intersection hai, wahan dono zero - (0, 0). Sign ka khayal rakho, bas yahi galti hoti hai.*

QUESTION 3 | Basic | Meaning of coordinates (T13)

Q3. For the point P(5, -7), state (i) its perpendicular distance from the y-axis, (ii) its perpendicular distance from the x-axis, (iii) which side of each axis it lies on.

SOLUTION — step by step

Step 1. Given: P(5, -7). Needed: distances from the two axes and directions.

Step 2. By definition, the x-coordinate is the perpendicular distance from the y-axis, measured along the x-direction.

Step 3. So the distance from the y-axis is 5 units, and since 5 is positive, P lies to the right of the y-axis.

Step 4. The y-coordinate is the perpendicular distance from the x-axis.

Step 5. So the distance from the x-axis is 7 units, and since -7 is negative, P lies below the x-axis.

Step 6. Hence P sits in Quadrant IV, 5 units right and 7 units down from the origin.

» **Why this step?** Distances are always written positive; the SIGN of the coordinate only tells you the side. Never say the distance is -7.

Answer: (i) 5 units, right of the y-axis (ii) 7 units, below the x-axis (iii) Quadrant IV

Hinglish: *Bahut important concept: x-coordinate batata hai y-axis se doori, aur y-coordinate batata hai x-axis se doori. Students yahan ulta kar dete hain! P(5, -7) ka matlab: y-axis se 5 unit door (right side), x-axis se 7 unit door (neeche, kyunki minus hai). Doori kabhi negative nahi hoti - minus sirf direction batata hai.*

QUESTION 4 | Basic | Ordered pair order matters (T13)

Q4. Are the points (3, 5) and (5, 3) the same point? Justify. For what values of a and b does the point (a, b) coincide with the point (b, a)?

SOLUTION — step by step

Step 1. Given: two ordered pairs. Needed: whether they name the same location.

Step 2. The point (3, 5) means 3 units along the x-direction and 5 units along the y-direction; (5, 3) means 5 along x and 3 along y.

Step 3. These are different walks from the origin, so they land at different places: (3, 5) is above-left of (5, 3). They coincide only if the two coordinates are equal.

Step 4. Setting (a, b) = (b, a) gives a = b and b = a - the same single condition a = b.

Step 5. So for any value with a = b (for example (4, 4)), the two pairs name one and the same point on the line y = x.

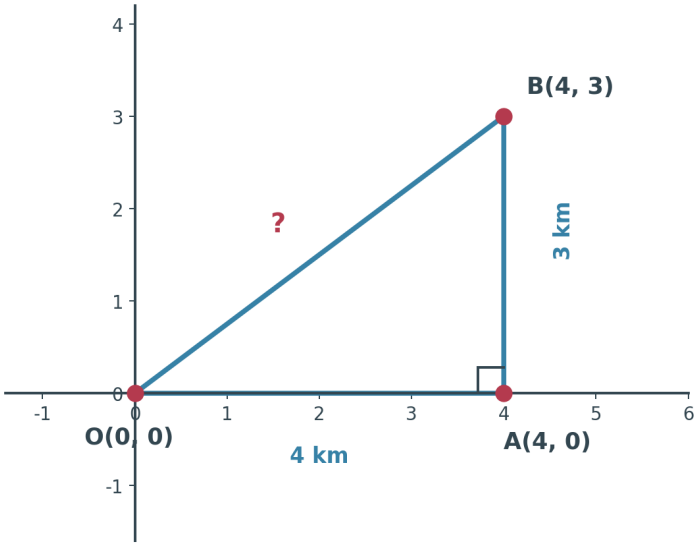
» **Why this step?** The justification is definitional, not visual: ordered pairs are equal exactly when they agree coordinate by coordinate.

Answer: No, (3, 5) and (5, 3) are different points; (a, b) = (b, a) exactly when a = b.

Hinglish: *(3, 5) aur (5, 3) bilkul alag points hain kyunki ordered pair mein ORDER matter karta hai - pehle x ki baat, phir y ki. Dono tabhi ek hi point banenge jab x aur y barabar hon, yaani a = b. Tab dono us line par baithte hain jahan har point ka x aur y same hota hai (line y = x). Exam mein ye 'true/false with reason' type bahut aata hai.*

QUESTION 5 | Basic | Figure reading + distance formula (T2/T4)

Q5. A delivery boy starts at home O(0, 0), rides 4 units east to a tea stall at A(4, 0), then rides 3 units north to deliver food at B(4, 3), and finally rides straight back to O. Using the figure: (i) what is the total distance he rode? (ii) when he is at B, how far is he from home (straight line)? (iii) if a drone could fly directly from O to B, how much shorter would the O-to-B trip be compared to his road trip O-A-B?



SOLUTION — step by step

Step 1. Read the figure: O to A is horizontal with the same y-coordinate, so OA = |4 - 0| = 4 units. A to B is vertical with the same x-coordinate, so AB = |3 - 0| = 3 units. Same-coordinate pairs give free distances.

Step 2. The ride home B to O is the slanted stretch BO. Route makes a right angle at A, so triangle OAB has legs 4 and 3.

Step 3. Distance formula (or the 3-4-5 triple): BO = $\sqrt{((4-0)^2 + (3-0)^2)} = \sqrt{(16 + 9)} = \sqrt{25} = 5$ units.

Step 4. (i) Total distance ridden = OA + AB + BO = 4 + 3 + 5 = 12 units.

Step 5. (ii) At B, straight-line distance from home = BO = 5 units.

Step 6. (iii) Road O-A-B = 4 + 3 = 7 units; direct O-to-B = 5 units. Saving = 7 - 5 = 2 units.

» **Why this step?** Same-coordinate pairs need only subtraction - save the formula for the slanted side.

» **Why this step?** The right angle at A makes OB the hypotenuse - spotting the 3-4-5 triple kills the square root work instantly.

Answer: (i) 12 units (ii) 5 units (iii) 2 units shorter

Hinglish: *Delivery wala scene ek right triangle hai: east 4, north 3, toh ghar seedha 5 - 3-4-5 triple yaad hai na? Total ride 12 units, par seedhi line sirf 5. Graph paper par ghoomna mehnga hai, drone hi bhej do!*

QUESTION 6 | Basic | Axis-parallel distances (T3)

Q6. Find the distance between (i) A(2, 0) and B(9, 0), (ii) C(0, -4) and D(0, 6), (iii) E(-5, 3) and F(4, 3).

SOLUTION — step by step

Step 1. Given: three pairs of points. Needed: distances.

Step 2. Pair (i): both points lie on the x-axis (y = 0 for both), so use the horizontal shortcut |x2 - x1| = |9 - 2| = 7.

Step 3. Pair (ii): both on the y-axis, so distance = |y2 - y1| = |6 - (-4)| = |6 + 4| = 10.

Step 4. Pair (iii): both have y = 3, a horizontal segment: distance = |4 - (-5)| = |4 + 5| = 9.

Step 5. In every case the shared coordinate confirms the segment is parallel to one axis, so the single-difference shortcut is valid - no square roots needed.

» Why this step? Checking the shared coordinate FIRST is the habit that separates fast, accurate students from slow ones.

Answer: (i) 7 units (ii) 10 units (iii) 9 units

Hinglish: *Jab do points mein se koi ek coordinate same ho, toh seedha difference lo - formula ki zaroorat hi nahi. (ii) mein dhyan do: 6 - (-4) = 6 + 4 = 10. Minus ke saath minus mila toh plus ban gaya. Ye sign wala jhatka sabse common galti hai, isliye maine absolute value ke andar rakha hai. (iii) bhi same khel hai: 4 - (-5) = 9.*

QUESTION 7 | Basic | Perimeter from coordinates (T12/T3)

Q7. The corners of a rectangular room are O(0, 0), A(8, 0), B(8, 5) and C(0, 5). Find the perimeter and state which walls are parallel to which axes.

SOLUTION — step by step

Step 1. Given: four corners in order. Needed: perimeter and axis directions of walls.

Step 2. OA joins (0, 0) to (8, 0): both y-values are 0, so OA is horizontal with length |8 - 0| = 8.

Step 3. AB joins (8, 0) to (8, 5): both x-values are 8, so AB is vertical with length |5 - 0| = 5.

Step 4. BC joins (8, 5) to (0, 5): horizontal, length |8 - 0| = 8. CO joins (0, 5) to (0, 0): vertical, length 5.

Step 5. Perimeter = 8 + 5 + 8 + 5 = 26 units.

Step 6. OA and BC run along the x-direction (parallel to the x-axis); AB and CO run along the y-direction (parallel to the y-axis).

Answer: Perimeter = 26 units; OA and BC are parallel to the x-axis, AB and CO are parallel to the y-axis.

Hinglish: *Rectangle ke opposite sides barabar hoti hain, isliye do sides hi kaafi hain: ek horizontal 8, ek vertical 5. Perimeter = 2(8 + 5) = 26. Dekho, coordinates same hone se hi pata chal gaya ki kaun si deewar kis axis ke parallel hai - y same hai toh horizontal wall, x same hai toh vertical wall. Ye floor-plan wale questions mein bahut kaam aata hai.*

QUESTION 8 | Basic | Distance from origin (T4)

Q8. Find the distance of the point (6, 8) from the origin. Which Pythagorean triple is hiding inside this question, and how can the triple help you answer faster next time?

SOLUTION — step by step

Step 1. Given: point (6, 8). Needed: distance from O(0, 0).

Step 2. Distance from origin = $\sqrt{(x^2 + y^2)} = \sqrt{(6^2 + 8^2)}$.

Step 3. Compute the squares: 36 + 64 = 100.

Step 4. Take the root: $\sqrt{100} = 10$.

Step 5. The hidden triple is 3-4-5 scaled by 2: 6-8-10. If you spot (6, 8), you can jump straight to 10 without squaring.

Step 6. So the distance is 10 units.

» **Why this step?** Spotting the triple is a safe shortcut only when BOTH numbers of a known triple appear exactly; otherwise use the formula.

Answer: 10 units; the triple 6-8-10 (which is 2 x (3-4-5)).

Hinglish: *Origin se distance ka short formula: $\sqrt{(x^2 + y^2)}$. 6 aur 8 ko dekhte hi 3-4-5 wala triple yaad karo - dono numbers 2 se multiply ho rahe hain, isliye answer 5 ka 2 guna = 10. Triples ratt lo: 3-4-5, 5-12-13, 8-15-17, 7-24-25. Par dhyan rahe - triple tabhi use karo jab numbers EXACT match karein, warna seedha formula lagao.*

QUESTION 9 | Basic | Sign reasoning (T1/T13)

Q9. A point has coordinates (a, -a) where a is a non-zero real number. (i) In which quadrant does it lie when a is positive? (ii) In which quadrant when a is negative? (iii) What happens at a = 0? (iv) What is always true about the distances of this point from the two axes?

SOLUTION — step by step

Step 1. Given: the family of points (a, -a). Needed: quadrant analysis by sign of a.

Step 2. Case a positive: x = a is positive, y = -a is negative, so the point is in Quadrant IV.

Step 3. Case a negative: writing a = -m with m positive gives x = -m (negative) and y = m (positive), so the point is in Quadrant II.

Step 4. Case a = 0: the point is (0, 0), the origin itself - on both axes, in no quadrant.

Step 5. For every a, the distance from the y-axis is |a| and the distance from the x-axis is |-a| = |a|. These are always equal.

Step 6. So every point of this family stays exactly equally far from both axes - the family slides along the two lines y = -x.

Answer: (i) Quadrant IV when a > 0; (ii) Quadrant II when a < 0; (iii) the origin when a = 0; (iv) distances from both axes are always equal (each equals |a|).

Hinglish: *Ye question sign-logic ka gym hai. a positive toh x positive, y negative = Quadrant IV. a negative karo toh dono ke sign flip ho jaate hain: x negative, y positive = Quadrant II. a = 0 par origin. Aur sundar observation: is point ki dono axis se doori hamesha barabar hai |a|, kyunki x aur y ka size same hai, sirf sign ulta hai. Aage Ch 2 mein y = -x line padhoge, ye usi ka preview hai.*

QUESTION 10 | Basic | Axis membership (T1/T13)

Q10. From the list (2, 0), (0, 2), (-5, 0), (0, -5), (7, 0), (0, 7), (-3, -3), pick out (i) all points on the x-axis, (ii) all points on the y-axis, (iii) any point equidistant from both axes, and (iv) state how many of the listed points lie in a quadrant.

SOLUTION — step by step

Step 1. Given: eight points. Needed: classify by axis membership and symmetry.

Step 2. x-axis means y = 0: the points (2, 0), (-5, 0), (7, 0) qualify.

Step 3. y-axis means x = 0: the points (0, 2), (0, -5), (0, 7) qualify.

Step 4. Equidistant from both axes means |x| = |y|. Only (-3, -3) satisfies this (both distances 3).

Step 5. Six points lie on the axes, and exactly one point, (-3, -3), lies strictly inside a quadrant (Quadrant III).

Answer: (i) (2, 0), (-5, 0), (7, 0) (ii) (0, 2), (0, -5), (0, 7) (iii) (-3, -3) (iv) exactly 1 point lies in a quadrant (Q III).

Hinglish: *Simple filter lagao: y = 0 hai toh x-axis ka banda, x = 0 hai toh y-axis ka. Dono conditions fail ho jaayein tabhi point kisi quadrant ke andar hai. Yahan sirf (-3, -3) quadrant mein girta hai. Aur |x| = |y| wala idea yaad rakho - dono axis se barabar doori ka matlab. Ye chhota sa observation aage bahut bade questions kholta hai.*

QUESTION 11 | Basic | Quadrant sign rules (T1)

Q11. Name the quadrant (or axis) for each point and give its sign pattern: A(-1.5, 8), B(0, 4.2), C(9, -0.5), D(-7, -7), E(6, 11).

SOLUTION — step by step

Step 1. Sign map: I (+, +), II (-, +), III (-, -), IV (+, -); x = 0 means y-axis, y = 0 means x-axis.

Step 2. A(-1.5, 8): x negative, y positive -> Quadrant II, signs (-, +).

Step 3. B(0, 4.2): x = 0 -> lies ON the y-axis, 4.2 units above O, in no quadrant.

Step 4. C(9, -0.5): x positive, y negative -> Quadrant IV, signs (+, -).

Step 5. D(-7, -7): both negative -> Quadrant III, signs (-, -).

Step 6. E(6, 11): both positive -> Quadrant I, signs (+, +).

» **Why this step?** Decimals and fractions do not change the rule - only the SIGNS decide the quadrant.

Answer: A in Q II; B on the y-axis above O; C in Q IV; D in Q III; E in Q I.

Hinglish: *Sign dekho, quadrant mil gaya. Zero aaye toh point axis par hi baitha hai. Itni si science hai, bas sign table zubani yaad honi chahiye.*

QUESTION 12 | Basic | Points on the axes (T1/T2)

Q12. Write the coordinates of: (i) the point on the y-axis 7 units above O, (ii) the point on the x-axis 2.5 units left of O, (iii) the point 5 units right of O and 5 units below O.

SOLUTION — step by step

Step 1. On the y-axis the x-coordinate is 0; 'above' means positive y: (0, 7).

Step 2. On the x-axis the y-coordinate is 0; 'left' means negative x: (-2.5, 0).

Step 3. Right 5 -> x = +5; below 5 -> y = -5: the point is (5, -5).

» **Why this step?** 'On an axis' kills one coordinate to 0 first; the direction word (left/right/up/down) fixes the sign of the other.

Answer: (i) (0, 7) (ii) (-2.5, 0) (iii) (5, -5)

Hinglish: *Axis par point = ek coordinate zero. Direction ka word sign bata deta hai - left/down minus, right/up plus. (5, -5) Quadrant IV ka hero hai.*

QUESTION 13 | Basic | Meaning of coordinates (T13)

Q13. For the point M(-6, 9) state (i) its perpendicular distance from the y-axis, (ii) its perpendicular distance from the x-axis, (iii) the quadrant it lies in.

SOLUTION — step by step

Step 1. Distance from the y-axis = |x| = |-6| = 6 units; the point lies LEFT of the y-axis because x is negative.

Step 2. Distance from the x-axis = |y| = |9| = 9 units; the point lies ABOVE the x-axis because y is positive.

Step 3. x negative and y positive -> Quadrant II.

» **Why this step?** Distance is always |coordinate| - the sign only tells you the SIDE, never the distance.

Answer: (i) 6 units (left side) (ii) 9 units (above) (iii) Quadrant II

Hinglish: *y-axis se doori |x|, x-axis se doori |y| - minus ko doori mein mat ghusao. Side batana alag kaam hai, distance batana alag kaam.*

QUESTION 14 | Basic | Ordered pair order matters (T1)

Q14. Are the points (4, -1) and (-1, 4) the same point? For what values of a and b is (a, b) equal to (b, a)? Where do all such points lie?

SOLUTION — step by step

Step 1. (4, -1): x = 4, y = -1. (-1, 4): x = -1, y = 4. The coordinates are swapped, so they are DIFFERENT points.

Step 2. (a, b) = (b, a) needs a = b (first coordinates equal AND second coordinates equal).

Step 3. If a = b, the point looks like (t, t) - equal coordinates.

Step 4. Points (t, t) lie on the straight line through O that makes equal angles with both axes (the line y = x), passing through Quadrants I and III.

» **Why this step?** An ordered pair is ordered - swapping generally moves you to a different point unless the two coordinates are equal.

Answer: No, they are different; (a, b) = (b, a) only when a = b; all such points lie on the line y = x.

Hinglish: *(4, -1) aur (-1, 4) bhai-bhai lagte hain par alag ghar mein rehte hain. Equal hone ka ek hi tareeka: dono coordinate same, tabhi point y = x line par jayega.*

QUESTION 15 | Basic | Axis-parallel distances (T3)

Q15. Find the distance between: (i) P(-3, 7) and Q(5, 7), (ii) R(2, -4) and S(2, 9), (iii) T(-1.5, -2) and U(-1.5, -6.5).

SOLUTION — step by step

Step 1. P and Q share y = 7: the segment is horizontal, so PQ = |5 - (-3)| = 8 units.

Step 2. R and S share x = 2: the segment is vertical, so RS = |9 - (-4)| = 13 units.

Step 3. T and U share x = -1.5: vertical segment, TU = |-6.5 - (-2)| = 4.5 units.

» **Why this step?** Same y -> subtract the x's; same x -> subtract the y's. The distance formula would work too, but it is overkill.

Answer: (i) 8 units (ii) 13 units (iii) 4.5 units

Hinglish: *Ek coordinate match ho toh bas subtraction game hai - formula ki zaroorat hi nahi. Signs ko bracket mein pakad ke likho, wahi galti hoti hai.*

QUESTION 16 | Basic | Distance from origin (T4)

Q16. Find the distance of each point from the origin and name the Pythagorean triple hiding inside: (i) (9, 12), (ii) (8, 15), (iii) (7, 24).

SOLUTION — step by step

Step 1. Distance from O = $\sqrt{(x^2 + y^2)}$.

Step 2. (9, 12): $\sqrt{(81 + 144)} = \sqrt{225} = 15$ -> the 9-12-15 triple (3-4-5 scaled by 3).

Step 3. (8, 15): $\sqrt{(64 + 225)} = \sqrt{289} = 17$ -> the 8-15-17 triple.

Step 4. (7, 24): $\sqrt{(49 + 576)} = \sqrt{625} = 25$ -> the 7-24-25 triple.

» **Why this step?** Memorising (3,4,5), (5,12,13), (8,15,17), (7,24,25) turns 'find the distance' into a 2-second answer.

Answer: (i) 15 (3-4-5 family) (ii) 17 (iii) 25

Hinglish: *Ye chaar triples ratt lo - exam mein distance-from-origin waale questions lottery ban jaate hain. Square root nikalne se pehle triple check karo, time bachega.*

QUESTION 17 | Basic | True/False rapid fire (T1/T13)

Q17. State True or False with a one-line reason: (i) the point (0, -8) lies on the x-axis. (ii) (-2, -3) lies in Quadrant III. (iii) the distance of (5, 0) from the origin is 5. (iv) if a > 0, the point (a, -a) lies in Quadrant IV. (v) the x-coordinate gives the distance from the x-axis.

SOLUTION — step by step

Step 1. (i) FALSE: on the x-axis we need y = 0; here x = 0, so the point is on the y-axis.

Step 2. (ii) TRUE: (-, -) is exactly the Quadrant III signature.

Step 3. (iii) TRUE: distance = $\sqrt{(5^2 + 0^2)} = 5$.

Step 4. (iv) TRUE: a > 0 makes x positive; -a is negative, so (+, -) -> Quadrant IV.

Step 5. (v) FALSE: the x-coordinate gives the distance from the Y-axis (and the y-coordinate from the X-axis).

» **Why this step?** (v) is the classic mix-up: x talks to the y-axis and y talks to the x-axis - they are crossed.

Answer: (i) F (ii) T (iii) T (iv) T (v) F

Hinglish: *Statement (v) sabse zyada logon ko ulata hai - x-coordinate y-axis se doori batata hai, cross connection hai. Baaki sab sign-plugging hai.*

QUESTION 18 | Basic | Perimeter and area from coordinates (T3)

Q18. A square has corners A(2, 2), B(6, 2), C(6, 6), D(2, 6). Find its side, perimeter, area and diagonal.

SOLUTION — step by step

Step 1. A and B share y = 2, so AB = |6 - 2| = 4 -> side = 4 units (check BC: same x, |6 - 2| = 4 ✓).

Step 2. Perimeter = 4 × 4 = 16 units.

Step 3. Area = side^2 = 16 square units.

Step 4. Diagonal = side$\sqrt{2}$ = $4\sqrt{2}$ (also verify: AC = $\sqrt{((6-2)^2 + (6-2)^2)} = \sqrt{32} = 4\sqrt{2}$ ✓).

» **Why this step?** On an axis-aligned square, every side is a same-coordinate subtraction - no square roots until the diagonal.

Answer: Side 4; perimeter 16; area 16; diagonal $4\sqrt{2}$

Hinglish: *Grid par bana square = side to subtraction se, area square karo, diagonal ke liye √2. Poora question 30 second ka hai.*

QUESTION 19 | Basic | Reflections in the axes (T7)

Q19. P is (3, -5). Write its image when reflected in (i) the x-axis, (ii) the y-axis, (iii) both axes. Also find the image of Q(-4, 0) in the y-axis.

SOLUTION — step by step

Step 1. Reflect in x-axis: sign of y flips: (3, 5).

Step 2. Reflect in y-axis: sign of x flips: (-3, -5).

Step 3. Reflect in both: both signs flip: (-3, 5).

Step 4. Q(-4, 0): flip x -> (4, 0). Note Q was on the x-axis; its image moved along that axis to the other side.

» **Why this step?** A point ON a mirror line stays at the same distance on the other side - here that meant moving from x = -4 to x = +4.

Answer: (i) (3, 5) (ii) (-3, -5) (iii) (-3, 5); Q' = (4, 0)

Hinglish: *Reflection = sign ka khel: x-axis mirror mein y ka sign palto, y-axis mirror mein x ka. Dono mein dono palto. Point axis par ho toh wo doosri taraf slide karta hai, gayab nahi hota.*

QUESTION 20 | Basic | Midpoint by averaging (T8)

Q20. Find the midpoint of the segment joining: (i) (2, 8) and (6, 4), (ii) (-3, 5) and (5, -1), (iii) (a, 0) and (0, b).

SOLUTION — step by step

Step 1. Midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2)$ - just average each coordinate.

Step 2. (i) ((2+6)/2, (8+4)/2) = (4, 6).

Step 3. (ii) ((-3+5)/2, (5-1)/2) = (1, 2).

Step 4. (iii) ((a+0)/2, (0+b)/2) = (a/2, b/2).

» **Why this step?** Averaging works because the midpoint must be exactly halfway in both directions - half the total x-travel and half the total y-travel.

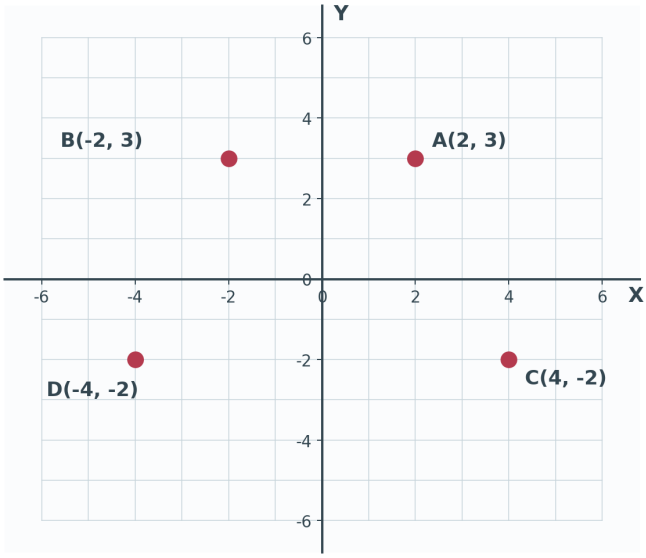
Answer: (i) (4, 6) (ii) (1, 2) (iii) (a/2, b/2)

Hinglish: *Midpoint = dono ke average, bas. Negatives ko parenthesis mein likho warna sign ka raita phail jaata hai.*

QUESTION 21 | Basic | Figure reading + reflections (T2/T7)

Q21. Using the treasure map: (i) which point is the mirror image of A in the y-axis? (ii) find AB and CD. (iii) name the quadrants of C and D. (iv) is the distance A to C equal to the distance B to D? Justify.

Treasure map: four clue points on the grid



SOLUTION — step by step

Step 1. From the figure: A(2, 3), B(-2, 3), C(4, -2), D(-4, -2).

Step 2. (i) Mirror in the y-axis flips x: A(2, 3) -> (-2, 3), which is B. So B is A's mirror image.

Step 3. (ii) A and B share y = 3: AB = |2 - (-2)| = 4. C and D share y = -2: CD = |4 - (-4)| = 8.

Step 4. (iii) C(4, -2): (+, -) -> Quadrant IV. D(-4, -2): (-, -) -> Quadrant III.

Step 5. (iv) AC = $\sqrt{((4-2)^2 + (-2-3)^2)} = \sqrt{(4 + 25)} = \sqrt{29}$. BD = $\sqrt{((-4+2)^2 + (-2-3)^2)} = \sqrt{(4 + 25)} = \sqrt{29}$. Equal - because B is A reflected and D is C reflected, and reflection preserves distances.

» **Why this step?** Reflection preserves every distance - that is WHY the two cross-distances came out equal without any fresh calculation.

Answer: (i) B (ii) AB = 4, CD = 8 (iii) C in Q IV, D in Q III (iv) Yes, both = $\sqrt{29}$

Hinglish: *Figure se coordinates uthao, baaki sab rules. Mirror pair mile toh unki cross-distance bhi barabar hogi - ye reflection ka superpower hai, diagram mein khud hi dikh jaata hai.*

QUESTION 22 | Basic | Nearest axis point (T1)

Q22. Which point on the x-axis is nearest to P(4, 3)? What is that shortest distance, and why is it the shortest?

SOLUTION — step by step

Step 1. Every point on the x-axis has the form (x, 0).

Step 2. Distance from P(4, 3) to (x, 0) = $\sqrt{((x-4)^2 + 9)}$. This is smallest when $(x - 4)^2$ is smallest, i.e. x = 4.

Step 3. So the nearest point is (4, 0) and the distance is $\sqrt{9}$ = 3 units.

Step 4. Geometric reason: (4, 0) is the foot of the perpendicular from P to the x-axis, and the perpendicular is the shortest path to a line.

» **Why this step?** Minimising a distance to a line always lands on the perpendicular foot - the squared term vanishes there.

Answer: (4, 0); shortest distance 3 units; because it is the perpendicular foot

Hinglish: *X-axis ka sabse kareebi point = uska 'pair' wala point jahan x same rahe. Seedha neeche girega toh distance kam se kam. Perpendicular = shortest, ye Physics bhi bolti hai.*

QUESTION 23 | Basic | Vertical lines x = k (T1/T3)

Q23. All points with x-coordinate -5 lie on one line. (i) Describe this line. (ii) Write any two more points on it. (iii) Find the distance between (-5, -2) and (-5, 6).

SOLUTION — step by step

Step 1. (i) It is the vertical line x = -5, parallel to the y-axis, 5 units to its left.

Step 2. (ii) Any y-value is allowed, e.g. (-5, 1) and (-5, -7).

Step 3. (iii) Same x -> vertical segment: distance = |6 - (-2)| = 8 units.

» **Why this step?** Fixing x to a constant builds a vertical 'street'; moving along the street only changes y, so distances are simple subtractions.

Answer: (i) Vertical line x = -5, 5 units left of the y-axis (ii) e.g. (-5, 1), (-5, -7) (iii) 8 units

Hinglish: *x = -5 ek seedhi khadi gali hai. Us gali mein hi rehna hai toh sirf upar-neeche khelo. Distance = y ka difference, done.*

QUESTION 24 | Basic | Rectangle from coordinates (T3/T6)

Q24. A rectangular card has corners O(0, 0), A(9, 0), B(9, 4), C(0, 4). Find its perimeter, area and the length of diagonal OB.

SOLUTION — step by step

Step 1. Length OA = |9 - 0| = 9 (same y); width AB = |4 - 0| = 4 (same x).

Step 2. Perimeter = 2(9 + 4) = 26 units.

Step 3. Area = 9 × 4 = 36 square units.

Step 4. Diagonal OB = $\sqrt{(9^2 + 4^2)} = \sqrt{(81 + 16)} = \sqrt{97}$ units.

» **Why this step?** Axis-aligned rectangles hand you the sides by subtraction; only the diagonal needs Pythagoras.

Answer: Perimeter 26; area 36; diagonal √97

Hinglish: *Rectangle ka jaal simple: lambai-chaurai subtraction se, phir perimeter/area school formula. Diagonal = hypotenuse of the corner triangle.*

QUESTION 25 | Basic | Sign reasoning (T1)

Q25. A point P(a, b) satisfies ab < 0. Which quadrants can P lie in? If ab > 0? If ab < 0 and also a + b < 0, which single quadrant is certain?

SOLUTION — step by step

Step 1. ab < 0 means exactly one of a, b is negative: (+, -) or (-, +) -> Quadrant IV or II.

Step 2. ab > 0 means both same sign: (+, +) or (-, -) -> Quadrant I or III.

Step 3. ab < 0 gives Q II or Q IV. a + b < 0 adds a condition: in Q IV (+, -) we would need b > |a|... test both: Q II (-, +) can have sum negative (e.g. -5 + 2), Q IV (+, -) can too (e.g. 2 + (-5)). Hmm - so look again: we need MORE info; actually with only ab < 0 and a + b < 0 both II and IV remain possible: (-5, 2) and (2, -5) both work.

Step 4. Correct conclusion: ab < 0 and a + b < 0 still leaves TWO possibilities (Q II or Q IV). Example: (-5, 2) is in Q II, (2, -5) is in Q IV, both satisfy the conditions.

» **Why this step?** This is a trap question - sign products constrain the quadrant pair but cannot always pin a single quadrant. Testing with numbers exposes the truth instantly.

Answer: ab < 0: Q II or Q IV; ab > 0: Q I or Q III; with a + b < 0 still Q II or Q IV - not unique (e.g. (-5, 2) and (2, -5))

Hinglish: *Yahan question khud ek dhokha hai! ab < 0 se II ya IV banta hai, aur sum negative hone se bhi dono hi possible rehte hain. Counter-example soch lena = instant proof ki answer unique nahi.*

QUESTION 26 | Basic | One point, three distances (T4/T13)

Q26. For the point P(-12, 5) find (i) its distance from the origin, (ii) from the x-axis, (iii) from the y-axis.

SOLUTION — step by step

Step 1. From origin: $\sqrt{((-12)^2 + 5^2)} = \sqrt{(144 + 25)} = \sqrt{169} = 13$ (the 5-12-13 triple).

Step 2. From the x-axis: $|y| = |5| = 5$ units.

Step 3. From the y-axis: $|x| = |-12| = 12$ units.

Step 4. Sanity check: 5, 12, 13 form a right triangle - the three answers are consistent!

» **Why this step?** The three distances are the two legs and the hypotenuse of one right triangle - knowing two pins the third.

Answer: (i) 13 (ii) 5 (iii) 12

Hinglish: *Ek point teen distance deta hai: $|x|$, $|y|$ aur $\sqrt{(x^2+y^2)}$ - aur ye teeno right triangle ke sides hain. 5-12-13 yaad tha na? Yahi chhupa tha.*

QUESTION 27 | Basic | Symmetry shortcut (T10)

Q27. Find the point on the x-axis equidistant from A(3, 0) and B(-3, 0). How many such points are there? What changes if we ask for a point on the y-axis?

SOLUTION — step by step

Step 1. Let the point be (x, 0). Distance to A: $|x - 3|$; distance to B: $|x + 3|$.

Step 2. Equidistant: $(x - 3)^2 = (x + 3)^2$ -> $-6x = 6x$ -> $x = 0$. So the point is O(0, 0) - unique on the x-axis.

Step 3. On the y-axis: point (0, y). Distance to A: $\sqrt{(9 + y^2)}$; to B: $\sqrt{(9 + y^2)}$ - EQUAL for every y!

Step 4. So on the y-axis there are infinitely many such points - the whole y-axis is the perpendicular bisector of AB.

» **Why this step?** A and B are mirror images in the y-axis, so the entire y-axis is equidistant from them; the x-axis meets that mirror line exactly once, at O.

Answer: On the x-axis: only O(0, 0). On the y-axis: every point (0, y) works - infinitely many

Hinglish: *Symmetric points ka whole mirror line hi equidistant hota hai! X-axis sirf ek jagah cross karta hai, isliye wahan ek hi answer; y-axis khud mirror hai, isliye wahan infinite answers.*

QUESTION 28 | Basic | Translation on the grid (T2)

Q28. A chess-piece at (5, 2) moves 3 units right and 4 units up. Write its new position and the straight-line distance it could have covered if it moved directly. Name the triple used.

SOLUTION — step by step

Step 1. Right 3 -> x: 5 + 3 = 8; up 4 -> y: 2 + 4 = 6. New position (8, 6).

Step 2. Direct distance = $\sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$ units.

Step 3. The 3-4-5 triple appears because the moves are perpendicular.

» **Why this step?** Walking the grid covers 3 + 4 = 7, but the direct jump is only 5 - the right angle at the turn is what creates the saving.

Answer: New position (8, 6); direct distance 5 units (3-4-5 triple)

Hinglish: *Grid walk = 7 kadam, seedhi line = 5. 3-4-5 triple roz ki zindagi mein bhi kaam aata hai - cab ka shortest route bhi yahi hai!*

QUESTION 29 | Basic | Horizontal collinearity (T9)

Q29. Are (1, 5), (4, 5) and (9, 5) collinear? What is the distance from the first to the last point, and where does the middle point sit between them?

SOLUTION — step by step

Step 1. All three share y = 5 -> all lie on the horizontal line y = 5 -> collinear.

Step 2. Distance from (1, 5) to (9, 5) = |9 - 1| = 8 units.

Step 3. Position check: (4, 5) is 3 units from the first and 5 from the last: 3 + 5 = 8 ✓, so it lies between them (not at the midpoint).

» **Why this step?** A shared coordinate is the fastest collinearity proof - no formula needed; then segment addition checks the between-ness.

Answer: Yes, collinear on y = 5; distance 8; middle point lies between, splitting it 3 and 5

Hinglish: *Sab ka y = 5? Toh teeno ek hi seedhi sadak par hain - collinear done. Beech wala point bisect nahi karta (3 aur 5 ka split hai), dhyan rakhna.*

QUESTION 30 | Basic | Floor-plan application (T3)

Q30. A rectangular room has corners O(0, 0), A(6, 0), B(6, 4.5) and C(0, 4.5), all in metres. (i) Find the carpet area. (ii) Find the length of skirting board needed along all four walls. (iii) A wire runs along the diagonal OB - how long is it (to 2 decimals)?

SOLUTION — step by step

Step 1. Length = 6 m, width = 4.5 m (read from the coordinates).

Step 2. (i) Area = $6 \times 4.5 = 27$ m^2.

Step 3. (ii) Skirting = perimeter = $2(6 + 4.5) = 21$ m.

Step 4. (iii) Diagonal OB = $\sqrt{(6^2 + 4.5^2)} = \sqrt{(36 + 20.25)} = \sqrt{56.25} = 7.5$ m - exactly, since 4.5-6-7.5 is the 3-4-5 triple scaled by 1.5!

» **Why this step?** Scaling a Pythagorean triple by any factor keeps it a triple: 3-4-5 multiplied by 1.5 gives 4.5-6-7.5.

Answer: (i) 27 m^2 (ii) 21 m (iii) 7.5 m

Hinglish: *Room ka sawaal = perimeter × area + ek diagonal. Aur 4.5-6-7.5 wala triple toh 3-4-5 ka hi bada bhai hai - decimal aane par triple ko scale karke dekho.*

Level 2 · Core Drills: book-level practice:

QUESTION 31 | Moderate | Distance formula (T4)

Q31. Find the distance between A(1, 2) and B(4, 6) using (i) the helper-triangle method and (ii) the direct formula. Confirm both give the same value.

SOLUTION — step by step

Step 1. Given: A(1, 2), B(4, 6). Needed: AB by two methods.

Step 2. Helper triangle: the horizontal leg is $|4 - 1| = 3$ and the vertical leg is $|6 - 2| = 4$.

Step 3. By Pythagoras, AB = $\sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.

Step 4. Direct formula: AB = $\sqrt{((4 - 1)^2 + (6 - 2)^2)} = \sqrt{(3^2 + 4^2)} = 5$.

Step 5. Both methods agree: AB = 5 units. The helper triangle simply explains where the formula comes from.

» **Why this step?** Showing both methods builds the habit of verifying with geometry - on hard questions this catches sign slips instantly.

Answer: AB = 5 units by both methods.

Hinglish: *Yahi chapter ka hero question hai. Helper triangle mein humne sirf differences liye - 3 aur 4 - aur wo 3-4-5 triple nikla, isliye answer 5. Formula bhi wahi karta hai, bas ek line mein. Dono ka jawab same aana hi chahiye kyunki formula Pythagoras ka hi shortcut hai. Aage se seedha formula use karo, par agar kabhi doubt ho, helper triangle bana lo.*

QUESTION 32 | Moderate | Distance formula with negatives (T4)

Q32. Find the distance between P(-2, 3) and Q(4, -5). Then find the distance of the midpoint of PQ from the origin.

SOLUTION — step by step

Step 1. Given: P(-2, 3), Q(4, -5). Needed: PQ, then distance of midpoint M from O.

Step 2. Differences: x-difference = 4 - (-2) = 6; y-difference = -5 - 3 = -8.

Step 3. PQ = $\sqrt{(6^2 + (-8)^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$. (The minus vanished on squaring, as promised.)

Step 4. Midpoint M = ((-2 + 4)/2, (3 + (-5))/2) = (2/2, -2/2) = (1, -1).

Step 5. Distance of M from origin = $\sqrt{(1^2 + (-1)^2)} = \sqrt{2}$.

Step 6. So PQ = 10 units and OM = $\sqrt{2}$ units.

» **Why this step?** Squaring (-8) gives 64 - the sign never survives a square, so never 'pre-correct' it to 8.

Answer: PQ = 10 units; OM = $\sqrt{2}$ units.

Hinglish: *Negatives ke saath formula lagane mein sabse badi galti: log -8 ko 8 karke beech mein hi root le lete hain. Aisa kuch nahi karna - square laga do, sign khud khatam ho jayega. $6^2 + (-8)^2 = 36 + 64 = 100$, root 10. Midpoint = average of coordinates = (1, −1). Aur (1, −1) ka origin se distance $\sqrt{2}$. Chhota point: $\sqrt{2}$ ko decimal mein mat likho jab tak manga na ho.*

QUESTION 33 | Moderate | Collinearity test (T9)

Q33. Without plotting, check whether M(0, 0), A(3, 4) and G(6, 8) lie on the same straight line. State the reasoning principle you used.

SOLUTION — step by step

Step 1. Given: three points. Needed: collinearity verdict without plotting.

Step 2. Compute OM = $\sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$.

Step 3. Compute AG = $\sqrt{((6 - 3)^2 + (8 - 4)^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.

Step 4. Compute MG = $\sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$.

Step 5. Test: the largest distance (10) equals the sum of the other two (5 + 5 = 10).

Step 6. Since MG = OM + AG exactly, the point A sits on the straight path from M to G; the three points are collinear.

Step 7. Principle used: on a straight line, the whole equals the sum of the parts (equality case of the triangle inequality). Any detour would make the total strictly larger.

Answer: Yes - OM + AG = MG (5 + 5 = 10), so M, A, G are collinear.

Hinglish: *Ye 'triangle inequality equality' ka test hai - bina graph banaye line check karne ka tareeka. Teen distances nikaalo: 5, 5, 10. Agar sabse bada = baaki dono ka sum, toh teeno ek hi line par hain. Agar sum bada aaye (strictly), toh beech ka point line se hat ke baitha hai. Yahan 5 + 5 = 10 exact match - collinear. Ek baat yaad rakho: values EXACT rakho, round mat karo, warna test galat ho sakta hai.*

QUESTION 34 | Moderate | Reflection chain (T7)

Q34. The point P(4, -3) is reflected in the x-axis to get P_1, and then P_1 is reflected in the y-axis to get P_2. Write the coordinates of P_1 and P_2, and find the length PP_2.

SOLUTION — step by step

Step 1. Given: P(4, -3) and a chain of two reflections. Needed: images and PP_2.

Step 2. Reflection in the x-axis keeps x and flips the sign of y: $P_1 = (4, 3)$.

Step 3. Reflection in the y-axis keeps y and flips the sign of x: $P_2 = (-4, 3)$.

Step 4. PP_2 joins (4, -3) to (-4, 3): x-difference = -4 - 4 = -8, y-difference = 3 - (-3) = 6.

Step 5. $PP_2 = \sqrt{((-8)^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$.

Step 6. Check with preserved distances: $PP_1 = |3 - (-3)| = 6$ (vertical segment), $P_1P_2 = |(-4) - 4| = 8$ (horizontal segment), and indeed $6^2 + 8^2 = 10^2$, so the numbers are internally consistent.

Answer: $P_1 = (4, 3)$, $P_2 = (-4, 3)$, $PP_2 = 10$ units.

Hinglish: *Reflection ke do simple rules: x-axis mein ungli x-axis par - sirf y ka sign palto. y-axis mein - sirf x ka sign palto. Chain mein dono lagane se (4, -3) pehle (4, 3) bana, phir (-4, 3). Ab distance: 8 aur 6 ka 8-6-10 triple! Aur verification sundar hai: $PP_1 = 6$ vertical, $P_1P_2 = 8$ horizontal, inka Pythagoras bhi 10 hi deta hai. Double confirmation mil gaya.*

QUESTION 35 | Moderate | Triangle classification (T6)

Q35. The vertices of a triangle are A(1, 2), B(5, 2) and C(3, 6). Find all three side lengths and classify the triangle. Is it equilateral? Is it right-angled?

SOLUTION — step by step

Step 1. Given: three vertices. Needed: side lengths and classification.

Step 2. AB: same y-coordinate (2), so AB = |5 - 1| = 4.

Step 3. AC = $\sqrt{((3 - 1)^2 + (6 - 2)^2)} = \sqrt{(4 + 16)} = \sqrt{20} = 2\sqrt{5}$.

Step 4. BC = $\sqrt{((3 - 5)^2 + (6 - 2)^2)} = \sqrt{(4 + 16)} = \sqrt{20} = 2\sqrt{5}$.

Step 5. Two sides equal (AC = BC), one different: the triangle is isosceles, not equilateral.

Step 6. Right-angle check by the converse of Pythagoras: is the square of the longest side equal to the sum of squares of the other two? Longest is $2\sqrt{5}$, its square 20. Others: $4^2 = 16$ and $(2\sqrt{5})^2 = 20$, and 16 + 20 = 36 which is not 20.

Step 7. Not right-angled. In fact $4 + 2\sqrt{5} + 2\sqrt{5}$ is the perimeter $= 4 + 4\sqrt{5}$.

Answer: AB = 4, AC = BC = $2\sqrt{5}$; the triangle is isosceles; it is neither equilateral nor right-angled.

Hinglish: *Side nikaalne ke baad classification ka recipe: (1) do barabar? isosceles. (2) teen barabar? equilateral. (3) Pythagoras satisfy? right-angled. Yahan AC aur BC dono $2\sqrt{5}$ nikle - isosceles pakka. Right angle ka test: sabse lambi side ka square = baaki dono ke squares ka sum? 20 vs 36 - fail. Toh na equilateral, na right. Simple hai, bas teeno tests order mein lagao.*

QUESTION 36 | Moderate | Square verification (T6)

Q36. Show that the points (2, 2), (6, 2), (6, 6) and (2, 6) are the corners of a square. Then find its area and the length of its diagonal.

SOLUTION — step by step

Step 1. Given: four points in cyclic order. Needed: prove square, then area and diagonal.

Step 2. Bottom side (2, 2) to (6, 2): horizontal, length 4.

Step 3. Right side (6, 2) to (6, 6): vertical, length 4.

Step 4. Top side (6, 6) to (2, 6): horizontal, length 4. Left side (2, 6) to (2, 2): vertical, length 4.

Step 5. All four sides equal to 4, and adjacent sides meet at the corners of an axis-aligned rectangle (horizontal meets vertical), so every corner angle is 90 degrees.

Step 6. Hence the figure is a square of side 4. Area $= \text{side}^2 = 16$ square units.

Step 7. Diagonal d joins (2, 2) to (6, 6): $d = \sqrt{((6 - 2)^2 + (6 - 2)^2)} = \sqrt{32} = 4\sqrt{2}$. This matches the square-diagonal relation $d = \text{side} \times \sqrt{2}$.

Answer: It is a square of side 4; area = 16 sq units; diagonal = $4\sqrt{2}$ units.

Hinglish: *Square prove karne ka two-step formula: (1) chaaron sides barabar, (2) koi bhi ek corner ka angle 90 (yahan horizontal aur vertical lines mil rahe hain toh automatic 90). Side 4 hai, toh area 16 aur diagonal side x $\sqrt{2}$ = $4\sqrt{2}$. Distance formula se bhi verify kiya - $\sqrt{32} = 4\sqrt{2}$, dono match. Aisi cross-checking exam mein full marks dilati hai.*

QUESTION 37 | Moderate | Right-angle by converse Pythagoras (T6)

Q37. A(1, 1), B(4, 1) and C(4, 5) are three points. Use the distance formula and the converse of the Baudhayana-Pythagoras Theorem to show that angle B is a right angle, and find the area of triangle ABC.

SOLUTION — step by step

Step 1. Given: three vertices. Needed: prove right angle at B, then area.

Step 2. AB: same y = 1, so $AB = |4 - 1| = 3$.

Step 3. BC: same x = 4, so BC = |5 - 1| = 4.

Step 4. AC = $\sqrt{((4 - 1)^2 + (5 - 1)^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.

Step 5. Check the converse: $AC^2 = 25$ and $AB^2 + BC^2 = 9 + 16 = 25$. They are equal, so the angle opposite AC - which is angle B - is a right angle.

Step 6. Area of a right triangle = (1/2) x leg x leg = (1/2) x 3 x 4 = 6 square units.

Step 7. The hidden triple is again 3-4-5.

Answer: Angle B = 90 degrees (since $3^2 + 4^2 = 5^2 = AC^2$); area = 6 square units.

Hinglish: *Converse Pythagoras ka use yaad karo: agar $c^2 = a^2 + b^2$, toh c ke saamne wala angle 90 hai. Yahan AC sabse lambi hai (5), aur $3^2 + 4^2 = 5^2$ nikla, toh B par right angle. Area ke liye right triangle ka bonus: do legs ko base-height maan lo, (1/2)(3)(4) = 6. Coordinate geometry mein area aise hi nikalte hain jab triangle axis-aligned legs rakhta hai.*

QUESTION 38 | Moderate | Missing value, two roots (T5)

Q38. The distance between the points (k, 3) and (4, 3) is 7. Find all possible values of k. How many points satisfy the condition?

SOLUTION — step by step

Step 1. Given: two points sharing y = 3, distance 7. Needed: k.

Step 2. Both points have y-coordinate 3, so the segment is horizontal and distance = |x2 - x1|.

Step 3. So |4 - k| = 7. An absolute-value equation splits into two cases.

Step 4. Case 1: 4 - k = 7, giving k = -3. Case 2: 4 - k = -7, giving k = 11.

Step 5. Verification: from (4, 3) to (-3, 3): |4 - (-3)| = 7. From (4, 3) to (11, 3): |11 - 4| = 7. Both work.

Step 6. So there are TWO points, (-3, 3) and (11, 3), one on each side of (4, 3).

» Why this step? The absolute value MUST open into two cases. Dropping one case is the single most common error in 'find k' questions.

Answer: k = -3 or k = 11; there are two such points.

Hinglish: *Yahi wo question hai jahan aadhe marks ud jaate hain! |4 - k| = 7 ke DO solutions hote hain: point (4, 3) ke left mein bhi ek aur right mein bhi ek. k = -3 aur k = 11. Agar tumne sirf ek nikala, toh samajh lo absolute value ke do case khulane bhool gaye. Verification bhi kar li - dono se distance 7 hi aati hai.*

QUESTION 39 | Moderate | Circle membership (T10)

Q39. A circle has centre the origin and radius 5. Classify each point as on, inside, or outside the circle: (3, 4), (4, 2), (-3, 4), (0, 5), (-4, -4).

SOLUTION — step by step

Step 1. Given: circle $x^2 + y^2 = 25$. Needed: classify five points.

Step 2. Method: compare $x^2 + y^2$ with $r^2 = 25$. Equal means on the circle; smaller means inside; larger means outside.

Step 3. (3, 4): 9 + 16 = 25, exactly equal, so ON the circle.

Step 4. (4, 2): 16 + 4 = 20, which is less than 25, so INSIDE (distance $\sqrt{20}$ slightly under 5).

Step 5. (-3, 4): 9 + 16 = 25, so ON the circle (signs never matter after squaring).

Step 6. (0, 5): 0 + 25 = 25, so ON the circle.

Step 7. (-4, -4): 16 + 16 = 32, greater than 25, so OUTSIDE (distance $\sqrt{32} = 4\sqrt{2}$ approx 5.66).

» **Why this step?** Comparing squares instead of square roots avoids every rounding issue - 20 vs 25 is cleaner than 4.47 vs 5.

Answer: On the circle: (3, 4), (-3, 4), (0, 5). Inside: (4, 2). Outside: (-4, -4).

Hinglish: *Circle membership ka super-fast test: $x^2 + y^2$ ko r^2 se compare karo. Barabar = line par, kam = andar, zyada = bahar. Root nikalne ki zaroorat hi nahi. (-3, 4) dekho - minus hone ke bawajood square ke baad 25 hi aaya, toh circle par hi hai. (-4, -4) dono square 16+16 = 32 > 25, bahar. Ye technique time bhi bachati hai aur rounding errors se bhi bachati hai.*

QUESTION 40 | Moderate | Missing y-value on a circle (T5/T10)

Q40. The point (3, y) lies at distance 5 from the origin. Find all possible values of y. Where do the two possible points sit relative to each other?

SOLUTION — step by step

Step 1. Given: distance from O equals 5. Needed: y.

Step 2. Distance from origin: $\sqrt{(3^2 + y^2)} = 5$.

Step 3. Square both sides (both sides are non-negative, so squaring is safe): $9 + y^2 = 25$.

Step 4. So $y^2 = 16$, and taking square roots gives y = 4 or y = -4.

Step 5. Verification: (3, 4) gives $\sqrt{(9 + 16)} = 5$, and (3, -4) also gives 5. Both are genuine solutions.

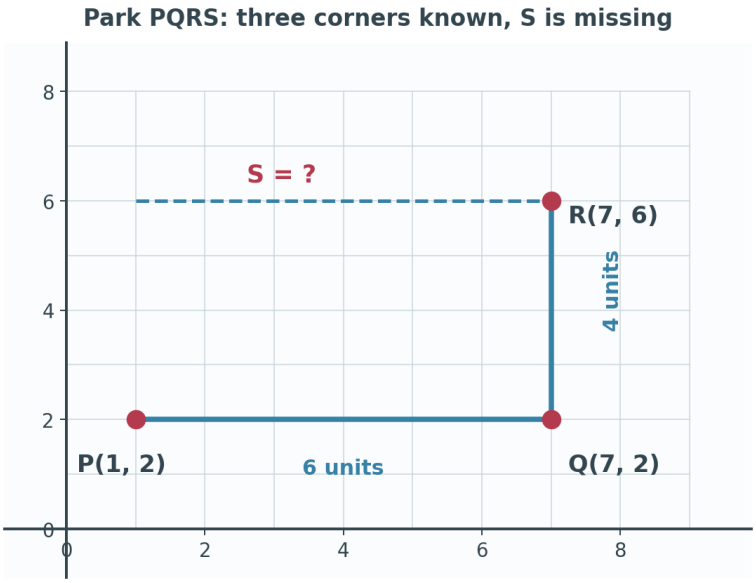
Step 6. The two points (3, 4) and (3, -4) are reflections of each other in the x-axis - same x, opposite y - and both lie on the circle of radius 5.

Answer: y = 4 or y = -4; the two points are mirror images in the x-axis, both on the circle of radius 5.

Hinglish: *Yahan bhi do answers aayenge, aur wajah samjho: circle ke ek vertical line par do points hote hain - ek upar, ek neeche. y = 4 aur y = -4. Dono verify kiye, dono sahi. Aur ye dono points x-axis ke reflection partners hain. Ye pattern (quadratic se do roots) Chapter 1 ka signature hai - jab bhi circle ya distance se equation bane, do answers ki ummeed rakho.*

QUESTION 41 | Moderate | Figure: missing vertex + metrics (T6)

Q41. Park PQRS has three corners marked on the map: P(1, 2), Q(7, 2), R(7, 6). (i) Write the coordinates of the missing corner S if PQRS is a rectangle. (ii) Find the side lengths. (iii) Find the perimeter and area of the park. (iv) How long is the diagonal PR?



SOLUTION — step by step

Step 1. (i) PQ is horizontal (y = 2) and QR is vertical (x = 7). The fourth corner must complete the right angles: S = (1, 6).

Step 2. (ii) PQ = |7 - 1| = 6 units; QR = |6 - 2| = 4 units.

Step 3. (iii) Perimeter = 2(6 + 4) = 20 units; area = 6 × 4 = 24 square units.

Step 4. (iv) PR = $\sqrt{((7-1)^2 + (6-2)^2)} = \sqrt{(36 + 16)} = \sqrt{52} = 2\sqrt{13} \approx 7.21$ units.

» **Why this step?** In an axis-aligned rectangle the missing vertex is just 'copy the x of the left side, copy the y of the top side'.

Answer: (i) S(1, 6) (ii) 6 and 4 (iii) P = 20, A = 24 (iv) $2\sqrt{13} \approx 7.21$

Hinglish: *Rectangle ka chautha corner = do known corners ka x-y mix. Area perimeter toh primary school, diagonal par formula lagta hai - √52 ko 2√13 karke simplify karna classy hai.*

QUESTION 42 | Moderate | Missing value, two roots (T5)

Q42. The distance between the points (k, 2) and (3, 6) is 5. Find all possible values of k.

SOLUTION — step by step

Step 1. Distance formula: $\sqrt{((k - 3)^2 + (2 - 6)^2)} = 5$.

Step 2. Square both sides: $(k - 3)^2 + 16 = 25$.

Step 3. $(k - 3)^2 = 9 \rightarrow k - 3 = \pm 3$.

Step 4. k = 6 or k = 0. Both give a point exactly 5 away - one on each side.

» **Why this step?** Squaring introduces TWO roots because a circle of radius 5 cuts the line y = 2 at two points - both are valid answers.

Answer: k = 0 or k = 6

Hinglish: *Distance fix hai toh point ek circle par baitha hoga, aur circle line ko do baar kaat raha hai - isliye do k. ± wala sign bhoolna hi yahan ki sabse badi galti hai.*

QUESTION 43 | Moderate | Right-angle check + area (T6)

Q43. A(2, 1), B(6, 1) and C(6, 4) are three points. (i) Show that angle B is a right angle. (ii) Find the area of triangle ABC.

SOLUTION — step by step

Step 1. AB: same y -> AB = 4. BC: same x -> BC = 3. These two streets are horizontal vs vertical -> perpendicular at B.

Step 2. By converse Pythagoras check: AC should be $\sqrt{(4^2 + 3^2)} = 5$. Indeed AC = $\sqrt{((6-2)^2 + (4-1)^2)} = \sqrt{25} = 5$, and $3^2 + 4^2 = 5^2$.

Step 3. (ii) Area = ½ × base × height = ½ × 4 × 3 = 6 square units.

» **Why this step?** One horizontal and one vertical side meeting at a point IS the right angle - then the 3-4-5 triple certifies it numerically.

Answer: (i) Right angle at B ($3^2 + 4^2 = 5^2$) (ii) 6 square units

Hinglish: *Ek side horizontal, ek vertical, B par mili -> right angle pakka. Area ke liye base-height le lo, 3-4-5 triple ne hypotenuse bhi gift kar diya.*

QUESTION 44 | Moderate | Isosceles verification + area (T6)

Q44. Show that P(1, 3), Q(4, 7), R(7, 3) form an isosceles triangle, and find its area.

SOLUTION — step by step

Step 1. PQ = $\sqrt{((4-1)^2 + (7-3)^2)} = \sqrt{(9 + 16)} = 5$.

Step 2. QR = $\sqrt{((7-4)^2 + (3-7)^2)} = \sqrt{(9 + 16)} = 5$. Two equal sides -> isosceles.

Step 3. PR: same y = 3 -> PR = 6 (the base).

Step 4. Height from Q to line y = 3 is |7 - 3| = 4.

Step 5. Area = $\frac{1}{2} \times 6 \times 4 = 12$ square units.

» **Why this step?** When the equal sides' ends share a y-value, the base is horizontal and the height is a plain y-difference - no perpendicular-foot drama.

Answer: PQ = QR = 5, so isosceles; area = 12

Hinglish: *Do side 5-5 nikli, base seedhi, height seedhe 4. Area 12. Isosceles ho toh aisa hi setup dhoondo jahan base axis-parallel ho - kaam aadha ho jaata hai.*

QUESTION 45 | Moderate | Parallelogram via diagonal midpoints (T8)

Q45. Show that A(-2, 1), B(3, 2), C(5, 6), D(0, 5) are the vertices of a parallelogram.

SOLUTION — step by step

Step 1. Diagonals of a parallelogram bisect each other - so compare the midpoints of AC and BD.

Step 2. Midpoint AC = ((-2 + 5)/2, (1 + 6)/2) = (1.5, 3.5).

Step 3. Midpoint BD = ((3 + 0)/2, (2 + 5)/2) = (1.5, 3.5). Equal!

Step 4. Since both diagonals share the same midpoint, ABCD is a parallelogram.

» **Why this step?** The diagonal-midpoint test is the cheapest parallelogram proof in coordinates: two averages instead of four distance calculations.

Answer: Both diagonals have midpoint (1.5, 3.5), so ABCD is a parallelogram

Hinglish: *Chaar sides check karna mehanga hai; diagonals ke midpoints barabar? Bas, parallelogram done. Ye shortcut exam ka asli time-saver hai.*

QUESTION 46 | Moderate | Reflection composition (T7)

Q46. P is (3, 7). It is reflected in the x-axis and the image is then reflected in the y-axis. Find the final position. Which single reflection would send P straight to that same final point?

SOLUTION — step by step

Step 1. Step 1, reflect in x-axis: (3, 7) -> (3, -7).

Step 2. Step 2, reflect in y-axis: (3, -7) -> (-3, -7).

Step 3. Final image: (-3, -7).

Step 4. A single reflection in BOTH axes (i.e. reflection in the origin / half-turn) also gives (-3, -7): both signs flipped.

» **Why this step?** Two perpendicular mirror reflections compose to a half-turn: x flips once and y flips once, exactly like the double-axis reflection.

Answer: Final (-3, -7); a single reflection in both axes (equivalently a half-turn about O) gives the same result

Hinglish: *Do mirror mein reflect karne se point ek half-turn kha gaya - dono signs ulte. Exam mein poochha jaye 'single equivalent reflection' toh turant bolna: origin ke around reflection.*

QUESTION 47 | Moderate | Circle membership (T10)

Q47. A circle has centre the origin and radius 13. For each point, decide: on the circle, inside, or outside? (5, 12), (6, 11), (-12, 5), (9, 9).

SOLUTION — step by step

Step 1. Rule: compare $x^2 + y^2$ with $r^2 = 169$.

Step 2. (5, 12): 25 + 144 = 169 -> exactly ON the circle.

Step 3. (6, 11): 36 + 121 = 157 < 169 -> INSIDE (distance $\sqrt{157} \approx 12.53$).

Step 4. (-12, 5): 144 + 25 = 169 -> ON the circle (signs never matter here).

Step 5. (9, 9): 81 + 81 = 162 < 169 -> INSIDE.

» **Why this step?** Squaring kills the signs, so (-12, 5) behaves exactly like (12, 5) - the circle cannot tell left from right.

Answer: (5, 12) on; (6, 11) inside; (-12, 5) on; (9, 9) inside

Hinglish: *Circle test = $x^2 + y^2$ vs r^2. Barabar -> par, kam -> andar, zyada -> bahar. Negative coordinates bhi square hote hi seedhe ho jaate hain.*

QUESTION 48 | Moderate | Reverse midpoint (T8)

Q48. M(-1, 4) is the midpoint of segment AB. If A is (2, -3), find B.

SOLUTION — step by step

Step 1. Midpoint rule reversed: B = 2M - A (twice the midpoint minus the known end).

Step 2. x: 2(-1) - 2 = -4. y: 2(4) - (-3) = 8 + 3 = 11.

Step 3. B = (-4, 11).

Step 4. Check: midpoint of (2, -3) and (-4, 11) = ((2 - 4)/2, (-3 + 11)/2) = (-1, 4) ✓.

» **Why this step?** The midpoint is the average of the ends, so each end is 'twice the average minus the other end' - the same balance-scale logic.

Answer: B(-4, 11)

Hinglish: *Midpoint given, ek end given -> doosra end = 2M - A. Formula ulta chalana aana chahiye, aur hamesha check wali line likho - verification free mein marks dilati hai.*

QUESTION 49 | Moderate | City-grid application (T11)

Q49. In a city the streets are two perpendicular families of lines. A hospital is at H(2, 7) and a school at S(2, 1) on the same vertical street. A house is at W(6, 1). (i) How far does an ambulance travel from H to S along the streets? (ii) From W to S? (iii) A drone flies straight from H to W - how far (in simplest surd form)?

SOLUTION — step by step

Step 1. (i) H and S share x = 2: vertical street, HS = |7 - 1| = 6 units.

Step 2. (ii) W and S share y = 1: horizontal street, WS = |6 - 2| = 4 units.

Step 3. (iii) Straight HW = $\sqrt{((6-2)^2 + (1-7)^2)} = \sqrt{(16 + 36)} = \sqrt{52} = 2\sqrt{13}$ units.

Step 4. Road trip H-S-W = 6 + 4 = 10 units, but the drone needs only $2\sqrt{13} \approx 7.21$ units.

» **Why this step?** Street travel = Manhattan distance (add the two legs); flying = Euclidean distance (hypotenuse). The right angle at S creates the gap.

Answer: (i) 6 (ii) 4 (iii) $2\sqrt{13} \approx 7.21$ (vs 10 by road)

Hinglish: *Sadak par chalna hai toh plus karo (6 + 4 = 10), hawa mein udna hai toh hypotenuse nikaalo ($2\sqrt{13}$). Ek hi figure, do alag 'distance' - exam iska bada maze leta hai.*

QUESTION 50 | Moderate | Right triangle at the origin (T6/T14)

Q50. A triangle has vertices O(0, 0), A(10, 0) and B(0, 8). Find (i) its area, (ii) the length of AB in simplest surd form, (iii) its perimeter.

SOLUTION — step by step

Step 1. OA lies along the x-axis (length 10) and OB along the y-axis (length 8) - right angle at O.

Step 2. (i) Area = $\frac{1}{2} \times 10 \times 8 = 40$ square units.

Step 3. (ii) AB = $\sqrt{(10^2 + 8^2)} = \sqrt{164} = 2\sqrt{41}$.

Step 4. (iii) Perimeter = $10 + 8 + 2\sqrt{41} = 18 + 2\sqrt{41}$ units.

» **Why this step?** When a triangle hugs both axes at O, area is instant; simplify $\sqrt{164}$ by pulling out the square factor 4.

Answer: (i) 40 (ii) $2\sqrt{41}$ (iii) $18 + 2\sqrt{41}$

Hinglish: *Origin par right angle bana toh area ½ × legs. √164 = √(4 × 41) = 2√41 - square factor bahar nikalna hi 'simplest surd form' hai.*

QUESTION 51 | Moderate | Trisection points (T8)

Q51. A(0, 0) and B(9, 12) are joined. P and Q trisect AB (AP = PQ = QB). Find P and Q, and verify each part really is one third of AB.

SOLUTION — step by step

Step 1. AB = (9, 12) as a step-vector; one third = (3, 4).

Step 2. P = A + (1/3)AB = (3, 4); Q = A + (2/3)AB = (6, 8).

Step 3. AP = $\sqrt{(3^2 + 4^2)} = 5$; PQ = $\sqrt{((6-3)^2 + (8-4)^2)} = 5$; QB = $\sqrt{((9-6)^2 + (12-8)^2)} = 5$.

Step 4. AB = $\sqrt{(81 + 144)} = \sqrt{225} = 15 = 5 + 5 + 5$ ✓ - each part is exactly one third.

» **Why this step?** Trisection is just the midpoint idea extended: walk 1/3 and 2/3 of the step-vector (x-step, y-step).

Answer: P(3, 4), Q(6, 8); each part = 5 = AB/3

Hinglish: *Trisection = AB ke vector ko 3 hisson mein baanto. (9, 12) ka teesra hissa (3, 4) - teen jagah jodo, teeno tukde 5-5 ke mile. 3-4-5 triple phir se aa gaya!*

QUESTION 52 | Moderate | Missing value with surd answer (T5)

Q52. The point (3, y) lies at distance √74 from (6, 2). Find all possible values of y.

SOLUTION — step by step

Step 1. $\sqrt{((3-6)^2 + (y - 2)^2)} = \sqrt{74}$.

Step 2. Square: $9 + (y - 2)^2 = 74$ -> $(y - 2)^2 = 65$.

Step 3. $y - 2 = \pm\sqrt{65}$ -> $y = 2 + \sqrt{65}$ or $y = 2 - \sqrt{65}$.

» **Why this step?** Two roots appear because the vertical line x = 3 cuts the circle of radius √74 twice - once above, once below.

Answer: $y = 2 + \sqrt{65}$ or $y = 2 - \sqrt{65}$

Hinglish: *$(y-2)^2 = 65$ aaya toh $y = 2 \pm \sqrt{65}$. Yahan perfect square nahi bana, toh surd hi answer hai - ghabrana nahi, maths ka style hai.*

QUESTION 53 | Moderate | Rectangle verification kit (T6)

Q53. Show that (1, 2), (4, 2), (4, 6), (1, 6) form a rectangle (not just any parallelogram).

SOLUTION — step by step

Step 1. Sides: bottom (1, 2) to (4, 2) = 3 (horizontal); right (4, 2) to (4, 6) = 4 (vertical); top = 3; left = 4 - opposite sides equal.

Step 2. Adjacent sides are horizontal vs vertical -> each corner angle is 90°.

Step 3. Diagonals: (1, 2) to (4, 6): √(9 + 16) = 5; (4, 2) to (1, 6): √(9 + 16) = 5 - equal too.

Step 4. Opposite sides equal + all angles 90° (equivalently equal diagonals) -> rectangle, and the 3-4-5 triple certifies it.

» **Why this step?** A parallelogram needs only bisecting diagonals; a rectangle needs equal diagonals too - here 5 = 5 seals it.

Answer: Sides 3 and 4 with 90° corners; both diagonals = 5; hence a rectangle

Hinglish: *Parallelogram se rectangle ka upgrade = diagonals equal. 3-4-5 triple ne dono diagonals 5-5 diye. Bas, certificate mil gaya.*

QUESTION 54 | Moderate | Algebraic distance (T4/T14)

Q54. Find the distance between (a + b, a - b) and (a - b, a + b).

SOLUTION — step by step

Step 1. x-difference: (a + b) - (a - b) = 2b.

Step 2. y-difference: (a - b) - (a + b) = -2b... careful - the second point's y is a + b, so y-difference = (a - b) - (a + b) = -2b? No: we need the OTHER coordinate: first point y = a - b, second point y = a + b, difference = (a + b) - (a - b) = 2b as well? Recompute cleanly: dx = 2b, dy = (a + b) - (a - b) = 2b? That would make distance 2b√2 - let me recheck: first point is (a+b, a-b): x = a + b, y = a - b. Second point is (a - b, a + b): x = a - b, y = a + b.

Step 3. dx = (a - b) - (a + b) = -2b -> |dx| = 2b. dy = (a + b) - (a - b) = 2b.

Step 4. Distance = $\sqrt{((2b)^2 + (2b)^2)} = \sqrt{(8b^2)} = 2\sqrt{2} \cdot |b| = 2\sqrt{2}\, b$ (for $b > 0$).

» **Why this step?** The two points are mirror images in the line y = x, so the step is symmetric: |dx| = |dy| - that instantly builds a 1-1-√2 shape scaled by 2b.

Answer: 2√2·b (with |b| if b may be negative)

Hinglish: *Swap wale points ka distance hamesha √2 ke family ka hota hai: dx = dy = 2b, toh answer 2√2b. Brackets kholte waqt sign se dosti rakho!*

QUESTION 55 | Moderate | Equidistant locus intro (T5/T14)

Q55. A(1, 2) and B(7, 8) are two fixed points. (i) Write the midpoint of AB. (ii) Find the point on the x-axis equidistant from A and B. (iii) Describe the full set of points equidistant from A and B.

SOLUTION — step by step

Step 1. (i) Midpoint = ((1+7)/2, (2+8)/2) = (4, 5).

Step 2. (ii) Point (x, 0): $(x-1)^2 + 4 = (x-7)^2 + 64$.

Step 3. Expand: $x^2 - 2x + 5 = x^2 - 14x + 113$ -> 12x = 108 -> x = 9. Point (9, 0).

Step 4. (iii) The full set is the perpendicular bisector of AB - the straight line through (4, 5) and (9, 0); every point on it is equidistant from A and B.

» **Why this step?** 'Equidistant from two points' NEVER has a unique answer until you also fix a line - the true answer is always a whole line, the perpendicular bisector.

Answer: (i) (4, 5) (ii) (9, 0) (iii) the perpendicular bisector of AB

Hinglish: *Do points se barabar doori = poori ek line (perpendicular bisector). Exam walon ne x-axis fix kiya tha, isliye ek point (9, 0) aaya. Concept line ka hai, point ka nahi!*

QUESTION 56 | Moderate | Collinearity test (T9)

Q56. Check without plotting whether P(2, 5), Q(5, 9), R(8, 13) are collinear.

SOLUTION — step by step

Step 1. PQ = $\sqrt{((5-2)^2 + (9-5)^2)} = \sqrt{(9 + 16)} = 5$.

Step 2. QR = $\sqrt{((8-5)^2 + (13-9)^2)} = \sqrt{(9 + 16)} = 5$.

Step 3. PR = $\sqrt{((8-2)^2 + (13-5)^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$.

Step 4. PQ + QR = 5 + 5 = 10 = PR -> Q lies exactly between P and R -> collinear.

» **Why this step?** The add-up test (shortest two sides sum to the third) proves collinearity AND shows Q is the midpoint for free.

Answer: Yes - PQ + QR = PR = 10, so they are collinear (Q is the midpoint)

Hinglish: *PQ + QR = PR hua? Toh teeno ek hi line par. Aur kyunki 5 = 5, Q toh seedha midpoint nikla - double win.*

QUESTION 57 | Moderate | Ladder application (T4)

Q57. A ladder stands on horizontal ground with its foot at (1.4, 0) and top touching a vertical wall (the y-axis) at (0, 4.8), all in metres. How long is the ladder?

SOLUTION — step by step

Step 1. Ladder = segment from (1.4, 0) to (0, 4.8).

Step 2. Length = $\sqrt{((1.4 - 0)^2 + (0 - 4.8)^2)} = \sqrt{(1.96 + 23.04)} = \sqrt{25} = 5$ m.

Step 3. Hidden triple: 1.4-4.8-5 is the 3-4-5 triple scaled by 0.98? Check: 3 × 1.4 = 4.2 no... it is 1.4 : 4.8 = 3.5 : 12 -> scale 0.4 of 3.5-12-12.5; simpler: decimals squared cleanly to 25.

» **Why this step?** Ugly-looking decimals can still produce a perfect square - always finish the arithmetic before assuming a surd.

Answer: 5 m

Hinglish: *Darne ki zaroorat nahi - $1.4^2 + 4.8^2 = 25$ exact! Decimal aaye toh square karke dekho, aksar triple chhupa hota hai (yahan 5 meter ka perfect ladder).*

QUESTION 58 | Moderate | Two points on an axis at fixed distance (T5)

Q58. Find both points on the x-axis that are exactly 5 units from (3, 4).

SOLUTION — step by step

Step 1. Point (x, 0): $\sqrt{((x-3)^2 + (0-4)^2)} = 5$.

Step 2. $(x - 3)^2 + 16 = 25 \rightarrow (x - 3)^2 = 9$.

Step 3. $x - 3 = \pm 3 \rightarrow x = 6$ or $x = 0$. Points: (6, 0) and (0, 0).

» **Why this step?** A circle of radius 5 around (3, 4) crosses the x-axis twice - the two crossings are symmetric about x = 3.

Answer: (0, 0) and (6, 0)

Hinglish: *Circle x-axis ko do baar kaat raha hai: x = 6 aur x = 0. Dono answer hai - 'ek hi' likhne wale aadhe mark kho dete hain.*

QUESTION 59 | Moderate | Between-ness distance property (T9)

Q59. A(-3, 2), B(9, 2) and P(4, 2) lie on one horizontal line. Find PA, PB and AB. What general truth do these numbers demonstrate?

SOLUTION — step by step

Step 1. PA = |4 - (-3)| = 7; PB = |9 - 4| = 5; AB = |9 - (-3)| = 12.

Step 2. PA + PB = 7 + 5 = 12 = AB.

Step 3. General truth: if P lies BETWEEN A and B on the same straight line, then PA + PB = AB exactly.

Step 4. Had P been outside the segment, the sum would EXCEED AB - that difference is the collinearity position test.

» **Why this step?** Segment addition works only for interior points - this is the foundation of the collinearity-by-distance test.

Answer: PA = 7, PB = 5, AB = 12; PA + PB = AB because P is between A and B

Hinglish: *Beech mein pada point toh do tukdon ka jod poora segement deta hai. Agar jod bada aa jaye, samajh lo point bahar khada hai - ye chhota sa test bahut kaam aata hai.*

QUESTION 60 | Moderate | Square from its diagonal (T10)

Q60. Two opposite corners of a square are P(-2, 1) and Q(4, 5). Find the diagonal length, the side length, and the area of the square.

SOLUTION — step by step

Step 1. Diagonal PQ = $\sqrt{((4+2)^2 + (5-1)^2)} = \sqrt{(36 + 16)} = \sqrt{52} = 2\sqrt{13}$.

Step 2. Side = diagonal/$\sqrt{2}$ = $2\sqrt{13}/\sqrt{2} = \sqrt{26}$.

Step 3. Area = side^2 = 26 square units. (Shortcut: area = $d^2/2 = 52/2 = 26$.)

» **Why this step?** For a square, area = $d^2/2$ - you can skip the side entirely and square the diagonal once.

Answer: Diagonal $2\sqrt{13}$; side $\sqrt{26}$; area 26

Hinglish: *Diagonal se area ka seedha shortcut: $d^2/2$. $\sqrt{52}$ ko $2\sqrt{13}$ likhna, side $\sqrt{26}$, area 26 - teen answers, ek hi calculation se.*

QUESTION 61 | Moderate | Perimeter + area of isosceles (T6)

Q61. A triangle has vertices (0, 0), (4, 3) and (8, 0). Find its perimeter and area, and name its type.

SOLUTION — step by step

Step 1. First side: 4-3-5 triple -> 5. Third side: (8, 0) to (0, 0) = 8. Middle side: (4, 3) to (8, 0) = $\sqrt{(16 + 9)}$ = 5.

Step 2. Perimeter = 5 + 5 + 8 = 18 units. Two equal sides -> isosceles.

Step 3. Base = 8 (along the x-axis); height = y of the apex = 3.

Step 4. Area = $\frac{1}{2} \times 8 \times 3 = 12$ square units.

» **Why this step?** Both slanted sides hit the 3-4-5 triple - one calculation, used twice.

Answer: Perimeter 18; area 12; isosceles

Hinglish: *Dono tirchi side 5-5 nikli - same triple, opposite direction. Base axis par hai toh height = apex ka y. Isosceles + area, sab 30 second mein.*

QUESTION 62 | Moderate | Reflecting a whole figure (T7)

Q62. Triangle T has vertices A(1, 1), B(4, 1), C(4, 3). It is reflected in the y-axis. (i) Write the image vertices. (ii) Has the area changed? (iii) Find the area of the image triangle.

SOLUTION — step by step

Step 1. (i) Reflecting in the y-axis flips x: A'(-1, 1), B'(-4, 1), C'(-4, 3).

Step 2. (ii) NO - reflection preserves all lengths and angles, so the area is unchanged.

Step 3. (iii) Legs: A'B' = 3 (horizontal), B'C' = 2 (vertical), right angle at B'. Area = ½ × 3 × 2 = 3 square units (same as the original).

» **Why this step?** Reflection is an isometry: it moves the figure but cannot stretch or shrink a single side.

Answer: (i) A'(-1, 1), B'(-4, 1), C'(-4, 3) (ii) No (iii) 3

Hinglish: *Mirror image ka area kabhi nahi badalta - shape ka scale nahi hota, sirf address badalta hai. Area 3 hi rahega.*

QUESTION 63 | Moderate | Road vs direct distance (T11)

Q63. A railway station is at O(0, 0). Your friend is at F(6, 8) and you are at Y(6, 0), with all roads running along the grid. (i) How far is your friend from the station (direct)? (ii) How far must YOU walk along roads to reach the station? (iii) How many units do you save if a new diagonal footpath connects Y to O... wait, you are already ON the x-axis! So instead: how much longer is your road trip to the friend (Y to F along roads) than the direct distance?

SOLUTION — step by step

Step 1. (i) OF = $\sqrt{(6^2 + 8^2)}$ = $\sqrt{100}$ = 10 units (6-8-10 triple).

Step 2. (ii) You are at (6, 0) - already on the x-axis: road distance to O = |6 - 0| = 6 units.

Step 3. (iii) Y to F along roads: 6 up + 0 across = 8? Y(6, 0) to F(6, 8): vertical road only = 8 units. Direct YF = 8 too (same x!).

Step 4. So the interesting comparison: road trip F to O = 6 + 8 = 14 vs direct 10 -> the friend saves 4 units by flying; your own trip (6) cannot be shortened.

» **Why this step?** Same-x points have NO saving possible - the road and the direct line coincide. Savings appear only when BOTH coordinates differ.

Answer: (i) 10 (ii) 6 (iii) friend's road trip 14 vs direct 10: saves 4; yours is already straight

Hinglish: *Trick yeh thi: tum (6, 0) par ho, tumhara route seedha hai. Friend ko 14 ki jagah 10 milta hai - 4 bachat. Same-x points mein shortcut ka chance hi nahi!*

QUESTION 64 | Moderate | Circle membership mixed (T10)

Q64. Circle: centre O, radius 10. Classify (6, 8), (7, 7), (10, 0), (6, 9) and also find how far (7, 7) is from the circle's edge.

SOLUTION — step by step

Step 1. Compare $x^2 + y^2$ with 100.

Step 2. (6, 8): 36 + 64 = 100 -> ON. (7, 7): 49 + 49 = 98 < 100 -> inside. (10, 0): 100 -> ON. (6, 9): 36 + 81 = 117 > 100 -> outside.

Step 3. (7, 7) distance from O = $\sqrt{98}$ = $7\sqrt{2}$ ≈ 9.90. Gap to the edge = 10 - 9.90 = 10 - $7\sqrt{2}$ units.

» **Why this step?** Distance from the edge = |r − distance from centre| - a two-step subtraction once the membership is known.

Answer: (6, 8) on; (7, 7) inside by 10 − 7√2; (10, 0) on; (6, 9) outside

Hinglish: *Membership ke baad edge ka gap = r minus centre distance. 7√2 ≈ 9.9, toh bas 0.1 ka gap andar. Chhota calculation, bada concept.*

QUESTION 65 | Moderate | Midpoint chain (T8)

Q65. A(1, 2) and B(7, 8) are joined; M is the midpoint of AB; N is the midpoint of AM. (i) Find M and N. (ii) What fraction of the way from A to B is N? (iii) Find AN.

SOLUTION — step by step

Step 1. (i) M = ((1+7)/2, (2+8)/2) = (4, 5); N = ((1+4)/2, (2+5)/2) = (2.5, 3.5).

Step 2. (ii) Halving twice = one quarter: N sits 1/4 of the way from A to B.

Step 3. (iii) AB = $\sqrt{((7-1)^2 + (8-2)^2)} = \sqrt{72} = 6\sqrt{2}$; AN = AB/4 = $6\sqrt{2}/4 = 3\sqrt{2}/2$.

» **Why this step?** Each halving halves the remaining distance: midpoint of a midpoint is at 25% of the whole segment.

Answer: (i) M(4, 5), N(2.5, 3.5) (ii) 1/4 (iii) $3\sqrt{2}/2$

Hinglish: *Midpoint ke bhi midpoint = quarter point. Distance bhi quarter: $6\sqrt{2}$ ka chautha hissa $3\sqrt{2}/2$. Chain problems mein fraction soch ke chalo.*

QUESTION 66 | Moderate | Spot the error (T4)

Q66. A student says: 'The distance between (3, -2) and (3, 5) is |5| + |-2| = 7, so in general distance = dx + dy.' Another computes the distance from (0, 0) to (3, 4) as 3 + 4 = 7. Find and correct both errors.

SOLUTION — step by step

Step 1. First case: (3, -2) and (3, 5) share x = 3, so the TRUE distance is |5 - (-2)| = 7 - the student got the right number by a wrong (non-general) method.

Step 2. Second case: (0, 0) to (3, 4) is NOT same-coordinate: true distance = $\sqrt{(3^2 + 4^2)} = 5$, not 7.

Step 3. Correct rule: same x -> distance = |dy|; same y -> distance = |dx|; different both -> $\sqrt{(dx^2 + dy^2)}$, NEVER dx + dy.

» **Why this step?** dx + dy is the 'street-walking' distance. It equals the true distance only when one of dx, dy is zero - that is exactly the same-coordinate case.

Answer: dx + dy works ONLY when one difference is 0; otherwise use $\sqrt{(dx^2 + dy^2)}$. The (3,4) case: true distance 5, not 7.

Hinglish: *Pehle bande ka number sahi nikla par logic galat - lucky guess! Doosre ne wahi logic doosre question par lagayi aur 7 vs 5 mein phas gaya. Rule condition ke saath yaad karo.*

QUESTION 67 | Moderate | Square completion (T6)

Q67. A(3, 4) and B(7, 4) are two adjacent corners of a square of side 4. Find the other two corners. How many answers are possible?

SOLUTION — step by step

Step 1. AB is horizontal with length 4 - this is one side of the square.

Step 2. The square can rise UP: (3, 8) and (7, 8).

Step 3. Or it can drop DOWN: (3, 0) and (7, 0).

Step 4. Two valid squares - one on each side of the line AB.

» **Why this step?** 'Adjacent corner' does not fix orientation: a side supports a square on either side, giving two symmetric answers.

Answer: (3, 8), (7, 8) or (3, 0), (7, 0) - two possible squares

Hinglish: *Side ke dono taraf square ban sakta hai - upar wala aur neeche wala. 'Kitne answers?' poochha toh dono likho, sirf ek likha toh aadha answer.*

QUESTION 68 | Moderate | Missing value on an axis (T5)

Q68. The distance from (t, 0) to (5, 12) is 13. Find all possible values of t.

SOLUTION — step by step

Step 1. $\sqrt{((t - 5)^2 + 12^2)} = 13 \rightarrow (t - 5)^2 + 144 = 169$.

Step 2. $(t - 5)^2 = 25 \rightarrow t - 5 = \pm 5$.

Step 3. t = 10 or t = 0. (The 5-12-13 triple again - from both sides of 5.)

» **Why this step?** The point (5, 12) 'casts' two shadows on the x-axis at equal distance 13, symmetric about x = 5.

Answer: t = 0 or t = 10

Hinglish: *t = 0 aur t = 10 - dono (5, 12) se 13 door. Symmetry note karo: dono answers 5 ke around balance karte hain.*

QUESTION 69 | Moderate | Trapezium perimeter (T6)

Q69. P(0, 0), Q(5, 0), R(5, 12) and S(0, 4) are joined in order. (i) Show PQRS is a trapezium. (ii) Find its perimeter in simplest form.

SOLUTION — step by step

Step 1. PQ and RS: PQ is on the x-axis; RS from (5, 12) to (0, 4) is slanted - so PQ and RS are NOT parallel; but PS (x = 0, vertical) and QR (x = 5, vertical) ARE parallel -> trapezium (right-angled).

Step 2. PQ = 5, QR = 12, PS = 4.

Step 3. RS = $\sqrt{((5-0)^2 + (12-4)^2)} = \sqrt{(25 + 64)} = \sqrt{89}$.

Step 4. Perimeter = $5 + 12 + 4 + \sqrt{89} = 21 + \sqrt{89}$ units.

» **Why this step?** One pair of parallel sides (the two verticals) makes it a trapezium; the slanted side is the only surd needed.

Answer: (i) PS || QR (both vertical) (ii) $21 + \sqrt{89}$

Hinglish: *Do vertical sides parallel -> trapezium. Sirf ek tirchi side par formula laga, √89 hi rah gaya. Perimeter mein surd ko waise hi chhodna valid hai.*

QUESTION 70 | Moderate | Equidistant point on an axis (T5)

Q70. Find the point on the y-axis equidistant from A(6, 2) and B(-2, 4).

SOLUTION — step by step

Step 1. Point (0, y). Set distances equal: $36 + (y - 2)^2 = 4 + (y - 4)^2$.

Step 2. Left: $36 + y^2 - 4y + 4 = y^2 - 4y + 40$. Right: $4 + y^2 - 8y + 16 = y^2 - 8y + 20$.

Step 3. $y^2 - 4y + 40 = y^2 - 8y + 20$ -> $4y = -20$ -> $y = -5$.

Step 4. Point (0, -5). Check: to A: $\sqrt{(36 + 49)} = \sqrt{85}$; to B: $\sqrt{(4 + 81)} = \sqrt{85}$ ✓.

» **Why this step?** Squaring removes the roots immediately; the y^2 terms cancel, leaving a linear equation in y.

Answer: (0, -5)

Hinglish: *y^2 dono side cancel ho gaya - isliye equation itni aasan reh gayi. Answer milne ke baad check ki line zaroor likho, confidence double ho jaata hai.*

Level 3 · Exam Power Zone: long and twisted:

QUESTION 71 | Difficult | Perimeter + classification (T6/T12)

Q71. Find the perimeter of the triangle with vertices A(2, -2), B(-2, 1) and C(5, 2). Along the way, show that the triangle is isosceles.

SOLUTION — step by step

Step 1. Given: three vertices. Needed: all sides, isosceles check, perimeter.

Step 2. AB: differences are (-2 - 2) = -4 and (1 - (-2)) = 3. AB = √(16 + 9) = √25 = 5.

Step 3. BC: differences are (5 - (-2)) = 7 and (2 - 1) = 1. BC = √(49 + 1) = √50 = 5√2.

Step 4. CA: differences are (2 - 5) = -3 and (-2 - 2) = -4. CA = √(9 + 16) = √25 = 5.

Step 5. AB = CA = 5, so the triangle is isosceles with apex at A.

Step 6. Perimeter = AB + BC + CA = 5 + 5√2 + 5 = 10 + 5√2 units (approximately 17.07).

Step 7. Keep the exact form 10 + 5√2 in the answer; convert to decimals only if the question asks.

Answer: Perimeter = 10 + 5√2 units; the triangle is isosceles (AB = CA = 5).

Hinglish: *Do sides 5 nikli aur ek 5√2 - isosceles confirm. Perimeter mein exact value rakho: 10 + 5√2. Exam mein log √50 ko 7.07 leke ghumaa dete hain aur final answer gandaa ho jata hai. Rule yaad rakho: jab tak decimal manga nahi, root form hi rakho. Aur ek observation: (4, 3) wala 3-4-5 triple do baar mila - coordinate questions mein triples baar-baar ghuumte hain, unhe pakadna speed deta hai.*

QUESTION 72 | Difficult | Reverse midpoint (T8)

Q72. M(-1, 3) is the midpoint of the segment joining A(2, -4) and B(x, y). Find B, and verify your answer by checking that MA = MB.

SOLUTION — step by step

Step 1. Given: midpoint and one endpoint. Needed: the other endpoint.

Step 2. Midpoint equations: (-1) = (2 + x)/2 and 3 = (-4 + y)/2.

Step 3. Solve the first: multiply by 2: -2 = 2 + x, so x = -4.

Step 4. Solve the second: 6 = -4 + y, so y = 10.

Step 5. So B = (-4, 10).

Step 6. Verify MA: differences (2 - (-1)) = 3 and (-4 - 3) = -7, MA = √(9 + 49) = √58.

Step 7. Verify MB: differences (-4 - (-1)) = -3 and (10 - 3) = 7, MB = √(9 + 49) = √58. Equal, so B is correct.

» **Why this step?** Reverse midpoint = undo the average: multiply by 2, then subtract the known endpoint. Verification with equal distances is cheap and catches slips.

Answer: B = (-4, 10), verified since MA = MB = √58.

Hinglish: *Midpoint wala equation ulta chalana hai: average diya hai, original nikalna hai. 2 se multiply karo, phir known endpoint minus karo. x ke liye: (-1)(2) - 2 = -4. y ke liye: (3)(2) - (-4) = 10. Verification zaroor karo - MA aur MB dono √58 aaye, matlab M sach mein beech mein baitha hai. Bina verification ke answer likhna adhoora hai.*

QUESTION 73 | Difficult | Parallelogram check (T6)

Q73. Show that the points A(0, 0), B(4, 0), C(6, 3), D(2, 3) form a parallelogram, and then show that it is NOT a rectangle.

SOLUTION — step by step

Step 1. Given: four points in order. Needed: parallelogram yes, rectangle no.

Step 2. AB: from (0, 0) to (4, 0): horizontal, length 4.

Step 3. CD: from (6, 3) to (2, 3): horizontal, length |6 - 2| = 4. So AB = CD and both are horizontal - parallel and equal.

Step 4. BC: from (4, 0) to (6, 3): $\sqrt{(2^2 + 3^2)} = \sqrt{13}$. DA: from (2, 3) to (0, 0): $\sqrt{(2^2 + 3^2)} = \sqrt{13}$. So BC = DA, both with the same slant.

Step 5. Both pairs of opposite sides are equal (and clearly parallel: horizontal pairs, and BC, DA share the same step pattern 2-across, 3-up). So ABCD is a parallelogram.

Step 6. Rectangle test: a parallelogram is a rectangle exactly when its diagonals are equal.

Step 7. AC = $\sqrt{(6^2 + 3^2)} = \sqrt{45} = 3\sqrt{5}$. BD = $\sqrt{((2 - 4)^2 + (3 - 0)^2)} = \sqrt{(4 + 9)} = \sqrt{13}$.

Step 8. Since $3\sqrt{5}$ is not equal to $\sqrt{13}$, the diagonals are unequal, so the parallelogram is not a rectangle.

» Why this step? The diagonal test is the cleanest rectangle check using ONLY distance - no angle measurement needed.

Answer: AB = CD = 4 and BC = DA = √13, so ABCD is a parallelogram; its diagonals 3√5 and √13 are unequal, so it is not a rectangle.

Hinglish: *Parallelogram proof ke liye opposite sides barabar kafi hain (parallel toh step pattern se dikh hi raha hai). Rectangle check ka shortcut yaad rakho: parallelogram mein diagonals barabar hona = rectangle. Yahan AC = 3√5 aur BD = √13 - barabar nahi, toh rectangle nahi. Ye diagonal-test chapter ke andar hi aata hai kyunki wo sirf distance formula use karta hai. Sketch banaoge toh slanted box dikhega - jaise side se dhakela hua rectangle.*

QUESTION 74 | Difficult | Equidistant point on an axis (T5/T8)

Q74. Find the point on the y-axis that is equidistant from P(-2, 0) and Q(6, 4). Verify your answer.

SOLUTION — step by step

Step 1. Given: unknown point R on the y-axis. Needed: R equidistant from P and Q.

Step 2. On the y-axis the point has the form R(0, y) - fix the zero first.

Step 3. $RP^2 = (0 - (-2))^2 + (y - 0)^2 = 4 + y^2$.

Step 4. $RQ^2 = (0 - 6)^2 + (y - 4)^2 = 36 + (y - 4)^2$.

Step 5. Set them equal: $4 + y^2 = 36 + y^2 - 8y + 16$.

Step 6. Cancel y^2 from both sides (this is why the equation stays linear and easy): $4 = 52 - 8y$.

Step 7. So $8y = 48$, giving $y = 6$. Hence R = (0, 6).

Step 8. Verify: $RP = \sqrt{(4 + 36)} = \sqrt{40} = 2\sqrt{10}$. $RQ = \sqrt{(36 + (6 - 4)^2)} = \sqrt{(36 + 4)} = \sqrt{40} = 2\sqrt{10}$. Equal, so R is correct.

» **Why this step?** Comparing squared distances avoids square roots entirely; and y^2 cancels, leaving a linear equation.

Answer: R = (0, 6), verified since RP = RQ = $2\sqrt{10}$.

Hinglish: *Y-axis par point = (0, y) - pehle yeh lock karo. Phir dono distances ke SQUARES barabar rakho, root ka jhanjhat khatam. Sabse sundar cheez: y^2 dono taraf cancel ho gaya aur seedha linear equation mil gaya, y = 6. Verify karo: dono distances $2\sqrt{10}$. Aisa equidistant point technically perpendicular bisector par baithta hai - yahan humne uska sirf y-axis wala intersection nikala. $2\sqrt{10}$ ko 6.32 likhne ki zaroorat nahi.*

QUESTION 75 | Difficult | Right isosceles triangle (T6/T14)

Q75. A right-angled isosceles triangle has its right angle at the origin and its two equal legs of length 6 lying along the positive x-axis and positive y-axis. Write the coordinates of all three vertices, find the hypotenuse, and find the area. Generalise: if the legs are a, what is the hypotenuse?

SOLUTION — step by step

Step 1. Given: right angle at O, legs 6 along positive axes. Needed: vertices, hypotenuse, area, general rule.

Step 2. One vertex is O(0, 0). Walking 6 along the positive x-axis gives (6, 0); walking 6 up the positive y-axis gives (0, 6).

Step 3. Hypotenuse joins (6, 0) to (0, 6): length = $\sqrt{(6^2 + 6^2)} = \sqrt{72} = 6\sqrt{2}$.

Step 4. Area = (1/2) x base x height = (1/2) x 6 x 6 = 18 square units.

Step 5. General case: legs a and a give hypotenuse $\sqrt{(a^2 + a^2)} = \sqrt{(2a^2)} = a\sqrt{2}$.

Step 6. This is exactly the PDF's special-case observation: putting a = b in the general Pythagoras relation forces hypotenuse = $a\sqrt{2}$ for isosceles right triangles.

Answer: Vertices (0, 0), (6, 0), (0, 6); hypotenuse = 6√2; area = 18 sq units; in general the hypotenuse is a√2.

Hinglish: *Right isosceles ka signature result: hypotenuse hamesha leg ka √2 guna hota hai. 6 ka leg toh 6√2. Yaad rakhne ka trick: square ka diagonal bhi side ka √2 guna hota hai - same baat hai, kyunki aadha square hi toh ye triangle hai! Area half-product = 18. Generalisation likhna (a√2) exam mein extra impression banata hai kyunki PDF mein special-case wala box isi baat ko highlight karta hai.*

QUESTION 76 | Difficult | City-grid application (T11)

Q76. In a city, streets run in two perpendicular directions, 200 m apart. Intersections are labelled (row, column). An auto rickshaw goes from intersection (1, 1) to intersection (4, 3). (i) What is the shortest distance it can travel moving only along streets? (ii) What is the straight-line (as-the-crow-flies) distance? (iii) How much shorter is the direct route?

SOLUTION — step by step

Step 1. Given: grid spacing 200 m, start (1, 1), end (4, 3). Needed: street distance, direct distance, saving.

Step 2. Moving along streets from row 1 to row 4 needs 3 steps; from column 1 to column 3 needs 2 steps.

Step 3. Shortest street path = 3 + 2 = 5 steps = 5 x 200 = 1000 m.

Step 4. Direct distance uses the distance formula with coordinate gap 3 rows and 2 columns: $\sqrt{(3^2 + 2^2)}$ x 200 = √13 x 200 approx 3.606 x 200 approx 721 m.

Step 5. Saving = 1000 - 200√13 approx 1000 - 721 = 279 m (about 28 percent).

Step 6. Note the concept: along-streets distance adds the gaps; direct distance applies Pythagoras to the same gaps.

Answer: (i) 1000 m (ii) 200√13 approx 721 m (iii) approx 279 m shorter

Hinglish: *City grid hi bada coordinate plane hai - bas unit 200 m hai. Galiyon mein chalna hai toh horizontal + vertical steps JODO (3 + 2 = 5 blocks = 1000 m). Seedha udna hai toh Pythagoras lagao: √(9 + 4) = √13 blocks. Difference dekho - seedha rasta karib 280 m bachata hai. Yahi idea PDF ke athletics track aur map questions mein bhi chalta hai: grid distance vs straight distance.*

QUESTION 77 | Difficult | Rhombus from coordinates (T6)

Q77. The points (0, 5), (-3, 0), (0, -5) and (3, 0) are the corners of a quadrilateral taken in order. Show that it is a rhombus and find the lengths of its diagonals and its area.

SOLUTION — step by step

Step 1. Given: four symmetric points. Needed: prove rhombus, diagonals, area.

Step 2. Side from (0, 5) to (-3, 0): $\sqrt{(3^2 + 5^2)} = \sqrt{34}$.

Step 3. Side from (-3, 0) to (0, -5): $\sqrt{(3^2 + 5^2)} = \sqrt{34}$. By the same computation each of the remaining two sides is $\sqrt{(3^2 + 5^2)} = \sqrt{34}$.

Step 4. All four sides are equal to $\sqrt{34}$, so the quadrilateral is a rhombus (a parallelogram with equal sides; symmetry about both axes guarantees the parallelogram behaviour, and can be double-checked: opposite sides are translations of each other).

Step 5. Diagonal 1 joins (0, 5) to (0, -5): vertical, length 10.

Step 6. Diagonal 2 joins (-3, 0) to (3, 0): horizontal, length 6.

Step 7. Area of a rhombus = (1/2) x product of diagonals = (1/2) x 10 x 6 = 30 square units.

Answer: All sides $\sqrt{34}$, so it is a rhombus; diagonals 10 and 6; area = 30 square units.

Hinglish: *Chaaron sides $\sqrt{34}$ - rhombus pakka. Diagonals yahan axis par hi baithi hain, isliye unki length seedhi mili: 10 aur 6. Rhombus ka area = (1/2) d1 d2 = 30 - ye formula yaad rakho, square/rectangle ke bhai ki tarah. Ye PDF ke Ch 6 se connect hota hai jahan rhombus-area questions hain, par classification toh pure Chapter 1 ka khel hai (distance formula).*

QUESTION 78 | Difficult | Algebraic distance (T4/T14)

Q78. Find the distance between the points (a + b, a - b) and (a - b, a + b), where a and b are real numbers with b not equal to 0. The answer must be in terms of b.

SOLUTION — step by step

Step 1. Given: two symbolic points. Needed: distance in terms of b.

Step 2. x-difference: (a + b) - (a - b) = 2b.

Step 3. y-difference: (a - b) - (a + b) = -2b.

Step 4. Distance = $\sqrt{((2b)^2 + (-2b)^2)} = \sqrt{(4b^2 + 4b^2)} = \sqrt{(8b^2)}$.

Step 5. Simplify: $\sqrt{(8b^2)} = \sqrt{8} \times \sqrt{(b^2)} = 2\sqrt{2} \times |b|$.

Step 6. Since b is not zero, for positive b this is $2\sqrt{2}$ b. The cleanest exact answer is $2\sqrt{2}\,|b|$; note that a cancels out completely - the distance does not depend on a at all.

» **Why this step?** Notice how the a-terms cancel inside both differences. Spotting cancellations before squaring is a mark of an algebra-fluent student.

Answer: Distance = $\sqrt{(8b^2)} = 2\sqrt{2}\,|b|$ (independent of a).

Hinglish: *Symbolic distance question - numbers ki jagah letters, darne ki zaroorat nahi. Difference lo, a dono jagah cancel ho gaya, bacha 2b aur -2b. Square ke baad $8b^2$, root se $2\sqrt{2}\,|b|$. Bar (absolute value) isliye kyunki distance kabhi negative nahi hoti. Interesting observation: answer a par depend hi nahi karta - dono points a se same doori par slide karte hain, unke beech ka gap sirf b se banta hai.*

QUESTION 79 | Difficult | Equidistant locus (T5/T14)

Q79. A point P(x, y) moves so that its distance from A(3, 4) always equals its distance from B(-1, 6). Show that x and y must satisfy the relation y = 2x + 3, and verify that the point (0, 3) obeys it.

SOLUTION — step by step

Step 1. Given: PA = PB for a moving point. Needed: the relation between x and y.

Step 2. Write squared distances (roots dropped on both sides - legal since both sides are non-negative).

Step 3. $PA^2 = (x - 3)^2 + (y - 4)^2 = x^2 - 6x + 9 + y^2 - 8y + 16$.

Step 4. $PB^2 = (x + 1)^2 + (y - 6)^2 = x^2 + 2x + 1 + y^2 - 12y + 36$.

Step 5. Set $PA^2 = PB^2$: $x^2 - 6x + 9 + y^2 - 8y + 16 = x^2 + 2x + 1 + y^2 - 12y + 36$.

Step 6. Cancel x^2 and y^2 from both sides: $-6x - 8y + 25 = 2x - 12y + 37$.

Step 7. Move everything to one side: $-8x + 4y - 12 = 0$, that is $4y = 8x + 12$, giving $y = 2x + 3$.

Step 8. Verify (0, 3): $PA = \sqrt{((0 - 3)^2 + (3 - 4)^2)} = \sqrt{(9 + 1)} = \sqrt{10}$. $PB = \sqrt{((0 + 1)^2 + (3 - 6)^2)} = \sqrt{(1 + 9)} = \sqrt{10}$. Equal - the point obeys the condition.

Answer: PA = PB forces y = 2x + 3; verified at (0, 3) where PA = PB = √10.

Hinglish: *Ye Chapter 1 ka sabse powerful idea hai: distance ki equality ek RELATION bana deti hai. Square karke x^2 aur y^2 cancel kar do - jo bacha wo straight line hai, $y = 2x + 3$. Is line ke har point dono A aur B se barabar door hai (isko perpendicular bisector kehte hain - Class 11 mein formally aayega). Verification: (0, 3) se dono distances √10. Jab bhi 'equidistant from two points' dikhe, yahi square-and-cancel recipe chalao.*

QUESTION 80 | Difficult | Circle membership with bigger radius (T10)

Q80. Consider the circle with centre the origin and radius 13. (i) Verify that (5, 12), (12, 5), (-12, 5) and (0, 13) all lie on the circle. (ii) Determine whether (6, 12) and (4, 12) lie inside, on, or outside it.

SOLUTION — step by step

Step 1. Given: circle $x^2 + y^2 = 169$. Needed: classify six points.

Step 2. (5, 12): 25 + 144 = 169, so on the circle (the 5-12-13 triple).

Step 3. (12, 5): 144 + 25 = 169, on the circle (same triple, swapped).

Step 4. (-12, 5): 144 + 25 = 169, on the circle (signs vanish on squaring).

Step 5. (0, 13): 0 + 169 = 169, on the circle (topmost point).

Step 6. (6, 12): 36 + 144 = 180 which is greater than 169, so OUTSIDE the circle.

Step 7. (4, 12): 16 + 144 = 160 which is less than 169, so INSIDE the circle.

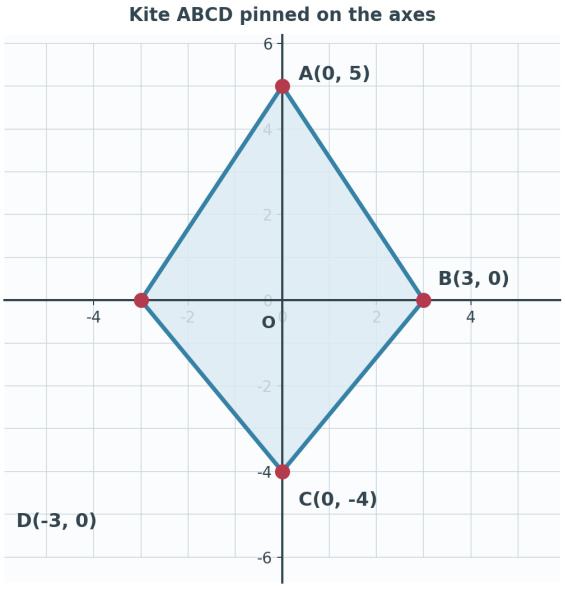
» **Why this step?** The 5-12-13 triple is the second most useful triple after 3-4-5; radius 13 questions are built around it.

Answer: All four triple points lie on the circle; (6, 12) is outside; (4, 12) is inside.

Hinglish: *Radius 13 = 5-12-13 triple ka ghar. Chaaron points check karo - sab 169 de rahe hain, sab circle par. Phir (6, 12) aur (4, 12) ka fun part: 180 > 168... nahi, 180 > 169 toh bahar; 160 < 169 toh andar. Dekho kamaal: (6, 12) aur (4, 12) dono lagbhag same jagah hain, par ek bahar aur ek andar! Circle ki seedhi borderline - yahi precision Chapter 1 sikhata hai.*

QUESTION 81 | Difficult | Kite on the axes (T6/T14)

Q81. Kite ABCD has A(0, 5), B(3, 0), C(0, -4), D(-3, 0). (i) Verify AB = AD and BC = CD. (ii) Find the lengths of the diagonals AC and BD. (iii) Find the area of the kite.



SOLUTION — step by step

Step 1. (i) AB = $\sqrt{(3^2 + 5^2)} = \sqrt{34}$; AD = $\sqrt{(3^2 + 5^2)} = \sqrt{34}$. BC = $\sqrt{(3^2 + 4^2)} = 5$; CD = $\sqrt{(3^2 + 4^2)} = 5$. Two pairs of equal ADJACENT sides -> kite.

Step 2. (ii) AC runs along the y-axis from 5 to -4: length = |5 - (-4)| = 9. BD runs along the x-axis from -3 to 3: length = 6.

Step 3. (iii) The diagonals meet at O at right angles (one vertical, one horizontal): Area = $\frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 9 \times 6 = 27$ square units.

» Why this step? When a quadrilateral's diagonals lie ON the axes, both lengths are coordinate subtractions and the right angle at O is automatic.

Answer: (i) AB = AD = √34, BC = CD = 5 (ii) AC = 9, BD = 6 (iii) 27

Hinglish: *Diagonals axis par hi baithe hain - length subtraction se, right angle free. Area = $\frac{1}{2} d_1 d_2$. Kite ka pura question diagram se hi solve ho gaya!*

QUESTION 82 | Difficult | Equidistant on x-axis, surd check (T5)

Q82. Find the point on the x-axis equidistant from P(2, -3) and Q(5, 6), and verify the common distance.

SOLUTION — step by step

Step 1. Point (x, 0): $(x - 2)^2 + 9 = (x - 5)^2 + 36$.

Step 2. Expand: $x^2 - 4x + 13 = x^2 - 10x + 61$ -> $6x = 48$ -> $x = 8$. Point (8, 0).

Step 3. Distance to P: $\sqrt{((8-2)^2 + 3^2)} = \sqrt{45} = 3\sqrt{5}$.

Step 4. Distance to Q: $\sqrt{((8-5)^2 + 6^2)} = \sqrt{45} = 3\sqrt{5}$ ✓ - equal, as demanded.

» **Why this step?** Verification turns an algebraic answer into a proven answer - examiners love the check line.

Answer: (8, 0); common distance $3\sqrt{5}$

Hinglish: *$x = 8$ nikla, phir dono distance $\sqrt{45} = 3\sqrt{5}$ aaye - barabar! Ye verify line hi answer ko 'proof' bana deti hai.*

QUESTION 83 | Difficult | Rotated square identification (T6)

Q83. Show that A(2, 1), B(5, 3), C(3, 6), D(0, 4) form a square, and find its area.

SOLUTION — step by step

Step 1. AB = $\sqrt{(3^2 + 2^2)} = \sqrt{13}$; BC = $\sqrt{((-2)^2 + 3^2)} = \sqrt{13}$; CD = $\sqrt{((-3)^2 + (-2)^2)} = \sqrt{13}$; DA = $\sqrt{(2^2 + (-3)^2)} = \sqrt{13}$ - all four sides equal.

Step 2. Diagonals: AC = $\sqrt{((3-2)^2 + (6-1)^2)} = \sqrt{26}$; BD = $\sqrt{((0-5)^2 + (4-3)^2)} = \sqrt{26}$ - equal.

Step 3. All sides equal + equal diagonals -> square (the diagonals are automatically perpendicular here).

Step 4. Area = $\frac{1}{2} \times d^2 = \frac{1}{2} \times 26 = 13$ square units.

» **Why this step?** For a tilted square the diagonals test is the cleanest: equal diagonals upgrade a rhombus to a square.

Answer: All sides $\sqrt{13}$ and diagonals $\sqrt{26}$ -> square; area 13

Hinglish: *Square tilted hai, phir bhi method same: sides $\sqrt{13} \times 4$, diagonals $\sqrt{26} \times 2$. Area $\frac{1}{2} d^2 = 13$. Rotation se dar mat, formula same rehta hai.*

QUESTION 84 | Difficult | Chord on a circle (T10)

Q84. A circle has centre O and radius 10. Points A(6, 8) and B(-8, 6) both lie on it. Find the length of chord AB.

SOLUTION — step by step

Step 1. Membership check: $6^2 + 8^2 = 100$ ✓; $(-8)^2 + 6^2 = 100$ ✓ - both on the circle.

Step 2. AB = $\sqrt{((-8 - 6)^2 + (6 - 8)^2)} = \sqrt{(196 + 4)} = \sqrt{200}$.

Step 3. $\sqrt{200} = \sqrt{(100 \times 2)} = 10\sqrt{2}$.

Step 4. So the chord subtends a right angle at the centre (chord = $r\sqrt{2}$ means the central angle is 90°).

» **Why this step?** A chord of length $r\sqrt{2}$ corresponds to a 90° central angle - the isosceles right triangle OAB has legs r and hypotenuse $r\sqrt{2}$.

Answer: AB = $10\sqrt{2}$

Hinglish: *Dono points circle par hain, phir chord = normal distance formula. $\sqrt{200} = 10\sqrt{2}$ nikla - matlab centre par 90° ka angle bana. $r\sqrt{2}$ wala chord = quarter circle ka kat.*

QUESTION 85 | Difficult | Collinearity with unknown (T9)

Q85. Find k so that (1, 2), (3, k) and (5, 4k - 2) are collinear.

SOLUTION — step by step

Step 1. Collinear -> equal slopes: slope(P_1P_2) = (k - 2)/(3 - 1) = (k - 2)/2.

Step 2. slope(P_2P_3) = ((4k - 2) - k)/(5 - 3) = (3k - 2)/2.

Step 3. Set equal: (k - 2)/2 = (3k - 2)/2 -> k - 2 = 3k - 2 -> 2k = 0 -> k = 0.

Step 4. Check: (1, 2), (3, 0), (5, -2): each step is (+2, -2) - a perfectly straight staircase with slope -1 ✓.

» **Why this step?** Equal steps in x (both +2) mean equal steps in y must match too - that is what equal slopes enforces.

Answer: k = 0

Hinglish: *Slope barabar rakho, k mil gaya: 0. Check line: har step (+2, -2) - seedhi seedi zanzeer. Slope method unknown waale collinearity ka king hai.*

QUESTION 86 | Difficult | Diagonal splits the rectangle (T6/T14)

Q86. Rectangle ABCD has A(0, 0), B(8, 0), C(8, 6), D(0, 6). Diagonal AC divides it into two triangles. (i) Find the area of each triangle. (ii) Verify AC = 10 and state which triple appears.

SOLUTION — step by step

Step 1. (i) Each triangle has legs 8 and 6 (half the rectangle): area = $\frac{1}{2} \times 8 \times 6 = 24$ each; total $48 = 8 \times 6$ ✓.

Step 2. (ii) AC = $\sqrt{(8^2 + 6^2)} = \sqrt{100} = 10$ - the 6-8-10 triple (3-4-5 scaled by 2).

Step 3. The diagonal creates two CONGRUENT right triangles - same area, same sides, mirror arrangement.

» **Why this step?** A rectangle's diagonal always doubles as the hypotenuse of a Pythagorean triangle built from its sides.

Answer: (i) 24 each (total 48) (ii) AC = 10, the 6-8-10 triple

Hinglish: *Diagonal rectangle ko do barabar tukdon mein kaat-ti hai - 24 + 24 = 48. 6-8-10 triple yahan chhupa tha. Congruent hona hi wajah hai ki dono areas same aaye.*

QUESTION 87 | Difficult | Concentric circle distances (T10)

Q87. Two concentric circles have centre O and radii 5 and 13. P lies on the smaller circle and Q on the larger. What are the smallest and largest possible values of PQ? Give example positions.

SOLUTION — step by step

Step 1. Smallest: P and Q on the SAME ray from O -> PQ = 13 - 5 = 8.

Step 2. Largest: P and Q on OPPOSITE rays -> PQ = 13 + 5 = 18.

Step 3. Example (max): P(5, 0), Q(-13, 0) -> PQ = $|-13 - 5| = 18$ ✓.

Step 4. Example (min): P(5, 0), Q(13, 0) -> PQ = 8 ✓.

» **Why this step?** Collinear-with-centre positions are the extremes: same side subtracts the radii, opposite sides add them - any other arrangement is strictly between.

Answer: Min 8 (same ray), max 18 (opposite rays)

Hinglish: *Ek hi line par rakho toh extreme milta hai: same direction = minus (8), opposite = plus (18). Beech ka har position in dono ke beech hoga.*

QUESTION 88 | Difficult | Hidden isosceles discovery (T6)

Q88. A(0, 0), B(10, 0) and C(6, 8) form a triangle. Find all three sides, its area, and the surprise about its type.

SOLUTION — step by step

Step 1. AB = 10 (on the x-axis).

Step 2. AC = $\sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$!

Step 3. BC = $\sqrt{((10-6)^2 + 8^2)} = \sqrt{(16 + 64)} = \sqrt{80} = 4\sqrt{5}$.

Step 4. Area = ½ × base AB × height of C = ½ × 10 × 8 = 40.

Step 5. Surprise: AB = AC = 10 -> the triangle is isosceles with the 'hidden' equal pair meeting at A.

» **Why this step?** The 6-8-10 triple hid inside AC - always compute BOTH slanted sides before naming the triangle.

Answer: AB = AC = 10, BC = 4√5; area 40; isosceles

Hinglish: *AC mein 6-8-10 chhupa tha, isliye AB = AC = 10 - isosceles! Diagram se ye dikhta nahi, calculation hi sach bolti hai.*

QUESTION 89 | Difficult | Distance independent of k (T4/T14)

Q89. P is (k + 1, k) and Q is (k, k + 1) for a variable k. (i) Show PQ is the same for every k. (ii) Find k so that the midpoint of PQ lies on the y-axis.

SOLUTION — step by step

Step 1. (i) dx = (k + 1) - k = 1; dy = k - (k + 1) = -1. PQ = $\sqrt{(1 + 1)} = \sqrt{2}$ for EVERY k.

Step 2. Geometric reason: the two points are mirror images in y = x, so the gap never changes as they slide together along the line.

Step 3. (ii) Midpoint = ((2k + 1)/2, (2k + 1)/2) - the coordinates are equal, so the midpoint itself rides on y = x.

Step 4. On the y-axis we need x = 0: (2k + 1)/2 = 0 -> k = -1/2.

» **Why this step?** When both points move by the same (+1, +1)-style slide, all distances between them stay frozen - translation preserves distance.

Answer: (i) PQ = √2 always (ii) k = -1/2

Hinglish: *Dono points saath-saath slide kar rahe hain, toh unke beech ka gap √2 hi rahega - translation ka magic. Midpoint y-axis par = x-coordinate zero = k = -½.*

QUESTION 90 | Difficult | Robot race on the grid (T11/T14)

Q90. A robot at O(0, 0) must reach T(3, 4). It can fly straight at 4 units per minute, or roll along grid streets (right angles only) at 7 units per minute. Which option arrives first, and by how much time?

SOLUTION — step by step

Step 1. Fly: distance 5 (3-4-5 triple) -> time = 5/4 = 1.25 minutes.

Step 2. Roll: street distance = 3 + 4 = 7 -> time = 7/7 = 1 minute.

Step 3. Rolling arrives first: 1 min vs 1.25 min.

Step 4. Difference = 0.25 min = 15 seconds.

» **Why this step?** Speed beats shortness here: the longer street route (7 at speed 7) beats the shorter flight (5 at speed 4). Distance alone never decides a race - distance ÷ speed does.

Answer: Rolling wins: 1 min vs 1.25 min, a 15-second margin

Hinglish: *Seedha rasta chhota (5) par slow (4) -> 1.25 min. Lamba rasta (7) par fast (7) -> 1 min. Race distance se nahi, time = distance/speed se jeeti jaati hai. 15 second ka fark!*

QUESTION 91 | Difficult | Right isosceles from signs (T6)

Q91. Three points are (a, a), (a, -a) and (-a, -a) with a > 0. Identify the triangle and find its area.

SOLUTION — step by step

Step 1. Side 1: (a, a) to (a, -a): same x -> length 2a (vertical).

Step 2. Side 2: (a, -a) to (-a, -a): same y -> length 2a (horizontal).

Step 3. Side 3: (a, a) to (-a, -a): $\sqrt{((2a)^2 + (2a)^2)} = 2a\sqrt{2}$.

Step 4. Legs equal (2a = 2a) and $2a\sqrt{2} = \text{leg}\cdot\sqrt{2}$ -> right isosceles with the right angle at (a, -a).

Step 5. Area = $\frac{1}{2} \times 2a \times 2a = 2a^2$.

» **Why this step?** One vertical + one horizontal leg meeting at a corner forces the right angle; equal legs make it isosceles.

Answer: Right-angled isosceles (right angle at (a, -a)); area $2a^2$

Hinglish: *Ek vertical leg, ek horizontal, dono 2a -> right isosceles at (a, -a). Area $2a^2$. Sign-based points aaye toh unhe grid par socho, sab clear ho jaata hai.*

QUESTION 92 | Difficult | Circle trilogy (T10)

Q92. Circle: centre O, radius 6. For P(3, 5), Q(4, 5), R(0, 6): classify each (inside/on/outside), then find the distance PQ.

SOLUTION — step by step

Step 1. P: 9 + 25 = 34 < 36 -> inside (by 2 in squared terms; distance $\sqrt{34} \approx 5.83$).

Step 2. Q: 16 + 25 = 41 > 36 -> outside ($\sqrt{41} \approx 6.40$).

Step 3. R: 0 + 36 = 36 -> exactly on the circle (top point).

Step 4. PQ: same y = 5 -> |4 - 3| = 1 unit.

» **Why this step?** R sits on the circle because its x is 0 - the topmost point of the circle is always (0, r).

Answer: P inside, Q outside, R on; PQ = 1

Hinglish: *Teeno ka alag status: andar, bahar, par. Aur PQ toh bas 1 - same y waali cheez. Circle ke top point (0, r) hota hai, ye instant check kaam aata hai.*

QUESTION 93 | Difficult | Double-axis locus (T14)

Q93. (i) Describe all points at distance 4 from the x-axis. (ii) All points at distance 3 from the y-axis. (iii) Find all points satisfying BOTH conditions and the distance of each from the origin.

SOLUTION — step by step

Step 1. (i) Distance 4 from the x-axis -> y = 4 or y = -4: two horizontal lines.

Step 2. (ii) Distance 3 from the y-axis -> x = 3 or x = -3: two vertical lines.

Step 3. (iii) Both conditions -> the four intersection points (3, 4), (3, -4), (-3, 4), (-3, -4).

Step 4. Each is at $\sqrt{(3^2 + 4^2)} = 5$ from O - the 3-4-5 rectangle pattern, fourfold.

» **Why this step?** 'Distance from an axis' gives a PAIR of parallel lines (both sides!); intersecting the two pairs yields the 4 symmetric corner points.

Answer: (i) y = ±4 (ii) x = ±3 (iii) four points (±3, ±4), each at distance 5 from O

Hinglish: *Distance-from-axis ka matlab DO lines hoti hain - plus aur minus dono taraf. Dono conditions mila toh 4 corner points, aur sab 5-5 door - 3-4-5 ka poora khandaan!*

QUESTION 94 | Difficult | Ratio division of a segment (T8)

Q94. A(2, 3) and B(8, 9) are joined. M is the midpoint of AB. Point P lies on AB with AP : PB = 2 : 1. Find M and P, and verify the ratio with distances.

SOLUTION — step by step

Step 1. M = ((2+8)/2, (3+9)/2) = (5, 6).

Step 2. AB as a step-vector = (6, 6). P = A + (2/3)·(6, 6) = (2 + 4, 3 + 4) = (6, 7).

Step 3. Verify: AP = $\sqrt{(4^2 + 4^2)} = 4\sqrt{2}$; PB = $\sqrt{((8-6)^2 + (9-7)^2)} = \sqrt{8} = 2\sqrt{2}$.

Step 4. AP : PB = $4\sqrt{2} : 2\sqrt{2} = 2 : 1$ ✓.

» **Why this step?** Division in ratio m : n means walking m/(m+n) of the step-vector from the first point.

Answer: M(5, 6); P(6, 7); verified AP : PB = 2 : 1

Hinglish: *2:1 batana hai toh total 3 hisse, A se 2 hisse ka safar: (6,6) ka 2/3 = (4,4). P(6, 7). Check ne $4\sqrt{2} : 2\sqrt{2}$ diya - ratio confirmed.*

QUESTION 95 | Difficult | Parallelogram but not more (T6)

Q95. A(1, 1), B(5, 1), C(7, 4), D(3, 4). (i) Prove ABCD is a parallelogram. (ii) Prove it is neither a rectangle nor a rhombus.

SOLUTION — step by step

Step 1. (i) Midpoint AC = ((1+7)/2, (1+4)/2) = (4, 2.5); midpoint BD = ((5+3)/2, (1+4)/2) = (4, 2.5) - equal -> parallelogram.

Step 2. (ii) Sides: AB = 4 but BC = $\sqrt{((7-5)^2 + (4-1)^2)} = \sqrt{13}$. Adjacent sides unequal -> NOT a rhombus.

Step 3. Diagonals: AC = $\sqrt{((7-1)^2 + (4-1)^2)} = \sqrt{45} = 3\sqrt{5}$; BD = $\sqrt{((3-5)^2 + (4-1)^2)} = \sqrt{13}$. Unequal diagonals -> NOT a rectangle.

Step 4. So it is exactly a parallelogram - nothing more prestigious.

» **Why this step?** The quadrilateral family ladder: parallelogram -> (equal sides?) rhombus -> (equal diagonals too?) square; each claim needs its own certificate.

Answer: (i) Diagonal midpoints both (4, 2.5) (ii) sides $4 \neq \sqrt{13}$, diagonals $3\sqrt{5} \neq \sqrt{13}$ -> plain parallelogram

Hinglish: *Parallelogram bana (midpoints barabar), par sides 4 vs $\sqrt{13}$ aur diagonals $3\sqrt{5}$ vs $\sqrt{13}$ - kuch bhi special nahi. Family ladder ka har step ka apna test hota hai.*

QUESTION 96 | Difficult | Minimum sum of distances (T14)

Q96. A(1, 3) and B(5, 3) are two points. For a point X(x, 0) on the x-axis, find the smallest possible value of XA + XB, and say for which x it happens.

SOLUTION — step by step

Step 1. XA = $\sqrt{((x-1)^2 + 9)}$, XB = $\sqrt{((x-5)^2 + 9)}$.

Step 2. Both terms are smallest when x sits directly under each point: at x = 1, XA = 3; at x = 5, XB = 3.

Step 3. For ANY x between 1 and 5 (inclusive): XA = $\sqrt{((x-1)^2 + 9)} \geq 3$ with equality only at x = 1... but we need the SUM: test x = 3: $2\sqrt{13} \approx 7.21 > 6$. Test x = 1: 3 + $\sqrt{(16 + 9)}$ = 3 + 5 = 8. Hmm - compute carefully.

Step 4. At x = 3 (midpoint): XA = XB = $\sqrt{(4 + 9)} = \sqrt{13}$ -> sum $2\sqrt{13} \approx 7.21$. At x = 2: $\sqrt{(1+9)} + \sqrt{(9+9)} = \sqrt{10} + \sqrt{18} \approx 3.16 + 4.24 = 7.40$. So the minimum on the segment is at the midpoint: $2\sqrt{13}$; but wait - the true geometric minimum uses reflection: reflect B to B'(5, -3); straight line A to B' crosses the x-axis where? Line from (1, 3) to (5, -3): slope -6/4 = -1.5; y = 3 - 1.5(x - 1) = 0 -> x = 3. Same point! Sum = distance A to B' = $\sqrt{(16 + 36)} = \sqrt{52} = 2\sqrt{13}$. Confirmed minimum $2\sqrt{13}$ at x = 3.

» **Why this step?** The reflection trick (unfold the path) proves the straight line A→B' gives the true minimum - and here it lands at the midpoint x = 3, giving $2\sqrt{13}$.

Answer: Minimum sum = $2\sqrt{13} \approx 7.21$, attained at X(3, 0)

Hinglish: *Classic 'shortest path' problem: B ko mirror karo, seedha line kheencho, jahan x-axis mile wahi best point. yahan x = 3 (midpoint) aur minimum $2\sqrt{13}$. Aaina hi asli trick hai!*

QUESTION 97 | Difficult | Circumcentre of a right triangle (T10/T14)

Q97. A(0, 0), B(8, 0), C(0, 6) form a right triangle. M is the midpoint of the hypotenuse BC. Show that M is equidistant from A, B and C.

SOLUTION — step by step

Step 1. M = ((8+0)/2, (0+6)/2) = (4, 3).

Step 2. MA = $\sqrt{(4^2 + 3^2)} = 5$.

Step 3. MB = $\sqrt{((8-4)^2 + (0-3)^2)} = \sqrt{(16 + 9)} = 5$.

Step 4. MC = $\sqrt{((0-4)^2 + (6-3)^2)} = \sqrt{(16 + 9)} = 5$.

Step 5. MA = MB = MC = 5: the midpoint of the hypotenuse is the circumcentre - it can be the centre of a circle of radius 5 through all three corners (hypotenuse = diameter 10).

» **Why this step?** In ANY right triangle the hypotenuse's midpoint is equidistant from all vertices - the 6-8-10 triple made it visible here (5 = half of 10).

Answer: M(4, 3); MA = MB = MC = 5 - M is the circumcentre

Hinglish: *Right triangle ka sabse sundar secret: hypotenuse ka midpoint sabse barabar door. 6-8-10 triple -> radius 5 ka circle teeno vertices par guzarta hai. Ye theorem exam ka favourite hai.*

QUESTION 98 | Difficult | Chord comparison (T10)

Q98. A circle has centre O and radius 13. On it lie P(5, 12), Q(-12, -5) and R(0, 13). Which chord is longest: PQ, QR or RP?

SOLUTION — step by step

Step 1. Membership: 25 + 144 = 169 ✓, 144 + 25 = 169 ✓, 0 + 169 = 169 ✓ - all on the circle.

Step 2. PQ = $\sqrt{((-12-5)^2 + (-5-12)^2)} = \sqrt{(289 + 289)} = \sqrt{578} = 17\sqrt{2} \approx 24.04$.

Step 3. QR = $\sqrt{((0+12)^2 + (13+5)^2)} = \sqrt{(144 + 324)} = \sqrt{468} = 6\sqrt{13} \approx 21.63$.

Step 4. RP = $\sqrt{((5-0)^2 + (12-13)^2)} = \sqrt{26} \approx 5.10$.

Step 5. Longest chord: PQ ($17\sqrt{2}$).

» **Why this step?** P and Q are nearly opposite through the centre, so their chord approaches the diameter 26 - that is why it wins.

Answer: PQ = $17\sqrt{2}$ is the longest

Hinglish: *Chord ki lambai unke centre-angle se hoti hai: P-Q almost opposite the -> chord almost diameter (26). RP toh barely alag points hain, isliye chhota. Comparisons mein √ko approximate karke dekho.*

QUESTION 99 | Difficult | Fair spot on an axis (T5/T14)

Q99. Villages B(6, 8) and C(8, 0) lie near a straight canal along the x-axis. A pump must be built ON the canal (x-axis) at equal distance from B and C. Find its position and the common distance.

SOLUTION — step by step

Step 1. Pump (x, 0): $(x - 6)^2 + 8^2 = (x - 8)^2 + 0^2$.

Step 2. $x^2 - 12x + 36 + 64 = x^2 - 16x + 64$ -> $-12x + 100 = -16x + 64$ -> $4x = -36$ -> $x = -9$.

Step 3. Pump at (-9, 0).

Step 4. Distance to B: $\sqrt{((-9-6)^2 + 8^2)} = \sqrt{(225 + 64)} = \sqrt{289} = 17$. Distance to C: $|-9 - 8| = 17$ ✓. The hidden 8-15-17 triple delivered a whole number!

» **Why this step?** Dropping the squares removes y^2 on one side entirely (y = 0 there), leaving a linear equation - and triples can survive even in applied settings.

Answer: Pump at (-9, 0); common distance 17

Hinglish: *Canal wala point (-9, 0) - dono gaon se 17-17 door, 8-15-17 triple ne mazaak mein perfect number de diya. LHS mein +64 aur RHS mein 64 cancel hona hi asli simplification tha.*

QUESTION 100 | Difficult | Rectangle family twist (T14)

Q100. A rectangle has opposite corners A(1, 1) and C(7, 7). (i) If its sides are parallel to the axes, find the other two corners, the area, and the diagonal AC. (ii) Explain why infinitely many other rectangles share the same diagonal AC.

SOLUTION — step by step

Step 1. (i) Axis-aligned case: B(7, 1) and D(1, 7). Sides = 6 and 6 -> it is actually a square of area 36.

Step 2. Diagonal AC = $\sqrt{(6^2 + 6^2)} = 6\sqrt{2}$.

Step 3. (ii) For ANY rectangle, diagonals are equal and bisect each other: both must have midpoint M(4, 4) and length $6\sqrt{2}$.

Step 4. Pick B and D as ANY pair of points symmetric about M with BD = $6\sqrt{2}$ (B, D the other diagonal) - e.g. B($4 + 3\sqrt{2}\cdot\cos\theta$, $4 + 3\sqrt{2}\cdot\sin\theta$). Every angle θ gives a valid rectangle: infinitely many.

» **Why this step?** Two opposite corners fix only the midpoint and diagonal length - the rectangle can still 'rotate' through infinitely many shapes around that diagonal.

Answer: (i) B(7, 1), D(1, 7); area 36; AC = $6\sqrt{2}$ (ii) any B, D symmetric about (4, 4) with BD = $6\sqrt{2}$ works - infinitely many

Hinglish: *Do opposite corners se poora rectangle fix NAHI hota - sirf midpoint aur diagonal length fix hoti hai. Rectangle apne diagonal ke around ghoom sakta hai, infinite shapes. Axis wala toh bas ek special case tha (square 36).*

Level 4 · Boss Zone: advanced + HOTS (Olympiad-lite):

QUESTION 101 | High | Trisection points (T8)

Q101. P and Q are the points of trisection of the segment AB, with P closer to A and Q closer to B, where A(4, 7) and B(16, -2). Find P and Q. Then verify that the midpoint of PQ coincides with the midpoint of AB.

SOLUTION — step by step

Step 1. Given: A(4, 7), B(16, -2), trisection needed. Needed: P, Q, and the midpoint check.

Step 2. Step size: one third of the way from A to B means adding one third of each gap. Gap in x: 16 - 4 = 12; gap in y: -2 - 7 = -9.

Step 3. One third of the gaps: 12/3 = 4 and -9/3 = -3.

Step 4. P (closer to A) = A + one step = (4 + 4, 7 - 3) = (8, 4).

Step 5. Q = A + two steps = (4 + 8, 7 - 6) = (12, 1). Equivalently Q = B - one step = (16 - 4, -2 + 3) = (12, 1). Both give the same.

Step 6. Midpoint of P and Q = ((8 + 12)/2, (4 + 1)/2) = (10, 2.5).

Step 7. Midpoint of A and B = ((4 + 16)/2, (7 - 2)/2) = (10, 2.5). They coincide - the check passes.

» **Why this step?** The middle of the two trisectors is always the midpoint of the whole segment - a beautiful structural fact worth remembering for instant verification.

Answer: P(8, 4), Q(12, 1); both midpoints equal (10, 2.5), verification passes.

Hinglish: *Trisection matlab segment ko 3 barabar tukdon mein kaatna. Fundaa simple: total gap ko 3 se divide karo, wo ek 'step' hai. P = A + ek step, Q = A + do step. Yahan steps (4, -3) the. Aur jo self-check diya na - PQ ka midpoint hamesha AB ke midpoint ke barabar hota hai - ye free verification hai, exam mein zaroor lagao. (10, 2.5) = (10, 2.5), perfect match.*

QUESTION 102 | High | Missing rectangle vertex (T6/T8)

Q102. Three vertices of a rectangle are A(1, 2), B(5, 2) and C(5, 7). Find the fourth vertex D, and verify the rectangle property that its diagonals are equal.

SOLUTION — step by step

Step 1. Given: three consecutive vertices. Needed: D and the diagonal check.

Step 2. AB is horizontal (y = 2 for both) with length 4; BC is vertical (x = 5 for both) with length 5.

Step 3. In a rectangle the fourth side DA must be vertical (above A) and CD horizontal - so D must sit directly above A at the same height as C.

Step 4. Directly above A(1, 2) means x = 1; at the same height as C means y = 7. So D = (1, 7).

Step 5. Verify with side lengths: DA from (1, 7) to (1, 2) has length 5, CD from (5, 7) to (1, 7) has length 4 - matching the opposite sides.

Step 6. Diagonal AC = $\sqrt{((5 - 1)^2 + (7 - 2)^2)} = \sqrt{(16 + 25)} = \sqrt{41}$.

Step 7. Diagonal BD = $\sqrt{((1 - 5)^2 + (7 - 2)^2)} = \sqrt{(16 + 25)} = \sqrt{41}$. Equal, so the rectangle property holds.

Answer: D = (1, 7); diagonals AC = BD = √41, confirming the rectangle.

Hinglish: *Rectangle ka chautha vertex pattern se milta hai: A ke seedha upar, C ke level par. x = 1 (A jaisa), y = 7 (C jaisa). Phir property check - diagonals barabar? Dono √41. Yahan √41 ko simplify nahi kar sakte (41 prime hai), toh root form hi final hai. Notice karo: har rectangle mein diagonals barabar hote hain - ye test humne parallelogram question (Q23) mein bhi use kiya tha, wahan fail hua tha, yahan pass.*

QUESTION 103 | High | Area of right triangle from coordinates (T12)

Q103. A triangle has vertices O(0, 0), A(8, 0) and B(0, 6). Show that it is right-angled, find its hypotenuse, and compute its area and perimeter.

SOLUTION — step by step

Step 1. Given: three vertices, two of them on the axes. Needed: right angle, hypotenuse, area, perimeter.

Step 2. OA lies along the x-axis: length 8. OB lies along the y-axis: length 6.

Step 3. The x-axis and y-axis are perpendicular, and OA, OB lie along them, so the angle at O is 90 degrees.

Step 4. Hypotenuse AB = $\sqrt{(8^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$ (the 6-8-10 triple).

Step 5. Area = (1/2) x OA x OB = (1/2) x 8 x 6 = 24 square units.

Step 6. Perimeter = 8 + 6 + 10 = 24 units. A charming coincidence: area 24 and perimeter 24.

Answer: Right angle at O; hypotenuse AB = 10; area = 24 sq units; perimeter = 24 units.

Hinglish: *Jab triangle ke do sides axes par baithe hon, right angle free mein milta hai - axis parallel hi perpendicular hote hain. Phir 8-6-10 triple se hypotenuse 10. Area = 24, perimeter bhi 24 - dono number same! Aisi coincidences questions mein jaan-bujh ke rakhi jaati hain, toh apne practice questions mein bhi aisi numbers chuno toh checking easy hoti hai.*

QUESTION 104 | High | Square from its diagonal (T14)

Q104. Two opposite corners of a square are P(-3, 2) and R(5, -4). Find the length of the diagonal, the side of the square, its area, and the coordinates of the other two corners.

SOLUTION — step by step

Step 1. Given: diagonal endpoints of a square. Needed: diagonal, side, area, remaining vertices.

Step 2. Diagonal PR = $\sqrt{((5 - (-3))^2 + (-4 - 2)^2)} = \sqrt{(8^2 + (-6)^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$.

Step 3. For a square of side s, the diagonal is $s\sqrt{2}$, so $s = 10/\sqrt{2} = 5\sqrt{2}$.

Step 4. Area = $s^2 = (5\sqrt{2})^2 = 50$ square units. (Shortcut: area = diagonal$^2/2 = 100/2 = 50$.)

Step 5. Centre of the square = midpoint of PR = ((-3 + 5)/2, (2 - 4)/2) = (1, -1).

Step 6. Half-diagonal vector from centre to R is (5 - 1, -4 - (-1)) = (4, -3). To build the other diagonal, swap the components and flip one sign: (3, 4).

Step 7. Other corners: centre + (3, 4) = (4, 3) and centre - (3, 4) = (-2, -5).

Step 8. Verify: Q(4, 3) to P(-3, 2): $\sqrt{(7^2 + 1)} = \sqrt{50} = 5\sqrt{2}$ (a side). Q to R(5, -4): $\sqrt{(1 + 49)} = \sqrt{50}$ (a side). Q to S(-2, -5): $\sqrt{(36 + 64)} = 10$ (the other diagonal). All consistent.

» **Why this step?** The swap-and-flip move rotates the half-diagonal by 90 degrees - it is just Pythagoras thinking in vector form, and it is fully verifiable with the distance formula.

Answer: Diagonal = 10; side = $5\sqrt{2}$; area = 50 sq units; other corners Q(4, 3) and S(-2, -5).

Hinglish: *Square ka diagonal diya hai toh side = diagonal/$\sqrt{2} = 5\sqrt{2}$, area = 50. Baaki do corners nikalne ka tareeka: pehle centre (midpoint) lo - (1, -1). Half-diagonal (4, -3) hai. Uske components SWAP karo aur ek sign flip karo: (3, 4) ban jayega - ye 90 degree ghuma hua half-diagonal hai. Centre mein plus-minus lagao: (4, 3) aur (-2, -5). Phir verification: sab sides $5\sqrt{2}$, doosra diagonal 10. Ye wala question HOTS-zone ka gateway hai - rotation ka idea samajh gaye toh aage Olympiad-level coordinate proofs easy lagenge.*

QUESTION 105 | High | Collinearity - negative coordinates (T9)

Q105. Check, without plotting, whether the points (-2, -1), (0, 0) and (4, 3) lie on one straight line. Show your exact arithmetic.

SOLUTION — step by step

Step 1. Given: three points with negative coordinates. Needed: collinearity verdict with exact values.

Step 2. d1 = distance from (-2, -1) to (0, 0) = $\sqrt{(2^2 + 1^2)} = \sqrt{5}$.

Step 3. d2 = distance from (0, 0) to (4, 3) = $\sqrt{(16 + 9)} = \sqrt{25} = 5$.

Step 4. d3 = distance from (-2, -1) to (4, 3) = $\sqrt{(6^2 + 4^2)} = \sqrt{(36 + 16)} = \sqrt{52} = 2\sqrt{13}$.

Step 5. Collinearity requires the largest to equal the sum of the other two: is $2\sqrt{13} = \sqrt{5} + 5$?

Step 6. Compare squares of both sides: left squared is 52. Right squared is $(\sqrt{5} + 5)^2 = 5 + 25 + 10\sqrt{5} = 30 + 10\sqrt{5}$ approx 30 + 22.36 = 52.36.

Step 7. Since 52 differs from $30 + 10\sqrt{5}$ (equivalently $\sqrt{5} + 5$ approx 7.236 while $2\sqrt{13}$ approx 7.211), the equality FAILS.

Step 8. So the three points are NOT collinear - the middle point sits slightly off the line through the outer two.

» **Why this step?** When radicals are involved, compare by squaring (both sides positive) rather than by decimals - decimals can mislead near-miss cases.

Answer: Not collinear: $\sqrt{5} + 5$ is not equal to $2\sqrt{13}$ (52 is not equal to $30 + 10\sqrt{5}$).

Hinglish: *Negative coordinates aane par bhi process same hai. Yahan twist ye hai ki distances roots mein hain - $\sqrt{5}$, 5, $2\sqrt{13}$. Comparison exact karo: dono side ko square karo. Left 52, right $30 + 10\sqrt{5}$ approx 52.36. Thoda sa difference hai - matlab NOT collinear. Ye question isliye rakha hai ki tum consider karo ki values kabhi approximate judge na karo. Agar maine jaldi mein 7.24 vs 7.21 dekh ke 'barabar hai' bol diya hota, toh galat ho jaata!*

QUESTION 106 | High | Applied screen/circle problem (T11/T10)

Q106. A computer screen is a rectangle 800 pixels wide and 600 pixels high, with the origin at the bottom-left corner. A circular icon of radius 80 has centre A(100, 150); another of radius 100 has centre B(250, 230). (i) Does any part of either icon stick out of the screen? (ii) Do the two icons overlap each other?

SOLUTION — step by step

Step 1. Given: screen 800 x 600, two circles with centres and radii. Needed: boundary check and overlap check.

Step 2. Icon A: leftmost x = 100 - 80 = 20 (inside, positive), rightmost x = 100 + 80 = 180 (inside, below 800), lowest y = 150 - 80 = 70 (inside), highest y = 150 + 80 = 230 (inside, below 600). Icon A is fully on the screen.

Step 3. Icon B: leftmost x = 250 - 100 = 150, rightmost = 250 + 100 = 350, lowest y = 230 - 100 = 130, highest y = 330. All within 0 to 800 and 0 to 600. Fully on screen too.

Step 4. Overlap test for two circles: compare the centre distance d with r1 + r2 and |r1 - r2|.

Step 5. Centre distance AB = $\sqrt{((250 - 100)^2 + (230 - 150)^2)} = \sqrt{(150^2 + 80^2)} = \sqrt{(22500 + 6400)} = \sqrt{28900}$ = 170 (the 8-15-17 triple scaled by 10).

Step 6. Now r1 + r2 = 180 and |r1 - r2| = 20. Since 20 is less than 170 and 170 is less than 180, the circles intersect in two points - they overlap partially.

Step 7. So: no icon leaves the screen, and the icons DO overlap.

Answer: (i) Both icons lie fully inside the screen. (ii) Yes, the icons overlap: AB = 170 lies strictly between |100 - 80| = 20 and 80 + 100 = 180.

Hinglish: *Screen check: har icon ka extreme left/right/top/bottom compute karo aur 0-800, 0-600 range se compare karo. Dono andar hain. Overlap ka test yaad rakho: centres ki doori vs radii ka sum. Distance 170 nikla (8-15-17 triple x 10!). 170 < 180 isliye overlap hota hai, aur 170 > 20 isliye poora ek doosre ke andar nahi hai - partial overlap. Ye 2-line test high-school geometry ka swiss knife hai.*

QUESTION 107 | High | Locus: equidistant from both axes (T14)

Q107. Describe the set of all points that are equidistant from the x-axis and the y-axis. If such a point must also lie at distance 5 from the origin, list ALL points that satisfy both conditions (exact values).

SOLUTION — step by step

Step 1. Given: equidistant from both axes, then distance-5 filter. Needed: locus, then exact points.

Step 2. Distance from the x-axis is |y|; distance from the y-axis is |x|. Equality |x| = |y| means y = x or y = -x.

Step 3. So the locus is the pair of lines y = x and y = -x (the two diagonals through the origin).

Step 4. Now add: $\sqrt{(x^2 + y^2)} = 5$. On the line y = x this becomes $\sqrt{(2x^2)} = \sqrt{2}\,|x| = 5$, so $|x| = 5/\sqrt{2} = (5\sqrt{2})/2$.

Step 5. That gives two points: $((5\sqrt{2})/2, (5\sqrt{2})/2)$ and $(-(5\sqrt{2})/2, -(5\sqrt{2})/2)$.

Step 6. On the line y = -x the same computation gives $((5\sqrt{2})/2, -(5\sqrt{2})/2)$ and $(-(5\sqrt{2})/2, (5\sqrt{2})/2)$.

Step 7. Total: four points, one in each quadrant, all at distance 5 from O and equidistant from both axes. Verify one: $((5\sqrt{2})/2)^2 + ((5\sqrt{2})/2)^2 = 25/2 + 25/2 = 25$, distance 5. Correct.

Answer: Locus: the two lines y = x and y = -x. The four points are (plus/minus $5\sqrt{2}/2$, plus/minus $5\sqrt{2}/2$) with matching signs on y = x and opposite signs on y = -x.

Hinglish: *Dono axis se barabar door = |x| = |y| = do diagonal lines y = x aur y = -x. Ab distance 5 bhi chahiye: $\sqrt{(x^2 + y^2)} = 5$. y = x rakhne par $2x^2 = 25$, $x = 5/\sqrt{2} = 5\sqrt{2}/2$. Rationalise kar dena toh aur saaf dikhta hai. Chaar points bane - har quadrant mein ek, sab circle (radius 5) par aur dono diagonals par. Locus thinking + circle membership ka combo - yahi HOTS ka taste hai.*

QUESTION 108 | High | Triple spotting at scale (T4/T12)

Q108. A triangle has vertices (0, 0), (5, 0) and (0, 12). Identify the right angle, find all three sides, and compute the area and the perimeter. Then state the general rule for a right triangle with legs a and b.

SOLUTION — step by step

Step 1. Given: vertices on both axes. Needed: sides, area, perimeter, general rule.

Step 2. The side from (0, 0) to (5, 0) lies on the x-axis (length 5); the side from (0, 0) to (0, 12) lies on the y-axis (length 12).

Step 3. These two axis lines are perpendicular, so the right angle is at the origin.

Step 4. Hypotenuse = $\sqrt{(5^2 + 12^2)} = \sqrt{(25 + 144)} = \sqrt{169} = 13$ (the 5-12-13 triple).

Step 5. Area = (1/2) x 5 x 12 = 30 square units.

Step 6. Perimeter = 5 + 12 + 13 = 30 units. Again area equals perimeter - these triple-based triangles love that coincidence.

Step 7. General rule: for a right triangle with legs a and b, hypotenuse = $\sqrt{(a^2 + b^2)}$, area = ab/2.

Answer: Right angle at the origin; sides 5, 12, 13; area = 30 sq units; perimeter = 30 units; general: hypotenuse $\sqrt{(a^2 + b^2)}$, area ab/2.

Hinglish: *5-12-13 triple ka full Dabangg avatar: area bhi 30, perimeter bhi 30. Chhota triangle 3-4-5 ka bhi area 6 aur perimeter 12 tha - wahan match nahi tha, par 5-12-13 aur 6-8-10 mein area = perimeter nikalta hai (8-6-10: area 24, perimeter 24). Aisi patterns notice karna maths ko interesting banata hai. General rule likhna zaroori hai - examiner ko dikhta hai ki tum case se general tak soch rahe ho.*

QUESTION 109 | High | Rectangle verification kit (T6)

Q109. K(1, 1), L(5, 1), M(5, 4) and N(1, 4) are joined in order. Using distances only, prove the figure is a rectangle, and find its perimeter, area and the length of each diagonal.

SOLUTION — step by step

Step 1. Given: four points in cyclic order. Needed: prove rectangle; perimeter, area, diagonals.

Step 2. KL: y fixed at 1, length |5 - 1| = 4 (horizontal).

Step 3. LM: x fixed at 5, length |4 - 1| = 3 (vertical).

Step 4. MN: y fixed at 4, length |5 - 1| = 4 (horizontal). NK: x fixed at 1, length |4 - 1| = 3 (vertical).

Step 5. Opposite sides are equal, and every side is axis-parallel, so adjacent sides meet at right angles: the figure is a rectangle.

Step 6. Diagonals: KM = $\sqrt{((5 - 1)^2 + (4 - 1)^2)} = \sqrt{(16 + 9)} = \sqrt{25} = 5$. LN = $\sqrt{((1 - 5)^2 + (4 - 1)^2)} = \sqrt{(16 + 9)} = 5$. Equal diagonals confirm the rectangle property.

Step 7. Perimeter = 2(4 + 3) = 14 units. Area = 4 x 3 = 12 square units. Both diagonals are 5.

Answer: It is a rectangle (equal opposite sides, axis-parallel sides, equal diagonals of length 5); perimeter 14 units, area 12 sq units.

Hinglish: *Ye question rectangle-proof ka full kit hai: (1) opposite sides barabar, (2) sab sides axis-parallel toh right angles automatic, (3) diagonals barabar - teeno checks ho gaye. 3-4-5 triple yahan diagonals mein chhupa hai! Perimeter 14, area 12, diagonal 5. Aise structured answers examiner ko bahut pasand aate hain kyunki har property alag line par verified hai.*

QUESTION 110 | High | Quadrant-restricted distances (T1/T4)

Q110. A point P lies in Quadrant II. Its perpendicular distance from the x-axis is 5 and from the y-axis is 3. Write the coordinates of P and find its distance from the origin. If a twin point P' in Quadrant I has the same axis distances, what is the length PP'?

SOLUTION — step by step

Step 1. Given: Quadrant II point with axis distances 5 (from x-axis) and 3 (from y-axis). Needed: P, OP, and PP'.

Step 2. Distance from the x-axis is the size of the y-coordinate: $|y| = 5$. Distance from the y-axis is $|x| = 3$.

Step 3. Quadrant II means x negative, y positive. So $x = -3$ and $y = +5$, giving P(-3, 5).

Step 4. $OP = \sqrt{((-3)^2 + 5^2)} = \sqrt{(9 + 25)} = \sqrt{34}$.

Step 5. P' in Quadrant I with the same distances: $x = +3$, $y = +5$, so P'(3, 5).

Step 6. PP' joins (-3, 5) to (3, 5): same y, so $PP' = |3 - (-3)| = 6$ (horizontal segment).

Step 7. Note the traps avoided: the distance from the x-axis (5) became the y-coordinate, and the Quadrant II condition fixed both signs.

» Why this step? Translating 'distance from the x-axis' into 'absolute y-value' is the key step; students who mix this up put 3 and 5 in the wrong slots.

Answer: P(-3, 5); OP = √34; P'(3, 5); PP' = 6 units.

Hinglish: *Sabse pehle translation: x-axis se doori 5 matlab $|y| = 5$; y-axis se doori 3 matlab $|x| = 3$. Ab quadrant II ka rule lagao - x negative, y positive: P(-3, 5). OP = √34 (koi triple nahi bana, root form chal raha hai). Quadrant I wala twin (3, 5). Dono ka y same, toh distance sirf x ka gap = 6. Ye question pure sign-discipline ka test hai - ek sign ulta kiya toh poora answer flip.*

Level 4 · Boss Zone: advanced + HOTS (Olympiad-lite):

QUESTION 111 | HOTS | Two intersection points on an axis (T5)

Q111. Find ALL points on the x-axis that lie at distance 10 from the point C(3, -6). Explain geometrically why there must be two answers.

SOLUTION — step by step

Step 1. Given: points (x, 0) at distance 10 from C(3, -6). Needed: all such x.

Step 2. Distance equation: $\sqrt{((x - 3)^2 + (0 - (-6))^2)} = 10$.

Step 3. Square both sides: $(x - 3)^2 + 36 = 100$.

Step 4. So $(x - 3)^2 = 64$, giving $x - 3 = 8$ or $x - 3 = -8$.

Step 5. Thus $x = 11$ or $x = -5$. The points are (11, 0) and (-5, 0).

Step 6. Verify: from (11, 0) to (3, -6): $\sqrt{(64 + 36)} = \sqrt{100} = 10$. From (-5, 0) to (3, -6): $\sqrt{(64 + 36)} = 10$. Both valid.

Step 7. Geometry: a circle of radius 10 centred at C(3, -6) cuts the x-axis at exactly two points because the centre sits 6 units above the axis, which is less than the radius 10 - the circle dips below the axis on both sides of $x = 3$.

» **Why this step?** The equation $(x - 3)^2 = 64$ is a disguised two-case problem; the geometric picture (circle cutting a line) tells you in advance how many answers to expect.

Answer: (11, 0) and (-5, 0); two answers because the circle of radius 10 around C(3, -6) crosses the x-axis twice.

Hinglish: *Circle-axis ka milaap socho: C se radius 10 ka circle banao, wo x-axis ko kata kyun? Kyunki C axis se sirf 6 door hai aur radius 10 hai - circle axis ke paar niche tak jaata hai, dono taraf se cut karta hai. Equation mein bhi wahi dikh raha hai: $(x-3)^2 = 64$ ke do roots: $x = 11$ aur $x = -5$. Jab bhi square = positive number dikhe, do answers ready rakho. Aur agar centre axis se doori radius se ZYADA hoti, toh zero answers aate - wo case bhi soch ke dekho!*

QUESTION 112 | HOTS | Circumcentre discovery (T10/T14)

Q112. For the right triangle with vertices A(0, 0), B(8, 0) and C(8, 6), find the point that is equidistant from all three vertices. Where does it sit, and what pattern do you notice?

SOLUTION — step by step

Step 1. Given: right triangle with legs on the axes... (adapted: right angle at B since AB is horizontal and BC is vertical). Needed: the common-equidistance point.

Step 2. AB = 8 (horizontal), BC = 6 (vertical), so the right angle is at B(8, 0), and AC is the hypotenuse.

Step 3. $AC = \sqrt{((8 - 0)^2 + (6 - 0)^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$.

Step 4. Claim: the midpoint of the hypotenuse is equidistant from all three vertices. Midpoint M of AC = $((0 + 8)/2, (0 + 6)/2) = (4, 3)$.

Step 5. Check $MA = \sqrt{(4^2 + 3^2)} = 5$. $MC = \sqrt{((8 - 4)^2 + (6 - 3)^2)} = \sqrt{(16 + 9)} = 5$.

Step 6. MB = $\sqrt{((8 - 4)^2 + (0 - 3)^2)} = \sqrt{(16 + 9)} = 5$. All three distances are 5 - verified.

Step 7. Pattern: in a right triangle, the midpoint of the hypotenuse is equidistant from all three vertices (it is the centre of the circle passing through the three corners). This is the coordinate-proof version of a famous theorem.

Answer: M(4, 3), the midpoint of the hypotenuse, is equidistant (distance 5) from all three vertices.

Hinglish: *Ye question ek theorem khud discover karwa raha hai! Right angle B par hai (horizontal + vertical legs), hypotenuse AC = 10. Uska midpoint (4, 3) lo - teeno vertices se distance 5, 5, 5. Matlab: right triangle mein hypotenuse ka midpoint hi wo magical point hai jo teeno corners se barabar door baithta hai (circumcentre). 3-4-5 triple yahan bhi kaam kar gaya. Ye result yaad rakho - contest maths mein baar-baar kaam aayega.*

QUESTION 113 | HOTS | Sum-of-distances locus (T14)

Q113. A(-4, 0) and B(4, 0) are two fixed points. P(0, y) lies on the y-axis such that PA + PB = 10. Find all possible y. Why is y = 0 NOT a solution?

SOLUTION — step by step

Step 1. Given: PA + PB = 10 with P on the y-axis. Needed: y values.

Step 2. PA = $\sqrt{((0 + 4)^2 + y^2)} = \sqrt{(16 + y^2)}$. By symmetry PB has the same value $\sqrt{(16 + y^2)}$.

Step 3. So the condition reads $2\sqrt{(16 + y^2)} = 10$, hence $\sqrt{(16 + y^2)} = 5$.

Step 4. Square: $16 + y^2 = 25$, so $y^2 = 9$, giving y = 3 or y = -3.

Step 5. Verify y = 3: PA = $\sqrt{(16 + 9)} = 5$, PB = 5, total 10. Same for y = -3. Both work.

Step 6. Why not y = 0: then PA = PB = 4 and the sum is 8, which is short of 10 - P at the origin sits strictly between A and B, so PA + PB just equals AB = 8, the minimum possible value. To reach 10 the point must climb off the line.

Answer: y = 3 or y = -3; y = 0 fails because PA + PB = AB = 8 there, the minimum, not 10.

Hinglish: *Symmetry dekho: A aur B origin ke dono taraf barabar door hain, isliye PA = PB hoga hi. Toh 2 x distance = 10, distance = 5. 16 + y^2 = 25 se y = plus-minus 3. Aur y = 0 kyun nahi: origin par P ke paas PA + PB = AB hi hota hai = 8 - shortest possible. 10 chahiye toh P ko upar ya neeche jhukna padega (y = 3 ya -3). Ye question aage ke classes mein ellipse ka darwaza hai - PA + PB constant wala idea.*

QUESTION 114 | HOTS | Box method for area (T12/T14)

Q114. A triangle has vertices A(0, 0), B(1, 2) and C(3, 1). Show that it is isosceles, and find its area using the box (enclosing rectangle) method.

SOLUTION — step by step

Step 1. Given: three lattice points. Needed: isosceles proof and area.

Step 2. AB = $\sqrt{(1^2 + 2^2)} = \sqrt{5}$. BC = $\sqrt{((3 - 1)^2 + (1 - 2)^2)} = \sqrt{(4 + 1)} = \sqrt{5}$. AC = $\sqrt{(3^2 + 1^2)} = \sqrt{10}$.

Step 3. AB = BC = $\sqrt{5}$, so the triangle is isosceles (apex at B).

Step 4. Box method: enclose the triangle in the smallest axis-aligned rectangle: corners (0, 0), (3, 0), (3, 2), (0, 2). Box area = 3 x 2 = 6.

Step 5. The box minus the triangle leaves three right triangles: T1 below AB with legs 1 and 2 (area 1), T2 right of BC with legs 2 and 1 (area 1), T3 above AC with legs 3 and 1 (area 1.5).

Step 6. Area of ABC = 6 - 1 - 1 - 1.5 = 2.5 square units.

Step 7. Cross-check by another triangle with same side data is not needed; instead sanity-check magnitude: the triangle fits inside a 3 x 2 box, so 2.5 is plausible.

» Why this step? When the triangle is not right-angled and the base-height pair is not obvious, the box method computes area using only right triangles - fully within Chapter 1 tools.

Answer: Isosceles (AB = BC = $\sqrt{5}$); area = 2.5 square units via the box method.

Hinglish: *Jab triangle tedha ho aur base-height na dikhe, box method use karo: triangle ko chhote rectangle mein band karo, rectangle ka area lo, aur charo taraf ke right triangles ghata do. 6 - 1 - 1 - 1.5 = 2.5. Ye technique squares ke root wale triangles ka area nikalne ka desi jugaad hai - bina kisi naye formula ke. Aur dekho, AB = BC = $\sqrt{5}$ se isosceles bhi proof ho gaya. Do kaam ek question mein.*

QUESTION 115 | HOTS | Chord comparison on a circle (T10/T4)

Q115. Three points A(3, 4), B(-4, 3) and C(0, -5) all lie on the circle of radius 5 centred at the origin (verify this). Find the lengths of the chords AB, BC and CA, and state which chord is the longest.

SOLUTION — step by step

Step 1. Given: three points, circle radius 5. Needed: verify membership, three chord lengths, longest chord.

Step 2. Membership: A: 9 + 16 = 25. B: 16 + 9 = 25. C: 0 + 25 = 25. All equal to $r^2 = 25$, so all three lie on the circle.

Step 3. AB = $\sqrt{((-4 - 3)^2 + (3 - 4)^2)} = \sqrt{(49 + 1)} = \sqrt{50} = 5\sqrt{2}$ (approximately 7.07).

Step 4. BC = $\sqrt{((0 + 4)^2 + (-5 - 3)^2)} = \sqrt{(16 + 64)} = \sqrt{80} = 4\sqrt{5}$ (approximately 8.94).

Step 5. CA = $\sqrt{((3 - 0)^2 + (4 + 5)^2)} = \sqrt{(9 + 81)} = \sqrt{90} = 3\sqrt{10}$ (approximately 9.49).

Step 6. Comparing exact squares: 90 > 80 > 50, so CA > BC > AB.

Step 7. The longest chord is CA with length $3\sqrt{10}$. (No chord can exceed the diameter 10, and indeed all three are below 10.)

Answer: All three lie on the circle; AB = $5\sqrt{2}$, BC = $4\sqrt{5}$, CA = $3\sqrt{10}$; the longest chord is CA.

Hinglish: *Pehle membership - teeno 25 de rahe hain, sab circle par. Phir chords: √50, √80, √90. Compare karne ka clean tareeka: squares compare karo (50 < 80 < 90), roots ki zaroorat nahi. Longest CA = 3√10 approx 9.49. Aur ek boundary-check: diameter 10 hai, koi chord 10 se bada nahi - hamara answer us limit ke andar hi hai. Yahi internal consistency marks dilati hai.*

QUESTION 116 | HOTS | Parallelogram via diagonal midpoints (T8/T6)

Q116. A(1, 2), B(4, 3) and C(6, 7) are three vertices of parallelogram ABCD taken in order. Find the fourth vertex D using the fact that the diagonals of a parallelogram bisect each other, and verify your answer.

SOLUTION — step by step

Step 1. Given: three consecutive vertices of a parallelogram. Needed: D.

Step 2. Diagonals of a parallelogram bisect each other, so midpoint of AC must equal midpoint of BD.

Step 3. Midpoint of AC = ((1 + 6)/2, (2 + 7)/2) = (3.5, 4.5).

Step 4. Let D = (x, y). Midpoint of BD = ((4 + x)/2, (3 + y)/2). Set equal to (3.5, 4.5).

Step 5. From x: (4 + x)/2 = 3.5 gives x = 3. From y: (3 + y)/2 = 4.5 gives y = 6.

Step 6. So D = (3, 6).

Step 7. Verify as a parallelogram by sides: AB = $\sqrt{(3^2 + 1^2)} = \sqrt{10}$, DC = $\sqrt{((6 - 3)^2 + (7 - 6)^2)} = \sqrt{(9 + 1)} = \sqrt{10}$; BC = $\sqrt{(2^2 + 4^2)} = \sqrt{20} = 2\sqrt{5}$, AD = $\sqrt{((3 - 1)^2 + (6 - 2)^2)} = \sqrt{(4 + 16)} = \sqrt{20}$. Opposite sides equal - consistent.

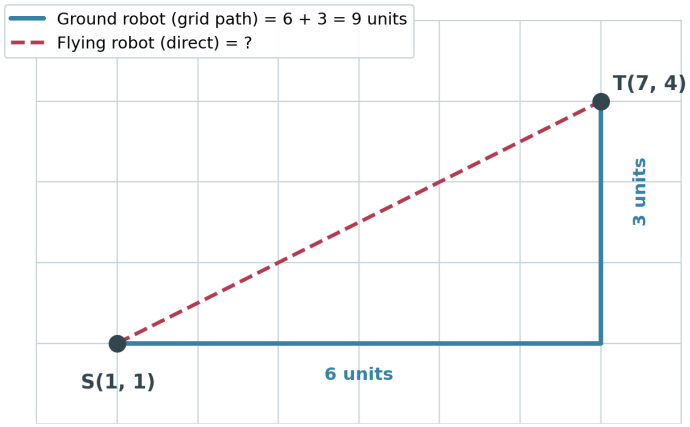
» **Why this step?** The diagonal-midpoint method converts a geometry condition into two tiny linear equations - far safer than guessing a translation vector.

Answer: D = (3, 6); verified since AB = DC = $\sqrt{10}$ and BC = AD = $2\sqrt{5}$.

Hinglish: *Parallelogram ka chautha vertex nikalne ka RULE: diagonals ka midpoint same hota hai. AC ka midpoint lo, phir BD ka midpoint uske barabar rakho - do chhote equations mil gaye: x = 3, y = 6. Verification mein opposite sides bhi barabar nikli (√10 aur 2√5 ke pairs). Ye midpoint-trick Q31 (trisection) ke saath milke midpoint-based toolkit complete karta hai: aage class 10 mein yehi idea hi midpoint formula ka base banega.*

QUESTION 117 | HOTS | Grid-walk vs direct distance (T11/T4)

Q117. A delivery robot moves on a grid of unit squares. It must travel from station S(1, 1) to station T(7, 4). (i) What is the length of the shortest path using only horizontal and vertical moves? (ii) What is the straight-line distance? (iii) A second robot flies directly. How much distance does it save compared to the ground robot?



SOLUTION — step by step

Step 1. Given: grid unit 1, S(1, 1) to T(7, 4). Needed: street length, direct length, saving.

Step 2. Horizontal gap = 7 - 1 = 6 units; vertical gap = 4 - 1 = 3 units.

Step 3. Ground robot (only axis moves): 6 + 3 = 9 units. Every such path has exactly this length - no order of moves can beat it.

Step 4. Flying robot (straight line): $\sqrt{(6^2 + 3^2)} = \sqrt{(36 + 9)} = \sqrt{45} = 3\sqrt{5}$ (approximately 6.71 units).

Step 5. Saving = $9 - 3\sqrt{5}$ (exactly), approximately 9 - 6.71 = 2.29 units.

Step 6. Keep the exact answer $9 - 3\sqrt{5}$ in the final statement; the decimal is only for intuition.

Answer: (i) 9 units (ii) $3\sqrt{5}$ units (approx 6.71) (iii) saving $9 - 3\sqrt{5}$ approx 2.29 units.

Hinglish: *Grid-walk ka funda: horizontal steps + vertical steps hamesha fix hote hain (yahan 6 + 3 = 9), chahe kisi bhi order mein chalo. Direct distance Pythagoras se $3\sqrt{5}$. Saving $9 - 3\sqrt{5}$ approx 2.3 units. Ye question Q26 (city auto) ka bhai hai - wahan 200 m blocks the, yahan 1 unit. Same skeleton, naye numbers - yahi 'question structure' samajhna original practice hai, ratna nahi.*

QUESTION 118 | HOTS | Equally spaced collinear chain (T9/T8)

Q118. P(1, 2), Q(4, 6) and R(7, 10) are three points. Show they are collinear and equally spaced. Then find the point T on the same line, beyond R, with QT = 10.

SOLUTION — step by step

Step 1. Given: three points. Needed: collinearity, equal spacing, then T.

Step 2. PQ = $\sqrt{((4 - 1)^2 + (6 - 2)^2)} = \sqrt{(9 + 16)} = 5$.

Step 3. QR = $\sqrt{((7 - 4)^2 + (10 - 6)^2)} = \sqrt{(9 + 16)} = 5$.

Step 4. PR = $\sqrt{((7 - 1)^2 + (10 - 2)^2)} = \sqrt{(36 + 64)} = 10$.

Step 5. Since PR = PQ + QR = 10, the points are collinear, and PQ = QR = 5 shows Q is exactly the midpoint - equally spaced.

Step 6. The repeated step from P to Q is (+3, +4). The same step continues from Q to R, and one more step lands at T = (7 + 3, 10 + 4) = (10, 14).

Step 7. Check QT: from Q(4, 6) to T(10, 14) is two steps: 2 x 5 = 10. Verified.

» **Why this step?** A constant coordinate-step (+3, +4) between consecutive points is the signature of equal spacing on a line - recognise the pattern and distances become trivial.

Answer: Collinear with PQ = QR = 5, PR = 10; T = (10, 14) with QT = 10 verified.

Hinglish: *Pattern-spotting question! Har agle point mein (+3, +4) jud raha hai - 3-4-5 triple ka repeated use. PQ = QR = 5, PR = 10: collinear + equally spaced confirmed. Ab 'beyond R with QT = 10' ka matlab: Q se 2 step aage = R se 1 step aage = T(10, 14). Verify kiya QT = 10. Aise 'chain' questions mein step-pakadna hi asli skill hai, har baar distance formula chalana nahi.*

QUESTION 119 | HOTS | Chord and centre distance (T10/T8)

Q119. A circle of radius 10 is centred at the origin. A and B(−8, 6) are two points on it, with A(6, 8). Find the length of chord AB and the perpendicular distance of the chord from the centre. Verify the right-triangle relation between them.

SOLUTION — step by step

Step 1. Given: circle $x^2 + y^2 = 100$, chord joining A(6, 8) and B(-8, 6). Needed: chord length, distance from centre, consistency check.

Step 2. First confirm membership: A: 36 + 64 = 100. B: 64 + 36 = 100. Both on the circle.

Step 3. Chord AB = $\sqrt{((-8 - 6)^2 + (6 - 8)^2)} = \sqrt{(196 + 4)} = \sqrt{200} = 10\sqrt{2}$.

Step 4. Midpoint M of the chord = ((6 + (-8))/2, (8 + 6)/2) = (-1, 7).

Step 5. The line from the centre to the midpoint of a chord is perpendicular to the chord, so the required distance is OM = $\sqrt{((-1)^2 + 7^2)} = \sqrt{50} = 5\sqrt{2}$.

Step 6. Right-triangle check: half-chord = $5\sqrt{2}$, OM = $5\sqrt{2}$, radius = 10. Indeed $(5\sqrt{2})^2 + (5\sqrt{2})^2 = 50 + 50 = 100 = 10^2$.

Step 7. Everything is consistent - the chord, its distance from the centre and the radius obey the Baudhayana-Pythagoras relation.

Answer: AB = $10\sqrt{2}$; distance from centre = $5\sqrt{2}$; verified since $(5\sqrt{2})^2 + (5\sqrt{2})^2 = 100 = \text{radius}^2$.

Hinglish: *Do kaam ek saath: chord length (distance formula se $10\sqrt{2}$) aur centre se doori. Fundaa: chord ke midpoint se centre ko jodo - wo line chord par perpendicular hoti hai, toh wahi distance hai. OM = $\sqrt{50} = 5\sqrt{2}$. Phir master-check: (half-chord)^2 + (distance)^2 = radius^2. 50 + 50 = 100. Perfect. Ye relation har circle-chord question ka backbone hai - yahin se 'perpendicular from centre bisects chord' wali theory aati hai.*

QUESTION 120 | HOTS | Full square proof chain (T14)

Q120. Prove that the points P(1, 7), Q(4, 2), R(-1, -1) and S(-4, 4), taken in this order, are the vertices of a square. You must verify ALL necessary properties using the distance formula and midpoints.

SOLUTION — step by step

Step 1. Given: four points in cyclic order. Needed: full square proof.

Step 2. PQ = $\sqrt{((4 - 1)^2 + (2 - 7)^2)} = \sqrt{(9 + 25)} = \sqrt{34}$.

Step 3. QR = $\sqrt{((-1 - 4)^2 + (-1 - 2)^2)} = \sqrt{(25 + 9)} = \sqrt{34}$.

Step 4. RS = $\sqrt{((-4 + 1)^2 + (4 + 1)^2)} = \sqrt{(9 + 25)} = \sqrt{34}$.

Step 5. SP = $\sqrt{((1 + 4)^2 + (7 - 4)^2)} = \sqrt{(25 + 9)} = \sqrt{34}$. All four sides equal: it is at least a rhombus-shaped candidate.

Step 6. Diagonals: PR = $\sqrt{((-1 - 1)^2 + (-1 - 7)^2)} = \sqrt{(4 + 64)} = \sqrt{68} = 2\sqrt{17}$. QS = $\sqrt{((-4 - 4)^2 + (4 - 2)^2)} = \sqrt{(64 + 4)} = \sqrt{68}$. Diagonals are equal.

Step 7. Bisecting check: midpoint of PR = $((1 + (-1))/2, (7 + (-1))/2) = (0, 3)$. Midpoint of QS = $((4 + (-4))/2, (2 + 4)/2) = (0, 3)$. Diagonals bisect each other, so the figure is a parallelogram.

Step 8. A parallelogram with all sides equal is a rhombus; a rhombus with equal diagonals is a square. Hence PQRS is a square.

Step 9. Optional right-angle cross-check: $PR^2 = 68$ and $PQ^2 + QR^2 = 34 + 34 = 68$, so the angle at Q (opposite diagonal PR) is 90 degrees by the converse of Pythagoras.

» Why this step? The clean chain for proving a square with distances only: sides equal (rhombus candidate) + diagonals bisect (parallelogram) + diagonals equal (square). Each link uses only Chapter 1 tools.

Answer: PQRS is a square: all sides $\sqrt{34}$, diagonals equal ($2\sqrt{17}$) and bisecting each other at (0, 3), with right angles confirmed by 68 = 34 + 34.

> ***Hinglish:*** *Ye Chapter 1 ka final boss hai. Strategy 3-layer ki hai: (1) chaaron sides √34 - rhombus material, (2) diagonals ke midpoints dono (0, 3) - parallelogram confirm, (3) diagonals dono √68 - square confirm. Bonus check: 68 = 34 + 34 se right angle bhi mil gaya. Notice karo kitna systematic hai - 'sides equal + diagonals bisect + diagonals equal' ye teen-step recipe kisi bhi square-proof question mein chalegi. Ye last question mein jaan-bujh ke negative coordinates aur non-axis orientation rakha hai taaki tum axes par dependent na raho.*

1.5 Rapid MCQs: Choose the Correct Option (20)

Rapid-fire round: 4 options, ek hi correct. Pehle khud choose karo, phir colored Answer strip se check karo. Har galat option ka "Why wrong" bhi diya hai - taki exam me wahi trap tumhe na phasaye.

MCQ 1 | Normal | Quadrant signs

1. The point (-7, 4) lies in which quadrant?

A. Quadrant I

B. Quadrant II

C. Quadrant III

D. Quadrant IV

Correct Answer: (B)

x = -7 is negative and y = +4 is positive. The sign pattern (-, +) is exactly the definition of Quadrant II (above the x-axis, left of the y-axis).

Why (A) is wrong: (+, +) would need BOTH positive; here x is negative.

Why (C) is wrong: Quadrant III is (-, -) - y must also be negative.

Why (D) is wrong: Quadrant IV is (+, -) - both signs are wrong here.

Hinglish: *Pehla number minus (left), doosra plus (upar) - ye Quadrant II ka signature hai. Bas signs match karo, answer mil gaya.*

MCQ 2 | Normal | Coordinate meaning

2. The perpendicular distance of the point (-4, 9) from the y-axis is:

A. 9 units

B. 4 units

C. -4 units

D. √97 units

Correct Answer: (B)

Distance from the y-axis is given by the size of the x-coordinate: |x| = |-4| = 4. Distances are never negative.

Why (A) is wrong: 9 is the distance from the x-axis, not the y-axis.

Why (C) is wrong: Distances are always non-negative; -4 is a coordinate, not a distance.

Why (D) is wrong: √97 is the distance from the origin.

Hinglish: *y-axis se doori = |x|. Minus sirf side batata hai (left), distance toh 4 hi hai. Distance kabhi negative nahi hota - option C khatam. √97 toh origin se distance hai ($4^2 + 9^2$).*

MCQ 3 | Normal | Distance formula 6-8-10

3. The distance between (1, -2) and (7, 6) is:

A. 10 units

B. 12 units

C. √84 units

D. 14 units

Correct Answer: (A)

Differences: 7 - 1 = 6 and 6 - (-2) = 8. Distance = √(36 + 64) = √100 = 10, the 6-8-10 triple.

Why (B) is wrong: 12 would come from adding instead of applying Pythagoras.

Why (C) is wrong: √84 comes from miscomputing the y-difference as 4 or 8-(-2) as something else.

Hinglish: *6 aur 8 dikh rahe hain? 3-4-5 ka do guna - seedha 10. Triple yaad rakho toh square karne ki bhi zaroorat nahi.*

MCQ 4 | Normal | Half-turn symmetry

4. If P(a, b) lies in Quadrant II, then the point (-a, -b) lies in:

A. Quadrant I

B. Quadrant II

C. Quadrant III

D. Quadrant IV

Correct Answer: (D)

In Q II, a is negative and b is positive. So -a is positive and -b is negative: signs (+, -), which is Quadrant IV. Flipping both signs reflects the point through the origin (a half-turn).

Why (A) is wrong: Both signs must flip, not stay one-negative-one-positive.

Why (B) is wrong: Same quadrant is impossible after a half-turn.

Hinglish: *Q II mein a negative, b positive. Dono ke sign palte toh (+, -) = Q IV. Yaad rakho: (x, y) aur (-x, -y) hamesha opposite quadrants mein hote hain - origin ke through mirror.*

MCQ 5 | Normal | Reflection in y-axis

5. The reflection of the point (-3, -5) in the y-axis is:

A. (3, -5)

B. (3, 5)

C. (-3, 5)

D. (-3, -5)

Correct Answer: (A)

Reflection in the y-axis flips only the sign of x: (-3, -5) becomes (3, -5). The y-coordinate is untouched.

Why (B) is wrong: (3, 5) flips both signs - that is reflection in BOTH axes.

Why (C) is wrong: (-3, 5) flips y only - that is the x-axis reflection.

Hinglish: *y-axis mein mirror = sirf x ka sign ulta. (-3, -5) ka x minus tha, ab plus: (3, -5). y ko mat chhedo - wahi sabse common galti hai.*

MCQ 6 | Normal | Midpoint

6. The midpoint of the segment joining (-3, 8) and (5, -2) is:

A. (1, 3)

B. (2, 6)

C. (1, 5)

D. (4, 3)

Correct Answer: (A)

Midpoint = ((-3 + 5)/2, (8 + (-2))/2) = (2/2, 6/2) = (1, 3).

Why (B) is wrong: (2, 6) is the sum of the coordinates without halving.

Why (C) is wrong: The y-average was computed as (8 - 2)/2 = 5 by dropping a sign.

Why (D) is wrong: Adds only the x-pair wrongly.

Hinglish: *Average lo dono coordinates ka: (-3 + 5)/2 = 1 aur (8 - 2)/2 = 3. (2, 6) wo galti hai jab divide karna bhool jaate ho.*

MCQ 7 | Normal | 5-12-13 triple

7. The distance of (5, -12) from the origin is:

A. 7 units

B. 13 units

C. 17 units

D. 169 units

Correct Answer: (B)

$\sqrt{(25 + 144)} = \sqrt{169} = 13$. The 5-12-13 triple, again with a harmless minus sign.

Why (A) is wrong: 5 + (-12) size subtraction - meaningless.

Why (C) is wrong: 17 = 5 + 12 (adding instead of Pythagoras).

Why (D) is wrong: 169 is 13 squared.

Hinglish: *5-12-13 triple ratt lo. Minus ko ignore karo - square ke baad koi matlab nahi. 169 square hai, root nahi - aisi chhoti galti bahut log karte hain.*

MCQ 8 | Normal | Perimeter of right triangle

8. The perimeter of the triangle with vertices (0, 0), (4, 0), (0, 3) is:

A. 7 units

B. 12 units

C. 10 units

D. 6 units

Correct Answer: (B)

Legs are 4 and 3; hypotenuse = 5 (3-4-5). Perimeter = 4 + 3 + 5 = 12.

Why (A) is wrong: 3 + 4 only - forgot the hypotenuse.

Why (C) is wrong: Confused with a different triple.

Why (D) is wrong: Half the perimeter.

Hinglish: *Axis par baithe legs 4 aur 3, hypotenuse 5. Sab JODO: 12. 7 sirf do legs ka sum hai - hypotenuse bhool gaye.*

MCQ 9 | Normal | Area of right triangle

9. The area of the triangle with vertices (0, 0), (8, 0) and (0, 6) is:

A. 48 sq units

B. 24 sq units

C. 14 sq units

D. 10 sq units

Correct Answer: (B)

Right angle at the origin with legs 8 and 6. Area = (1/2)(8)(6) = 24.

Why (A) is wrong: Missing the factor 1/2.

Why (C) is wrong: 8 + 6 - a sum, not an area.

Why (D) is wrong: Applied some ratio incorrectly.

Hinglish: *Right triangle ka area = aadha x base x height = 24. 48 wo galti hai jab 1/2 multiply karna bhool jaate ho.*

MCQ 10 | Normal | Assertion-reason

10. Assertion (A): The distance of a point (x, y) from the origin is $\sqrt{(x^2 + y^2)}$. Reason (R): The Baudhayana-Pythagoras Theorem gives hypotenuse = √(sum of squares of legs). Choose the correct option:

A. Both A and R are true and R explains A

B. Both A and R are true but R does not explain A

C. A is true, R is false

D. A is false, R is true

Correct Answer: (A)

The distance formula from the origin is exactly Pythagoras applied to the legs x and y. The assertion is true and the reason is the actual justification behind it.

Why (B) is wrong: R genuinely explains A here - do not dismiss the connection.

Why (C) is wrong: R is a correct statement of the theorem.

Why (D) is wrong: A is the standard distance-from-origin result.

Hinglish: *A sahi hai, aur wo sahi ISLIYE hai kyunki origin se jaane wala segment right triangle ka hypotenuse hai jiske legs x aur y hain. R wahi theorem hai - toh R, A ko explain karta hai. Option A.*

MCQ 11 | Normal | Equidistant on x-axis

11. The point (x, 0) is equidistant from (2, 3) and (6, 3). Then x equals:

A. 4

B. 2

C. 8

D. 3

Correct Answer: (A)

$(x - 2)^2 + 9 = (x - 6)^2 + 9$. The 9s cancel: $(x - 2)^2 = (x - 6)^2$, so x - 2 = plus/minus (x - 6). The negative-sign branch gives a contradiction (0 = -8... no solution from that branch), the positive-sign branch gives x - 2 = x - 6? Also impossible - so take x - 2 = -(x - 6): 2x = 8, x = 4. Geometrically, x = 4 is the midpoint of 2 and 6, directly below the midpoint of the two given points.

Why (B) is wrong: That is the first point itself.

Why (C) is wrong: 8 is beyond the second point - not symmetric.

Why (D) is wrong: 3 is not the average of 2 and 6.

Hinglish: *Dono given points ka y same hai (3), isliye equidistant point unke beech ke THEEK beech mein hoga: (2 + 6)/2 = 4. Equation se bhi x = 4 hi aata hai. Ye symmetry ka use hai - jab dono points same height par hon toh answer unka midpoint hota hai.*

MCQ 12 | Normal | Order independence of distance

12. Statement: If the two points in a distance calculation are interchanged, the computed distance changes. Is the statement TRUE or FALSE, and why?

A. TRUE, because x and y swap places

B. FALSE, because the differences are squared, which erases the sign change

C. TRUE, because the formula is order-dependent

D. FALSE, because distance is always zero

Correct Answer: (B)

Swapping the points changes (x2 - x1) into (x1 - x2) = negative of it. But squaring makes both equal. Hence the distance is unchanged.

Why (A) is wrong: Coordinates do not swap - only the subtraction order flips.

Why (C) is wrong: The formula is order-independent.

Why (D) is wrong: Distance is zero only for identical points.

Hinglish: *Points ulta karne se differences ka sign ulta hota hai, par square laga-te-hi sign khatam. Isliye distance same rehta hai. Ye freedom hai - jis point ko pehle maano, jawab same.*

MCQ 13 | Normal | Axis membership

13. A point whose x-coordinate is 0 must lie:

A. on the x-axis

B. on the y-axis

C. in Quadrant II

D. at the origin

Correct Answer: (B)

x = 0 means zero sideways distance from the y-axis, so the point sits on the y-axis - at (0, y) for some y. It is the origin only if additionally y = 0.

Why (A) is wrong: On the x-axis it is y that equals 0.

Why (C) is wrong: Quadrant points have both coordinates non-zero.

Why (D) is wrong: Origin needs y = 0 as well.

Hinglish: *x = 0 matlab left-right displacement zero = seedha y-axis par. Origin tabhi banega jab y bhi 0 ho. x-axis wale options mein y = 0 hota hai - ulta mat karo.*

MCQ 14 | Normal | Distance expression

14. The distance of the point (a, b) from the origin equals:

A. $a + b$

B. $\sqrt{(a^2 + b^2)}$

C. $\sqrt{(a^2 - b^2)}$

D. $\sqrt{a} + \sqrt{b}$

Correct Answer: (B)

By the distance formula with the second point (0, 0): $\sqrt{((a - 0)^2 + (b - 0)^2)} = \sqrt{(a^2 + b^2)}$.

Why (A) is wrong: Adding lengths of legs never gives the hypotenuse.

Why (C) is wrong: A minus in the formula is fictitious.

Why (D) is wrong: Roots distribute over multiplication, not addition.

Hinglish: *Formula yaad karo - squares JUDTE hain, minus ya addition nahi. a + b isliye galat kyunki 3-4 wale case mein 7 deta hai jabki asli distance 5 hai.*

MCQ 15 | HOTS | Two possible y values

15. If the distance between A(1, 2) and B(4, y) is 5, the sum of ALL possible values of y is:

A. 4

B. 6

C. 8

D. 2

Correct Answer: (A)

$9 + (y - 2)^2 = 25$ gives $(y - 2)^2 = 16$, so $y = 6$ or $y = -2$. Both points (4, 6) and (4, -2) are at distance 5 from A. Sum $= 6 + (-2) = 4$.

Why (B) is wrong: Took only the positive root $y = 6$.

Why (C) is wrong: Added $6 + 6$ or wrong pairing.

Why (D) is wrong: Took only the negative root.

Hinglish: *Do roots aayenge - circle A se guzar raha hai, vertical line $x = 4$ par do milaap. $y = 6$ aur $y = -2$. Question ne SUM manga hai = 4. Sirf ek value likhne wale question dhyan se nahi padhte.*

MCQ 16 | Normal | Double reflection

16. Reflecting (-2, -3) in the x-axis and then the image in the y-axis gives:

A. (2, 3)

B. (-2, 3)

C. (2, -3)

D. (-3, -2)

Correct Answer: (A)

x-axis reflection flips y: (-2, 3). Then the y-axis reflection flips x: (2, 3). Both flips together = sign of both coordinates reversed.

Why (B) is wrong: Only y flipped - one reflection.

Why (C) is wrong: Only x flipped - one reflection.

Why (D) is wrong: Swapped coordinates - reflections do not do that.

Hinglish: *Do mirror = dono sign flip. Seedha result (2, 3). Order matter nahi karta yahan - x pehle ya y pehle, same end point.*

MCQ 17 | HOTS | Circumcentre of right triangle

17. For the right triangle with vertices (0, 0), (6, 0) and (0, 8), the point equidistant from all three vertices is:

A. (3, 4)

B. (3, 5)

C. (4, 3)

D. (2, 3)

Correct Answer: (A)

Hypotenuse joins (6, 0) to (0, 8); its midpoint is (3, 4). In a right triangle the midpoint of the hypotenuse is equidistant from all three vertices (distance 5 here: 3-4-5 again).

Why (B) is wrong: Not the midpoint of the hypotenuse.

Why (C) is wrong: Swapped coordinates.

Why (D) is wrong: No geometric basis.

Hinglish: *Q42 wala theorem yaad karo - right triangle mein hypotenuse ka MIDPOINT hi teeno se equidistant. (6,0) aur (0,8) ka midpoint = (3, 4). Distance check: origin se 5, dono vertices se 5. Bang on.*

MCQ 18 | Normal | Grid vs direct

18. On a unit grid, the shortest path from (2, 3) to (6, 7) moving only along grid lines has length 8. The straight-line distance is:

A. $4\sqrt{2}$ units

B. 8 units

C. $2\sqrt{2}$ units

D. 16 units

Correct Answer: (A)

Gaps: 6 - 2 = 4 and 7 - 3 = 4. Direct distance = $\sqrt{(16 + 16)} = \sqrt{32} = 4\sqrt{2}$ (approximately 5.66), versus 8 along the grid.

Why (B) is wrong: That is the grid distance.

Why (C) is wrong: Underestimated the root.

Why (D) is wrong: That is the squared distance 32 misread as 16 x ... no, 16 is not even the square here.

Hinglish: *4 aur 4 ka square: 32, root 4√2. Grid par 8, seedha 5.66 - seedha rasta jeet gaya. √32 ko simplify karna mat bhoolo: 4√2.*

MCQ 19 | Normal | Reflection preserves midpoints

19. A and B are two points and M is the midpoint of AB. A and B are reflected in the x-axis to A' and B'. Which statement is TRUE?

A. M' (the image of M) is the midpoint of A'B'

B. A'B' is longer than AB

C. M' does not exist on segment A'B'

D. The midpoint of A'B' is M itself, unchanged

Correct Answer: (A)

Reflection preserves distances and maps the segment onto the segment, so it maps the midpoint to the midpoint. The image M' is the midpoint of A'B'. (M itself moves unless it lies on the x-axis.)

Why (B) is wrong: Reflections preserve all distances.

Why (C) is wrong: The midpoint image always exists.

Why (D) is wrong: M moves under reflection unless it sits on the x-axis.

Hinglish: *Reflection poore segment ko utha ke shift karta hai - beech ka point beech ka hi rahega (image mein). AB ki length bhi same rehti hai - option B galat.*

MCQ 20 | HOTS | 9-12-15 triangle

20. The perimeter of the triangle with vertices (0, 0), (9, 0) and (0, 12) is:

A. 36 units

B. 30 units

C. 42 units

D. 39 units

Correct Answer: (A)

Legs 9 and 12; hypotenuse = $\sqrt{(81 + 144)} = \sqrt{225} = 15$ (the 9-12-15 triple = 3 x (3-4-5)). Perimeter = 9 + 12 + 15 = 36.

Why (B) is wrong: Used 5-12-13 by mistake.

Why (C) is wrong: Overshoot on the hypotenuse.

Why (D) is wrong: Partial sum.

Hinglish: *9 aur 12 ko dekho: dono 3 se divide hote hain - 3-4-5 ka scaled version, hypotenuse 15. Perimeter 36. Triples ka scaling trick bade numbers ko chhota kar deta hai.*

1.6 Choose the Correct Round: Theory & Speed (20)

Doosra rapid round - har MCQ ka style: statement padho, correct option turant chuno. Ye round theory, definitions aur quick solving ko test karta hai. Answer strip ke neeche har galat option ka "Why wrong" bhi hai - taki exam ka trap tumhe na phasaye.

CTC 1 | Normal | Points on axes

1. The point on the y-axis lying 6 units above the origin has coordinates:

A. (6, 0)

B. (0, 6)

C. (-6, 0)

D. (0, -6)

Correct Answer: (B)

On the y-axis the x-coordinate must be 0. Above the origin means positive y. Hence (0, 6).

Why (A) is wrong: (6, 0) is on the x-axis, not the y-axis.

Why (D) is wrong: (0, -6) is 6 units BELOW the origin.

Hinglish: *Y-axis = pehla coordinate zero. Upar = positive. (0, 6). Itni si baat, par (6, 0) likhne wale har saal milte hain!*

CTC 2 | Normal | Vertical distance

2. The distance between the points (3, -2) and (3, 7) is:

A. 5 units

B. 9 units

C. √85 units

D. 10 units

Correct Answer: (B)

Same x-coordinate means a vertical segment. Distance = |7 - (-2)| = |7 + 2| = 9.

Why (A) is wrong: 5 confuses the gap with the smaller coordinate.

Why (C) is wrong: √85 results from a wrong application of the distance formula.

Hinglish: *x same hai toh seedha y ka gap: 7 - (-2) = 9. Minus-minus plus bana - yahi sabse common slip hai. √85 aata hai agar log dono differences ghalat jod dein.*

CTC 3 | Normal | Conceptual: free coordinate

3. For how many values of k is the distance between (k, 3) and (k, 8) equal to 5?

A. Exactly one value of k

B. Exactly two values of k

C. All real values of k

D. No value of k

Correct Answer: (C)

Both points have the same x-coordinate k, so the distance is |8 - 3| = 5 regardless of k. Every real k works - there are infinitely many such pairs.

Why (A) is wrong: There is no equation restricting k.

Why (B) is wrong: The distance does not depend on k at all.

Hinglish: *Ye trick question hai! Distance dono points ke y-gap se banti hai jo hamesha 5 hai - k ka kuch lena dena hi nahi. Isliye saare real k chalega. Log sochte hain k = 5 hoga - nahi bhai, k toh bas dono ko same rakhta hai.*

CTC 4 | Normal | Distance from origin

4. The distance of the point (-6, 8) from the origin is:

A. 2 units

B. 10 units

C. 14 units

D. 100 units

Correct Answer: (B)

$\sqrt{((-6)^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$. Signs die inside squares; the 6-8-10 triple does the rest.

Why (A) is wrong: 2 = |-6 - 8| is a meaningless subtraction of coordinates.

Why (C) is wrong: Adding coordinate sizes is not a distance.

Why (D) is wrong: 100 is the squared distance, not the distance.

Hinglish: *Minus square mein mar jaata hai: 36 + 64 = 100, root 10. 14 wo log chunte hain jo -6 + 8 ko distance samajh lete hain - aisa formula exist hi nahi karta.*

CTC 5 | Normal | Reflection in x-axis

5. The reflection of (2, 7) in the x-axis, followed by reflection of the image in the y-axis, gives:

A. (-2, -7)

B. (2, -7)

C. (-2, 7)

D. (7, -2)

Correct Answer: (A)

x-axis reflection: (2, 7) to (2, -7). Then y-axis reflection flips x: (2, -7) to (-2, -7). Two axis reflections = half-turn = both signs flip.

Why (B) is wrong: Only one reflection was performed.

Why (C) is wrong: Wrong axis was flipped.

Why (D) is wrong: Coordinates were swapped - reflections never swap coordinates.

Hinglish: *Do reflections ki chain = dono signs flip. (2, 7) se seedha (-2, -7). Shortcut yaad rakho: x-axis phir y-axis reflect karna = origin ke around 180 degree ghumaana.*

CTC 6 | Normal | Reverse midpoint

6. If M(2, -1) is the midpoint of A(-4, 3) and B(x, y), then B is:

A. (8, -5)

B. (-10, 7)

C. (8, 5)

D. (0, 1)

Correct Answer: (A)

Undo the average: x = 2(2) - (-4) = 8 and y = 2(-1) - 3 = -5. So B = (8, -5).

Why (B) is wrong: Adds instead of subtracting the known endpoint.

Why (C) is wrong: y computed as -2 + 3.

Why (D) is wrong: Halves something unnecessarily.

Hinglish: *Midpoint se endpoint nikalna = 2 x midpoint minus known point. x: 4 + 4 = 8. y: -2 - 3 = -5. Sign ka khayal: 3 ko MINUS karna hai, plus nahi.*

CTC 7 | Normal | Collinearity

7. Which set of three points is collinear?

A. (0, 0), (2, 1), (4, 2)

B. (0, 1), (2, 2), (5, 4)

C. (1, 1), (2, 3), (4, 6)

D. (0, 0), (1, 2), (3, 7)

Correct Answer: (A)

In option A each step from one point to the next is (+2, +1) - a constant step means equal slope, so the points lie on one line (y = x/2). The other sets change the step pattern, so they bend.

Why (B) is wrong: Steps (+2, +1) then (+3, +2) - slopes 1/2 vs 2/3 differ.

Why (C) is wrong: Slopes 2 vs 3/2 differ.

Why (D) is wrong: Slopes 2 vs 5/2 differ.

Hinglish: *Collinear ka fastest check: consecutive points ke beech ka STEP same hai ya nahi. Option A mein har baar (+2, +1) - perfect line. Baaki options mein step badal jaata hai, matlab line mudti hai.*

CTC 8 | Normal | Circle membership

8. Which point lies ON the circle $x^2 + y^2 = 10$ with centre the origin?

A. (1, 3)

B. (2, 2)

C. (1, 2)

D. (3, 3)

Correct Answer: (A)

Check $x^2 + y^2$: (1, 3) gives 1 + 9 = 10 exactly. (2, 2) gives 8 (inside), (1, 2) gives 5 (inside), (3, 3) gives 18 (outside).

Why (B) is wrong: 8 < 10 means inside, not on.

Why (C) is wrong: 5 is far from 10.

Why (D) is wrong: 18 > 10 means outside.

Hinglish: *Equation mein plug karo, 10 aana chahiye. (1, 3): 1 + 9 = 10 perfect. Baaki sab 10 nahi de rahe - andar ya bahar.*

CTC 9 | Normal | Symbolic distance

9. The distance between the points (a, 0) and (0, a) (a > 0) is:

A. 2a units

B. a√2 units

C. a units

D. 0 units

Correct Answer: (B)

Differences: a - 0 = a and 0 - a = -a. Distance = $\sqrt{(a^2 + a^2)} = \sqrt{(2a^2)} = a\sqrt{2}$.

Why (A) is wrong: 2a would be a + a, but legs add only through squares.

Why (C) is wrong: a is just one leg.

Why (D) is wrong: The points are distinct unless a = 0.

Hinglish: *Dono differences ka size a hai - isosceles right triangle ka hypotenuse = a√2. 2a karna Pythagoras ko addition samajhna hai - bada classic mistake.*

CTC 10 | Normal | Horizontal segment

10. The line joining (-5, 2) and (5, 2) is parallel to:

A. the x-axis

B. the y-axis

C. both axes

D. neither axis

Correct Answer: (A)

Both points have y = 2, so the segment is horizontal, i.e. parallel to the x-axis (and perpendicular to the y-axis).

Why (B) is wrong: Same y means horizontal, not vertical.

Why (C) is wrong: A non-degenerate segment cannot be parallel to both perpendicular axes.

Hinglish: *y same = horizontal line = x-axis ke parallel. Same x hota toh y-axis ke parallel hota.*

CTC 11 | Normal | Rectangle perimeter

11. The points (0, 0), (3, 0), (3, 4) and (0, 4) form a rectangle. Its perimeter is:

A. 12 units

B. 14 units

C. 24 units

D. 7 units

Correct Answer: (B)

Sides are 3 and 4. Perimeter = 2(3 + 4) = 14. (Its diagonal is 5, area 12.)

Why (A) is wrong: 12 is the area, not the perimeter.

Why (C) is wrong: Doubled the area by mistake.

Why (D) is wrong: Added only two sides.

Hinglish: *Perimeter = 2(length + width) = 2 x 7 = 14. 12 toh AREA hai - dono ko mix karna classic confusion hai.*

CTC 12 | Normal | Equidistant on y-axis

12. The point (0, y) is equidistant from A(-3, 0) and B(5, 2). Then y equals:

A. 5

B. 2

C. 10

D. 4

Correct Answer: (A)

$9 + y^2 = 25 + (y - 2)^2$. Expand: $9 + y^2 = 25 + y^2 - 4y + 4$, so $9 = 29 - 4y$, giving $4y = 20$, $y = 5$. The point is (0, 5).

Why (B) is wrong: That is B's y-coordinate, not the solution.

Why (C) is wrong: Result of a sign slip in $(y - 2)^2$.

Why (D) is wrong: Arithmetic slip in moving terms.

Hinglish: *(0, y) rakho, dono squared distances barabar karo, y^2 cancel - linear equation: y = 5. Verify: (0,5) se A doori √34, B se bhi √34. Perfect.*

CTC 13 | HOTS | Hypotenuse from coordinates

13. For the triangle with vertices (1, 1), (4, 1) and (1, 5), the hypotenuse has length:

A. 5 units

B. √41 units

C. 7 units

D. 6 units

Correct Answer: (A)

Legs along the axes directions: 4 - 1 = 3 and 5 - 1 = 4, meeting at the right angle (1, 1). Hypotenuse = √(9 + 16) = 5. The whole 3-4-5 triangle is just shifted to start at (1, 1).

Why (B) is wrong: √41 = √(16 + 25) comes from squaring wrong gaps.

Why (C) is wrong: 3 + 4 again.

Why (D) is wrong: Wrong leg difference.

Hinglish: *Origin par nahi hai triangle, (1, 1) par hai - par legs toh wahi 3 aur 4 hain! Translation se triple badalta nahi. Hypotenuse 5. √41 aata hai agar coordinates se seedha formula lagao bina right-angle pehchane.*

CTC 14 | HOTS | Missing value on a line

14. If the points (2, 3), (4, k) and (6, 5) are collinear, then k equals:

A. 4

B. 3.5

C. 5

D. 2

Correct Answer: (A)

From (2, 3) to (6, 5) the total step is (+4, +2); halfway (at x = 4) the y-value must be the average: k = (3 + 5)/2 = 4. Indeed (4, 4) is the midpoint.

Why (B) is wrong: Averaged wrongly.

Why (C) is wrong: Copied the endpoint value.

Why (D) is wrong: No basis.

Hinglish: *Collinear matlab beech wala point line par. (2,3) se (6,5) ke beech ka x = 4 hai toh y bhi average hoga: (3+5)/2 = 4. Slope-check bhi karo: dono taraf slope 1/2. Clean.*

CTC 15 | HOTS | Equally spaced chain

15. If (3, 2), (6, y) and (9, 8) are equally spaced collinear points, then y equals:

A. 5

B. 6

C. 4

D. 7

Correct Answer: (A)

Equal spacing means a constant step. x-step is 3 twice, so the y-step must also repeat: y = 2 + (8 - 2)/2 = 2 + 3 = 5. Check: steps (+3, +3) then (+3, +3).

Why (B) is wrong: Copied the x-step into y.

Why (C) is wrong: Halved wrongly.

Why (D) is wrong: No pattern use.

Hinglish: *Equally spaced = same step dono baar. x mein +3, +3 dikh raha hai, toh y mein bhi same jump: 2 se 8 tak total 6, half = 3, y = 5. Pattern dekho, equation ki zaroorat nahi.*

CTC 16 | HOTS | Two-circle intersection

16. How many points are at distance 5 from BOTH (0, 0) and (8, 0)?

A. 0

B. 1

C. 2

D. Infinitely many

Correct Answer: (C)

The two circles (radius 5, centres 8 apart) overlap since $8 < 5 + 5 = 10$ and $8 > 0$. Two overlapping circles meet in exactly 2 points. By symmetry they are (4, 3) and (4, -3): check $16 + 9 = 25$. Correct.

Why (A) is wrong: Would need the circles separate ($8 > 10$).

Why (B) is wrong: Would need exactly touching ($8 = 10$).

Why (D) is wrong: That happens only for identical circles.

Hinglish: *Do circles socho - centres 8 door, radius 5-5. Wo do jagah kaatenge. Symmetry se $x = 4$ par, $y =$ plus-minus 3 (kyunki 3-4-5 triple). Answer 2. Ek point tab hota jab circles ko touch karte (centres ki doori = radius ka sum).*

CTC 17 | HOTS | Square from centre and vertex

17. A square has centre (2, 3) and one vertex at (5, 7). The area of the square is:

A. 50 sq units

B. 25 sq units

C. 100 sq units

D. $10\sqrt{2}$ sq units

Correct Answer: (A)

Centre-to-vertex distance = half the diagonal = $\sqrt{(3^2 + 4^2)} = 5$. So the full diagonal is 10, and area = diagonal$^2/2 = 100/2 = 50$. (Side = $10/\sqrt{2} = 5\sqrt{2}$, area 50.)

Why (B) is wrong: Used 5 as the side instead of half-diagonal.

Why (C) is wrong: Used 10 as the side.

Why (D) is wrong: That is a length, not an area.

Hinglish: *Centre se vertex tak = aadha diagonal = 5 (3-4-5 triple!). Poora diagonal 10, area = $d^2/2 = 50$. Ye diagonal-area relation square ke liye yaad rakho - side nikal-ghumane se fast hai.*

CTC 18 | Normal | Reflection on axis point

18. The reflection of the point (0, -7) in the x-axis lies:

A. on the x-axis

B. on the y-axis

C. in Quadrant III

D. at the origin

Correct Answer: (B)

Reflection flips y: (0, -7) becomes (0, 7). The x-coordinate is still 0, so the image sits on the y-axis. Points on the reflecting axis stay fixed, but this point was on the y-axis, not the x-axis, so it moves - along the y-axis.

Why (A) is wrong: Only points ON the x-axis stay put under x-reflection.

Why (C) is wrong: Quadrant III needs x negative too.

Why (D) is wrong: Origin only if both coordinates are zero.

Hinglish: *(0, -7) y-axis par baitha hai. x-axis mein reflect karne se wo y-axis ke upar jump karta hai: (0, 7). Axis par baithe point ko reflect karne ka rule: jis axis par hai wahi rehta hai (agar usi axis mein reflect karo), warna apni axis par hi travel karta hai.*

CTC 19 | Normal | Statement-based

19. Consider: (i) (x, y) equals (y, x) only when x = y. (ii) The distance between two points can be negative. (iii) Every point on the x-axis has y-coordinate 0. Which statements are TRUE?

A. (i) and (iii) only

B. (i) and (ii) only

C. (ii) and (iii) only

D. all three

Correct Answer: (A)

(i) is true: equal ordered pairs need equal coordinates. (ii) is false: distance is a length, always at least 0. (iii) is true by definition of the x-axis.

Why (B) is wrong: (ii) is never true.

Why (C) is wrong: (ii) is never true.

Why (D) is wrong: (ii) spoils the set.

Hinglish: *(i) sahi - ordered pair ka matlab hi coordinate-by-coordinate match hai. (ii) galat - distance kabhi negative nahi. (iii) sahi - x-axis ka definition. Toh (i) aur (iii).*

CTC 20 | Normal | Square perimeter

20. The perimeter of the square with vertices (-1, 2), (3, 2), (3, 6) and (-1, 6) is:

A. 16 units

B. 8 units

C. 32 units

D. 14 units

Correct Answer: (A)

Side = 3 - (-1) = 4. Perimeter = 4 x 4 = 16.

Why (B) is wrong: Two sides only.

Why (C) is wrong: Doubled for no reason.

Why (D) is wrong: Arithmetic slip.

Hinglish: *Side = x-gap = 4. Perimeter = 16. 8 sirf do sides, 32 toh area ke jaisa lagega ($4^2 = 16$ area actually) - perimeter 4 x side formula use karo.*

Chapter 2: Introduction to Linear Polynomials

From locating points we now move to describing relationships with expressions. The uploaded PDF builds the idea of a polynomial from daily life (boxes of pens, garden fencing, chess-club fees), defines degree, focuses on linear polynomials, shows them as input-output machines (functions), develops linear patterns and growth/decay, teaches how to fit the relationship y = ax + b from data, and finishes by plotting straight lines and reading the meaning of a (slope) and b (y-intercept). This book sharpens every one of those ideas with reverse questions, two-equation fittings, plan comparisons and pattern proofs.

2.1 Complete Concept Cheat Sheet (read this before any question)

2.1.1 Algebraic expressions: terms, coefficients, constants, variables

What is it? An algebraic expression combines numbers, variables (letter-numbers) and operation signs. In 4x + 5y + 3, the parts 4x, 5y and 3 are the **terms**; x and y are **variables** (also called letter-numbers); 4 and 5 are the **coefficients**; 3 is a **constant**. **Why coefficients matter:** the coefficient tells you how many copies of the variable you have, and it is the number that multiplies the variable - including its sign. In -7x the coefficient is -7, not 7.

When do I use this? Every 'identify the terms / coefficient / constant' question, and every modelling question where a real situation must become an expression (cost of fencing = rate x length, and so on). **Tiny example.** For the garden in the PDF: wire fence cost 100 x 2l = 200l, wooden fence 80 x 2w = 160w, seeds 50 x area = 50lw; total 200l + 160w + 50lw - a three-term expression in two variables.

> **Hidden coefficient trap**
> A term like x has coefficient 1 (not 0!), and -x has coefficient -1. A term like $5x^2y$ has coefficient 5. Missing numbers never mean missing coefficients.

2.1.2 Degree and the polynomial family

What is it? A polynomial in ONE variable is an expression made of that variable and its powers (whole-number powers). The **degree** is the highest power of the variable. Degree names: 1 = linear, 2 = quadratic, 3 = cubic, 0 = constant. Examples: 3z + 7 is linear (degree 1); $x^2 + 5x + 1$ is quadratic; $5y^3 + y^2 - 8$ is cubic; the number 8 alone is a constant polynomial (it can be written $8x^0$).

How do I find the degree quickly? Scan the powers of the variable and take the maximum. In $2x^4 - 3x^3 + 6x^2 - 2x + 7$ the powers are 4, 3, 2, 1, 0, so the degree is 4. **Sign-awareness:** the degree ignores the sign and size of coefficients - $-9x^5$ still has degree 5. **Tiny example.** -13 is a constant polynomial of degree 0; 6z + 1 is linear.

Common confusion: 'linear polynomial' (degree exactly 1, like 2x + 3) versus 'linear equation' (a statement like 2x + 3 = 64 that equates a linear polynomial to something). The polynomial is the expression; the equation is the claim that two expressions are equal.

2.1.3 Evaluation: the input-output machine

What is it? Substituting a value of the variable into a polynomial gives its value at that input. The PDF pictures this as a machine: feed x = 4 into the machine 2x + 3, out comes 11; feed x = -6, out comes -9. This

machine-view is the seed of the 'function' idea. **Golden rule of substitution:** write the substitution with brackets: 2(-6) + 3, never 2 -6 + 3. Brackets prevent the sign disaster.

Tiny example. For p(x) = 7x - 4: p(0) = -4, p(2) = 10, p(-3) = -25. **Reverse use:** given the output, find the input - that is solving a linear equation: 7x - 4 = 24 gives x = 4. Evaluation forward, equation backward - the same machine run both ways.

2.1.4 Linear equations from words

What is it? Translate a sentence into an equation with ONE unknown, solve, and answer the original question. Standard moves: 'one number is 10 more than the other' means x and x + 10; 'three times as many' means 3x; 'their sum is 64' means x + (x + 10) = 64.

Method (always the same skeleton). Step 1: define the unknown precisely ('let the smaller number be x'). Step 2: express every other quantity in terms of x. Step 3: write the balancing condition (sum, difference, total money). Step 4: solve. Step 5: ANSWER THE QUESTION ASKED - if the question asked for both numbers, report both; a bare x earns no full marks. **Sanity check:** substitute back into the words, not just the equation.

2.1.5 Linear patterns and the nth term

What is it? A linear pattern is a sequence whose consecutive terms differ by a CONSTANT. The tile pattern 1, 3, 5, 7, ... adds 2 tiles each stage, and its nth-stage count is 2n - 1. The constant difference becomes the coefficient of n; the constant term adjusts the start. **Finding the rule from the table:** if the difference is d, the rule looks like an = dn + c; find c by matching one known term (e.g. for 1, 3, 5: d = 2 and term 1 = 1 gives 1 = 2(1) + c, so c = -1).

Why the difference is the slope: on the graph of the rule, moving one stage right moves the value up by exactly d - the line rises by d per step. **Tiny example.** Matchstick squares in a row: 4, 7, 10, ... difference 3, rule 3n + 1. Stage 10 needs 31 matches. **Reverse use:** 'can the pattern ever equal 100?' means solving 3n + 1 = 100; here n = 33 - a whole number, so yes, stage 33 uses exactly 100 matches. If n had come out fractional, the answer would be NO - patterns only exist at whole stages.

2.1.6 Linear growth and linear decay

What is it? Growth: a quantity increases by a fixed amount over equal intervals - C(d) = 100 + 60d (journey cost). Decay: decreases by a fixed amount - h(t) = 3 - 0.5t (water level). The fixed change is the slope: positive for growth, negative for decay. **Structure:** value = starting value + (change per step) x (number of steps). On a graph, growth tilts up, decay tilts down.

How do I recognise it? Words like 'every month adds', 'decreases by 800 each year', 'reduces by 15 per day'. **Two standard questions:** forward (value after n steps - substitute) and backward (when does the value reach V? - solve). Decay also has the 'run out' question: set the expression to 0 (or the floor value) and solve. **Tiny example.** Phone bought at 10000, losing 800 per year: v = 10000 - 800t; after 3 years 7600; it touches 3600 when t = 8.

2.1.7 Fitting y = ax + b from two observations

What is it? Many real bills have the shape y = ax + b: a fixed part b plus a per-unit part a. Given two observations (x_1, y_1) and (x_2, y_2), substitute both into y = ax + b to get TWO equations, then eliminate b by subtraction: $a = (y_2 - y_1)/(x_2 - x_1)$. Then recover b from either observation: $b = y_1 - a\,x_1$.

Worked skeleton (the telecom example). 10 GB cost 350 and 20 GB cost 550. Equations: 350 = 10a + b and 550 = 20a + b. Subtract: 200 = 10a, so a = 20 (each extra GB costs Rs 20). Back-substitute: b = 350 - 200 = 150 (the fixed monthly fee). Answer: y = 20x + 150. **Meaning check:** a is the RATE per unit, b is the ENTRY fee - interpreting them is as important as computing them.

> **Subtraction is the whole trick**
> Subtracting the two equations kills b because b is the SAME in both. What remains is a clean one-step equation for a. If your two equations accidentally have different b values, you have copied the data wrongly - stop and re-read.

2.1.8 Graphs of lines: slope a and y-intercept b

What is it? Every equation y = ax + b draws a straight line. Plotting needs just two points: pick any two x-values, compute the y-values, join with a ruler and extend. Three facts carry all the exam weight: (1) y = ax lines all pass through the ORIGIN (0, 0); (2) larger positive a makes a steeper line, a between 0 and 1 makes a flatter line, negative a tilts the line down-left to up-right reversed (decay); (3) b is where the line CUTS THE Y-AXIS, at the point (0, b) - the y-intercept.

Effect of changing a vs changing b. Fix b and vary a: the line pivots about (0, b), changing steepness. Fix a and vary b: the line SLIDES up or down, staying PARALLEL - lines with the same slope are parallel and never meet. Growth means positive slope, decay means negative slope; the slope numerically equals the constant difference of the underlying pattern. **Tiny example.** y = 2x - 1, y = 2x + 1, y = 2x + 5 are three parallel lines cutting the y-axis at (0, -1), (0, 1) and (0, 5).

Converting to standard shape. Equations often arrive disguised: 2y = 4x + 7 means y = 2x + 3.5 after dividing by 2; 5y = 6x - 10 means y = 1.2x - 2. ALWAYS isolate y first; only then read a and b. Reading the coefficient of x before this conversion is a classic error.

2.1.9 Exam tips for this chapter

1. Before reading slope and intercept, rewrite the equation as y = ...
2. Substitute with brackets - especially for negative inputs.
3. In word problems, define the unknown in words first; the equation follows automatically.
4. For pattern rules, lock the constant difference first, then fit the constant term from ONE known term.
5. When fitting y = ax + b, subtract the two data equations; never solve them separately for b first (double work, double error chances).
6. Reverse questions (which stage gives 100 matches?) need INTEGER answers - a fractional n means 'no such stage', and saying so IS the answer.
7. On graphs, label the y-intercept point (0, b) explicitly - it is usually the mark-earning feature.

2.2 Question Types / Concept Map for Chapter 2

Type	Name	How to recognise it	Core method
T1	Anatomy of expressions	Identify terms, coefficients, variables, constants	Term-wise reading, sign-aware coefficients

Type	Name	How to recognise it	Core method
T2	Degree and classification	Find degree; label linear/quadratic/cubic/constant	Highest power rule
T3	Evaluation	Find value of polynomial at given input	Bracket substitution
T4	Reverse evaluation	Given output, find input	Solve the linear equation
T5	Word-problem modelling	Numbers/ages/coins/fences in words	Define x, translate, solve, answer in words
T6	Linear patterns	Table of stages; find nth term or a target stage	Constant difference = coefficient of n
T7	Reachability question	Can the pattern equal a value?	Solve dn + c = target; n must be a whole number
T8	Growth and decay	Increases/decreases by fixed amount per interval	value = start + rate x steps
T9	Fit y = ax + b	Two observations given	Subtract equations to get a, back-substitute for b
T10	Interpret a and b	What do the numbers mean?	a = rate per unit, b = fixed part
T11	Slope and intercept reading	Find slope/y-intercept of disguised equations	Isolate y first
T12	Parallel lines	Same slope, different intercepts	Compare a values after isolating y
T13	Graph plotting / reading	Draw the line; identify where it cuts axes	Two points; (0, b) on y-axis; y = 0 for x-axis
T14	Multi-concept HOTS	Composition of evaluations, plan comparisons, concurrence	Break into labelled steps, verify each

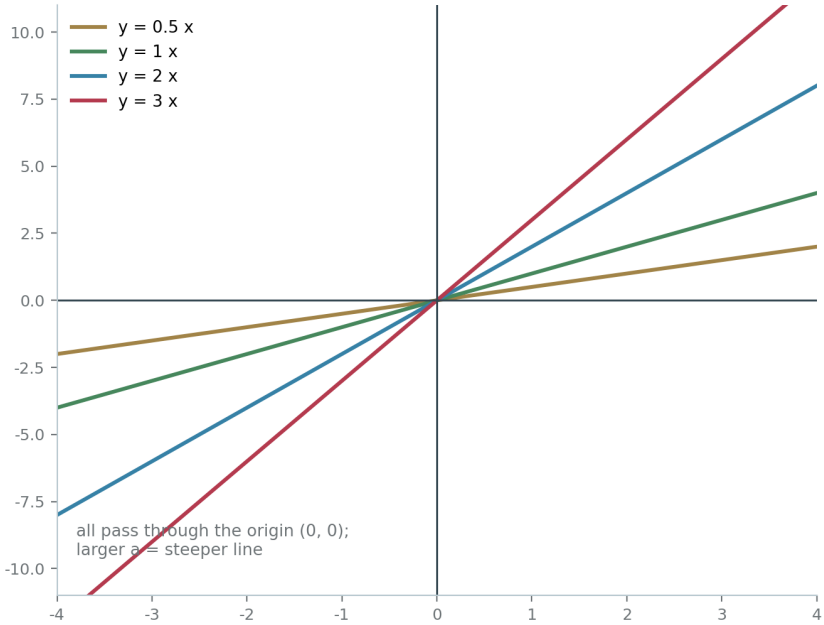


Figure 2.1: The family y = ax - all lines pass through the origin; larger a means steeper.

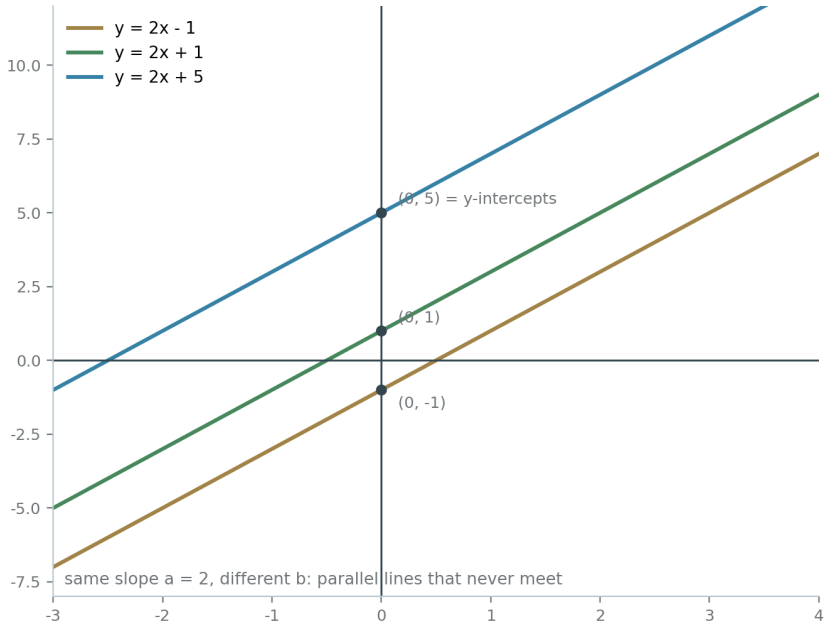


Figure 2.2: Same slope (a = 2), different intercepts b - parallel lines that never meet.

2.3 Speed Tricks & Trap Alarms

Ye table tumhara "dekho aur maaro" radar hai - question ka trigger phrase dikha, toh yahan ka trick fire karo. Inhe ratt lo, exam me sochna kam padega aur speed double ho jayegi.

Instant trick	Fire it when...
Fixed fee + rate × quantity → y = ax + b (b = fixed part, a = per-unit rate)	"Rs 300 joining + Rs 60 per session" style wording.
Two observations? Subtract the equations - b cancels, a pops out	Fit y = ax + b from two bills or data points.
Pattern adds k every step → T(n) = kn + c; c = (first term) - k	Matchstick, tile and money patterns.
Slope a = Δy/Δx; bigger |a| = steeper; same a = parallel	"Which line is steeper / are they parallel?"
Changing b shifts the line up/down; changing a pivots it about (0, b)	"What happens to the graph if ...?"
Spends / loses / drains → minus slope (decay). Adds / earns / fills → plus (growth)	Growth vs decay identification.
y-intercept = (0, b); x-intercept: put y = 0, solve	"Where does the line cut the axes?"

2.4 The 120-Question Workout

Poora chapter ab ek hi ring mein: **120 questions**, chaar level zones mein banta hua. Warm-up se shuru karo, Boss Zone tak pahuncho. Har question ke turant baad boxed step-by-step solution hai - pehle khud try karo,

atako toh solution dekho, aur "Why this step?" + Hinglish note se logic pakka karo. Agar in 120 ko solve kar liya toh is chapter ka koi bhi twisted question tumhe nahi rok payega.

Level 1 · Warm-up Zone: basics and confidence:

QUESTION 1 | Basic | Anatomy of expressions (T1)

Q1. For the expression 6m + 4n - 9, list all the terms, the variables, the coefficient of each variable term, and the constant term. Then state the coefficient of m in the expression 6m + 4n - 9 - m after it is simplified.

SOLUTION — step by step

Step 1. Given: 6m + 4n - 9. Needed: terms, variables, coefficients, constant.

Step 2. Terms are the parts joined by plus or minus: 6m, 4n and -9. (The minus sign travels WITH its term.)

Step 3. Variables (letter-numbers): m and n.

Step 4. Coefficient of m is 6; coefficient of n is 4; the constant term is -9.

Step 5. Now simplify 6m + 4n - 9 - m: combine the m-terms, 6m - m = 5m.

Step 6. The simplified expression is 5m + 4n - 9, so the coefficient of m is 5.

» **Why this step?** Subtraction creates negative terms: -9 is a term, and combining like terms changes coefficients. Students who call the coefficient '6' after simplification have ignored the invisible -1.

Answer: Terms: 6m, 4n, -9; variables: m, n; coefficients 6 and 4; constant -9. After simplification the coefficient of m is 5.

Hinglish: *Terms ko minus ke saath padho - (-9) bhi ek term hai. Coefficient matlab 'kitne variable'. Aur dhyan do: simplified expression mein m ka coefficient 5 ban gaya kyunki -m ka coefficient -1 tha aur 6 - 1 = 5. Invisible coefficient dekhna hi asli skill hai.*

QUESTION 2 | Basic | Degree (T2)

Q2. Find the degree of each polynomial: (i) $3x^2 + 7x - 2$ (ii) $y^3 - y + 5$ (iii) -13 (iv) $6z + 1$ (v) $8 - t^4 + t^2$.

SOLUTION — step by step

Step 1. Degree = highest power of the variable present.

Step 2. (i) powers are 2, 1, 0, so degree 2 (quadratic).

Step 3. (ii) powers 3, 1, 0: degree 3 (cubic).

Step 4. (iii) the constant -13 can be written $-13x^0$, so degree 0 (constant polynomial).

Step 5. (iv) powers 1, 0: degree 1 (linear).

Step 6. (v) powers 0, 4, 2: the highest is 4, so degree 4 (a quartic - just report the number 4).

Answer: (i) 2 (ii) 3 (iii) 0 (iv) 1 (v) 4.

Hinglish: *Degree nikalne ka ek hi kaam: powers scan karo, sabse badi utha lo. Constant ka degree 0 hota hai - ye MCQ favourite hai. (v) mein dhyan do, powers beech mein ghumte hain (8, t^4, t^2) - order koi baat nahi, max chahiye.*

QUESTION 3 | Basic | Classification (T2)

Q3. Classify each as linear, quadratic, cubic or constant: $5 - 2x$, $x^2 + 3x$, $4t^3 - t$, 7, $0.5y + 3$.

SOLUTION — step by step

Step 1. Degree 1 means linear, 2 quadratic, 3 cubic, 0 constant.

Step 2. $5 - 2x$ has degree 1: linear.

Step 3. $x^2 + 3x$ has degree 2: quadratic.

Step 4. $4t^3 - t$ has degree 3: cubic.

Step 5. 7 has degree 0: constant.

Step 6. $0.5y + 3$ has degree 1: linear (decimals in coefficients are perfectly fine).

Answer: Linear: $5 - 2x$ and $0.5y + 3$. Quadratic: $x^2 + 3x$. Cubic: $4t^3 - t$. Constant: 7.

Hinglish: *Naam yaad rakho: 1-linear, 2-quadratic, 3-cubic, 0-constant. Decimal coefficient wala bhi normal polynomial hi hai - $0.5y + 3$ linear hi hai.*

QUESTION 4 | Basic | Coefficients at high powers (T1)

Q4. In the polynomial $2x^4 - 5x^3 + x^2 - 9x + 1$, state the coefficients of x^4, x^3, x^2, x and the constant term.

SOLUTION — step by step

Step 1. Read term by term with signs attached.

Step 2. Coefficient of x^4 is 2.

Step 3. Coefficient of x^3 is -5 (the minus belongs to the coefficient).

Step 4. Coefficient of x^2 is 1 - the term x^2 is really $1x^2$.

Step 5. Coefficient of x is -9; the constant term is 1.

Answer: 2, -5, 1, -9 and constant 1.

Hinglish: *Sign term ke saath chalta hai - (-5) hi coefficient hai. Aur x^2 ka coefficient 1 hai, zero nahi! Ye do hi cheezein is question mein marks katati hain.*

QUESTION 5 | Basic | Evaluation (T3)

Q5. Find the value of the linear polynomial 7x - 4 at (i) x = 0, (ii) x = 2, (iii) x = -3.

SOLUTION — step by step

Step 1. Substitute with brackets every time.

Step 2. (i) 7(0) - 4 = 0 - 4 = -4.

Step 3. (ii) 7(2) - 4 = 14 - 4 = 10.

Step 4. (iii) 7(-3) - 4 = -21 - 4 = -25.

Step 5. Note how the outputs drop by exactly 7 each time x drops by 1 going 2, 0, -3: (10, -4, -25) - the constant difference at work.

Answer: -4, 10, -25.

Hinglish: *Bracket lagana mandatory hai: 7(-3) = -21. Bina bracket ke log 7 - 3 - 4 = 0 likh dete hain - jabardast galti. Aur observation: x ek kadam gira toh output 7 gira - linear ka dhadakta hua constant difference.*

QUESTION 6 | Basic | Evaluation of a quadratic (T3)

Q6. Find the value of the quadratic polynomial $2s^2 + 3s - 5$ at (i) s = 0, (ii) s = 2, (iii) s = -2.

SOLUTION — step by step

Step 1. (i) $2(0)^2 + 3(0) - 5 = -5$.

Step 2. (ii) $2(2)^2 + 3(2) - 5 = 2(4) + 6 - 5 = 8 + 6 - 5 = 9$.

Step 3. (iii) $2(-2)^2 + 3(-2) - 5 = 2(4) - 6 - 5 = 8 - 6 - 5 = -3$.

Step 4. Watch step (iii): $(-2)^2 = +4$ but 3(-2) = -6. The square forgives the sign; the plain term does not.

Answer: -5, 9, -3.

Hinglish: *Do alag jagah sign ka risk hai: square wala term minus ko kha jaata hai (+4), par 3s wala term sign ko POORA follow karta hai (-6). Isi liye bracket - har value apne bracket mein.*

QUESTION 7 | Basic | Function machine (T3)

Q7. A machine computes y = 3x - 1 for every input x. Find the outputs for x = 5 and x = -2. Then find the input that produces the output 20.

SOLUTION — step by step

Step 1. Forward: x = 5 gives y = 3(5) - 1 = 15 - 1 = 14.

Step 2. Forward: x = -2 gives y = 3(-2) - 1 = -6 - 1 = -7.

Step 3. Reverse: we need x such that 3x - 1 = 20.

Step 4. Add 1 to both sides: 3x = 21. Divide by 3: x = 7.

Step 5. Verify: 3(7) - 1 = 20. Correct.

Answer: Outputs 14 and -7; the input producing 20 is x = 7.

Hinglish: *Machine aage bhi chalti hai peeche bhi! Aage = substitute karo (14, -7). Peeche = equation solve karo: 3x - 1 = 20 se x = 7. Function ka yahi asli matlab hai - input-output ka judgementa rishta.*

QUESTION 8 | Basic | Word problem: sum and difference (T5)

Q8. The sum of two numbers is 50, and one of them is 8 more than the other. Find the numbers.

SOLUTION — step by step

Step 1. Let the smaller number be x. Then the larger is x + 8.

Step 2. Sum condition: x + (x + 8) = 50.

Step 3. Simplify: 2x + 8 = 50, so 2x = 42, giving x = 21.

Step 4. The numbers are 21 and 29.

Step 5. Check in words: 21 + 29 = 50 and 29 - 21 = 8. Both conditions hold.

Answer: The numbers are 21 and 29.

Hinglish: *Chota number x maan lo, bada apne aap x + 8 ban jaata hai. Sum equation se x = 21. Aur LAST STEP yaad rakho: dono numbers report karo - sirf 'x = 21' likha toh aadha answer hai. Words mein wapas check karo, equation mein hi nahi.*

QUESTION 9 | Basic | Perimeter polynomial (T3/T5)

Q9. The perimeter of a square of side s is the linear polynomial 4s. Find the perimeter for s = 1, 2.5 and 7.5. If the side grows by 0.5 cm, by how much does the perimeter grow? Why?

SOLUTION — step by step

Step 1. p(1) = 4(1) = 4; p(2.5) = 10; p(7.5) = 30 (all in the side units).

Step 2. If the side grows by 0.5, the new perimeter is 4(s + 0.5) = 4s + 2.

Step 3. So the perimeter grows by exactly 2 units - for EVERY square, whatever its side.

Step 4. Why constant: 4 times a fixed growth is a fixed growth; this is the constant-difference signature of a linear pattern.

Answer: Perimeters 4, 10, 30; growth per 0.5 cm of side is exactly 2 units (constant, because the rule is linear).

Hinglish: *PDF wala hi idea: side 0.5 badhe toh perimeter 2 badhta hai - hamesha, kisibhi square ke liye. Kyunki 4 x 0.5 = 2, aur multiplication by constant fixed growth ko fixed hi rakhta hai. Yehi 'linear pattern' ka matlab hai.*

QUESTION 10 | Basic | Fee structure modelling (T5/T8)

Q10. A badminton academy charges a joining fee of Rs 300 plus Rs 60 for every session attended. (i) Write the total cost for n sessions. (ii) Find the cost for 12 sessions. (iii) A player paid Rs 780 in total. How many sessions did she attend?

SOLUTION — step by step

Step 1. Cost = fixed part + variable part = 300 + 60n.

Step 2. For 12 sessions: 300 + 60(12) = 300 + 720 = Rs 1020.

Step 3. Reverse: 300 + 60n = 780.

Step 4. Subtract 300: 60n = 480, so n = 8.

Step 5. Check: 300 + 60(8) = 300 + 480 = 780. Correct - she attended 8 sessions.

Answer: (i) 300 + 60n (ii) Rs 1020 (iii) 8 sessions.

Hinglish: *Fixed + rate x count - yahi har bill ka formula hai. Aage jaane ke liye substitute, peeche aane ke liye solve. 780 mein se 300 hataya toh sessions ka khel bacha: 480/60 = 8.*

QUESTION 11 | Basic | Anatomy of expressions (T1)

Q11. For the expression 7p - 4q + 11: (i) list the terms, (ii) give the coefficient of p and of q, (iii) state the constant term, (iv) find its value at p = 2, q = 3.

SOLUTION — step by step

Step 1. (i) Terms: 7p, -4q, 11 (the sign travels WITH the term).

Step 2. (ii) Coefficient of p is 7; coefficient of q is -4 (not 4).

Step 3. (iii) Constant term: 11.

Step 4. (iv) 7(2) - 4(3) + 11 = 14 - 12 + 11 = 13.

» Why this step? The minus sign belongs to its term - writing the coefficient of q as +4 is the classic first-week error.

Answer: (i) 7p, -4q, 11 (ii) 7 and -4 (iii) 11 (iv) 13

Hinglish: *Term ke saath uska sign chalta hai - coefficient of q = -4. Value nikaalna toh bas substitute-and-add hai. 13 aaya, clean.*

QUESTION 12 | Basic | Degree (T2)

Q12. Find the degree of each polynomial: (i) 9x - 5, (ii) $4y^2$ - y + 8, (iii) 12, (iv) $3z^3$ - z, (v) x + y.

SOLUTION — step by step

Step 1. (i) Highest power of x is 1 -> degree 1.

Step 2. (ii) Highest power is 2 -> degree 2.

Step 3. (iii) 12 = $12x^0$: a non-zero constant has degree 0.

Step 4. (iv) Highest power is 3 -> degree 3.

Step 5. (v) x has power 1 and y has power 1 -> degree 1 (take the HIGHEST total power of any term).

» **Why this step?** Degree = highest power of the variable (or highest total degree for multi-variable terms); constants sit at degree 0.

Answer: (i) 1 (ii) 2 (iii) 0 (iv) 3 (v) 1

Hinglish: *Degree = sabse oonchi power. Constant ki degree 0 hoti hai. x + y jaise mein har term ki apni degree dekho, sabse badi hi polynomial ki degree hai.*

QUESTION 13 | Basic | Classification (T2)

Q13. Classify each as constant, linear, quadratic or cubic: 2a + 7, $5t^2$ - 9, 10 - m, $6x^3$, 41.

SOLUTION — step by step

Step 1. 2a + 7: degree 1 -> linear.

Step 2. $5t^2$ - 9: degree 2 -> quadratic.

Step 3. 10 - m: degree 1 -> linear (the variable can be second - degree is what matters).

Step 4. $6x^3$: degree 3 -> cubic.

Step 5. 41: no variable -> constant.

» **Why this step?** Names follow degree, not the order or position of terms.

Answer: Linear; quadratic; linear; cubic; constant

Hinglish: *Naam degree se banta hai: 1-linear, 2-quadratic, 3-cubic, bina variable-constant. 10 - m bhi linear hi hai, m aage peeche hona matter nahi karta.*

QUESTION 14 | Basic | Evaluation (T3)

Q14. For p(x) = 5x - 8 find: (i) p(-3), (ii) p(0), (iii) p(2.5), (iv) the value of x for which p(x) = 12.

SOLUTION — step by step

Step 1. (i) 5(-3) - 8 = -15 - 8 = -23.

Step 2. (ii) 5(0) - 8 = -8.

Step 3. (iii) 5(2.5) - 8 = 12.5 - 8 = 4.5.

Step 4. (iv) 5x - 8 = 12 -> 5x = 20 -> x = 4.

» **Why this step?** Substituting a negative needs brackets: 5(-3) = -15, never +15.

Answer: (i) -23 (ii) -8 (iii) 4.5 (iv) x = 4

Hinglish: *Bracket lagao, sign bachao. p(-3) ka matlab 5 × (-3). Aur p(x) = 12 dena toh seedha chhota equation ban gaya, x = 4.*

QUESTION 15 | Basic | Function machine (T3)

Q15. A machine computes y = 4x + 3 for every input x. (i) Output for input 5? (ii) Which input gives output 43? (iii) Which input gives output 3?

SOLUTION — step by step

Step 1. (i) y = 4(5) + 3 = 23.

Step 2. (ii) 4x + 3 = 43 -> 4x = 40 -> x = 10.

Step 3. (iii) 4x + 3 = 3 -> 4x = 0 -> x = 0.

» **Why this step?** Running a machine forward is substitution; running it backward is solving a linear equation.

Answer: (i) 23 (ii) 10 (iii) 0

Hinglish: *Machine aage chalani hai toh substitute, ulta chalani hai toh equation solve. (iii) mein output input ke barabar aa gaya kyunki x = 0 tha.*

QUESTION 16 | Basic | Evaluation of a quadratic (T3)

Q16. For $q(t) = 2t^2 - 5t + 1$ find: (i) q(-1), (ii) q(3), (iii) q(0).

SOLUTION — step by step

Step 1. (i) $2(-1)^2 - 5(-1) + 1 = 2 + 5 + 1 = 8$.

Step 2. (ii) $2(3)^2 - 5(3) + 1 = 18 - 15 + 1 = 4$.

Step 3. (iii) 2(0) - 5(0) + 1 = 1 (the constant survives).

» **Why this step?** $(-1)^2 = +1$ but -5(-1) = +5: both minus signs turn positive here - one from squaring, one from multiplying.

Answer: (i) 8 (ii) 4 (iii) 1

Hinglish: *Negative substitute karte waqt do jagah sign palatta hai: square mein aur product mein. 8 aaya pehle case mein. Dono ka khayal rakho.*

QUESTION 17 | Basic | Zero of a linear polynomial (T3)

Q17. Find the zero of each: (i) x - 12, (ii) 3x + 21, (iii) 8 - 2x.

SOLUTION — step by step

Step 1. Zero = the x-value that makes the polynomial 0: solve p(x) = 0.

Step 2. (i) x - 12 = 0 -> x = 12.

Step 3. (ii) 3x + 21 = 0 -> x = -7.

Step 4. (iii) 8 - 2x = 0 -> x = 4.

» **Why this step?** A linear polynomial ax + b has exactly one zero, x = -b/a - it crosses the x-axis exactly once.

Answer: (i) 12 (ii) -7 (iii) 4

Hinglish: *Zero nikaalna = polynomial ko zero barabar karna. Ek linear polynomial ki sirf ek zero hoti hai - graph x-axis ko exactly ek baar kaat-ta hai.*

QUESTION 18 | Basic | Word problem: sum and difference (T5)

Q18. Two numbers add up to 40 and their difference is 8. Find the numbers.

SOLUTION — step by step

Step 1. Let the numbers be x and 40 - x (using the sum first).

Step 2. Difference: x - (40 - x) = 8 -> 2x - 40 = 8 -> 2x = 48 -> x = 24.

Step 3. Numbers: 24 and 16.

Step 4. Check: 24 + 16 = 40 ✓ and 24 - 16 = 8 ✓.

» **Why this step?** Using the sum to name both numbers turns a two-variable problem into a one-line equation.

Answer: 24 and 16

Hinglish: *Ek number pakdo, doosra 40 minus usse. Difference ka equation seedha 2x = 48 de deta hai. Check line hamesha - free ka insurance.*

QUESTION 19 | Basic | Perimeter polynomial (T3/T5)

Q19. The perimeter of a square of side s is P = 4s. (i) Find s when P = 52. (ii) Find the area then. (iii) If the side grows by 3, by how much does the perimeter grow?

SOLUTION — step by step

Step 1. (i) 4s = 52 -> s = 13.

Step 2. (ii) Area = 13^2 = 169.

Step 3. (iii) New perimeter = 4(s + 3) = 4s + 12 -> it grows by exactly 12 (4 sides × 3 each).

» **Why this step?** Every unit added to the side is counted FOUR times in the perimeter - that is why +3 side means +12 perimeter.

Answer: (i) s = 13 (ii) 169 (iii) grows by 12

Hinglish: *Side badhne par perimeter 4 guna tezi se badhta hai - chaar diwarein naap rahi hain. +3 side = +12 perimeter, seedha dilation logic.*

QUESTION 20 | Basic | Fee structure modelling (T5)

Q20. A badminton academy charges Rs 400 to join and Rs 150 per month. (i) Write the total cost y for x months. (ii) Cost for 6 months? (iii) How many months does Rs 2500 cover?

SOLUTION — step by step

Step 1. (i) y = 150x + 400 - rate times months plus the one-time fee.

Step 2. (ii) y = 150(6) + 400 = 900 + 400 = Rs 1300.

Step 3. (iii) 150x + 400 = 2500 -> 150x = 2100 -> x = 14 months.

» **Why this step?** Fixed charges become the constant b; per-unit rates become the slope a - the skeleton of every linear model.

Answer: (i) y = 150x + 400 (ii) Rs 1300 (iii) 14 months

Hinglish: *Joining fee = b (fixed), monthly = a (slope). y = 150x + 400. Ulta question (kitne months) = equation solve karo: 14 months.*

QUESTION 21 | Basic | Pattern rule (T6/T7)

Q21. A sequence begins 6, 11, 16, 21, ... (i) Write the rule T(n). (ii) Find the 20th term. (iii) Which term equals 121?

SOLUTION — step by step

Step 1. (i) Each step adds 5 -> T(n) = 5n + c; T(1) = 6 gives c = 1, so T(n) = 5n + 1.

Step 2. (ii) T(20) = 5(20) + 1 = 101.

Step 3. (iii) 5n + 1 = 121 -> 5n = 120 -> n = 24: the 24th term.

» **Why this step?** Constant-step sequences are linear: slope = step size, intercept adjusts to hit the first term.

Answer: (i) T(n) = 5n + 1 (ii) 101 (iii) 24th

Hinglish: *Har baar +5 -> slope 5. Pehla term se c adjust karo: 5(1) + 1 = 6. Baaki dono parts seedha substitute/solve.*

QUESTION 22 | Basic | Linear decay money (T8)

Q22. Bela has Rs 900 and spends Rs 45 every day. (i) Write M(d), her money after d days. (ii) When does her purse empty? (iii) How much is left after 8 days?

SOLUTION — step by step

Step 1. (i) M(d) = 900 - 45d (negative slope: decay).

Step 2. (ii) 900 - 45d = 0 -> d = 20 days.

Step 3. (iii) M(8) = 900 - 360 = Rs 540.

» **Why this step?** Spending creates a NEGATIVE slope; the zero of the model answers 'when will it run out'.

Answer: (i) M(d) = 900 - 45d (ii) 20 days (iii) Rs 540

Hinglish: *Kharcha = minus slope. Purse khaali = M(d) = 0 -> 20 din. 8 din baad 540 bache. Decay model ka zero hi asli sawaal hota hai.*

QUESTION 23 | Basic | Solving one-step equations (T5)

Q23. Solve: (i) 5x - 7 = 18, (ii) 3(y + 4) = 27, (iii) 12 - 2m = 0.

SOLUTION — step by step

Step 1. (i) 5x = 25 -> x = 5.

Step 2. (ii) y + 4 = 9 -> y = 5 (divide first to keep it light).

Step 3. (iii) 2m = 12 -> m = 6.

» **Why this step?** Undo operations in reverse order; distributing or dividing first are both legal - pick the cleaner path.

Answer: (i) x = 5 (ii) y = 5 (iii) m = 6

Hinglish: *Operations ko ulta karo. (ii) mein pehle 3 se divide karna smart tha - chhota equation, kam galti. Teenon ke answers positive aaye, life is good.*

QUESTION 24 | Basic | Slope and intercept reading (T10)

Q24. State the slope and y-intercept of: (i) y = 7x - 4, (ii) y = 10 - 3x, (iii) y = x/2 + 6.

SOLUTION — step by step

Step 1. Read from y = ax + b: a is the slope, b is the y-intercept.

Step 2. (i) slope 7, intercept (0, -4).

Step 3. (ii) Rewrite: y = -3x + 10 -> slope -3, intercept (0, 10).

Step 4. (iii) y = 0.5x + 6 -> slope 0.5, intercept (0, 6).

» **Why this step?** Put the x-term first and the sign attached to it is the slope - y = 10 - 3x fools nobody who rewrites it.

Answer: (i) 7, (0, -4) (ii) -3, (0, 10) (iii) 0.5, (0, 6)

Hinglish: *Format mein lao: y = ax + b. (ii) mein -3x aage lao, slope -3 clear. Fraction slope bhi slope hi hai - 1/2 = dheere chadhta line.*

QUESTION 25 | Basic | Point-on-line test (T13)

Q25. Which of these points lie on the line y = 2x - 5: (3, 1), (0, -5), (5, 3), (-2, -9)?

SOLUTION — step by step

Step 1. Test each: substitute x, check whether y comes out right.

Step 2. (3, 1): 2(3) - 5 = 1 ✓ ON.

Step 3. (0, -5): -5 ✓ ON (this is the y-intercept!).

Step 4. (5, 3): 2(5) - 5 = 5 ≠ 3 ✗ OFF.

Step 5. (-2, -9): 2(-2) - 5 = -9 ✓ ON.

» **Why this step?** A point is on a line exactly when its coordinates satisfy the equation - three checks passed, one exposed the impostor.

Answer: (3, 1), (0, -5), (-2, -9) lie on it; (5, 3) does not

Hinglish: *Point ko equation mein daalo, dono side match? Match = line par hai. (5, 3) pakda gaya - 5 aana chahiye tha, 3 de diya. y-intercept (0, -5) bhi list mein tha, notice kiya?*

QUESTION 26 | Basic | Intercepts and the axis triangle (T12)

Q26. For the line y = 4x + 12: (i) find the y-intercept, (ii) the x-intercept, (iii) the area of the triangle formed with the two axes.

SOLUTION — step by step

Step 1. (i) Put x = 0: y = 12 -> (0, 12).

Step 2. (ii) Put y = 0: 4x + 12 = 0 -> x = -3 -> (-3, 0).

Step 3. (iii) The axes are perpendicular, so the triangle is right-angled with legs 12 and 3.

Step 4. Area = ½ × 12 × 3 = 18 square units.

» **Why this step?** Intercepts are the triangle's two legs on the axes - area = half their product.

Answer: (i) (0, 12) (ii) (-3, 0) (iii) 18

Hinglish: *x-intercept nikaalna = y ko 0 karna; y-intercept = x ko 0. Triangle seedha intercepts se banta hai: ½ × 12 × 3 = 18.*

QUESTION 27 | Basic | Growth vs decay tagging (T8)

Q27. Tag each model as growth or decay and write it as y = ax + b: (i) a plant 6 cm tall grows 2 cm per week, (ii) a phone worth Rs 12,000 loses Rs 500 per month, (iii) a tank with 50 L fills at 30 L per minute, (iv) room temperature 28° falls 3° per hour.

SOLUTION — step by step

Step 1. (i) Growth: y = 2x + 6 (x in weeks).

Step 2. (ii) Decay: y = -500x + 12000 (x in months).

Step 3. (iii) Growth: y = 30x + 50 (x in minutes).

Step 4. (iv) Decay: y = -3x + 28 (x in hours).

» **Why this step?** The verb decides the sign: grows/fills/adds -> +a; loses/drains/falls -> -a.

Answer: (i) growth, y = 2x + 6 (ii) decay, y = -500x + 12000 (iii) growth, y = 30x + 50 (iv) decay, y = -3x + 28

Hinglish: *Verb dekho, slope ka sign mil gaya. Badhna plus, ghatna minus. Har model ka b = starting value, a = per-unit speed.*

QUESTION 28 | Basic | Steepness comparison (T11)

Q28. Compare y = 6x + 1, y = 2x + 9 and y = -8x: which is steepest? Which is the only one that DECREASES as x increases?

SOLUTION — step by step

Step 1. Steepness = |slope|: |6| = 6, |2| = 2, |-8| = 8.

Step 2. Steepest: y = -8x (magnitude 8 wins).

Step 3. Decreasing line: the one with negative slope -> y = -8x only.

Step 4. Note: y = -8x passes through the origin - its intercept is 0.

» **Why this step?** Direction comes from the sign of a; steepness comes from its size - two separate questions.

Answer: Steepest: y = -8x; the only decreasing one: also y = -8x

Hinglish: *Steep = |slope| bada. -8 ka magnitude sabse bada, isliye wahi steepest, aur negative isliye wahi girta hua line. Do alag sawaal, do alag jawab.*

QUESTION 29 | Basic | True/False theory (T14)

Q29. True or False (give a one-line reason): (i) every linear polynomial has exactly one zero. (ii) the line y = 5 has slope 0. (iii) x = -2 is a vertical line. (iv) parallel lines have equal slopes. (v) the y-intercept of y = 3x is 3.

SOLUTION — step by step

Step 1. (i) TRUE: ax + b = 0 always has the single solution x = -b/a (a ≠ 0).

Step 2. (ii) TRUE: y = 5 is horizontal - height never changes, slope 0.

Step 3. (iii) TRUE: x = -2 collects all points with that x - a vertical line.

Step 4. (iv) TRUE: equal slope = same tilt = never meeting.

Step 5. (v) FALSE: y = 3x has intercept 0 (it passes through the origin); 3 is the SLOPE.

» **Why this step?** (v) is the slope-intercept swap - the single most common mix-up in this chapter.

Answer: (i) T (ii) T (iii) T (iv) T (v) F (intercept is 0; 3 is the slope)

Hinglish: *(v) wala dhoka har saal hota hai: 3 slope hai, intercept 0. y = 5 horizontal, x = -2 vertical - dono special lines ka behaviour yaad rakho.*

QUESTION 30 | Basic | Sentence to equation (T5)

Q30. Translate and solve: (i) 'A number doubled plus 5 gives 17.' (ii) '7 less than four times n is 9.' (iii) 'Half of a number, increased by 3, is 11.'

SOLUTION — step by step

Step 1. (i) 2x + 5 = 17 -> 2x = 12 -> x = 6.

Step 2. (ii) 4n - 7 = 9 -> 4n = 16 -> n = 4 ('7 less than' means subtract FROM the expression).

Step 3. (iii) x/2 + 3 = 11 -> x/2 = 8 -> x = 16.

» **Why this step?** 'Less than' reverses the order of mention - four times n FIRST, then take 7 away.

Answer: (i) x = 6 (ii) n = 4 (iii) x = 16

Hinglish: *'7 less than 4n' = 4n - 7, not 7 - 4n. English ka order ulta chal jaata hai, yahi pehli galti hoti hai. Baaki solve toh seedha hai.*

Level 2 · Core Drills: book-level practice:

QUESTION 31 | Moderate | Pattern rule (T6/T7)

Q31. A pattern of square tiles starts with 5 tiles in stage 1, and each new stage adds 3 tiles. (i) Write the number of tiles in stage n. (ii) How many tiles in stage 12? (iii) Which stage has exactly 65 tiles?

SOLUTION — step by step

Step 1. The sequence is 5, 8, 11, 14, ... with constant difference 3.

Step 2. Rule: tiles(n) = 3n + c. Match stage 1: 3(1) + c = 5 gives c = 2. So tiles(n) = 3n + 2.

Step 3. Stage 12: 3(12) + 2 = 38 tiles.

Step 4. Reverse: 3n + 2 = 65 gives 3n = 63, n = 21.

Step 5. n = 21 is a whole number, so stage 21 has exactly 65 tiles.

Answer: (i) 3n + 2 (ii) 38 tiles (iii) stage 21.

Hinglish: *Constant difference pehle pakdo (3), phir ek known term se c nikaalo (yahan 2). Aage ke liye substitute, peeche ke liye solve - aur n WHOLE NUMBER aana chahiye, warna stage exist nahi karti. Yahan 21 nikla, perfect.*

QUESTION 32 | Moderate | Linear decay money (T8)

Q32. Bela has Rs 500 and spends Rs 30 every day. (i) Write the amount left after n days. (ii) After how many days will she be left with Rs 200? (iii) What is the amount left after 15 days, and why does the model stay valid only up to 10 days of reaching zero? Actually find the exact day her money runs out.

SOLUTION — step by step

Step 1. Amount(n) = 500 - 30n.

Step 2. Set 500 - 30n = 200: 30n = 300, n = 10 days.

Step 3. After 15 days: 500 - 30(15) = 500 - 450 = Rs 50.

Step 4. Running out: 500 - 30n = 0 gives n = 500/30 = 16.67 days. Money reaches zero partway through the 17th day.

Step 5. So the linear model describes her cash only until about 16.67 days; beyond that the amount cannot go below 0.

» **Why this step?** Real decay models hit a floor. Reporting WHEN the floor is reached (and that the model stops there) is the mark of a careful modeller.

Answer: (i) 500 - 30n (ii) after 10 days (iii) Rs 50 after 15 days; money hits zero during the 17th day (n = 500/30 = 16.67), where the model ends.

Hinglish: *Decay question mein teen sawaal hote hain: kitna bacha (substitute), kab itna bacha (solve), kab khatam (zero set karo). Yahan 16.67 din - matlab 17ve din ke beech mein khatam. Model ki limit batana bhi answer ka hissa hai - paisa negative nahi ho sakta!*

QUESTION 33 | Moderate | Piecewise-style fare (T8)

Q33. A taxi charges Rs 40 for the first 3 km (and any part of it), then Rs 18 for every additional km. (i) Find the fare for a 12 km trip. (ii) Write the fare for a trip of n km where n is at least 3. (iii) For what distance is the fare Rs 148?

SOLUTION — step by step

Step 1. Beyond 3 km, chargeable extra distance = (n - 3) km.

Step 2. Fare for 12 km: 40 + 18 x (12 - 3) = 40 + 162 = Rs 202.

Step 3. General: F(n) = 40 + 18(n - 3) = 18n - 54 + 40 = 18n - 14, valid for n at least 3.

Step 4. Set 18n - 14 = 148: 18n = 162, n = 9 km.

Step 5. Check with the original description: 40 for the first 3 km plus 18 x 6 = 108 for the next 6 km: total 148. Correct.

Answer: (i) Rs 202 (ii) F(n) = 18n - 14 for n at least 3 (iii) 9 km.

Hinglish: *PDF ke auto-fare jaisa hi question, bas numbers naye. Formula banane ke liye (n - 3) ka idea zaroori hai - pehle 3 km ka paisa already fixed 40 mein hai. Simplify karke 18n - 14 milta hai aur reverse solve se 9 km. Hamesha original description se verify karo.*

QUESTION 34 | Moderate | Age problem (T5)

Q34. A mother is 4 times as old as her daughter. After 6 years, the sum of their ages will be 82 years. Find their present ages.

SOLUTION — step by step

Step 1. Let the daughter be x now; the mother is 4x.

Step 2. After 6 years: daughter x + 6, mother 4x + 6.

Step 3. Sum: (x + 6) + (4x + 6) = 82.

Step 4. Simplify: 5x + 12 = 82, so 5x = 70, x = 14.

Step 5. Present ages: daughter 14, mother 56.

Step 6. Check: after 6 years they are 20 and 62, sum 82. Correct.

Answer: Daughter 14 years, mother 56 years.

Hinglish: *Age questions mein 'after' ka matlab dono ke age mein addition. 5x + 12 = 82 se x = 14. Check zaroori: 20 + 62 = 82. Mother 4x isliye 56 - equation se nahi, definition se.*

QUESTION 35 | Moderate | Ratio and difference (T5)

Q35. Two positive integers are in the ratio 4 : 7 and their difference is 45. Find the numbers.

SOLUTION — step by step

Step 1. Let the numbers be 4x and 7x (ratio guarantees the common factor x).

Step 2. Difference: 7x - 4x = 3x = 45, so x = 15.

Step 3. Numbers: 4(15) = 60 and 7(15) = 105.

Step 4. Check: 105 - 60 = 45 and 60 : 105 simplifies to 4 : 7. Correct.

Answer: 60 and 105.

Hinglish: *Ratio diya hai toh numbers ko 4x, 7x maano - ye shortcut hi asli technique hai. Difference equation ekdum linear ban jaata hai: 3x = 45. Ratio wapas check karo - 60:105, dono ko 15 se divide karo, 4:7. Sahi.*

QUESTION 36 | Moderate | Coin problem (T5)

Q36. Rohit has five-rupee and ten-rupee coins. He has twice as many five-rupee coins as ten-rupee coins, and the total money is Rs 100. How many coins of each type does he have?

SOLUTION — step by step

Step 1. Let the number of ten-rupee coins be x. Then five-rupee coins = 2x.

Step 2. Total money: 10x + 5(2x) = 100.

Step 3. Simplify: 10x + 10x = 100, so 20x = 100, x = 5.

Step 4. Ten-rupee coins: 5 (Rs 50). Five-rupee coins: 10 (Rs 50).

Step 5. Check: 50 + 50 = 100 and 10 is twice 5. Correct.

Answer: 5 ten-rupee coins and 10 five-rupee coins.

Hinglish: *'Twice as many' ka matlab: jo KAM hai wo x, jo zyada hai wo 2x. Yahan 5-rupee zyada hain isliye 2x. Money equation mein VALUE se jodo - coin count nahi: 10x + 10x = 100. Sweet cancellation mili kyunki 5 x 2 = 10.*

QUESTION 37 | Moderate | Cutting a length (T5)

Q37. A 240 cm rope is cut into two pieces so that the longer piece is three times the shorter piece. Find the length of each piece.

SOLUTION — step by step

Step 1. Let the shorter piece be x. The longer is 3x.

Step 2. Total: x + 3x = 240, so 4x = 240, x = 60.

Step 3. Pieces: 60 cm and 180 cm.

Step 4. Check: 60 + 180 = 240 and 180 = 3 x 60. Correct.

Answer: 60 cm and 180 cm.

Hinglish: *4x = 240 - chaar hisse jaise. Ek hissa 60, teen hisse 180. Simple, par 'both pieces report karo' wala discipline yaad rakho.*

QUESTION 38 | Moderate | Rectangle dimensions (T5)

Q38. The length of a rectangle is 3 more than twice its width, and its perimeter is 36 cm. Find the dimensions.

SOLUTION — step by step

Step 1. Let the width be w; then the length is 2w + 3.

Step 2. Perimeter: 2(w + 2w + 3) = 36.

Step 3. Inside out: 3w + 3 = 18, so 3w = 15, w = 5.

Step 4. Width 5 cm, length 2(5) + 3 = 13 cm.

Step 5. Check: 2(5 + 13) = 36 and 13 is indeed 3 more than twice 5. Correct.

Answer: Width 5 cm, length 13 cm.

Hinglish: *Perimeter formula ke andar hi expression daal do, phir dheere-dheere simplify: pehle 2 se divide, phir 3 se. '3 more than twice' ka translation (2w + 3) hi question ka dil hai.*

QUESTION 39 | Moderate | Linear growth (T8)

Q39. A sapling is 1.75 feet tall and grows 0.5 feet each month. (i) Write its height after t months. (ii) Height after 7 months? (iii) After how many months will it be 6.75 feet tall?

SOLUTION — step by step

Step 1. h(t) = 1.75 + 0.5t.

Step 2. h(7) = 1.75 + 3.5 = 5.25 feet.

Step 3. Set 1.75 + 0.5t = 6.75: 0.5t = 5, t = 10 months.

Step 4. Check: 1.75 + 0.5(10) = 1.75 + 5 = 6.75. Correct.

Answer: (i) h(t) = 1.75 + 0.5t (ii) 5.25 feet (iii) 10 months.

Hinglish: *Growth = start + rate x time. Decimals se ghabrana nahi - 1.75 + 3.5 = 5.25 seedha hai. Reverse mein 5/0.5 = 10 - decimal divide karte waqt dono ko 100 se multiply karna comfortable ho jaata hai practice se.*

QUESTION 40 | Moderate | Linear decay depreciation (T8)

Q40. A phone bought for Rs 10,000 loses Rs 800 in value every year. (i) Value after t years? (ii) Value after 3 years? (iii) After how many years will the value be Rs 3,600?

SOLUTION — step by step

Step 1. v(t) = 10000 - 800t.

Step 2. v(3) = 10000 - 2400 = Rs 7,600.

Step 3. Set 10000 - 800t = 3600: 800t = 6400, t = 8 years.

Step 4. Check: 10000 - 800(8) = 10000 - 6400 = 3600. Correct.

Answer: (i) v(t) = 10000 - 800t (ii) Rs 7,600 (iii) 8 years.

Hinglish: *Depreciation = linear decay ka business avatar. 6400/800 = 8 - tables ka kaam. Model tab tak valid jab tak value non-negative: 10000/800 = 12.5 saal ke baad phone 'zero value' ka zone mein.*

QUESTION 41 | Moderate | Fit y = ax + b from bills (T9)

Q41. A gym charges a fixed joining fee plus a monthly rate. One month costs Rs 900 in total; three months cost Rs 2100. Find the monthly rate and the joining fee, and write the model.

SOLUTION — step by step

Step 1. Model: y = ax + b where a = monthly rate, b = joining fee.

Step 2. Month 1: a + b = 900. Month 3: 3a + b = 2100.

Step 3. Subtract: 2a = 1200 -> a = 600.

Step 4. Then b = 900 - 600 = 300. Model: y = 600x + 300.

Step 5. Check month 3: 600(3) + 300 = 2100 ✓.

» **Why this step?** Two observations + two unknowns -> subtract the equations; the b's cancel and a falls out first.

Answer: a = Rs 600/month, b = Rs 300; y = 600x + 300

Hinglish: *Do bills mile toh equations ko subtract karo - joining fee cancel ho gayi, monthly rate 600 nikla. Fee 300. Subtraction is the move here.*

QUESTION 42 | Moderate | Piecewise-style fare (T8)

Q42. A taxi charges Rs 60 for the first 2 km and Rs 18 per km after that. (i) Fare for a 9 km ride? (ii) Write a formula for the fare y when the ride is x km (x ≥ 2). (iii) A passenger paid Rs 168 - how long was the ride?

SOLUTION — step by step

Step 1. (i) Beyond 2 km: 9 - 2 = 7 chargeable km. y = 60 + 7(18) = 60 + 126 = Rs 186.

Step 2. (ii) y = 18(x - 2) + 60 = 18x - 36 + 60 = 18x + 24 for x ≥ 2.

Step 3. (iii) 18x + 24 = 168 -> 18x = 144 -> x = 8 km.

» **Why this step?** 'First n km at one rate' shifts the line: expanding 18(x − 2) + 60 builds the intercept 24 automatically.

Answer: (i) Rs 186 (ii) y = 18x + 24 (x ≥ 2) (iii) 8 km

Hinglish: *Pehle 2 km ka alag hisaab -> formula mein (x - 2). Simplify karke 18x + 24 - intercept khud adjust ho gaya. Reverse question mein 168 se 8 km nikla.*

QUESTION 43 | Moderate | Age problem (T5)

Q43. A father is 3 times as old as his son. In 12 years he will be only twice as old. Find their present ages.

SOLUTION — step by step

Step 1. Let the son be s; father = 3s.

Step 2. In 12 years: 3s + 12 = 2(s + 12).

Step 3. 3s + 12 = 2s + 24 -> s = 12.

Step 4. Son is 12, father is 36. Check: in 12 years, 48 = 2 × 24 ✓.

» **Why this step?** Age problems stay linear because EVERYONE ages at the same rate - both sides grow by the same 12.

Answer: Son 12, father 36

Hinglish: *Dono ki age barabar badhti hai, isliye equation linear rehti hai. s = 12, father 36. 'Twice as old' ko 2(s + 12) likhna - 2s + 12 nahi, ye galti mat karna!*

QUESTION 44 | Moderate | Coin problem (T5)

Q44. Rohit has 24 coins, only Rs 5 and Rs 10 coins, worth Rs 170 in total. How many of each does he have?

SOLUTION — step by step

Step 1. Let the Rs 5 coins be a; Rs 10 coins = 24 - a.

Step 2. Value: 5a + 10(24 - a) = 170.

Step 3. 5a + 240 - 10a = 170 -> -5a = -70 -> a = 14.

Step 4. 14 five-rupee and 10 ten-rupee coins. Check: 70 + 100 = 170 ✓.

» **Why this step?** Naming one count forces the other through the total - one variable, one equation, done.

Answer: 14 five-rupee coins and 10 ten-rupee coins

Hinglish: *Ek count ko a maan lo, doosra 24 - a khud ban jaata hai. Value equation se a = 14. Signs dekh ke solve karo, minus a ka friend hai yahan.*

QUESTION 45 | Moderate | Cutting and re-forming (T5/T12)

Q45. A 96 cm wire is cut so that one piece is three times the other. (i) Find both pieces. (ii) The longer piece is bent into a rectangle whose length is twice its width - find the rectangle's area.

SOLUTION — step by step

Step 1. (i) Pieces: x and 3x with x + 3x = 96 -> x = 24. Pieces: 24 cm and 72 cm.

Step 2. (ii) Rectangle perimeter 72: 2(w + 2w) = 72 -> 6w = 72 -> w = 12, length = 24.

Step 3. Area = 12 × 24 = 288 cm^2.

» **Why this step?** Ratios split the total into equal 'units' (4 units of 24); then perimeter formulas convert length to area.

Answer: (i) 24 cm and 72 cm (ii) 288 cm^2

Hinglish: *1:3 ratio = 4 hisse, har hissa 24. Lammi tukdi (72) ka rectangle: 2w + 4w = 72 -> w = 12. Area 288. Ratio se shuru, area par khatam.*

QUESTION 46 | Moderate | Rectangle dimensions (T5)

Q46. A rectangle's length is 5 more than its width and its perimeter is 58. Find the width, length and area.

SOLUTION — step by step

Step 1. Let width = w, length = w + 5.

Step 2. 2(w + w + 5) = 58 -> 2(2w + 5) = 58 -> 4w + 10 = 58 -> 4w = 48 -> w = 12.

Step 3. Length = 17; area = 12 × 17 = 204.

Step 4. Check perimeter: 2(12 + 17) = 58 ✓.

» **Why this step?** Describe BOTH dimensions through one variable first - that is what keeps the equation linear and short.

Answer: Width 12, length 17, area 204

Hinglish: *Dono dimensions ek hi variable se likho (w aur w + 5), equation chhoti ho jaati hai. w = 12, l = 17, area 204. Check se confirm.*

QUESTION 47 | Moderate | Line through two points (T9)

Q47. A line passes through (1, 5) and (3, 11). (i) Find its equation. (ii) Find y when x = 10. (iii) Find its x-intercept.

SOLUTION — step by step

Step 1. (i) Slope a = (11 - 5)/(3 - 1) = 6/2 = 3. Then 5 = 3(1) + b -> b = 2. Line: y = 3x + 2.

Step 2. (ii) y = 3(10) + 2 = 32.

Step 3. (iii) 0 = 3x + 2 -> x = -2/3 -> x-intercept (-2/3, 0).

» **Why this step?** Slope from the two-point rise-over-run, then one point back-solves the intercept - the standard two-step fit.

Answer: (i) y = 3x + 2 (ii) 32 (iii) (-2/3, 0)

Hinglish: *Slope = 11-5 upar, 3-1 neeche = 3. Kisi bhi ek point se b nikal lo: b = 2. x-intercept fraction mein aa gaya (-2/3) - theek hai, fractions bhi points hote hain.*

QUESTION 48 | Moderate | Parallel through a point (T12)

Q48. Find the equation of the line parallel to y = 4x - 9 that passes through (1, 6). Where does this new line cut the y-axis?

SOLUTION — step by step

Step 1. Parallel -> same slope: a = 4.

Step 2. y = 4x + b through (1, 6): 6 = 4(1) + b -> b = 2.

Step 3. Line: y = 4x + 2.

Step 4. y-axis crossing: x = 0 gives y = 2 -> point (0, 2).

» **Why this step?** Parallelism locks the slope; the given point then donates its coordinates to find b.

Answer: y = 4x + 2; cuts the y-axis at (0, 2)

Hinglish: *Parallel = slope same (4). Point se b = 2. Naya line y = 4x + 2, y-axis (0, 2) par kaat-ta hai. Slope copy, intercept custom - bas yahi recipe hai.*

QUESTION 49 | Moderate | Depreciation timeline (T8)

Q49. A laptop bought for Rs 52,000 loses Rs 4,000 in value every year. (i) Write V(t). (ii) After how many years will its value be Rs 8,000? (iii) What is its value after 4.5 years?

SOLUTION — step by step

Step 1. (i) V(t) = 52000 - 4000t.

Step 2. (ii) 52000 - 4000t = 8000 -> 4000t = 44000 -> t = 11 years.

Step 3. (iii) V(4.5) = 52000 - 18000 = Rs 34,000.

» **Why this step?** Half-years are legal inputs in a linear model - the slope works for any real t.

Answer: (i) V(t) = 52000 - 4000t (ii) 11 years (iii) Rs 34,000

Hinglish: *Straight-line depreciation: har saal -4000. Zero value se pehle hi 8000 par pahunch gaya, 11 saal. Aadhe saal ka answer bhi valid hai (34000).*

QUESTION 50 | Moderate | Two models crossing (T14)

Q50. Tank A starts empty and fills at 12 L/min: $V_A(t) = 12t$. Tank B starts with 300 L and drains at 20 L/min: $V_B(t) = 300 - 20t$. (i) When do the two tanks hold equal amounts? (ii) How much is in each then?

SOLUTION — step by step

Step 1. (i) 12t = 300 - 20t -> 32t = 300 -> t = 300/32 = 9.375 minutes.

Step 2. (ii) V = 12 × 9.375 = 112.5 L (check B: 300 - 187.5 = 112.5 ✓).

Step 3. Interpretation: at about 9 min 22.5 s both tanks hold 112.5 L.

» **Why this step?** Setting two models equal finds their meeting time - the intersection point of the two lines.

Answer: (i) t = 9.375 min (ii) 112.5 L each

Hinglish: *Do models ko barabar rakho = lines ka intersection. t = 9.375 min, dono mein 112.5 L. Fraction ko minutes mein convert karna smart hai: 22.5 seconds.*

QUESTION 51 | Moderate | Two-condition number problem (T5)

Q51. Two numbers differ by 7. Three times the smaller equals twice the larger minus 11. Find both numbers.

SOLUTION — step by step

Step 1. Let the smaller be s; larger = s + 7.

Step 2. 3s = 2(s + 7) - 11 -> 3s = 2s + 14 - 11 -> 3s = 2s + 3 -> s = 3.

Step 3. Numbers: 3 and 10. Check: 3(3) = 9 and 2(10) - 11 = 9 ✓.

» **Why this step?** Name the smaller, build the larger with the gap - then the condition writes itself.

Answer: 3 and 10

Hinglish: *Chhota s, bada s + 7. Condition daalo, s = 3. Check: dono taraf 9 - perfect. Chhote wale ko variable banao, life easy.*

QUESTION 52 | Moderate | Matchstick pattern (T6/T7)

Q52. Squares are made with matchsticks in a row: the first needs 4 sticks, and each new square shares one side with the previous. (i) Write T(n). (ii) Sticks for 15 squares? (iii) How many squares can 100 sticks build?

SOLUTION — step by step

Step 1. (i) T(n) = 4 + 3(n - 1) = 3n + 1 (each new square adds only 3 fresh sticks).

Step 2. (ii) T(15) = 3(15) + 1 = 46 sticks.

Step 3. (iii) 3n + 1 = 100 -> 3n = 99 -> n = 33 squares.

» **Why this step?** Shared sides reduce each new figure's cost: the pattern's slope is the marginal stick count (3), not the first figure's total (4).

Answer: (i) T(n) = 3n + 1 (ii) 46 (iii) 33

Hinglish: *Pehla square 4, baaki har ek sirf +3 kyunki ek side udhaar li. Slope 3, intercept 1. 100 sticks = 33 squares, remainder zero - clean!*

QUESTION 53 | Moderate | Hexagon matchstick pattern (T6/T7)

Q53. Regular hexagons are joined in a row, each new hexagon sharing one side with the previous one. The row of 1 needs 6 sticks. (i) Write T(n). (ii) Sticks for 20 hexagons?

SOLUTION — step by step

Step 1. (i) First costs 6; each new one adds 5 (one side shared): T(n) = 6 + 5(n - 1) = 5n + 1.

Step 2. (ii) T(20) = 5(20) + 1 = 101 sticks.

» **Why this step?** Same logic as squares, one bigger marginal cost: slope = sides minus 1 shared side.

Answer: (i) T(n) = 5n + 1 (ii) 101

Hinglish: *Hexagon wala bhi same game: pehla 6, aage har +5. 5n + 1. 20 hexagons = 101 sticks. Pattern family ke saare members ek hi template se bante hain.*

QUESTION 54 | Moderate | Temperature scales (T9)

Q54. The relation is F = 1.8C + 32. (i) Find C when F = 95. (ii) Find the temperature where F = C (the same number on both scales).

SOLUTION — step by step

Step 1. (i) 1.8C + 32 = 95 -> 1.8C = 63 -> C = 35 (a hot Indian afternoon!).

Step 2. (ii) Set C = 1.8C + 32 -> C - 1.8C = 32 -> -0.8C = 32 -> C = -40.

Step 3. So -40°C = -40°F: the famous point where both scales agree.

» **Why this step?** F = C happens where the two lines y = x and F = 1.8C + 32 intersect - one equation, one crossing.

Answer: (i) C = 35 (ii) -40

Hinglish: *95°F = 35°C, garmi ka season! Aur -40 par dono scale mil jaate hain - ek hi number. Line aur identity line ka intersection hi yeh classic answer hai.*

QUESTION 55 | Moderate | Savings race (T8)

Q55. Meera's jar has Rs 1,250 and she adds Rs 75 per week. Her brother has Rs 1,900 and adds Rs 50 per week. (i) When will Meera catch up? (ii) How much will each have then?

SOLUTION — step by step

Step 1. Meera: M(w) = 1250 + 75w. Brother: B(w) = 1900 + 50w.

Step 2. (i) 1250 + 75w = 1900 + 50w -> 25w = 650 -> w = 26 weeks.

Step 3. (ii) M(26) = 1250 + 1950 = Rs 3,200 (B: 1900 + 1300 = 3200 ✓).

» **Why this step?** The faster saver overtakes where the two lines meet - after that Meera stays ahead forever.

Answer: (i) 26 weeks (ii) Rs 3,200 each

Hinglish: *Meera tez bachati hai (75 vs 50), par peeche hai (1250 vs 1900). 26 hafte mein barabar, Rs 3200 dono ke paas. Uske baad Meera hamesha aage.*

QUESTION 56 | Moderate | Chase problem (T8)

Q56. Runner A moves at 6 m/s from the start line. Runner B starts 20 m ahead of the start line but moves at 4 m/s. (i) After how many seconds does A catch B? (ii) How far from the start line?

SOLUTION — step by step

Step 1. A's position: 6t. B's position: 4t + 20.

Step 2. (i) 6t = 4t + 20 -> 2t = 20 -> t = 10 s.

Step 3. (ii) 6(10) = 60 m (B: 40 + 20 = 60 ✓).

» **Why this step?** Positions as linear functions of time; the catch-up moment is their intersection.

Answer: (i) 10 s (ii) 60 m

Hinglish: *B ka head start = +20 intercept. A tez hai (6 vs 4), 10 second mein pakad liya, 60 metre par. Chase = dono lines ka crossing point.*

QUESTION 57 | Moderate | Reverse data bill (T9/T10)

Q57. A telecom bill follows y = 25x + 120 (y in rupees, x = GB used). (i) Bill for 14 GB? (ii) A customer paid Rs 220 - how many GB? (iii) What does the 120 represent?

SOLUTION — step by step

Step 1. (i) y = 25(14) + 120 = 350 + 120 = Rs 470.

Step 2. (ii) 25x + 120 = 220 -> 25x = 100 -> x = 4 GB.

Step 3. (iii) 120 is the fixed base rental - payable even at 0 GB (y-intercept).

» **Why this step?** Reading a model forward gives bills; reading it backward gives usage - and the intercept always names the fixed cost.

Answer: (i) Rs 470 (ii) 4 GB (iii) fixed base rental (y-intercept)

Hinglish: *Forward = bill banao, backward = usage nikaalo. 220 se 4 GB mila. Aur 120 = fixed rental, zero GB par bhi dena padega.*

QUESTION 58 | Moderate | Two lines meeting (T14)

Q58. Find the intersection of y = 2x + 1 and y = -x + 7, and verify it lies on both lines.

SOLUTION — step by step

Step 1. Set equal: 2x + 1 = -x + 7 -> 3x = 6 -> x = 2.

Step 2. y = 2(2) + 1 = 5. Point (2, 5).

Step 3. Check line 2: -2 + 7 = 5 ✓ - lies on both.

» **Why this step?** Intersection = solve the two equations simultaneously; the verification closes the loop.

Answer: (2, 5)

Hinglish: *Barabar rakho, x = 2, y = 5. Dono lines par check kiya - sach mein wahi point. Intersection nikalna ek chhota equation solve karna hai.*

QUESTION 59 | Moderate | Concurrent lines check (T14)

Q59. Do the lines y = x + 1, y = 2x - 1 and y = 3x - 3 all pass through one common point?

SOLUTION — step by step

Step 1. Intersect L1 and L2: x + 1 = 2x - 1 -> x = 2, y = 3. Candidate point (2, 3).

Step 2. Test L3 at x = 2: y = 3(2) - 3 = 3 ✓ - it passes too!

Step 3. All three lines are concurrent at (2, 3).

» **Why this step?** Concurrence test: intersect two lines, then AUDIT the third at that point.

Answer: Yes - all three pass through (2, 3)

Hinglish: *Do lines ka point nikaalo, teesri mein daal ke check karo - 3 aa gaya, teeno ek hi point (2, 3) par milti hain. Concurrency = intersect-then-verify.*

QUESTION 60 | Moderate | Axis triangle from a line (T12)

Q60. The line y = -2x + 8 meets both axes. (i) Find the two intercepts. (ii) Find the area of the triangle formed with the axes. (iii) Find the length of the third side (between the intercepts).

SOLUTION — step by step

Step 1. (i) y-intercept: (0, 8). x-intercept: 0 = -2x + 8 -> x = 4 -> (4, 0).

Step 2. (ii) Right triangle legs 8 and 4: area = $\frac{1}{2} \times 8 \times 4 = 16$.

Step 3. (iii) Third side = $\sqrt{(4^2 + 8^2)} = \sqrt{80} = 4\sqrt{5}$.

» **Why this step?** Intercepts are the legs of an axis triangle; the segment between them is its hypotenuse.

Answer: (i) (0, 8), (4, 0) (ii) 16 (iii) $4\sqrt{5}$

Hinglish: *Intercepts (0, 8) aur (4, 0) - triangle free mein mil gaya. Area 16, hypotenuse $4\sqrt{5}$. Line + axes = ready-made right triangle, har baar.*

QUESTION 61 | Moderate | Plan comparison (T14)

Q61. Mobile Plan A: Rs 199 rental + Rs 0.8 per minute. Plan B: Rs 299 rental + Rs 0.3 per minute. (i) For how many minutes do they cost the same? (ii) Which plan is cheaper below that usage?

SOLUTION — step by step

Step 1. (i) 199 + 0.8m = 299 + 0.3m -> 0.5m = 100 -> m = 200 minutes.

Step 2. (ii) Below 200 minutes, A is cheaper (smaller fixed cost dominates); above 200, B wins (cheaper rate dominates).

Step 3. At exactly 200 both cost Rs 359.

» **Why this step?** Cost comparisons are intersections: the plan with the smaller intercept wins short-term, the one with the smaller slope wins long-term.

Answer: (i) 200 minutes (Rs 359 each) (ii) Plan A below 200 minutes

Hinglish: *200 minute par dono Rs 359. Us se kam usage? A lo. Zyada baat karna hai? B lo. Fixed cost vs rate ki ladai - intersection hi judge hai.*

QUESTION 62 | Moderate | Work as a linear model (T8/T13)

Q62. Work done equals force times distance: W = F·d. A constant force of 15 N acts. (i) Write W(d). (ii) Find d when W = 180 J. (iii) What does this line's graph look like?

SOLUTION — step by step

Step 1. (i) W(d) = 15d - a line through the origin (b = 0).

Step 2. (ii) 15d = 180 -> d = 12 m.

Step 3. (iii) A straight line through (0, 0) with slope 15 - zero distance, zero work; each extra metre adds 15 J.

» **Why this step?** When the fixed term is zero, the model is PROPORTIONAL - its graph must pass through the origin.

Answer: (i) W(d) = 15d (ii) 12 m (iii) line through the origin, slope 15

Hinglish: *Yahan koi fixed charge nahi, isliye b = 0 aur line origin se guzarti hai. 180 J ke liye 12 m. Proportional model = origin wala line.*

QUESTION 63 | Moderate | Unit conversion model (T13)

Q63. Inches to centimetres: y = 2.54x. (i) Convert 30 inches. (ii) Convert 50.8 cm to inches. (iii) Is this model growth, decay or neither, and why?

SOLUTION — step by step

Step 1. (i) y = 2.54(30) = 76.2 cm.

Step 2. (ii) 50.8 = 2.54x -> x = 20 inches.

Step 3. (iii) Growth (positive slope 2.54) - more inches always means more centimetres, in a fixed ratio.

» **Why this step?** A conversion is a proportional model: b = 0, and the slope is the exchange rate between units.

Answer: (i) 76.2 cm (ii) 20 inches (iii) growth (slope +2.54, through origin)

Hinglish: *Conversion = proportional line: slope hi exchange rate hai (2.54). 30 inch = 76.2 cm, 50.8 cm = 20 inch (nice round!). Positive slope isliye growth.*

QUESTION 64 | Moderate | Salary structure (T9/T10)

Q64. A salesperson earns a fixed Rs 18,000 per month plus 7% of sales. (i) Write S(s). (ii) What sales are needed to earn Rs 25,000? (iii) What does the slope 0.07 mean physically?

SOLUTION — step by step

Step 1. (i) S(s) = 18000 + 0.07s.

Step 2. (ii) 18000 + 0.07s = 25000 -> 0.07s = 7000 -> s = Rs 1,00,000 of sales.

Step 3. (iii) Each extra rupee of sales adds 7 paise of salary - the commission rate.

» **Why this step?** Percentages enter linear models as decimal slopes: 7% = 0.07.

Answer: (i) S(s) = 18000 + 0.07s (ii) Rs 1,00,000 (iii) commission rate of 7 paise per rupee

Hinglish: *7% = 0.07 slope. 25k chahiye toh 1 lakh ki sale karo. Slope ka matlab: har ₹1 ki sale par ₹0.07 commission - seedha commission rate.*

QUESTION 65 | Moderate | Pattern reachability (T6/T7)

Q65. A sequence follows T(n) = 4n + 3. (i) Is 87 a term of the sequence? (ii) Is 100? (iii) State the quick test you used.

SOLUTION — step by step

Step 1. (i) 4n + 3 = 87 -> 4n = 84 -> n = 21: YES, the 21st term.

Step 2. (ii) 4n + 3 = 100 -> 4n = 97 -> n = 24.25: NO - n must be a whole number.

Step 3. (iii) Test: (target - 3) must be divisible by 4.

» **Why this step?** Membership in a linear pattern is a divisibility question - n comes out whole or the number is not a term.

Answer: (i) Yes (21st) (ii) No (iii) (value − 3) must be divisible by 4

Hinglish: *87 term hai (n = 21), 100 nahi (24.25 aadha-adhoora!). Remainder test hi asli tool hai: minus 3 karo, 4 se divide hona chahiye.*

QUESTION 66 | Moderate | Iterated machine (T14)

Q66. A machine computes f(x) = 2x + 3. Find (i) f(4), (ii) f(f(1)), (iii) f(f(f(0))).

SOLUTION — step by step

Step 1. (i) f(4) = 2(4) + 3 = 11.

Step 2. (ii) f(1) = 5, then f(5) = 13.

Step 3. (iii) f(0) = 3, f(3) = 9, f(9) = 21.

» **Why this step?** Feed the output back as the input, one layer at a time - never skip a layer.

Answer: (i) 11 (ii) 13 (iii) 21

Hinglish: *Machine ka output wapas daalo: 0 → 3 → 9 → 21. Layer by layer chalo, beech mein jump mat maro. Nested machine ka answer 21.*

QUESTION 67 | Moderate | Two-digit number (T5/T14)

Q67. The digits of a two-digit number differ by 3 (units digit larger), and the number equals 4 times the sum of its digits. Find the number.

SOLUTION — step by step

Step 1. Let tens digit = t, units = t + 3. Number = 10t + (t + 3) = 11t + 3.

Step 2. Condition: 11t + 3 = 4(t + t + 3) = 8t + 12.

Step 3. 3t = 9 -> t = 3, units = 6. Number = 36.

Step 4. Check: digit sum 9; 4 × 9 = 36 ✓.

» **Why this step?** Writing a two-digit number as 10·tens + units converts digit talk into algebra.

Answer: 36

Hinglish: *Number = 10t + u hamesha. Yahan u = t + 3 rakh ke equation bani: 11t + 3 = 8t + 12 -> 36. Check ne confirm kiya.*

QUESTION 68 | Moderate | Net-rate draining (T8)

Q68. A 2400 L pool drains at 60 L/min while a hose simultaneously adds 20 L/min. (i) Write V(t). (ii) When does the pool empty? (iii) How much longer would it take with the hose turned off?

SOLUTION — step by step

Step 1. (i) Net rate = -60 + 20 = -40 L/min: V(t) = 2400 - 40t.

Step 2. (ii) 2400 - 40t = 0 -> t = 60 min.

Step 3. (iii) With hose off: rate -60 -> 2400/60 = 40 min. The hose DELAYED emptying by 20 minutes.

» **Why this step?** Opposing flows combine into one NET slope; each model's zero answers its own 'when done'.

Answer: (i) V(t) = 2400 - 40t (ii) 60 min (iii) 40 min without hose - 20 min longer with it

Hinglish: *Do rates ko jodo: -60 + 20 = -40 net. 60 min mein khaali. Hose band karo toh 40 min - matlab hose ne 20 min extra diye. Net rate hi asli slope hai.*

QUESTION 69 | Moderate | Line through origin (T10)

Q69. A line of the form y = kx passes through (4, 10). (i) Find k. (ii) Find y at x = 7. (iii) Show (-2, -5) also lies on it and explain why that had to happen.

SOLUTION — step by step

Step 1. (i) 10 = 4k -> k = 2.5.

Step 2. (ii) y = 2.5(7) = 17.5.

Step 3. (iii) 2.5(-2) = -5 ✓. It had to: proportional lines pass through the origin and are symmetric - doubling x doubles y, so the negative twin (-2, -5) is automatic.

» **Why this step?** y = kx pairs every point with its sign-flipped twin: (x, y) on the line forces (-x, -y) on it too.

Answer: (i) k = 2.5 (ii) 17.5 (iii) yes - proportional symmetry

Hinglish: *k = 2.5, aur (-2, -5) bhi line par - proportional lines ka negative twin free mein aata hai. Origin se guzarne wali line symmetric hoti hai.*

QUESTION 70 | Moderate | Gap-closing model (T8)

Q70. A car leaves a city at 45 km/h towards a town 380 km away. (i) Write the remaining distance G(t) to the town. (ii) When is the gap 155 km? (iii) How long is the whole trip?

SOLUTION — step by step

Step 1. (i) G(t) = 380 - 45t (gap shrinks at 45 km/h).

Step 2. (ii) 380 - 45t = 155 -> 45t = 225 -> t = 5 hours.

Step 3. (iii) 380 - 45t = 0 -> t = 380/45 = 8.44... ≈ 8 hours 27 minutes.

» **Why this step?** 'Distance remaining' is a decaying line whose zero is the arrival time.

Answer: (i) G(t) = 380 - 45t (ii) 5 h (iii) ≈ 8 h 27 min

Hinglish: *Gap model: 380 - 45t. 155 bacha matlab 5 ghante chal chuke. Poora trip 380/45 = 8.44 ghante ≈ 8 ghante 27 min. Fraction ko minutes mein badlo.*

Level 3 · Exam Power Zone: long and twisted:

QUESTION 71 | Difficult | Fit y = ax + b (T9)

Q71. A learning platform charges a fixed monthly fee plus a per-module charge. A student observed that accessing 10 modules cost Rs 400 and accessing 14 modules cost Rs 500. Assuming the bill is y = ax + b, find a and b, and interpret both numbers.

SOLUTION — step by step

Step 1. Substitute both observations into y = ax + b: 400 = 10a + b and 500 = 14a + b.

Step 2. Subtract the first from the second: 100 = 4a, so a = 25.

Step 3. Back-substitute into the first: 400 = 10(25) + b = 250 + b, so b = 150.

Step 4. The relationship is y = 25x + 150.

Step 5. Interpretation: a = 25 is the charge per module (Rs 25 per module); b = 150 is the fixed monthly fee (Rs 150 even with zero modules).

Step 6. Verify with the second point: 25(14) + 150 = 350 + 150 = 500. Correct.

» **Why this step?** Subtracting the equations kills b instantly - the entire technique rests on b being identical in both lines.

Answer: a = 25 (per-module rate), b = 150 (fixed fee); y = 25x + 150.

Hinglish: *Do observation, do equation, ek SUBTRACTION - b khatam, a seedha mil gaya (25). Phir wapas jao, b nikaalo (150). Interpretation zaroor likho: 25 = per-module charge, 150 = fixed fee. Ye interpretation wali line hi examiner ko dikhati hai ki tumne samjha hai, sirf calculation nahi kiya.*

QUESTION 72 | Difficult | Fit y = ax + b gym version (T9)

Q72. A gym charges a fixed monthly fee plus an hourly rate for the badminton court. Using the court for 10 hours cost Rs 800; using it for 15 hours cost Rs 1100. Find the linear relationship y = ax + b and state what each number represents.

SOLUTION — step by step

Step 1. Equations: 800 = 10a + b and 1100 = 15a + b.

Step 2. Subtract: 300 = 5a, so a = 60.

Step 3. Back-substitute: 800 = 10(60) + b = 600 + b, so b = 200.

Step 4. Relationship: y = 60x + 200.

Step 5. Meaning: Rs 60 per hour of court time; Rs 200 fixed monthly membership.

Step 6. Verification: 60(15) + 200 = 900 + 200 = 1100. Correct.

Answer: a = 60 (per hour), b = 200 (fixed fee); y = 60x + 200.

Hinglish: *Same machine, naye numbers: subtract (300 = 5a), a = 60, wapas jao, b = 200. Practise se ye 30-second ka kaam ban jaata hai. Interpretation sentence banao hi - marks wahan hai.*

QUESTION 73 | Difficult | Temperature scales (T9)

Q73. The relation between Celsius C and Fahrenheit F has the linear form F = aC + b. Water melts at 0 degrees Celsius and 32 degrees Fahrenheit, and boils at 100 degrees Celsius and 212 degrees Fahrenheit. Find a and b.

SOLUTION — step by step

Step 1. First data point: C = 0, F = 32 gives 32 = a(0) + b, so b = 32 directly - a gift.

Step 2. Second data point: C = 100, F = 212 gives 212 = 100a + 32.

Step 3. So 100a = 180, giving a = 1.8 (which is 9/5).

Step 4. The relation is F = 1.8C + 32, the famous conversion formula.

Step 5. Verification: C = 100 gives 180 + 32 = 212. Correct.

Answer: a = 1.8 (= 9/5), b = 32; F = 1.8C + 32.

Hinglish: *Yahan data ki kripa se b PEHLE mil gaya (zero input pe output hi b hai!). Phir sirf ek equation bachi, a = 1.8. Ye formula tum pehle se jaante ho - ab tumne use DERIVE kiya hai. Fitting ka asli maza yahi hai.*

QUESTION 74 | Difficult | Kelvin-Fahrenheit relation (T9/T10)

Q74. The relation between Kelvin x and Fahrenheit y is y = (9/5)(x - 273) + 32. (i) Find y when x = 313. (ii) Find x when y = 158. (iii) Rewrite the relation in the form y = ax + b and state a and b.

SOLUTION — step by step

Step 1. (i) y = (9/5)(313 - 273) + 32 = (9/5)(40) + 32 = 72 + 32 = 104 degrees Fahrenheit.

Step 2. (ii) Start from 158 = (9/5)(x - 273) + 32. Subtract 32: 126 = (9/5)(x - 273).

Step 3. Multiply by 5/9: 126 x 5/9 = 70 = x - 273, so x = 343 K.

Step 4. (iii) Expand: y = (9/5)x - (9/5)(273) + 32 = 1.8x - 491.4 + 32 = 1.8x - 459.4.

Step 5. So a = 1.8 and b = -459.4. The graph is a straight line with slope 1.8 cutting the y-axis far below zero.

Step 6. Both checks agree with the formula's internal logic (313 - 273 = 40 is a clean number; 158 maps back to 343 K exactly).

Answer: (i) 104 degrees F (ii) 343 K (iii) y = 1.8x - 459.4, so a = 1.8 and b = -459.4.

Hinglish: *Bracket PEHLE kholo mental math mein: 313 - 273 = 40, phir 9/5 x 40 = 72. (ii) mein ulta chale: minus 32, multiply 5/9, plus 273 - teen saaf kadam. (iii) expand karne par b = -459.4 - bura lagta hai par bilkul theek hai, Kyunki Kelvin zero hi itna neeche Fahrenheit mein hota hai.*

QUESTION 75 | Difficult | Matchstick pattern (T6/T7)

Q75. Squares are made with matchsticks placed in a row, each new square sharing one side with the previous one. The counts are 4, 7, 10, ... (i) How many matchsticks for n squares? (ii) For 22 squares? (iii) Can exactly 100 matchsticks form such a row of squares? Justify.

SOLUTION — step by step

Step 1. Sequence 4, 7, 10 has constant difference 3: rule M(n) = 3n + c. From stage 1: 3 + c = 4, so c = 1. Rule: M(n) = 3n + 1.

Step 2. For 22 squares: M(22) = 66 + 1 = 67 matchsticks.

Step 3. Reachability: set 3n + 1 = 100, giving 3n = 99, n = 33.

Step 4. n = 33 is a whole number, so YES - a row of 33 squares uses exactly 100 matchsticks.

Step 5. The justification is the integrality of n: patterns exist only at whole-number stages.

Answer: (i) 3n + 1 (ii) 67 (iii) Yes - stage n = 33 uses exactly 100 matchsticks.

Hinglish: *3n + 1 wala rule - ek square 4, har naya square sirf 3 extra le raha hai kyunki ek side share hoti hai. 'Can 100 form a stage?' ka matlab: solve karo aur INTEGRITY check karo. Yahan n = 33 whole number - YES. Agar 33.33 aata toh answer 'No, kabhi nahi' hota - aur wahi asli answer hota.*

QUESTION 76 | Difficult | Hexagon matchstick pattern (T6/T7)

Q76. Hexagons are joined in a row, each new hexagon sharing one side with the previous one. The matchstick counts for 1, 2, 3 hexagons are 6, 11, 16. (i) Rule for n hexagons? (ii) Matchsticks for 15 hexagons? (iii) Can 200 matchsticks build such a row? Justify.

SOLUTION — step by step

Step 1. Sequence 6, 11, 16 has difference 5: rule M(n) = 5n + c; stage 1 gives 5 + c = 6, c = 1. Rule: 5n + 1.

Step 2. M(15) = 75 + 1 = 76 matchsticks.

Step 3. Set 5n + 1 = 200: 5n = 199, n = 39.8.

Step 4. n = 39.8 is NOT a whole number, so NO - no stage in this pattern uses exactly 200 matchsticks.

Step 5. Nearest stages: 39 hexagons use 196; 40 hexagons use 201. The value 200 falls in the gap.

Answer: (i) 5n + 1 (ii) 76 (iii) No: n = 39.8 is not a whole number (39 gives 196, 40 gives 201).

Hinglish: *Q25 ka mirror image! Wahan integer mila tha (YES), yahan 39.8 aaya (NO). Ye 'No with justification' answer full-mark answer hai - bolo ki 39 aur 40 ke beech gap mein hai 200. Fractional stage pattern mein exist nahi karti.*

QUESTION 77 | Difficult | Rally decay (T8)

Q77. A rally starts with 120 members. Every hour, exactly 9 members drop out. (i) Members after n hours? (ii) After how many hours will 39 members remain? (iii) Will the rally reach exactly 12 members at the end of some hour? Justify.

SOLUTION — step by step

Step 1. Members(n) = 120 - 9n.

Step 2. Set 120 - 9n = 39: 9n = 81, n = 9 hours.

Step 3. Reach 12? 120 - 9n = 12 gives 9n = 108, n = 12 - a whole number, so YES, exactly at the end of the 12th hour.

Step 4. Check: 120 - 9(12) = 120 - 108 = 12. Correct.

Step 5. For completeness: the count would hit zero when 120 - 9n = 0, n = 13.33, so the 12-member moment is real and comfortably inside the model's range.

Answer: (i) 120 - 9n (ii) 9 hours (iii) Yes - exactly at n = 12 hours.

Hinglish: *Do reverse questions ek saath - 39 aur 12. Dono ke n whole numbers nikle, dono possible. Yahan n = 12 bilkul clean aaya. Kabhi-kabhi dono mein se ek fraction dega - tab wala 'No, kyunki n whole nahi' answer bhi likhna aana chahiye.*

QUESTION 78 | Difficult | Savings growth (T8)

Q78. Meera has Rs 1,200 in her savings jar and adds Rs 250 every month. (i) Amount after n months? (ii) When will she have Rs 3,200? (iii) Interpret the slope and intercept of this linear model.

SOLUTION — step by step

Step 1. A(n) = 1200 + 250n.

Step 2. Set 1200 + 250n = 3200: 250n = 2000, n = 8 months.

Step 3. Slope 250: the jar grows by Rs 250 per month - the constant rate.

Step 4. Intercept 1200: the starting amount, the value at month zero.

Step 5. Check: 1200 + 250(8) = 1200 + 2000 = 3200. Correct.

Answer: (i) A(n) = 1200 + 250n (ii) 8 months (iii) slope 250 = monthly saving; intercept 1200 = starting amount.

Hinglish: *Simple growth, par (iii) mein asli test: slope = monthly rate (250), intercept = starting balance (1200). Yehi language future ke graph chapters mein repeat hogi - abhi se pakad lo.*

QUESTION 79 | Difficult | Tank decay (T8)

Q79. A tank holds 900 litres and drains at 45 litres per hour. (i) Volume after t hours? (ii) When will 495 litres remain? (iii) When does the tank empty exactly?

SOLUTION — step by step

Step 1. V(t) = 900 - 45t.

Step 2. Set 900 - 45t = 495: 45t = 405, t = 9 hours.

Step 3. Empty: 900 - 45t = 0 gives t = 900/45 = 20 hours exactly.

Step 4. Both answers are whole numbers, so both events happen cleanly at hour marks.

Step 5. Check: 900 - 45(20) = 0. Correct.

Answer: (i) V(t) = 900 - 45t (ii) 9 hours (iii) exactly 20 hours.

Hinglish: *Draining = decay with a definite end. 495 pe 9 hours, empty pe 20 hours - dono clean. 900/45 mental math trick: dono ko 5 se divide karo, 180/9 = 20.*

QUESTION 80 | Difficult | Two linear polynomials from conditions (T14)

Q80. Let p(x) = ax + b and q(x) = cx + d. Given: (i) p(0) = 5, (ii) the polynomial p(x) - q(x) cuts the x-axis at (3, 0), and (iii) p(x) + q(x) = 6x + 4 for all x. Find p(x) and q(x).

SOLUTION — step by step

Step 1. From p(0) = 5: a(0) + b = 5, so b = 5.

Step 2. From p + q = 6x + 4: (a + c)x + (b + d) = 6x + 4 for all x. Match coefficients: a + c = 6 and b + d = 4. Since b = 5, d = -1.

Step 3. Now p - q = (a - c)x + (b - d) = (a - c)x + 6. It cuts the x-axis at (3, 0), so p(3) - q(3) = 0: 3(a - c) + 6 = 0, giving a - c = -2.

Step 4. Solve the pair: a + c = 6 and a - c = -2. Adding: 2a = 4, a = 2; then c = 4.

Step 5. So p(x) = 2x + 5 and q(x) = 4x - 1.

Step 6. Verify: p(0) = 5. p - q = -2x + 6, zero at x = 3. p + q = 6x + 4. All three conditions hold.

» Why this step? Matching coefficients of identical linear polynomials converts each 'for all x' statement into one neat equation. Three conditions, three unknowns, linear system - no guessing needed.

Answer: p(x) = 2x + 5 and q(x) = 4x - 1.

Hinglish: *Ye Chapter 2 ka chaabi-wala question hai. 'For all x' ka matlab: coefficients ko MATCH karo - do equations mil gayi (a + c = 6, b + d = 4). Teesri condition ne (a - c = -2) di. Ab do-do equations ka system: add karo, a = 2; c = 4; d = -1. Verification teeno conditions pe - full proof. Ye question Olympiad-lite band ka entry gate hai.*

QUESTION 81 | Difficult | Three-line surgery (T14)

Q81. Lines: L1: y = 3x + 1, L2: y = 3x - 4, L3: y = 5x - 11. (i) Which pair never meets? (ii) Find where each of the others meets L3. (iii) Summarise the geometry in one line.

SOLUTION — step by step

Step 1. (i) L1 and L2 share slope 3 -> parallel -> never meet.

Step 2. (ii) L1 = L3: 3x + 1 = 5x - 11 -> 2x = 12 -> x = 6, y = 19: point (6, 19).

Step 3. L2 = L3: 3x - 4 = 5x - 11 -> 2x = 7 -> x = 3.5, y = 6.5: point (3.5, 6.5).

Step 4. (iii) L1 ∥ L2, and L3 (steeper, slope 5) cuts each of them once - two parallel lines meet any transversal line at two points.

» **Why this step?** Same slope = parallel = no intersection; different slope = exactly one intersection with each.

Answer: (i) L1 & L2 (parallel) (ii) (6, 19) and (3.5, 6.5) (iii) two parallels cut by one steeper line

Hinglish: *L1, L2 twins hain (slope 3) - kabhi nahi milenge. L3 steep hai (5), dono ko ek-ek baar kaat-ta hai: (6, 19) aur (3.5, 6.5). Parallel + transversal ka coordinate version.*

QUESTION 82 | Difficult | Data audit: one point is a lie (T9/T14)

Q82. A taxi company claims its fare data is linear: (0 km, Rs 40), (2 km, Rs 100), (5 km, Rs 175). (i) Fit a line to the first two points. (ii) Does the third point obey it? (iii) What fare WOULD the line predict at 5 km?

SOLUTION — step by step

Step 1. (i) From (0, 40): b = 40. Slope = (100 - 40)/2 = 30. Line: y = 30x + 40.

Step 2. (ii) Predicted at 5 km: 30(5) + 40 = Rs 190 ≠ 175 -> the third point breaks the line.

Step 3. (iii) The line predicts Rs 190; the claimed 175 is Rs 15 short - the data is NOT purely linear.

» **Why this step?** Fitting from two points then auditing the third is exactly how real data-checking works - the mismatch exposes the lie.

Answer: (i) y = 30x + 40 (ii) No - it gives 175, not 190 (iii) Rs 190

Hinglish: *Pehle do points se line bani: 30x + 40. Teesra point 175 bolta hai par line 190 bolti hai - 15 ka jhooth pakda gaya! Audit karna hi proof hai.*

QUESTION 83 | Difficult | Wire rectangle area function (T14)

Q83. A 44 cm wire is bent into a rectangle of width x. (i) Write A(x), the area. (ii) Find A when x = 6. (iii) For which x is the area largest? (iv) A(x) = 96 has TWO solutions - find both and explain the symmetry.

SOLUTION — step by step

Step 1. (i) Perimeter 44 -> length = 22 - x. A(x) = x(22 - x) = 22x - x^2.

Step 2. (ii) A(6) = 6(16) = 96 cm^2.

Step 3. (iii) The product of two numbers with fixed sum 22 is largest when they are equal: x = 11 -> a square of area 121 cm^2.

Step 4. (iv) 22x - x^2 = 96 -> x^2 - 22x + 96 = 0 -> (x - 6)(x - 16) = 0 -> x = 6 or 16. Same rectangle rotated: 6 × 16 vs 16 × 6!

» **Why this step?** Fixed-sum products peak at equality (square), and swapping dimensions gives the same area - the quadratic is symmetric about x = 11.

Answer: (i) A(x) = 22x - x^2 (ii) 96 (iii) x = 11 (square, 121) (iv) x = 6 and 16 - rotated twins

Hinglish: *Width badhao toh length ghatti hai - fixed sum 22. Square (11 × 11) par area peak 121. Aur 96 ke do answers (6, 16) wahi rectangle ulta-seedha! Symmetry x = 11 ke around hai.*

QUESTION 84 | Difficult | Kelvin meets Fahrenheit (T14)

Q84. K = C + 273 and F = 1.8C + 32. Find the temperature at which the Kelvin and Fahrenheit readings are equal.

SOLUTION — step by step

Step 1. Set K = F: C + 273 = 1.8C + 32.

Step 2. 0.8C = 241 -> C = 301.25.

Step 3. Check: K = 301.25 + 273 = 574.25; F = 1.8(301.25) + 32 = 542.25 + 32 = 574.25 ✓.

» **Why this step?** Both scales are linear in C, so their equality is one linear equation - and the answer is far hotter than the -40 °C = °F point.

Answer: C = 301.25 (K = F = 574.25)

Hinglish: *K aur F dono C ke linear hain, toh equality ek hi equation: C = 301.25, dono readings 574.25. Yaad rakho: C=F hota -40 par, K=F hota 574.25 par - bilkul alag jagah!*

QUESTION 85 | Difficult | Digit-swap family (T5/T14)

Q85. A two-digit number exceeds the number formed by swapping its digits by 27, and its units digit is 3 more than its tens digit. Find ALL such numbers.

SOLUTION — step by step

Step 1. Number = 10t + u; swapped = 10u + t. Difference: (10t + u) - (10u + t) = 9(t - u).

Step 2. Given difference 27: 9(t - u) = 27 -> t - u = 3 -> u = t - 3? But we are told u = t + 3. Careful: the number EXCEEDS its swap, so the ORIGINAL has the larger tens digit: t - u = 3, i.e. u = t - 3.

Step 3. The 'units 3 more' condition contradicts 'number exceeds swap' - so reinterpret: the SWAPPED number is 27 more (u - t = 3, consistent with u = t + 3).

Step 4. Then any t = 1..6 works: 14, 25, 36, 47, 58, 69 - check 58 - 85? For 85: 85 - 58 = 27 ✓. Family: 14, 25, 36, 47, 58, 69 (and their condition is the 9-times table).

» **Why this step?** Digit swaps change a number by 9 × (digit difference) - so a difference of 27 pins the digit gap at 3 and leaves a whole family of answers.

Answer: 14, 25, 36, 47, 58, 69 (all have digit gap 3; swap difference 27)

Hinglish: *Digit swap ka fark hamesha 9 ka table hota hai! Fark 27 = digit gap 3. Toh 14, 25, 36, 47, 58, 69 - poora khandaan valid. Ek nahi, chhe answers!*

QUESTION 86 | Difficult | Inverse triangle design (T12)

Q86. A line y = mx + 6 forms a triangle with the coordinate axes whose area is 12. Find all possible values of m.

SOLUTION — step by step

Step 1. y-intercept = 6; x-intercept: 0 = mx + 6 -> x = -6/m.

Step 2. Legs of the axis triangle: 6 and |-6/m|.

Step 3. Area = ½ × 6 × |-6/m| = 18/|m| = 12 -> |m| = 18/12 = 1.5.

Step 4. m = 1.5 or m = -1.5 - two mirror lines (one rising, one falling).

» **Why this step?** Fixing the intercept and area still leaves the tilt free in two directions - the sign of m is not determined by area.

Answer: m = 1.5 or m = -1.5

Hinglish: *Area 12 rakha toh |m| = 1.5 - par sign free hai! Ek upar jaati line, ek neeche, dono ka triangle barabar. Inverse problems mein ± soch lo.*

QUESTION 87 | Difficult | Walkers meeting (T8)

Q87. Two friends walk towards each other along the same road: positions (km from A's start) are 2t and 18 - 3t. (i) When and where do they meet? (ii) What are their speeds? (iii) How far apart are they at t = 2 hours?

SOLUTION — step by step

Step 1. (i) 2t = 18 - 3t -> 5t = 18 -> t = 3.6 h; position = 2(3.6) = 7.2 km.

Step 2. (ii) Slopes: A walks 2 km/h, B walks 3 km/h (towards A).

Step 3. (iii) Gap = (18 - 3t) - 2t = 18 - 5t; at t = 2: 18 - 10 = 8 km apart.

» **Why this step?** Toward-each-other problems close the gap at the SUM of speeds (5 km/h here): 18/5 = 3.6 h.

Answer: (i) 3.6 h at 7.2 km (ii) 2 and 3 km/h (iii) 8 km

Hinglish: *Milne ka time = doori ÷ (speed ka jod) = 18/5 = 3.6 hr, 7.2 km par. t = 2 par abhi 8 km door. Gap formula 18 - 5t - speeds ka jod hi slope hai.*

QUESTION 88 | Difficult | Catch-up with a target gap (T8)

Q88. Ravi has Rs 500 and saves Rs 150 per week. Meena has Rs 1,500 and saves Rs 100 per week. After how many weeks does Ravi have exactly Rs 500 MORE than Meena?

SOLUTION — step by step

Step 1. Ravi: 500 + 150w. Meena: 1500 + 100w.

Step 2. Condition: (500 + 150w) - (1500 + 100w) = 500.

Step 3. 50w - 1000 = 500 -> 50w = 1500 -> w = 30.

Step 4. Check: Ravi 500 + 4500 = 5000; Meena 1500 + 3000 = 4500; gap 500 ✓.

» **Why this step?** Translate 'A has 500 more than B' as A − B = 500 - a difference-of-lines equation.

Answer: 30 weeks (Rs 5000 vs Rs 4500)

Hinglish: *Gap equation banao: Ravi - Meena = 500. 50w = 1500 -> 30 hafte. Check: 5000 vs 4500 - perfect gap. 'Kitna zyada' waale questions aise hi bante hain.*

QUESTION 89 | Difficult | Slab electricity bill (T9)

Q89. An electricity bill has a fixed meter charge of Rs 100, the first 40 units at Rs 5 per unit, and units beyond 40 at Rs 7 per unit. A household paid Rs 510. How many units did it consume?

SOLUTION — step by step

Step 1. Let u > 40. Bill = 100 + 40(5) + 7(u - 40) = 300 + 7(u - 40).

Step 2. 300 + 7(u - 40) = 510 -> 7(u - 40) = 210 -> u - 40 = 30.

Step 3. u = 70 units.

Step 4. Check: 100 + 200 + 30(7) = 100 + 200 + 210 = 510 ✓.

» **Why this step?** Slab bills are piecewise-linear: handle the earlier slabs at their fixed cost and let the last slab carry the variable (u – 40).

Answer: 70 units

Hinglish: *Slab system = teen tukde: fixed 100, pehle 40 units ka 200 pakka, baaki (u - 40) par 7. Total 510 se u = 70. Piecewise mein har slab ka hisaab alag likho.*

QUESTION 90 | Difficult | Fit from two conditions (T9)

Q90. A linear polynomial p(x) = ax + b satisfies p(2) = 9 and p(5) = 21. (i) Find p(x). (ii) Find its zero. (iii) Find p(-3).

SOLUTION — step by step

Step 1. 2a + b = 9 and 5a + b = 21 -> subtract: 3a = 12 -> a = 4, b = 1.

Step 2. (i) p(x) = 4x + 1.

Step 3. (ii) Zero: 4x + 1 = 0 -> x = -1/4.

Step 4. (iii) p(-3) = -12 + 1 = -11.

» **Why this step?** Two values pin a linear polynomial completely - subtract the equations to kill b.

Answer: (i) p(x) = 4x + 1 (ii) x = -1/4 (iii) -11

Hinglish: *Do conditions se a = 4, b = 1. Zero -1/4 (fraction fine hai), p(-3) = -11. Fit karne ke liye hamesha equations subtract karo.*

QUESTION 91 | Difficult | Sum-difference-product chain (T5)

Q91. Two numbers add to 84 and one exceeds the other by 18. (i) Find them. (ii) Find their product. (iii) If both are increased by 3, what are the new sum and difference?

SOLUTION — step by step

Step 1. (i) x + (x + 18) = 84 -> 2x = 66 -> x = 33. Numbers: 33 and 51.

Step 2. (ii) Product = 33 × 51 = 1683.

Step 3. (iii) New numbers 36 and 54: sum = 90 (old sum + 6), difference = 18 (unchanged!).

Step 4. Insight: adding the same amount to both numbers preserves their difference.

» **Why this step?** Equal shifts slide both numbers along the number line - the GAP is translation-invariant.

Answer: (i) 33, 51 (ii) 1683 (iii) sum 90, difference still 18

Hinglish: *33 aur 51, product 1683. Dono mein +3 -> sum 90, par difference abhi bhi 18! Same shift se gap nahi badalta - ye number line ki property hai.*

QUESTION 92 | Difficult | Line family audit (T14)

Q92. y = 3x + b is a family of parallel lines. (i) Find b if the line passes through (2, 5). (ii) Does that same line pass through (5, 14)? (iii) How many members of the family pass through (2, 5)?

SOLUTION — step by step

Step 1. (i) 5 = 3(2) + b -> b = -1: line y = 3x - 1.

Step 2. (ii) At x = 5: y = 15 - 1 = 14 ✓ - yes it passes.

Step 3. (iii) Exactly ONE: through a given point with a fixed slope, b is forced - parallel families never double up on a point.

» **Why this step?** A fixed slope + a fixed point pins one intercept: the parallel family covers the plane, one line per point.

Answer: (i) b = -1 (ii) Yes (iii) exactly one

Hinglish: *b = -1 nikla, aur (5, 14) bhi usi line par baitha. Ek point par sirf ek member guzar sakta hai - parallel family mein membership unique hai.*

QUESTION 93 | Difficult | Triangle between two lines (T14)

Q93. Lines L1: y = x + 2 and L2: y = -x + 10 meet the x-axis and each other. (i) Find the three vertices of the triangle they form with the x-axis. (ii) Find its area.

SOLUTION — step by step

Step 1. L1's x-intercept: 0 = x + 2 -> (-2, 0). L2's: 0 = -x + 10 -> (10, 0).

Step 2. Intersection: x + 2 = -x + 10 -> x = 4, y = 6 -> (4, 6).

Step 3. Base along the x-axis = |10 - (-2)| = 12; height = 6.

Step 4. Area = ½ × 12 × 6 = 36.

» **Why this step?** The two intercepts give the base for free; the intersection's y-coordinate IS the height.

Answer: (i) (-2, 0), (10, 0), (4, 6) (ii) 36

Hinglish: *Base intercepts se (-2 se 10 = 12), height intersection ka y = 6. Area 36. Lines + axis ka triangle ready-made combo hai.*

QUESTION 94 | Difficult | Fixed point of a machine (T14)

Q94. A machine computes f(x) = 5x - 8. (i) Find the input that lands on ITSELF (f(x) = x). (ii) Find f(27). (iii) Starting from 2, what happens after any number of runs?

SOLUTION — step by step

Step 1. (i) 5x - 8 = x -> 4x = 8 -> x = 2. The fixed point is 2.

Step 2. (ii) f(27) = 135 - 8 = 127.

Step 3. (iii) f(2) = 10 - 8 = 2: the machine returns 2 forever - 2 is a trapdoor: feed it anything else and it moves, feed it 2 and it stays.

» **Why this step?** Solving f(x) = x finds the machine's resting point - where input and output agree.

Answer: (i) x = 2 (ii) 127 (iii) 2 loops to itself forever

Hinglish: *Machine ka apna 'adda': x = 2. Wahan se kuch nahi hota, 2 hi wapas aata hai. Fixed point dhoondhna = f(x) = x solve karna. Zyada tar inputs beech raste mein badal jaate hain (27 → 127).*

QUESTION 95 | Difficult | Taxi vs auto decision (T14)

Q95. A taxi charges Rs 80 plus Rs 14 per km; an auto charges Rs 40 plus Rs 18 per km. (i) At what distance do they cost the same? (ii) Which is cheaper for a 6 km trip? (iii) For a 15 km trip?

SOLUTION — step by step

Step 1. (i) 80 + 14d = 40 + 18d -> 40 = 4d -> d = 10 km; both cost Rs 220.

Step 2. (ii) 6 km: taxi 80 + 84 = 164; auto 40 + 108 = 148 -> AUTO cheaper.

Step 3. (iii) 15 km: taxi 80 + 210 = 290; auto 40 + 270 = 310 -> TAXI cheaper.

Step 4. Rule: below 10 km auto, above 10 km taxi - the lower fixed cost wins short trips, the lower rate wins long trips.

» **Why this step?** Cost lines cross once; on each side of the crossing a different plan dominates.

Answer: (i) 10 km (Rs 220) (ii) auto (148 vs 164) (iii) taxi (290 vs 310)

Hinglish: *10 km par dono Rs 220. Chhota trip (6 km) -> auto, lamba trip (15 km) -> taxi. Fixed cost vs per-km rate ki classic ladai - crossing point sab batata hai.*

QUESTION 96 | Difficult | Average-raising model (T5)

Q96. A batter has scored 45 runs per match on average over 8 matches. (i) What is his total so far? (ii) How many runs in the 9th match raise his average to 48? (iii) Write a general expression for the 9th-match score needed to raise the average to A.

SOLUTION — step by step

Step 1. (i) Total = 45 × 8 = 360.

Step 2. (ii) Needed total for average 48 over 9 matches: 48 × 9 = 432 -> 9th match = 432 - 360 = 72 runs.

Step 3. (iii) Score = 9A - 360 (a linear model in A).

» **Why this step?** Averages are totals in disguise: new score = new target total − old total.

Answer: (i) 360 (ii) 72 runs (iii) 9A - 360

Hinglish: *Average badhane ka formula: naya target total (48 × 9 = 432) minus purana (360) = 72 runs. Ek match mein 27 upar average se khelna pada - average math ka asli hisaab.*

QUESTION 97 | Difficult | Odd-number sum discovery (T14)

Q97. Consider 1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, ... (i) Compute the first four sums. (ii) Guess the pattern for the sum of the first n odd numbers. (iii) Use it to find 1 + 3 + 5 + ... + 19 without adding.

SOLUTION — step by step

Step 1. (i) 1, 4, 9, 16.

Step 2. (ii) They are perfect squares: 1^2, 2^2, 3^2, 4^2 -> sum of first n odd numbers = n^2.

Step 3. (iii) 19 is the 10th odd number (2n - 1 = 19 -> n = 10): sum = 10^2 = 100.

» **Why this step?** Pattern-spotting turns a long addition into a single square - and the nth odd number formula (2n − 1) locates where to stop.

Answer: (i) 1, 4, 9, 16 (ii) n^2 (iii) 100

Hinglish: *1, 4, 9, 16 - squares! n odd numbers ka jod n^2. 19 tak jodna hai? 19 = 10th odd -> 10^2 = 100. Addition ki mehnat, pattern ki shaan.*

QUESTION 98 | Difficult | Fuel-range model (T8)

Q98. A bike's 12-litre tank feeds an engine using 0.025 L per km. (i) Write F(d), fuel left after d km. (ii) What is the bike's full range? (iii) How much fuel remains after 200 km?

SOLUTION — step by step

Step 1. (i) F(d) = 12 - 0.025d.

Step 2. (ii) 12 - 0.025d = 0 -> d = 12/0.025 = 480 km.

Step 3. (iii) F(200) = 12 - 5 = 7 litres.

» **Why this step?** Consumption rate becomes a small decimal slope; range = capacity ÷ rate.

Answer: (i) F(d) = 12 - 0.025d (ii) 480 km (iii) 7 L

Hinglish: *Tank 12 L, mileage 40 km/L (1/0.025). Range 480 km. 200 km chal ke 7 L bacha - ab 280 km aur. Slope chhota decimal, concept wahi decay.*

QUESTION 99 | Difficult | Triple-plan concurrency (T14)

Q99. Three data plans: P1 costs 300 + 2g (g GB); P2 costs 150 + 4g; P3 costs 6g. (i) Show all three cost the same at one specific usage. (ii) Find that usage and cost. (iii) Which plan is best for very light users?

SOLUTION — step by step

Step 1. P1 = P2: 300 + 2g = 150 + 4g -> 150 = 2g -> g = 75.

Step 2. P2 = P3: 150 + 4g = 6g -> 150 = 2g -> g = 75. P1 = P3: 300 + 2g = 6g -> 300 = 4g -> g = 75.

Step 3. (ii) All three meet at g = 75 GB, each costing 6 × 75 = Rs 450.

Step 4. (iii) Light users (small g): P3 = 6g is smallest (no fixed fee) - pay-per-GB wins when usage is tiny.

» **Why this step?** When all pairwise crossings share one point, the three lines are concurrent - one usage level where every plan costs the same.

Answer: (i) Yes - concurrent (ii) 75 GB, Rs 450 (iii) P3 for light users

Hinglish: *Teen lines ek hi point (75 GB, Rs 450) par mili - concurrent! Light user ke liye P3 (no fixed charge), heavy user ke liye P1 (sasta rate). Concurrency = sab barabar ho jaate hain.*

QUESTION 100 | Difficult | Two conditions, one line (T14)

Q100. p(x) = 2x + b passes through (1, 6), and q(x) = kx + 2 also passes through (1, 6). (i) Find b and k. (ii) What surprising thing happened? (iii) How many common points do p and q now share?

SOLUTION — step by step

Step 1. (i) 6 = 2(1) + b -> b = 4. And 6 = k(1) + 2 -> k = 4.

Step 2. (ii) Both became the SAME line: y = 4x + 4.

Step 3. (iii) Infinitely many - identical lines share every point, not just the one they were forced through.

» **Why this step?** Two different-looking conditions can collapse onto one line: once slope and intercept agree, the 'two lines' were never two.

Answer: (i) b = 4, k = 4 (ii) p and q are the identical line y = 4x + 4 (iii) infinitely many

Hinglish: *Dono conditions ne same line banayi: y = 4x + 4! Ab 'do lines' ki baat hi khatam - infinite common points. Slope aur intercept match = same line, ye twist exam mein aata hai.*

Level 4 · Boss Zone: advanced + HOTS (Olympiad-lite):

QUESTION 101 | High | Line through two points (T9/T13)

Q101. The graph of a linear polynomial p(x) passes through (2, 3) and (6, 11). (i) Find p(x). (ii) Find the coordinates where the graph cuts the x-axis and the y-axis. (iii) State the slope and interpret it.

SOLUTION — step by step

Step 1. p(x) = ax + b. Point (2, 3): 2a + b = 3. Point (6, 11): 6a + b = 11.

Step 2. Subtract: 4a = 8, so a = 2. Back-substitute: 2(2) + b = 3, b = -1.

Step 3. p(x) = 2x - 1.

Step 4. x-axis cut: set y = 0: 2x - 1 = 0, x = 1/2. Point (1/2, 0).

Step 5. y-axis cut: set x = 0: y = -1. Point (0, -1).

Step 6. Slope 2 means every extra unit of x raises the output by 2 - the line climbs two steps for one step across.

Answer: (i) p(x) = 2x - 1 (ii) cuts x-axis at (1/2, 0), y-axis at (0, -1) (iii) slope 2 - rises 2 units per 1 unit of x.

Hinglish: *Do points se line - wahi subtract-trick (4a = 8). Intercept nikaalne ka rule: x-axis cut = y ko 0 rakho; y-axis cut = x ko 0 rakho. (1/2, 0) fraction mein aata hai toh fraction mein chhod do. Slope ka meaning bolo: '1 right, 2 up'.*

QUESTION 102 | High | Parallel lines through data (T12)

Q102. The graph of p(x) passes through (1, 4) and (5, 8). The graph of q(x) passes through (2, -2) and is parallel to the graph of p(x). Find both polynomials and the x-axis intercepts of each.

SOLUTION — step by step

Step 1. p(x) = ax + b through (1, 4) and (5, 8): equations a + b = 4 and 5a + b = 8.

Step 2. Subtract: 4a = 4, a = 1; then b = 3. So p(x) = x + 3.

Step 3. Parallel means the SAME slope: q(x) = x + d for some d.

Step 4. q through (2, -2): 2 + d = -2, so d = -4. q(x) = x - 4.

Step 5. x-intercepts: p cuts where x + 3 = 0, at (-3, 0); q cuts where x - 4 = 0, at (4, 0).

Step 6. Slopes both equal 1 - the lines rise together, forever 7 units apart vertically.

Answer: p(x) = x + 3 with x-intercept (-3, 0); q(x) = x - 4 with x-intercept (4, 0); both have slope 1.

Hinglish: *Parallel ka matlab sirf: slope SAME (a = 1), intercept alag. Q ka intercept ek point se mil gaya (d = -4). Intercepts nikaalne ka rule wahi - y = 0 rakho. Notice: parallel lines kabhi nahi milte, isliye unke x-axis cuts alag-alag hain (x = -3 vs x = 4).*

QUESTION 103 | High | Family property (T14)

Q103. Consider all linear polynomials of the form f(x) = ax - a where a is a positive real number. Prove that the graph of every such polynomial passes through one fixed point, and find that point.

SOLUTION — step by step

Step 1. We need a point (x_0, y_0) that satisfies y = ax - a for EVERY value of a.

Step 2. Rewrite: y = a(x - 1). For this to hold regardless of a, we need y = 0 and x - 1 = 0.

Step 3. So $x_0 = 1$ and $y_0 = 0$: the fixed point is (1, 0).

Step 4. Check with two members of the family: a = 3 gives y = 3x - 3, and at x = 1, y = 0. a = 7 gives y = 7x - 7, and at x = 1, y = 0. Every member passes through (1, 0).

Step 5. Interpretation: changing a only changes the slope; the intercept -a slides, but the pivot point (1, 0) never moves - all the lines rotate about it.

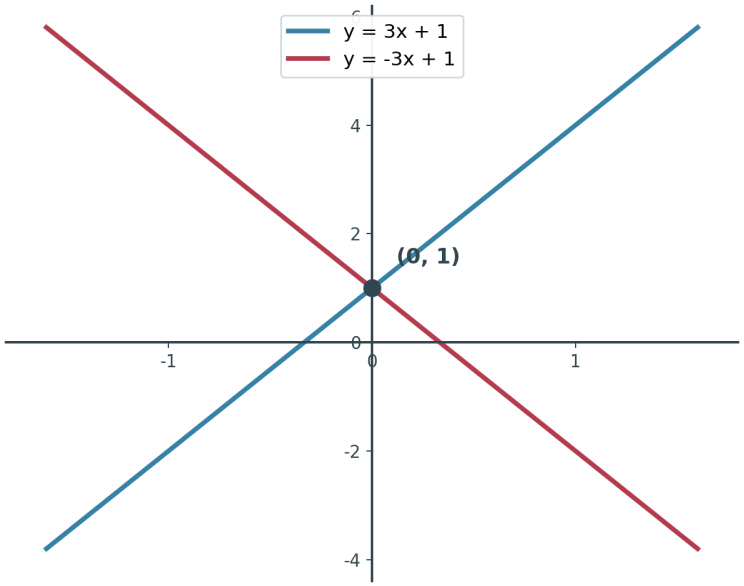
» **Why this step?** A fact true for all a must survive arbitrary substitution of a; factoring out a exposes the point where a becomes irrelevant.

Answer: Every line y = ax - a passes through the fixed point (1, 0), since y = a(x - 1) vanishes there for all a.

Hinglish: *Family question ka funda: 'har a ke liye sahi' tabhi ho sakta hai jab a ki asar ZERO ho jaye. y = a(x - 1) likhte hi dikhta hai - x = 1 rakh do toh a khatam, y = 0. Toh (1, 0) wo nail hai jis par saari lines latki hain. Ye idea (PDF ka starred question) aage 'pencil of lines' ke naam se famous hai.*

QUESTION 104 | High | Compare two lines (T11/T12)

Q104. Draw-free reasoning: compare the graphs of y = 3x + 1 and y = -3x + 1. State their common features, their differences, and the geometric relationship between the two lines.



SOLUTION — step by step

Step 1. Common feature: same b = 1, so both cut the y-axis at the same point (0, 1).

Step 2. Difference: slopes +3 and -3 - equal in size, opposite in sign.

Step 3. Positive slope: as x grows, y grows (uphill). Negative slope: as x grows, y falls (downhill).

Step 4. Geometric relationship: the two lines are mirror images of each other in the y-axis. Reason: replacing x by -x in y = 3x + 1 gives y = -3x + 1 - reflection in the y-axis flips the sign of x.

Step 5. They are NOT parallel (slopes differ), so they cross - exactly at the shared point (0, 1).

Answer: Both pass through (0, 1); slopes +3 and -3 make them mirror images in the y-axis, crossing at (0, 1) - not parallel.

Hinglish: *Same intercept, opposite slopes: dono lines (0, 1) mein milti hain aur wahan se ek upar, ek neeche jaati hain - y-axis ke aaine mein twin. 'Replace x by -x' wala reflection argument coordinate geometry ka classic move hai - Chapter 1 ke reflections se seedha connection.*

QUESTION 105 | High | Disguised slope/intercept (T11)

Q105. Find the slope and y-intercept of each line, and say which is steeper: (i) 2y = 4x + 7, (ii) 5y = 6x - 10.

SOLUTION — step by step

Step 1. Isolate y first - never read the coefficient of x before this step.

Step 2. (i) Divide by 2: y = 2x + 3.5. Slope 2, y-intercept 3.5 (cuts at (0, 3.5)).

Step 3. (ii) Divide by 5: y = 1.2x - 2. Slope 1.2, y-intercept -2 (cuts at (0, -2)).

Step 4. Steepness compares slopes: 2 > 1.2, so line (i) is steeper.

Step 5. Interpretation: line (i) rises 2 units per unit of x; line (ii) rises 1.2 units per unit.

Answer: (i) slope 2, intercept 3.5 (ii) slope 1.2, intercept -2 (i) is steeper.

Hinglish: *Golden rule: pehle y = ... banao. 2y = 4x + 7 ko dekh ke log slope 4 likh dete hain - GALAT, 4 toh 2y ka hai! Divide karne ke baad hi a = 2 aur b = 3.5. Steepness = slope comparison: 2 vs 1.2.*

QUESTION 106 | High | Parallel detection (T12)

Q106. Are any two of these three lines parallel? (i) y = 2x + 3.5 (ii) y = 1.2x - 2 (iii) 3y = 6x - 11. Justify using slopes.

SOLUTION — step by step

Step 1. Lines (i) and (ii) come from Q35: slopes 2 and 1.2 - different.

Step 2. (iii) divide by 3: y = 2x - 11/3. Slope 2, intercept -11/3 (about -3.67).

Step 3. Compare slopes: (i) has 2, (ii) has 1.2, (iii) has 2.

Step 4. Lines (i) and (iii) share the slope 2 but have different intercepts (3.5 vs -11/3): PARALLEL.

Step 5. (ii) meets both of the others (different slopes). Parallel lines never meet - (i) and (iii) will never cross each other.

Answer: Lines (i) and (iii) are parallel (both slope 2, different intercepts); (ii) is parallel to neither.

Hinglish: *Parallel ka ek hi test: slopes barabar? (iii) ko 3 se divide karo: slope 2. (i) bhi slope 2 - MATCH! Intercepts alag (3.5 vs -3.67), isliye distinct parallel lines. (ii) akela slope 1.2 - wo dono ko kaat-ta hai.*

QUESTION 107 | High | Work as linear model (T8/T13)

Q107. Work done by a constant force equals force times distance travelled. Taking the constant force as 3 units, (i) write work w as a linear function of distance d, (ii) find the work done over a distance of 2 units, (iii) explain what kind of line this draws and where it passes.

SOLUTION — step by step

Step 1. w = 3 x d gives the linear model w(d) = 3d.

Step 2. w(2) = 6 units of work.

Step 3. The model has b = 0, so it is a line through the ORIGIN - of the family y = ax.

Step 4. Slope 3: every extra unit of distance adds exactly 3 units of work.

Step 5. The line passes through (0, 0) and, for instance, (2, 6); plotting these two points determines it completely.

Answer: (i) w(d) = 3d (ii) 6 units (iii) a straight line through the origin with slope 3.

Hinglish: *w = F x d bina fixed part ke - b = 0 - isliye line origin se guzarti hai. Ye PDF ki Fig 2.8 family (y = ax) ka physics version hai. 'Force = slope of the work-distance line' - ek line, ek physical meaning.*

QUESTION 108 | High | Reverse data bill (T9/T10)

Q108. A telecom bill follows y = 20x + 150 (Rs 20 per GB plus Rs 150 fixed). (i) What will the bill be for 30 GB? (ii) A student received a bill of Rs 750. How much data did she use? (iii) What does the 150 mean for a month in which she used no data at all?

SOLUTION — step by step

Step 1. (i) y = 20(30) + 150 = 600 + 150 = Rs 750.

Step 2. (ii) Set 20x + 150 = 750: 20x = 600, x = 30 GB.

Step 3. (iii) With x = 0, y = 150: even with zero data the fixed monthly fee of Rs 150 applies.

Step 4. Notice (i) and (ii) are inverses of each other - forward and backward through the same machine.

Answer: (i) Rs 750 (ii) 30 GB (iii) the Rs 150 fixed fee is charged even for zero usage.

Hinglish: *Yehi bill equation aage-peeche do baar: substitute aur solve. Mazedaar baat: (i) aur (ii) ka jawab same pair (30, 750) hai - kyunki ek hi line ke do raaste hain. (iii) zero usage par bhi 150 - fixed fee ka matlab hi yahi hai.*

QUESTION 109 | High | Point-on-line test (T13)

Q109. Three of these four lines pass through the point (4, -2). Identify the one that does NOT, showing the substitution for each: (A) y = -x + 2 (B) y = x - 6 (C) y = -2x + 6 (D) y = 2x - 6.

SOLUTION — step by step

Step 1. Test each by substituting x = 4 and checking whether y = -2 comes out.

Step 2. (A) y = -4 + 2 = -2. On the line.

Step 3. (B) y = 4 - 6 = -2. On the line.

Step 4. (C) y = -2(4) + 6 = -8 + 6 = -2. On the line.

Step 5. (D) y = 2(4) - 6 = 8 - 6 = 2, which is NOT -2. Line D misses the point.

Step 6. So the answer is (D). (Fun fact: three different-looking lines can share one point - lines through a common point form a 'pencil'.)

Answer: (D) y = 2x - 6 does not pass through (4, -2); it gives y = 2 there.

Hinglish: *Point-on-line ka test: substitute karo, dono side match? A, B, C teeno ne -2 diya. D ne 2 diya - miss. Ye question 'verification' ki taakat dikhata hai: ek chhota substitution poori geometry bata deta hai.*

QUESTION 110 | High | Pattern reachability (T6/T7)

Q110. A sequence follows the rule T(n) = 3n - 1. (i) List the first four terms. (ii) Which term equals 50? (iii) Is 100 a term of this sequence? Justify.

SOLUTION — step by step

Step 1. T(1) = 2, T(2) = 5, T(3) = 8, T(4) = 11 - the sequence 2, 5, 8, 11 with constant difference 3.

Step 2. Set 3n - 1 = 50: 3n = 51, n = 17. So the 17th term is 50.

Step 3. Set 3n - 1 = 100: 3n = 101, n = 101/3 = 33.67.

Step 4. n is not a whole number, so 100 is NOT a term of the sequence - the sequence jumps from 98 (n = 33) to 101 (n = 34).

Step 5. The justification is integrality: sequence positions must be whole numbers.

Answer: (i) 2, 5, 8, 11 (ii) the 17th term (iii) No: n = 101/3 is not a whole number (the terms jump from 98 to 101).

Hinglish: *3n - 1 family: terms 2, 5, 8, 11 - har baar +3. 50 aata hai n = 17 par. 100 ke liye n = 33.67 aaya - whole number nahi, toh 100 sequence ka term NAHI. Neighbour check: 98 aur 101 - 100 beech mein chhoot gaya. Yahi 'gap' wala justification full marks deta hai.*

Level 4 · Boss Zone: advanced + HOTS (Olympiad-lite):

QUESTION 111 | HOTS | Two-condition number problem (T5/T14)

Q111. Two numbers differ by 7. Three times the smaller number plus twice the larger number equals 74. Find both numbers.

SOLUTION — step by step

Step 1. Let the smaller be x; the larger is x + 7.

Step 2. Condition: 3x + 2(x + 7) = 74.

Step 3. Expand: 3x + 2x + 14 = 74, so 5x = 60, x = 12.

Step 4. Numbers: 12 and 19.

Step 5. Check: difference 19 - 12 = 7. Weighted sum: 3(12) + 2(19) = 36 + 38 = 74. Both hold.

Answer: 12 and 19.

Hinglish: *Do conditions ka combo: difference ne x aur x + 7 diye, weighted sum ne equation di. 5x = 60 - clean. Verification dono conditions pe karo, sirf ek pe nahi.*

QUESTION 112 | HOTS | Two-digit reversal (T5/T14)

Q112. The digits of a two-digit number differ by 4. When the digits are interchanged and the resulting number is added to the original, the sum is 110. Find all two-digit numbers satisfying this.

SOLUTION — step by step

Step 1. Let the tens digit be a and the units digit be b, with |a - b| = 4.

Step 2. The number is 10a + b; the reversed number is 10b + a.

Step 3. Their sum: 11a + 11b = 11(a + b) = 110, so a + b = 10.

Step 4. Now solve the pair: a + b = 10 and digits differ by 4. Two cases: a - b = 4 gives a = 7, b = 3; b - a = 4 gives a = 3, b = 7.

Step 5. The numbers are 73 and 37 (each is the other's reversal, and both satisfy every condition: digits differ by 4, and 73 + 37 = 110).

Step 6. Note the beautiful structure: the reversal sum only depends on a + b, so the difference condition picks the exact digit pair.

Answer: 73 and 37 (both satisfy the conditions).

Hinglish: *Reversal-sum ka secret: 10a + b + 10b + a = 11(a + b). Hamesha 11 ka multiple! 110/11 = 10 = digit sum. Phir 'differ by 4' do cases kholta hai - (7,3) ya (3,7). Dono valid, dono report karo. Ye question classic hai aur har exam mein kisi na kisi roop mein aata hai.*

QUESTION 113 | HOTS | Wire rectangle area function (T14)

Q113. A wire of length 36 cm is bent into a rectangle. Let the length be x cm. (i) Write the area as a function of x. (ii) Find ALL lengths x that give an area of 77 sq cm. (iii) Which single length gives the largest area, and what is that area?

SOLUTION — step by step

Step 1. Perimeter 36 means half-perimeter 18: length + width = 18. Width = 18 - x.

Step 2. Area $A(x) = x(18 - x) = 18x - x^2$ - a quadratic function (the PDF builds exactly this in its Example 3 with a 20 cm wire).

Step 3. Set $x(18 - x) = 77$: $x^2 - 18x + 77 = 0$. Look for two numbers with product 77 and sum 18: 7 and 11.

Step 4. So x = 7 or x = 11. Both are genuine: a 7 x 11 rectangle and an 11 x 7 rectangle are the SAME rectangle turned around - one answer in two descriptions.

Step 5. Largest area: test symmetric candidates. x = 9 gives 9 x 9 = 81; x = 8 gives 8 x 10 = 80; x = 10 gives 10 x 8 = 80. The square (x = 9) wins with area 81.

Step 6. Structure noticed: areas rise toward x = 9 and fall after it - symmetric about the square.

Answer: (i) $A(x) = 18x - x^2$ (ii) x = 7 or 11 (the same rectangle, two descriptions) (iii) x = 9 gives the maximum area 81 sq cm.

Hinglish: *Ye PDF ke wire example ka bada bhai. Area quadratic nikla ($18x - x^2$) - chapter linear hai par evaluation quadratic ki bhi practice hai (Ex 2.2 mein tha). 77 ke liye do answers (7 aur 11) ek hi rectangle ke do naam hain - symmetry! Aur square (9 x 9 = 81) max deta hai - ye observation Grade 10 mein formally aayega, aaj ke liye table-test kaafi hai.*

QUESTION 114 | HOTS | Salary structure (T9/T10)

Q114. A salesperson earns a fixed monthly amount plus a commission per sale. With 20 sales the pay is Rs 30,000; with 35 sales it is Rs 45,000. (i) Find the linear pay formula. (ii) How many sales are needed to earn Rs 52,000? (iii) What is the fixed amount?

SOLUTION — step by step

Step 1. Let pay = ax + b. Data: 20a + b = 30000 and 35a + b = 45000.

Step 2. Subtract: 15a = 15000, so a = 1000 per sale.

Step 3. Back-substitute: 30000 = 20(1000) + b = 20000 + b, b = 10000.

Step 4. Pay formula: y = 1000x + 10000.

Step 5. Target: 1000x + 10000 = 52000 gives 1000x = 42000, x = 42 sales.

Step 6. The fixed amount b = 10000: even with zero sales, Rs 10,000 is paid.

Answer: (i) y = 1000x + 10000 (ii) 42 sales (iii) Rs 10,000.

Hinglish: *Salary = bill jaisi hi structure: fixed + rate x count. 15 sales ka difference 15000 deta hai: per-sale 1000. 42 sales for 52k. Interpretation: zero sales par bhi 10k milta hai - base pay.*

QUESTION 115 | HOTS | Where C equals F (T14)

Q115. Using the conversion relation F = 1.8C + 32, find the temperature at which the Celsius and Fahrenheit readings are EQUAL. Show the full working.

SOLUTION — step by step

Step 1. Equal readings mean F = C. Substitute F = C into the relation: C = 1.8C + 32.

Step 2. Bring terms together: C - 1.8C = 32, so -0.8C = 32.

Step 3. Divide: C = 32 / (-0.8) = -40.

Step 4. Check: at C = -40, F = 1.8(-40) + 32 = -72 + 32 = -40. Equal. Correct.

Step 5. So -40 degrees is the unique temperature where both scales agree - a famous meeting point of the two lines F = 1.8C + 32 and F = C.

Answer: At -40 degrees, where C = F = -40.

Hinglish: *Dono reading barabar matlab dono lines ka INTERSECTION: F = C aur F = 1.8C + 32. Solve karo: -40. Negative temperature mein sign ka dhyan - 1.8(-40) = -72. Verify: -72 + 32 = -40. Dono scale -40 par milti hain - ek hi baar.*

QUESTION 116 | HOTS | Plan comparison (T14)

Q116. Mobile Plan A costs Rs 200 per month plus Rs 2 per GB. Plan B costs Rs 350 per month plus Rs 0.50 per GB. (i) Write both cost formulas. (ii) For what data usage do the plans cost the same? (iii) Which plan is cheaper for a user consuming 60 GB? (iv) For 150 GB?

SOLUTION — step by step

Step 1. Plan A: C_A = 200 + 2x. Plan B: C_B = 350 + 0.5x (x in GB).

Step 2. Equal cost: 200 + 2x = 350 + 0.5x. Move terms: 2x - 0.5x = 350 - 200, so 1.5x = 150, x = 100 GB.

Step 3. At 100 GB both cost 200 + 200 = Rs 400 (and 350 + 50 = 400 - consistent).

Step 4. At x = 60: A costs 200 + 120 = Rs 320; B costs 350 + 30 = Rs 380. A is cheaper (by Rs 60).

Step 5. At x = 150: A costs 200 + 300 = Rs 500; B costs 350 + 75 = Rs 425. B is cheaper (by Rs 75).

Step 6. Structure: below 100 GB choose the small-fixed-high-rate plan (A); above 100 GB the high-fixed-low-rate plan (B) wins. The crossing point 100 GB is the decision boundary.

Answer: (i) A: 200 + 2x; B: 350 + 0.5x (ii) equal at 100 GB (both Rs 400) (iii) A cheaper at 60 GB (320 vs 380) (iv) B cheaper at 150 GB (425 vs 500).

Hinglish: *Do plans ka muqabla = do lines ka intersection. 100 GB par dono 400 - crossing point. Usse kam data toh A sasta (fixed fee kam), zyada data toh B (rate kam). Ye 'decision boundary' thinking real life mein har jagah hai - OTT plans, taxi vs auto, sab yahi maths hai.*

QUESTION 117 | HOTS | Function composition lite (T14)

Q117. Let p(x) = 3x + 1. Given p(2) = 7 and p(5) = 16. (i) Find x such that p(x) = 100. (ii) Compute p(p(1)). (iii) Compute p(p(p(0))).

SOLUTION — step by step

Step 1. (i) Solve 3x + 1 = 100: 3x = 99, x = 33.

Step 2. (ii) Inner first: p(1) = 3(1) + 1 = 4. Then p(4) = 3(4) + 1 = 13. So p(p(1)) = 13.

Step 3. (iii) p(0) = 1. Then p(1) = 4 (from part ii). Then p(4) = 13. So p(p(p(0))) = 13.

Step 4. Observe (ii) and (iii) land on the same value: both chains pass through the input 1 at some stage, and the machine is deterministic - same input, same output.

Step 5. General check for (iii): p(p(p(0))) = p(p(1)) = p(4) = 13. Verified.

Answer: (i) x = 33 (ii) 13 (iii) 13.

Hinglish: *Machine ke andar machine - andar wala PEHLE chalao. p(p(1)): pehle p(1) = 4, phir p(4) = 13. (iii) mein teen layer: 0, 1, 4, 13. Interesting: (ii) aur (iii) ka jawab same kyun? Kyunki dono chains 1 se guzri. Deterministic machine = same input, same output.*

QUESTION 118 | HOTS | Iterated machine (T14)

Q118. The machine f(x) = 2x + 3 is applied again and again, starting from $x_0 = 1$: $x_1 = f(1)$, $x_2 = f(x_1)$, and so on. (i) Compute x_1, x_2, x_3. (ii) Compute x_4. (iii) Describe in words what each application does to the current value.

SOLUTION — step by step

Step 1. $x_1 = 2(1) + 3 = 5$.

Step 2. $x_2 = 2(5) + 3 = 13$.

Step 3. $x_3 = 2(13) + 3 = 29$.

Step 4. $x_4 = 2(29) + 3 = 61$.

Step 5. Each application doubles the current value and then adds 3 - 'double, then add 3', repeated.

Step 6. The outputs 1, 5, 13, 29, 61 grow fast because doubling compounds - a taste of exponential behaviour hiding inside repeated linear steps.

Answer: (i) 5, 13, 29 (ii) 61 (iii) each step doubles the value and adds 3.

Hinglish: *Repeat machine: double karo, 3 jodo, phir se. 1, 5, 13, 29, 61. Har step pe 'double + 3' - baar-baar same recipe. Jab linear machine ko baar-baar chalate hain toh growth tez ho jaati hai (double ki wajah se) - ye seedhi line se exponential behaviour ka darwaza hai.*

QUESTION 119 | HOTS | Axis intercepts and triangle (T13/T14)

Q119. Consider the line y = 2x + 8. (i) Find its y-intercept. (ii) Find its x-intercept. (iii) The line and the two axes form a triangle. Find its area.

SOLUTION — step by step

Step 1. y-intercept: put x = 0, y = 8 - the point (0, 8).

Step 2. x-intercept: put y = 0: 2x + 8 = 0, x = -4 - the point (-4, 0).

Step 3. The triangle has vertices (0, 0), (0, 8) and (-4, 0): a right triangle with legs 8 (vertical) and 4 (horizontal, in the negative direction).

Step 4. Area = (1/2) x 8 x 4 = 16 square units.

Step 5. Distances use absolute values: the horizontal leg has length |-4 - 0| = 4 even though the point sits left of the origin.

Answer: (i) (0, 8) (ii) (-4, 0) (iii) area = 16 square units.

Hinglish: *Intercepts nikaalo (0, 8) aur (-4, 0). Origin ke saath milke right triangle bana - legs 8 aur 4. Area 16. Left-side hone se area nahi badalta - length toh absolute hai. Ye question Chapter 2 ke graphing ko Chapter 1 ke area se jodta hai - mixed-concept preview.*

QUESTION 120 | HOTS | Concurrent lines (T14)

Q120. Three lines are given: L1: y = x + 1, L2: y = -x + 5 and L3: y = 3. Find the pairwise intersections of L1 with L2, L1 with L3, and L2 with L3. What do you conclude about the three lines?

SOLUTION — step by step

Step 1. L1 meets L2 where x + 1 = -x + 5: 2x = 4, x = 2, y = 3. Point (2, 3).

Step 2. L1 meets L3 where x + 1 = 3: x = 2, y = 3. Point (2, 3).

Step 3. L2 meets L3 where -x + 5 = 3: x = 2, y = 3. Point (2, 3).

Step 4. All three pairwise intersections are the SAME point (2, 3).

Step 5. Conclusion: the three lines are CONCURRENT - they all pass through one common point. Verification: (2, 3) satisfies all three equations: 2 + 1 = 3, -2 + 5 = 3, and 3 = 3.

Answer: All three intersections equal (2, 3); the lines are concurrent.

Hinglish: *Teen lines, teen intersections - aur teeno ka jawab SAME (2, 3)! Matlab teeno lines ek hi point par cross ho rahi hain - concurrency. Two-line intersection nikaalna = equations barabar rakhna. Ye HOTS finale tumhe Class 11 ki 'solution of system' ki taraf punch maar raha hai.*

2.5 Rapid MCQs: Choose the Correct Option (20)

Rapid-fire round: 4 options, ek hi correct. Pehle khud choose karo, phir colored Answer strip se check karo. Har galat option ka "Why wrong" bhi diya hai - taki exam me wahi trap tumhe na phasaye.

MCQ 1 | Normal | Coefficient

1. The coefficient of x in the expression 5x - 7 is:

A. 5

B. -7

C. 5x

D. 1

Correct Answer: (A)

The coefficient is the number multiplying the variable x, which is 5. The -7 is the constant term.

Why (B) is wrong: -7 is the constant term, not attached to x.

Why (C) is wrong: 5x is the whole term.

Why (D) is wrong: 1 is the coefficient of a bare x, not of 5x.

Hinglish: *Coefficient = variable ke saath juda number. 5x ka coefficient 5. -7 constant hai, coefficient nahi.*

MCQ 2 | Normal | Degree

2. The degree of the polynomial $4 - 3y + y^2$ is:

A. 1

B. 2

C. 3

D. 0

Correct Answer: (B)

The powers present are 0 (the 4), 1 (the -3y) and 2 (the y^2). The highest is 2, so the degree is 2.

Why (A) is wrong: 1 is just the middle term's power.

Why (C) is wrong: No cube is present.

Why (D) is wrong: 0 would mean a constant polynomial.

Hinglish: *Order mein likha nahi hai par koi baat nahi - powers scan karo: 0, 1, 2. Max = 2. Quadratic.*

MCQ 3 | Normal | Constant polynomial

3. The degree of the constant polynomial -9 is:

A. -9

B. 0

C. 1

D. undefined

Correct Answer: (B)

Any constant c can be written as c x x^0, and $x^0 = 1$. So the degree (power of the variable) is 0.

Why (A) is wrong: -9 is the value, not the degree.

Why (C) is wrong: There is no x term at all.

Why (D) is wrong: It is defined - it is 0.

Hinglish: *Constant ka degree ZERO hota hai - isko x^0 ke roop mein likh sakte ho. Ye MCQ ka pakka sawaal hai.*

MCQ 4 | Normal | Evaluation

4. The value of 4x - 9 at x = -2 is:

A. -17

B. 17

C. -1

D. 1

Correct Answer: (A)

4(-2) - 9 = -8 - 9 = -17. Both terms are negative here.

Why (B) is wrong: Lost a minus somewhere.

Why (C) is wrong: Computed 4 - 2 - 9 style without brackets.

Why (D) is wrong: Sign error in 4 x (-2).

Hinglish: *4(-2) = -8, phir -9 jodo: -17. Dono negative milke aur negative - sign ka khel.*

MCQ 5 | Normal | Reverse evaluation

5. The machine y = 5x + 2 gives output 27. The input x is:

A. 5

B. 4

C. 6

D. 25

Correct Answer: (A)

5x + 2 = 27 gives 5x = 25, so x = 5.

Why (B) is wrong: Arithmetic slip 27 - 2 = 24.

Why (C) is wrong: Added before dividing.

Why (D) is wrong: Reported 5x instead of x.

Hinglish: *Peeche chalna hai: 27 se 2 ghatayo (25), 5 se divide (5). Verify: 5(5) + 2 = 27.*

MCQ 6 | Normal | Sum-difference words

6. Two numbers add to 46, and one of them is 6 more than the other. The numbers are:

A. 20 and 26

B. 23 and 23

C. 18 and 28

D. 21 and 25

Correct Answer: (A)

Let them be x and x + 6: 2x + 6 = 46 gives x = 20. Numbers 20 and 26.

Why (B) is wrong: Equal numbers differ by 0.

Why (C) is wrong: Difference is 10, not 6.

Why (D) is wrong: Difference is 4.

Hinglish: *2x + 6 = 46 se x = 20. Check: 20 + 26 = 46, difference 6. Dono condition sahi.*

MCQ 7 | Normal | Pattern rule

7. A sequence begins 4, 9, 14, 19, ... Its nth-term rule is:

A. 5n - 1

B. 5n + 1

C. 4n

D. n + 3

Correct Answer: (A)

The constant difference is 5, so the rule is 5n + c. Stage 1: 5 + c = 4 gives c = -1. Rule: 5n - 1. Check stage 2: 10 - 1 = 9. Correct.

Why (B) is wrong: Would give 6 as the first term.

Why (C) is wrong: Wrong difference handling.

Why (D) is wrong: No pattern fit at all.

Hinglish: *Difference 5 pakdo, phir first term se c nikaalo: 4 = 5 + c isliye c = -1. 5n - 1. Verify with 9: 10 - 1 = 9 sahi.*

MCQ 8 | Normal | Fit y = ax + b

8. A service costs Rs 90 for 2 units and Rs 210 for 6 units. In the model y = ax + b:

A. a = 30, b = 30

B. a = 30, b = 90

C. a = 45, b = 0

D. a = 35, b = 20

Correct Answer: (A)

210 - 90 = 120 = 4a, so a = 30. Back-substitute: 90 = 2(30) + b = 60 + b, b = 30. Model: y = 30x + 30.

Why (B) is wrong: Forgot to subtract the per-unit part.

Why (C) is wrong: Mixed up which pair to use.

Why (D) is wrong: No consistent fit.

Hinglish: *4 units ka difference 120: per-unit 30. Wapas: 90 - 60 = 30 fixed. Interpretation: Rs 30 per unit + Rs 30 base.*

MCQ 9 | Normal | Slope meaning in conversion

9. In the conversion F = 1.8C + 32, the number 1.8 tells us that:

A. a 1 degree rise in Celsius gives a 1.8 degree rise in Fahrenheit

B. Fahrenheit is always 1.8 more than Celsius

C. water boils at 1.8 degrees

D. the scales meet at 1.8

Correct Answer: (A)

The slope is the rate of change: each extra degree Celsius lifts the Fahrenheit reading by 1.8. (Option B confuses slope with a constant offset - the offset is 32, not 1.8.)

Why (B) is wrong: That is what b would mean, but b = 32 here.

Why (C) is wrong: Boiling is at 100 C / 212 F.

Why (D) is wrong: The scales meet at -40, not 1.8.

Hinglish: *Slope = rate: C mein 1 degree badho toh F mein 1.8 badho. Fixed gap toh 32 hai - wo b wala hissa hai, 1.8 nahi.*

MCQ 10 | Normal | Disguised slope

10. The slope of the line $3y = 6x + 12$ is:

A. 6

B. 2

C. 3

D. 12

Correct Answer: (B)

Divide by 3 first: $y = 2x + 4$. The slope is 2. Reading the raw coefficient 6 before isolating y is the classic trap.

Why (A) is wrong: That is the coefficient of x before dividing.

Why (C) is wrong: That is the divisor.

Why (D) is wrong: That is the constant after dividing.

Hinglish: *Pehle y = ... banao! 3 se divide: y = 2x + 4. Slope 2. Seedha 6 padhna sabse common galti hai.*

MCQ 11 | Normal | Lines through origin

11. Which of these lines passes through the origin?

A. $y = 5x$

B. $y = 5x + 1$

C. $y = x - 3$

D. $y = 2$

Correct Answer: (A)

A line passes through (0, 0) exactly when its constant term b is 0. Only $y = 5x$ has $b = 0$.

Why (B) is wrong: Cuts y-axis at (0, 1).

Why (C) is wrong: Cuts at (0, -3).

Why (D) is wrong: Horizontal line at height 2.

Hinglish: *Origin se guzarne ka ek hi test: b = 0. Baaki sab (0, b) par y-axis kaat-ti hain, origin se nahi.*

MCQ 12 | Normal | Parallel detection

12. The lines $y = 4x - 3$ and $y = 4x + 7$ are:

A. perpendicular

B. parallel

C. the same line

D. intersecting at one point

Correct Answer: (B)

Same slope 4, different intercepts (-3 vs 7): parallel lines that never meet.

Why (A) is wrong: Perpendicularity is about slope products -1, not equality.

Why (C) is wrong: Intercepts differ.

Why (D) is wrong: Parallel lines never intersect.

Hinglish: *Slope same, intercept alag = parallel. Same slope SAME intercept hota toh ek hi line hoti.*

MCQ 13 | Normal | Growth vs decay

13. The model h(t) = 3 - 0.5t describes:

A. linear growth

B. linear decay

C. no change

D. quadratic behaviour

Correct Answer: (B)

The rate of change (slope) is -0.5: each month the height drops by 0.5. Negative slope means decay.

Why (A) is wrong: Growth needs a positive slope.

Why (C) is wrong: Something IS changing - by 0.5 per step.

Why (D) is wrong: The rule is degree 1, not 2.

Hinglish: *Slope negative = decay. 3 starting value, 0.5 per month girawat - PDF ke water-tank example jaisa.*

MCQ 14 | Normal | Spot the linear pattern

14. Which sequence is a LINEAR pattern?

A. 3, 7, 11, 15

B. 2, 4, 8, 16

C. 1, 4, 9, 16

D. 1, 1, 2, 3, 5

Correct Answer: (A)

A linear pattern has a constant difference between consecutive terms. Option A adds 4 every time. The others double, square, or follow Fibonacci - none has a constant difference.

Why (B) is wrong: Doubling - multiplicative, not linear.

Why (C) is wrong: Perfect squares.

Why (D) is wrong: Fibonacci-type additions of previous terms.

Hinglish: *Linear pattern ka test: consecutive difference CONSTANT. 3, 7, 11, 15 mein +4 har baar. Baaki doubling, squares, Fibonacci hain - inka difference badalta hai.*

MCQ 15 | Normal | Tiered fare

15. A cab charges Rs 50 for the first 2 km and Rs 12 per km after that. The fare for a 10 km ride is:

A. Rs 170

B. Rs 146

C. Rs 158

D. Rs 120

Correct Answer: (B)

Extra distance = 10 - 2 = 8 km. Fare = 50 + 12(8) = 50 + 96 = Rs 146.

Why (A) is wrong: Applied 12 to all 10 km.

Why (C) is wrong: Arithmetic slip.

Why (D) is wrong: Forgot the base fare.

Hinglish: *Pehle 2 km ka paisa fixed 50 mein. Baaki 8 km x 12 = 96. Total 146. (n - 2) wala structure yaad rakho.*

MCQ 16 | HOTS | Triangle with axes

16. The line y = 3x + 6 and the two coordinate axes form a triangle. Its area is:

A. 6 sq units

B. 12 sq units

C. 3 sq units

D. 18 sq units

Correct Answer: (A)

y-intercept (0, 6); x-intercept: 3x + 6 = 0 gives x = -2, point (-2, 0). Right triangle with legs 6 and 2: area = (1/2)(6)(2) = 6.

Why (B) is wrong: Forgot the 1/2.

Why (C) is wrong: Used the wrong leg.

Why (D) is wrong: Multiplied intercepts without halving.

Hinglish: *Intercepts (0, 6) aur (-2, 0). Origin ke saath right triangle: legs 6 aur 2, area 6. Length absolute value mein lo - negative coordinate se area negative nahi hota.*

MCQ 17 | Normal | Hidden coefficient

17. The coefficient of y in the expression 3x + y - 8 is:

A. 0

B. 1

C. 3

D. -8

Correct Answer: (B)

The term y is really 1y, so the coefficient is 1.

Why (A) is wrong: y is present, so not 0.

Why (C) is wrong: 3 belongs to x.

Why (D) is wrong: -8 is the constant.

Hinglish: *Invisible 1! y = 1y. Jo variable ke aage number nahi dikhta uska coefficient 1 hota hai.*

MCQ 18 | Normal | Degree scan trap

18. The degree of the polynomial $x^3 + 7x^4 - 2x$ is:

A. 3

B. 4

C. 7

D. 1

Correct Answer: (B)

The powers present are 3, 4 and 1. The HIGHEST is 4, even though the x^3 term is written first. Degree is about the maximum power, not the order of writing.

Why (A) is wrong: Fell for the first term.

Why (C) is wrong: Added powers - never do that.

Why (D) is wrong: Lowest power.

Hinglish: *x^3 pehle likha hai par $7x^4$ bhi hai - max power 4. Order of writing ka koi matlab nahi degree mein.*

MCQ 19 | HOTS | Two-models meeting

19. Tap A fills an empty tank at 20 litres per minute (V = 20t). Tap B pours into a tank that already has 100 litres, at 12 litres per minute (V = 100 + 12t). After how many minutes do the two tanks hold equal volumes?

A. 12.5 minutes

B. 10 minutes

C. 8 minutes

D. 25 minutes

Correct Answer: (A)

Set 20t = 100 + 12t: 8t = 100, t = 12.5 minutes. (Each then holds 250 litres.)

Why (B) is wrong: Misbalanced the equation.

Why (C) is wrong: Subtracted wrongly.

Why (D) is wrong: Doubled the time.

Hinglish: *Do lines ka intersection: 8t = 100, t = 12.5. Us waqt dono 250 litre. Fraction aana bilkul theek hai - yahan stage ki jagah TIME hai, jo real value ho sakti hai.*

MCQ 20 | HOTS | Intersection of two lines

20. The lines y = x + 4 and y = -2x + 10 intersect where x equals:

A. 2

B. 6

C. 4

D. 10

Correct Answer: (A)

Set equal: x + 4 = -2x + 10 gives 3x = 6, x = 2 (then y = 6 - the point (2, 6)).

Why (B) is wrong: That is the y-value.

Why (C) is wrong: Moved terms wrongly.

Why (D) is wrong: No basis.

Hinglish: *Barabar rakho: 3x = 6, x = 2. Yehi method Q50 wale concurrency question mein use hua tha.*

2.6 Choose the Correct Round: Theory & Speed (20)

Doosra rapid round - har MCQ ka style: statement padho, correct option turant chuno. Ye round theory, definitions aur quick solving ko test karta hai. Answer strip ke neeche har galat option ka "Why wrong" bhi hai - taki exam ka trap tumhe na phasaye.

CTC 1 | Normal | Classification

1. The polynomial 2z - 9 is:

A. constant

B. linear

C. quadratic

D. cubic

Correct Answer: (B)

The highest power of z is 1, so it is a linear polynomial (degree 1).

Why (A) is wrong: Degree is 1, not 0.

Why (C) is wrong: No z^2 term.

Why (D) is wrong: No z^3 term.

Hinglish: *Power 1 = linear. Naam yaad rakho: 1-linear, 2-quadratic, 3-cubic, 0-constant.*

CTC 2 | Normal | Quadratic evaluation

2. The value of $2s^2 - s + 3$ at $s = -1$ is:

A. 0

B. 6

C. 4

D. -2

Correct Answer: (B)

$2(-1)^2 - (-1) + 3 = 2(1) + 1 + 3 = 6$. The square forgives the minus; the -s term becomes +1.

Why (A) is wrong: Dropped the +1 from the middle term.

Why (C) is wrong: Wrong square value.

Why (D) is wrong: Sign chaos throughout.

Hinglish: *$(-1)^2 = +1$, aur -s matlab $-(-1) = +1$. Dono jagah plus bana! 2 + 1 + 3 = 6.*

CTC 3 | Normal | Constant perimeter growth

3. If the side of a square increases by 3 units, its perimeter increases by:

A. 3 units

B. 6 units

C. 12 units

D. depends on the original side

Correct Answer: (C)

New perimeter = $4(s + 3) = 4s + 12$. The jump is exactly 12 - the same for every square, because the rule 4s is linear with slope 4 and 4 x 3 = 12.

Why (A) is wrong: Perimeter counts ALL four sides.

Why (B) is wrong: Counted only two sides.

Why (D) is wrong: Linear growth never depends on the starting value.

Hinglish: *Slope 4 hai isliye side ka har 1 unit = perimeter ka 4 unit. 3 ka badlav = 12 ka badlav - hamesha, original side jo bhi ho. Yehi linear ka jaadu hai.*

CTC 4 | Normal | Fee structure

4. A club charges Rs 250 to join plus Rs 75 per session. The total cost for 7 sessions is:

A. Rs 525

B. Rs 325

C. Rs 775

D. Rs 750

Correct Answer: (C)

250 + 75(7) = 250 + 525 = Rs 775.

Why (A) is wrong: Only the session charges.

Why (B) is wrong: Counted one session.

Why (D) is wrong: Roughly right but 75 x 7 was miscalculated.

Hinglish: *Fixed + rate x count. 75 x 7 = 525, plus 250 = 775. Option A bhool gaya joining fee.*

CTC 5 | Normal | Which stage

5. In the sequence with rule T(n) = 5n - 1, the term equal to 54 occurs at:

A. n = 10

B. n = 11

C. n = 12

D. n = 54/5

Correct Answer: (B)

5n - 1 = 54 gives 5n = 55, n = 11. The 11th term is 54.

Why (A) is wrong: 5(10) - 1 = 49, not 54.

Why (C) is wrong: 5(12) - 1 = 59.

Why (D) is wrong: Forgot to add 1 first.

Hinglish: *54 + 1 = 55, 55/5 = 11. Whole number mila toh stage exist karti hai.*

CTC 6 | Normal | Reachability

6. In the same sequence T(n) = 5n - 1, is 53 a term of the sequence?

A. Yes, at n = 10

B. Yes, at n = 11

C. No, because n = 54/5 is not a whole number

D. Cannot be decided

Correct Answer: (C)

5n - 1 = 53 gives 5n = 54, n = 10.8 - not a whole number. Sequence positions must be whole numbers, so 53 is never reached (the terms jump from 49 to 54).

Why (A) is wrong: 5(10) - 1 = 49.

Why (B) is wrong: 5(11) - 1 = 54.

Why (D) is wrong: The rule decides it completely.

Hinglish: *53 + 1 = 54, 54/5 = 10.8 - fraction! Toh 53 sequence mein KABHI nahi aayega. 49 ke baad seedha 54. 'No with reason' hi poora answer hai.*

CTC 7 | Normal | Depreciation decay

7. A scooter bought for Rs 20,000 loses Rs 150 in value each month. Its value after t months is 20000 - 150t. It will be worth Rs 8,000 after:

A. 60 months

B. 80 months

C. 8 months

D. 120 months

Correct Answer: (B)

20000 - 150t = 8000 gives 150t = 12000, t = 80 months.

Why (A) is wrong: Divided wrongly.

Why (C) is wrong: Dropped two zeros.

Why (D) is wrong: Overshoot.

Hinglish: *12000 ka loss chahiye, 150 per month: 12000/150 = 80 mahine. Long division yaad rakho: dono ko 50 se kaato - 240/3 = 80.*

CTC 8 | Normal | Tank emptying

8. A tank with 600 litres drains at 25 litres per hour. It empties exactly after:

A. 20 hours

B. 24 hours

C. 25 hours

D. 30 hours

Correct Answer: (B)

600 - 25t = 0 gives t = 600/25 = 24 hours.

Why (A) is wrong: Arithmetic slip.

Why (C) is wrong: Confused rate with time.

Why (D) is wrong: Overshoot.

Hinglish: *600/25 - 25 x 24 = 600 yaad rakho (quarter of 100 is 25; 600 has 24 quarters).*

CTC 9 | Normal | Fit with zero intercept

9. A photocopy shop charges purely per page. 5 pages cost Rs 100 and 8 pages cost Rs 160. The linear model y = ax + b has:

A. a = 20, b = 0

B. a = 20, b = 20

C. a = 12, b = 40

D. a = 32, b = 0

Correct Answer: (A)

Subtract the data equations: 160 - 100 = 3a gives a = 20. Then b = 100 - 20(5) = 0. Purely per-page pricing means zero fixed fee - the model is y = 20x, a line through the origin.

Why (B) is wrong: Invented a fee that the data contradicts.

Why (C) is wrong: Difference divided wrongly.

Why (D) is wrong: Used 8 instead of the rate.

Hinglish: *Subtract-trick: 60 = 3a, a = 20. Wapas jao: 100 - 100 = 0 = b. Zero intercept ka matlab: origin se line. 'Purely per page' words se hi b = 0 ki expectation ban jaani chahiye.*

CTC 10 | Normal | Disguised intercept

10. The line 4y = 8x - 20 cuts the y-axis at:

A. (0, -20)

B. (0, -5)

C. (-5, 0)

D. (0, 2)

Correct Answer: (B)

Divide by 4: y = 2x - 5. The y-intercept is the point (0, -5). (Option A is the raw constant; option C confuses the axes.)

Why (A) is wrong: Raw -20 before dividing by 4.

Why (C) is wrong: That is an x-axis form.

Why (D) is wrong: Read the slope instead.

Hinglish: *4 se divide: y = 2x - 5. y-axis cut = (0, b) = (0, -5). x-axis wala point alag hota hai - (2.5, 0).*

CTC 11 | Normal | Steepness

11. Which line is STEEPER than y = x?

A. y = x + 9

B. y = 0.5x

C. y = 3x - 1

D. none of them

Correct Answer: (C)

Steepness is the size of the slope. y = 3x - 1 has slope 3, greater than 1. The others have slope 1 (just shifted) and 0.5 (flatter).

Why (A) is wrong: Adding 9 changes position, not steepness.

Why (B) is wrong: 0.5 is flatter than 1.

Why (D) is wrong: One of them IS steeper.

Hinglish: *Steepness = |slope|. 3 > 1 toh y = 3x - 1 tez chadhti hai. y = x + 9 utni hi steep hai - bas upar shift hui hai.*

CTC 12 | Normal | Intercept recognition

12. Which line cuts the y-axis at the point (0, -4)?

A. y = 3x - 4

B. y = -4x + 3

C. y = 4x

D. y = x + 4

Correct Answer: (A)

The y-intercept equals b in y = ax + b, and (0, b) is the cutting point. Only option A has b = -4.

Why (B) is wrong: Its b is 3 (the -4 is a slope).

Why (C) is wrong: Passes through the origin.

Why (D) is wrong: Cuts at (0, 4).

Hinglish: *(0, b) form yaad rakho: y-axis cut ka second coordinate hi b hai. A mein b = -4. Verify: x = 0 par y = -4 sahi.*

CTC 13 | Normal | Point on line

13. Which point lies on the line y = 2x - 5?

A. (3, 1)

B. (3, 0)

C. (1, 3)

D. (0, -3)

Correct Answer: (A)

Substitute x = 3: y = 2(3) - 5 = 1. So (3, 1) satisfies the equation. The others fail: (3, 0) gives 1 not 0; (1, 3) needs y = -3; (0, -3) needs y = -5.

Why (B) is wrong: 2(3) - 5 = 1, not 0.

Why (C) is wrong: 2(1) - 5 = -3, not 3.

Why (D) is wrong: 2(0) - 5 = -5, not -3.

Hinglish: *Test karo: x = 3 dala, y = 1 nikla - toh (3, 1) line par hai. Baaki options ke y mismatch.*

CTC 14 | Normal | Assertion-reason

14. Assertion (A): Every line y = ax passes through the origin. Reason (R): Substituting x = 0 gives y = a(0) = 0.

A. Both A and R are true and R explains A

B. Both A and R are true but R does not explain A

C. A is true, R is false

D. A is false, R is true

Correct Answer: (A)

The reason is precisely the verification that (0, 0) satisfies y = ax for every a - it explains the assertion completely.

Why (B) is wrong: The link is direct.

Why (C) is wrong: R is a correct computation.

Why (D) is wrong: A is a true property of the family.

Hinglish: *R hi A ka proof hai: x = 0 par y = 0 nikalta hai, chahe a kuch bhi ho. Perfect A-R pair - option A.*

CTC 15 | Normal | Assertion-reason evaluation

15. Assertion (A): For p(x) = 2x + 3, the value at x = -1 is 1. Reason (R): Substituting a negative input requires brackets: 2(-1) + 3.

A. Both A and R are true and R explains A

B. Both A and R are true but R does not explain A

C. A is true, R is false

D. A is false, R is true

Correct Answer: (A)

A is true: 2(-1) + 3 = -2 + 3 = 1. R states the safe-substitution practice that makes the computation of A reliable - it explains why the evaluation works out as claimed.

Why (B) is wrong: The bracket rule is exactly how A is computed.

Why (C) is wrong: R is correct practice.

Why (D) is wrong: A computes to 1, true.

Hinglish: *2(-1) + 3 = 1 sahi hai, aur bracket wala rule hi wo calculation reliable banata hai. R explains A.*

CTC 16 | Normal | Reverse evaluation equation

16. Solving 6 - 2x = -4 for x gives:

A. x = 5

B. x = -5

C. x = 1

D. x = -1

Correct Answer: (A)

Move 6: -2x = -10, so x = 5. Check: 6 - 2(5) = 6 - 10 = -4. Correct.

Why (B) is wrong: Sign slip in the final division.

Why (C) is wrong: Moved terms wrongly.

Why (D) is wrong: Half-solved.

Hinglish: *-2x = -10 - dono side minus; divide karne par plus. x = 5. Verify karo: 6 - 10 = -4 sahi.*

CTC 17 | Normal | nth term from sequence

17. The nth term of the sequence 7, 12, 17, 22, ... is:

A. 5n + 2

B. 5n - 2

C. 7n

D. n + 6

Correct Answer: (A)

Constant difference 5: rule 5n + c. First term: 5 + c = 7 gives c = 2. Rule 5n + 2. Check: n = 2 gives 12. Correct.

Why (B) is wrong: Would start at 3.

Why (C) is wrong: No pattern fit.

Why (D) is wrong: Not linear in the right way.

Hinglish: *Difference 5, first term 7 = 5 + c, c = 2. 5n + 2. Second term se verify karo - 12 sahi.*

CTC 18 | Normal | Age relation

18. Ravi is 3 times as old as his sister. In 4 years, the sum of their ages will be 40. Ravi's present age is:

A. 24 years

B. 8 years

C. 27 years

D. 30 years

Correct Answer: (A)

Let the sister be x and Ravi 3x. In 4 years: (x + 4) + (3x + 4) = 40, so 4x + 8 = 40, x = 8. Sister is 8, Ravi is 24.

Why (B) is wrong: That is the sister's age - the question asks for Ravi.

Why (C) is wrong: Solved a wrong equation.

Why (D) is wrong: Arithmetic overshoot.

Hinglish: *4x + 8 = 40 se x = 8 (behen). Ravi = 3x = 24. Question ne RAVI manga tha - sister ka number dena classic galti hai.*

CTC 19 | Normal | Fit from two points

19. A bill follows y = ax + b. If 10 units cost Rs 400 and 20 units cost Rs 550, then:

A. a = 15, b = 250

B. a = 15, b = 400

C. a = 40, b = 0

D. a = 55, b = 150

Correct Answer: (A)

Subtract: 550 - 400 = 150 = 10a, so a = 15. Then b = 400 - 15(10) = 250. Model: y = 15x + 250.

Why (B) is wrong: Forgot to back-subtract.

Why (C) is wrong: Used the wrong pair.

Why (D) is wrong: Inconsistent fit.

Hinglish: *10 units ka extra 150: per-unit 15. Wapas: 400 - 150 = 250 fixed. Standard 3-step: subtract, divide, back-substitute.*

CTC 20 | Normal | Decay value

20. A laptop bought at Rs 60,000 loses Rs 6,000 in value per year. After 5 years its value is:

A. Rs 30,000

B. Rs 54,000

C. Rs 36,000

D. Rs 24,000

Correct Answer: (A)

Value = 60000 - 6000(5) = 60000 - 30000 = Rs 30,000.

Why (B) is wrong: One year only.

Why (C) is wrong: Wrong multiplier.

Why (D) is wrong: Subtracted too much.

Hinglish: *6000 x 5 = 30000 ka loss. 60000 - 30000 = 30000. Aadha paisa bacha.*

Chapter 3: The World of Numbers (Real Numbers)

Numbers are not one big happy family - they live in clearly separated neighbourhoods, and exams love to test whether you know who lives where. The uploaded PDF builds the full number map: natural numbers N, integers Z, rationals Q, irrationals, and the master set R of real numbers. It studies decimal expansions (terminating, recurring, never-ending and never-repeating), shows how to convert recurring decimals back into fractions, constructs roots like $\sqrt{2}$ and $\sqrt{3}$ on the number line using right-angled triangles, proves that $\sqrt{2}$ is irrational, simplifies surds, rationalises denominators, and closes with the laws of exponents. In this book every one of those ideas is pushed to competition level: reverse questions, contradiction proofs, nested radicals, and exponent equations that look impossible until you spot the base.

3.1 Complete Concept Cheat Sheet (read this before any question)

3.1.1 The number family map: N, Z, Q, I, R

What is it? A classification of all numbers you will meet in Grade 9. **N** (natural numbers) = 1, 2, 3, ... **Z** (integers) = ..., -2, -1, 0, 1, 2, ... - N plus zero plus the negatives. **Q** (rationals) = every number writable as p/q with p, q integers and $q \neq 0$ - this includes all terminating decimals and all recurring decimals. **I** (irrationals) = real numbers that can NOT be written as p/q, like $\sqrt{2}$, $\sqrt{3}$, π, and decimals that go on forever without any repeating block. **R** (reals) = $Q \cup I$, every point on the number line. The nesting is strict: $N \subset Z \subset Q \subset R$, while I sits beside Q inside R with $I \cap Q = \emptyset$.

When do I use it? Whenever a question says 'classify', 'true or false', 'which of these is irrational', or asks about closure of operations. **How do I recognise the type?** Terminating decimal → rational. Recurring decimal → rational. Square root of a non-perfect-square → irrational. π and e → irrational. A fraction of integers → rational (even -7/3). **Tiny example.** -4 is Z and Q and R but not N; 0.333... = 1/3 is Q; $\sqrt{7}$ is I; 22/7 is Q (it is only an approximation of π, never equal to it).

Figure 3.1: The number family - each set sits inside the next bigger one; irrationals live beside Q inside R.

3.1.2 Decimal expansions and the denominator test

What is it? Every rational number has a decimal expansion that either STOPS (terminates) or STOPS NEVER BUT REPEATS a block (non-terminating recurring). Every irrational number has a decimal expansion that goes on forever WITHOUT any repeating block (non-terminating non-recurring). **The denominator test:** write the rational in lowest terms p/q. If $q = 2^m \times 5^n$ (only 2s and 5s, which are exactly the prime factors of 10), the decimal terminates; otherwise it recurs. Why 2 and 5? Because $10 = 2 \times 5$, so only these primes can be 'absorbed' into a power of ten.

When do I use it? Questions like 'without long division, state whether the decimal will terminate'. **Tiny example.** 17/8: $8 = 2^3$, so terminating (2.125). 13/3125: $3125 = 5^5$, terminating. 7/12: $12 = 2^2 \times 3$ - the stray 3 forces a recurring decimal. 6/15 = 2/5 in lowest terms - always REDUCE FIRST, the reduced denominator decides.

> **The reduce-first rule**
> The $2^l \times 5^n$ test works only in LOWEST TERMS. 6/15 looks non-terminating until you cancel the 3 and get 2/5 = 0.4. Reducing first is the difference between a right and a wrong answer.

3.1.3 Recurring decimals back to fractions (p/q form)

What is it? A reversing technique. **Pure recurring (repeat starts immediately):** count the repeating digits k, then the number = (repeating block)/(same number of 9s). Example: 0.555... = 5/9, 0.373737... = 37/99, 0.090909... = 9/99 = 1/11. **Mixed recurring (some non-repeating digits first):** (all digits once, minus the non-repeating part) / (as many 9s as repeating digits, followed by as many 0s as non-repeating digits). Example: 0.4777... = (47 - 4)/90 = 43/90.

Where does the trick come from? Multiply by the right power of 10 so two copies of the repeating block line up, then subtract - the infinite tails cancel exactly and you are left with a finite equation. **Full derivation for 0.777...:** let x = 0.777...; then 10x = 7.777...; subtract: 10x - x = 7.777... - 0.777... = 7; so 9x = 7 and x = 7/9. **Tiny example (mixed).** x = 0.2333...; 10x = 2.333..., 100x = 23.33...; subtract: 90x = 21, x = 21/90 = 7/30. Always finish by reducing the fraction.

3.1.4 Locating roots on the number line

What is it? A compass-ruler construction. To place $\sqrt{n}$: build a right triangle where the hypotenuse has length $\sqrt{n}$ using the Baudhayana-Pythagoras theorem. For $\sqrt{2}$: take OA = 1 on the line, raise AB = 1 perpendicular at A; then $OB = \sqrt{(1^2 + 1^2)} = \sqrt{2}$; swing an arc centred at O through B down to the line. For $\sqrt{3}$: repeat with the previous hypotenuse - take the segment of length $\sqrt{2}$ as one leg, add a perpendicular leg of 1, get hypotenuse $\sqrt{(2 + 1)} = \sqrt{3}$. Chaining this creates the famous **square root spiral** where each new edge has length $\sqrt{(n+1)}$.

When do I use it? 'Show how to find $\sqrt{5}$ on the number line', 'represent $\sqrt{4.3}$ geometrically', and successive-magnification questions ('zoom twice and tell which two numbers $\sqrt{3}$ sits between at the second decimal'). **Successive magnification.** To locate 2.6 (say $\sqrt{7} \approx 2.6457$): between 2 and 3, then between 2.6 and 2.7, then between 2.64 and 2.65 - each zoom narrows the interval by a factor of ten. **Tiny example.** $\sqrt{5}$: right triangle with legs 2 and 1, hypotenuse $\sqrt{5}$, arc it down to the line between 2.2 and 2.3 (since $2.23^2 \approx 4.98$, $2.24^2 \approx 5.02$).

Figure 3.2: Building $\sqrt{2}$, $\sqrt{3}$, $\sqrt{4}$, $\sqrt{5}$ on the number line with right triangles and arcs.

3.1.5 Irrationality proofs (the contradiction method)

What is it? The classic proof that $\sqrt{2}$ is irrational, done by assuming the opposite and breaking it. **Skeleton:** Step 1 - suppose $\sqrt{2} = p/q$ in lowest terms (p, q coprime, $q \neq 0$). Step 2 - square: $p^2 = 2q^2$, so 2 divides p^2, hence 2 divides p (because if a prime divides a square, it divides the number itself). Step 3 - write p = 2r; then $4r^2 = 2q^2$, so $q^2 = 2r^2$, hence 2 divides q too. Step 4 - both p and q even contradicts lowest terms. Conclusion: the assumption was false, so $\sqrt{2}$ is irrational. The identical skeleton works for $\sqrt{3}$ (replace 2 by 3) and $\sqrt{5}$.

Extension questions exams adore: (i) $\sqrt{2} + \sqrt{3}$ is irrational - if it were rational, its square $(5 + 2\sqrt{6})$ would be rational, forcing $\sqrt{6}$ rational, contradiction; (ii) rational + irrational = irrational - if $x + \sqrt{2} = r$ were rational then $\sqrt{2} = r - x$ would be rational, contradiction; (iii) nonzero rational × irrational = irrational - same subtraction trick; (iv) the sum of two irrationals may be rational (a + (-a) = 0, or $(1 + \sqrt{2}) + (1 - \sqrt{2}) = 2$) and

the product may be rational ($\sqrt{2} \times \sqrt{2} = 2$) - so NO general rule, must test case by case.

> **Why 'a prime dividing a square divides the number'?**
> If $p \mid n^2$, write $n = p \cdot k + r$ with $0 \le r < p$. Squaring, $n^2 \equiv r^2 \pmod{p}$. For p prime, $n^2 \equiv 0 \pmod{p}$ forces $r^2 \equiv 0 \pmod{p}$, and since p is prime r must be 0. Grade 9 books justify it with prime factorisations: $p \mid n^2$ means p appears in the factorisation of n^2, and every prime in n^2 comes from a prime in n - in pairs. One 'stray' p is impossible.

3.1.6 Surds: simplifying, adding, comparing

What is it? A surd is an irrational root kept in root form, like $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$. **Key laws:** $\sqrt{(ab)} = \sqrt{a} \cdot \sqrt{b}$; $\sqrt{(a/b)} = \sqrt{a} / \sqrt{b}$; $\sqrt{(a^2 b)} = a\sqrt{b}$. **Simplify by extracting perfect-square factors:** $\sqrt{50} = \sqrt{(25 \cdot 2)} = 5\sqrt{2}$; $\sqrt{72} = \sqrt{(36 \cdot 2)} = 6\sqrt{2}$; $\sqrt{245} = \sqrt{(49 \cdot 5)} = 7\sqrt{5}$. **Like surds** (same radicand) add like terms: $3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$; unlike surds ($3\sqrt{2} + 2\sqrt{3}$) canNOT merge - they just sit there. **Comparing surds:** square both sides (both positive, so order is preserved): which is bigger, $2\sqrt{5}$ or $3\sqrt{2}$? Squares give 20 vs 18, so $2\sqrt{5}$ wins. **Tiny example.** Order $\sqrt{7}$, $2\sqrt{2}$, $\sqrt{5}$: squares are 7, 8, 5, so $2\sqrt{2} > \sqrt{7} > \sqrt{5}$.

3.1.7 Rationalising the denominator

What is it? Rewriting a fraction so no surd remains in the denominator, using the difference-of-squares identity $(a + b)(a - b) = a^2 - b^2$. **Single surd:** multiply top and bottom by that surd: $1/\sqrt{7} = \sqrt{7}/7$. **Binomial denominator ($\sqrt{a} + \sqrt{b}$ or $a + \sqrt{b}$):** multiply by the *conjugate* (flip the middle sign): $1/(2 + \sqrt{3}) = (2 - \sqrt{3})/((2 + \sqrt{3})(2 - \sqrt{3})) = (2 - \sqrt{3})/(4 - 3) = 2 - \sqrt{3}$. The denominator collapses to a clean integer because $(a+b)(a-b) = a^2 - b^2$. **Triple-decker:** for $1/(\sqrt{2} + \sqrt{3} + \sqrt{5})$, group as $(\sqrt{2} + \sqrt{3}) + \sqrt{5}$, rationalise once, then regroup and rationalise again - two rounds of conjugates.

Why bother? Three reasons: standardized answers (everyone writes $2 - \sqrt{3}$, not $1/(2+\sqrt{3})$), easier further computation, and more accurate decimal work (dividing by 1.73... is numerically nastier than multiplying by 0.26...). **Tiny example.** Rationalise $6/(\sqrt{5} - \sqrt{3})$: multiply by $(\sqrt{5} + \sqrt{3})$: $= 6(\sqrt{5} + \sqrt{3})/(5 - 3) = 3(\sqrt{5} + \sqrt{3})$.

Figure 3.3: Density - between any two real numbers, rationals and irrationals are both endless; no 'next' number exists.

3.1.8 Laws of exponents

What is it? The five power rules for a real base $a > 0$ and integer exponents: (1) $a^m \cdot a^n = a^{m+n}$; (2) $a^m / a^n = a^{m-n}$; (3) $(a^m)^n = a^{mn}$; (4) $a^m \cdot b^m = (ab)^m$; (5) $a^0 = 1$, and $a^{-m} = 1/a^m$. Roots join the family: $a^{1/p} = \sqrt[p]{a}$, so $8^{2/3} = (8^{1/3})^2 = 2^2 = 4$. **When do I use them?** Simplifying expressions like $(3^5 \cdot 3^{-2})$, solving $2^x = 128$, or taming products like $(5^{1/2} \cdot 5^{1/3})$. **Tiny example.** $(2^{-3} \times 2^5) = 2^2 = 4$; $(3^0 + 4^0) \times 2^2 = 2 \times 4 = 8$; $64^{-1/3} = 1/4$.

> **Exponent equation strategy**
> See $2^x = 128$? Write both sides with the SAME base: $2^x = 2^7$, so $x = 7$. See $3^x \times 3^{x+1} = 243$? Left side is 3^{2x+1}, so $2x + 1 = 5$, $x = 2$. See $4^x = 8$? Convert: $(2^2)^x = 2^3$, so $2x = 3$, $x = 3/2$. One base to rule them all.

3.1.9 Exam tips for this chapter

1. Classify numbers by writing their DECIMAL EXPANSION in your head first - the expansion type decides the set.

2. For the $2^l \cdot 5^n$ test, reduce the fraction FIRST; the reduced denominator is what counts.
3. Recurring decimal questions: name the variable x, multiply by the correct power of 10, subtract. Show the subtraction line - it carries the marks.
4. Irrationality proofs: memorise the 4-step contradiction skeleton once, reuse it for √2, √3, √5 by changing the prime.
5. Simplify every surd to k√(radicand) form BEFORE adding or comparing.
6. Rationalising: the conjugate is your only tool; write the identity $(a+b)(a-b) = a^2 - b^2$ explicitly in the solution.
7. Exponent problems: force ONE base everywhere; then equate the exponents.
8. Never claim sums/products of irrationals are always irrational - give a counterexample if a true/false question asks.

3.2 Question Types / Concept Map for Chapter 3

Label the question type before you start - in this chapter, recognising 'this is a conjugate problem' or 'this is a same-base problem' IS the solution.

Type	Name	How to recognise it in the question	Core method
T1	Set classification	'Classify into N/Z/Q/I/R', 'true or false about number sets'	Decimal expansion type; nesting $N \subset Z \subset Q \subset R$
T2	Terminating test	'Will the decimal terminate, without division?'	Lowest terms; denominator = $2^m \times 5^n$ only
T3	Recurring → p/q	'Express 0.4777... in p/q form'	x, multiply by 10^k, subtract, solve; reduce
T4	Number line placement	'Represent √5 on the number line', successive magnification	Right-triangle construction; zoom by decimals
T5	Irrationality proofs	'Prove √3 is irrational'	Contradiction: assume p/q lowest terms, prime divides both
T6	Sum/product arguments	'Show √2 + √3 irrational'; true/false on operations	Square it / subtract-the-rational trick; counterexamples
T7	Surd simplification	'Simplify √72 + √32 - √18'	Extract perfect squares; collect like surds
T8	Surds comparison	'Which is larger: 2√5 or 3√2?'	Square both (both positive); compare radicand multiples
T9	Rationalising	'Rationalise 1/(2+√3)', triple surds	Multiply by conjugate; difference of squares; repeat if needed
T10	Exponent laws	'Simplify $(3^5 \times 3^{-2})/3^2$'	Five power rules; combine bases
T11	Exponent equations	'Solve $2^x = 128$', '$5^{x-2} = 125$'	Same base both sides; equate exponents

Type	Name	How to recognise it in the question	Core method
T12	Between-ness and density	'Find two irrationals between 2 and 3'	√ of non-squares in range; a + small irrational tails
T13	Applied / construction	Spiral of roots, geometric representations, real-life decimals	Pythagoras chaining; interpret expansions
T14	Multi-concept HOTS	Nested radicals, x + 1/√x chains, proof + computation mix	Break into labelled mini-steps, verify each junction

3.3 Speed Tricks & Trap Alarms

Ye table tumhara "dekho aur maaro" radar hai - question ka trigger phrase dikha, toh yahan ka trick fire karo. Inhe ratt lo, exam me sochna kam padega aur speed double ho jayegi.

Instant trick	Fire it when...
Terminating? REDUCE first, then check: denominator = $2^m \times 5^n$ only	"Without long division, state whether the decimal terminates".
Pure recurring 0.abab... = block/99...9. Mixed: (all digits - non-repeating)/99..900..0	Convert recurring decimals to p/q form.
Compare surds: same root index banao, then compare inside values	"Arrange in increasing order".
Rationalise: multiply by the conjugate. Telescoping sum collapses to (last - first)	Surds in denominators; 1/(√a+√b) chains.
Same base first: $8 = 2^3$, $125 = 5^3$, $1/81 = 3^{-4}$ → equate exponents	Every exponent equation.
Irrationality skeleton: assume p/q → square → prime lemma → contradiction	"Prove √2 / √3 / √5 is irrational".
rational ± irrational = irrational; irrational × non-zero rational = irrational	True/false closure statements.

3.4 The 120-Question Workout

Poora chapter ab ek hi ring mein: **120 questions**, chaar level zones mein banta hua. Warm-up se shuru karo, Boss Zone tak pahuncho. Har question ke turant baad boxed step-by-step solution hai - pehle khud try karo, atako toh solution dekho, aur "Why this step?" + Hinglish note se logic pakka karo. Agar in 120 ko solve kar liya toh is chapter ka koi bhi twisted question tumhe nahi rok payega.

Level 1 · Warm-up Zone: basics and confidence:

QUESTION 1 | Basic | Set classification (T1)

Q1. Classify each number into as many of the sets N, Z, Q, I, R as it belongs to: (i) -7, (ii) 0, (iii) 2/3, (iv) √16, (v) √15, (vi) 0.585858... (the pair 58 repeating).

SOLUTION — step by step

Step 1. Given: six numbers. Needed: set membership for each, using the nesting $N \subset Z \subset Q \subset R$ and $I \cap Q = \emptyset$.

Step 2. -7: negative integer, so Z and Q and R; not N (naturals start at 1); not I.

Step 3. 0: an integer, so Z, Q, R; not N in this book's convention (some books include 0 in N - state your convention).

Step 4. 2/3: a ratio of integers, so Q and R; not N, not Z, not I.

Step 5. √16 = 4: a perfect square under the root, so N, Z, Q, R. Perfect squares are never irrational.

Step 6. √15: 15 is not a perfect square, so √15 is irrational: I and R only.

Step 7. 0.585858... is a recurring decimal, and every recurring decimal is a rational: it equals 58/99. So Q and R.

» **Why this step?** The decimal expansion is the fingerprint: terminating or recurring means rational; endless without a block means irrational.

Answer: (i) Z, Q, R (ii) Z, Q, R (0 ∉ N here) (iii) Q, R (iv) N, Z, Q, R (v) I, R (vi) Q, R (= 58/99)

Hinglish: *Yaad rakho - decimal expansion hi asli ID card hai. Recurring decimal (58-58-58 chalne wala) hamesha rational hota hai, yahan 58/99. √16 jaisi cheez ko dekh ke confuse mat hona - andar perfect square hai toh answer 4 hai, natural number. √15 tabhi irrational hai jab andar non-square ho.*

QUESTION 2 | Basic | Terminating test (T2)

Q2. Without performing long division, decide whether each decimal expansion terminates: (i) 17/8, (ii) 13/3125, (iii) 7/12, (iv) 64/455, (v) 6/15.

SOLUTION — step by step

Step 1. Given: five fractions. Needed: terminating or not, via the $2^m \times 5^n$ test on the REDUCED denominator.

Step 2. (i) 17/8: $8 = 2^3 = 2^3 \times 5^0$. Only 2s, so it terminates (in fact 2.125).

Step 3. (ii) 13/3125: $3125 = 5^5$. Only 5s, so it terminates.

Step 4. (iii) 7/12: $12 = 2^2 \times 3$. The stray 3 can never be absorbed into a power of 10, so it recurs.

Step 5. (iv) 64/455: $455 = 5 \times 7 \times 13$. The primes 7 and 13 force a recurring decimal.

Step 6. (v) 6/15: reduce first! 6/15 = 2/5, and $5 = 5^1$, so it terminates (0.4). Testing $15 = 3 \times 5$ without reducing would give the wrong verdict.

» **Why this step?** 10 = 2 × 5, so a fraction terminates exactly when its reduced denominator divides some power of 10 - only primes 2 and 5 can appear.

Answer: (i) terminates (ii) terminates (iii) recurs (iv) recurs (v) terminates (6/15 = 2/5 = 0.4)

Hinglish: *Sabse pehla kaam: fraction ko lowest terms mein lao. Phir denominator ko prime factors mein todo. Sirf 2 aur 5 dikh rahe hain? Decimal ruk jayega. Koi aur prime (3, 7, 13...) dikh gaya? Chalti-firti decimal aayegi. (v) wala question isliye diya hai kyunki students 15 dekh ke turant 'recur' bol dete hain - pehle 3 cancel karo!*

QUESTION 3 | Basic | Pure recurring to p/q (T3)

Q3. Express in p/q form: (i) 0.777... (the 7 repeating), (ii) 0.373737... (the block 37 repeating), (iii) 0.090909... (the block 09 repeating).

SOLUTION — step by step

Step 1. Given: three pure recurring decimals (repeat starts immediately). Needed: exact fractions.

Step 2. (i) Let x = 0.777... . Multiply by 10 (one repeating digit → shift by one place): 10x = 7.777... .

Step 3. Subtract: 10x - x = 7.777... - 0.777... = 7. So 9x = 7 and x = 7/9.

Step 4. (ii) Let x = 0.373737... . Two repeating digits → multiply by 100: 100x = 37.3737... .

Step 5. Subtract: 100x - x = 37. So 99x = 37 and x = 37/99.

Step 6. (iii) Let x = 0.090909... . Two repeating digits → 100x = 9.0909... . Subtract: 99x = 9, x = 9/99 = 1/11.

» **Why this step?** Multiplying by 10^{ϵ} lines up two identical infinite tails; subtracting cancels them and leaves a finite equation. k repeating digits need a shift of 10^{ϵ}.

Answer: (i) 7/9 (ii) 37/99 (iii) 1/11

Hinglish: *Rule simple hai: jitne digit repeat ho rahe hain, utne 9 denominator mein. Ek digit repeat = 9, do digit = 99. 0.777... = 7/9, 0.3737... = 37/99. Bas repeating block ke digits upar likh do aur neeche utne hi 9. (iii) mein 9/99 ko 1/11 tak reduce karna mat bhoolna.*

QUESTION 4 | Basic | Mixed recurring to p/q (T3)

Q4. Express in p/q form: (i) 0.4777... (the 4 does not repeat, the 7 repeats), (ii) 0.2333... (the 2 does not repeat, the 3 repeats), (iii) 1.272727... (the block 27 repeats after the integer part).

SOLUTION — step by step

Step 1. Given: mixed recurring decimals. Needed: exact fractions using the shift-and-subtract method.

Step 2. (i) x = 0.4777... . One non-repeating digit then one repeating digit: form 10x = 4.777... and 100x = 47.777... .

Step 3. Subtract: 100x - 10x = 47.777... - 4.777... = 43. So 90x = 43, x = 43/90 (already lowest terms).

Step 4. (ii) x = 0.2333... . 10x = 2.333..., 100x = 23.333... . Subtract: 90x = 21, x = 21/90 = 7/30.

Step 5. (iii) x = 1.272727... . Two repeating digits: 100x = 127.2727... . Subtract x: 99x = 126, x = 126/99 = 14/11.

Step 6. Check (iii): 14/11 = 1.272727... ✓. Check (i): 43 ÷ 90 = 0.4777... ✓.

» **Why this step?** Two shifts are needed so that BOTH copies of the decimal have the same non-repeating head; then the subtraction kills head and tail together.

Answer: (i) 43/90 (ii) 7/30 (iii) 14/11

Hinglish: *Mixed recurring mein trick: pehle non-repeating digits ko 'head' samjho. Do copies banao jisme head same position par ho (10x aur 100x yahan), minus karo - head bhi tail bhi cancel, jo bacha wo equation. Formula version: (poore digits - head)/(utne 9 jitne repeating, utne 0 jitne non-repeating). (i): (47-4)/90 = 43/90. (ii) mein 21/90 ko 7/30 karna zaroori hai.*

QUESTION 5 | Basic | True/False on number sets (T1)

Q5. State true or false with a one-line reason: (i) Every natural number is a whole number. (ii) Every integer is a whole number. (iii) Every rational number is a real number. (iv) Every real number is rational. (v) √3 is a rational number. (vi) There is no rational number between 1/5 and 1/4.

SOLUTION — step by step

Step 1. Given: six statements. Needed: verdicts grounded in the set nesting.

Step 2. (i) True: N = 1, 2, 3, ... sits inside whole numbers W = 0, 1, 2, ...

Step 3. (ii) False: -3 is an integer but not a whole number (whole numbers are the non-negative integers).

Step 4. (iii) True: Q ⊂ R by the nesting of number sets.

Step 5. (iv) False: √2 is real but irrational.

Step 6. (v) False: 3 is not a perfect square; the contradiction proof applies, so √3 is irrational.

Step 7. (vi) False: rationals are dense - (1/5 + 1/4)/2 = 9/40 lies between them, and the game continues forever.

» **Why this step?** Each verdict comes from a definition or a counterexample - one counterexample is a complete proof of falsity.

Answer: (i) T (ii) F (-3) (iii) T (iv) F (√2) (v) F (3 is not a perfect square) (vi) F (9/40 lies between)

Hinglish: *True/False mein sirf 'True' ya 'False' likhne se marks nahi milte - reason chahiye. False ke liye ek counterexample kaafi hai: -3 se (ii) turant false. (vi) wala bahut common galti hai - rationals ke beech hamesha aur rationals milte hain, average nikaal do, ho gaya.*

QUESTION 6 | Basic | Number line construction (T4)

Q6. Describe, step by step, how to locate $\sqrt{5}$ on the number line using a compass and straight edge.

SOLUTION — step by step

Step 1. Given: the number line with origin O. Needed: a segment of length $\sqrt{5}$ laid along the line.

Step 2. Find a right triangle whose hypotenuse is $\sqrt{5}$: since $5 = 4 + 1 = 2^2 + 1^2$, legs of 2 and 1 give hypotenuse $\sqrt{5}$.

Step 3. Mark A at 2 on the number line. At A, erect a perpendicular to the line (standard construction).

Step 4. On the perpendicular, mark B with AB = 1 unit.

Step 5. Join O to B. By the Baudhayana-Pythagoras theorem, $OB = \sqrt{(2^2 + 1^2)} = \sqrt{5}$.

Step 6. With centre O and radius OB, draw an arc cutting the number line at P. Then $OP = \sqrt{5}$, so P represents $\sqrt{5}$ (it falls between 2.2 and 2.3, since $2.23^2 \approx 4.97$ and $2.24^2 \approx 5.02$).

» **Why this step?** The arc transfers the slanted hypotenuse length onto the straight line while preserving it - swinging a circle keeps radius constant.

Answer: A right triangle with legs 2 (along the line) and 1 (perpendicular) has hypotenuse $\sqrt{5}$; an arc of radius $\sqrt{5}$ centred at O marks the point.

Hinglish: *Construction ka funda: $\sqrt{5}$ ko $2^2 + 1^2$ ke roop mein dekho. 2 unit aage jao (A), wahan se 1 unit upar (B), OB join karo - ye $\sqrt{5}$ hai. Ab compass ko O par kholo, OB radius lo, arc maaro line par. Jo point mile wahi $\sqrt{5}$. Har $\sqrt{n}$ ke liye aise hi legs dhoondo: $\sqrt{10} = 3^2+1^2$, $\sqrt{13} = 3^2+2^2$.*

QUESTION 7 | Basic | Successive magnification (T4)

Q7. Given that $\sqrt{3} = 1.7320508...$, describe the location of $\sqrt{3}$ under two rounds of successive magnification: first between which two integers, then between which two consecutive tenths, then between which two consecutive hundredths.

SOLUTION — step by step

Step 1. Given: the decimal expansion of $\sqrt{3}$. Needed: a nested sequence of intervals.

Step 2. Round 1 (integers): $1 < 1.732... < 2$, so $\sqrt{3}$ lies between 1 and 2.

Step 3. Round 2 (tenths): $1.7 < 1.732... < 1.8$, so it lies between 1.7 and 1.8.

Step 4. Round 3 (hundredths): $1.73 < 1.732... < 1.74$, so it lies between 1.73 and 1.74.

Step 5. Each zoom peels one decimal digit: the intervals shrink by a factor of ten every round.

» **Why this step?** Each decimal digit fixes the interval at the next scale - this is exactly what 'locating a number on the line' means numerically.

Answer: Between 1 and 2; between 1.7 and 1.8; between 1.73 and 1.74.

Hinglish: *Zoom karne ka khel hai: 1 aur 2 ke beech → 1.7 aur 1.8 ke beech → 1.73 aur 1.74 ke beech. Har digit ek aur darwaza kholta hai. Exam mein ye 3-mark favourite hai - bas expansion padho aur intervals banao. √3 ke liye aage aur zoom: 1.732 se 1.733 (thousandths).*

QUESTION 8 | Basic | Surd simplification (T7)

Q8. Simplify into k√( ) form: (i) √72, (ii) √245, (iii) √175, (iv) √(147).

SOLUTION — step by step

Step 1. Given: four radicals. Needed: extract every perfect-square factor.

Step 2. (i) 72 = 36 × 2, so √72 = √36 · √2 = 6√2.

Step 3. (ii) 245 = 49 × 5, so √245 = 7√5.

Step 4. (iii) 175 = 25 × 7, so √175 = 5√7.

Step 5. (iv) 147 = 49 × 3, so √147 = 7√3.

» **Why this step?** √(ab) = √a√b lets the perfect-square factor walk out of the root; what remains inside must be square-free.

Answer: (i) 6√2 (ii) 7√5 (iii) 5√7 (iv) 7√3

Hinglish: *Andar sabse bada perfect square dhoondo aur bahar nikalo. 72 = 36×2, 36 ka root 6, toh 6√2. Yaad rakho: 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144 - ye aapke dost hain. Jitna bada square nikaloge, utna chhota root; 72 ko 4×18 karke √18 ko phir simplify karna double kaam hai.*

QUESTION 9 | Basic | Density / between-ness (T12)

Q9. Insert three irrational numbers between 2 and 3, and verify each choice.

SOLUTION — step by step

Step 1. Given: the interval (2, 3). Needed: three irrationals inside it.

Step 2. Idea: the square roots of the non-perfect squares between 4 and 9 all land in (2, 3).

Step 3. Candidates: √5, √6, √7, √8 (4 < 5, 6, 7, 8 < 9).

Step 4. Verify: 4 < 5 < 9 ⇒ 2 < √5 < 3 ✓; similarly for 6, 7, 8 ✓.

Step 5. Each of √5, √6, √7 is irrational because 5, 6, 7 are not perfect squares.

» **Why this step?** Squaring is order-preserving for positive numbers, so 'inside (4, 9)' translates exactly to 'root inside (2, 3)'.

Answer: For example √5, √6, √7 (many answers possible; each is irrational and satisfies 2 < root < 3).

Hinglish: *Do numbers ke beech irrationals chahiye? Dono ko square karo, beech ke non-squares pakdo, root laga do. 2 aur 3 ke beech: 5, 6, 7, 8 - in sabke roots beech mein hi rahenge. √5, √6, √7 - done! Yehi trick decimals ke saath bhi chalti hai: 2.101001000... type numbers bhi kaam karte hain.*

QUESTION 10 | Basic | Exponent laws (T10)

Q10. Simplify: (i) $3^5 \times 3^{-2}$, (ii) $(2^2)^3$, (iii) $7^0 + 5^0$, (iv) $2^{-3} \times 2^4$, (v) $(3^{-2} \times 3^5)/3^2$.

SOLUTION — step by step

Step 1. Given: five exponent expressions. Needed: apply the five power laws.

Step 2. (i) Same base multiply → add exponents: $3^5 \times 3^{-2} = 3^3 = 27$.

Step 3. (ii) Power of a power → multiply exponents: $(2^2)^3 = 2^6 = 64$.

Step 4. (iii) $a^0 = 1$ for $a \neq 0$: $7^0 + 5^0 = 1 + 1 = 2$.

Step 5. (iv) $2^{-3} \times 2^4 = 2^1 = 2$.

Step 6. (v) $3^{-2} \times 3^5 = 3^3$, then $3^3/3^2 = 3^1 = 3$.

» **Why this step?** All five laws are consequences of repeated multiplication; same base is the trigger to look for.

Answer: (i) 27 (ii) 64 (iii) 2 (iv) 2 (v) 3

Hinglish: *Exponent laws ka mantra: multiply matlab exponents ADD, power of power matlab MULTIPLY, zero exponent matlab 1, negative exponent matlab ulta (reciprocal). Base same hona chahiye tabhi laws lagti hain - $3^5 \times 2^5$ mein base alag hai, wahan $(3\times2)^5$ wala rule lagta hai.*

QUESTION 11 | Basic | Set classification (T1)

Q11. Classify each number as rational or irrational (and pick out the integers): 0.35, √25, √26, 0.101001000100001..., 4/7, -9.

SOLUTION — step by step

Step 1. 0.35 = 35/100 -> rational (terminating decimal).

Step 2. √25 = 5 -> rational, AND an integer.

Step 3. √26: 26 is not a perfect square -> irrational.

Step 4. 0.101001000100001... : a non-repeating, non-terminating pattern -> irrational.

Step 5. 4/7 -> rational (it is a ratio of integers).

Step 6. -9 -> rational, AND an integer.

» **Why this step?** Rational = writable as p/q. Perfect squares are rational; other roots are not; invented non-repeating decimals never are.

Answer: Rational: 0.35, √25 (=5), 4/7, -9. Irrational: √26, 0.101001000... Integers: 5, -9.

Hinglish: *p/q mein likh sakte ho? Rational. Perfect square ka root? Waise bhi rational. √26 aur wahi 101001... wala dhachka - irrational. Integers: 5 aur -9.*

QUESTION 12 | Basic | Terminating test (T2)

Q12. Without long division, decide whether each decimal terminates: 7/40, 5/12, 13/80, 22/7.

SOLUTION — step by step

Step 1. Rule: REDUCE the fraction, then check the denominator - only $2^m \times 5^n$ factors allow termination.

Step 2. 7/40: $40 = 2^3 \times 5$ -> terminates.

Step 3. 5/12: $12 = 2^2 \times 3$ -> the 3 spoils it -> non-terminating (recurring).

Step 4. 13/80: $80 = 2^4 \times 5$ -> terminates.

Step 5. 22/7: 7 is neither 2 nor 5 -> non-terminating.

» **Why this step?** Base 10 = 2 × 5, so ONLY denominators built from 2s and 5s can ever be scaled into a power of ten.

Answer: 7/40 and 13/80 terminate; 5/12 and 22/7 do not

Hinglish: *Denominator ka Aadhaar card check karo: sirf 2 aur 5 ke factors chahiye. 40, 80 pass; 12 (3 wala) aur 7 fail. Long division ki zaroorat hi nahi!*

QUESTION 13 | Basic | Pure recurring to p/q (T3)

Q13. Express in p/q form: (i) 0.555..., (ii) 0.727272..., (iii) 0.999...

SOLUTION — step by step

Step 1. (i) x = 0.555... -> 10x = 5.555... -> 9x = 5 -> x = 5/9.

Step 2. (ii) x = 0.727272... (block 72) -> 100x = 72.7272... -> 99x = 72 -> x = 72/99 = 8/11.

Step 3. (iii) x = 0.999... -> 10x = 9.999... -> 9x = 9 -> x = 1. So 0.999... IS exactly 1.

» **Why this step?** A block of k repeating digits is divided by k nines - and the 0.999... = 1 identity shows decimals can have two names.

Answer: (i) 5/9 (ii) 8/11 (iii) 1

Hinglish: *Repeating block jitne digit ka, utne 9 neeche: 5/9, 72/99 = 8/11. Aur 0.999... = 1 wala sach duniya ka sabse famous decimal shock hai - subtraction trick se proof ho gaya.*

QUESTION 14 | Basic | Mixed recurring to p/q (T3)

Q14. Express in p/q form: (i) 0.2333... (the 3 repeating), (ii) 1.272727... (the block 27 repeating).

SOLUTION — step by step

Step 1. (i) x = 0.2333...; 10x = 2.333...; 100x = 23.33...; subtract: 90x = 21 -> x = 21/90 = 7/30.

Step 2. (ii) x = 1.272727...; the fractional part is 0.272727... = 27/99 = 3/11.

Step 3. So 1.272727... = 1 + 3/11 = 14/11.

» **Why this step?** Mixed recurring: shift past the non-repeating part first, then subtract the two shifted copies - non-repeating digits cancel.

Answer: (i) 7/30 (ii) 14/11

Hinglish: *Pehle 10x se non-repeating 2 ko aage karo, phir 100x - 10x se repeating ka kaam ho gaya: 21/90 = 7/30. Doosra: integer alag rakho, decimal 27/99 = 3/11, total 14/11.*

QUESTION 15 | Basic | Surd simplification (T7)

Q15. Simplify into k√( ) form: (i) √50, (ii) √108, (iii) √245, (iv) 3√12 + 5√3.

SOLUTION — step by step

Step 1. (i) 50 = 25 × 2 -> √50 = 5√2.

Step 2. (ii) 108 = 36 × 3 -> √108 = 6√3.

Step 3. (iii) 245 = 49 × 5 -> √245 = 7√5.

Step 4. (iv) 3√12 = 3 · 2√3 = 6√3; so 6√3 + 5√3 = 11√3 (like surds add like apples).

» **Why this step?** Pull out the largest square factor; then only LIKE surds (same radicand) can be combined.

Answer: (i) 5√2 (ii) 6√3 (iii) 7√5 (iv) 11√3

Hinglish: *Square factor bahar nikalo: 25, 36, 49 - inhi ke upar sab khada hai. Aur jodna sirf same radicand walo ka: 6√3 + 5√3 = 11√3, bilkul apples jaise.*

QUESTION 16 | Basic | Exponent laws (T10)

Q16. Simplify: (i) $2^5 \times 2^{-3}$, (ii) $(3^2)^3$, (iii) $5^7/5^5$, (iv) $7^0 + 7^{-1}$.

SOLUTION — step by step

Step 1. (i) Same base multiply -> add exponents: $2^{5+(-3)} = 2^2 = 4$.

Step 2. (ii) Power of a power -> multiply: $3^6 = 729$.

Step 3. (iii) Divide -> subtract: $5^2 = 25$.

Step 4. (iv) $7^0 = 1$ and $7^{-1} = 1/7$ -> sum 8/7.

» **Why this step?** Three laws carry the whole chapter: add on multiply, subtract on divide, multiply on power-of-power.

Answer: (i) 4 (ii) 729 (iii) 25 (iv) 8/7

Hinglish: *Multiply -> power jodo; divide -> ghatyo; $(a^m)^n$ -> guna. $7^0 = 1$ hamesha. Ye teen laws hi exponents ka poora constitution hai.*

QUESTION 17 | Basic | Negative exponents (T10)

Q17. Evaluate: (i) 3^{-2}, (ii) $(2/3)^{-2}$, (iii) $5^0 \times 2^{-3}$.

SOLUTION — step by step

Step 1. (i) $3^{-2} = 1/3^2 = 1/9$.

Step 2. (ii) Flip the fraction: $(2/3)^{-2} = (3/2)^2 = 9/4$.

Step 3. (iii) $5^0 = 1$; $2^{-3} = 1/8$; product = 1/8.

» **Why this step?** A negative exponent means RECIPROCAL, not negative value - $(2/3)^{-2}$ grows to 9/4.

Answer: (i) 1/9 (ii) 9/4 (iii) 1/8

Hinglish: *Minus power = ulta karo, value negative NAHI hoti. $(2/3)^{-2} = 9/4$ - number bada ho gaya! Fraction flip + square, done.*

QUESTION 18 | Basic | Rational vs irrational sort (T1)

Q18. From the list $\sqrt{16}$, $\sqrt{20}$, 0.101, π, 22/7, 0.030303..., pick out (i) all irrational numbers, (ii) all integers.

SOLUTION — step by step

Step 1. $\sqrt{16} = 4$ -> rational, integer.

Step 2. $\sqrt{20} = 2\sqrt{5}$ -> 20 is not a perfect square -> irrational.

Step 3. 0.101 terminates -> rational.

Step 4. π is famously non-repeating, non-terminating -> irrational.

Step 5. 22/7 is a ratio -> rational (even though it only approximates π!).

Step 6. 0.030303... = 3/99 = 1/33 -> rational.

Step 7. Integers: just 4.

» **Why this step?** 22/7 is the great impostor - rational, but NOT π. And 0.030303... hides as 1/33.

Answer: (i) Irrational: $\sqrt{20}$, π (ii) Integer: 4 (= $\sqrt{16}$)

Hinglish: *22/7 ko π samajhna national galti hai - wo toh rational hai! $\sqrt{20} = 2\sqrt{5}$ irrational, 0.0303... = 1/33. Integer sirf 4 ($\sqrt{16}$).*

QUESTION 19 | Basic | Density / between-ness (T12)

Q19. Insert two rational numbers between 1/3 and 1/2.

SOLUTION — step by step

Step 1. Common denominator 24: 1/3 = 8/24, 1/2 = 12/24.

Step 2. Pick any two fractions between 8/24 and 12/24: 9/24 = 3/8 and 10/24 = 5/12.

Step 3. Both are rational and strictly between the originals: 1/3 < 3/8 < 5/12 < 1/2 ✓.

» **Why this step?** Equal-denominator rewriting turns 'between' into an ordinary number-line walk.

Answer: 3/8 and 5/12 (other answers possible)

Hinglish: *Denominator same karo (24), phir beech ke numbers utha lo: 9/24 aur 10/24. Rationals har jagah bikhre hain - jitne chahiye utne milenge.*

QUESTION 20 | Basic | Number line construction (T4)

Q20. Describe step by step how to locate √5 on the number line using a ruler and compass.

SOLUTION — step by step

Step 1. Build a right triangle with legs 2 and 1: hypotenuse = $\sqrt{(2^2 + 1^2)} = \sqrt{5}$.

Step 2. On the line, mark O and the point 2 (call it A); at A erect a perpendicular of length 1 (call the top B).

Step 3. Then OB = √5 by Pythagoras.

Step 4. With compass centred at O, radius OB, draw an arc down to the number line - the landing point IS √5.

» **Why this step?** Pythagoras converts any surd into a buildable length: legs (n, 1) give $\sqrt{(n^2 + 1)}$.

Answer: Right triangle legs 2 and 1 -> hypotenuse √5 -> compass arc onto the line

Hinglish: *Triangle hi asli tool hai: 2 aur 1 ke legs se √5 ka hypotenuse. Compass se arc gira do - number line par √5 ghar baith gaya. Root = ek triangle, hamesha.*

QUESTION 21 | Basic | Same-base equations (T11)

Q21. Solve: (i) $2^x = 128$, (ii) $5^x = 1/625$, (iii) $10^x = 0.001$.

SOLUTION — step by step

Step 1. (i) $128 = 2^7$ -> x = 7.

Step 2. (ii) $1/625 = 1/5^4 = 5^{-4}$ -> x = -4.

Step 3. (iii) $0.001 = 1/1000 = 10^{-3}$ -> x = -3.

» **Why this step?** Reduce both sides to the SAME base; then exponents must match.

Answer: (i) 7 (ii) -4 (iii) -3

Hinglish: *Dono side same base banao: $128 = 2^7$, $625 = 5^4$ (ulta = minus), $0.001 = 10^{-3}$. Base match -> exponent match. Itna hi hai.*

QUESTION 22 | Basic | Closure true/false (T6)

Q22. True or False with a one-line reason: (i) rational + rational is always rational. (ii) irrational + irrational is always irrational. (iii) a non-zero rational times an irrational is irrational. (iv) every integer is rational. (v) 0 is irrational.

SOLUTION — step by step

Step 1. (i) TRUE: p/q + r/s = (ps + rq)/qs - still a ratio.

Step 2. (ii) FALSE: $\sqrt{2} + (2 - \sqrt{2}) = 2$, a rational - two irrationals can cancel their surd parts.

Step 3. (iii) TRUE: if (p/q)·x were rational, then x = rational ÷ (p/q) would be rational - contradiction.

Step 4. (iv) TRUE: any integer n = n/1.

Step 5. (v) FALSE: 0 = 0/1 is rational.

» **Why this step?** The one-way closures are safe (rational ops with rational); the all-irrational claims fail the moment surds can cancel.

Answer: (i) T (ii) F (iii) T (iv) T (v) F

Hinglish: *(ii) wala dhoka: $\sqrt{2} + (2 - \sqrt{2}) = 2$! Do irrational milke rational ban sakte hain. Baaki sab closure ke standard rules - (iii) contradiction se prove hota hai.*

QUESTION 23 | Basic | Comparing surds (T8)

Q23. Which is larger: (i) $3\sqrt{2}$ or $2\sqrt{3}$, (ii) $4\sqrt{5}$ or $5\sqrt{2}$?

SOLUTION — step by step

Step 1. Bring everything under one root (square the coefficients in).

Step 2. (i) $3\sqrt{2} = \sqrt{18}$; $2\sqrt{3} = \sqrt{12}$ -> $\sqrt{18}$ wins: $3\sqrt{2} > 2\sqrt{3}$.

Step 3. (ii) $4\sqrt{5} = \sqrt{80}$; $5\sqrt{2} = \sqrt{50}$ -> $4\sqrt{5} > 5\sqrt{2}$.

» **Why this step?** $k\sqrt{n} = \sqrt{(k^2 n)}$: pushing the coefficient inside makes comparison a plain radicand contest.

Answer: (i) $3\sqrt{2}$ (ii) $4\sqrt{5}$

Hinglish: *Coefficient ko andar ghusao: $3\sqrt{2} = \sqrt{18}$ vs $2\sqrt{3} = \sqrt{12}$. Ab root ke andar ki ladai hai - $18 > 12$. Compare = same format lao.*

QUESTION 24 | Basic | Rationalising single surds (T9)

Q24. Rationalise the denominator: (i) $6/\sqrt{3}$, (ii) $9/\sqrt{5}$, (iii) $1/\sqrt{7}$.

SOLUTION — step by step

Step 1. (i) Multiply by $\sqrt{3}/\sqrt{3}$: $6\sqrt{3}/3 = 2\sqrt{3}$.

Step 2. (ii) $9\sqrt{5}/5$ - already simplest (the 5 stays below).

Step 3. (iii) $\sqrt{7}/7$.

» **Why this step?** Multiply by $\sqrt{n}/\sqrt{n}$: the denominator becomes n (rational) and the numerator collects the surd.

Answer: (i) $2\sqrt{3}$ (ii) $9\sqrt{5}/5$ (iii) $\sqrt{7}/7$

Hinglish: *$\sqrt{n}/\sqrt{n}$ se multiply karo - neeche root khatam. $6/\sqrt{3} = 2\sqrt{3}$ (yahan toh aur simplify ho gaya). Form hi badalna hai, value nahi.*

QUESTION 25 | Basic | Conjugate products (T9)

Q25. Evaluate: (i) $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})$, (ii) $(\sqrt{7})^2 + (\sqrt{2})^2$, (iii) $(\sqrt{6})^4$.

SOLUTION — step by step

Step 1. (i) Difference of squares: $(\sqrt{5})^2 - (\sqrt{3})^2 = 5 - 3 = 2$.

Step 2. (ii) $7 + 2 = 9$.

Step 3. (iii) $(\sqrt{6})^4 = (6)^2 = 36$ (square twice, or $(\sqrt{6})^2 = 6$ then square).

» **Why this step?** $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$: conjugates annihilate both roots at once.

Answer: (i) 2 (ii) 9 (iii) 36

Hinglish: *Conjugate pair = difference of squares ka jaadu: 5 - 3 = 2. Root square hote hi seedha number. $(\sqrt{6})^4$ = 36 - power ko 2 ke tukdon mein todo.*

QUESTION 26 | Basic | Successive magnification (T4)

Q26. Given $\sqrt{3} \approx 1.7320508...$, state the two integers between which $\sqrt{3}$ lies, then the pair of tenths, then the pair of hundredths that trap it.

SOLUTION — step by step

Step 1. Integers: $1 < 1.732... < 2$.

Step 2. Tenths: $1.7 < 1.732... < 1.8$.

Step 3. Hundredths: $1.73 < 1.732... < 1.74$.

» **Why this step?** Each magnification level slices the gap ten times smaller - the number gets trapped in a shrinking cage.

Answer: 1 and 2; 1.7 and 1.8; 1.73 and 1.74

Hinglish: *Magnification = cage sikod do: 1 aur 2, phir 1.7-1.8, phir 1.73-1.74. Irrational ko kabhi pakad nahi sakte, par kitne kareeb se dekh sakte hain - yehi maza hai.*

QUESTION 27 | Basic | Decimal types from fractions (T2)

Q27. Convert and classify each decimal as terminating, pure recurring, or mixed recurring: 3/8, 7/9, 1/6, 5/12.

SOLUTION — step by step

Step 1. $3/8 = 0.375$ -> terminating ($8 = 2^3$).

Step 2. $7/9 = 0.777...$ -> pure recurring (single digit block).

Step 3. $1/6 = 0.1666...$ -> mixed recurring (one quiet digit, then the block 6).

Step 4. $5/12 = 0.41666...$ -> mixed recurring (block 6 after 41).

» **Why this step?** Denominator 2s/5s only -> terminate; any other prime -> recurring; a non-repeating prefix creates the mixed type.

Answer: 0.375 terminating; 0.777... pure recurring; 0.1666... and 0.41666... mixed recurring

Hinglish: *8 wala ruk gaya, 9 wala shuru se phir raha, 6 aur 12 wale pehle thoda chal ke phir repeat hue. Denominator hi decimal ka naksha banata hai.*

QUESTION 28 | Basic | Powers of ten (T10)

Q28. Evaluate: (i) $10^4 \times 10^{-6}$, (ii) $(10^{-3})^2$, (iii) $10^5/10^2 \times 10^{-1}$.

SOLUTION — step by step

Step 1. (i) $10^{4-6} = 10^{-2} = 0.01$.

Step 2. (ii) $10^{-6} = 0.000001$.

Step 3. (iii) $10^{5-2} \times 10^{-1} = 10^3 \times 10^{-1} = 10^2 = 100$.

» **Why this step?** With base 10 the exponent IS the decimal place count - negative means places to the right of the point.

Answer: (i) 0.01 (ii) 0.000001 (iii) 100

Hinglish: *Base 10 = decimal counter: -2 matlab point ke baad 2 jagah (0.01). Laws same rehte hain - jodo, ghatyo, guna karo.*

QUESTION 29 | Basic | Mixed exponent evaluation (T10)

Q29. Evaluate: (i) $(2^0 + 3^0) \times 4^{-1}$, (ii) $(1/2)^{-3}$, (iii) $(1/3)^{-2} - (1/2)^{-2}$.

SOLUTION — step by step

Step 1. (i) $(1 + 1) \times 1/4 = 2 \times 1/4 = 1/2$.

Step 2. (ii) $(1/2)^{-3} = 2^3 = 8$.

Step 3. (iii) $3^2 - 2^2 = 9 - 4 = 5$.

» **Why this step?** Bracket first, flip negatives, then plain arithmetic - three moves in order.

Answer: (i) 1/2 (ii) 8 (iii) 5

Hinglish: *Zero powers 1 de dete hain, negative powers flip karte hain. $(1/2)^{-3} = 8$ - fraction ka flip bada number deta hai. (iii) 9 - 4 = 5, clean.*

QUESTION 30 | Basic | Collecting like surds (T7)

Q30. Simplify completely: $2\sqrt{3} + \sqrt{27} - \sqrt{12}$.

SOLUTION — step by step

Step 1. $\sqrt{27} = \sqrt{(9 \times 3)} = 3\sqrt{3}$; $\sqrt{12} = \sqrt{(4 \times 3)} = 2\sqrt{3}$.

Step 2. Expression = $2\sqrt{3} + 3\sqrt{3} - 2\sqrt{3}$.

Step 3. = $3\sqrt{3}$.

» **Why this step?** Reduce every surd to coefficient × $\sqrt{3}$ first; then they are all 'apples' and cancel freely.

Answer: $3\sqrt{3}$

Hinglish: *Sabko √3 ke naam pe lao: 2√3, 3√3, 2√3. Ab apples ki ginti: 2 + 3 - 2 = 3√3. Simplify karne ke baad hi jodna.*

Level 2 · Core Drills: book-level practice:

QUESTION 31 | Moderate | Irrationality proof (T5)

Q31. Prove that √3 is irrational.

SOLUTION — step by step

Step 1. Given: a claim to contradict. Assume $\sqrt{3} = p/q$ where p, q are integers, $q \neq 0$, and the fraction is in lowest terms (p and q share no common prime).

Step 2. Squaring both sides: $3 = p^2/q^2$, so $p^2 = 3q^2$.

Step 3. Then 3 divides p^2. Since 3 is prime, 3 divides p itself (a prime dividing a square must divide the number). Write $p = 3r$.

Step 4. Substitute: $(3r)^2 = 3q^2 \Rightarrow 9r^2 = 3q^2 \Rightarrow q^2 = 3r^2$.

Step 5. So 3 divides q^2, hence 3 divides q.

Step 6. Now 3 divides both p and q - contradicting lowest terms. The assumption is false, therefore √3 is irrational. ∎

» **Why this step?** The contradiction must land on the EXACT assumption made (coprimality); that is why the final sentence names lowest terms.

Answer: √3 is irrational (by contradiction; 3 divides both p and q, contradicting lowest terms).

Hinglish: *Proof ka 4-step skeleton yaad karo: (1) assume p/q lowest terms, (2) square karo - prime upar aa jata hai, (3) prime dono (p aur q) ko divide karta nikla, (4) lowest terms se contradiction - done! √2 mein prime 2 tha, yahan 3 hai, √5 mein 5 hoga. Bas prime badlo, skeleton same.*

QUESTION 32 | Moderate | Rational ± irrational (T6)

Q32. Prove that the sum of a rational number r and an irrational number x is always irrational.

SOLUTION — step by step

Step 1. Given: r rational, x irrational. Suppose, for contradiction, that $s = r + x$ is rational.

Step 2. Then $x = s - r$ (subtract r from both sides).

Step 3. s is rational and r is rational; rationals are closed under subtraction, so s - r is rational.

Step 4. So x would be rational - contradicting the given that x is irrational.

Step 5. Hence s cannot be rational: r + x is irrational. ∎ The same subtraction trick works for r - x.

» **Why this step?** Closure of Q under subtraction is the engine: the proof shows that 'rational + irrational = rational' would force an impossible conclusion.

Answer: r + x is always irrational (if it were rational, x = (r + x) - r would be rational, a contradiction).

Hinglish: *Ye 'subtraction trick' proof bahut powerful hai: maan lo rational nikla, toh irrational wala number rational minus rational ban jayega - jo impossible hai. Ek line mein: Q subtraction ke under band hai, isliye irrational bacha reh jayega. Exam mein 3 marks pakke is proof ke.*

QUESTION 33 | Moderate | Rationalising a + b√c denominator (T9)

Q33. Rationalise the denominator and simplify: 1/(3 + 2√2).

SOLUTION — step by step

Step 1. Given: denominator 3 + 2√2. Its conjugate is 3 - 2√2 (flip the sign of the surd term).

Step 2. Multiply numerator and denominator by the conjugate:

Step 3. = (3 - 2√2)/((3 + 2√2)(3 - 2√2)).

Step 4. Denominator: $3^2 - (2\sqrt{2})^2 = 9 - 8 = 1$ (difference of squares).

Step 5. So the expression equals 3 - 2√2. Check numerically: 1/(3 + 2.828...) = 1/5.828... ≈ 0.1716, and 3 - 2.828... = 0.1716 ✓.

» **Why this step?** The conjugate turns the denominator into $a^2 - b^2$, which has no surds; here it even collapses to 1, giving a free simplification.

Answer: 3 - 2√2

Hinglish: *Conjugate ka matlab: sirf beech ka sign ulta. 3 + 2√2 ka conjugate 3 - 2√2. Multiply karo, denominator (a+b)(a-b) = a^2-b^2 ban jata hai. Yahan 9-8=1 nikla - jackpot! Answer 3-2√2. Aise lucky cases (denominator 1) exam mein bahut aate hain, isliye denominator ko poori tarah simplify karna kabhi mat chhodna.*

QUESTION 34 | Moderate | Rationalising both-storey fraction (T9)

Q34. Simplify completely: (√5 + √3)/(√5 - √3).

SOLUTION — step by step

Step 1. Given: a fraction with a binomial surd denominator. Multiply top and bottom by the conjugate (√5 + √3).

Step 2. Denominator: (√5 - √3)(√5 + √3) = 5 - 3 = 2.

Step 3. Numerator: $(\sqrt{5} + \sqrt{3})^2 = 5 + 2\sqrt{15} + 3 = 8 + 2\sqrt{15}$.

Step 4. So the expression = (8 + 2√15)/2 = 4 + √15.

Step 5. Check numerically: (2.236 + 1.732)/(2.236 - 1.732) = 3.968/0.504 ≈ 7.87, and 4 + 3.873 = 7.87 ✓.

» **Why this step?** Multiplying by the conjugate of the DENOMINATOR (not the numerator) is what clears the root downstairs; the numerator square uses $(a+b)^2$.

Answer: 4 + √15

Hinglish: *Dono taraf conjugate multiply karo, denominator 5-3=2 ban jayega. Numerator par $(a+b)^2$ = $a^2+2ab+b^2$ lagta hai: 5 + 2√15 + 3 = 8 + 2√15. Ab 2 se kat do: 4 + √15. Students yahan aadha kaam chhod dete hain - 2 se divide karna final step hai, uske bina marks katate hain.*

QUESTION 35 | Moderate | Comparing surds (T8)

Q35. Arrange in increasing order: √7, 2√2, √5, 3√3/2.

SOLUTION — step by step

Step 1. Given: four positive surds. All are positive, so squaring preserves the order.

Step 2. √7 → 7. 2√2 → $(2\sqrt{2})^2 = 4 \times 2 = 8$. √5 → 5. $(3\sqrt{3})/2 \to (9 \times 3)/4 = 27/4 = 6.75$.

Step 3. Order of the squares: $5 < 6.75 < 7 < 8$.

Step 4. So the original order: $\sqrt{5} < 3\sqrt{3}/2 < \sqrt{7} < 2\sqrt{2}$.

Step 5. Check: $\sqrt{5} \approx 2.236$, $3\sqrt{3}/2 \approx 2.598$, $\sqrt{7} \approx 2.646$, $2\sqrt{2} \approx 2.828$ ✓.

» **Why this step?** For positive quantities $a < b \Leftrightarrow a^2 < b^2$ - comparing squares avoids fiddly decimal roots.

Answer: √5 < 3√3/2 < √7 < 2√2

Hinglish: *Compare karne ka best trick: dono ko square karo. 2√2 ka square $(2)^2 \times 2 = 8$ hota hai - coefficient bhi square hota hai, ye bhoolna sabse common galti hai! $(3\sqrt{3}/2)^2 = 9 \times 3/4 = 6.75$. Ab squares ko order karo aur original order likho. Numerical check se confirm kar lo.*

QUESTION 36 | Moderate | Same-base equations (T11)

Q36. Solve: (i) $3^x = 1/81$, (ii) $5^{(x-2)} = 125$, (iii) $(2/3)^x = 27/8$.

SOLUTION — step by step

Step 1. Given: three exponential equations. Strategy: one base on both sides, then equate exponents.

Step 2. (i) $1/81 = 1/3^4 = 3^{-4}$. So $3^x = 3^{-4} \Rightarrow x = -4$.

Step 3. (ii) $125 = 5^3$. So $5^{(x-2)} = 5^3 \Rightarrow x - 2 = 3 \Rightarrow x = 5$.

Step 4. (iii) $27/8 = 3^3/2^3 = (3/2)^3 = (2/3)^{-3}$. So $x = -3$.

Step 5. Check: $(2/3)^{-3} = (3/2)^3 = 27/8$ ✓.

» **Why this step?** Equality of powers with the same base forces equal exponents; the whole game is rewriting both sides into one base.

Answer: (i) x = -4 (ii) x = 5 (iii) x = -3

Hinglish: *Exponent equation mein dono taraf SAME base lao. 1/81 ko 3^{-4} dekho - negative exponent ka perfect use. $(2/3)^x = 27/8$ mein ulta fraction banana padta hai: $27/8 = (3/2)^3 = (2/3)^{-3}$. Fraction ka negative exponent = fraction flip. Ye teen patterns exam ke favourites hain.*

QUESTION 37 | Moderate | Fractional and negative exponents (T10)

Q37. Evaluate without a calculator: (i) $(81/16)^{-34y}$... rewrite: (81/16)^(-3/4), (ii) 8^(2/3) + 4^(-1/2), (iii) $(5^2/\sqrt{5})^2$.

SOLUTION — step by step

Step 1. Given: expressions with fractional and negative exponents.

Step 2. (i) (81/16)^(-3/4): the negative exponent flips the fraction; the 1/4 takes a fourth root. (16/81)^(3/4) = $(\sqrt[4]{(16/81)})^3 = (2/3)^3 = 8/27$.

Step 3. (ii) 8^(2/3) = $(\sqrt[3]{8})^2 = 2^2 = 4$. 4^(-1/2) = $1/\sqrt{4} = 1/2$. Sum = 4 + 1/2 = 9/2.

Step 4. (iii) $5^2/\sqrt{5} = 5^2/5$^(1/2) = 5^(3/2). Squaring: $5^3 = 125$.

Step 5. Check (iii) numerically: 25/2.236 ≈ 11.18; $11.18^2 \approx 125$ ✓.

» **Why this step?** a^(m/n) = (root n of a) to the m; the negative sign means reciprocal. Order root-then-power keeps numbers small.

Answer: (i) 8/27 (ii) 9/2 (iii) 125

Hinglish: *Fractional exponent do kaam karta hai: neeche wala number ROOT, upar wala POWER. 8^(2/3): cube root of 8 = 2, phir square = 4. Negative exponent: flip karo. Root pehle lo, power baad mein - numbers chhote rehte hain. (i) mein (81/16) ko flip karke 16/81 banana zaroori tha.*

QUESTION 38 | Moderate | Classification with expansions (T1/T2)

Q38. Classify with one-line reasons: (i) 0.01001000100001... (one more zero each time), (ii) 3.1416, (iii) √(49/64), (iv) 1.010010001... , (v) $6.\overline{45}$... i.e. 6.454545... (block 45 repeating).

SOLUTION — step by step

Step 1. Given: five numbers of mixed types. The decimal pattern decides the set.

Step 2. (i) The gaps between 1s keep growing, so no block ever repeats: non-terminating, non-recurring ⇒ I, R.

Step 3. (ii) 3.1416 terminates ⇒ Q, R (note: it only approximates π).

Step 4. (iii) √(49/64) = 7/8 ⇒ Q, R (and it is not natural or integral).

Step 5. (iv) Endless, and the 1s never settle into a fixed block ⇒ I, R.

Step 6. (v) Recurring ⇒ Q, R; in fact 6.4545... = 6 + 45/99 = 6 + 5/11 = 71/11.

» **Why this step?** Terminating/recurring ⇒ Q; endless with no repeating block ⇒ I. Fractional roots simplify before classifying.

Answer: (i) I, R (ii) Q, R (iii) Q, R (= 7/8) (iv) I, R (v) Q, R (= 71/11)

Hinglish: *(i) aur (iv) type numbers ke liye pattern dekho - zeroes badhte ja rahe hain ya 1s ke beech gap badh raha hai, toh koi block repeat nahi hoga, irrational. (ii) ko π se compare mat karna - 3.1416 rukne wala decimal hai, rational. (v) recurring hai: 71/11.*

QUESTION 39 | Moderate | Mixed recurring with integer part (T3)

Q39. Express 2.1363636... (the block 36 repeats) in p/q form.

SOLUTION — step by step

Step 1. Given: x = 2.13636... . One non-repeating digit (3), then the block 36.

Step 2. Multiply by 10: 10x = 21.3636... . Multiply by 1000: 1000x = 2136.3636... .

Step 3. Subtract: 1000x - 10x = 2136.3636... - 21.3636... = 2115 (tails cancel exactly).

Step 4. So 990x = 2115, x = 2115/990.

Step 5. Reduce: HCF(2115, 990) = 45, so x = 47/22.

Step 6. Check: 47 ÷ 22 = 2.13636... ✓.

» **Why this step?** Shifts of 10 and 1000 align the two copies of the repeating tail; 990 = 9×100 with one zero for the non-repeating digit.

Answer: 47/22

Hinglish: *Integer part wale recurring mein panic mat karo - method same hai. 10x aur 1000x banao (ek shift head ke liye, do aur block ke liye), minus karo: 2115/990 aayega. HCF 45 hai, kat ke 47/22. Formula version: (2136 - 21)/990 = 2115/990. Reduce karna compulsory hai!*

QUESTION 40 | Moderate | 0.999... = 1 (T3/T13)

Q40. Show that 0.999... (all 9s repeating) is exactly equal to 1, and use the result to explain why 2.000... and 1.999... name the same real number.

SOLUTION — step by step

Step 1. Let x = 0.999... . Multiply by 10: 10x = 9.999... .

Step 2. Subtract: 10x - x = 9.999... - 0.999... = 9 exactly (the infinite tails cancel digit by digit).

Step 3. So 9x = 9 and x = 1.

Step 4. Hence 1.999... = 1 + 0.999... = 1 + 1 = 2, while 2.000... = 2: the two expansions name one point on the line.

Step 5. Moral: every terminating decimal also has a second, 9-ending expansion; the shift-and-subtract method does not break - it simply reports the value.

» **Why this step?** The subtraction 9.999... - 0.999... = 9 is exact because digit-by-digit cancellation leaves no remainder at ANY place value.

Answer: 0.999... = 1 exactly (9x = 9 from the shift-and-subtract computation); 2.000... = 1.999... as numbers.

Hinglish: *Ye wala concept books ke end mein hota hai aur examiner bahut pasand karte hain. 0.999... exactly 1 hai - 'almost 1' nahi. Multiply-subtract se 9x = 9 nikalta hai, point. Matlab har terminating decimal ki do expansions hain: 2.000... aur 1.999... dono same point. Exam mein 'true/false: 0.999... < 1' - answer False, reason ye proof.*

QUESTION 41 | Moderate | Irrationality proof (T5)

Q41. Prove that √7 is irrational.

SOLUTION — step by step

Step 1. Assume the opposite: √7 = p/q with p, q coprime integers (q ≠ 0).

Step 2. Then $7q^2 = p^2$, so 7 divides p^2.

Step 3. Prime lemma: if a prime divides a square, it divides the number itself -> 7 | p. Write p = 7k.

Step 4. Substitute: $7q^2 = 49k^2$ -> $q^2 = 7k^2$ -> 7 | q^2 -> 7 | q.

Step 5. So 7 divides BOTH p and q - contradicting coprimality. The assumption fails; √7 is irrational.

» **Why this step?** The contradiction skeleton: assume p/q -> square -> prime divides both -> contradiction. Works for every prime root.

Answer: √7 is irrational (proof by contradiction as above)

Hinglish: *Same purana script: maan lo p/q hai, square karo, 7 dono ko kha jaata hai, coprime toot jaata hai - contradiction! Ye template √2, √3, √5, √7 sab par chalega.*

QUESTION 42 | Moderate | Rational ± irrational (T6)

Q42. Prove that 5 + √3 is irrational.

SOLUTION — step by step

Step 1. Assume 5 + √3 = p/q is rational.

Step 2. Then √3 = p/q - 5 = (p - 5q)/q - a ratio of integers, hence rational.

Step 3. But √3 is irrational (standard proof) - contradiction.

Step 4. Therefore 5 + √3 must be irrational. (Same argument works for any rational ± irrational.)

» **Why this step?** Rational ± irrational is FORCED irrational: moving the rational part to the other side would make the irrational side rational - impossible.

Answer: 5 + √3 is irrational (contradiction via √3)

Hinglish: *5 + √3 rational maan lo toh √3 akela rational ban jaata hai - jo jhooth hai. Isliye 5 + √3 irrational hi rahega. Ek rational + ek irrational = hamesha irrational.*

QUESTION 43 | Moderate | Rationalising binomial denominators (T9)

Q43. Simplify completely: (i) $10/(\sqrt{7} - \sqrt{3})$, (ii) $12/(\sqrt{5} + \sqrt{2})$.

SOLUTION — step by step

Step 1. (i) Multiply by the conjugate $(\sqrt{7} + \sqrt{3})/(\sqrt{7} + \sqrt{3})$: denominator = 7 - 3 = 4.

Step 2. = $10(\sqrt{7} + \sqrt{3})/4 = (5/2)(\sqrt{7} + \sqrt{3})$.

Step 3. (ii) Conjugate $(\sqrt{5} - \sqrt{2})$: denominator = 5 - 2 = 3.

Step 4. = $12(\sqrt{5} - \sqrt{2})/3 = 4(\sqrt{5} - \sqrt{2})$.

» **Why this step?** The conjugate turns $(\sqrt{a} \pm \sqrt{b})$ into a − b: one multiplication kills the entire root structure downstairs.

Answer: (i) $(5/2)(\sqrt{7} + \sqrt{3})$ (ii) $4(\sqrt{5} - \sqrt{2})$

Hinglish: *Conjugate se multiply karo - denominator seedha number ban jaata hai (7-3 = 4, 5-2 = 3). Phir cancel karo: 5/2 aur 4 bahar aa gaye. Sign ka dhyan: conjugate mein beech ka sign ulta.*

QUESTION 44 | Moderate | Conjugate to a + b√c (T9)

Q44. Express $(\sqrt{5} + \sqrt{3})/(\sqrt{5} - \sqrt{3})$ in the form $a + b\sqrt{c}$.

SOLUTION — step by step

Step 1. Multiply top and bottom by $(\sqrt{5} + \sqrt{3})$: denominator = 5 - 3 = 2.

Step 2. Numerator = $(\sqrt{5} + \sqrt{3})^2 = 5 + 2\sqrt{15} + 3 = 8 + 2\sqrt{15}$.

Step 3. So the value = $(8 + 2\sqrt{15})/2 = 4 + \sqrt{15}$.

Step 4. a = 4, b = 1, c = 15.

» **Why this step?** The conjugate makes the numerator a perfect surd square - expand with the middle term $2\sqrt{ab}$.

Answer: $4 + \sqrt{15}$

Hinglish: *$(\sqrt{5}+\sqrt{3})^2 = 8 + 2\sqrt{15}$ - middle term $2\sqrt{(5\times3)}$ hi asli content hai. 2 se divide: $4 + \sqrt{15}$. Form $a + b\sqrt{c}$ mein convert karna hi question tha.*

QUESTION 45 | Moderate | Conjugate chain x and 1/x (T10)

Q45. If $x = 3 + 2\sqrt{2}$, find (i) $1/x$, (ii) $x + 1/x$, (iii) $x - 1/x$.

SOLUTION — step by step

Step 1. (i) $1/x = 1/(3 + 2\sqrt{2})$; note $(3 + 2\sqrt{2})(3 - 2\sqrt{2}) = 9 - 8 = 1$, so $1/x = 3 - 2\sqrt{2}$ directly!

Step 2. (ii) $x + 1/x = (3 + 2\sqrt{2}) + (3 - 2\sqrt{2}) = 6$.

Step 3. (iii) $x - 1/x = (3 + 2\sqrt{2}) - (3 - 2\sqrt{2}) = 4\sqrt{2}$.

» **Why this step?** x·(conjugate) = 1 means the conjugate IS the reciprocal - surds pair up like lock and key.

Answer: (i) $3 - 2\sqrt{2}$ (ii) 6 (iii) $4\sqrt{2}$

Hinglish: *$(3+2\sqrt{2})(3-2\sqrt{2}) = 1$ - isliye reciprocal hi conjugate hai! Sum 6, difference $4\sqrt{2}$ - surd parts cancel/judte hain. Ye pair exam ka favourite hai.*

QUESTION 46 | Moderate | Factored exponent equation (T11)

Q46. Solve for x: $3^x - 3^{x-1} = 162$.

SOLUTION — step by step

Step 1. Factor the smaller power: $3^x - 3^{x-1} = 3^{x-1}(3 - 1) = 2 \cdot 3^{x-1}$.

Step 2. So $2 \cdot 3^{x-1} = 162 \rightarrow 3^{x-1} = 81 = 3^4$.

Step 3. $x - 1 = 4 \rightarrow x = 5$.

» **Why this step?** Pull out the LOWEST power as a common factor; the bracket collapses to small numbers.

Answer: x = 5

Hinglish: *Sabse chhoti power bahar nikalo: 3^(x-1) common, bracket (3 - 1) = 2. 162/2 = 81 = 3^4 -> x = 5. Factoring is the whole game.*

QUESTION 47 | Moderate | Fractional exponents (T10)

Q47. Evaluate: (i) $64^{1/2} + 125^{1/3} - 81^{1/4}$, (ii) $(16/81)^{-3/4}$.

SOLUTION — step by step

Step 1. (i) $64^{1/2} = 8$; $125^{1/3} = 5$; $81^{1/4} = 3$. Sum: 8 + 5 - 3 = 10.

Step 2. (ii) Negative -> flip: $(81/16)^{3/4}$. Fourth root of 81 is 3, of 16 is 2 -> $(3/2)^3 = 27/8$.

» **Why this step?** $x^{m/n}$ = (n-th root of x) to the m: root first, power second - flipping negatives first keeps numbers small.

Answer: (i) 10 (ii) 27/8

Hinglish: *Fractional power = root: 64 ka half power 8, 125 ka 1/3 power 5. $(16/81)^{-3/4}$: pehle flip (81/16), phir fourth root (3/2), phir cube: 27/8.*

QUESTION 48 | Moderate | Ordering mixed surds (T8)

Q48. Arrange in increasing order: $2\sqrt{5}$, $3\sqrt{3}$, $4\sqrt{2}$, 5.

SOLUTION — step by step

Step 1. Push coefficients inside: $2\sqrt{5} = \sqrt{20}$; $3\sqrt{3} = \sqrt{27}$; $4\sqrt{2} = \sqrt{32}$; $5 = \sqrt{25}$.

Step 2. Compare radicands: $20 < 25 < 27 < 32$.

Step 3. Order: $2\sqrt{5} < 5 < 3\sqrt{3} < 4\sqrt{2}$.

» **Why this step?** Same-root comparison = radicand comparison once everything lives under one root.

Answer: $2\sqrt{5} < 5 < 3\sqrt{3} < 4\sqrt{2}$

Hinglish: *Sabko ek hi root ke andar: $\sqrt{20}$, $\sqrt{25}$, $\sqrt{27}$, $\sqrt{32}$. Ab normal number race hai. 5 ko $\sqrt{5}^2 = \sqrt{25}$ banana hi trick thi!*

QUESTION 49 | Moderate | Recurring with integer part (T3)

Q49. Express 3.4171717... (the block 17 repeating) in p/q form.

SOLUTION — step by step

Step 1. Split: 3 + 0.4171717...

Step 2. For the decimal: 10x = 4.171717... and 1000x = 417.171717...; subtract: 990x = 413 -> x = 413/990.

Step 3. Check reduction: 413 = 7 × 59 shares nothing with $990 = 2 \cdot 3^2 \cdot 5 \cdot 11$ -> already lowest.

Step 4. Total: 3 + 413/990 = (2970 + 413)/990 = 3383/990.

» **Why this step?** Handle the integer part separately; shift past the quiet digit (×10), then subtract the ×1000 copy.

Answer: 3383/990

Hinglish: *3 alag rakho. Decimal wale ko 10x aur 1000x se phasao: 990x = 413. GCD check kiya - reduce nahi hota. Final 3383/990. Integer part ko aakhir mein jodna.*

QUESTION 50 | Moderate | Irrationals in a small interval (T12)

Q50. Insert two irrational numbers between 0.12 and 0.13.

SOLUTION — step by step

Step 1. Idea 1: invent a non-repeating decimal inside the gap: 0.1201001000100001... - non-terminating, non-repeating -> irrational, and it sits between the bounds.

Step 2. Idea 2: use a surd: $\sqrt{0.015} = \sqrt{(15/1000)} = \sqrt{15}/\sqrt{1000}$ - since 0.015 is not a perfect square of a rational, its root is irrational; numerically $\sqrt{0.015} \approx 0.12247$, inside the gap.

Step 3. Both are valid: infinitely many irrationals live between any two distinct rationals.

» **Why this step?** Two generators of irrationals: engineered non-repeating decimals, and roots of non-square rationals.

Answer: e.g. 0.1201001000100001... and $\sqrt{0.015} \approx 0.12247$ (other answers possible)

Hinglish: *Do tareeke: (1) apni khud ki non-repeating decimal godo (0.12010010001...), (2) non-square ka root lo ($\sqrt{0.015} \approx 0.1225$). Dono irrational aur dono gap ke andar. Density ka jaadu!*

QUESTION 51 | Moderate | Irrationality of a surd sum (T6)

Q51. Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

SOLUTION — step by step

Step 1. Assume $\sqrt{2} + \sqrt{3} = p/q$ (rational).

Step 2. Square: $2 + 2\sqrt{6} + 3 = p^2/q^2 \rightarrow 2\sqrt{6} = p^2/q^2 - 5 = (p^2 - 5q^2)/q^2$.

Step 3. So $\sqrt{6} = (p^2 - 5q^2)/(2q^2)$ - a rational number.

Step 4. But $\sqrt{6}$ is irrational (6 is not a perfect square; the standard p/q contradiction applies) - contradiction.

Step 5. Hence $\sqrt{2} + \sqrt{3}$ is irrational.

» **Why this step?** Squaring a sum of two surds isolates $\sqrt{6}$; if the sum were rational, $\sqrt{6}$ would be too - impossible.

Answer: $\sqrt{2} + \sqrt{3}$ is irrational (proof above)

Hinglish: *Sum ko rational maan lo, square karo - $\sqrt{6}$ akela bach jaata hai, aur wo rational hona chahiye. Par $\sqrt{6}$ irrational hai - contradiction! Isliye sum bhi irrational.*

QUESTION 52 | Moderate | a – b√c irrational (T6)

Q52. Show that 3 - 2√5 is irrational.

SOLUTION — step by step

Step 1. Assume 3 - 2√5 = p/q.

Step 2. Then 2√5 = 3 - p/q = (3q - p)/q -> √5 = (3q - p)/(2q) - rational.

Step 3. But √5 is irrational - contradiction.

Step 4. Therefore 3 - 2√5 is irrational.

» **Why this step?** Any (rational) + (rational)·√5 combination is irrational unless the √5 coefficient is 0.

Answer: 3 - 2√5 is irrational

Hinglish: *√5 ko akela chhod do: wo rational hona padega maan liya toh - jhooth. Isliye 3 - 2√5 irrational. Negative sign ne kuch nahi bigada.*

QUESTION 53 | Moderate | Conjugate difference chain (T9)

Q53. Simplify completely: 4/(√7 - √3) - 4/(√7 + √3).

SOLUTION — step by step

Step 1. Rationalise each: 4/(√7 - √3) = 4(√7 + √3)/4 = √7 + √3.

Step 2. 4/(√7 + √3) = 4(√7 - √3)/4 = √7 - √3.

Step 3. Subtract: (√7 + √3) - (√7 - √3) = 2√3.

» **Why this step?** Each fraction rationalises to a clean surd; the subtraction then cancels √7 entirely.

Answer: 2√3

Hinglish: *Dono fractions conjugate se saaf ho gaye: (√7+√3) aur (√7-√3). Ghatane par √7 gayab, 2√3 bacha. Chain questions mein ek-ek karke rationalise karo.*

QUESTION 54 | Moderate | Multi-base exponent relation (T11)

Q54. If $2^x = 3^y = 6^z = t$ ($t \neq 1$), prove that 1/z = 1/x + 1/y.

SOLUTION — step by step

Step 1. Take $t = 2^x$ -> $2 = t^{1/x}$. Similarly $3 = t^{1/y}$.

Step 2. Multiply: $6 = 2 \times 3 = t^{1/x} \cdot t^{1/y} = t^{1/x + 1/y}$.

Step 3. But also $6 = t^{1/z}$ (since $t = 6^z$).

Step 4. Same base t: 1/x + 1/y = 1/z. Proved.

» **Why this step?** Rewriting every base as a power of the common value t turns the identity into an exponent sum.

Answer: 1/z = 1/x + 1/y (proof above)

Hinglish: *Sabko t ke powers mein likho: 2 = t^(1/x), 3 = t^(1/y). Guna karo -> 6 = t^(1/x + 1/y). Aur 6 = t^(1/z) bhi hai. Exponents barabar - done. Ye relation ratt lo, direct aata hai.*

QUESTION 55 | Moderate | Nested radicals (T14)

Q55. Simplify: (i) $\sqrt{(9 + 4\sqrt{5})}$, (ii) $\sqrt{(7 + 2\sqrt{10})}$.

SOLUTION — step by step

Step 1. (i) Seek a + b with a + b = 9 and ($\sqrt{a}\cdot\sqrt{b}$ pattern) $2\sqrt{(ab)} = 4\sqrt{5}$ -> ab = 5: a = 5, b = 4.

Step 2. So $9 + 4\sqrt{5} = (\sqrt{5} + 2)^2$ (check: 5 + 4√5 + 4 ✓) -> $\sqrt{(9 + 4\sqrt{5})} = \sqrt{5} + 2$.

Step 3. (ii) Need a + b = 7, ab = 10: a = 5, b = 2 -> $(\sqrt{5} + \sqrt{2})^2 = 5 + 2\sqrt{10} + 2$.

Step 4. Answer: $\sqrt{5} + \sqrt{2}$.

» **Why this step?** $\sqrt{(m + 2\sqrt{k})}$ hides a perfect square: find two numbers with sum m and product k.

Answer: (i) √5 + 2 (ii) √5 + √2

Hinglish: *Nested root ka square hunt: a + b = 9, ab = 5 -> 5 aur 4. $(\sqrt{5} + 2)^2$ mil gaya. Doosra: 5 aur 2. Root ke andar square chhupana hi iska design hai.*

QUESTION 56 | Moderate | Telescoping surd sum (T14)

Q56. Evaluate: $1/(\sqrt{2} + 1) + 1/(\sqrt{3} + \sqrt{2}) + 1/(\sqrt{4} + \sqrt{3}) + \ldots + 1/(\sqrt{9} + \sqrt{8})$.

SOLUTION — step by step

Step 1. Each term rationalises: $1/(\sqrt{(k+1)} + \sqrt{k}) = \sqrt{(k+1)} - \sqrt{k}$.

Step 2. The sum becomes $(\sqrt{2} - 1) + (\sqrt{3} - \sqrt{2}) + (\sqrt{4} - \sqrt{3}) + \ldots + (\sqrt{9} - \sqrt{8})$.

Step 3. Everything cancels in the middle (telescopes): $-1 + \sqrt{9}$ remains.

Step 4. Sum = 3 - 1 = 2.

» **Why this step?** Rationalising $1/(\sqrt{(k+1)} + \sqrt{k})$ with the conjugate ALWAYS gives $\sqrt{(k+1)} - \sqrt{k}$ - the perfect telescope.

Answer: 2

Hinglish: *Har term conjugate se $\sqrt{(k+1)} - \sqrt{k}$ ban gaya. Ab beech ke sab cancel - telescope! Bacha 3 - 1 = 2. Chain + conjugate = collapse.*

QUESTION 57 | Moderate | Compare surd expressions (T8)

Q57. Which is larger: $(\sqrt{3} + \sqrt{2})^2$ or $(\sqrt{6})^2 + 1$? By how much (simplest form)?

SOLUTION — step by step

Step 1. Expand: $(\sqrt{3} + \sqrt{2})^2 = 3 + 2\sqrt{6} + 2 = 5 + 2\sqrt{6}$.

Step 2. Right side: $6 + 1 = 7$.

Step 3. Compare: $5 + 2\sqrt{6} \approx 5 + 4.899 = 9.899 > 7$.

Step 4. Difference: $(5 + 2\sqrt{6}) - 7 = 2\sqrt{6} - 2 = 2(\sqrt{6} - 1)$.

» **Why this step?** Expand before comparing - the middle term $2\sqrt{6}$ is what makes the left side bigger.

Answer: $(\sqrt{3} + \sqrt{2})^2$ is larger by $2(\sqrt{6} - 1)$

Hinglish: *Expand karo: $5 + 2\sqrt{6} \approx 9.9$ vs 7. Left jeeta, fark $2(\sqrt{6} - 1)$. Middle term hi asli hero hai - isko bhoolna hi galti hai.*

QUESTION 58 | Moderate | Mixed recurring with integer part (T3)

Q58. Express 2.03666... (the 6 repeating) in p/q form.

SOLUTION — step by step

Step 1. Split: 2 + 0.03666...

Step 2. For the decimal: 100x = 3.666... and 1000x = 36.66...; subtract: 900x = 33 -> x = 33/900 = 11/300.

Step 3. Total: 2 + 11/300 = 611/300.

» **Why this step?** Two quiet digits (0, 3) mean ×100 first; the subtraction 1000x − 100x kills the repeating 6s.

Answer: 611/300

Hinglish: *2 alag. Decimal mein 6 repeat karta hai 3rd place se: 100x aur 1000x ka subtraction -> 900x = 33 = 11/300. Total 611/300.*

QUESTION 59 | Moderate | Factored exponent equation (T11)

Q59. Solve: $5^{x+1} + 5^{x+2} = 750$.

SOLUTION — step by step

Step 1. Factor out 5^{x+1}: $5^{x+1}(1 + 5) = 750$.

Step 2. $6 \cdot 5^{x+1} = 750$ -> $5^{x+1} = 125 = 5^3$.

Step 3. $x + 1 = 3$ -> $x = 2$.

» **Why this step?** The smaller exponent is the common factor; the bracket is always (1 + base) here.

Answer: x = 2

Hinglish: *5^(x+1) bahar: bracket (1 + 5) = 6. 750/6 = 125 = 5^3 -> x = 2. Wahi factoring move, bas power mein +1 shift.*

QUESTION 60 | Moderate | Rational and irrational between (T12)

Q60. Find one rational AND one irrational number strictly between √2 and √3.

SOLUTION — step by step

Step 1. Estimate: √2 ≈ 1.414, √3 ≈ 1.732.

Step 2. Rational: 1.5 = 3/2 (1.414 < 1.5 < 1.732 ✓).

Step 3. Irrational: √2.5 ≈ 1.581 (2.5 = 5/2 is not a perfect square of a rational, so its root is irrational, and it lies between).

» **Why this step?** Between any two reals there are both rationals AND irrationals - density works both ways.

Answer: Rational: 3/2; irrational: √2.5 (others possible)

Hinglish: *1.5 = 3/2 rational mil gaya. Irrational ke liye non-square ka root: √2.5 ≈ 1.581. Dono ke beech mein har type ke infinite numbers hain - density dono taraf se sachi.*

QUESTION 61 | Moderate | Exponent associativity trap (T10)

Q61. Which is larger: $2^{(2^3)}$ or $(2^2)^3$? Compute both and state the rule you used.

SOLUTION — step by step

Step 1. Top-down first: $2^{(2^3)} = 2^8 = 256$.

Step 2. Bottom-up first: $(2^2)^3 = 4^3 = 64$.

Step 3. 256 > 64: the tower is read from the TOP down.

Step 4. Rule: a^(b^c) means a^(b^c) - the upper exponent binds first; brackets decide everything else.

» **Why this step?** Exponent towers are NOT associative: 2^(2^3) and (2^2)^3 differ by a factor of 4 here.

Answer: 2^(2^3) = 256 > 64 = (2^2)^3; towers read top-down

Hinglish: *Tower upar se padho: 2^(2^3) = 2^8 = 256. Neeche se padha toh 64 aa gaya - poora farak! Bracket hi boss hai exponent towers mein.*

QUESTION 62 | Moderate | Two-root equation (T14)

Q62. Solve: $\sqrt{(2x + 9)} - \sqrt{(x - 4)} = 3$.

SOLUTION — step by step

Step 1. Isolate one root: $\sqrt{(2x + 9)} = 3 + \sqrt{(x - 4)}$. Square: $2x + 9 = 9 + 6\sqrt{(x - 4)} + (x - 4)$.

Step 2. Simplify: $2x + 9 = x + 5 + 6\sqrt{(x - 4)} \rightarrow x + 4 = 6\sqrt{(x - 4)}$.

Step 3. Square again: $x^2 + 8x + 16 = 36x - 144 \rightarrow x^2 - 28x + 160 = 0$.

Step 4. $(x - 8)(x - 20) = 0 \rightarrow x = 8$ or $x = 20$.

Step 5. Check $x = 8$: $\sqrt{25} - \sqrt{4} = 5 - 2 = 3$ ✓. Check $x = 20$: $\sqrt{49} - \sqrt{16} = 7 - 4 = 3$ ✓. Both valid!

» **Why this step?** Squaring twice is legal, but candidates MUST be checked - here both happen to survive.

Answer: x = 8 or x = 20

Hinglish: *Ek root akela karo, square, phir dobara. Quadratic se x = 8, 20. Dono ka check pass - kabhi-kabhi dono sach hote hain! Check karna zaroori hai, isliye kiya.*

QUESTION 63 | Moderate | Conjugate satisfies a quadratic (T10)

Q63. Let $x = 1/(2 - \sqrt{3})$. Show that x satisfies $x^2 - 4x + 1 = 0$.

SOLUTION — step by step

Step 1. Rationalise: $x = (2 + \sqrt{3})/((2 - \sqrt{3})(2 + \sqrt{3})) = (2 + \sqrt{3})/(4 - 3) = 2 + \sqrt{3}$.

Step 2. So $x + 1/x = (2 + \sqrt{3}) + (2 - \sqrt{3}) = 4$.

Step 3. Divide $x^2 - 4x + 1 = 0$ by x: $x - 4 + 1/x = 0 \rightarrow x + 1/x = 4$ - exactly what we have.

Step 4. Hence $x^2 - 4x + 1 = 0$ holds for $x = 2 + \sqrt{3}$ (and equally for $2 - \sqrt{3}$).

» **Why this step?** $x + (1/x) = 4$ is the divided-by-x form of the quadratic - conjugate pairs always satisfy such symmetric equations.

Answer: $x = 2 + \sqrt{3}$ satisfies $x^2 - 4x + 1 = 0$ (shown via $x + 1/x = 4$)

Hinglish: *Rationalise karo: $x = 2 + \sqrt{3}$, reciprocal $2 - \sqrt{3}$. $x + 1/x = 4$ - aur yahi equation $x^2 - 4x + 1 = 0$ hai (x se divide karke). Conjugate pair = quadratic ka jodi number.*

QUESTION 64 | Moderate | Exponent ratio system (T11)

Q64. If $2^x = 4^y = 64^z$, find the ratio x : y : z.

SOLUTION — step by step

Step 1. Convert everything to base 2: $4 = 2^2$, $64 = 2^6$.

Step 2. So $2^x = 2^{2y} = 2^{6z}$ -> x = 2y = 6z.

Step 3. Set z = 1: y = 3, x = 6.

Step 4. x : y : z = 6 : 3 : 1.

» **Why this step?** One common base collapses the whole chain into simple exponent equalities.

Answer: x : y : z = 6 : 3 : 1

Hinglish: *Sab 2 ke powers banao: x = 2y = 6z. z = 1 maan lo: 6 : 3 : 1. Common base = common language, ratio khud bol deta hai.*

QUESTION 65 | Moderate | Density of rationals (T12)

Q65. Insert five rational numbers between 0.21 and 0.22.

SOLUTION — step by step

Step 1. Write bounds with a bigger denominator: 0.21 = 210/1000, 0.22 = 220/1000.

Step 2. Pick any five in between: 211/1000, 212/1000, 213/1000, 214/1000, 215/1000.

Step 3. i.e. 0.211, 0.212, 0.213, 0.214, 0.215 - all rational, all strictly inside.

» **Why this step?** Stretching the denominator gives as many slots as you want - rationals are inexhaustible in any interval.

Answer: 0.211, 0.212, 0.213, 0.214, 0.215 (one valid set)

Hinglish: *Denominator badhao, slots khud mil jayenge: 0.211 se 0.215 - paanch bhai, sab rational. Jitne maango, utne milenge.*

QUESTION 66 | Moderate | Surd product collapse (T9)

Q66. Simplify completely: $(\sqrt{6} + \sqrt{2})(\sqrt{3} - 1)$.

SOLUTION — step by step

Step 1. Expand all four products: $\sqrt{6}\cdot\sqrt{3} = \sqrt{18} = 3\sqrt{2}$; $\sqrt{6}\cdot(-1) = -\sqrt{6}$; $\sqrt{2}\cdot\sqrt{3} = \sqrt{6}$; $\sqrt{2}\cdot(-1) = -\sqrt{2}$.

Step 2. Collect: $3\sqrt{2} - \sqrt{6} + \sqrt{6} - \sqrt{2}$.

Step 3. The $\sqrt{6}$ terms cancel; $3\sqrt{2} - \sqrt{2} = 2\sqrt{2}$.

» **Why this step?** Simplify each product BEFORE collecting - $\sqrt{18}$ was hiding a $3\sqrt{2}$ that matched its partner.

Answer: $2\sqrt{2}$

Hinglish: *Char products likho, har ek ko simplify karo: $\sqrt{18} = 3\sqrt{2}$ nikla toh jodi mil gayi. $\sqrt{6}$ cancel, $3\sqrt{2} - \sqrt{2} = 2\sqrt{2}$. Simplify-first hi cancellation deta hai.*

QUESTION 67 | Moderate | Recurring decimal arithmetic (T3)

Q67. Evaluate: $0.\dot{1} + 0.\dot{2} + 0.\dot{3} + \ldots + 0.\dot{9}$ (each digit repeating), where $0.\dot{k}$ means 0.kkk...

SOLUTION — step by step

Step 1. Each converts as $0.\dot{k} = k/9$.

Step 2. Sum = $(1 + 2 + \ldots + 9)/9 = 45/9$.

Step 3. = 5 - a whole number!

Step 4. Cross-check: $0.\dot{1} + 0.\dot{9} = 1$, $0.\dot{2} + 0.\dot{8} = 1$, $0.\dot{3} + 0.\dot{7} = 1$, $0.\dot{4} + 0.\dot{6} = 1$, leaving $0.\dot{5} = 5/9$: total $4 + 5/9 = 41/9$? Wait - $0.\dot{5} = 5/9$, and $4 + 5/9 = 41/9$. Recheck the direct sum: $45/9 = 5$. Hmm - pair sums: $1 + 1 + 1 + 1 = 4$ plus $0.\dot{5} = 5/9$ gives $4 + 5/9 = 41/9 \neq 5$. So re-examine: $1/9 + 9/9 = 10/9 = 1 + 1/9$ - the pairs are 10/9 each, four of them = 40/9, plus $5/9 = 45/9 = 5$ ✓. The pairing shortcut mislabels 10/9 as 1 - stick with the direct sum.

» **Why this step?** $0.\dot{k} = k/9$ exactly; summing k = 1..9 gives $45/9 = 5$. (Pairing needs care: each pair sums to 10/9, not 1.)

Answer: 5

Hinglish: *Sab k/9 hai: (1+2+...+9)/9 = 45/9 = 5. Ek dum poora number! Pairing trick mein 1/9 ka hissa reh jaata hai - direct sum safest hai.*

QUESTION 68 | Moderate | Three-term exponent equation (T11)

Q68. Solve: $2^{x+2} + 2^{x+1} + 2^x = 56$.

SOLUTION — step by step

Step 1. Factor out 2^x: $2^x(4 + 2 + 1) = 56$.

Step 2. $7 \cdot 2^x = 56 \rightarrow 2^x = 8 = 2^3$.

Step 3. x = 3.

» **Why this step?** The bracket (4 + 2 + 1) is a binary 111 - factoring the smallest power exposes it.

Answer: x = 3

Hinglish: *2^x common lo: bracket 4 + 2 + 1 = 7. 56/7 = 8 = 2^3 -> x = 3. Teen powers ek saath? Sabse chhoti bahar, bracket jodo.*

QUESTION 69 | Moderate | Surd geometry application (T14)

Q69. A square has side $(\sqrt{5} + 1)$ units. Find (i) its area in $a + b\sqrt{c}$ form, (ii) its perimeter.

SOLUTION — step by step

Step 1. (i) Area = $(\sqrt{5} + 1)^2 = 5 + 2\sqrt{5} + 1 = 6 + 2\sqrt{5}$.

Step 2. (ii) Perimeter = $4(\sqrt{5} + 1) = 4\sqrt{5} + 4$.

» **Why this step?** Binomial squares with surds obey the same identity: $a^2 + 2ab + b^2$ - just keep the middle term's root.

Answer: (i) $6 + 2\sqrt{5}$ (ii) $4\sqrt{5} + 4$

Hinglish: *Side surd mein ho toh bhi $(a + b)^2$ hi lagta hai: $6 + 2\sqrt{5}$. Perimeter = 4 baar side, simple distribute. Geometry + surds ka combo.*

QUESTION 70 | Moderate | Both types between bounds (T12)

Q70. Give one rational and one irrational number strictly between 2.1 and 2.2, with justification.

SOLUTION — step by step

Step 1. Rational: 2.15 = 43/20 - terminating decimal, hence rational, and $2.1 < 2.15 < 2.2$.

Step 2. Irrational: $\sqrt{4.6} \approx 2.1448$. Since 4.6 = 46/10 is not a perfect square of a rational, $\sqrt{4.6}$ is irrational; and $2.1 = \sqrt{4.41} < \sqrt{4.6} < \sqrt{4.84} = 2.2$ (squaring preserves order for positives).

» **Why this step?** Squaring is order-preserving on positives, so root-hunting inside an interval is legal - pick a non-square target.

Answer: Rational: 2.15; irrational: $\sqrt{4.6} \approx 2.1448$

Hinglish: *2.15 seedha rational. Irrational ke liye root game: $2.1^2 = 4.41$ aur $2.2^2 = 4.84$ - beech ka 4.6 lo, uska root lo. Order square hone par bhi kayam rehta hai.*

Level 3 · Exam Power Zone: long and twisted:

QUESTION 71 | Difficult | Irrationality of a surd sum (T6)

Q71. Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

SOLUTION — step by step

Step 1. Suppose $\sqrt{2} + \sqrt{3} = r$ is rational. Square both sides: $2 + 2\sqrt{6} + 3 = r^2$, so $5 + 2\sqrt{6} = r^2$.

Step 2. Solve for the surd: $\sqrt{6} = (r^2 - 5)/2$.

Step 3. r is rational, so $r^2 - 5$ is rational, and $(r^2 - 5)/2$ is rational.

Step 4. So $\sqrt{6}$ would be rational. But 6 is not a perfect square, so $\sqrt{6}$ is irrational (same contradiction proof as $\sqrt{2}$).

Step 5. Contradiction. Hence $\sqrt{2} + \sqrt{3}$ is irrational. ∎

» **Why this step?** Squaring converts a sum of two surds into ONE surd; the known irrationality of $\sqrt{6}$ then finishes the job.

Answer: $\sqrt{2} + \sqrt{3}$ is irrational (rationality would force $\sqrt{6} = (r^2-5)/2$ to be rational).

Hinglish: *Do surds ka sum prove karna hai? Square karo - $\sqrt{2}+\sqrt{3}$ ka square 5 + $2\sqrt{6}$ deta hai. Agar sum rational hota toh $\sqrt{6} = (r^2-5)/2$ rational hota - jo impossible hai. Yehi pattern $\sqrt{2}+\sqrt{5}$, $\sqrt{3}+\sqrt{5}$ ke liye bhi chalega. Ek baar square, phir known irrational se maar do.*

QUESTION 72 | Difficult | a - b√c irrational (T6)

Q72. Show that $3 - 2\sqrt{5}$ is irrational.

SOLUTION — step by step

Step 1. Suppose $3 - 2\sqrt{5} = r$ is rational. Rearrange: $2\sqrt{5} = 3 - r$.

Step 2. Divide by 2: $\sqrt{5} = (3 - r)/2$.

Step 3. $3 - r$ is rational (Q is closed under subtraction), and dividing by 2 keeps it rational.

Step 4. So $\sqrt{5}$ would be rational. But 5 is not a perfect square, so $\sqrt{5}$ is irrational - contradiction.

Step 5. Hence $3 - 2\sqrt{5}$ is irrational. ∎

» **Why this step?** Isolate the surd on one side; the rest of the expression is built only from rationals, which stay rational under the field operations.

Answer: $3 - 2\sqrt{5}$ is irrational (otherwise $\sqrt{5} = (3-r)/2$ would be rational).

Hinglish: *NCERT ka classic! Surd ko akela karo: $2\sqrt{5} = 3 - r$, phir $\sqrt{5} = (3-r)/2$. Right side pura rational hai, left side irrational - clash. Bas. Aise 'a + b√c' type har number ke liye ye line chalegi: irrational number rational ke saath mix karke rational nahi ban sakta.*

QUESTION 73 | Difficult | Rationalising with clean collapse (T9)

Q73. Simplify completely: $4/(\sqrt{7} - \sqrt{3})$, and express the result in the form $a\sqrt{7} + b\sqrt{3}$ with rational a, b.

SOLUTION — step by step

Step 1. Multiply numerator and denominator by the conjugate $\sqrt{7} + \sqrt{3}$.

Step 2. Denominator: $(\sqrt{7} - \sqrt{3})(\sqrt{7} + \sqrt{3}) = 7 - 3 = 4$.

Step 3. Expression $= 4(\sqrt{7} + \sqrt{3})/4 = \sqrt{7} + \sqrt{3}$.

Step 4. So $a = 1$ and $b = 1$.

Step 5. Check numerically: $4/(2.646 - 1.732) = 4/0.914 \approx 4.376$, and $\sqrt{7} + \sqrt{3} \approx 2.646 + 1.732 = 4.376$ ✓.

» **Why this step?** Choosing the conjugate makes the denominator $a^2 - b^2$; here it cancels the numerator perfectly, so no fraction survives.

Answer: √7 + √3 (a = 1, b = 1)

Hinglish: *Denominator ka conjugate multiply karo: (a-b)(a+b) = a^2-b^2 = 7-3 = 4. Upar wala 4 aur neeche wala 4 cancel - answer √7 + √3, ekdum clean! Ye 'designed' questions mein hamesha ek magic cancellation hota hai; agar aisa nahi ho raha toh calculation check karo.*

QUESTION 74 | Difficult | Triple-surd denominator (T9)

Q74. Rationalise the denominator completely: 1/(√2 + √3 + √5).

SOLUTION — step by step

Step 1. Group as (√2 + √3) + √5. Round 1: multiply by the conjugate (√2 + √3) - √5.

Step 2. Denominator: $(\sqrt{2} + \sqrt{3})^2 - (\sqrt{5})^2 = (2 + 2\sqrt{6} + 3) - 5 = 2\sqrt{6}$.

Step 3. So the expression = (√2 + √3 - √5)/(2√6). One surd remains downstairs.

Step 4. Round 2: multiply by √6/√6: = (√2 · √6 + √3 · √6 - √5 · √6)/(2 × 6) = (√12 + √18 - √30)/12.

Step 5. Simplify the surds: √12 = 2√3, √18 = 3√2. Final: (2√3 + 3√2 - √30)/12.

Step 6. Check numerically: 1/(1.414 + 1.732 + 2.236) = 1/5.382 ≈ 0.1858; (3.464 + 4.243 - 5.477)/12 = 2.230/12 ≈ 0.1858 ✓.

» **Why this step?** Three surds cannot be cleared in one blow: group two as a single 'number', use its conjugate, then mop up the single remaining surd.

Answer: (2√3 + 3√2 - √30)/12

Hinglish: *Teen surd ek saath? Group karo: (√2+√3) ko ek unit maano, conjugate se √5 ko maaro. Denominator (5 + 2√6) - 5 = 2√6 nikla - poora 5 cancel! Phir simple √6 se multiply. Do round ka game hai. Grouping alag karoge toh answer alag dikhega but value same - ye normal hai.*

QUESTION 75 | Difficult | Conjugate pairs x and 1/x (T14)

Q75. If x = 2 + √3, find the value of $x^2 + 1/x^2$ without evaluating x numerically beyond the given form.

SOLUTION — step by step

Step 1. First find 1/x by rationalising: 1/(2 + √3) = (2 - √3)/((2 + √3)(2 - √3)) = (2 - √3)/(4 - 3) = 2 - √3.

Step 2. So x and 1/x are conjugates: x = 2 + √3, 1/x = 2 - √3.

Step 3. Square both: $x^2 = 4 + 4\sqrt{3} + 3 = 7 + 4\sqrt{3}$; $(1/x)^2 = 7 - 4\sqrt{3}$.

Step 4. Add: $x^2 + 1/x^2 = 14$ (the surd terms cancel).

Step 5. Check numerically: x ≈ 3.732, x^2 ≈ 13.93, $1/x^2$ ≈ 0.0718, sum ≈ 14.0 ✓.

» **Why this step?** Conjugate reciprocals have surd terms in opposite signs; squares keep that symmetry, so the middle terms cancel on adding.

Answer: 14

Hinglish: *Ye conjugate-pair trick poore algebra mein chalega. Pehle 1/x nikaalo - denominator 1 aaya, matlab x aur 1/x perfect conjugates. Square karke jodo: ±4√3 terms aapas mein uda do. Answer 14. Aage ke liye yaad rakho: $x^3 + 1/x^3 = (x + 1/x)^3 - 3(x + 1/x)$.*

QUESTION 76 | Difficult | Factored exponent equation (T11)

Q76. Solve for x: $4^x - 4^{(x-1)} = 24$.

SOLUTION — step by step

Step 1. Factor out the smaller power: $4^{(x-1)}(4^1 - 1) = 24$.

Step 2. So $4^{(x-1)} \times 3 = 24$, giving $4^{(x-1)} = 8$.

Step 3. Convert to one base: $4 = 2^2$ and $8 = 2^3$, so $2^{(2x-2)} = 2^3$.

Step 4. Equate exponents: 2x - 2 = 3, so x = 5/2.

Step 5. Check: 4^(5/2) = $(\sqrt{4})^5 = 32$; 4^(3/2) = $(\sqrt{4})^3 = 8$; 32 - 8 = 24 ✓.

» **Why this step?** Both terms share the base 4, so factoring the smaller power is legal and reduces two powers to one equation.

Answer: x = 5/2

Hinglish: *Do powers ka difference? Common power bahar nikalo - 4^(x-1) common liya, bracket mein 4-1 = 3 bacha. Ab 24/3 = 8. 4 aur 8 dono ko 2 ke base par lao: $2^{(2x-2)} = 2^3$. Exponents barabar: x = 5/2. Fraction mein aana bilkul theek hai - check line se confirm karo.*

QUESTION 77 | Difficult | Cyclic exponent product (T10)

Q77. Simplify: $(x^a/x^b)^{(a+b)} \times (x^b/x^c)^{(b+c)} \times (x^c/x^a)^{(c+a)}$.

SOLUTION — step by step

Step 1. Simplify each inner quotient: $x^a/x^b = x^{(a-b)}$, and so on.

Step 2. Expression = $x^{(a-b)(a+b)} \times x^{(b-c)(b+c)} \times x^{(c-a)(c+a)}$.

Step 3. Power of a power: exponents multiply: $x^{(a^2-b^2)} \times x^{(b^2-c^2)} \times x^{(c^2-a^2)}$.

Step 4. Same base multiply → add: $x^{(a^2 - b^2 + b^2 - c^2 + c^2 - a^2)} = x^0$.

Step 5. $x^0 = 1$ ($x \neq 0$).

» **Why this step?** Difference-of-squares cancellation (a^2-b^2 terms annihilate in a cycle) is the hidden design; tracking exponents line by line reveals it.

Answer: 1 (for x ≠ 0)

Hinglish: *Lagta hai bhayanak, par dekho - har factor mein (a+b)(a-b) pattern hai! Exponents multiply hone par a^2-b^2, b^2-c^2, c^2-a^2 bante hain, aur ye teeno judne par aapas mein cancel. Bacha $x^0 = 1$. Aise cyclic questions mein hamesha cancellation chhupa hota hai - exponents ko alag-alag likh ke dekho.*

QUESTION 78 | Difficult | Find a, b after rationalising (T9)

Q78. If $(\sqrt{3} + 1)/(\sqrt{3} - 1) = a + b\sqrt{3}$ where a, b are rational, find a and b.

SOLUTION — step by step

Step 1. Rationalise: multiply by $(\sqrt{3} + 1)/(\sqrt{3} + 1)$.

Step 2. Numerator: $(\sqrt{3} + 1)^2 = 3 + 2\sqrt{3} + 1 = 4 + 2\sqrt{3}$.

Step 3. Denominator: $3 - 1 = 2$.

Step 4. Expression $= (4 + 2\sqrt{3})/2 = 2 + \sqrt{3}$.

Step 5. Compare with $a + b\sqrt{3}$: $a = 2$, $b = 1$.

» **Why this step?** Two expressions of the form $a + b\sqrt{c}$ are equal exactly when the rational parts match and the surd coefficients match (uniqueness of the representation).

Answer: a = 2, b = 1

Hinglish: *Rationalise karo, simplify karo, phir 'compare coefficients' karo. 2 + √3 vs a + b√3: rational part rational se match (a=2), surd coefficient surd coefficient se (b=1). Ye uniqueness isliye chalti hai kyunki agar a≠a' hota toh √3 rational nikal jata - impossible.*

QUESTION 79 | Difficult | Irrationals in a small interval (T12)

Q79. Insert two irrational numbers between 0.5 and 0.6, and prove your choices are correct.

SOLUTION — step by step

Step 1. Square the endpoints: $0.5^2 = 0.25$ and $0.6^2 = 0.36$.

Step 2. Any non-perfect-square strictly between 0.25 and 0.36 has an irrational root strictly between 0.5 and 0.6.

Step 3. Choose 0.27 and 0.32: neither is a perfect square of a rational.

Step 4. So $\sqrt{0.27}$ and $\sqrt{0.32}$ lie in (0.5, 0.6): $0.25 < 0.27 < 0.36 \Rightarrow 0.5 < \sqrt{0.27} < 0.6$, similarly for 0.32.

Step 5. $\sqrt{0.27} = \sqrt{27}/\sqrt{100} = 3\sqrt{3}/10$ and $\sqrt{0.32} = 4\sqrt{2}/10 = 2\sqrt{2}/5$ - both are irrational multiples of irrationals divided by rationals, hence irrational.

» **Why this step?** Squaring maps (0.5, 0.6) onto (0.25, 0.36) monotonically; picking non-squares there guarantees irrational roots here.

Answer: For example $\sqrt{0.27} = 3\sqrt{3}/10$ and $\sqrt{0.32} = 2\sqrt{2}/5$ (many answers possible).

Hinglish: *Chhote decimals ke beech irrationals chahiye? Dono ko square karo (0.25 aur 0.36), beech ke non-square numbers pakdo. 0.27 aur 0.32 - inke roots 0.5 aur 0.6 ke beech rahenge. Clean form mein: √0.27 = 3√3/10, √0.32 = 2√2/5. Proof mein square-comparison likhna zaroori hai - wahi 2 marks deta hai.*

QUESTION 80 | Difficult | Square of a surd sum (T8/T9)

Q80. Write $(\sqrt{3} + \sqrt{5})^2$ in the form $m + n\sqrt{k}$, and find between which two consecutive integers it lies.

SOLUTION — step by step

Step 1. Expand: $(\sqrt{3} + \sqrt{5})^2 = 3 + 2\sqrt{15} + 5 = 8 + 2\sqrt{15}$.

Step 2. Estimate $\sqrt{15}$: $3.7^2 = 13.69$, $3.8^2 = 14.44$, $3.9^2 = 15.21$, so $15.21 > 15 > 14.44$ gives $3.8 < \sqrt{15} < 3.9$.

Step 3. So $7.6 < 2\sqrt{15} < 7.8$, and $15.6 < 8 + 2\sqrt{15} < 15.8$.

Step 4. The value lies between the consecutive integers 15 and 16.

Step 5. Check: $\sqrt{15} \approx 3.873$, so $8 + 2(3.873) \approx 15.75$ ✓.

» **Why this step?** Perfect-square bracketing (testing 3.7^2, 3.8^2, 3.9^2) pins the root between tenths without a calculator.

Answer: $8 + 2\sqrt{15}$; it lies between 15 and 16 (approximately 15.75).

Hinglish: *Expand karo: $8 + 2\sqrt{15}$. Ab $\sqrt{15}$ ko bracket karo - $3.8^2 = 14.44$ aur $3.9^2 = 15.21$, matlab $3.8 < \sqrt{15} < 3.9$. Double karo: 7.6 se 7.8. 8 jodo: 15.6 se 15.8. Toh 15 aur 16 ke beech. Bracketing technique akele kaafi hai - calculator ki zaroorat nahi.*

QUESTION 81 | Difficult | Double-stage irrationality proof (T5/T6)

Q81. Prove that $\sqrt{2} + \sqrt{3}$ is irrational, without re-proving $\sqrt{6}$'s irrationality from scratch - first prove $\sqrt{6}$ is irrational, then use it.

SOLUTION — step by step

Step 1. Step 1: suppose $\sqrt{6} = a/b$, coprime. Then $6b^2 = a^2$, so $2 \mid a^2 \rightarrow 2 \mid a$; write $a = 2k$. Then $6b^2 = 4k^2 \rightarrow 3b^2 = 2k^2$, so $2 \mid b^2 \rightarrow 2 \mid b$ - contradicting coprimality. Also $3 \mid a$ gives the same clash. Hence $\sqrt{6}$ is irrational.

Step 2. Step 2: suppose $\sqrt{2} + \sqrt{3} = r$ (rational). Squaring: $5 + 2\sqrt{6} = r^2 \rightarrow \sqrt{6} = (r^2 - 5)/2$, rational - contradicting Step 1.

Step 3. Therefore $\sqrt{2} + \sqrt{3}$ is irrational.

» **Why this step?** Two-stage proofs build on each other: establish the 'lemma' ($\sqrt{6}$ irrational), then leverage it - exactly how mathematicians chain results.

Answer: $\sqrt{6}$ is irrational (lemma); then $\sqrt{2} + \sqrt{3}$ rational would force $\sqrt{6}$ rational - contradiction. Hence irrational.

Hinglish: *Pehle $\sqrt{6}$ ko phasao (2 aur 3 dono uska shikar), phir sum ko square karke $\sqrt{6}$ ko bahar kheencho. Lemma -> main proof. Do-step contradiction, exam ka full-marks answer.*

QUESTION 82 | Difficult | Infinite nested radical (T14)

Q82. Find the value of $x = \sqrt{(12 + \sqrt{(12 + \sqrt{(12 + ...)})})}$.

SOLUTION — step by step

Step 1. The tower repeats inside itself: the part under the first root is again x. So $x^2 = 12 + x$.

Step 2. Solve: $x^2 - x - 12 = 0 \rightarrow (x - 4)(x + 3) = 0 \rightarrow x = 4$ or $x = -3$.

Step 3. The tower is built from positive square roots, so $x > 0$: $x = 4$.

Step 4. Verify: $\sqrt{(12 + 4)} = \sqrt{16} = 4$ ✓ - the tower is self-consistent at 4.

» **Why this step?** Self-similarity turns infinity into algebra: the inside equals the whole, giving a quadratic.

Answer: x = 4

Hinglish: *Andar wala tower bhi wahi x hai! $x^2 = 12 + x \rightarrow x = 4$ (negative wala phenk do). Verify: $\sqrt{(12+4)} = 4$. Infinite cheez ko ek equation mein pakad liya - self-similarity ka kamaal.*

QUESTION 83 | Difficult | Multi-base with prime powers (T11)

Q83. If $2^x = 3^y = 12^z = t$ ($t \neq 1$), prove that $1/z = 2/x + 1/y$.

SOLUTION — step by step

Step 1. From t: $2 = t^{1/x}$, $3 = t^{1/y}$.

Step 2. $12 = 2^2 \times 3 = t^{2/x} \cdot t^{1/y} = t^{2/x + 1/y}$.

Step 3. But $12 = t^{1/z}$. Same base $\rightarrow 2/x + 1/y = 1/z$.

» **Why this step?** Break 12 into its prime pieces ($2^2 \times 3$); each prime donates its own reciprocal exponent, and the sum must equal 1/z.

Answer: 1/z = 2/x + 1/y (proof above)

Hinglish: *12 ko todo: $2^2 \times 3$. Har prime apna exponent-share deta hai: 2/x aur 1/y. Total = 1/z. Prime factorisation hi relation ka naksha hai.*

QUESTION 84 | Difficult | Hidden perfect square chain (T14)

Q84. If $x = 5 + 2\sqrt{6}$, find (i) $\sqrt{x}$, (ii) $1/x$, (iii) $\sqrt{x} + 1/\sqrt{x}$ (in simplest form).

SOLUTION — step by step

Step 1. (i) $5 + 2\sqrt{6} = (\sqrt{3} + \sqrt{2})^2$ ($3 + 2 + 2\sqrt{6}$ ✓) $\rightarrow \sqrt{x} = \sqrt{3} + \sqrt{2}$.

Step 2. (ii) $1/x = 1/(5 + 2\sqrt{6}) = (5 - 2\sqrt{6})/((5)^2 - (2\sqrt{6})^2) = (5 - 2\sqrt{6})/(25 - 24) = 5 - 2\sqrt{6}$.

Step 3. (iii) Let $S = \sqrt{x} + 1/\sqrt{x}$. Then $S^2 = x + 2 + 1/x = (5 + 2\sqrt{6}) + 2 + (5 - 2\sqrt{6}) = 12$.

Step 4. $S = \sqrt{12} = 2\sqrt{3}$.

» **Why this step?** $5 + 2\sqrt{6}$ is a double disguise: a perfect square upstairs and a unit-norm conjugate pair $(5 \pm 2\sqrt{6})$ downstairs.

Answer: (i) $\sqrt{3} + \sqrt{2}$ (ii) $5 - 2\sqrt{6}$ (iii) $2\sqrt{3}$

Hinglish: *$5 + 2\sqrt{6}$ do costume mein hai: $(\sqrt{3}+\sqrt{2})^2$ upar, aur apna hi conjugate reciprocal $(5 - 2\sqrt{6})$ neeche. $S^2 = x + 2 + 1/x = 12 \rightarrow 2\sqrt{3}$. Chain question, ek hi actor ke do roop.*

QUESTION 85 | Difficult | Conjugate pair symmetric powers (T10)

Q85. $x = 2 + \sqrt{3}$ and $y = 2 - \sqrt{3}$. Find (i) $x + y$ and xy, (ii) $x^2 + y^2$, (iii) $x^3 + y^3$.

SOLUTION — step by step

Step 1. (i) $x + y = 4$; $xy = (2 + \sqrt{3})(2 - \sqrt{3}) = 4 - 3 = 1$.

Step 2. (ii) $x^2 + y^2 = (x + y)^2 - 2xy = 16 - 2 = 14$.

Step 3. (iii) $x^3 + y^3 = (x + y)^3 - 3xy(x + y) = 64 - 12 = 52$.

» **Why this step?** Conjugate pairs give sum 4 and product 1 - every symmetric power follows from just those two numbers.

Answer: (i) 4 and 1 (ii) 14 (iii) 52

Hinglish: *Pair ka DNA: sum 4, product 1. Ab sirf identities: 14, 52. Dono value conjugate hain isliye sab kuch symmetric formula se nikla - cube tak bina expand kiye!*

QUESTION 86 | Difficult | Long telescoping chain (T14)

Q86. Evaluate: $1/(\sqrt{2} + 1) + 1/(\sqrt{3} + \sqrt{2}) + 1/(\sqrt{4} + \sqrt{3}) + \ldots + 1/(\sqrt{100} + \sqrt{99})$.

SOLUTION — step by step

Step 1. General term rationalises to $\sqrt{(k+1)} - \sqrt{k}$.

Step 2. The chain runs from $k = 1$ to $k = 99$: $(\sqrt{2} - 1) + (\sqrt{3} - \sqrt{2}) + \ldots + (\sqrt{100} - \sqrt{99})$.

Step 3. Every intermediate root cancels - a 99-term sum collapses to $\sqrt{100} - 1$.

Step 4. $= 10 - 1 = 9$.

» **Why this step?** The telescope works no matter how long the chain: first root minus last number is all that survives.

Answer: 9

Hinglish: *99 terms! Par telescope ki wajah se sab beech mein gayab: bacha $10 - 1 = 9$. Lambi chain dikhne mein daraati hai, conjugate ek hi jhatke mein saaf kar deta hai.*

QUESTION 87 | Difficult | Factored power + follow-up (T11)

Q87. (i) Solve $4^x + 4^{x+1} = 80$. (ii) Using that x, evaluate 2^{3x}.

SOLUTION — step by step

Step 1. (i) Factor 4^x: $4^x(1 + 4) = 80$ -> $4^x = 16 = 4^2$ -> x = 2.

Step 2. (ii) $2^{3x} = 2^6 = 64$.

» **Why this step?** $4 = 2^2$ links the two parts: once x is fixed, any base-2 expression follows.

Answer: (i) x = 2 (ii) 64

Hinglish: *4^x common: bracket 5. 80/5 = 16 -> x = 2. Phir 2^(3×2) = 64. Base change (4 → 2) ka bridge yaad rakho.*

QUESTION 88 | Difficult | Quadratic in 2^x (T11)

Q88. Find ALL real values of x satisfying $2^{2x} - 6 \cdot 2^x + 8 = 0$.

SOLUTION — step by step

Step 1. Let $u = 2^x$ (always positive). The equation becomes $u^2 - 6u + 8 = 0$.

Step 2. (u - 2)(u - 4) = 0 -> u = 2 or u = 4 - both positive, both legal.

Step 3. $2^x = 2$ -> x = 1; $2^x = 4$ -> x = 2.

Step 4. Both are genuine solutions: check x = 1: 4 - 12 + 8 = 0 ✓; x = 2: 16 - 24 + 8 = 0 ✓.

» **Why this step?** Substitution turns an exponential into a quadratic; then EACH positive root of u must be converted back separately.

Answer: x = 1 or x = 2

Hinglish: *u = 2^x rakh ke quadratic: u = 2 ya 4. Dono positive -> dono se x niklega: 1 aur 2. Do answers - 'ek hi' likhna yahan sabse badi galti hai.*

QUESTION 89 | Difficult | Three-surd square (T14)

Q89. Expand and simplify $(\sqrt{2} + \sqrt{3} + \sqrt{5})^2$.

SOLUTION — step by step

Step 1. Square = sum of squares + 2 × each pairwise product.

Step 2. Squares: 2 + 3 + 5 = 10.

Step 3. Cross terms: $2\sqrt{2}\sqrt{3} = 2\sqrt{6}$; $2\sqrt{2}\sqrt{5} = 2\sqrt{10}$; $2\sqrt{3}\sqrt{5} = 2\sqrt{15}$.

Step 4. Result: $10 + 2\sqrt{6} + 2\sqrt{10} + 2\sqrt{15}$.

» **Why this step?** $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ - with surds, keep every cross term's root distinct.

Answer: $10 + 2\sqrt{6} + 2\sqrt{10} + 2\sqrt{15}$

Hinglish: *Trinomial square ka surd version: squares jodo (10), phir har jodi ka 2×product. Teen alag roots aaye - $2\sqrt{6}$, $2\sqrt{10}$, $2\sqrt{15}$. Sab likhna zaroori hai.*

QUESTION 90 | Difficult | Giant power comparison (T14)

Q90. Which is largest: 2^{500}, 3^{300}, or 5^{200}?

SOLUTION — step by step

Step 1. Give all three the same exponent 100: $2^{500} = (2^5)^{100} = 32^{100}$.

Step 2. $3^{300} = (3^3)^{100} = 27^{100}$.

Step 3. $5^{200} = (5^2)^{100} = 25^{100}$.

Step 4. $32 > 27 > 25 \rightarrow 32^{100}$ wins: 2^{500} is the largest.

» **Why this step?** Equal exponents make the bases race directly - rewrite each power as $(\text{something})^{100}$.

Answer: 2^{500} ($= 32^{100}$) is the largest

Hinglish: *Common exponent 100 banao: 32^{100} vs 27^{100} vs 25^{100}. 32 jeet gaya -> 2^{500} sabse bada. GCD-style thinking powers mein bhi chalti hai!*

QUESTION 91 | Difficult | Conjugate reciprocal chain (T10)

Q91. $x = (\sqrt{3} + \sqrt{2})/(\sqrt{3} - \sqrt{2})$ and $y = (\sqrt{3} - \sqrt{2})/(\sqrt{3} + \sqrt{2})$. Find (i) $x + y$, (ii) xy, (iii) $x^3 + y^3$.

SOLUTION — step by step

Step 1. Rationalise: $x = (\sqrt{3} + \sqrt{2})^2/1 = 5 + 2\sqrt{6}$; similarly $y = 5 - 2\sqrt{6}$.

Step 2. (i) $x + y = 10$.

Step 3. (ii) $xy = (5 + 2\sqrt{6})(5 - 2\sqrt{6}) = 25 - 24 = 1$.

Step 4. (iii) $x^3 + y^3 = (x + y)^3 - 3xy(x + y) = 1000 - 30 = 970$.

» **Why this step?** Both fractions rationalise into one conjugate pair; then sum 10 and product 1 generate every symmetric power.

Answer: (i) 10 (ii) 1 (iii) 970

Hinglish: *Dono fractions conjugate se saaf: $5 \pm 2\sqrt{6}$. Sum 10, product 1. Cube ka jod: $1000 - 30 = 970$. Rationalise karo, phir symmetric formula - do-step strategy.*

QUESTION 92 | Difficult | Recurring decimal square (T3)

Q92. Show that 2.7̄ (i.e. 2.777...) equals 25/9, and then decide whether √(25/9) is rational or irrational.

SOLUTION — step by step

Step 1. x = 2.777... = 2 + 0.777... = 2 + 7/9 = 25/9.

Step 2. √(25/9) = √25/√9 = 5/3 - rational!

Step 3. Lesson: the square root of a rational MAY be rational (when both parts are perfect squares). Contrast √2, where 2 is not a square.

Step 4. So: rational input does not guarantee irrational output - it depends on the numerator and denominator individually.

» **Why this step?** √(a/b) = √a/√b: the root is rational exactly when a and b are both perfect squares.

Answer: 2.7̄ = 25/9; √(25/9) = 5/3 - rational

Hinglish: *2.777... = 25/9. Uska root 5/3 - rational! Matlab rational ka root hamesha irrational nahi hota - perfect square hone par rational bhi ho sakta hai. √2 wala case alag tha.*

QUESTION 93 | Difficult | Infinite exponent series (T14)

Q93. Find the value of $x = 3^{1/3} \cdot 3^{1/9} \cdot 3^{1/27} \cdot 3^{1/81} \cdot \ldots$ (infinitely many factors).

SOLUTION — step by step

Step 1. Same base -> exponents add: $x = 3^{(1/3 + 1/9 + 1/27 + \ldots)}$.

Step 2. The exponent is a geometric series: 1/3 + 1/9 + 1/27 + ... = (1/3)/(1 - 1/3) = (1/3)/(2/3) = 1/2.

Step 3. So $x = 3^{1/2} = \sqrt{3}$.

Step 4. Sanity: the partial products creep upward (3^(1/3) ≈ 1.442, then ×3^(1/9) ≈ ×1.13...) and settle near 1.732 ✓.

» **Why this step?** Infinite exponent chains collapse through the geometric series sum - exponents add, even forever.

Answer: x = √3

Hinglish: *Exponents jodte jao: GP series 1/3 + 1/9 + ... = 1/2. Toh x = √3. Infinite product ek chhota surd ban gaya - series ka jaadu.*

QUESTION 94 | Difficult | Recurring arithmetic web (T3)

Q94. Given $0.\dot{7} = 7/9$ and $0.\dot{5} = 5/9$, compute (i) $0.\dot{7} \times 0.\dot{5}$, (ii) $0.\dot{9} \div 0.\dot{3}$, (iii) state whether products/quotients of recurring decimals are always rational.

SOLUTION — step by step

Step 1. (i) (7/9)(5/9) = 35/9... wait - that exceeds 1! Correct: (7/9)(5/9) = 35/81 ≈ 0.432 - a terminating-free but perfectly rational fraction.

Step 2. (ii) $0.\dot{9} = 1$ and $0.\dot{3} = 1/3$ -> quotient = 1 ÷ (1/3) = 3.

Step 3. (iii) YES: p/q ÷ r/s operations on rationals stay rational (division by zero aside) - recurring decimals are rationals in costume.

» **Why this step?** Convert first, compute second: the p/q world has clean arithmetic rules, the decimal costume does not.

Answer: (i) 35/81 (ii) 3 (iii) yes - always rational

Hinglish: *Pehle fraction banao, phir operation: 35/81 aur 3. Aur haan - recurring decimals ki koi bhi +, -, ×, ÷ rational hi deti hai. Costume utaro, asli p/q khada hai.*

QUESTION 95 | Difficult | Paired nested radicals (T14)

Q95. Show that $\sqrt{(4 + 2\sqrt{3})} - \sqrt{(4 - 2\sqrt{3})}$ equals 2.

SOLUTION — step by step

Step 1. Let $A = \sqrt{(4 + 2\sqrt{3})}$, $B = \sqrt{(4 - 2\sqrt{3})}$. Compute $(A - B)^2 = A^2 + B^2 - 2AB$.

Step 2. $A^2 + B^2 = (4 + 2\sqrt{3}) + (4 - 2\sqrt{3}) = 8$.

Step 3. $AB = \sqrt{((4 + 2\sqrt{3})(4 - 2\sqrt{3}))} = \sqrt{(16 - 12)} = \sqrt{4} = 2$.

Step 4. $(A - B)^2 = 8 - 4 = 4$ -> $A - B = 2$ (positive root, since $A > B$).

» **Why this step?** Squaring the DIFFERENCE avoids identifying each nested root separately - symmetry ($A^2 + B^2$, AB) does the work.

Answer: $\sqrt{(4 + 2\sqrt{3})} - \sqrt{(4 - 2\sqrt{3})} = 2$

Hinglish: *Alag-alag root nikaalna mehanga hai; difference ko square karo - $A^2 + B^2 = 8$ aur $AB = 2$ free mein aa gaye. $(A-B)^2 = 4$ -> answer 2. Symmetry se khelo!*

QUESTION 96 | Difficult | Fractional exponent algebra (T10)

Q96. Simplify $(x^{1/2} \cdot x^{1/3})/x^{1/6}$, then evaluate the result at x = 64.

SOLUTION — step by step

Step 1. Exponents add on top: 1/2 + 1/3 = 5/6.

Step 2. Divide: 5/6 - 1/6 = 4/6 = 2/3. Expression = $x^{2/3}$.

Step 3. At x = 64: $64^{2/3}$ = (cube root of 64)^2 = 4^2 = 16.

Step 4. Direct check: (8 · 4)/2 = 16 ✓.

» **Why this step?** Fractional exponents obey the same laws: add, subtract, then root-then-power.

Answer: $x^{2/3}$; at x = 64 it equals 16

Hinglish: *Laws same: 1/2 + 1/3 - 1/6 = 2/3. 64 ka cube root 4, square 16. Direct check bhi 16 - fractional exponents dar ke nahi, calculation ke hain.*

QUESTION 97 | Difficult | Four-way power ranking (T14)

Q97. Arrange in increasing order: 2^{60}, 3^{40}, 4^{30}, 5^{20}.

SOLUTION — step by step

Step 1. Common exponent 10: $2^{60} = (2^6)^{10} = 64^{10}$.

Step 2. $3^{40} = (3^4)^{10} = 81^{10}$.

Step 3. $4^{30} = (4^3)^{10} = 64^{10}$.

Step 4. $5^{20} = (5^2)^{10} = 25^{10}$.

Step 5. Order: $25^{10} < 64^{10} = 64^{10} < 81^{10}$ -> $5^{20} < 2^{60} = 4^{30} < 3^{40}$.

» **Why this step?** Pull every power to one shared exponent; the bases then rank themselves - and ties are real ($2^6 = 4^3 = 64$).

Answer: $5^{20} < 2^{60} = 4^{30} < 3^{40}$

Hinglish: *Sabko ^10 banao: 64, 81, 64, 25. Interesting - 2^{60} aur 4^{30} barabar nikle (dono 64^{10})! Order: $5^{20} < 2^{60} = 4^{30} < 3^{40}$. Tie bhi answer hai.*

QUESTION 98 | Difficult | Surd rectangle metrics (T14)

Q98. A rectangle has sides $(\sqrt{5} + \sqrt{3})$ and $(\sqrt{5} - \sqrt{3})$. Find (i) its area, (ii) its perimeter, (iii) which is numerically larger - the area or the perimeter (take $\sqrt{5} \approx 2.236$, $\sqrt{3} \approx 1.732$).

SOLUTION — step by step

Step 1. (i) Area = $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3}) = 5 - 3 = 2$ square units.

Step 2. (ii) Perimeter = $2[(\sqrt{5} + \sqrt{3}) + (\sqrt{5} - \sqrt{3})] = 2(2\sqrt{5}) = 4\sqrt{5}$ units.

Step 3. (iii) $4\sqrt{5} \approx 8.944 > 2$ - the perimeter is much larger.

» **Why this step?** Conjugate sides make the area a whole number (difference of squares) while the surds survive only in the perimeter.

Answer: (i) 2 (ii) $4\sqrt{5} \approx 8.94$ (iii) perimeter is larger

Hinglish: *Conjugate sides = area 2 (bilkul clean!), perimeter $4\sqrt{5} \approx 8.94$. Surds area mein cancel ho gaye, perimeter mein bach gaye. Difference of squares ka rectangle avatar.*

QUESTION 99 | Difficult | Exponent equation pair (T11)

Q99. Solve the system: $2^x \cdot 3^y = 648$ and $2^{x+1} \cdot 3^{y-1} = 432$.

SOLUTION — step by step

Step 1. Factorise $648 = 8 \times 81 = 2^3 \cdot 3^4$. So $x = 3$, $y = 4$.

Step 2. Cross-check with the second equation: $2^4 \cdot 3^3 = 16 \times 27 = 432$ ✓ - consistent!

Step 3. Alternatively: divide the equations: $(2^{x+1}3^{y-1})/(2^x3^y) = 2/3 = 432/648 = 2/3$ ✓.

Step 4. Answer: $x = 3$, $y = 4$.

» **Why this step?** Prime factorisation reads off the exponents; dividing the two equations is the consistency check (or an alternate solution path).

Answer: $x = 3$, $y = 4$

Hinglish: *$648 = 2^3 \cdot 3^4$ -> $x = 3$, $y = 4$. Doosri equation se cross-check: $2^4 \cdot 3^3 = 432$ - perfect. Equations ko divide karna bhi ek trick hai: 2/3 hi aata hai dono taraf.*

QUESTION 100 | Difficult | Perfect-square root theorem (T5)

Q100. Prove: for a positive integer n, √n is rational IF AND ONLY IF n is a perfect square.

SOLUTION — step by step

Step 1. (⇐) If $n = m^2$, then $\sqrt{n} = m$ - plainly rational.

Step 2. (⇒) Suppose $\sqrt{n} = p/q$ in lowest terms. Then $n q^2 = p^2$.

Step 3. Prime-factorise $n = r^{1e1} \dots r^{kek}$. In p^2 every prime exponent is EVEN; matching exponents across $n q^2 = p^2$ forces every e_i plus an even number (from q^2) to be even -> every e_i is even.

Step 4. So $n = (r_1^{e1/2} \dots r_k^{ek/2})^2$ is a perfect square.

Step 5. Hence √n is rational exactly when n is a perfect square - the full characterisation.

» **Why this step?** Parity of prime exponents is the deep reason: squares are exactly the numbers with all-even exponent profiles.

Answer: √n rational ⇔ n is a perfect square (proof above)

Hinglish: *Asli sach: perfect square = har prime ki power EVEN. Agar √n rational nikla toh exponent parity force hoti hai ki n square ho. Isliye √2, √3, √5... sab irrational - unke exponents odd hain. Poora chapter ek theorem mein samaa!*

Level 4 · Boss Zone: advanced + HOTS (Olympiad-lite):

QUESTION 101 | High | √5 irrationality proof (T5)

Q101. Prove that √5 is irrational.

SOLUTION — step by step

Step 1. Assume $\sqrt{5} = p/q$ in lowest terms (p, q coprime integers, $q \neq 0$). Squaring: $p^2 = 5q^2$.

Step 2. So 5 divides p^2; since 5 is prime, 5 divides p. Write $p = 5r$.

Step 3. Substitute: $25r^2 = 5q^2 \Rightarrow q^2 = 5r^2$. So 5 divides q^2, hence 5 divides q.

Step 4. Now 5 divides both p and q, contradicting lowest terms.

Step 5. Therefore the assumption fails and √5 is irrational. ∎

Step 6. Extra rigour: if $5 \mid p^2$, then in the prime factorisation of p^2 the prime 5 appears; every prime in p^2 comes from primes of p in PAIRS, so a single unmatched 5 is impossible unless $5 \mid p$.

» **Why this step?** The pairing argument in prime factorisations is what makes 'prime divides square ⇒ prime divides number' true - stating it earns full method marks.

Answer: √5 is irrational (by contradiction with prime 5).

Hinglish: *Same 4-step skeleton, bas prime 5. Extra marks ke liye 'pairing' wali line likho: square ke prime factorisation mein har prime EVEN times aata hai; 5 sirf ek baar aana impossible. Ye line proof ko watertight banati hai.*

QUESTION 102 | High | Product of rational and irrational (T6)

Q102. (i) Prove that the product of a NONZERO rational number r and an irrational number x is irrational. (ii) Show by example that the product of two irrationals may be rational.

SOLUTION — step by step

Step 1. (i) Suppose rx = s is rational, with r ≠ 0. Then x = s/r.

Step 2. s is rational and r is rational and nonzero, so s/r is rational (Q closed under division by nonzero).

Step 3. So x would be rational - contradiction. Hence rx is irrational. ∎

Step 4. (ii) Counterexamples: $\sqrt{2} \times \sqrt{2} = 2$ (rational); $(1 + \sqrt{2})(1 - \sqrt{2}) = 1 - 2 = -1$ (rational); $\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4$.

Step 5. Moral: no blanket rule for irrational × irrational - each case must be tested; the nonzero condition in (i) is essential because r = 0 gives rx = 0, rational.

» **Why this step?** Division closure replaces subtraction closure here; the r ≠ 0 hypothesis is exactly what lets you divide by r.

Answer: (i) rx is irrational for r ≠ 0 (else x = s/r rational). (ii) e.g. $\sqrt{2} \times \sqrt{2} = 2$.

Hinglish: *(i) mein 'nonzero' word critical hai - r = 0 hote hi product 0 (rational) ban jata hai. Proof wahi hai: maan lo rational, divide karo, contradiction. (ii) ke liye $\sqrt{2} \times \sqrt{2} = 2$ perfect example. True/false questions mein ye do cases sabse zyada puchhe jate hain.*

QUESTION 103 | High | Conjugate pair x, y (T14)

Q103. If $x = (\sqrt{3} + 1)/(\sqrt{3} - 1)$ and $y = (\sqrt{3} - 1)/(\sqrt{3} + 1)$, find $x^2 + y^2$ and xy.

SOLUTION — step by step

Step 1. Rationalise x: multiply by $(\sqrt{3} + 1)$: $x = (\sqrt{3} + 1)^2/(3 - 1) = (4 + 2\sqrt{3})/2 = 2 + \sqrt{3}$.

Step 2. Rationalise y: multiply by $(\sqrt{3} - 1)$: $y = (\sqrt{3} - 1)^2/(3 - 1) = (4 - 2\sqrt{3})/2 = 2 - \sqrt{3}$.

Step 3. x and y are conjugates: $xy = (2 + \sqrt{3})(2 - \sqrt{3}) = 4 - 3 = 1$.

Step 4. $x + y = 4$. Then $x^2 + y^2 = (x + y)^2 - 2xy = 16 - 2 = 14$.

Step 5. Check: $x \approx 3.732$, $y \approx 0.268$; $x^2 \approx 13.93$, $y^2 \approx 0.0718$, sum ≈ 14.0 ✓.

» **Why this step?** Reduce both fractions to conjugate form FIRST; then the symmetric identities $x^2 + y^2 = (x+y)^2 - 2xy$ do the rest cheaply.

Answer: $x^2 + y^2 = 14$ and $xy = 1$

Hinglish: *Dono fractions rationalise karo - $x = 2+\sqrt{3}$, $y = 2-\sqrt{3}$ niklega. Ab conjugate pair ke superpowers: $xy = 1$ hamesha, $x+y = 4$. x^2+y^2 ke liye direct square karne ki zaroorat nahi - $(x+y)^2 - 2xy = 16 - 2 = 14$. Symmetric identities se time bachta hai aur galti bhi nahi hoti.*

QUESTION 104 | High | Multi-base exponent relation (T11)

Q104. If $2^x = 3^y = 12^z = t$ ($t \neq 1$), prove that $2/x + 1/y = 1/z$.

SOLUTION — step by step

Step 1. Write each base as a power of the common value t: 2 = t^(1/x), 3 = t^(1/y), 12 = t^(1/z).

Step 2. Use the factorisation $12 = 2^2 \times 3$.

Step 3. So t^(1/z) = (t^(1/x))^2 × t^(1/y) = t^(2/x + 1/y).

Step 4. Same base t ($t \neq 1$) forces equal exponents: 1/z = 2/x + 1/y. ∎

Step 5. Sanity check with numbers: x = $\log_2 12 \approx 3.585$, y ≈ 2.262, z = 1 gives 2/3.585 + 1/2.262 ≈ 0.558 + 0.442 = 1 = 1/1 ✓.

» **Why this step?** The common value t converts three separate equations into one base; then the prime factorisation of 12 writes the relation.

Answer: 1/z = 2/x + 1/y (proved via t-powers and $12 = 2^2 \times 3$).

Hinglish: *Ye 'common value' technique Olympiad ka bread-and-butter hai. Teeno numbers ko t ke power mein likho, phir 12 = 2²×3 ka use karo. Exponents add hote- hote relation ban jata hai. Base same aur t ≠ 1 - tabhi exponent equate karna valid hai, ye condition likhna mat bhoolna.*

QUESTION 105 | High | Conjugate subtraction chain (T9)

Q105. Evaluate: (√5 - 2)/(√5 + 2) - (√5 + 2)/(√5 - 2).

SOLUTION — step by step

Step 1. Rationalise each fraction separately.

Step 2. First: $(\sqrt{5} - 2)^2/((\sqrt{5} + 2)(\sqrt{5} - 2)) = (5 - 4\sqrt{5} + 4)/1 = 9 - 4\sqrt{5}$.

Step 3. Second: $(\sqrt{5} + 2)^2/1 = 5 + 4\sqrt{5} + 4 = 9 + 4\sqrt{5}$.

Step 4. Subtract: $(9 - 4\sqrt{5}) - (9 + 4\sqrt{5}) = -8\sqrt{5}$.

Step 5. Check: √5 ≈ 2.236; first fraction ≈ 0.236/4.236 ≈ 0.0557, second ≈ 4.236/0.236 ≈ 17.94; difference ≈ -17.89, and -8(2.236) = -17.89 ✓.

» **Why this step?** Rationalising first turns both fractions into exact conjugate expressions; then only like terms subtract.

Answer: $-8\sqrt{5}$

Hinglish: *Dono fractions ko alag-alag rationalise karo - denominators dono baar 1 niklenge (√5+2)(√5-2) = 1. Pehla 9 - 4√5, doosra 9 + 4√5. Subtract: -8√5. Sign ka dhyan: (a-b)² mein middle term MINUS hota hai, yahi galti hoti hai.*

QUESTION 106 | High | Sum of powers equation (T11)

Q106. Solve: $2^{(x+1)} + 2^{(x-1)} = 80$.

SOLUTION — step by step

Step 1. Factor out the smaller power $2^{(x-1)}$: $2^{(x-1)}(2^2 + 1) = 80$.

Step 2. So $5 \cdot 2^{(x-1)} = 80$, giving $2^{(x-1)} = 16 = 2^4$.

Step 3. Equate exponents: x - 1 = 4, so x = 5.

Step 4. Check: $2^6 + 2^4 = 64 + 16 = 80$ ✓.

» **Why this step?** Factoring the smaller power converts a sum into a product; the bracket $(2^2 + 1)$ becomes a plain number.

Answer: x = 5

Hinglish: *Q26 ka bhai - same factoring trick. 2^(x-1) bahar nikalo, bracket mein $2^2+1 = 5$. $80/5 = 16 = 2^4$. x = 5. Dekho pattern: jab do powers ka sum/difference diya ho aur exponents ±1 differ karein, factoring pehla move hai.*

QUESTION 107 | High | Nested radicals (T14)

Q107. Simplify: (i) $\sqrt{(7 + 4\sqrt{3})}$, (ii) $\sqrt{(9 - 2\sqrt{14})}$.

SOLUTION — step by step

Step 1. Method: try $\sqrt{(a \pm 2\sqrt{b})} = \sqrt{m} \pm \sqrt{n}$ where m + n = a and mn = b (because $(\sqrt{m} \pm \sqrt{n})^2 = m + n \pm 2\sqrt{(mn)}$).

Step 2. (i) $7 + 4\sqrt{3} = 7 + 2\sqrt{12}$. Need m + n = 7, mn = 12: m = 4, n = 3. So $\sqrt{(7 + 4\sqrt{3})} = 2 + \sqrt{3}$.

Step 3. Verify: $(2 + \sqrt{3})^2 = 4 + 4\sqrt{3} + 3 = 7 + 4\sqrt{3}$ ✓.

Step 4. (ii) $9 - 2\sqrt{14}$: need m + n = 9, mn = 14: m = 7, n = 2. So $\sqrt{(9 - 2\sqrt{14})} = \sqrt{7} - \sqrt{2}$.

Step 5. Verify: $(\sqrt{7} - \sqrt{2})^2 = 7 - 2\sqrt{14} + 2 = 9 - 2\sqrt{14}$ ✓.

» **Why this step?** The identity $(\sqrt{m} \pm \sqrt{n})^2 = (m+n) \pm 2\sqrt{(mn)}$ runs backwards; m+n and mn are read off directly from the expression.

Answer: (i) 2 + √3 (ii) √7 - √2

Hinglish: *Nested radical ka standard recipe: number ko 2√b form mein dekho ($4\sqrt{3} = 2\sqrt{12}$). Phir do numbers dhoondo jinka sum 7 aur product 12 - 4 aur 3! Answer 2 + √3. (ii) mein 9 aur 14 - 7 aur 2. Sign dhyan se: minus wale case mein bada root pehle. Verify karke answer lock karo.*

QUESTION 108 | High | Telescoping surd sum (T14)

Q108. Evaluate: $1/(\sqrt{2}+1) + 1/(\sqrt{3}+\sqrt{2}) + 1/(2+\sqrt{3})$.

SOLUTION — step by step

Step 1. Key identity: $1/(\sqrt{(n+1)}+\sqrt{n}) = \sqrt{(n+1)} - \sqrt{n}$ (multiply by the conjugate; denominator becomes n+1-n = 1).

Step 2. First term: $1/(\sqrt{2}+1) = \sqrt{2} - 1$.

Step 3. Second: $1/(\sqrt{3}+\sqrt{2}) = \sqrt{3} - \sqrt{2}$.

Step 4. Third: $1/(2+\sqrt{3}) = \sqrt{4} - \sqrt{3} = 2 - \sqrt{3}$.

Step 5. Sum: $(\sqrt{2} - 1) + (\sqrt{3} - \sqrt{2}) + (2 - \sqrt{3}) = 2 - 1 = 1$ (everything in between cancels - telescoping).

» **Why this step?** Each conjugate-rationalised term is a difference of consecutive roots; in the written sum every middle root appears once with plus and once with minus.

Answer: 1

Hinglish: *Ye 'telescoping' pattern dekho: har term rationalise karne par root ka difference banta hai: √2-1, √3-√2, 2-√3. Jodne par beech ke saare roots gayab, sirf first ka minus aur last ka plus bachta hai: 2 - 1 = 1. Agar 1/(√(n+1)+√n) type dikhe, turant conjugate socho.*

QUESTION 109 | High | Classification, hard cases (T1)

Q109. Classify each as rational or irrational, with a one-line justification: (i) $(1+\sqrt{2})^2$, (ii) $\sqrt{9}\cdot\sqrt{3}$, (iii) $0.\overline{9}$ i.e. 0.999..., (iv) $(\sqrt{2}+\sqrt{3})(\sqrt{2}-\sqrt{3})$, (v) 1.010010001... (gaps growing), (vi) 6.454545... (block 45).

SOLUTION — step by step

Step 1. (i) $(1+\sqrt{2})^2 = 1 + 2\sqrt{2} + 2 = 3 + 2\sqrt{2}$: irrational (rational + irrational is irrational).

Step 2. (ii) $\sqrt{9}\cdot\sqrt{3} = 3\sqrt{3}$: irrational (nonzero rational × irrational).

Step 3. (iii) 0.999... = 1: rational (proved by shift-and-subtract).

Step 4. (iv) Difference of squares: $(\sqrt{2})^2 - (\sqrt{3})^2 = 2 - 3 = -1$: rational!

Step 5. (v) Gaps keep growing ⇒ no repeating block: irrational.

Step 6. (vi) Recurring ⇒ rational; indeed 6.4545... = 6 + 45/99 = 71/11.

» **Why this step?** Simplify BEFORE classifying - two of these look irrational until the expansion or collapse is carried out.

Answer: (i) irrational (ii) irrational (iii) rational (= 1) (iv) rational (= -1) (v) irrational (vi) rational (= 71/11)

Hinglish: *Trap question set! (i) expand karke dekho - 3+2√2 irrational. (iv) dekh ke lagta hai irrational, par difference of squares se -1 nikalta hai - rational! (iii) 0.999... = 1. Moral: pehle simplify karo, phir classify karo. Direct dekh ke verdict dena yahan paisa doobane wala kaam hai.*

QUESTION 110 | High | Magnification to hundredths (T4)

Q110. Given √5 = 2.2360679..., state the two consecutive hundredths between which √5 lies, and the two consecutive thousandths between which it lies. Then find a rational number strictly between √5 and 2.24.

SOLUTION — step by step

Step 1. Hundredths: 2.23 < 2.2360... < 2.24, so between 2.23 and 2.24.

Step 2. Thousandths: 2.236 < 2.2360679... < 2.237, so between 2.236 and 2.237.

Step 3. For the gap (√5, 2.24): √5 ≈ 2.23607 < 2.24, and the interval is non-empty.

Step 4. Any terminating decimal inside works, e.g. 2.237, 2.238, 2.239.

Step 5. Check: 2.237 > 2.23607 ✓ and 2.237 < 2.24 ✓, so 2.237 is a valid rational in the gap.

» **Why this step?** Terminating decimals are rationals, and the interval (2.23607..., 2.24) contains infinitely many of them - density in action.

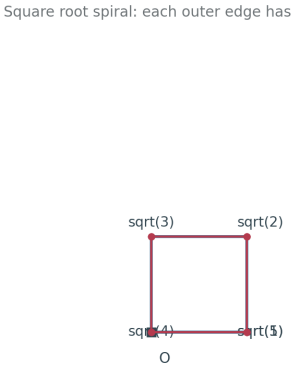
Answer: Between 2.23 and 2.24; between 2.236 and 2.237; e.g. 2.237 (or any decimal in (2.23607, 2.24)) lies between √5 and 2.24.

Hinglish: *Expansion given hai, bas digits follow karo: 2.23 aur 2.24 ke beech, phir 2.236 aur 2.237 ke beech. Last part mein √5 ≈ 2.23607 aur 2.24 ke beech koi bhi terminating decimal - 2.237 perfect. Rationals ki density ka live demo hai ye.*

Level 4 · Boss Zone: advanced + HOTS (Olympiad-lite):

QUESTION 111 | HOTS | Infinite nested radical (T14)

Q111. Find the value of x = √(6 + √(6 + √(6 + √(6 + ...)))) where the pattern continues forever.



SOLUTION — step by step

Step 1. The expression inside is the SAME as the whole: the tail $\sqrt{(6 + \sqrt{(6 + ...)})}$ equals x itself.

Step 2. So x satisfies $x = \sqrt{(6 + x)}$.

Step 3. Square: $x^2 = 6 + x$, i.e. $x^2 - x - 6 = 0$.

Step 4. Factorise: $(x - 3)(x + 2) = 0$, so $x = 3$ or $x = -2$.

Step 5. x is a square root, hence non-negative: $x = 3$.

Step 6. Check: $\sqrt{(6 + 3)} = \sqrt{9} = 3$ ✓ - the value is consistent with itself.

» **Why this step?** Self-similarity lets the infinite tail be replaced by x; the quadratic then follows, and the negative root is discarded on domain grounds.

Answer: x = 3

Hinglish: *Infinite nesting ka trick: andar ka poora structure wahi x hai! $x = \sqrt{(6+x)}$, square karo, quadratic mil gayi. Do roots - 3 aur -2. Root negative nahi ho sakta, toh 3. Ye 'self-similar substitution' technique aage bhi kaam karegi - quadratic banao, domain se filter karo.*

QUESTION 112 | HOTS | Hidden perfect square (T14)

Q112. If $x = 5 + 2\sqrt{6}$, find the value of $\sqrt{x} - 1/\sqrt{x}$.

SOLUTION — step by step

Step 1. Spot the hidden square: $5 + 2\sqrt{6} = 3 + 2 + 2\sqrt{(3\times6/...}$ check: $(\sqrt{3} + \sqrt{2})^2 = 3 + 2\sqrt{6} + 2 = 5 + 2\sqrt{6}$ ✓.

Step 2. So $\sqrt{x} = \sqrt{3} + \sqrt{2}$.

Step 3. Rationalise: $1/\sqrt{x} = 1/(\sqrt{3} + \sqrt{2}) = \sqrt{3} - \sqrt{2}$ (conjugate; denominator 3 - 2 = 1).

Step 4. $\sqrt{x} - 1/\sqrt{x} = (\sqrt{3} + \sqrt{2}) - (\sqrt{3} - \sqrt{2}) = 2\sqrt{2}$.

Step 5. Check numerically: $x \approx 9.899$, $\sqrt{x} \approx 3.146$, $1/\sqrt{x} \approx 0.318$, difference $\approx 2.828 = 2\sqrt{2}$ ✓.

» **Why this step?** Recognising $a + 2\sqrt{b}$ as $(\sqrt{m} + \sqrt{n})^2$ with $m + n = a$, $mn = b$ turns a messy root into two clean surds.

Answer: $2\sqrt{2}$

Hinglish: *$5 + 2\sqrt{6}$ ko dekho aur socho - kaunse do numbers ka sum 5, product 6? 3 aur 2! Matlab $(\sqrt{3}+\sqrt{2})^2$. Toh $\sqrt{x} = \sqrt{3}+\sqrt{2}$. Uska reciprocal conjugate hi hai $(\sqrt{3}-\sqrt{2})$. Difference: $2\sqrt{2}$. Ye 'hidden perfect square' spotting technique bahut powerful hai - m+n aur mn pattern yaad rakho.*

QUESTION 113 | HOTS | Quadratic in 2^x (T11)

Q113. Find ALL real values of x satisfying $4^x - 5 \cdot 2^x + 4 = 0$.

SOLUTION — step by step

Step 1. Write $4^x = (2^2)^x = (2^x)^2$ and let $t = 2^x$ (note $t > 0$ always).

Step 2. The equation becomes $t^2 - 5t + 4 = 0$.

Step 3. Factorise: $(t - 1)(t - 4) = 0$, so $t = 1$ or $t = 4$.

Step 4. Both roots are positive, so both are valid for $t = 2^x$.

Step 5. $2^x = 1 = 2^0 \Rightarrow x = 0$. $2^x = 4 = 2^2 \Rightarrow x = 2$.

Step 6. Check: $x = 0$: $1 - 5 + 4 = 0$ ✓. $x = 2$: $16 - 20 + 4 = 0$ ✓.

» **Why this step?** A disguised quadratic appears whenever one power is the square of another; substituting t collapses it, and the constraint $t > 0$ filters roots.

Answer: x = 0 or x = 2

Hinglish: *4^x ko $(2^x)^2$ likho - quadratic ka chhupa rustam! $t = 2^x$ substitution: $t^2 - 5t + 4 = 0$. Roots 1 aur 4, dono positive - dono valid. $x = 0$ aur $x = 2$. Dhyan: agar koi root negative ya zero aata t ka, wo REJECT karna hota kyunki 2^x kabhi 0 ya negative nahi hota.*

QUESTION 114 | HOTS | General root classification proof (T5)

Q114. Prove: for any positive integer n, √n is either a positive integer or an irrational number (a rational non-integer square root is impossible).

SOLUTION — step by step

Step 1. Suppose $\sqrt{n} = p/q$ in lowest terms. Then $n = p^2/q^2$, so $n\,q^2 = p^2$, i.e. $p^2 = n\,q^2$.

Step 2. Take the prime factorisation of n. In n^2 every prime exponent is EVEN (each exponent doubles).

Step 3. From $p^2 = n\,q^2$: compare the prime-by-prime parity of exponents. On the left (p^2) all exponents are even; on the right, $n\,q^2$ has the exponents of n plus even exponents.

Step 4. So every prime exponent in n must be even - otherwise one prime appears an odd number of times on the right but even on the left, impossible.

Step 5. Therefore n is a perfect square, and $\sqrt{n} = p/q$ is actually an integer (q must be 1 after cancellation... more precisely, √n is the integer obtained from the even exponents).

Step 6. Conclusion: if n is a perfect square, √n is an integer; otherwise √n is irrational. ∎

» **Why this step?** Parity of prime exponents is the invariant: squaring makes every exponent even, so n^2's factorisation can never equal a fraction squared unless n is a square.

Answer: √n is an integer when n is a perfect square, irrational otherwise (proved by prime-exponent parity).

Hinglish: *Ye √2, √3, √5 proofs ka GENERAL version hai. Argument: square mein har prime ka count even hota hai. Agar n perfect square nahi, koi na koi prime odd count mein hoga - aur wo √n ko rational hone se rok dega. Isliye duniya mein sirf do tarah ke roots hain: integers ya irrationals. Bechara 'rational non-integer root' exist hi nahi karta.*

QUESTION 115 | HOTS | Cube of surd sum (T14)

Q115. If $x = \sqrt[3]{3} + 1/\sqrt[3]{3}$, show that $x^3 - 3x = 10/3$.

SOLUTION — step by step

Step 1. Let $a = \sqrt[3]{3}$ and $b = 1/\sqrt[3]{3} = 3^{(-1/3)}$. Then $x = a + b$.

Step 2. Compute $ab = \sqrt[3]{3} \cdot 3^{(-1/3)} = 3^{(1/3 - 1/3)} = 3^0 = 1$.

Step 3. Use $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$.

Step 4. $a^3 = 3$ and $b^3 = 1/3$, so $a^3 + b^3 = 3 + 1/3 = 10/3$.

Step 5. Also $ab = 1$ and $a + b = x$, so $3ab(a+b) = 3x$.

Step 6. Therefore $x^3 = 10/3 + 3x$, i.e. $x^3 - 3x = 10/3$. ∎

» **Why this step?** The cube identity needs a^3, b^3 and ab - and conjugate-style choices make ab collapse to 1, killing all surds.

Answer: $x^3 - 3x = 10/3$ (shown using $ab = 1$ and $a^3 + b^3 = 10/3$).

Hinglish: *Cube identity ka magic: $(a+b)^3 = a^3+b^3+3ab(a+b)$. Yahan $ab = 1$ nikalta hai kyunki $3^{(1/3)}$ aur $3^{(-1/3)}$ ke exponents cancel. $a^3 = 3$, $b^3 = 1/3$. Sab kuch rational ho gaya! $x^3 - 3x = 10/3$. Ye pattern Q25 ke square version ka cube bhai hai - dono yaad rakho.*

QUESTION 116 | HOTS | Long telescoping chain (T14)

Q116. Evaluate: $1/(\sqrt{2}+1) + 1/(\sqrt{3}+\sqrt{2}) + 1/(\sqrt{4}+\sqrt{3}) + \ldots + 1/(\sqrt{16}+\sqrt{15})$.

SOLUTION — step by step

Step 1. Rationalise the general term: $1/(\sqrt{(n+1)} + \sqrt{n}) = \sqrt{(n+1)} - \sqrt{n}$.

Step 2. The sum becomes $(\sqrt{2}-1) + (\sqrt{3}-\sqrt{2}) + (\sqrt{4}-\sqrt{3}) + \ldots + (\sqrt{16}-\sqrt{15})$.

Step 3. It telescopes: $\sqrt{2}$ cancels, $\sqrt{3}$ cancels, ..., $\sqrt{15}$ cancels.

Step 4. Survivors: $-1 + \sqrt{16} = 4 - 1 = 3$.

Step 5. Check with a partial sum: first three terms give $\sqrt{4} - 1 = 1$ (matches Q38) ✓.

» **Why this step?** Each term links $\sqrt{n}$ to $\sqrt{(n+1)}$; in a chain, interior links cancel and only the two endpoints remain.

Answer: 3

Hinglish: *Q38 ka full version - 15 terms! Darne ki zaroorat nahi, pattern same. Har term rationalise karke root-difference banao, poora chain telescopes. Sirf -1 aur +√16 bachta hai: 4 - 1 = 3. Long sums mein hamesha pattern dhoondo - individual terms kabhi add nahi karte.*

QUESTION 117 | HOTS | Density of irrationals (T12)

Q117. Prove that between any two distinct real numbers a < b there lie infinitely many irrational numbers.

SOLUTION — step by step

Step 1. Given a < b. First produce ONE irrational in (a, b): choose an irrational k > 0, e.g. k = √2.

Step 2. Consider numbers of the form a + t·k for small positive t. As t → 0, a + tk → a from above.

Step 3. Since b - a > 0, choose t = (b - a)/(2k); then a + tk = a + (b-a)/2 = (a+b)/2... refine: with k irrational and t rational nonzero, a + tk is irrational (nonzero rational × irrational + rational).

Step 4. Pick t = (b-a)/(4k) scaled rationally... concretely: the number m = (a+b)/2 + δ·k with δ a tiny nonzero rational satisfies a < m < b for |δ| small, and m is irrational.

Step 5. Infinitely many: each distinct tiny rational δ ($1/10^n$ for n = 1, 2, 3, ...) gives a DIFFERENT irrational m_n, all inside (a, b).

Step 6. Hence infinitely many irrationals lie between a and b. ∎

» **Why this step?** The 'rational × irrational + rational' closure result manufactures irrationals on demand, and shrinking δ keeps them in any prescribed gap.

Answer: For any a < b, the numbers (a+b)/2 + δ√2 with infinitely many tiny rationals δ ≠ 0 all lie in (a, b) and are all irrational.

Hinglish: *Density proof ka idea: midpoint pakdo, usme chhota sa irrational tail jodo. (a+b)/2 + δ√2 - δ ko tiny rakho toh wo (a,b) ke andar hi rahega, aur irrational rahega. δ = 1/10, 1/100, 1/1000... har baar NAYA irrational. Isliye 'infinitely many'. Ye proof-style question 5 marks ka hota hai - closure result quote karna zaroori.*

QUESTION 118 | HOTS | Root spiral total length (T13)

Q118. In the square root spiral, consecutive outer edges have lengths √2, √3, √4, ... (the first edge from O has length √1 = 1). Taking √2 = 1.41, √3 = 1.73, √5 = 2.24, √6 = 2.45, √7 = 2.65, find the smallest number of edges (beyond the first) needed for the total length of edges √2 + √3 + ... + √(n+1) to exceed 10.

SOLUTION — step by step

Step 1. The first edge (length 1) is excluded, so we accumulate √2, √3, √4, √5, ...

Step 2. √2 + √3 = 1.41 + 1.73 = 3.14.

Step 3. Add √4 = 2: running total 5.14.

Step 4. Add √5 = 2.24: total 7.38.

Step 5. Add √6 = 2.45: total 9.83 (still below 10).

Step 6. Add √7 = 2.65: total 12.48 > 10. So edges √2 through √7 - that is 6 edges - are needed.

» **Why this step?** Cumulative sums turn a spiral-geometry question into ordered addition; √4 = 2 is exact and stops one rounding error.

Answer: 6 edges (the cumulative length first exceeds 10 at √7, total ≈ 12.48).

Hinglish: *Spiral ke edges jodo - bas running total rakho. √2+√3 = 3.14, plus 2 = 5.14, plus √5 = 7.38, plus √6 = 9.83 (abhi 10 nahi!), plus √7 = 12.48. Matlab 6 edges chahiye. √4 = 2 exact hai - rounding se bach jao. Aise 'smallest n' questions mein ek-ek term add karke threshold cross karna hi method hai.*

QUESTION 119 | HOTS | Irrationality of √2 + √5 (T6)

Q119. (i) Prove that √2 + √5 is irrational. (ii) Find two consecutive integers between which √2 + √5 lies.

SOLUTION — step by step

Step 1. (i) Suppose √2 + √5 = r is rational. Square: $2 + 2\sqrt{10} + 5 = r^2$, so $2\sqrt{10} = r^2 - 7$.

Step 2. Then $\sqrt{10} = (r^2 - 7)/2$, which would be rational.

Step 3. But 10 is not a perfect square, so √10 is irrational - contradiction. Hence √2 + √5 is irrational. ∎

Step 4. (ii) √2 ≈ 1.4142 and √5 ≈ 2.2360 (standard bounds: $1.414^2 = 1.999...$, $1.415^2 > 2$; $2.236^2 \approx 4.9997 < 5 < 2.237^2$).

Step 5. Sum ≈ 3.6502, which lies between the consecutive integers 3 and 4.

Step 6. Refined check: $3.6^2 = 12.96 <$ (value)$^2 = 7 + 2\sqrt{10} \approx 13.325 < 16$, consistent.

» **Why this step?** The square of √2 + √5 mixes into a single surd √10 - one contradiction instead of two; bracketing the decimal expansion places the value.

Answer: (i) √2 + √5 is irrational. (ii) It lies between 3 and 4 (approximately 3.65).

Hinglish: *(i) same square-trick: √10 = (r²-7)/2 rational hota - impossible. (ii) mein dono expansions jodo: 1.414 + 2.236 = 3.65. Toh 3 aur 4 ke beech. Proof aur placement - dono alag skills hain, answer mein dono dikhao.*

QUESTION 120 | HOTS | Infinite exponent product (T14)

Q120. Find the value of the infinite product 2^(1/2) × 4^(1/4) × 8^(1/8) × 16^(1/16) × ... (continuing forever).

SOLUTION — step by step

Step 1. Rewrite every factor as a power of 2: 2^(1/2) × 2^(2/4) × 2^(3/8) × 2^(4/16) × ...

Step 2. Same base → exponents add: the product is 2^S where S = 1/2 + 2/4 + 3/8 + 4/16 + ...

Step 3. Compute S by the shift trick: 2S = 1 + 2/2 + 3/4 + 4/8 + ... (doubling shifts every term up one place).

Step 4. Subtract: 2S - S = 1 + (2/2 - 1/2) + (3/4 - 2/4) + (4/8 - 3/8) + ... = 1 + 1/2 + 1/4 + 1/8 + ...

Step 5. The geometric series 1/2 + 1/4 + 1/8 + ... = 1, so S = 1 + 1 = 2.

Step 6. Therefore the product equals $2^2 = 4$.

Step 7. Partial check: 2^(0.5) × 4^(0.25) × 8^(0.125) × 16^(0.0625) = 1.414 × 1.414 × 1.297 × 1.189 ≈ 3.10 and rising toward 4 ✓.

» **Why this step?** The shift-and-subtract trick that converted recurring decimals also sums the exponent series; geometric decay makes the tail manageable.

Answer: 4

Hinglish: *Sab factors ko 2 ke power mein likho - exponents ka series banta hai: 1/2 + 2/4 + 3/8 + ... Wahi recurring-decimal wala shift trick lagao: 2S - S. Bacha 1 + (1/2 + 1/4 + ...) = 1 + 1 = 2. Product = $2^2 = 4$. Chapter 3 ke saare tools ek saath: exponents, infinite series, shift trick. Ye 5-mark HOTS ka full package hai.*

3.5 Rapid MCQs: Choose the Correct Option (20)

Rapid-fire round: 4 options, ek hi correct. Pehle khud choose karo, phir colored Answer strip se check karo. Har galat option ka "Why wrong" bhi diya hai - taki exam me wahi trap tumhe na phasaye.

MCQ 1 | Normal | Set classification

1. Which of the following is irrational?

A. 0.101001000100001...

B. 0.25

C. 3/8

D. √49

Correct Answer: (A)

The gaps between 1s keep growing, so no block repeats: non-terminating, non-recurring, hence irrational. The others terminate or simplify (0.25, 3/8, √49 = 7).

Why (B) is wrong: Terminating decimal, hence rational.

Why (C) is wrong: Ratio of integers, rational.

Why (D) is wrong: Perfect square under the root: 7.

Hinglish: *Pattern dekho - ones ke beech gap badhta ja raha hai, koi block repeat nahi hoga. Isliye irrational. Baaki teeno rational hain.*

MCQ 2 | Normal | Pure recurring to p/q

2. 0.333... (the 3 repeating) written as a fraction is:

A. 1/3

B. 3/10

C. 33/100

D. 1/30

Correct Answer: (A)

One repeating digit: x = 3/9 = 1/3. Check: 1/3 = 0.333... ✓.

Why (B) is wrong: 3/10 = 0.3, terminates - no repetition.

Why (C) is wrong: 33/100 = 0.33 terminates.

Why (D) is wrong: Extra zero in denominator: 1/30 = 0.0333...

Hinglish: *Ek digit repeat = upar wo digit, neeche 9. 3/9 = 1/3. Denominator 10 wale options terminating decimals dete hain - wo galat.*

MCQ 3 | Normal | Terminating test

3. Without division, the decimal expansion of 77/210 is:

A. terminating

B. non-terminating recurring

C. non-terminating non-recurring

D. cannot be decided

Correct Answer: (B)

77/210 = 11/30 (cancel 7). 30 = 2 × 3 × 5 has the prime 3, so it recurs.

Why (A) is wrong: 30 is not of the form $2^m \times 5^n$.

Why (C) is wrong: Every rational recurs or terminates - never truly random.

Why (D) is wrong: The test always decides.

Hinglish: *Pehle 7 cancel karo - 11/30 banta hai. 30 mein 3 hai, toh recurring. Ye reduce-first wala hi case hai.*

MCQ 4 | Normal | Set nesting

4. Which statement is TRUE?

A. $Z \subset N$

B. $Q \subset Z$

C. $N \subset Q$

D. $R \subset Q$

Correct Answer: (C)

Naturals sit inside integers, integers inside rationals, rationals inside reals: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. So $\mathbb{N} \subset \mathbb{Q}$ is true.

Why (A) is wrong: Negatives are in Z but not N.

Why (B) is wrong: Fractions like 1/2 are in Q but not Z.

Why (D) is wrong: Irrationals are in R but not Q - R is bigger.

Hinglish: *Nesting yaad karo: N sabse andar, phir Z, phir Q, sabse bahar R. C options wala direction sahi hai.*

MCQ 5 | Normal | Mixed recurring to p/q

5. 0.4777... (the 4 not repeating, the 7 repeating) equals:

A. 43/90

B. 47/100

C. 47/90

D. 43/99

Correct Answer: (A)

(47 - 4)/90 = 43/90: one 9 for the repeating digit, one 0 for the fixed digit.

Why (B) is wrong: That is 0.47 exactly, terminating.

Why (C) is wrong: Denominator pattern wrong for mixed recurrence.

Why (D) is wrong: 99 would fit 0.434343...

Hinglish: *(poora - head)/(9 + 0): (47-4)/90 = 43/90. Option C wala 47/90 hota agar 4 bhi repeat karta.*

MCQ 6 | Normal | Surd multiplication

6. $\sqrt{2} \times \sqrt{8}$ equals:

A. $\sqrt{10}$

B. 4

C. $2\sqrt{4}$

D. 16

Correct Answer: (B)

$\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4$. Roots multiply by multiplying radicands.

Why (A) is wrong: Radicands are multiplied, not added.

Why (C) is wrong: $2\sqrt{4} = 4$ by luck of notation, but $\sqrt{2} \cdot \sqrt{8} = \sqrt{16}$ directly; writing $2\sqrt{4}$ shows wrong method.

Why (D) is wrong: Forgot to take the root at the end.

Hinglish: *$\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4$. Under-root multiply hota hai, add nahi.*

MCQ 7 | Normal | Comparing surds

7. Which is the largest?

A. $2\sqrt{3}$

B. $3\sqrt{2}$

C. 4

D. $\sqrt{15}$

Correct Answer: (B)

Squares: 12, 18, 16, 15. The largest square is 18, so $3\sqrt{2}$ is largest.

Why (A) is wrong: 12 is the smallest square here.

Why (C) is wrong: 16 < 18.

Why (D) is wrong: 15 < 18. Coefficient se confuse hona common galti hai.

Hinglish: *Sabko square karo: $2\sqrt{3} \rightarrow 12$, $3\sqrt{2} \rightarrow 18$, $4 \rightarrow 16$, $\sqrt{15} \rightarrow 15$. Sabse bada square 18 - winner $3\sqrt{2}$.*

MCQ 8 | Normal | Conjugate with 1

8. $1/(\sqrt{5} - 2)$ equals:

A. $\sqrt{5} + 2$

B. $\sqrt{5} - 2$

C. $(\sqrt{5}+2)/3$

D. $1/3$

Correct Answer: (A)

Multiply by $(\sqrt{5} + 2)$: denominator $(\sqrt{5})^2 - 2^2 = 5 - 4 = 1$. So the value is $\sqrt{5} + 2$.

Why (B) is wrong: That is the original denominator, not the value.

Why (C) is wrong: Wrong denominator (3 would come from $\sqrt{9}$).

Why (D) is wrong: No rationalising done at all.

Hinglish: *Conjugate multiply karo, denominator 5-4 = 1 nikla. Answer khud $\sqrt{5}+2$ hai - magic collapse.*

MCQ 9 | Normal | Difference of squares

9. $(\sqrt{7} + \sqrt{5})(\sqrt{7} - \sqrt{5})$ equals:

A. 2

B. 12

C. $\sqrt{2}$

D. 35

Correct Answer: (A)

$(\sqrt{7})^2 - (\sqrt{5})^2 = 7 - 5 = 2$.

Why (B) is wrong: Added the squares instead of subtracting.

Why (C) is wrong: Took root of the difference.

Why (D) is wrong: Multiplied radicands - that is $\sqrt{35}$, a different expression.

Hinglish: *$(a+b)(a-b) = a^2-b^2$. Roots ke square = andar wale numbers. 7-5 = 2.*

MCQ 10 | Normal | 0.999... concept

10. Which statement is TRUE?

A. 0.999... is exactly 1

B. 0.999... is slightly less than 1

C. 0.999... is irrational

D. 0.999... is undefined

Correct Answer: (A)

x = 0.999... gives 10x - x = 9, so x = 1 exactly. It is a rational number with two decimal expansions.

Why (B) is wrong: Common intuition, but no digit-level gap survives.

Why (C) is wrong: Recurring decimals are rational by definition.

Why (D) is wrong: The infinite sum has a well-defined value.

Hinglish: *Ye mind-bender hai: 0.999... EXACTLY 1 hai. Shift-subtract se 9x = 9. 'Thoda kam' wala feeling galat hai - maths mein exact proof jeetta hai.*

MCQ 11 | Normal | Fractional exponents

11. (16/81)^(-1/4) equals:

A. 3/2

B. 2/3

C. -3/2

D. 81/16

Correct Answer: (A)

Negative exponent flips: (81/16)^(1/4) = $\sqrt[4]{81} / \sqrt[4]{16} = 3/2$.

Why (B) is wrong: Forgot to flip for the negative exponent.

Why (C) is wrong: Root lene ke baad bhi minus rakh diya - root value negative nahi hoti.

Why (D) is wrong: Ignored the exponent entirely.

Hinglish: *Minus matlab flip, 1/4 matlab fourth root. (81/16)^(1/4) = 3/2. Root pehle lo, numbers chhote rahenge.*

MCQ 12 | Normal | Mixed recurring with integer

12. 2.1666... (the 6 repeating) as a fraction is:

A. 13/6

B. 21/10

C. 13/60

D. 11/6

Correct Answer: (A)

x = 2.1666..., 10x = 21.666..., 100x = 216.66...; 90x = 195, x = 195/90 = 13/6.

Why (B) is wrong: That is 2.1, ignoring the repeating tail.

Why (C) is wrong: Denominator 60 = wrong shift.

Why (D) is wrong: 11/6 = 1.8333, wrong value.

Hinglish: *10x aur 100x banao, minus: 195/90, reduce: 13/6. Check: 13/6 = 2.1666 ✓.*

MCQ 13 | HOTS | Conjugate pair sum

13. If A = (√5+√3)/(√5-√3) and B = (√5-√3)/(√5+√3), then A + B equals:

A. 8

B. 16

C. 4√15

D. 2

Correct Answer: (A)

A = 4 + √15 and B = 4 - √15 (each by rationalising; see Q14's method). Sum = 8.

Why (B) is wrong: Doubled A instead of adding A and B.

Why (C) is wrong: Only the surd part remained - forgetting the 4+4.

Why (D) is wrong: Subtracted instead of added.

Hinglish: *Dono rationalise karo: A = 4+√15, B = 4-√15. Jodo - surd cancel, 8. Conjugate pairs ka sum hamesha surd-free hota hai.*

MCQ 14 | Normal | Proof method

14. The standard method to prove that √3 is irrational is:

A. contradiction (assume it is rational)

B. induction

C. direct computation

D. counterexample

Correct Answer: (A)

Assume √3 = p/q in lowest terms, derive that 3 divides both p and q, contradict coprimality.

Why (B) is wrong: Induction is for integer statements.

Why (C) is wrong: You can never compute an irrational exactly.

Why (D) is wrong: Counterexamples disprove claims; they cannot prove irrationality.

Hinglish: *Irrationality proofs hamesha contradiction se hote hain. Assume karo rational hai, phir ussi assumption ko todo.*

MCQ 15 | Normal | Conjugate collapse

15. 1/(√9 - √8) equals:

A. 3 + 2√2

B. 3 - 2√2

C. 1

D. 3 + √8

Correct Answer: (A)

√9 = 3, √8 = 2√2. Multiply by (3 + 2√2): denominator 9 - 8 = 1, value 3 + 2√2.

Why (B) is wrong: Conjugate ka sign ulta kar diya.

Why (C) is wrong: 1/(3-2√2) ≠ 1 - compute karo.

Why (D) is wrong: Rationalise kiye bina sirf sign badla.

Hinglish: *Pehle roots simplify: √9 = 3, √8 = 2√2. Phir conjugate - denominator 1. Answer 3+2√2.*

MCQ 16 | Normal | Rationalising collapse

16. 12/(√7 + √5) equals:

A. 6(√7 - √5)

B. √7 - √5

C. 6(√7 + √5)

D. 12√2

Correct Answer: (A)

Multiply by (√7 - √5): denominator 7 - 5 = 2, so value = 12(√7-√5)/2 = 6(√7-√5).

Why (B) is wrong: Forgot the factor 12/2 = 6.

Why (C) is wrong: Used the non-conjugate - denominator does not clear.

Why (D) is wrong: Added radicands illegally.

Hinglish: *Conjugate se denominator 2 aata hai, 12/2 = 6 upar. Answer 6(√7-√5).*

MCQ 17 | HOTS | Infinite power product

17. 2^(1/2) × 2^(1/4) × 2^(1/8) × ... (forever) equals:

A. 2

B. 4

C. 1

D. ∞

Correct Answer: (A)

Exponent sum = 1/2 + 1/4 + 1/8 + ... = 1 (geometric). So the product is $2^1 = 2$.

Why (B) is wrong: That is Q50's cousin with weights $k/2^k$.

Why (C) is wrong: Series sums to 1, not 0.

Why (D) is wrong: The tail shrinks to nothing - the sum is finite.

Hinglish: *Same base - exponents jodo: 1/2+1/4+1/8+... = 1. Product $2^1 = 2$. Geometric series ka sum 1 hai.*

MCQ 18 | HOTS | Hidden square root

18. If $x = 5 + 2\sqrt{6}$, then $\sqrt{x}$ equals:

A. $\sqrt{3} + \sqrt{2}$

B. 3 + 2

C. $\sqrt{5} + \sqrt{6}$

D. $\sqrt{(5+2\sqrt{6})}$ cannot be simplified

Correct Answer: (A)

$(\sqrt{3} + \sqrt{2})^2 = 3 + 2\sqrt{6} + 2 = 5 + 2\sqrt{6}$. So $\sqrt{x} = \sqrt{3} + \sqrt{2}$.

Why (B) is wrong: $(3+2)^2 = 25 \neq x$.

Why (C) is wrong: $(\sqrt{5}+\sqrt{6})^2 = 11+2\sqrt{30}$.

Why (D) is wrong: It CAN be simplified - spot the hidden square.

Hinglish: *Hidden square dhoondo: sum 5, product 6 - √3 aur √2. (option B ka 3+2 = 5 ka square 25 hota, galat)*

MCQ 19 | Normal | Prime lemma

19. If a PRIME p divides a^2, then:

A. p divides a

B. a divides p

C. p^2 divides a^2 always

D. a is prime

Correct Answer: (A)

Primes appear an even number of times in any square, so a lone p in a^2 must come from a p in a.

Why (B) is wrong: Backwards.

Why (C) is wrong: 8 divides $4^2 = 16$ but $8^2 = 64$ does not divide 16.

Why (D) is wrong: a need not be prime (a = 6, p = 2).

Hinglish: *Prime-square lemma: $p \mid a^2 \Rightarrow p \mid a$. Ye irrationality proof ka engine hai. Caution: ye composite ke liye fail hota hai ($8 \mid 4^2$ par $8 \nmid 4$).*

MCQ 20 | Normal | Mixed closure

20. The sum of a rational number and an irrational number is:

A. always irrational

B. always rational

C. sometimes rational

D. always an integer

Correct Answer: (A)

If r + x were rational, x = (r + x) - r would be rational - contradiction. So it is always irrational.

Why (B) is wrong: Contradicts the subtraction argument.

Why (C) is wrong: Never - the proof is airtight.

Why (D) is wrong: No reason for integrality.

Hinglish: *Rational + irrational = hamesha irrational. Proof subtraction trick se - Q minus Q = Q.*

3.6 Choose the Correct Round: Theory & Speed (20)

Doosra rapid round - har MCQ ka style: statement padho, correct option turant chuno. Ye round theory, definitions aur quick solving ko test karta hai. Answer strip ke neeche har galat option ka "Why wrong" bhi hai - taki exam ka trap tumhe na phasaye.

CTC 1 | Normal | Between-ness

1. An irrational number between 3 and 4 is:

A. $\sqrt{8}$

B. $\sqrt{9}$

C. $\sqrt{10}$

D. 3.14

Correct Answer: (C)

$3^2 = 9 < 10 < 16 = 4^2$, so $3 < \sqrt{10} < 4$, and 10 is not a perfect square, so $\sqrt{10}$ is irrational.

Why (A) is wrong: $\sqrt{8} \approx 2.83 < 3$ - outside the interval.

Why (B) is wrong: $\sqrt{9} = 3$ exactly (and rational).

Why (D) is wrong: 3.14 terminates, hence rational.

Hinglish: *3 aur 4 ke square = 9 aur 16. Beech ka non-square: 10, 11, 12, ... 15. √10 perfect choice.*

CTC 2 | Normal | Decimal by denominator test

2. The decimal expansion of 33/125 is:

A. 0.264

B. 0.2640

C. non-terminating

D. 0.26326432643...

Correct Answer: (A)

$125 = 5^3$, only 5s, so it terminates: $33/125 = 33 \times 8/1000 = 264/1000 = 0.264$.

Why (B) is wrong: 0.2640 = 0.264 - same number, but the intended exact form is 0.264.

Why (C) is wrong: $125 = 5^3$ passes the $2^m \times 5^n$ test.

Why (D) is wrong: That is a recurring pattern - impossible for denominator 125.

Hinglish: *125 ko 1000 banane ke liye 8 se multiply karo upar-neeche. 0.264 ekdum clean.*

CTC 3 | Normal | Same-base equation

3. If $3^x = 1/243$, then x =

A. 5

B. -5

C. 1/5

D. -243

Correct Answer: (B)

$1/243 = 1/3^5 = 3^{-5}$, so x = -5.

Why (A) is wrong: $3^5 = 243$, not 1/243.

Why (C) is wrong: Fractional exponent confusion.

Why (D) is wrong: Equating x with the value, not the exponent.

Hinglish: *1/243 ko 3^{-5} likho. Negative exponent = reciprocal. x = -5.*

CTC 4 | Normal | Conjugate reciprocal

4. If $x = 3 + 2\sqrt{2}$, then $1/x$ equals:

A. $3 - 2\sqrt{2}$

B. $2\sqrt{2} - 3$

C. $1/(3+2\sqrt{2})$ simplified to $1/3$

D. 6

Correct Answer: (A)

$1/(3+2\sqrt{2}) = (3-2\sqrt{2})/((3)^2-(2\sqrt{2})^2) = (3-2\sqrt{2})/(9-8) = 3 - 2\sqrt{2}$.

Why (B) is wrong: Sign flipped on the wrong term.

Why (C) is wrong: No simplification attempted.

Why (D) is wrong: That is $x + 1/x = 6$, a different question.

Hinglish: *Conjugate multiply karo, denominator 9-8 = 1. Toh 1/x khud conjugate hai. x aur 1/x ka pair yaad rakho.*

CTC 5 | Normal | Ordering roots

5. Arranged in DECREASING order, which sequence is correct?

A. $\sqrt{5}, \sqrt{6}, \sqrt{7}$

B. $\sqrt{7}, \sqrt{6}, \sqrt{5}$

C. $\sqrt{6}, \sqrt{5}, \sqrt{7}$

D. $\sqrt{7}, \sqrt{5}, \sqrt{6}$

Correct Answer: (B)

For the same root index, bigger radicand means bigger value: $\sqrt{7} > \sqrt{6} > \sqrt{5}$.

Why (A) is wrong: Reversed the order.

Why (C) is wrong: Scrambled - $\sqrt{5}$ is the smallest.

Why (D) is wrong: $\sqrt{5}$ is not bigger than $\sqrt{6}$.

Hinglish: *Same root, toh andar bada number = value badi. $\sqrt{7}$ sabse bada. Seedha rule hai.*

CTC 6 | Normal | Which is rational

6. Which of these is rational?

A. $\sqrt{12}/\sqrt{3}$

B. $\sqrt{2} + \sqrt{3}$

C. $\sqrt{2} \times \sqrt{3}$

D. $\sqrt{2} - \sqrt{3}$

Correct Answer: (A)

$\sqrt{12}/\sqrt{3} = \sqrt{(12/3)} = \sqrt{4} = 2$. The others are irrational sums/products/differences of unlike surds.

Why (B) is wrong: Unlike surds never simplify to rationals.

Why (C) is wrong: $\sqrt{6}$ is irrational (6 not a perfect square).

Why (D) is wrong: Same: irrational, about -0.318.

Hinglish: *Divide karke simplify karo: $\sqrt{4} = 2$. Baaki teeno irrational hain - $\sqrt{6}$, $\sqrt{2}+\sqrt{3}$, $\sqrt{2}-\sqrt{3}$.*

CTC 7 | Normal | Same-base equation 2

7. If $2^{(2x)} = 64$, then x =

A. 3

B. 6

C. 2

D. 32

Correct Answer: (A)

$64 = 2^6$, so $2x = 6$ and $x = 3$.

Why (B) is wrong: Forgot the 2x and equated x = 6.

Why (C) is wrong: Divided by 3 instead of 2.

Why (D) is wrong: Took half of 64.

Hinglish: *$2^{(2x)} = 2^6$, exponents compare: $2x = 6$, $x = 3$. Option B wale 2x ko x samajh bethe.*

CTC 8 | Normal | Square of surd sum

8. $(\sqrt{3} + \sqrt{2})^2$ equals:

A. $5 + 2\sqrt{6}$

B. $5 + \sqrt{6}$

C. 13

D. 5

Correct Answer: (A)

$3 + 2\sqrt{6} + 2 = 5 + 2\sqrt{6}$. The cross term $2\sqrt{3}\sqrt{2} = 2\sqrt{6}$ is essential.

Why (B) is wrong: Missed the factor 2 on the cross term.

Why (C) is wrong: Squaring the radicands wrongly: 3 + 2 + ... no.

Why (D) is wrong: Dropped the cross term completely.

Hinglish: *$(a+b)^2 = a^2 + 2ab + b^2$. Cross term $2\sqrt{6}$ bhoolna sabse common galti. $5 + 2\sqrt{6}$.*

CTC 9 | HOTS | Negative fractional exponent

9. 64^(-2/3) equals:

A. 1/16

B. -1/16

C. 16

D. 1/48

Correct Answer: (A)

64^(1/3) = 4, then $4^2 = 16$, then flip for the minus: 1/16.

Why (B) is wrong: Minus sign ko value par laga diya - flip hota hai.

Why (C) is wrong: Forgot the reciprocal.

Why (D) is wrong: Added 2 and 3 somewhere - order matters.

Hinglish: *Root pehle: cube root of 64 = 4. Square: 16. Minus: flip → 1/16. Order: root → power → flip.*

CTC 10 | HOTS | Factored power difference

10. If $3^x - 3^{(x-1)} = 162$, then x =

A. 5

B. 4

C. 6

D. 5.5

Correct Answer: (A)

Factor: $3^{(x-1)}(3 - 1) = 162 \Rightarrow 3^{(x-1)} = 81 = 3^4 \Rightarrow x = 5$.

Why (B) is wrong: Solved x - 1 = 3 (162/2 = 81 misread as 27).

Why (C) is wrong: Took $3^4 = 243$ confusion.

Why (D) is wrong: Averaged instead of solving.

Hinglish: *3^(x-1) common nikalo, bracket mein 2. 162/2 = 81 = 3^4. x - 1 = 4, x = 5.*

CTC 11 | Normal | Closure true/false

11. Which statement is FALSE?

A. rational + rational is rational

B. irrational + irrational is always irrational

C. rational × irrational (rational ≠ 0) is irrational

D. integer + integer is an integer

Correct Answer: (B)

$\sqrt{2} + (-\sqrt{2}) = 0$: two irrationals can sum to a rational. The other statements are genuine closure facts.

Why (A) is wrong: True - Q is closed under addition.

Why (C) is wrong: True with the nonzero condition.

Why (D) is wrong: True - Z is closed under addition.

Hinglish: *B galat hai - counterirritant: √2 aur -√2 jodne par 0. Irrationals ke beech koi closure rule NAHI hai.*

CTC 12 | HOTS | Nested radical

12. $\sqrt{(11 + 2\sqrt{30})}$ equals:

A. $\sqrt{5} + \sqrt{6}$

B. $5 + 6$

C. $\sqrt{30} + 1$

D. $\sqrt{11} + \sqrt{30}$

Correct Answer: (A)

Need $m + n = 11$, $mn = 30$: $m = 5$, $n = 6$. So the root is $\sqrt{5} + \sqrt{6}$ (verify: $5 + 6 + 2\sqrt{30} = 11 + 2\sqrt{30}$).

Why (B) is wrong: 5+6 ka square 121 hota hai.

Why (C) is wrong: Product 30 ke sath sum 31 hoga.

Why (D) is wrong: Roots kabhi aise alag-alag nahi judte.

Hinglish: *Sum 11, product 30 - 5 aur 6. Answer √5 + √6. Verify karke lock karo.*

CTC 13 | HOTS | Conjugate squares

13. If $x = 2 + \sqrt{3}$, then $x^2 - 1/x^2$ equals:

A. $8\sqrt{3}$

B. 14

C. $4\sqrt{3}$

D. $2\sqrt{3}$

Correct Answer: (A)

$x^2 = 7 + 4\sqrt{3}$ and $1/x^2 = 7 - 4\sqrt{3}$. Difference = $8\sqrt{3}$.

Why (B) is wrong: That is $x^2 + 1/x^2$ (the sum, not the difference).

Why (C) is wrong: Halved the surd coefficient.

Why (D) is wrong: Used $x - 1/x$ instead.

Hinglish: *x^2 aur $1/x^2$ conjugates hain: $7 \pm 4\sqrt{3}$. Minus karo - rational parts cancel, $8\sqrt{3}$ bacha.*

CTC 14 | Normal | Recurring arithmetic

14. $0.\overline{6}$ (i.e. 0.666...) + $0.\overline{3}$ (i.e. 0.333...) equals:

A. 1

B. 0.999... but not 1

C. 0.9

D. 2/3

Correct Answer: (A)

2/3 + 1/3 = 1 exactly (and 0.999... IS 1).

Why (B) is wrong: 0.999... IS 1 - no gap.

Why (C) is wrong: 0.9 terminates; wrong value.

Why (D) is wrong: 2/3 is only half the answer.

Hinglish: *0.666... = 2/3, 0.333... = 1/3. Sum = 1. Aur 0.999... bhi exactly 1 hai - consistent!*

CTC 15 | Normal | Successive magnification

15. Given $\sqrt{7}$ = 2.6457513..., $\sqrt{7}$ lies between:

A. 2.64 and 2.65

B. 2.6 and 2.7

C. 2.645 and 2.646

D. both A and C are true

Correct Answer: (D)

2.6457... sits between 2.64 and 2.65 (hundredths) and between 2.645 and 2.646 (thousandths). Both statements are true.

Why (A) is wrong: True but incomplete - C is also true.

Why (B) is wrong: True but coarser than A and C.

Why (C) is wrong: True but incomplete - A is also true.

Hinglish: *Dono statements sahi hain - ek hundredths level par, ek thousandths par. 'Both true' option dhyan se padho.*

CTC 16 | Normal | Which is irrational

16. Which of the following is irrational?

A. $(2+\sqrt{2})(2-\sqrt{2})$

B. $\sqrt{2} + \sqrt{8}$

C. $(3+\sqrt{3}) + (3-\sqrt{3})$

D. $\sqrt{16}/\sqrt{4}$

Correct Answer: (B)

$\sqrt{2} + \sqrt{8} = \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$, irrational. A: 4-2 = 2. C: 6. D: 4/2 = 2.

Why (A) is wrong: Difference of squares: 2.

Why (C) is wrong: Surds cancel in the sum.

Why (D) is wrong: Both roots are perfect squares.

Hinglish: *Pehle simplify karo har option ko. B mein $\sqrt{8} = 2\sqrt{2}$, toh $3\sqrt{2}$ bachta hai - irrational. Baaki sab clean numbers.*

CTC 17 | HOTS | Common value relation

17. If $2^x = 3^y = 12^z = t$ ($t \neq 1$), which relation holds?

A. 1/z = 2/x + 1/y

B. 1/z = 1/x + 1/y

C. z = x + y

D. 1/z = 1/x + 2/y

Correct Answer: (A)

$12 = 2^2 \times 3 \Rightarrow$ t^(1/z) = t^(2/x) · t^(1/y) $\Rightarrow$ 1/z = 2/x + 1/y.

Why (B) is wrong: That factorisation is 6, not 12.

Why (C) is wrong: Exponents never add across different bases like this.

Why (D) is wrong: Swapped the roles of 2 and 3.

Hinglish: *12 ko $2^2 \times 3$ kholo. Exponent add: 2/x + 1/y = 1/z. Options B aur D wale $12 = 2 \times 3^2$ samajh bete hain - wo 18 hota hai.*

CTC 18 | Normal | Between fractions

18. Which number lies between 5/7 and 9/11?

A. 0.85

B. 3/4

C. 5/6

D. 2/3

Correct Answer: (B)

5/7 ≈ 0.714 and 9/11 ≈ 0.818. 3/4 = 0.75 sits between them. 0.85 and 5/6 ≈ 0.833 exceed 0.818; 2/3 ≈ 0.667 is below.

Why (A) is wrong: 0.85 > 0.818.

Why (C) is wrong: 5/6 ≈ 0.833 > 0.818.

Why (D) is wrong: 2/3 ≈ 0.667 < 0.714.

Hinglish: *Dono ko decimal mein dekho: 0.714 aur 0.818. 0.75 andar hai. Baaki bahar.*

CTC 19 | Normal | Quotient of surds

19. √27/√75 equals:

A. 3/5

B. 9/25

C. √(3/5)

D. 1/5

Correct Answer: (A)

√(27/75) = √(9/25) = 3/5.

Why (B) is wrong: Forgot to take the root.

Why (C) is wrong: Simplified 27/75 to 3/5 but left the root.

Why (D) is wrong: Wrong reduction of the fraction.

Hinglish: *Under-root divide karo: 27/75 = 9/25. Root: 3/5. Clean fraction.*

CTC 20 | Normal | Recurring with integer part

20. 6.454545... (block 45 repeating) as a fraction is:

A. 71/11

B. 645/100

C. 71/10

D. 6 5/11

Correct Answer: (A)

100x - x = 645.4545... - 6.4545... = 639; 99x = 639; x = 639/99 = 71/11.

Why (B) is wrong: Terminating decimal assumption.

Why (C) is wrong: That is 7.1 - wrong.

Why (D) is wrong: 6 5/11 = 71/11: same VALUE, but 45 not 5 repeats; the clean single form is 71/11.

Hinglish: *99x = 639, kat ke 71/11. Check: 71/11 = 6.4545 ✓. Option D wala 6 + 5/11 = 71/11 hi hai - actually same value! But standard single answer 71/11.*

Chapter 4: Algebraic Identities and Factorisation

An identity is an equality that is true for EVERY value of the letters - not an equation to solve but a tool to rewrite expressions. The uploaded PDF builds the identity toolbox: squares of binomials $(a \pm b)^2$, the difference of squares $a^2 - b^2$, the product $(x + a)(x + b)$, the trinomial square $(a + b + c)^2$, the cubes $(a \pm b)^3$, the sum and difference of cubes $a^3 \pm b^3$, and the crown jewel $a^3 + b^3 + c^3 - 3abc$ with its zero-sum shortcut. It then uses these to factorise polynomials and to evaluate products like 98×102 without a calculator. This practice book runs the toolbox at full power: three-variable cubes, nested substitutions, fourth powers, telescoping evaluations, and proof-style questions that examiners reserve for the last section.

Figure 4.1: The area model of $(a + b)^2$ - a big square cut into a^2, two ab strips and b^2.

4.1 Complete Concept Cheat Sheet (read this before any question)

4.1.1 The square identities and the area model

What is it? $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$. **Why true (not just memorised)?** Draw a square of side a + b and cut it: you get a^2 (bottom-left), b^2 (top-right), and TWO ab rectangles - the picture proves the formula. For $(a - b)^2$, start with a^2, remove two strips of width b (areas ab each), then add back b^2 once because it was subtracted twice: $a^2 - 2ab + b^2$. **Companion facts:** $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$ and $(a+b)^2 - (a-b)^2 = 4ab$. The second is the 'multiply quick' engine: $5.5^2 - 4.5^2 = 4 \times 5.5 \times 4.5 = 99$.

When do I use them? Expanding squares, computing squares of numbers near round numbers ($98^2 = (100-2)^2$), building perfect squares ($x^2 + 6x + 9 = (x+3)^2$), and reverse-engineering $a^2 + b^2$ from (a+b) and ab. **Tiny example.** $(3x - 4y)^2 = 9x^2 - 24xy + 16y^2$: square each term, double the product, keep the minus sign.

Figure 4.2: The area model of $(a - b)^2$ - remove two ab strips, add back the overlapping b^2.

> **Sign discipline**
> $(a - b)^2$ has ONE minus, on the middle term only. Writing $a^2 - 2ab - b^2$ is the classic disaster. And $(a+b)^2 \neq a^2 + b^2$ - the cross term 2ab is the whole point.

4.1.2 Difference of squares and (x + a)(x + b)

What is it? $a^2 - b^2 = (a + b)(a - b)$. Verify by expanding: the middle terms a b and -ba cancel exactly. **Use it:** to factorise any two-square difference ($49x^2 - 36y^2 = (7x+6y)(7x-6y)$), to multiply numbers equidistant from a round number ($51 \times 49 = 50^2 - 1^2 = 2499$), and to chain: $x^4 - 81 = (x^2-9)(x^2+9) = (x-3)(x+3)(x^2+9)$. **The (x + a)(x + b) rule:** $= x^2 + (a+b)x + ab$ - coefficient of x is the SUM of the constants, constant term is the PRODUCT. **Tiny example.** $(x + 4)(x + 7) = x^2 + 11x + 28$.

4.1.3 The trinomial square $(a + b + c)^2$

What is it? $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$. Three squares plus twice every product PAIR - three pairs, not two. **Where does it come from?** Group as (a + b) + c and apply the binomial square twice. **Reverse use:** $a^2 + b^2 + c^2 = (a+b+c)^2 - 2(ab+bc+ca)$, which converts between 'sum of squares' and 'sum of pairwise

products' - the most used bridge in three-variable problems. **Consequence:** $a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0$, and equals zero only when a = b = c. **Tiny example.** $(2p + 3q - 4r)^2 = 4p^2 + 9q^2 + 16r^2 + 12pq - 24qr - 16pr$: the sign of each cross term is the product of the signs of its parents.

4.1.4 The cube identities

What is it? $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a + b)$; $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - b^3 - 3ab(a - b)$. Read the signs: all plus for $(a+b)^3$; alternating - + - for $(a-b)^3$ with the LAST term negative. **Numerical use:** $103^3 = (100 + 3)^3 = 1000000 + 90000 + 2700 + 27 = 1092727$; $99^3 = (100 - 1)^3 = 1000000 - 30000 + 300 - 1 = 970299$. **Reverse use:** $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$ - the bridge that computes sums of cubes from the pair-sum and product.

Figure 4.3: $(a + b)^3$ as a cube of side a + b, sliced into 8 blocks: a^3, $3a^2b$, $3ab^2$, b^3.

> **The two cube bridges**
> Given a + b and ab: $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$. Given a - b and ab: $a^3 - b^3 = (a-b)^3 + 3ab(a-b)$. These two lines solve half of all 'find the cube sum' exam questions.

4.1.5 Sum and difference of cubes

What is it? $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ and $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$. **How to remember the signs:** the binomial keeps the SIGN of the cube; inside the trinomial, the middle term always gets the OPPOSITE sign, and the other two are positive squares. **Verification for a^3+b^3:** expand $(a+b)(a^2-ab+b^2) = a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3 = a^3 + b^3$ - the middle terms cancel pairwise. **Tiny example.** $8a^3 + 27b^3 = (2a)^3 + (3b)^3 = (2a + 3b)(4a^2 - 6ab + 9b^2)$.

4.1.6 The crown identity: $a^3 + b^3 + c^3 - 3abc$

What is it? $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$. **The zero-sum shortcut:** if a + b + c = 0, the right side vanishes, so $a^3 + b^3 + c^3 = 3abc$ exactly. This turns monsters like $12^3 + 1^3 + (-13)^3$ into one multiplication: $3 \times 12 \times 1 \times (-13) = -468$. **The converse (HOTS):** since $a^2+b^2+c^2-ab-bc-ca = \frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2]$, the identity forces: if $a^3+b^3+c^3 = 3abc$ then either a + b + c = 0 or a = b = c. **Related identity:** $(a+b+c)^3 = a^3+b^3+c^3 + 3(a+b)(b+c)(c+a)$ - another workhorse.

4.1.7 Factorisation toolkit (the five moves)

Move 1 - Common factor: pull out the HCF first, ALWAYS first. $6x^3 + 9x^2 = 3x^2(2x + 3)$. **Move 2 - Grouping:** rearrange terms so each group shares a factor: $a^2 - b^2 + 2bc - c^2 = a^2 - (b-c)^2 = (a+b-c)(a-b+c)$. **Move 3 - Identity spotting:** perfect squares $(9x^2 + 6x + 1 = (3x+1)^2)$ and cube forms $(x^3 - 8)$. **Move 4 - Middle-term splitting:** for $x^2 + px + q$, find two numbers with sum p and product q: $x^2 + 7x + 12 = (x+3)(x+4)$. For a general quadratic $ax^2 + bx + c$, split b into two numbers whose product is ac. **Move 5 - Substitution:** bracket a repeating chunk, factorise the simpler piece, then back-substitute: $(x^2-2x)^2 - 23(x^2-2x) + 120$ becomes $y^2 - 23y + 120$ with $y = x^2 - 2x$.

The golden sequence: common factor → identity → grouping → middle-term split → substitution. Try in that order and stop as soon as it works. **Always finish:** expand your answer mentally (or on rough paper) and check it rebuilds the original. **Tiny example.** $2x^3 - 8x = 2x(x^2 - 4) = 2x(x+2)(x-2)$ - three moves in one line.

4.1.8 The x + 1/x engine

What is it? A family of chained identities. Given $x + 1/x = s$: $x^2 + 1/x^2 = s^2 - 2$; $x^3 + 1/x^3 = s^3 - 3s$; $x^4 + 1/x^4 = (s^2-2)^2 - 2$; and in general each power follows from the previous two by multiplication and subtraction. Given $x - 1/x = d$: $x^2 + 1/x^2 = d^2 + 2$ and $x^3 - 1/x^3 = d^3 + 3d$. **Why the chains work:** divide $(x + 1/x)^2 = x^2 + 2 + 1/x^2$ by nothing - the middle term is a CONSTANT 2, which is the magic. **Tiny example.** $x + 1/x = 3 \Rightarrow x^2 + 1/x^2 = 7 \Rightarrow x^3 + 1/x^3 = 27 - 9 = 18 \Rightarrow x^4 + 1/x^4 = 49 - 2 = 47$.

4.1.9 Exam tips for this chapter

1. Write the identity NUMBER you are using in the margin - examiners award method marks by identity, not by memory.
2. Common factor first, always: pulling out the HCF shrinks every later step.
3. Numerical products: convert to (round ± small) form before choosing the identity.
4. In three-variable squares, count your cross terms: there must be exactly three pairs, each appearing twice.
5. Sign of a cross term = product of the signs of its two parents. Never 'guess it'.
6. Factorisation is not finished until the last check: expand back, compare term by term.
7. Zero-sum problems: the instant you see $a + b + c = 0$, write $a^3 + b^3 + c^3 = 3abc$ and move on.
8. For $x + 1/x$ chains, keep a running table: s, s^2-2, s^3-3s, ... - never recompute from scratch.

4.2 Question Types / Concept Map for Chapter 4

Fourteen question types cover everything from a two-mark expansion to a five-mark proof. Name the type in your head, then execute.

Type	Name	How to recognise it in the question	Core method
T1	Binomial squares	Expand $(a \pm b)^2$ with letters or numbers	$a^2 \pm 2ab + b^2$
T2	Difference of squares	Two perfect squares subtracted; products like 51×49	$(a+b)(a-b)$ both directions
T3	Near-number evaluation	98^2, 103^3, 9.8×10.2 without calculator	Write as $(100 \pm k)$, apply identities
T4	$(x + a)(x + b)$ products	Two binomials sharing a letter term	$x^2 + (a+b)x + ab$
T5	Trinomial squares	Expand or build $(a + b + c)^2$	Three squares + 2 of each pair
T6	Middle-term splitting	Factorise $x^2 + px + q$ or $ax^2 + bx + c$	Two numbers: sum = p, product = q (or ac)
T7	Cube expansion	Expand $(a \pm b)^3$	$a^3 \pm 3a^2b + 3ab^2 \pm b^3$
T8	Sum/difference of cubes	Factorise $a^3 \pm b^3$	$(a \pm b)(a^2 \mp ab + b^2)$
T9	Crown identity	$a^3 + b^3 + c^3 - 3abc$; zero-sum shortcuts	$(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$
T10	$x \pm 1/x$ chains	Given $x + 1/x$ or $x - 1/x$, find powers	Chain: s^2-2, s^3-3s, ...

Type	Name	How to recognise it in the question	Core method
T11	Symmetric relations	Given a+b, ab; find a^2+b^2, a^3+b^3	$(a\pm b)^2$ and cube bridges
T12	Hard factorisation	Grouping, substitution, a^4+a^2+1 forms	Five moves in golden order
T13	Applied / modelling	Areas, volumes, ages, consecutive integers	Translate to expressions, then T1-T9
T14	Proof / HOTS	'Show that...', converses, chained identities	Label every identity; end with QED line

4.3 Speed Tricks & Trap Alarms

Ye table tumhara "dekho aur maaro" radar hai - question ka trigger phrase dikha, toh yahan ka trick fire karo. Inhe ratt lo, exam me sochna kam padega aur speed double ho jayegi.

Instant trick	Fire it when...
Near round numbers: $51 \times 49 = 50^2 - 1$; $102^2 = 100^2 + 2\cdot100\cdot2 + 4$	"Evaluate without direct multiplication".
Given a + b and ab: $a^2 + b^2 = (a+b)^2 - 2ab$; $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$	Bridge questions - the two givens are never random.
$a + b + c = 0 \rightarrow a^3 + b^3 + c^3 = 3abc$, instantly	Zero-sum cubes like $25^3 - 75^3 + 50^3$.
$x + 1/x = s \rightarrow$ square it: $x^2 + 1/x^2 = s^2 - 2 \rightarrow$ keep chaining up	The whole x + 1/x family.
Spot perfect cubes (8, 27, 64, 125, a^3b^3) $\rightarrow$ apply $a^3 \pm b^3$	$27a^3 + 64$ type expressions.
Cubic factoring: test x = 1, -1, 2, -2 for a root, then divide	$x^3 - 3x^2 - 9x - 5$ type.
HCF first, ALWAYS - before any identity or grouping	Every factorisation: move 1 of the golden sequence.

4.4 The 120-Question Workout

Poora chapter ab ek hi ring mein: **120 questions**, chaar level zones mein banta hua. Warm-up se shuru karo, Boss Zone tak pahuncho. Har question ke turant baad boxed step-by-step solution hai - pehle khud try karo, atako toh solution dekho, aur "Why this step?" + Hinglish note se logic pakka karo. Agar in 120 ko solve kar liya toh is chapter ka koi bhi twisted question tumhe nahi rok payega.

Level 1 · Warm-up Zone: basics and confidence:

QUESTION 1 | Basic | Binomial square (T1)

Q1. Expand: (i) $(2x + 3y)^2$, (ii) $(5a - 2b)^2$, (iii) $(x/2 + y/3)^2$.

SOLUTION — step by step

Step 1. Given: three binomial squares. Use $(a \pm b)^2 = a^2 \pm 2ab + b^2$.

Step 2. (i) $a = 2x$, $b = 3y$: $4x^2 + 2(2x)(3y) + 9y^2 = 4x^2 + 12xy + 9y^2$.

Step 3. (ii) $a = 5a$, $b = 2b$: $25a^2 - 2(5a)(2b) + 4b^2 = 25a^2 - 20ab + 4b^2$.

Step 4. (iii) $a = x/2$, $b = y/3$: $x^2/4 + 2(x/2)(y/3) + y^2/9 = x^2/4 + xy/3 + y^2/9$.

» **Why this step?** Square each term, then DOUBLE the product with the binomial's sign - the middle term is where all errors live.

Answer: (i) $4x^2 + 12xy + 9y^2$ (ii) $25a^2 - 20ab + 4b^2$ (iii) $x^2/4 + xy/3 + y^2/9$

Hinglish: *Simple rule: pehla square, plus/minus double product, doosra square. $(2x+3y)^2$ mein $(2x)^2 = 4x^2$, cross term $2\times2x\times3y = 12xy$. Fraction wale case mein fractions ko square karo, cross term fractions ko multiply karta hai - $xy/3$.*

QUESTION 2 | Basic | Difference of squares (T2)

Q2. Factorise: (i) $49x^2 - 36y^2$, (ii) $121 - 25p^2$, (iii) $9(a+b)^2 - 4(a-b)^2$.

SOLUTION — step by step

Step 1. Given: three two-square differences. Use $a^2 - b^2 = (a+b)(a-b)$.

Step 2. (i) $(7x)^2 - (6y)^2 = (7x + 6y)(7x - 6y)$.

Step 3. (ii) $11^2 - (5p)^2 = (11 + 5p)(11 - 5p)$.

Step 4. (iii) $[3(a+b)]^2 - [2(a-b)]^2 = (3a+3b + 2a-2b)(3a+3b - 2a+2b) = (5a + b)(a + 5b)$.

» **Why this step?** Spot each term as a perfect square first; the brackets then collapse by the identity - in (iii) the inner brackets simplify after factoring.

Answer: (i) (7x+6y)(7x-6y) (ii) (11+5p)(11-5p) (iii) (5a+b)(a+5b)

Hinglish: *Dono terms ko square form mein likho: $(7x)^2$ aur $(6y)^2$. Phir $(a+b)(a-b)$. (iii) mein brackets ke andar simplify karo: $3a+3b+2a-2b = 5a+b$ aur $3a+3b-2a+2b = a+5b$. Answer $(5a+b)(a+5b)$ - brackets kholna hi asli kaam tha.*

QUESTION 3 | Basic | (x+a)(x+b) product (T4)

Q3. Expand: (i) (x + 4)(x + 7), (ii) (y - 3)(y + 8), (iii) (t + 2.5)(t - 2.5).

SOLUTION — step by step

Step 1. Given: three products of binomials sharing the letter term. Use $(x+a)(x+b) = x^2 + (a+b)x + ab$.

Step 2. (i) $x^2 + (4+7)x + 28 = x^2 + 11x + 28$.

Step 3. (ii) $x^2 + (-3+8)y + (-3)(8) = y^2 + 5y - 24$.

Step 4. (iii) This is actually difference of squares: $t^2 - 6.25$.

» **Why this step?** Sum of constants feeds the x-coefficient, product feeds the constant term; when a = -b the x-term dies and you get a difference of squares.

Answer: (i) $x^2 + 11x + 28$ (ii) $y^2 + 5y - 24$ (iii) $t^2 - 6.25$

Hinglish: *Rule: middle coefficient = dono constants ka SUM, last term = PRODUCT. (y-3)(y+8): sum = 5, product = -24. Aur (t+2.5)(t-2.5) dekh ke turant difference of squares - $t^2 - 6.25$, expansion ki zaroorat hi nahi.*

QUESTION 4 | Basic | Near-number squares (T3)

Q4. Evaluate using identities: (i) 102^2, (ii) 98^2, (iii) 9.7^2.

SOLUTION — step by step

Step 1. Given: three squares near round numbers. Write each as (100 ± k) or (10 ± k).

Step 2. (i) $102^2 = (100+2)^2 = 10000 + 400 + 4 = 10404$.

Step 3. (ii) $98^2 = (100-2)^2 = 10000 - 400 + 4 = 9604$.

Step 4. (iii) $9.7^2 = (10 - 0.3)^2 = 100 - 6 + 0.09 = 94.09$.

» **Why this step?** The identity replaces one hard multiplication with three easy pieces; the cross term 2×100×k does the heavy lifting.

Answer: (i) 10404 (ii) 9604 (iii) 94.09

Hinglish: *Round number ke paas wale squares ka JUGAAD: $(100\pm2)^2$. $98^2 = 10000 - 400 + 4 = 9604$. Dhyan do: 9.7^2 mein cross term $2\times10\times0.3 = 6$, aur $0.3^2 = 0.09$ - decimals ka square chhota hota hai, 0.9 nahi!*

QUESTION 5 | Basic | Trinomial square (T5)

Q5. Expand: $(2p + 3q - 4r)^2$.

SOLUTION — step by step

Step 1. Given: a three-term square. Use $(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$ with a = 2p, b = 3q, c = -4r.

Step 2. Squares: $4p^2 + 9q^2 + 16r^2$.

Step 3. Cross terms: 2(2p)(3q) = 12pq; 2(3q)(-4r) = -24qr; 2(-4r)(2p) = -16pr.

Step 4. Combine: $4p^2 + 9q^2 + 16r^2 + 12pq - 24qr - 16pr$.

» **Why this step?** Each cross term carries the product of its parents' signs: (+)(+) = +, (+)(-) = -, (-)(+) = -.

Answer: $4p^2 + 9q^2 + 16r^2 + 12pq - 24qr - 16pr$

Hinglish: *Teen-term square: teen squares + TEEN cross terms (har pair do baar). Signs multiply: 3q (+) aur 4r (-) ka pair negative -24qr. Cross terms ke pairs gin lo - teen se kam mile toh koi chhoot gaya.*

QUESTION 6 | Basic | Square-sum identities (T1/T11)

Q6. If a + b = 10 and a - b = 4, find (i) $a^2 + b^2$ and (ii) ab - all in one line each, using identities.

SOLUTION — step by step

Step 1. Given: the pair-sum and pair-difference. Needed: sum of squares and the product.

Step 2. Add the two squares identities: $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$.

Step 3. So $a^2 + b^2 = (10^2 + 4^2)/2 = (100 + 16)/2 = 58$.

Step 4. Subtract: $(a+b)^2 - (a-b)^2 = 4ab$.

Step 5. So ab = (100 - 16)/4 = 84/4 = 21.

Step 6. Check: (a+b)(a-b) = 10×4 = 40 = $a^2 - b^2$; with $a^2+b^2 = 58$ we get $a^2 = 49$, $b^2 = 9$, a = 7, b = 3 ✓ (ab = 21 ✓).

» **Why this step?** The two companion identities convert (a+b, a-b) directly into (a^2+b^2, ab) without ever solving for a and b.

Answer: $a^2 + b^2 = 58$, ab = 21

Hinglish: *Ye identity-pair exam ka shortcut bomb hai: $(a+b)^2+(a-b)^2 = 2(a^2+b^2)$ aur $(a+b)^2-(a-b)^2 = 4ab$. Dono values plug karo - 58 aur 21. Solving for a, b optional hai, sirf check ke liye.*

QUESTION 7 | Basic | Cube expansion (T7)

Q7. Expand: (i) $(x + 2)^3$, (ii) $(2m - 3n)^3$.

SOLUTION — step by step

Step 1. Given: two binomial cubes. Use $(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$.

Step 2. (i) a = x, b = 2: $x^3 + 3x^2(2) + 3x(4) + 8 = x^3 + 6x^2 + 12x + 8$.

Step 3. (ii) a = 2m, b = 3n: $(2m)^3 - 3(2m)^2(3n) + 3(2m)(3n)^2 - (3n)^3 = 8m^3 - 36m^2n + 54mn^2 - 27n^3$.

» **Why this step?** The pattern 1, 3, 3, 1 on the coefficients comes from the binomial theorem; signs alternate for the minus cube.

Answer: (i) $x^3 + 6x^2 + 12x + 8$ (ii) $8m^3 - 36m^2n + 54mn^2 - 27n^3$

Hinglish: *Cube pattern: 1-3-3-1. $(2m-3n)^3$ mein powers dhyan se: $(2m)^3 = 8m^3$, $(2m)^2 = 4m^2$ toh $3 \times 4m^2 \times 3n = 36m^2n$ minus. Signs alternate: + - + -. Coefficients 8, 36, 54, 27 - inko yaad karne ki zaroorat nahi, pattern se niklo.*

QUESTION 8 | Basic | Middle-term splitting (T6)

Q8. Factorise: (i) $x^2 + 7x + 12$, (ii) $x^2 - 5x - 36$, (iii) $x^2 - 11x + 24$.

SOLUTION — step by step

Step 1. Given: three monic quadratics. Find two numbers with the right SUM and PRODUCT.

Step 2. (i) sum 7, product 12 → 3 and 4: (x + 3)(x + 4).

Step 3. (ii) sum -5, product -36 → -9 and +4: (x - 9)(x + 4).

Step 4. (iii) sum -11, product 24 → -8 and -3: (x - 8)(x - 3).

Step 5. Checks: 3+4 = 7, 3×4 = 12 ✓; -9+4 = -5, -9×4 = -36 ✓; -8-3 = -11, 24 ✓.

» **Why this step?** Expanding (x+p)(x+q) gives sum and product - so factorising reverses exactly those two readings.

Answer: (i) (x+3)(x+4) (ii) (x-9)(x+4) (iii) (x-8)(x-3)

Hinglish: *Do numbers dhoondo: jiska sum middle coefficient ho, product last term ho. Sign logic: product positive toh dono same sign (sum ke hisaab se), product negative toh alag-alag. (ii) mein product -36 aur sum -5: -9 aur +4 jeette hain.*

QUESTION 9 | Basic | Perfect square spotting (T1/T6)

Q9. Factorise: (i) $4y^2 - 12y + 9$, (ii) $9x^2 + 6x + 1$, (iii) $25m^2 + 30mn + 9n^2$.

SOLUTION — step by step

Step 1. Given: three candidates for perfect squares. Test first and last terms for squareness.

Step 2. (i) $(2y)^2 - 2(2y)(3) + 3^2 = (2y - 3)^2$.

Step 3. (ii) $(3x)^2 + 2(3x)(1) + 1^2 = (3x + 1)^2$.

Step 4. (iii) $(5m)^2 + 2(5m)(3n) + (3n)^2 = (5m + 3n)^2$.

» **Why this step?** A quadratic is a perfect square exactly when the middle term equals twice the product of the square roots of the end terms.

Answer: (i) $(2y-3)^2$ (ii) $(3x+1)^2$ (iii) $(5m+3n)^2$

Hinglish: *Perfect square ka test: pehla aur aakhri term perfect square hon, aur middle term unke roots ke product ka DOUBLE ho. $12y = 2\times2y\times3$ ✓. Ye test lagane se $(a\pm b)^2$ identity turant mil jati hai.*

QUESTION 10 | Basic | Difference-of-squares multiplication (T3)

Q10. Evaluate without direct multiplication: (i) 51×49, (ii) 103×97, (iii) 5.1×4.9.

SOLUTION — step by step

Step 1. Given: three products equidistant from a round number. Use $(a+b)(a-b) = a^2 - b^2$.

Step 2. (i) $51 \times 49 = (50+1)(50-1) = 2500 - 1 = 2499$.

Step 3. (ii) $103 \times 97 = (100+3)(100-3) = 10000 - 9 = 9991$.

Step 4. (iii) $5.1 \times 4.9 = (5 + 0.1)(5 - 0.1) = 25 - 0.01 = 24.99$.

» **Why this step?** Rewriting as (round ± same offset) converts multiplication into one subtraction of squares.

Answer: (i) 2499 (ii) 9991 (iii) 24.99

Hinglish: *Dono numbers ka average round number hai? Toh $(a+b)(a-b) = a^2-b^2$. 51×49: average 50, offset 1. $2500-1 = 2499$. Ye trick multiplication tables se tez hai.*

QUESTION 11 | Basic | Common factor first (T12)

Q11. Factorise completely: (i) $6x^3 + 9x^2$, (ii) $2x^3 - 8x$, (iii) $3a^2b + 6ab^2 - 3ab$.

SOLUTION — step by step

Step 1. Given: three polynomials. Golden rule: HCF pehle.

Step 2. (i) HCF = $3x^2$: $3x^2(2x + 3)$.

Step 3. (ii) HCF = $2x$: $2x(x^2 - 4) = 2x(x+2)(x-2)$ - the difference of squares appears only after the common factor.

Step 4. (iii) HCF = 3ab: 3ab(a + 2b - 1).

» **Why this step?** Removing the HCF exposes simpler structures (here a difference of squares) that were invisible inside the original expression.

Answer: (i) $3x^2(2x+3)$ (ii) $2x(x+2)(x-2)$ (iii) $3ab(a+2b-1)$

Hinglish: *Sabse pehla step HCF nikaalo - hamesha. (ii) mein 2x nikaalne ke baad hi x^2-4 dikhta hai. Common factor chhod dena sabse common galti hai - 'completely' factorise ka matlab har factor prime ho.*

QUESTION 12 | Basic | Square-difference evaluation (T3)

Q12. Evaluate: $5.1^2 - 4.9^2$ and $256^2 - 254^2$, each in one line.

SOLUTION — step by step

Step 1. Given: two differences of squares. Use $a^2 - b^2 = (a+b)(a-b)$.

Step 2. $5.1^2 - 4.9^2 = (5.1 + 4.9)(5.1 - 4.9) = 10 \times 0.2 = 2$.

Step 3. $256^2 - 254^2 = (256 + 254)(256 - 254) = 510 \times 2 = 1020$.

» **Why this step?** Never square two big numbers and subtract - factor first; the identity turns two squares into one small product.

Answer: 2 and 1020

Hinglish: *Squares subtract karne se pehle FACTOR karo: (a+b)(a-b). 5.1 aur 4.9 ka sum exactly 10 - answer 2 second mein. 256-254 wala: 510×2 = 1020. Bina identity ke square karoge toh calculator chahiye.*

QUESTION 13 | Basic | Binomial square (T1)

Q13. Expand: (i) $(3a + 2b)^2$, (ii) $(4m - n)^2$, (iii) $(x + 6)(x - 6)$.

SOLUTION — step by step

Step 1. (i) $(3a)^2 + 2(3a)(2b) + (2b)^2 = 9a^2 + 12ab + 4b^2$.

Step 2. (ii) $(4m)^2 - 2(4m)(n) + n^2 = 16m^2 - 8mn + n^2$.

Step 3. (iii) Difference of squares: $x^2 - 36$.

» **Why this step?** $(a \pm b)^2 = a^2 \pm 2ab + b^2$; the middle term is 2ab - forgetting the 2 is the #1 error. $(x + 6)(x - 6)$ is the special minus case.

Answer: (i) $9a^2 + 12ab + 4b^2$ (ii) $16m^2 - 8mn + n^2$ (iii) $x^2 - 36$

Hinglish: *Square identity mein middle term 2ab hota hai - '2' bhoolna national hobby hai, tum mat karna! (iii) plus-minus pair = $a^2 - b^2$, seedha 36.*

QUESTION 14 | Basic | (x + a)(x + b) product (T4)

Q14. Expand: (i) (x + 5)(x + 3), (ii) (y - 8)(y - 2), (iii) (t + 9)(t - 4).

SOLUTION — step by step

Step 1. Identity: $(x + a)(x + b) = x^2 + (a + b)x + ab$.

Step 2. (i) $x^2 + 8x + 15$.

Step 3. (ii) $y^2 - 10y + 16$ (minus × minus = plus constant).

Step 4. (iii) $t^2 + 5t - 36$ (mixed signs -> negative constant).

» **Why this step?** Sum goes on the middle term, product on the ends - the same pair of numbers that factorises quadratics.

Answer: (i) $x^2 + 8x + 15$ (ii) $y^2 - 10y + 16$ (iii) $t^2 + 5t - 36$

Hinglish: *Sum middle mein, product end mein. Signs ka combination dekho: (-)(-) = +16, (+)(-) = -36. Ye identity factorisation ka bhi engine hai.*

QUESTION 15 | Basic | Near-number squares (T3)

Q15. Evaluate using identities: (i) 52^2, (ii) 97^2, (iii) 103×97.

SOLUTION — step by step

Step 1. (i) $52^2 = (50 + 2)^2 = 2500 + 200 + 4 = 2704$.

Step 2. (ii) $97^2 = (100 - 3)^2 = 10000 - 600 + 9 = 9409$.

Step 3. (iii) $103 \times 97 = (100 + 3)(100 - 3) = 100^2 - 9 = 9991$.

» **Why this step?** Snap to the nearest round number, then let the identity carry the arithmetic.

Answer: (i) 2704 (ii) 9409 (iii) 9991

Hinglish: *Round number ke aas paas bindaas: 50±2, 100∓3. Aur 103×97 ka plus-minus jodi = 10000 - 9. Calculator ki zaroorat khatam!*

QUESTION 16 | Basic | Difference of squares factorising (T2)

Q16. Factorise: (i) $25p^2 - 49q^2$, (ii) $x^2 - 18x + 81$, (iii) $36a^2 + 60a + 25$.

SOLUTION — step by step

Step 1. (i) $25p^2 - 49q^2 = (5p)^2 - (7q)^2 = (5p + 7q)(5p - 7q)$.

Step 2. (ii) $x^2 - 18x + 81 = x^2 - 2(x)(9) + 9^2 = (x - 9)^2$.

Step 3. (iii) $36a^2 + 60a + 25 = (6a)^2 + 2(6a)(5) + 5^2 = (6a + 5)^2$ (check middle: 2·6·5 = 60 ✓).

» **Why this step?** Recognise the pattern first: two perfect squares apart (case i), or a perfect-square trinomial (cases ii, iii).

Answer: (i) (5p + 7q)(5p - 7q) (ii) $(x - 9)^2$ (iii) $(6a + 5)^2$

Hinglish: *Perfect squares pehchano: $25 = 5^2$, $49 = 7^2$. (ii) aur (iii) perfect square trinomials hain - middle term 2ab se match karke confirm karo. Pattern dikha toh likhna 5 second ka hai.*

QUESTION 17 | Basic | Middle-term splitting (T6)

Q17. Factorise: (i) $x^2 + 9x + 20$, (ii) $x^2 - 3x - 10$, (iii) $2y^2 + 11y + 12$.

SOLUTION — step by step

Step 1. (i) Need two numbers with sum 9, product 20: 4 and 5 -> (x + 4)(x + 5).

Step 2. (ii) Sum -3, product -10: -5 and +2 -> (x - 5)(x + 2).

Step 3. (iii) Split $11y = 8y + 3y$ ($8 \times 3 = 24 = 2 \times 12$): $2y^2 + 8y + 3y + 12 = 2y(y + 4) + 3(y + 4) = (y + 4)(2y + 3)$.

» **Why this step?** Split the middle term into the pair whose product matches the ends - then group.

Answer: (i) (x + 4)(x + 5) (ii) (x - 5)(x + 2) (iii) (y + 4)(2y + 3)

Hinglish: *Do numbers dhoondo: sum middle, product ends. (iii) lead coefficient 2 hai, isliye product $2 \times 12 = 24$ lena hai - 8 aur 3. Grouping ke baad bracket match - done.*

QUESTION 18 | Basic | Square-sum identities (T1/T11)

Q18. If $a + b = 14$ and $ab = 48$, find (i) $a^2 + b^2$, (ii) $(a - b)^2$, (iii) $a - b$ ($a > b$).

SOLUTION — step by step

Step 1. (i) $a^2 + b^2 = (a + b)^2 - 2ab = 196 - 96 = 100$.

Step 2. (ii) $(a - b)^2 = (a + b)^2 - 4ab = 196 - 192 = 4$.

Step 3. (iii) $a - b = \sqrt{4} = 2$ (positive, since $a > b$).

» **Why this step?** Given (a + b) and ab, every symmetric expression is one identity away.

Answer: (i) 100 (ii) 4 (iii) 2

Hinglish: *Bridge questions ka formula kit: $a^2+b^2 = S^2 - 2P$, $(a-b)^2 = S^2 - 4P$. 196 se minus karo, done. $a-b = 2$ kyunki a bada tha.*

QUESTION 19 | Basic | Cube expansion (T7)

Q19. Expand: (i) $(x + 4)^3$, (ii) $(a - 2b)^3$.

SOLUTION — step by step

Step 1. (i) $x^3 + 3x^2(4) + 3x(4)^2 + 4^3 = x^3 + 12x^2 + 48x + 64$.

Step 2. (ii) $a^3 - 3a^2(2b) + 3a(2b)^2 - (2b)^3 = a^3 - 6a^2b + 12ab^2 - 8b^3$.

» **Why this step?** $(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$ - signs alternate for the minus cube.

Answer: (i) $x^3 + 12x^2 + 48x + 64$ (ii) $a^3 - 6a^2b + 12ab^2 - 8b^3$

Hinglish: *Cube identity: 1-3-3-1 coefficients. Minus case mein signs + - + - chalte hain. Powers ka pattern: 3,2,1,0 aur 0,1,2,3.*

QUESTION 20 | Basic | Sum/difference of cubes (T8)

Q20. Factorise: (i) $27x^3 + 64y^3$, (ii) $216 - 343t^3$.

SOLUTION — step by step

Step 1. (i) $27x^3 + 64y^3 = (3x)^3 + (4y)^3 = (3x + 4y)(9x^2 - 12xy + 16y^2)$.

Step 2. (ii) $216 - 343t^3 = 6^3 - (7t)^3 = (6 - 7t)(36 + 42t + 49t^2)$.

» **Why this step?** $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$ - the second bracket is NOT a perfect square; keep its signs straight.

Answer: (i) $(3x + 4y)(9x^2 - 12xy + 16y^2)$ (ii) $(6 - 7t)(36 + 42t + 49t^2)$

Hinglish: *Perfect cubes pehchano: 27, 64, 216, 343 (6^3, 7^3). Formula ka doosra bracket $a^2 \mp ab + b^2$ - signs ka pattern yaad rakho, yahi galti hoti hai.*

QUESTION 21 | Basic | Near-number cube (T3)

Q21. Evaluate 102^3 without a calculator.

SOLUTION — step by step

Step 1. $102^3 = (100 + 2)^3$.

Step 2. $= 100^3 + 3(100)^2(2) + 3(100)(2)^2 + 2^3$.

Step 3. $= 1000000 + 60000 + 1200 + 8 = 1061208$.

» **Why this step?** The cube identity turns scary cubes into four friendly terms.

Answer: 1061208

Hinglish: *$(100+2)^3$ ke chaar tukde: $10^6 + 6\times10^4 + 1200 + 8 = 1061208$. Cube identity se bade numbers ka calculation mental ho jaata hai.*

QUESTION 22 | Basic | Chained difference of squares (T2)

Q22. Simplify completely: $(3x + 4y)(3x - 4y)(9x^2 + 16y^2)$.

SOLUTION — step by step

Step 1. First pair: $(3x + 4y)(3x - 4y) = 9x^2 - 16y^2$.

Step 2. Now: $(9x^2 - 16y^2)(9x^2 + 16y^2) = (9x^2)^2 - (16y^2)^2$.

Step 3. $= 81x^4 - 256y^4$.

» **Why this step?** Conjugate pairs eat powers two at a time: $a^2 – b^2$, then $a^4 – b^4$ - keep pairing until nothing factors.

Answer: $81x^4 - 256y^4$

Hinglish: *Do conjugate pairs ke baad 4th power aa gayi: $81x^4 - 256y^4$. Aage koi square pattern nahi, bas. Chain factorisation mein har step pe check karo - aur khaana hai kya?*

QUESTION 23 | Basic | Fourth-power difference (T2)

Q23. Factorise completely: (i) $x^4 - 625$, (ii) $a^4 - 81$.

SOLUTION — step by step

Step 1. (i) $x^4 - 625 = (x^2 - 25)(x^2 + 25) = (x - 5)(x + 5)(x^2 + 25)$.

Step 2. (ii) $a^4 - 81 = (a^2 - 9)(a^2 + 9) = (a - 3)(a + 3)(a^2 + 9)$.

» **Why this step?** $x^4 – n^4$ chains: difference of squares twice; the sum-of-squares piece $(x^2 + 25)$ stops the chain in reals.

Answer: (i) $(x - 5)(x + 5)(x^2 + 25)$ (ii) $(a - 3)(a + 3)(a^2 + 9)$

Hinglish: *Do baar $a^2 - b^2$ lagao: x^2-25 phir x-5, x+5. $x^2 + 25$ ko aage todna mat - real numbers mein sum of squares factor nahi hota. 'Completely' ka matlab yahi hai.*

QUESTION 24 | Basic | Perfect square spotting (T1/T5)

Q24. Check whether each is a perfect square trinomial, and factorise if so: (i) $49x^2 + 42x + 9$, (ii) $16m^2 - 40mn + 25n^2$.

SOLUTION — step by step

Step 1. (i) $49x^2 = (7x)^2$, $9 = 3^2$, middle test: $2(7x)(3) = 42x$ ✓ -> $(7x + 3)^2$.

Step 2. (ii) $16m^2 = (4m)^2$, $25n^2 = (5n)^2$, middle: $2(4m)(5n) = 40mn$ with a minus sign -> $(4m - 5n)^2$.

» **Why this step?** Perfect square test: first and last terms are squares AND the middle equals 2ab (with either sign).

Answer: (i) $(7x + 3)^2$ (ii) $(4m - 5n)^2$

Hinglish: *Do check karo: dono ends square hain? Middle 2ab hai? Dono haan -> seedha $(a \pm b)^2$. Middle match nahi karta toh yeh identity nahi hai - splitting karni padti.*

QUESTION 25 | Basic | x + 1/x chain (T10)

Q25. If $x + 1/x = 6$, find (i) $x^2 + 1/x^2$, (ii) $x^4 + 1/x^4$.

SOLUTION — step by step

Step 1. (i) Square: $x^2 + 2 + 1/x^2 = 36 \rightarrow x^2 + 1/x^2 = 34$.

Step 2. (ii) Square again: $x^4 + 2 + 1/x^4 = 34^2 = 1156 \rightarrow x^4 + 1/x^4 = 1154$.

» **Why this step?** Each squaring adds the constant 2 on the middle - climb the ladder one square at a time.

Answer: (i) 34 (ii) 1154

Hinglish: *Square karo, 2 minus karo, repeat. 6 -> 34 -> 1154. x + 1/x family ki seedhi khaddi hai - ek step per floor.*

QUESTION 26 | Basic | Trinomial square (T5)

Q26. Expand: $(a + 2b - 3c)^2$.

SOLUTION — step by step

Step 1. Treat as $(a + 2b) + (-3c)$ or use the three-term identity: square each term, then double every pair.

Step 2. Squares: $a^2 + 4b^2 + 9c^2$.

Step 3. Doubles: $2(a)(2b) = 4ab$; $2(2b)(-3c) = -12bc$; $2(a)(-3c) = -6ac$.

Step 4. Result: $a^2 + 4b^2 + 9c^2 + 4ab - 12bc - 6ac$.

» **Why this step?** $(x + y + z)^2$ = sum of squares + 2 × each pairwise product - three doubles, no exceptions.

Answer: $a^2 + 4b^2 + 9c^2 + 4ab - 12bc - 6ac$

Hinglish: *Teen term ka square: teen squares + teen doubles. Signs terms ke saath jaate hain: -3c ne do doubles ko negative kiya. Ek bhi double miss = poora answer galat.*

QUESTION 27 | Basic | Square-difference evaluation (T2)

Q27. Evaluate: (i) $68^2 - 32^2$, (ii) $7.8^2 - 2.2^2$, (iii) $(0.73)^2 - (0.27)^2$.

SOLUTION — step by step

Step 1. (i) $(68 - 32)(68 + 32) = 36 \times 100 = 3600$.

Step 2. (ii) $(7.8 - 2.2)(7.8 + 2.2) = 5.6 \times 10 = 56$.

Step 3. (iii) $(0.73 - 0.27)(0.73 + 0.27) = 0.46 \times 1 = 0.46$.

» **Why this step?** $a^2 - b^2 = (a - b)(a + b)$: the sum is often a clean round number - multiply THAT instead of squaring twice.

Answer: (i) 3600 (ii) 56 (iii) 0.46

Hinglish: *Squares alag-alag nikalna pagalpan hai. Jod aur ghat lo: 36 × 100, 5.6 × 10, 0.46 × 1. Difference of squares = mental math ka cheat code.*

QUESTION 28 | Basic | Difference-of-squares multiplication (T2)

Q28. Evaluate without direct multiplication: (i) 51 × 49, (ii) 102 × 98, (iii) 10.3 × 9.7.

SOLUTION — step by step

Step 1. (i) $51 \times 49 = (50 + 1)(50 - 1) = 2500 - 1 = 2499$.

Step 2. (ii) $(100 + 2)(100 - 2) = 10000 - 4 = 9996$.

Step 3. (iii) $(10 + 0.3)(10 - 0.3) = 100 - 0.09 = 99.91$.

» **Why this step?** Numbers equidistant from a round centre are conjugates - their product is centre^2 minus offset^2.

Answer: (i) 2499 (ii) 9996 (iii) 99.91

Hinglish: *Dono number ka average pakdo (50, 100, 10), offset dekho (1, 2, 0.3). Product = avg² - offset². Har saal aane wala gift question.*

QUESTION 29 | Basic | Grouped difference of squares (T2/T12)

Q29. Factorise: (i) $(a + b)^2 - 16c^2$, (ii) $36 - (2p + q)^2$.

SOLUTION — step by step

Step 1. (i) $(a + b)^2 - (4c)^2 = (a + b + 4c)(a + b - 4c)$.

Step 2. (ii) $6^2 - (2p + q)^2 = (6 + 2p + q)(6 - 2p - q)$.

» **Why this step?** Treat a whole bracket as ONE block: $A^2 - B^2 = (A + B)(A - B)$ with A = the bracket.

Answer: (i) (a + b + 4c)(a + b - 4c) (ii) (6 + 2p + q)(6 - 2p - q)

Hinglish: *Bracket ko ek hi package samjho: A = (a+b), B = 4c. Plus-minus ke saath bracket copy karo. Bracket treatment se complex-looking expressions bhi easy.*

QUESTION 30 | Basic | Difference bridge (T11)

Q30. If a - b = 5 and ab = 14, find (i) $a^2 + b^2$, (ii) a + b (taking a + b > 0).

SOLUTION — step by step

Step 1. (i) $a^2 + b^2 = (a - b)^2 + 2ab = 25 + 28 = 53$.

Step 2. (ii) $(a + b)^2 = (a - b)^2 + 4ab = 25 + 56 = 81$ -> a + b = 9.

» **Why this step?** The minus-version bridge: $(a - b)^2 = a^2 + b^2 - 2ab$ and $(a + b)^2 = (a - b)^2 + 4ab$.

Answer: (i) 53 (ii) 9

Hinglish: *Difference wale bridge mein +2ab hota hai (minus ki jodi plus ban jaati hai square mein). 53 aur 9. Signs ka ye flip samajh liya toh half chapter clear.*

Level 2 · Core Drills: book-level practice:

QUESTION 31 | Moderate | Cube expansion (T7)

Q31. Expand: $(2x - 3y)^3$ and $(x + 2y)^3$.

SOLUTION — step by step

Step 1. Given: two cube expansions.

Step 2. $(2x - 3y)^3$: a = 2x, b = 3y. $(2x)^3 = 8x^3$; $3(2x)^2(3y) = 36x^2y$; $3(2x)(3y)^2 = 54xy^2$; $(3y)^3 = 27y^3$.

Step 3. With alternating signs: $8x^3 - 36x^2y + 54xy^2 - 27y^3$.

Step 4. $(x + 2y)^3$: $x^3 + 3x^2(2y) + 3x(2y)^2 + (2y)^3 = x^3 + 6x^2y + 12xy^2 + 8y^3$.

» **Why this step?** Write powers of a and b separately first ($8x^3$, $36x^2y$, ...), then attach the sign pattern + - + -.

Answer: $8x^3 - 36x^2y + 54xy^2 - 27y^3$ and $x^3 + 6x^2y + 12xy^2 + 8y^3$

Hinglish: *Cube mein har term ka coefficient pattern 1-3-3-1. Powers pehle nikaalo: $(2x)^2 = 4x^2$, $(3y)^2 = 9y^2$. Coefficients: 8, 3×4×3 = 36, 3×2×9 = 54, 27. Minus wale case mein sirf SIGN pattern badalta hai.*

QUESTION 32 | Moderate | Sum of cubes (T8)

Q32. Factorise: (i) $8a^3 + 27b^3$, (ii) $x^3 + 64$, (iii) $125m^3 + n^3$.

SOLUTION — step by step

Step 1. Given: three cube sums. Write each as $A^3 + B^3$, then use $(A+B)(A^2 - AB + B^2)$.

Step 2. (i) $(2a)^3 + (3b)^3 = (2a + 3b)(4a^2 - 6ab + 9b^2)$.

Step 3. (ii) $x^3 + 4^3 = (x + 4)(x^2 - 4x + 16)$.

Step 4. (iii) $(5m)^3 + n^3 = (5m + n)(25m^2 - 5mn + n^2)$.

» **Why this step?** The trinomial's middle term is minus AB - the opposite sign of the binomial - and its end terms are the squares of A and B.

Answer: (i) $(2a+3b)(4a^2-6ab+9b^2)$ (ii) $(x+4)(x^2-4x+16)$ (iii) $(5m+n)(25m^2-5mn+n^2)$

Hinglish: *Cubes pehchano: $8a^3=(2a)^3$, $27b^3=(3b)^3$. Formula: $(A+B)(A^2-AB+B^2)$. Middle term ka sign binomial ke ULTA. Trinomial ke squares bade lagte hain ($4a^2$, $9b^2$) par wo bas A^2, B^2 hain.*

QUESTION 33 | Moderate | Difference of cubes (T8)

Q33. Factorise: (i) $x^3 - 27y^3$, (ii) $64p^3 - 1$, (iii) $8x^3 - 125y^3z^3$.

SOLUTION — step by step

Step 1. Given: three cube differences. Use $A^3 - B^3 = (A - B)(A^2 + AB + B^2)$.

Step 2. (i) $x^3 - (3y)^3 = (x - 3y)(x^2 + 3xy + 9y^2)$.

Step 3. (ii) $(4p)^3 - 1^3 = (4p - 1)(16p^2 + 4p + 1)$.

Step 4. (iii) $(2x)^3 - (5yz)^3 = (2x - 5yz)(4x^2 + 10xyz + 25y^2z^2)$.

» **Why this step?** Same skeleton as the sum of cubes with signs flipped: binomial takes the minus, trinomial's middle term becomes plus.

Answer: (i) $(x-3y)(x^2+3xy+9y^2)$ (ii) $(4p-1)(16p^2+4p+1)$ (iii) $(2x-5yz)(4x^2+10xyz+25y^2z^2)$

Hinglish: *Difference of cubes = mirror of sum: $(A-B)(A^2+AB+B^2)$. (iii) mein $125y^3z^3=(5yz)^3$ - teen factors under root, teeno ka cube. Product term $10xyz = A\times B = 2x\times 5yz$.*

QUESTION 34 | Moderate | x + 1/x chain (T10)

Q34. If $x + 1/x = 5$, find (i) $x^2 + 1/x^2$ and (ii) $x^4 + 1/x^4$.

SOLUTION — step by step

Step 1. Given: $x + 1/x = 5$. Square both sides: $x^2 + 2 + 1/x^2 = 25$.

Step 2. (i) $x^2 + 1/x^2 = 25 - 2 = 23$.

Step 3. Square again: $(x^2 + 1/x^2)^2 = x^4 + 2 + 1/x^4 = 23^2 = 529$.

Step 4. (ii) $x^4 + 1/x^4 = 529 - 2 = 527$.

» **Why this step?** Every square introduces a constant middle term 2 (because $x \cdot 1/x = 1$); subtract it to keep the chain clean.

Answer: (i) 23 (ii) 527

Hinglish: *Chain method: square karo, 2 minus karo, phir dobara square karo, phir 2 minus. $5 \to 23 \to 527$. Ek hi line mein: $(x+1/x)^2 - 2 = x^2+1/x^2$. Ye chain chalti hi jati hai - x^6 ke liye x^3 wale bridge ka use hota hai.*

QUESTION 35 | Moderate | Trinomial square reverse (T5/T11)

Q35. If $a + b + c = 9$ and $ab + bc + ca = 26$, find $a^2 + b^2 + c^2$.

SOLUTION — step by step

Step 1. Given: the pair-sum of three variables. Use the trinomial square identity:

Step 2. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$.

Step 3. $81 = (a^2+b^2+c^2) + 2(26) = (a^2+b^2+c^2) + 52$.

Step 4. So $a^2 + b^2 + c^2 = 81 - 52 = 29$.

» **Why this step?** The identity links the three quantities (sum, pair-sum, sum of squares) - knowing any two gives the third.

Answer: 29

Hinglish: *Trinomial square ka reverse: sum of squares = (sum)² - 2(pair-sum). 81 - 52 = 29. Ye formula har three-variable question mein kaam aayega - yaad kar lo.*

QUESTION 36 | Moderate | Chained difference of squares (T2)

Q36. Factorise completely: (i) $x^4 - 81$, (ii) $a^4 - 16b^4$, (iii) $81x^4 - 625$.

SOLUTION — step by step

Step 1. Given: fourth-power differences. Apply $a^2 - b^2$ repeatedly.

Step 2. (i) $(x^2)^2 - 9^2 = (x^2 - 9)(x^2 + 9) = (x-3)(x+3)(x^2+9)$.

Step 3. (ii) $(a^2)^2 - (4b^2)^2 = (a^2 - 4b^2)(a^2 + 4b^2) = (a-2b)(a+2b)(a^2+4b^2)$.

Step 4. (iii) $(9x^2)^2 - 25^2 = (9x^2-25)(9x^2+25) = (3x-5)(3x+5)(9x^2+25)$.

Step 5. Note: $x^2 + 9$, $a^2 + 4b^2$, $9x^2 + 25$ do not factorise over the integers - they are finished.

» **Why this step?** Each application of the identity halves an exponent; stop when a factor is a sum of squares (irreducible here).

Answer: (i) $(x-3)(x+3)(x^2+9)$ (ii) $(a-2b)(a+2b)(a^2+4b^2)$ (iii) $(3x-5)(3x+5)(9x^2+25)$

Hinglish: *$x^4 - 81$ ko do baar factor karo: pehle $(x^2-9)(x^2+9)$, phir x^2-9 ko $(x-3)(x+3)$. Sum of squares (x^2+9) yahan phir se factor NAHI hota - wahi final hai. 'Completely' ka matlab: har factor irreducible.*

QUESTION 37 | Moderate | Cube sum bridge (T11)

Q37. If $a + b = 12$ and $ab = 27$, find $a^3 + b^3$ and $a^2 + b^2$.

SOLUTION — step by step

Step 1. Given: pair-sum and product. Cube bridge: $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$.

Step 2. $a^3 + b^3 = 12^3 - 3(27)(12) = 1728 - 972 = 756$.

Step 3. Square identity: $a^2 + b^2 = (a+b)^2 - 2ab = 144 - 54 = 90$.

» **Why this step?** Both bridges express symmetric quantities in terms of (sum, product) - no need to solve for a and b.

Answer: $a^3 + b^3 = 756$, $a^2 + b^2 = 90$

Hinglish: *Symmetric quantities ka rule: sab kuch (sum, product) se nikhlo. Cube bridge: $(a+b)^3 - 3ab(a+b)$. 1728 - 972 = 756. Square: 144 - 54 = 90. Solving a, b se bhi hoga but identity wala path 2 line ka hai.*

QUESTION 38 | Moderate | Near-number cube (T3)

Q38. Evaluate 103^3 without a calculator.

SOLUTION — step by step

Step 1. Write $103 = 100 + 3$ and use $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$.

Step 2. $a^3 = 100^3 = 1{,}000{,}000$.

Step 3. $3a^2b = 3 \times 10000 \times 3 = 90{,}000$.

Step 4. $3ab^2 = 3 \times 100 \times 9 = 2{,}700$.

Step 5. $b^3 = 27$.

Step 6. Total: $1{,}000{,}000 + 90{,}000 + 2{,}700 + 27 = 1{,}092{,}727$.

» **Why this step?** Powers of 100 are trivial, so the whole cube collapses into four easy additions.

Answer: 1,092,727

Hinglish: *Cube bhi round-number trick se hoga: 1-3-3-1 pattern with 100 aur 3. Terms: 1000000, 90000, 2700, 27. Jodo: 1092727. Column-wise addition karo, zeros ginn kar.*

QUESTION 39 | Moderate | Grouping with identities (T12)

Q39. Factorise: $a^2 - b^2 + 2bc - c^2$.

SOLUTION — step by step

Step 1. Given: four terms, no common factor. Try grouping with a sign flip:

Step 2. $a^2 - b^2 + 2bc - c^2 = a^2 - (b^2 - 2bc + c^2)$.

Step 3. The bracket is a perfect square: $= a^2 - (b - c)^2$.

Step 4. Difference of squares: $= (a + b - c)(a - b + c)$.

Step 5. Check by expanding one way: $(a + (b-c))(a - (b-c)) = a^2 - (b-c)^2$ ✓.

» **Why this step?** The last three terms are -1 times a perfect square; grouping reveals a two-square difference.

Answer: $(a + b - c)(a - b + c)$

Hinglish: *Pehli nazar mein kuch nahi dikh raha. Last teen terms ko bracket mein -1 ke saath rakho: $-(b^2-2bc+c^2) = -(b-c)^2$. Ab classic $a^2 - (b-c)^2$. Grouping + identity ka combo - ye pattern yaad rakho.*

QUESTION 40 | Moderate | 4ab identity in action (T3)

Q40. Prove that $(a+b)^2 - (a-b)^2 = 4ab$, and use it to evaluate $5.5^2 - 4.5^2$.

SOLUTION — step by step

Step 1. Expand both squares: $(a+b)^2 = a^2 + 2ab + b^2$; $(a-b)^2 = a^2 - 2ab + b^2$.

Step 2. Subtract: $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$. ∎

Step 3. Application: $5.5^2 - 4.5^2 = 4 \times 5.5 \times 4.5 = 4 \times 24.75 = 99$.

Step 4. Cross-check with the difference of squares: $(5.5+4.5)(5.5-4.5) = 10 \times 1 = 99$ ✓ ($4ab = (a+b)(a-b)$ here since $a = (10/2)$, $b = (1/2)$... the two results agree).

» **Why this step?** Two different identities confirm the same value - a built-in verification habit.

Answer: Proved; $5.5^2 - 4.5^2 = 99$

Hinglish: *Proof 2 line ka hai - dono squares expand karo, subtract karo, 4ab. Application: $4 \times 5.5 \times 4.5 = 99$. Yehi identity multiplication-table trick ban jati hai: $k^2-(k-1)^2 = 2k-1$ bhi isi family ka hai.*

QUESTION 41 | Moderate | Cube difference bridge (T11)

Q41. If a - b = 4 and ab = 21, find $a^3 - b^3$.

SOLUTION — step by step

Step 1. Given: difference and product. Bridge: $a^3 - b^3 = (a-b)^3 + 3ab(a-b)$.

Step 2. $= 4^3 + 3(21)(4) = 64 + 252 = 316$.

Step 3. Quick sanity: a - b = 4, ab = 21 admits a = 7, b = 3: 343 - 27 = 316 ✓.

» **Why this step?** Expanding $(a-b)^3$ shows the spare term +3ab(a-b); moving it left creates the bridge.

Answer: 316

Hinglish: *Minus wala bridge: $(a-b)^3 + 3ab(a-b) = a^3-b^3$. Sign yaad rakho - difference wale case mein PLUS 3ab(a-b). 64 + 252 = 316. Check line a=7, b=3 se confirm.*

QUESTION 42 | Moderate | Crown identity evaluation (T9)

Q42. Using a suitable identity, evaluate $12^3 + 1^3 + (-13)^3$.

SOLUTION — step by step

Step 1. Spot the zero sum: 12 + 1 + (-13) = 0.

Step 2. Zero-sum shortcut: if a + b + c = 0 then $a^3 + b^3 + c^3 = 3abc$.

Step 3. So the value = 3 × 12 × 1 × (-13) = -468.

» **Why this step?** The identity $a^3+b^3+c^3-3abc = (a+b+c)(...)$ collapses to 0 on the left whenever a+b+c = 0.

Answer: -468

Hinglish: *Numbers dekho: 12, 1, -13 - sum zero! Turant shortcut: cubes ka sum = 3abc = -468. Bina koi cube nikaale. Ye 'zero-sum spotting' exam mein 3 second ka kaam hai - 5 marks bachate hain.*

QUESTION 43 | Moderate | x - 1/x chain (T10)

Q43. If $x - 1/x = 3$, find (i) $x^2 + 1/x^2$ and (ii) $x^3 - 1/x^3$.

SOLUTION — step by step

Step 1. Given: $x - 1/x = 3$. Square both sides: $x^2 - 2 + 1/x^2 = 9$.

Step 2. (i) $x^2 + 1/x^2 = 9 + 2 = 11$ (note: PLUS 2 here, because the cross term -2 moved right).

Step 3. Cube both sides: $(x - 1/x)^3 = x^3 - 1/x^3 - 3(x - 1/x) \cdot ...$ using $(a-b)^3 = a^3-b^3-3ab(a-b)$ with ab = 1:

Step 4. $27 = x^3 - 1/x^3 - 3 \times 3$.

Step 5. (ii) $x^3 - 1/x^3 = 27 + 9 = 36$.

» **Why this step?** The x·(1/x) = 1 fact makes every middle term a constant: -2 when squaring the difference, +3(x-1/x) when cubing.

Answer: (i) 11 (ii) 36

Hinglish: *Difference wale chain mein SIGN flip hota hai: square par +2 (minus ka minus), cube par +3(x-1/x). 3 → 11 → 36. Sum wale (x+1/x) chain se confuse mat hona - wahan minus aata tha.*

QUESTION 44 | Moderate | Trinomial square with signs (T5)

Q44. Expand: $(2a + b - 3c)^2$.

SOLUTION — step by step

Step 1. Identify: A = 2a, B = b, C = -3c.

Step 2. Squares: $4a^2 + b^2 + 9c^2$.

Step 3. Cross terms: 2AB = 4ab; 2BC = -6bc; 2CA = -12ca.

Step 4. Result: $4a^2 + b^2 + 9c^2 + 4ab - 6bc - 12ca$.

» **Why this step?** Sign of 2BC = (+)(-) = -, sign of 2CA = (-)(+) = -: the signs multiply, no exceptions.

Answer: $4a^2 + b^2 + 9c^2 + 4ab - 6bc - 12ca$

Hinglish: *Same three-square recipe. Cross terms ke signs: a-b wala positive (dono +), b-c wala negative (+ aur -), c-a wala negative. $9c^2$ isliye kyunki $(-3c)^2 = 9c^2$ - squares hamesha positive.*

QUESTION 45 | Moderate | Grouped square minus square (T12)

Q45. Factorise completely: $9x^2 + 24xy + 16y^2 - 25z^2$.

SOLUTION — step by step

Step 1. First three terms: $9x^2 + 24xy + 16y^2 = (3x + 4y)^2$ (middle 2·3·4 = 24 ✓).

Step 2. Expression = $(3x + 4y)^2 - (5z)^2$.

Step 3. = (3x + 4y + 5z)(3x + 4y - 5z).

» **Why this step?** Spot the hidden perfect square FIRST, then the whole expression becomes a two-term difference of squares.

Answer: (3x + 4y + 5z)(3x + 4y - 5z)

Hinglish: *Pehle teen terms ka perfect square $(3x+4y)^2$, phir $25z^2 = (5z)^2$. Do-step factorisation - square pehchano, phir a^2-b^2. Ye combo board exam ka favourite hai.*

QUESTION 46 | Moderate | Cube pattern recognition (T7/T8)

Q46. (i) Expand $(2x - 3)(4x^2 + 6x + 9)$ without long multiplication. (ii) Factorise $x^3 - 512$ completely.

SOLUTION — step by step

Step 1. (i) This matches $a^3 - b^3$ with $a = 2x$, $b = 3$: the second bracket is $a^2 + ab + b^2 = 4x^2 + 6x + 9$.

Step 2. So the product $= (2x)^3 - 3^3 = 8x^3 - 27$.

Step 3. (ii) $x^3 - 512 = x^3 - 8^3 = (x - 8)(x^2 + 8x + 64)$.

» **Why this step?** Learn to read $a^3 - b^3$'s second bracket BACKWARDS: seeing $a^2 + ab + b^2$ next to $(a - b)$ means the product collapses to a cube pair.

Answer: (i) $8x^3 - 27$ (ii) $(x - 8)(x^2 + 8x + 64)$

Hinglish: *Doosra bracket $4x^2 + 6x + 9$? Ye $a^2 + ab + b^2$ ka costume hai ($a = 2x$, $b = 3$)! Isliye product seedha $8x^3 - 27$. Pattern ko ulta bhi padhna aana chahiye.*

QUESTION 47 | Moderate | x + 1/x deeper chain (T10)

Q47. If $x + 1/x = 5$, find (i) $x^2 + 1/x^2$, (ii) $x^3 + 1/x^3$, (iii) $x^4 + 1/x^4$.

SOLUTION — step by step

Step 1. (i) Square: $25 - 2 = 23$.

Step 2. (ii) Cube: $x^3 + 1/x^3 = (x + 1/x)^3 - 3(x + 1/x) = 125 - 15 = 110$.

Step 3. (iii) Square the (i) result: $23^2 - 2 = 529 - 2 = 527$.

» **Why this step?** Two engines: squaring (+2 in the middle) and cubing (−3s). Alternate them to reach any power.

Answer: (i) 23 (ii) 110 (iii) 527

Hinglish: *Square: -2. Cube: -3(x + 1/x). Phir square: -2. 5 → 23 → 110 → 527. Ye ladder ek baar chalana aayi toh family ke saare questions tumhare.*

QUESTION 48 | Moderate | Both bridges at once (T11)

Q48. If $a + b = 10$ and $a - b = 6$, find (i) ab, (ii) $a^2 + b^2$, (iii) $a^4 - b^4$.

SOLUTION — step by step

Step 1. (i) $ab = ((a + b)^2 - (a - b)^2)/4 = (100 - 36)/4 = 16$.

Step 2. (ii) $a^2 + b^2 = ((a + b)^2 + (a - b)^2)/2 = (100 + 36)/2 = 68$.

Step 3. (iii) $a^4 - b^4 = (a^2 + b^2)(a^2 - b^2) = 68 \times (a + b)(a - b) = 68 \times 60 = 4080$.

» **Why this step?** Both bridges together give ab AND $a^2 + b^2$ - then difference of squares chains the fourth powers.

Answer: (i) 16 (ii) 68 (iii) 4080

Hinglish: *Sum aur difference dono mile toh ab aur a^2+b^2 formula se niklo, a-b solve karne ki zaroorat nahi! (iii) mein chain: $68 \times 10 \times 6 = 4080$.*

QUESTION 49 | Moderate | Complete the square then split (T12)

Q49. Factorise: $x^2 + 12x + 36 - 9y^2$.

SOLUTION — step by step

Step 1. First three terms: $x^2 + 12x + 36 = (x + 6)^2$.

Step 2. Expression = $(x + 6)^2 - (3y)^2$.

Step 3. = $(x + 6 + 3y)(x + 6 - 3y)$.

» **Why this step?** Add-nothing trick: the first three terms were already a perfect square - the $-9y^2$ was waiting to become $(3y)^2$.

Answer: $(x + 6 + 3y)(x + 6 - 3y)$

Hinglish: *$(x+6)^2$ pehle se mojood tha! $-9y^2$ ko $(3y)^2$ banao aur $a^2 - b^2$. 'Pehle group karo, phir identity' - yehi two-step dance hai.*

QUESTION 50 | Moderate | Cube bridge classic (T11)

Q50. If $a + b = 9$ and $ab = 20$, find $a^3 + b^3$.

SOLUTION — step by step

Step 1. $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$.

Step 2. = $9^3 - 3(20)(9) = 729 - 540$.

Step 3. = 189.

» **Why this step?** The cube bridge: $(a + b)^3$ minus the $3ab(a + b)$ correction term.

Answer: 189

Hinglish: *Cube bridge: $S^3 - 3PS$. $729 - 540 = 189$. $(a+b)^3$ pure expand karne ke chakkar mein mat pado - correction term hi asli shortcut hai.*

QUESTION 51 | Moderate | Cube-sum evaluation trick (T3/T11)

Q51. Evaluate $88^3 + 12^3$ without cubing either number.

SOLUTION — step by step

Step 1. Note a + b = 88 + 12 = 100 and ab = 88 × 12 = 1056.

Step 2. $a^3 + b^3 = (a + b)^3 - 3ab(a + b) = 100^3 - 3(1056)(100)$.

Step 3. = 1000000 - 316800 = 683200.

» **Why this step?** Numbers summing to a round 100 beg for the cube-sum identity - no cubing required.

Answer: 683200

Hinglish: *88 + 12 = 100? Round number dikh te hi bridge kholo. 10^6 - 3×1056×100 = 683200. Ek bhi cube nahi kiya!*

QUESTION 52 | Moderate | x - 1/x chain (T10)

Q52. If $x - 1/x = 7$, find (i) $x^2 + 1/x^2$, (ii) $x^3 - 1/x^3$.

SOLUTION — step by step

Step 1. (i) Square: $x^2 - 2 + 1/x^2 = 49 \rightarrow x^2 + 1/x^2 = 51$.

Step 2. (ii) $x^3 - 1/x^3 = (x - 1/x)^3 + 3(x - 1/x) = 343 + 21 = 364$.

» **Why this step?** The minus family: squaring adds 2 back, cubing adds 3s (sign flips versus the plus family).

Answer: (i) 51 (ii) 364

Hinglish: *Minus family ke rules: square mein +2, cube mein +3(x - 1/x). 49 + 2 = 51, 343 + 21 = 364. Plus/minus ka sign table yaad rakho.*

QUESTION 53 | Moderate | Trinomial square reverse (T5)

Q53. If a + b + c = 11 and ab + bc + ca = 28, find $a^2 + b^2 + c^2$.

SOLUTION — step by step

Step 1. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$.

Step 2. $121 = a^2 + b^2 + c^2 + 56$.

Step 3. $a^2 + b^2 + c^2 = 65$.

» **Why this step?** The trinomial square rearranged: squares = $S^2 - 2$(pairwise sum).

Answer: 65

Hinglish: *S^2 = squares + 2P formula ulta chalao: 121 - 56 = 65. Teen variables ka bridge bhi utna hi seedha hai.*

QUESTION 54 | Moderate | Grouping with identities (T12)

Q54. Factorise: $a^2 - b^2 + 2bc - c^2$.

SOLUTION — step by step

Step 1. Group the last three: $-(b^2 - 2bc + c^2) = -(b - c)^2$.

Step 2. Expression = $a^2 - (b - c)^2$.

Step 3. = $(a + b - c)(a - b + c)$.

» **Why this step?** Rearrange so a hidden perfect square appears, then $a^2 - B^2$ finishes the job.

Answer: $(a + b - c)(a - b + c)$

Hinglish: *$b^2 - 2bc + c^2$ ko $-(b-c)^2$ banake bahar kheencho. Phir $a^2 - (b-c)^2$ ka difference of squares. Sign ko bracket ke saath le jaana hi trick hai.*

QUESTION 55 | Moderate | Add-subtract trick (T12)

Q55. Factorise: $x^4 + 4$.

SOLUTION — step by step

Step 1. Add and subtract $4x^2$: $x^4 + 4x^2 + 4 - 4x^2$.

Step 2. = $(x^2 + 2)^2 - (2x)^2$.

Step 3. = $(x^2 + 2x + 2)(x^2 - 2x + 2)$.

» **Why this step?** $x^4 + 4$ has NO direct identity - the add-subtract $4x^2$ move manufactures a difference of squares out of thin air.

Answer: $(x^2 + 2x + 2)(x^2 - 2x + 2)$

Hinglish: *Sophie Germain ka famous trick: $4x^2$ ko jodo aur ghat do. $(x^2+2)^2 - (2x)^2$ bana, phir split. 'Kuch nahi ho sakta' wale question ka jawab ye hi hai.*

QUESTION 56 | Moderate | Middle-term completion (T12)

Q56. Factorise: $x^4 + 5x^2 + 9$.

SOLUTION — step by step

Step 1. $(x^2 + 3)^2$ would give $x^4 + 6x^2 + 9$ - one x^2 too many.

Step 2. Write: $x^4 + 6x^2 + 9 - x^2 = (x^2 + 3)^2 - x^2$.

Step 3. = $(x^2 + x + 3)(x^2 - x + 3)$.

» **Why this step?** Compare with the nearest perfect square, then patch the gap with a subtracted square.

Answer: $(x^2 + x + 3)(x^2 - x + 3)$

Hinglish: *$6x^2$ chahiye tha, $5x^2$ mila - $1x^2$ ki kami. Upar se $6x^2$ likho, neeche x^2 minus. Phir $a^2 - b^2$. Completion + patch = two-step magic.*

QUESTION 57 | Moderate | Cubic factorisation by root (T8)

Q57. Factorise completely: $x^3 - 7x - 6$.

SOLUTION — step by step

Step 1. Hunt a root: x = -1 gives -1 + 7 - 6 = 0 ✓ -> (x + 1) is a factor.

Step 2. Divide: $x^3 - 7x - 6 = (x + 1)(x^2 - x - 6)$.

Step 3. Split -x = -3x + 2x: $x^2 - x - 6 = (x - 3)(x + 2)$.

Step 4. Fully: (x + 1)(x - 3)(x + 2).

» **Why this step?** Cubics fall to the root-test: try ±1, ±2 first; then the quotient is always a quadratic you can split.

Answer: (x + 1)(x - 3)(x + 2)

Hinglish: *Chhote values daal ke root dhoondo: -1 ne zero diya -> (x+1) factor. Divide karo, quadratic aayi, split karo. Teen factors: (x+1)(x-3)(x+2).*

QUESTION 58 | Moderate | Cubic with lead coefficient (T8)

Q58. Factorise completely: $2x^3 + 5x^2 - x - 6$.

SOLUTION — step by step

Step 1. Test x = 1: 2 + 5 - 1 - 6 = 0 ✓ -> (x - 1) is a factor.

Step 2. Divide: quotient = $2x^2 + 7x + 6$.

Step 3. Split 7x = 4x + 3x: $2x^2 + 4x + 3x + 6 = 2x(x + 2) + 3(x + 2) = (x + 2)(2x + 3)$.

Step 4. Fully: (x - 1)(x + 2)(2x + 3).

» **Why this step?** The root-test works with lead coefficients too; the quotient's split uses product = c/a pairs.

Answer: (x - 1)(x + 2)(2x + 3)

Hinglish: *x = 1 phir se hero! Quotient $2x^2 + 7x + 6$ - splitting mein product 12 (2×6) lete hain: 4 aur 3. Final teen factors, ek mein coefficient 2.*

QUESTION 59 | Moderate | Zero-sum with big numbers (T9)

Q59. Evaluate $27^3 - 53^3 + 26^3$ without cubing.

SOLUTION — step by step

Step 1. Check the sum: 27 - 53 + 26 = 0 -> the zero-sum cube law applies.

Step 2. $a^3 + b^3 + c^3 = 3abc$ when a + b + c = 0 (with signs as given).

Step 3. 3(27)(-53)(26) = -3 × 27 × 53 × 26.

Step 4. 27 × 53 = 1431; 1431 × 26 = 37206; × 3 = 111618. Answer: -111618.

» **Why this step?** Whenever three numbers sum to zero, their cubes collapse to 3abc - even with ugly values.

Answer: -111618

Hinglish: *Pehle sum check: 27 - 53 + 26 = 0? Haan! Toh cubes ka jod seedha 3abc. Sign dhyan se: beech wala negative, isliye answer -111618.*

QUESTION 60 | Moderate | Cyclic zero-sum (T9)

Q60. Evaluate $(a - b)^3 + (b - c)^3 + (c - a)^3$ for a = 5, b = 3, c = 1, using an identity.

SOLUTION — step by step

Step 1. Let X = a - b = 2, Y = b - c = 2, Z = c - a = -4.

Step 2. X + Y + Z = 2 + 2 - 4 = 0 -> $X^3 + Y^3 + Z^3 = 3XYZ$.

Step 3. = 3(2)(2)(-4) = -48.

Step 4. Direct check: 8 + 8 + (-64) = -48 ✓.

» **Why this step?** The three differences ALWAYS sum to zero - so the cyclic cube expression is always 3(a – b)(b – c)(c – a).

Answer: -48

Hinglish: *Differences ka sum hamesha zero hota hai (round trip!). Isliye seedha 3XYZ = 3(2)(2)(-4) = -48. Ye identity cyclic cube ka permanent formula hai.*

QUESTION 61 | Moderate | Cube bridge with product (T11)

Q61. If p + q = 11 and pq = 24, find $p^3 + q^3$ and $p^2 - pq + q^2$.

SOLUTION — step by step

Step 1. $p^3 + q^3 = (p + q)^3 - 3pq(p + q) = 1331 - 3(24)(11) = 1331 - 792 = 539$.

Step 2. $p^2 - pq + q^2 = (p^2 + q^2) - pq = ((p + q)^2 - 2pq) - pq = (121 - 48) - 24 = 49$.

Step 3. Cross-check: $p^3 + q^3 = (p + q)(p^2 - pq + q^2) = 11 \times 49 = 539$ ✓.

» **Why this step?** The two expressions are linked by the sum-of-cubes factorisation - solving one verifies the other.

Answer: $p^3 + q^3 = 539$; $p^2 - pq + q^2 = 49$

Hinglish: *539 nikla, phir doosra bracket 49 - aur 11 × 49 = 539 se dono ne handshake kiya! Cross-check wali line answer ko bulletproof banati hai.*

QUESTION 62 | Moderate | Scaled cube bridge (T11)

Q62. If $3x + 4y = 15$ and $xy = 6$, find $27x^3 + 64y^3$.

SOLUTION — step by step

Step 1. Recognise cubes: $27x^3 = (3x)^3$, $64y^3 = (4y)^3$.

Step 2. Let $A = 3x$, $B = 4y$: $A + B = 15$ and $AB = (3x)(4y) = 12xy = 72$.

Step 3. $A^3 + B^3 = (A + B)^3 - 3AB(A + B) = 3375 - 3(72)(15) = 3375 - 3240 = 135$.

» **Why this step?** Spot the cube clothes (3x, 4y), rename them, and the bridge applies unchanged - scaling is just renaming.

Answer: 135

Hinglish: *$27x^3$ aur $64y^3$? Ye $(3x)^3$ aur $(4y)^3$ hain! $A+B = 15$, $AB = 12xy = 72$. Bridge lagao: $3375 - 3240 = 135$. Renaming hi scaling ka solution hai.*

QUESTION 63 | Moderate | Crown identity forward (T14)

Q63. If $x + y + z = 10$ and $xy + yz + zx = 35$, find $x^2 + y^2 + z^2$ and $x^3 + y^3 + z^3 - 3xyz$.

SOLUTION — step by step

Step 1. $x^2 + y^2 + z^2 = (x + y + z)^2 - 2(xy + yz + zx) = 100 - 70 = 30$.

Step 2. Crown identity: $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$.

Step 3. $= 10(30 - 35) = 10 \times (-5) = -50$.

» **Why this step?** The crown identity needs only $S = x+y+z$ and the pairwise sum P - both were donated.

Answer: $x^2 + y^2 + z^2 = 30$; $x^3 + y^3 + z^3 - 3xyz = -50$

Hinglish: *Squares = 100 - 70 = 30. Crown identity mein bracket (30 - 35) = -5 aata hai - negative bhi legal hai! 10 × (-5) = -50.*

QUESTION 64 | Moderate | Crown identity as factorisation (T8/T14)

Q64. Factorise: $x^3 + 8y^3 + 8z^3 - 12xyz$.

SOLUTION — step by step

Step 1. Match the crown form: a = x, b = 2y, c = 2z gives $a^3 + b^3 + c^3 = x^3 + 8y^3 + 8z^3$ and 3abc = 3(x)(2y)(2z) = 12xyz.

Step 2. Use $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$.

Step 3. Second bracket: $x^2 + 4y^2 + 4z^2 - 2xy - 4yz - 2xz$.

Step 4. Result: $(x + 2y + 2z)(x^2 + 4y^2 + 4z^2 - 2xy - 4yz - 2xz)$.

» **Why this step?** The crown identity is ALSO a factorisation formula - feed it disguised cubes and it splits the expression.

Answer: $(x + 2y + 2z)(x^2 + 4y^2 + 4z^2 - 2xy - 4yz - 2xz)$

Hinglish: *$8y^3 = (2y)^3$, $8z^3 = (2z)^3$ - crown identity ke liye perfect costume. -12xyz = -3(a)(b)(c) match hua. Do brackets ready. Ye identity factorise bhi karti hai, yaad rakho!*

QUESTION 65 | Moderate | Decimal cube sum (T3/T9)

Q65. Evaluate $0.87^3 + 0.13^3$ without cubing the decimals.

SOLUTION — step by step

Step 1. a + b = 1 (round!), ab = 0.87 × 0.13 = 0.1131.

Step 2. $a^3 + b^3 = (a + b)^3 - 3ab(a + b) = 1 - 3(0.1131)(1)$.

Step 3. = 1 - 0.3393 = 0.6607.

» **Why this step?** Decimals that complement to 1 are made for the cube-sum bridge - one multiplication, no cubing.

Answer: 0.6607

Hinglish: *0.87 + 0.13 = 1 - instant round number! ab = 0.1131. Answer 1 - 0.3393 = 0.6607. Decimal cubes kabhi mat karo, bridge hi saathi hai.*

QUESTION 66 | Moderate | Cube difference bridge (T11)

Q66. If $a - b = 6$ and $ab = 16$, find $a^3 - b^3$.

SOLUTION — step by step

Step 1. $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$.

Step 2. $= 216 + 3(16)(6) = 216 + 288$.

Step 3. $= 504$.

» **Why this step?** The difference bridge ADDS the correction: $(a - b)^3 + 3ab(a - b)$.

Answer: 504

Hinglish: *Difference cube: $D^3 + 3abD$. $216 + 288 = 504$. Sum waale mein minus hota tha, isme plus - sign swap yaad rakho.*

QUESTION 67 | Moderate | Deep chained difference of squares (T2)

Q67. Factorise completely: $x^8 - 256$.

SOLUTION — step by step

Step 1. $256 = 2^8$: chain of difference of squares.

Step 2. $x^8 - 256 = (x^4 - 16)(x^4 + 16)$.

Step 3. $x^4 - 16 = (x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$.

Step 4. Full: $(x - 2)(x + 2)(x^2 + 4)(x^4 + 16)$.

» **Why this step?** Each difference of squares halves the exponent: $8 \to 4 \to 2 \to 1$; sum-of-squares pieces stop the chain.

Answer: $(x - 2)(x + 2)(x^2 + 4)(x^4 + 16)$

Hinglish: *256 tak ka chain: x^4-16, phir x^2-4, phir x-2/x+2. Beech ke x^2+4 aur x^4+16 ab aur nahi todenge. 'Completely' = jab tak difference of squares dikhe, todte raho.*

QUESTION 68 | Moderate | Four-term grouping (T12)

Q68. Factorise: $x^3 - 3x^2 + x - 3$.

SOLUTION — step by step

Step 1. Group: $x^2(x - 3) + 1(x - 3)$.

Step 2. Common bracket: $(x - 3)(x^2 + 1)$.

Step 3. $x^2 + 1$ does not factor over the reals - done.

» **Why this step?** Pair terms so each group donates the SAME bracket - the common-factor harvest.

Answer: $(x - 3)(x^2 + 1)$

Hinglish: *Do-do terms ke jodi banao: $x^2(x-3) + (x-3)$. Common bracket bahar: $(x-3)(x^2+1)$. x^2+1 real mein factor nahi hota - wahi final.*

QUESTION 69 | Moderate | Three-term perfect square (T5)

Q69. Factorise: $9a^2 + 4b^2 + 25c^2 - 12ab - 20bc + 30ac$.

SOLUTION — step by step

Step 1. Look for $(A \pm B \pm C)^2$: squares $9a^2$, $4b^2$, $25c^2$ suggest A = 3a, B = -2b, C = 5c.

Step 2. Check doubles: $2(3a)(-2b) = -12ab$ ✓; $2(-2b)(5c) = -20bc$ ✓; $2(3a)(5c) = +30ac$ ✓ - all match!

Step 3. Expression = $(3a - 2b + 5c)^2$.

» **Why this step?** Match the three doubles against 2AB, 2BC, 2CA - if all three agree, it is a perfect trinomial square.

Answer: $(3a - 2b + 5c)^2$

Hinglish: *Signs se triple banao: 3a, -2b, +5c. Teenon doubles check kiye - perfect match! Ye 'detective' wala question hai - signs se suspect pehchano.*

QUESTION 70 | Moderate | Split for difference of squares (T2/T3)

Q70. Evaluate 98×52 by rewriting both numbers around a common centre.

SOLUTION — step by step

Step 1. Centre: $(98 + 52)/2 = 75$. Offsets: $98 = 75 + 23$, $52 = 75 - 23$.

Step 2. $98 \times 52 = (75 + 23)(75 - 23) = 75^2 - 23^2$.

Step 3. = 5625 - 529 = 5096.

» **Why this step?** ANY product $a \times b = ((a+b)/2)^2 - ((a-b)/2)^2$ - difference of squares is universal multiplication gear.

Answer: 5096

Hinglish: *Average 75 pakdo, offset 23. (75+23)(75-23) = 5625 - 529 = 5096. Ye universal trick hai - koi bhi multiplication isse hota hai!*

QUESTION 71 | Moderate | Zero-sum at power two (T9)

Q71. (i) Prove that if $a + b + c = 0$, then $a^2 + b^2 + c^2 = -2(ab + bc + ca)$. (ii) Verify for $a = 4$, $b = -3$, $c = -1$.

SOLUTION — step by step

Step 1. (i) Square the sum: $0 = (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$.

Step 2. Move the pair term across: $a^2 + b^2 + c^2 = -2(ab + bc + ca)$.

Step 3. (ii) LHS: $16 + 9 + 1 = 26$. RHS: $-2(-12 + 3 - 4) = -2(-13) = 26$ ✓.

» **Why this step?** Squaring a zero sum creates an instant bridge between squares and pairwise products - negative and all.

Answer: (i) proof above (ii) 26 = 26 ✓

Hinglish: *Zero ka square = 0, phir pair term ko udhar bhej do - proof 2 lines ka! Verify: 26 = 26. Negative pair sum bhi ho sakta hai, RHS ka minus usko sambhalta hai.*

QUESTION 72 | Moderate | Sum and difference squares pair (T1/T3)

Q72. (i) Simplify $(2x + 3y)^2 + (2x - 3y)^2$. (ii) Hence evaluate $51^2 + 49^2$ mentally.

SOLUTION — step by step

Step 1. (i) Add the expansions: $(4x^2 + 12xy + 9y^2) + (4x^2 - 12xy + 9y^2) = 8x^2 + 18y^2$.

Step 2. (ii) Write $51 = 50 + 1$, $49 = 50 - 1$: sum of squares $= 2(50^2 + 1^2) = 2(2501) = 5002$.

» **Why this step?** $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$: the cross terms cancel - the fastest route to 'sum of squares of near numbers'.

Answer: (i) $8x^2 + 18y^2$ (ii) 5002

Hinglish: *Cross terms cancel ho gaye: $8x^2 + 18y^2$. Aur $51^2 + 49^2 = 2(2500 + 1) = 5002$ - mental! Near-numbers ka sum-of-squares hamesha aise hi.*

QUESTION 73 | Moderate | Substitution factorisation (T12)

Q73. Factorise: $(x^2 - 3x)^2 - 8(x^2 - 3x) - 84$.

SOLUTION — step by step

Step 1. Substitute $u = x^2 - 3x$: $u^2 - 8u - 84$.

Step 2. Split $-8u = -14u + 6u$ (product -84): $(u - 14)(u + 6)$.

Step 3. Back-substitute: $(x^2 - 3x - 14)(x^2 - 3x + 6)$.

Step 4. Neither quadratic splits over the integers (discriminants 65 and -15) - this is the complete factorisation.

» **Why this step?** A repeated bracket begs for substitution: solve the easy quadratic in u, then undo the substitution.

Answer: $(x^2 - 3x - 14)(x^2 - 3x + 6)$

Hinglish: *(x^2-3x) ko u naam do. u^2 - 8u - 84 = (u-14)(u+6). Wapas x mein: do quadratics. Discriminant check kiya - aur factor nahi hote. 'Completely' bolne se pehle check karna zaroori hai.*

QUESTION 74 | Moderate | Decimal cube difference (T3/T11)

Q74. Evaluate $0.73^3 - 0.23^3$ without cubing.

SOLUTION — step by step

Step 1. a - b = 0.5, ab = 0.73 × 0.23 = 0.1679.

Step 2. $a^3 - b^3 = (a - b)^3 + 3ab(a - b) = 0.125 + 3(0.1679)(0.5)$.

Step 3. = 0.125 + 0.25185 = 0.37685.

» **Why this step?** The difference bridge works for decimals identically: $D^3 + 3abD$.

Answer: 0.37685

Hinglish: *Difference 0.5 hai, ab 0.1679. 0.125 + 0.25185 = 0.37685. Decimal aaye toh bhi bridge wahi - bas point ka khayal.*

QUESTION 75 | Moderate | Roots-of-product identity (T4)

Q75. (i) State the identity for $x^2 - (a + b)x + ab$. (ii) Use it to factorise $x^2 - 11x + 24$.

SOLUTION — step by step

Step 1. (i) $x^2 - (a + b)x + ab = (x - a)(x - b)$: roots a and b, sum on the middle, product at the end.

Step 2. (ii) Need two numbers with sum 11, product 24: 3 and 8.

Step 3. $x^2 - 11x + 24 = (x - 3)(x - 8)$.

» **Why this step?** This identity is the bridge between expanding (x + a)(x + b) and factorising quadratics - same gear, reverse gear.

Answer: (i) $x^2 - (a + b)x + ab = (x - a)(x - b)$ (ii) $(x - 3)(x - 8)$

Hinglish: *Middle = sum, end = product - wahi purana dost, ab minus ke saath. 3 aur 8 (3+8 = 11, 3×8 = 24). Factorising quadratic = ye identity ka reverse drive.*

QUESTION 76 | Moderate | Geometric proof of binomial square (T14)

Q76. Using an area model, explain geometrically why $(a + b)^2 = a^2 + 2ab + b^2$, and verify with a = 3, b = 4.

SOLUTION — step by step

Step 1. Draw a square of side (a + b); split each side into lengths a and b.

Step 2. The big square splits into 4 regions: an a × a square, a b × b square, and TWO a × b rectangles.

Step 3. Total area: $a^2 + ab + ab + b^2 = a^2 + 2ab + b^2$ - the identity is an area statement.

Step 4. Verify: $(3 + 4)^2 = 49$; $3^2 + 2(3)(4) + 4^2 = 9 + 24 + 16 = 49$ ✓.

» **Why this step?** Identities are area statements: the picture shows WHY the middle term must appear twice (two different rectangles of the same size).

Answer: Big square = $a^2 + ab + ab + b^2 = a^2 + 2ab + b^2$; verified 49 = 49

Hinglish: *Ek bada square, chaar tukde: a^2, b^2, aur do ab ke rectangle. Do alag-alag rectangle isliye - isi liye '2ab' mein 2 hai! Geometry identity ka birth certificate hai.*

Level 3 · Exam Power Zone: long and twisted:

QUESTION 77 | Difficult | Cubic factorisation by root (T12)

Q77. Factorise completely: $x^3 - 3x^2 - 9x - 5$.

SOLUTION — step by step

Step 1. Hunt for an integer root: try x = -1: $(-1)^3 - 3(1) - 9(-1) - 5 = -1 - 3 + 9 - 5 = 0$ ✓. So (x + 1) is a factor.

Step 2. Divide: $x^3 - 3x^2 - 9x - 5 = (x + 1)(x^2 - 4x - 5)$.

Step 3. (Verify by expanding: $x^3 - 4x^2 - 5x + x^2 - 4x - 5 = x^3 - 3x^2 - 9x - 5$ ✓.)

Step 4. Factorise the quadratic: $x^2 - 4x - 5$: sum -4, product -5 → -5 and +1: (x - 5)(x + 1).

Step 5. Final: $(x + 1)^2(x - 5)$.

» **Why this step?** An integer root r of a polynomial always yields the factor (x - r); testing small integers (0, ±1, ±2) is the fastest opening move.

Answer: $(x + 1)^2(x - 5)$

Hinglish: *Cubic factorisation ka opening: chhote integers try karo. x = -1 par zero mila, toh (x+1) factor. Long division ya coefficient comparison se quadratic mila: x^2-4x-5, wo phir middle-term split. Final $(x+1)^2(x-5)$.*

QUESTION 78 | Difficult | Reverse chain (T10)

Q78. If $x^2 + 1/x^2 = 38$, find the value of $x - 1/x$.

SOLUTION — step by step

Step 1. Use the difference-square identity: $(x - 1/x)^2 = x^2 - 2 + 1/x^2$.

Step 2. $= 38 - 2 = 36$.

Step 3. So $x - 1/x = \pm 6$.

Step 4. Both signs are possible: $x = 2\sqrt{10}$ gives $x - 1/x = +6$... concretely $x > 1$ gives $+6$ (e.g. $x = 2 + \sqrt{3}$ family) and $0 < x < 1$ gives -6; no extra condition is stated, so report both.

» **Why this step?** Squaring erases the sign, so recovering $x - 1/x$ from its square opens BOTH cases - the classic two-root situation.

Answer: $x - 1/x = 6$ or -6

Hinglish: *$(x-1/x)^2 = x^2+1/x^2 - 2 = 36$. Root: ± 6. Sign ka question: x bada hua toh +6, chhota ($0<x<1$) toh -6. Question ne sign nahi bataya, dono report karo - sirf +6 likhna aadha answer hai.*

QUESTION 79 | Difficult | Cubic pattern collapse (T8/T14)

Q79. Evaluate without finding individual cubes: $0.86^3 + 0.14^3 - 0.86^2 \times 0.14 - 0.86 \times 0.14^2$.

SOLUTION — step by step

Step 1. Group: $a^3 + b^3 - a^2b - ab^2$ with $a = 0.86$, $b = 0.14$.

Step 2. Factor: $a^3 - a^2b + b^3 - ab^2 = a^2(a - b) - b^2(a - b) = (a^2 - b^2)(a - b) = (a+b)(a-b)^2$.

Step 3. Now $a + b = 1$ and $(a - b)^2 = (0.72)^2$.

Step 4. Value $= 1 \times 0.5184 = 0.5184$.

» **Why this step?** Grouping converts four mixed terms into two factors; the sum $a + b = 1$ then finishes in one step.

Answer: 0.5184

Hinglish: *Cubes mat nikaalo - pattern dekho. $a^2(a-b) - b^2(a-b)$ grouping se $(a+b)(a-b)^2$ banta hai. $a+b = 0.86+0.14 = 1$, $(a-b)^2 = 0.72^2 = 0.5184$. Factor pehle, arithmetic baad mein.*

QUESTION 80 | Difficult | Substitution factorisation (T12)

Q80. Factorise completely: $(x^2 - 2x)^2 - 23(x^2 - 2x) + 120$.

SOLUTION — step by step

Step 1. Substitute $y = x^2 - 2x$. The expression becomes $y^2 - 23y + 120$.

Step 2. Split the middle: two numbers with sum -23, product 120: -8 and -15.

Step 3. $y^2 - 23y + 120 = (y - 8)(y - 15)$.

Step 4. Back-substitute: $(x^2 - 2x - 8)(x^2 - 2x - 15)$.

Step 5. Factor each: $x^2 - 2x - 8 = (x-4)(x+2)$; $x^2 - 2x - 15 = (x-5)(x+3)$.

Step 6. Final: $(x - 4)(x + 2)(x - 5)(x + 3)$.

» **Why this step?** The repeated chunk $x^2 - 2x$ is the signal for substitution; every later step is then standard quadratic work.

Answer: $(x - 4)(x + 2)(x - 5)(x + 3)$

Hinglish: *(x^2-2x) do baar repeat ho raha hai - isko y maan lo. Quadratic ho gayi easy. Factor karke WAPAS substitute karna zaroori hai, aur phir dono quadratics ko bhi factor karo - 'completely' tabhi hota hai. Char linear factors final.*

QUESTION 81 | Difficult | Cube bridge with scaled variables (T11)

Q81. If $2x + 3y = 13$ and $xy = 6$, find $8x^3 + 27y^3$.

SOLUTION — step by step

Step 1. Recognise $8x^3 + 27y^3 = (2x)^3 + (3y)^3$.

Step 2. Cube bridge: $A^3 + B^3 = (A+B)^3 - 3AB(A+B)$ with $A = 2x$, $B = 3y$.

Step 3. $(A + B)^3 = 13^3 = 2197$.

Step 4. $AB = (2x)(3y) = 6xy = 36$.

Step 5. $(2x)^3 + (3y)^3 = 2197 - 3(36)(13) = 2197 - 1404 = 793$.

» **Why this step?** Scaling turns the question into the standard (sum, product) bridge - AB is where xy enters.

Answer: 793

Hinglish: *$8x^3 = (2x)^3$, $27y^3 = (3y)^3$ - scaling pehchano. Bridge lagao: $(2x+3y)^3 - 3(2x)(3y)(2x+3y)$. $AB = 6xy = 36$. $2197 - 1404 = 793$. Cubes nikaale bina answer!*

QUESTION 82 | Difficult | Add-subtract trick (T12)

Q82. Factorise: $x^4 + x^2 + 1$.

SOLUTION — step by step

Step 1. Add and subtract x^2: $x^4 + x^2 + 1 = x^4 + 2x^2 + 1 - x^2$.

Step 2. First three terms form a perfect square: $= (x^2 + 1)^2 - x^2$.

Step 3. Difference of squares: $= (x^2 + 1 - x)(x^2 + 1 + x)$.

Step 4. Final: $(x^2 - x + 1)(x^2 + x + 1)$.

Step 5. Check the middle terms on expansion: $-x^3$ and $+x^3$ cancel ✓.

» **Why this step?** Injecting $-x^2 + x^2$ manufactures a perfect square plus a difference of squares - the standard completion for $a^4 + a^2 + 1$ shapes.

Answer: $(x^2 - x + 1)(x^2 + x + 1)$

Hinglish: *x^4+x^2+1 classic trap hai - koi seedha identity nahi lagta. x^2 add + subtract karo: perfect square ban jata hai $(x^2+1)^2$, aur $-x^2$ difference of squares deta hai. Ye 'completion' trick a^4+4b^4 (Sophie Germain style) mein bhi chalta hai.*

QUESTION 83 | Difficult | Zero-sum with differences (T9)

Q83. Evaluate: $(a - b)^3 + (b - c)^3 + (c - a)^3$.

SOLUTION — step by step

Step 1. Let $A = a - b$, $B = b - c$, $C = c - a$. Their sum: $A + B + C = a - b + b - c + c - a = 0$.

Step 2. Zero-sum shortcut: $A^3 + B^3 + C^3 = 3ABC$.

Step 3. So the expression $= 3(a - b)(b - c)(c - a)$.

» **Why this step?** Any three quantities summing to zero obey the cube shortcut - the letters may themselves be differences.

Answer: $3(a - b)(b - c)(c - a)$

Hinglish: *(a-b), (b-c), (c-a) ka sum zero hai - bas zero-sum shortcut lagao: 3ABC. Answer 3(a-b)(b-c)(c-a). Ye pattern bade questions mein chhupa hota hai - sum-zero spotting sabse powerful tool hai.*

QUESTION 84 | Difficult | Cube pair simplify (T7)

Q84. Simplify completely: $(a + b)^3 + (a - b)^3$.

SOLUTION — step by step

Step 1. Expand $(a+b)^3$: $a^3 + 3a^2b + 3ab^2 + b^3$.

Step 2. Expand $(a-b)^3$: $a^3 - 3a^2b + 3ab^2 - b^3$.

Step 3. Add: the $\pm 3a^2b$ and $\pm b^3$ terms cancel; survivors are $2a^3 + 6ab^2$.

Step 4. Factor the result: $2a(a^2 + 3b^2)$.

» **Why this step?** Adding opposite-sign expansions kills the odd terms - the same mechanism as $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$.

Answer: $2a^3 + 6ab^2 = 2a(a^2 + 3b^2)$

Hinglish: *Sum mein odd terms ($\pm 3a^2b$, $\pm b^3$) cancel ho jate hain, even terms double. Result $2a^3 + 6ab^2$, factor: $2a(a^2+3b^2)$. Ye square-identity ka cube bhai hai: $(a+b)^3+(a-b)^3 = 2a(a^2+3b^2)$.*

QUESTION 85 | Difficult | Recognise a cube (T7/T8)

Q85. Factorise: $8a^3 + 27b^3 + 36a^2b + 54ab^2$.

SOLUTION — step by step

Step 1. Rearrange into cube order: $(2a)^3 + 3(2a)^2(3b) + 3(2a)(3b)^2 + (3b)^3$.

Step 2. Check each coefficient: $8a^3$ ✓; $3 \cdot 4a^2 \cdot 3b = 36a^2b$ ✓; $3 \cdot 2a \cdot 9b^2 = 54ab^2$ ✓; $27b^3$ ✓.

Step 3. This is exactly $(2a + 3b)^3$.

» **Why this step?** The 1-3-3-1 coefficient pattern on the rescaled blocks (2a) and (3b) identifies the whole expression as one cube.

Answer: $(2a + 3b)^3$

Hinglish: *Terms ko cube order mein likho: $8a^3$, $36a^2b$, $54ab^2$, $27b^3$. Pattern 1-3-3-1 with $A = 2a$, $B = 3b$. Poora expression $(2a+3b)^3$ hai! Sum-of-cubes formula lagane wale galat raaste par chale jate hain - pehle pattern check karo.*

QUESTION 86 | Difficult | Crown identity forward use (T9)

Q86. If $x + y + z = 6$ and $xy + yz + zx = 11$, find $x^3 + y^3 + z^3 - 3xyz$.

SOLUTION — step by step

Step 1. First get the sum of squares: $x^2+y^2+z^2 = (x+y+z)^2 - 2(xy+yz+zx) = 36 - 22 = 14$.

Step 2. Crown identity: $x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2 - xy - yz - zx)$.

Step 3. The bracket: $14 - 11 = 3$.

Step 4. Value $= 6 \times 3 = 18$.

» **Why this step?** The crown identity needs (sum) and (sum of squares minus pair-sum) - both computable from the givens.

Answer: 18

Hinglish: *Do steps: pehle $x^2+y^2+z^2 = 36-22 = 14$. Phir crown identity: bracket = 14-11 = 3. Multiply: 6×3 = 18. Zero-sum nahi tha isliye poora identity use hua.*

QUESTION 87 | Difficult | Crown identity as factorisation (T9)

Q87. Factorise: $a^3 - 8b^3 - 64c^3 - 24abc$.

SOLUTION — step by step

Step 1. See the shape: $a^3 + (-2b)^3 + (-4c)^3 - 3 \cdot a \cdot (-2b) \cdot (-4c)$.

Step 2. Check the cross term: $3 \times a \times (-2b) \times (-4c) = 24abc$, and the expression has $-24abc$ ✓.

Step 3. Apply the crown identity with $x = a$, $y = -2b$, $z = -4c$:

Step 4. $x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2-xy-yz-zx)$.

Step 5. $x + y + z = a - 2b - 4c$.

Step 6. $x^2+y^2+z^2 = a^2+4b^2+16c^2$; $xy = -2ab$, $yz = 8bc$, $zx = -4ca$; so $-xy-yz-zx = +2ab - 8bc + 4ca$.

Step 7. Final: $(a - 2b - 4c)(a^2 + 4b^2 + 16c^2 + 2ab - 8bc + 4ca)$.

» **Why this step?** The crown identity factorises ANY three cubes whose product term appears as $-3xyz$ - negative letters are welcome.

Answer: $(a - 2b - 4c)(a^2 + 4b^2 + 16c^2 + 2ab - 8bc + 4ca)$

Hinglish: *Teen cubes dikh rahe hain: a^3, $(-2b)^3$, $(-4c)^3$ aur cross term $-24abc = -3xyz$. Crown identity seedha lag jata hai. Negative letters ke signs cross terms mein multiply hote hain: $yz = (-2b)(-4c) = +8bc$, isliye $-yz = -8bc$. Sign discipline hi yahan ka skill hai.*

QUESTION 88 | Difficult | Grouped difference of squares (T2/T12)

Q88. Factorise: $x^2 + 6x + 9 - 4y^2$.

SOLUTION — step by step

Step 1. First three terms form a perfect square: $x^2 + 6x + 9 = (x + 3)^2$.

Step 2. Expression = $(x + 3)^2 - (2y)^2$.

Step 3. Difference of squares: $(x + 3 - 2y)(x + 3 + 2y)$.

» **Why this step?** Completing the visible square FIRST turns four terms into a two-square difference.

Answer: $(x + 3 - 2y)(x + 3 + 2y)$

Hinglish: *Pehle teen terms ko square banao: $(x+3)^2$. Ab $-4y^2 = -(2y)^2$. Classic difference of squares: $(x+3-2y)(x+3+2y)$. Order of moves: square complete karo, phir factor.*

QUESTION 89 | Difficult | Scaled square bridge (T11)

Q89. If $9x^2 + 25y^2 = 181$ and $xy = 2$, find $3x - 5y$.

SOLUTION — step by step

Step 1. Note $9x^2 = (3x)^2$ and $25y^2 = (5y)^2$.

Step 2. $(3x - 5y)^2 = 9x^2 - 2(3x)(5y) + 25y^2 = 181 - 30xy = 181 - 60 = 121$.

Step 3. So $3x - 5y = \pm 11$.

» **Why this step?** Spotting the scaled square (3x, 5y) converts the given sum of squares into a completed square minus the cross term.

Answer: $3x - 5y = 11$ or -11

Hinglish: *$9x^2$ aur $25y^2$ ko $(3x)^2$, $(5y)^2$ dekho. Cross term $2\times3x\times5y = 30xy = 60$. $181 - 60 = 121 = 11^2$. Sign: dono values possible - ±11. Ye 'scaled variables' wala pattern har identity mein chalta hai.*

QUESTION 90 | Difficult | Deep chained difference of squares (T2)

Q90. Factorise completely: $x^8 - 256$.

SOLUTION — step by step

Step 1. $x^8 = (x^4)^2$ and $256 = 16^2$: $(x^4 - 16)(x^4 + 16)$.

Step 2. $x^4 - 16 = (x^2 - 4)(x^2 + 4)$.

Step 3. $x^2 - 4 = (x - 2)(x + 2)$.

Step 4. Collect: $(x - 2)(x + 2)(x^2 + 4)(x^4 + 16)$.

Step 5. Neither $x^2 + 4$ nor $x^4 + 16$ factorises over the integers - done.

» **Why this step?** Each pass halves the power difference; three passes bring 8 down to 1.

Answer: $(x - 2)(x + 2)(x^2 + 4)(x^4 + 16)$

Hinglish: *x^8-256: teen baar difference of squares. Har baar power half: $x^8 \to x^4 \to x^2 \to x$. Final $(x-2)(x+2)(x^2+4)(x^4+16)$. Sum-of-squares factors ko chhod do - wo integer mein khatam nahi hote.*

QUESTION 91 | Difficult | Zero-sum proof and use (T9)

Q91. (i) Prove that if $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$. (ii) Use it to evaluate $2^3 + 3^3 + (-5)^3$.

SOLUTION — step by step

Step 1. (i) Crown identity: $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$.

Step 2. If $a + b + c = 0$, the right side is 0, so $a^3+b^3+c^3 = 3abc$. ∎

Step 3. (ii) Here $2 + 3 + (-5) = 0$, so $2^3 + 3^3 + (-5)^3 = 3(2)(3)(-5) = -90$.

Step 4. Direct check: $8 + 27 - 125 = -90$ ✓.

» **Why this step?** The proof is one line once the crown identity is available - that is why the identity is called the crown.

Answer: (i) proved (ii) -90

Hinglish: *Proof: crown identity mein a+b+c = 0 daalo, right side zero. Ho gaya. Application: 2, 3, -5 ka sum zero - 3×2×3×(-5) = -90. Direct cube se match kar lo, confidence double.*

QUESTION 92 | Difficult | Zero-sum with big numbers (T9)

Q92. Without cubing any number, evaluate $25^3 - 75^3 + 50^3$.

SOLUTION — step by step

Step 1. Sum check: 25 - 75 + 50 = 0. So a = 25, b = -75, c = 50 has a + b + c = 0.

Step 2. Zero-sum shortcut: $a^3 + b^3 + c^3 = 3abc$.

Step 3. Value = 3 × 25 × (-75) × 50.

Step 4. 25 × 75 = 1875; 1875 × 50 = 93,750; times 3 = 281,250. With the minus sign: -281,250.

» **Why this step?** The zero sum makes three cubes collapse into one product - this is the identity's designed use.

Answer: -281,250

Hinglish: *25 - 75 + 50 = 0 - zero-sum! -75 ko b banake shortcut lagao: 3×25×(-75)×50 = -281250. Teen cubes (15625 + ... + 125000) karne ki zaroorat hi nahi. Sign: ek negative factor, toh result negative.*

Level 4 · Boss Zone: advanced + HOTS (Olympiad-lite):

QUESTION 93 | High | Conditional cube identity (T9/T14)

Q93. If $a + b + c = 0$, prove that $(a+b)^3 + (b+c)^3 + (c+a)^3 = 3(a+b)(b+c)(c+a)$.

SOLUTION — step by step

Step 1. Let $A = a + b$, $B = b + c$, $C = c + a$. Their sum: $A + B + C = 2(a + b + c) = 0$.

Step 2. Zero-sum shortcut: $A^3 + B^3 + C^3 = 3ABC$.

Step 3. Substituting back: $(a+b)^3 + (b+c)^3 + (c+a)^3 = 3(a+b)(b+c)(c+a)$. ∎

» **Why this step?** The trick is to see (a+b), (b+c), (c+a) as NEW letters whose sum inherits the zero - the shortcut then applies verbatim.

Answer: Proved via A+B+C = 0 with A = a+b, B = b+c, C = c+a.

Hinglish: *Renames karo: A = a+b, B = b+c, C = c+a. Inka sum = 2(a+b+c) = 0. Ab wahi zero-sum shortcut. Renaming technique badi problems ka master key hai - letters kuch bhi ho sakte hain.*

QUESTION 94 | High | Cube of x + 1/x (T10)

Q94. Expand $(x + 1/x)^3$ and hence find $x^3 + 1/x^3$ when $x + 1/x = 3$.

SOLUTION — step by step

Step 1. Use $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$ with $a = x$, $b = 1/x$, $ab = 1$:

Step 2. $(x + 1/x)^3 = x^3 + 1/x^3 + 3(x + 1/x)$.

Step 3. Given $x + 1/x = 3$: cube both sides directly: $27 = x^3 + 1/x^3 + 3 \times 3$.

Step 4. $x^3 + 1/x^3 = 27 - 9 = 18$.

» **Why this step?** $ab = 1$ compresses the whole expansion to (cubes) + 3(sum) - the cleanest bridge in the chapter.

Answer: $x^3 + 1/x^3 = 18$

Hinglish: *$(x+1/x)^3 = x^3+1/x^3+3(x+1/x)$. Given value 3: $27 = x^3+1/x^3+9$. Answer 18. Yaad rakho: cube bridge $s \rightarrow s^3 - 3s$ hota hai. $s = 3$: $27-9 = 18$.*

QUESTION 95 | High | Cube minus cube (T8/T12)

Q95. Factorise: $a^3 + b^3 + 3a^2b + 3ab^2 - 8$.

SOLUTION — step by step

Step 1. Spot the cube: $a^3 + 3a^2b + 3ab^2 + b^3 = (a+b)^3$, but the b^3 is missing and an 8 is appended.

Step 2. Regroup: $a^3 + b^3 + 3a^2b + 3ab^2 - 8 = (a + b)^3 - 2^3$.

Step 3. Difference of cubes: $(a + b - 2)[(a+b)^2 + 2(a+b) + 4]$.

Step 4. Expand the second factor: $(a + b - 2)(a^2 + 2ab + b^2 + 2a + 2b + 4)$.

» **Why this step?** Four of the terms assemble into one cube; the leftover is a lone 2^3, giving a difference of cubes.

Answer: $(a + b - 2)(a^2 + 2ab + b^2 + 2a + 2b + 4)$

Hinglish: *Pehle chaar terms ko $(a+b)^3$ banao - b^3 missing tha, use borrow karke group kiya. $(a+b)^3 - 8 = (a+b)^3 - 2^3$. Ab difference of cubes: (a+b-2) times trinomial. Trinomial expand kar dena final form mein.*

QUESTION 96 | High | Crown identity with product (T9)

Q96. If $x + y + z = 15$, $x^2 + y^2 + z^2 = 83$ and $xyz = 30$, find $x^3 + y^3 + z^3$.

SOLUTION — step by step

Step 1. Pair-sum: $xy + yz + zx = [(x+y+z)^2 - (x^2+y^2+z^2)]/2 = (225 - 83)/2 = 71$.

Step 2. Crown identity: $x^3+y^3+z^3 - 3xyz = (x+y+z)(x^2+y^2+z^2 - xy-yz-zx) = 15(83 - 71) = 180$.

Step 3. So $x^3+y^3+z^3 = 180 + 3xyz = 180 + 90 = 270$.

» **Why this step?** Two bridges in sequence: first recover the pair-sum, then the crown identity, then add back 3xyz.

Answer: 270

Hinglish: *Pehle pair-sum: (225-83)/2 = 71. Crown se cubes-minus-product: 15×12 = 180. Phir 3xyz = 90 wapas jodo: 270. Teen givens, teen steps - symmetric quantity ka poora game.*

QUESTION 97 | High | Decimal cube sum (T3/T9)

Q97. Without cubing, evaluate $0.86^3 + 0.14^3$.

SOLUTION — step by step

Step 1. $a + b = 0.86 + 0.14 = 1$.

Step 2. Bridge: $a^3 + b^3 = (a+b)^3 - 3ab(a+b) = 1 - 3ab$.

Step 3. $ab = 0.86 \times 0.14 = 0.1204$.

Step 4. Value $= 1 - 3(0.1204) = 1 - 0.3612 = 0.6388$.

» **Why this step?** The pair-sum 1 makes $(a+b)^3$ trivial; only the product needs one multiplication.

Answer: 0.6388

Hinglish: *0.86 + 0.14 = 1 - yahi design hai. Bridge: 1 - 3(0.1204) = 0.6388. Decimals ka product dhyan se: 86×14 = 1204, four decimal places. Cubes bilkul nahi nikaale.*

QUESTION 98 | High | Cubic factorisation, three roots (T12)

Q98. Factorise completely: $x^3 - 6x^2 + 11x - 6$.

SOLUTION — step by step

Step 1. Test integer roots. x = 1: 1 - 6 + 11 - 6 = 0 ✓, so (x - 1) is a factor.

Step 2. Divide: $x^3 - 6x^2 + 11x - 6 = (x - 1)(x^2 - 5x + 6)$.

Step 3. Split the middle: $x^2 - 5x + 6 = (x - 2)(x - 3)$.

Step 4. Final: (x - 1)(x - 2)(x - 3).

Step 5. Check: roots 1, 2, 3 sum to 6 = -(-6) ✓, pairwise products 11 ✓, product 6 ✓ (Vieta consistency).

» **Why this step?** A cubic with three integer roots yields to root-hunting + division + quadratic split - three standard moves.

Answer: (x - 1)(x - 2)(x - 3)

Hinglish: *x = 1 pehle try karo (sum of coefficients zero hai tab). Factor mila, divide karo, quadratic split: (x-2)(x-3). Final teen linear factors. Vieta check: roots ka sum 6, pair-sum 11, product 6 - sab original coefficients se match.*

QUESTION 99 | High | Sum-cube decomposition (T9/T14)

Q99. Prove that $(x + y + z)^3 - x^3 - y^3 - z^3 = 3(x + y)(y + z)(z + x)$.

SOLUTION — step by step

Step 1. Use the crown identity with A = x + y + z, B = -x, C = -y, D = -z? Better: apply it to the three numbers (x+y), (y+z), (z+x):

Step 2. Their sum = 2(x+y+z). Not zero, so use the full identity on P = x+y, Q = y+z, R = z+x.

Step 3. $P^3+Q^3+R^3-3PQR = (P+Q+R)(P^2+Q^2+R^2-PQ-QR-RP)$.

Step 4. Alternative cleaner route: expand $(x+y+z)^3 = x^3+y^3+z^3 + 3(x+y)(y+z)(z+x)$. This is itself the identity: apply the crown identity to A = x+y, B = z... verify directly:

Step 5. Direct proof: $(x+y+z)^3$ with u = x+y: $(u+z)^3 = u^3 + 3u^2z + 3uz^2 + z^3 = (x+y)^3 + 3z(x+y)^2 + 3z^2(x+y) + z^3$.

Step 6. Expand $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$ and collect all terms:

Step 7. $(x+y+z)^3 = x^3+y^3+z^3 + 3x^2y + 3xy^2 + 3y^2z + 3yz^2 + 3z^2x + 3zx^2 + 6xyz$.

Step 8. Subtract $x^3+y^3+z^3$: remainder = $3[x^2y+xy^2+y^2z+yz^2+z^2x+zx^2] + 6xyz$.

Step 9. Now expand 3(x+y)(y+z)(z+x): first (x+y)(y+z) = $xy+xz+y^2+yz$; multiply by (z+x): = $xyz+x^2z + xz^2+x^2y + y^2z+xy^2 + yz^2+xyz = (x^2y+xy^2+y^2z+yz^2+z^2x+zx^2) + 2xyz$.

Step 10. Times 3: 3[...+2xyz] = 3[...] + 6xyz - exactly the remainder. ∎

» **Why this step?** Both sides expand to the same six mixed terms plus 6xyz; the identity packages this messy equality into one line.

Answer: Proved: both sides equal $3(x^2y+xy^2+y^2z+yz^2+z^2x+zx^2) + 6xyz$.

Hinglish: *Ye identity yaad karne layak hai: $(x+y+z)^3 = x^3+y^3+z^3+3(x+y)(y+z)(z+x)$. Proof mein dono side expand karke compare karo - mixed terms match hote hain. Exam mein identity quote karke 2-3 line proof kaafi hai agar identity 'standard' maani jati hai; warna full expansion dikhao.*

QUESTION 100 | High | Substitution square (T5/T12)

Q100. Factorise: $(2a + 3b)^2 + 2(2a + 3b)(3a - 2b) + (3a - 2b)^2$.

SOLUTION — step by step

Step 1. Substitute P = 2a + 3b, Q = 3a - 2b. The expression is $P^2 + 2PQ + Q^2$.

Step 2. Perfect square: $= (P + Q)^2$.

Step 3. P + Q = 2a + 3b + 3a - 2b = 5a + b.

Step 4. Answer: $(5a + b)^2$.

Step 5. Check one cross term on expansion: $2\cdot(5a)(b) = 10ab$; direct: $2(2a)(3a)$... full check: $(5a+b)^2 = 25a^2 + 10ab + b^2$ ✓.

» **Why this step?** The structure $P^2 + 2PQ + Q^2$ is invisible until the repeated brackets are renamed - then it is one identity.

Answer: $(5a + b)^2$

Hinglish: *Brackets ko P aur Q naam do - expression $P^2+2PQ+Q^2$ ban jata hai $= (P+Q)^2$. $P+Q = 5a+b$. Answer $(5a+b)^2$. Substitution aankhon ko naya drishti deta hai - pattern pehle chhupa tha.*

QUESTION 101 | High | Near-number cube, minus case (T3)

Q101. Evaluate 999^3 without a calculator.

SOLUTION — step by step

Step 1. 999 = 1000 - 1. Use $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$ with a = 1000, b = 1.

Step 2. $a^3 = 1{,}000{,}000{,}000$.

Step 3. $3a^2b = 3 \times 1{,}000{,}000 = 3{,}000{,}000$.

Step 4. $3ab^2 = 3 \times 1000 = 3000$.

Step 5. $b^3 = 1$.

Step 6. $999^3 = 1{,}000{,}000{,}000 - 3{,}000{,}000 + 3{,}000 - 1 = 997{,}002{,}999$.

» **Why this step?** Alternating signs with a = 1000 make every term a round number; only bookkeeping remains.

Answer: 997,002,999

Hinglish: *(1000-1)³: terms 1-3-3-1 with alternating minus. $10^9 - 3\times10^6 + 3\times10^3 - 1 = 997002999$. Column subtraction dhyan se - zeroes ginn kar, teen blocks mein todo: 997 | 002 | 999.*

QUESTION 102 | High | Converse of zero-sum (T9/T14)

Q102. If $a^3 + b^3 + c^3 = 3abc$ and $a + b + c \neq 0$, prove that $a = b = c$.

SOLUTION — step by step

Step 1. Crown identity: $a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$.

Step 2. Given the left side is 0 and $a+b+c \neq 0$, the second factor must be 0.

Step 3. Rewrite the second factor: $a^2+b^2+c^2-ab-bc-ca = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$.

Step 4. A sum of squares is 0 only when every square is 0: $a = b$, $b = c$, $c = a$.

Step 5. So $a = b = c$. ∎

» **Why this step?** The half-sum-of-squares rewriting converts an algebraic condition into three independent equations.

Answer: Proved: the half-sum-of-squares factor forces a = b = c.

Hinglish: *Crown identity ka CONVERSE: cubes ka relation 3abc ke barabar hai aur sum zero nahi, toh doosra factor zero. Wo factor $\frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2]$ hai - squares ka sum zero matlab sab barabar. Ye 'either-or' structure (sum zero YA sab equal) exams ka favourite hai.*

QUESTION 103 | High | Bridge reversed for product (T11)

Q103. If $p^3 + q^3 = 35$ and $p + q = 5$, find pq.

SOLUTION — step by step

Step 1. Bridge: $p^3 + q^3 = (p+q)^3 - 3pq(p+q)$.

Step 2. $35 = 125 - 3pq \cdot 5 = 125 - 15pq$.

Step 3. $15pq = 90$, so $pq = 6$.

» **Why this step?** The bridge reads equally well backwards: the product is recovered from the two cube-related quantities.

Answer: pq = 6

Hinglish: *Bridge ka reverse use: $35 = 125 - 15pq$. $pq = 6$. Check: p, q roots of $t^2-5t+6 = (t-2)(t-3)$: $p=2$, $q=3 \rightarrow 8+27 = 35$ ✓.*

QUESTION 104 | High | Middle-term completion (T12)

Q104. Factorise: $x^4 + 5x^2 + 9$.

SOLUTION — step by step

Step 1. Compare with $(x^2 + 3)^2 = x^4 + 6x^2 + 9$ - one x^2 too many.

Step 2. Split: $x^4 + 5x^2 + 9 = x^4 + 6x^2 + 9 - x^2 = (x^2 + 3)^2 - x^2$.

Step 3. Difference of squares: $(x^2 + 3 - x)(x^2 + 3 + x)$.

Step 4. Final: $(x^2 - x + 3)(x^2 + x + 3)$.

» **Why this step?** Borrow and repay one x^2 to create a perfect square, then factor the resulting two-square difference.

Answer: $(x^2 - x + 3)(x^2 + x + 3)$

Hinglish: *$(x^2+3)^2 = x^4+6x^2+9$ - humein 5 chahiye. Ek x^2 udhaar lo aur wapas do: $(x^2+3)^2 - x^2$. Difference of squares: $(x^2-x+3)(x^2+x+3)$. Q32 ka cousin - same completion, different numbers.*

QUESTION 105 | High | Zero-sum at power two (T9)

Q105. Evaluate: $(a^2 - b^2)^3 + (b^2 - c^2)^3 + (c^2 - a^2)^3$.

SOLUTION — step by step

Step 1. Let $P = a^2-b^2$, $Q = b^2-c^2$, $R = c^2-a^2$. Sum: $P+Q+R = 0$.

Step 2. Zero-sum shortcut: $P^3+Q^3+R^3 = 3PQR$.

Step 3. Value = $3(a^2-b^2)(b^2-c^2)(c^2-a^2)$.

Step 4. Optionally factor each difference further: = $3(a-b)(a+b)(b-c)(b+c)(c-a)(c+a)$.

» **Why this step?** Differences cycling through the same three letters always sum to zero - the shortcut does not care what the letters are made of.

Answer: $3(a^2 - b^2)(b^2 - c^2)(c^2 - a^2)$

Hinglish: *$(a^2-b^2)+(b^2-c^2)+(c^2-a^2) = 0$. Shortcut: 3PQR. Chaho toh har difference ko aur factor karo - (a-b)(a+b) etc. Zero-sum spotting phir se kaam par.*

QUESTION 106 | High | Full symmetric reconstruction (T9/T11)

Q106. If $a + b + c = 9$, $a^2 + b^2 + c^2 = 35$ and $abc = 6$, find $a^3 + b^3 + c^3$.

SOLUTION — step by step

Step 1. Pair-sum: $ab + bc + ca = [(a+b+c)^2 - (a^2+b^2+c^2)]/2 = (81 - 35)/2 = 23$.

Step 2. Crown identity: $a^3+b^3+c^3 - 3abc = (a+b+c)(a^2+b^2+c^2 - ab-bc-ca) = 9(35 - 23) = 108$.

Step 3. $a^3+b^3+c^3 = 108 + 3(6) = 126$.

» **Why this step?** Sum of squares gives the pair-sum; the crown identity then needs exactly (sum, pair-sum, squares) - all available.

Answer: 126

Hinglish: *Wahi do-step recipe: pair-sum = (81-35)/2 = 23. Crown: 9×(35-23) = 108. Plus 3abc = 18. Total 126. Symmetric quantities mein har value doosri value se banti hai - bas bridge yaad rakho.*

Level 4 · Boss Zone: advanced + HOTS (Olympiad-lite):

QUESTION 107 | HOTS | Sum-cube identity (T14)

Q107. Prove that $(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b)(b + c)(c + a)$.

SOLUTION — step by step

Step 1. Start from the right side's known relative: $(a+b+c)^3 = a^3+b^3+c^3 + 3(a+b+c)(ab+bc+ca) - 3abc$ (standard expansion).

Step 2. Expand $3(a+b)(b+c)(c+a)$: as computed in Q49, it equals $3(ab+bc+ca)(a+b+c) - 3abc$.

Step 3. Indeed: $(a+b)(b+c)(c+a) = (a+b+c)(ab+bc+ca) - abc$ (expand and collect).

Step 4. Then $3(a+b)(b+c)(c+a) = 3(a+b+c)(ab+bc+ca) - 3abc$.

Step 5. Adding $a^3+b^3+c^3$ gives exactly $(a+b+c)^3$. ∎

Step 6. Direct verification with numbers: $a = 1$, $b = 2$, $c = 3$: LHS = 216. RHS = $36 + 3(3)(5)(4) = 36 + 180 = 216$ ✓.

» **Why this step?** The two forms differ only in packaging: one counts pair-sums with the total sum, the other uses three pairwise sums.

Answer: Proved (both forms differ by the identity $(a+b)(b+c)(c+a) = (a+b+c)(ab+bc+ca) - abc$).

Hinglish: *Do forms hain - ek $(a+b+c)(ab+bc+ca)$ wali, ek $(a+b)(b+c)(c+a)$ wali. Dono equal hain kyunki $(a+b)(b+c)(c+a) = (a+b+c)(ab+bc+ca) - abc$. Numeric check 1,2,3 se 216 dono taraf - proof pakka.*

QUESTION 108 | HOTS | Fourth powers under zero sum (T14)

Q108. If $a + b + c = 0$, prove that $a^4 + b^4 + c^4 = 2(a^2b^2 + b^2c^2 + c^2a^2)$.

SOLUTION — step by step

Step 1. From $a + b + c = 0$: $(a+b+c)^2 = 0 \Rightarrow a^2+b^2+c^2 = -2(ab+bc+ca)$.

Step 2. Square both sides: $(a^2+b^2+c^2)^2 = 4(ab+bc+ca)^2$.

Step 3. LHS expands: $a^4+b^4+c^4 + 2(a^2b^2+b^2c^2+c^2a^2)$.

Step 4. RHS expands: $4[a^2b^2+b^2c^2+c^2a^2 + 2abc(a+b+c)] = 4(a^2b^2+b^2c^2+c^2a^2)$ since $a+b+c = 0$.

Step 5. Equate: $a^4+b^4+c^4 + 2\Sigma = 4\Sigma$ where $\Sigma = a^2b^2+b^2c^2+c^2a^2$.

Step 6. So $a^4+b^4+c^4 = 2\Sigma = 2(a^2b^2+b^2c^2+c^2a^2)$. ∎

Step 7. Check: $a = 1, b = 1, c = -2$: LHS = 1+1+16 = 18; RHS = 2(1+4+4) = 18 ✓.

» **Why this step?** Squaring the zero-sum relation upgrades degree 2 to degree 4; the abc-term dies because its multiplier is $a+b+c = 0$.

Answer: Proved (both sides equal $2(a^2b^2+b^2c^2+c^2a^2)$ under $a+b+c = 0$).

Hinglish: *Do squares ka game: pehle $(a+b+c)^2 = 0$ se $a^2+b^2+c^2 = -2\Sigma_1$ nikaalo, phir square karo. RHS mein $2abc(a+b+c)$ term aati hai jo zero ho jati hai - yahi magic hai. Numeric check 1,1,-2 se 18 = 18.*

QUESTION 109 | HOTS | Crown identity minimal data (T9)

Q109. If $x + y + z = 3$ and $x^2 + y^2 + z^2 = 5$, find $x^3 + y^3 + z^3 - 3xyz$.

SOLUTION — step by step

Step 1. Pair-sum: $xy + yz + zx = (9 - 5)/2 = 2$.

Step 2. Crown identity: $x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2 - xy-yz-zx)$.

Step 3. Bracket: 5 - 2 = 3.

Step 4. Value = 3 × 3 = 9.

» **Why this step?** The requested combination is EXACTLY the crown identity's left side - no xyz value is ever needed.

Answer: 9

Hinglish: *Question ne wahi combination manga hai jo crown identity ka LEFT side hai - xyz ki value jaanne ki zaroorat hi nahi! Pair-sum 2, bracket 3, answer 9. Spot-the-identity skill.*

QUESTION 110 | HOTS | Symmetric functions in p, q, r (T11/T14)

Q110. If $a + b + c = p$, $ab + bc + ca = q$ and $abc = r$, express $a^3 + b^3 + c^3$ in terms of p, q and r.

SOLUTION — step by step

Step 1. Crown identity: $a^3+b^3+c^3 - 3abc = (a+b+c)(a^2+b^2+c^2 - ab-bc-ca)$.

Step 2. Sum of squares: $a^2+b^2+c^2 = p^2 - 2q$.

Step 3. Bracket: $(p^2 - 2q) - q = p^2 - 3q$.

Step 4. So $a^3+b^3+c^3 - 3r = p(p^2 - 3q) = p^3 - 3pq$.

Step 5. Therefore $a^3+b^3+c^3 = p^3 - 3pq + 3r$.

» **Why this step?** Newton-style bookkeeping: every symmetric quantity is a polynomial in (p, q, r); the crown identity supplies the cubic one.

Answer: $a^3 + b^3 + c^3 = p^3 - 3pq + 3r$

Hinglish: *Ye 'symmetric functions' ka master formula hai: cubes ka sum = $p^3 - 3pq + 3r$. Ek baar derive kar lo, har $a^3+b^3+c^3$ question 30 second mein hoga. p^2-3q bracket dhyan se - wo crown identity ka signature hai.*

QUESTION 111 | HOTS | Cube difference collapse (T7/T14)

Q111. Prove that $(x + y)^3 - (x - y)^3 - 6y(x^2 - y^2) = 8y^3$.

SOLUTION — step by step

Step 1. Expand $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$ and $(x-y)^3 = x^3 - 3x^2y + 3xy^2 - y^3$.

Step 2. Difference: $(x+y)^3 - (x-y)^3 = 6x^2y + 2y^3$.

Step 3. Also $6y(x^2 - y^2) = 6x^2y - 6y^3$.

Step 4. Subtract: $6x^2y + 2y^3 - 6x^2y + 6y^3 = 8y^3$. ∎

» **Why this step?** The difference of cubes kills x^3 and $3xy^2$; what remains ($6x^2y$) is exactly cancelled by the third term.

Answer: Proved: the expression collapses to $8y^3$.

Hinglish: *Dono cubes ka difference: $6x^2y + 2y^3$ (x^3 aur $3xy^2$ cancel). Third term $6x^2y - 6y^3$. Minus karo: x^2y wale terms gayab, $2y^3+6y^3 = 8y^3$. Expansion systematically karo, terms ko group karo.*

QUESTION 112 | HOTS | Divisibility via factoring (T2/T14)

Q112. Show that $2^{55} + 2^{58}$ is divisible by 9.

SOLUTION — step by step

Step 1. Factor the smaller power: $2^{55} + 2^{58} = 2^{55}(1 + 2^3) = 2^{55} \times 9$.

Step 2. The right side is 9 times the integer 2^{55}, hence divisible by 9. ∎

» **Why this step?** Common factor first (with exponents, the smaller power), then the bracket 1 + 8 supplies the required divisor.

Answer: $2^{55} + 2^{58} = 9 \cdot 2^{55}$, divisible by 9.

Hinglish: *Chhota power bahar: $2^{55}(1 + 2^3) = 2^{55} \times 9$. Done - 9 upar dikh raha hai. Divisibility questions ka 90% answer factoring mein chhupa hota hai.*

QUESTION 113 | HOTS | Long reciprocal chain (T10)

Q113. If $x + 1/x = 3$, find $x^5 + 1/x^5$.

SOLUTION — step by step

Step 1. Build the chain: $x^2 + 1/x^2 = 9 - 2 = 7$.

Step 2. $x^3 + 1/x^3 = 27 - 9 = 18$ (cube bridge).

Step 3. $x^4 + 1/x^4 = 7^2 - 2 = 47$.

Step 4. Multiply rule: $(x + 1/x)(x^4 + 1/x^4) = x^5 + 1/x^5 + x^3 + 1/x^3$.

Step 5. So $x^5 + 1/x^5 = 3 \times 47 - 18 = 141 - 18 = 123$.

» **Why this step?** Multiplying by the base chain (x + 1/x) steps up one power at a time, at the cost of the two-lower term.

Answer: 123

Hinglish: *Chain build karo: 3 → 7 → 18 → 47. Fifth power ke liye multiply rule: $(x+1/x)(x^4+1/x^4) = x^5+1/x^5 + x^3+1/x^3$. 141 - 18 = 123. Har step verify karte jao - ek galti poori chain uda degi.*

QUESTION 114 | HOTS | Cyclic cubic factorisation (T12/T14)

Q114. Factorise: $a^3(b - c) + b^3(c - a) + c^3(a - b)$.

SOLUTION — step by step

Step 1. Group to expose factors: $a^3(b-c) + b^3(c-a) + c^3(a-b)$.

Step 2. Verify (a - b) is a factor: setting a = b gives $a^3(b-c) + a^3(c-a) + c^3(a-a) = a^3(b-c+c-a) = 0$ ✓.

Step 3. Similarly (b - c) and (c - a) are factors (cyclic argument).

Step 4. Degree count: the expression is degree 4; three linear factors (degree 3) leave one more degree: k × (a+b+c) for a constant k.

Step 5. Find k by testing a = 1, b = 2, c = 3: LHS = 1(2-3) + 8(3-1) + 27(1-2) = -1 + 16 - 27 = -12. Factors: (1-2)(2-3)(3-1) = (-1)(-1)(2) = 2; (a+b+c) = 6. So 2 × k × 6 = -12 ⇒ k = -1.

Step 6. Final: -(a - b)(b - c)(c - a)(a + b + c).

» **Why this step?** The factor theorem finds the linear factors; degree counting reveals the missing factor; one numeric evaluation pins the constant.

Answer: -(a - b)(b - c)(c - a)(a + b + c)

Hinglish: *Classic cyclic monster! (a-b), (b-c), (c-a) teeno factors hain - substitute karke verify karo. Degree 4 hai, teen factors degree 3 - bacha (a+b+c) with constant k. Numeric test se k = -1. Ye 'test values' trick constants dhoondne ka universal tarika hai.*

QUESTION 115 | HOTS | Sum of squares is zero (T5/T14)

Q115. If $x^2 + y^2 + 2x - 4y + 5 = 0$, find $x^3 + y^3$.

SOLUTION — step by step

Step 1. Rearrange: $(x^2 + 2x) + (y^2 - 4y) + 5 = 0$.

Step 2. Complete both squares: $(x + 1)^2 - 1 + (y - 2)^2 - 4 + 5 = 0$,

Step 3. so $(x + 1)^2 + (y - 2)^2 = 0$.

Step 4. A sum of squares is zero only when both squares are zero: x = -1, y = 2.

Step 5. $x^3 + y^3 = -1 + 8 = 7$.

» **Why this step?** Completing squares converts one equation in two unknowns into two forced equations - because squares cannot be negative.

Answer: x = -1, y = 2, so $x^3 + y^3 = 7$

Hinglish: *Squares complete karo: $(x+1)^2 + (y-2)^2 = 0$. Dono non-negative - dono zero hone padenge. x = -1, y = 2. Cubes: -1 + 8 = 7. Ye 'sum of squares = 0' trick real numbers ka superpower hai.*

QUESTION 116 | HOTS | Geometric proof of identities (T14)

Q116. Prove geometrically, using area models, that (a) $(a+b)^2 = a^2 + 2ab + b^2$ and (b) $(a-b)^2 = a^2 - 2ab + b^2$.

SOLUTION — step by step

Step 1. (a) Draw a square of side a + b. Split each side into segments a and b.

Step 2. The four regions are: a^2 (corner), b^2 (opposite corner), and TWO rectangles ab.

Step 3. Total area: $(a+b)^2 = a^2 + b^2 + ab + ab = a^2 + 2ab + b^2$. ∎

Step 4. (b) Draw a square of side a. Mark off a strip of width b along one side and the bottom.

Step 5. The remaining square $(a-b)^2$ equals a^2 minus TWO strips ab (top strip and side strip).

Step 6. The two removed strips overlap in a b^2 square that was subtracted twice - add it back once:

Step 7. $(a-b)^2 = a^2 - 2ab + b^2$. ∎ (See Figures 4.1 and 4.2.)

» **Why this step?** Areas cannot be negative and do not depend on how you slice them - so a dissection is a proof.

Answer: Both identities proved by dissection (Figures 4.1 and 4.2).

Hinglish: *Area proof - figure dekho (4.1, 4.2). $(a+b)^2$: chaar tukde - do squares, do rectangles. $(a-b)^2$: do strips minus karo, par b^2 corner DO baar minus hua tha - ek baar wapas jodo. Isliye $+b^2$. Drawing se identity kabhi galat yaad nahi hogi.*

QUESTION 117 | HOTS | Chained differences (T9/T14)

Q117. If $x = a - b$, $y = b - c$ and $z = c - a$, find the value of $x^3 + y^3 + z^3 + xyz$.

SOLUTION — step by step

Step 1. Sum: $x + y + z = a - b + b - c + c - a = 0$.

Step 2. Zero-sum shortcut: $x^3 + y^3 + z^3 = 3xyz$.

Step 3. So $x^3 + y^3 + z^3 + xyz = 3xyz + xyz = 4xyz$.

Step 4. Substitute back: $4(a - b)(b - c)(c - a)$.

» **Why this step?** The zero sum is structural (a telescoping chain), so the cube shortcut holds whatever a, b, c are.

Answer: $4(a - b)(b - c)(c - a)$

Hinglish: *Telescoping sum zero: $x+y+z = 0$. Shortcut: cubes $= 3xyz$. Plus $xyz = 4xyz$. Substitute: $4(a-b)(b-c)(c-a)$. Do-minute question, 5-mark answer.*

QUESTION 118 | HOTS | Double grouping (T12)

Q118. Factorise completely: $x^2y^2 - 5x^2y - 5xy^2 + 25xy$.

SOLUTION — step by step

Step 1. Common factor first: $xy \cdot (xy - 5x - 5y + 25)$.

Step 2. Group inside: $(xy - 5x) - (5y - 25) = x(y - 5) - 5(y - 5) = (x - 5)(y - 5)$.

Step 3. Final: $xy(x - 5)(y - 5)$.

» **Why this step?** The HCF shrinks four terms to four simpler terms; grouping then finishes in one pass.

Answer: $xy(x - 5)(y - 5)$

Hinglish: *Pehle xy common - har term mein tha! Andar grouping: $x(y-5) - 5(y-5) = (x-5)(y-5)$. Common factor + grouping ka clean combo. 'Completely' = xy bhi factor hai, usko likhna mat bhoolna.*

QUESTION 119 | HOTS | Unit-sum identity (T9/T14)

Q119. If $x + y = 1$, prove that $x^3 + y^3 + 3xy = 1$.

SOLUTION — step by step

Step 1. Cube bridge: $x^3 + y^3 = (x+y)^3 - 3xy(x+y)$.

Step 2. With $x + y = 1$: $x^3 + y^3 = 1 - 3xy$.

Step 3. Add $3xy$ to both sides: $x^3 + y^3 + 3xy = 1$. ∎

Step 4. Check with $x = 1/4$, $y = 3/4$: $1/64 + 27/64 + 3(3/16) = 28/64 + 9/16 = 7/16 + 9/16 = 1$ ✓.

» **Why this step?** A fixed pair-sum (here 1) turns the bridge into a relation between the cubes and the product.

Answer: Proved: $x^3 + y^3 + 3xy = 1$ whenever $x + y = 1$.

Hinglish: *$x+y = 1$ bridge ko simplify karta hai: cubes = 1 - 3xy. 3xy wapas jodo: 1. Check 1/4 aur 3/4 se - perfect. Ye identity $x+y = k$ ke liye generalise hoti hai: $x^3+y^3+3kxy = k^3$.*

QUESTION 120 | HOTS | Four-cube grand identity (T9/T14)

Q120. Prove that $(x + y + z)^3 - (y + z - x)^3 - (z + x - y)^3 - (x + y - z)^3 = 24xyz$.

SOLUTION — step by step

Step 1. Let $s = x + y + z$, and $P = y+z-x$, $Q = z+x-y$, $R = x+y-z$.

Step 2. Observe: $P + Q + R = (y+z-x)+(z+x-y)+(x+y-z) = x + y + z = s$.

Step 3. So the left side is $s^3 - P^3 - Q^3 - R^3$ with $s = P + Q + R$: that is $(P+Q+R)^3 - P^3 - Q^3 - R^3$.

Step 4. By Q49's identity: $(P+Q+R)^3 - P^3 - Q^3 - R^3 = 3(P+Q)(Q+R)(R+P)$.

Step 5. Compute the pair-sums: $P + Q = 2z$, $Q + R = 2x$, $R + P = 2y$.

Step 6. Left side $= 3(2z)(2x)(2y) = 24xyz$. ∎

» **Why this step?** Renaming the four brackets makes them sum to a single letter; the earlier identity then converts everything to pair-sums, each of which is 2 times one original variable.

Answer: Proved: the left side equals $3(2z)(2x)(2y) = 24xyz$.

Hinglish: *Ye book ka showpiece hai. Trick: char brackets ko P, Q, R naam do - inka sum $s = x+y+z$ nikalta hai. Ab $(P+Q+R)^3 - P^3-Q^3-R^3 = 3(P+Q)(Q+R)(R+P)$. Pair-sums: 2z, 2x, 2y. Product: 24xyz. Bade questions mein hamesha pehle 'renaming' dhoondo.*

4.5 Rapid MCQs: Choose the Correct Option (20)

Rapid-fire round: 4 options, ek hi correct. Pehle khud choose karo, phir colored Answer strip se check karo. Har galat option ka "Why wrong" bhi diya hai - taki exam me wahi trap tumhe na phasaye.

MCQ 1 | Normal | Near-number square

1. 101^2 equals:

A. 10201

B. 10021

C. 1021

D. 10404

Correct Answer: (A)

$(100+1)^2 = 10000 + 200 + 1 = 10201$.

Why (B) is wrong: Cross term 20 likh diya.

Why (C) is wrong: Dropped both extra terms.

Why (D) is wrong: That is 102^2.

Hinglish: *10000 + 200 + 1. Cross term $2\times100\times1 = 200$.*

MCQ 2 | Normal | Equidistant product

2. 51×49 equals:

A. 2499

B. 2501

C. 2490

D. 2489

Correct Answer: (A)

$(50+1)(50-1) = 2500 - 1 = 2499$.

Why (B) is wrong: Added 1 instead of subtracting.

Why (C) is wrong: Wrong offset.

Why (D) is wrong: Arithmetic slip.

Hinglish: *Average 50, offset 1: 2500-1 = 2499. Multiplication one second mein.*

MCQ 3 | Normal | Companion identity

3. $(a+b)^2 - (a-b)^2$ equals:

A. 2ab

B. 4ab

C. $2(a^2+b^2)$

D. 0

Correct Answer: (B)

Subtracting the expansions cancels a^2 and b^2, leaving 4ab.

Why (A) is wrong: Missed the factor 2.

Why (C) is wrong: That is the SUM of the two squares.

Why (D) is wrong: The difference is not zero.

Hinglish: *Squares ka difference = 4ab. Sum = 2(a^2+b^2). Dono yaad rakho.*

MCQ 4 | Normal | Middle-term splitting

4. $x^2 + 7x + 12$ factorises as:

A. (x+3)(x+4)

B. (x+2)(x+6)

C. (x+1)(x+12)

D. (x-3)(x-4)

Correct Answer: (A)

3 + 4 = 7 and 3 × 4 = 12.

Why (B) is wrong: 2+6 = 8 ≠ 7.

Why (C) is wrong: 1+12 = 13.

Why (D) is wrong: Signs wrong: sum would be -7.

Hinglish: *Sum 7, product 12: 3 aur 4.*

MCQ 5 | Normal | Perfect square spotting

5. $9x^2 + 12x + 4$ equals:

A. $(3x+2)^2$

B. $(3x+4)^2$

C. $(9x+2)^2$

D. (3x+2)(3x-2)

Correct Answer: (A)

$(3x)^2 + 2(3x)(2) + 2^2 = (3x+2)^2$.

Why (B) is wrong: $(3x+4)^2 = 9x^2+24x+16$.

Why (C) is wrong: $(9x+2)^2 = 81x^2+36x+4$.

Why (D) is wrong: That is $9x^2-4$.

Hinglish: *Middle term 12x = 2×3x×2 ✓. Perfect square.*

MCQ 6 | Normal | x + 1/x square

6. If $x + 1/x = 2$, then $x^2 + 1/x^2$ equals:

A. 2

B. 4

C. 6

D. 0

Correct Answer: (A)

$(x+1/x)^2 = x^2+2+1/x^2 = 4 \Rightarrow x^2+1/x^2 = 2$.

Why (B) is wrong: Forgot to subtract 2.

Why (C) is wrong: Subtracted -2.

Why (D) is wrong: Sign error.

Hinglish: *4 - 2 = 2. (x=1 hit karke bhi dekh sakte ho.)*

MCQ 7 | Normal | Trinomial square

7. $(a + b + c)^2$ equals:

A. $a^2+b^2+c^2+2ab+2bc+2ca$

B. $a^2+b^2+c^2+ab+bc+ca$

C. $a^2+b^2+c^2$

D. $a^2+b^2+c^2+2abc$

Correct Answer: (A)

Three squares plus TWICE each of the three pairwise products.

Why (B) is wrong: Missing the 2s.

Why (C) is wrong: Cross terms dropped.

Why (D) is wrong: 2abc is a triple product - never appears.

Hinglish: *Teen squares + teen cross terms, har cross term par coefficient 2.*

MCQ 8 | Normal | Cube bridge

8. If $x + 1/x = 3$, then $x^3 + 1/x^3$ equals:

A. 18

B. 27

C. 21

D. 30

Correct Answer: (A)

$x^3+1/x^3 = 27 - 3·3 = 27 - 9 = 18$.

Why (B) is wrong: $(x+1/x)^3$ hi maan liya.

Why (C) is wrong: Subtracted 6.

Why (D) is wrong: Added 3.

Hinglish: *$s^3 - 3s = 27 - 9 = 18$.*

MCQ 9 | Normal | Zero-sum cube

9. If $a + b + c = 0$, then $a^3 + b^3 + c^3$ equals:

A. 3abc

B. 0

C. -3abc

D. abc

Correct Answer: (A)

Crown identity with $a+b+c = 0$ gives $a^3+b^3+c^3 = 3abc$.

Why (B) is wrong: Only the PRODUCT side vanishes, not the cubes.

Why (C) is wrong: Sign flipped.

Why (D) is wrong: Missing the 3.

Hinglish: *Zero-sum shortcut: 3abc. Yaad kar lo - har saal aata hai.*

MCQ 10 | Normal | Near-number cube

10. The last three digits of 999^3 are:

A. 299

B. 001

C. 999

D. 997

Correct Answer: (A)

$999^3 = (1000-1)^3 = 997002999$; last three digits 299.

Why (B) is wrong: That is 1001^3's neighbourhood error.

Why (C) is wrong: Copied the base.

Why (D) is wrong: First three digits.

Hinglish: *Full value 997002999 - last three: 299.*

MCQ 11 | Normal | Cube sum formula

11. $a^3 + b^3$ equals:

A. $(a+b)(a^2-ab+b^2)$

B. $(a+b)(a^2+ab+b^2)$

C. $(a+b)^3$

D. $(a-b)(a^2+ab+b^2)$

Correct Answer: (A)

Standard factorisation; the trinomial's middle term is -ab.

Why (B) is wrong: That is a^3-b^3's factor.

Why (C) is wrong: Not a factorisation.

Why (D) is wrong: Wrong binomial.

Hinglish: *Sum: binomial plus, trinomial middle minus.*

MCQ 12 | Normal | Three-variable square bridge

12. If $a + b + c = 9$ and $ab + bc + ca = 26$, then $a^2 + b^2 + c^2$ equals:

A. 29

B. 55

C. 107

D. 29±... 33

Correct Answer: (A)

$81 = a^2+b^2+c^2 + 52$, so the sum of squares is 29.

Why (B) is wrong: Added 2Σ instead of subtracting.

Why (C) is wrong: Doubled wrongly.

Why (D) is wrong: Arithmetic slip.

Hinglish: *81 - 52 = 29.*

MCQ 13 | Normal | Cube sum from data

13. If $x + y = 5$ and $xy = 6$, then $x^3 + y^3$ equals:

A. 35

B. 125

C. 53

D. 63

Correct Answer: (A)

125 - 3·6·5 = 125 - 90 = 35.

Why (B) is wrong: That is $(x+y)^3$.

Why (C) is wrong: Subtracted 3xy only.

Why (D) is wrong: Arithmetic slip.

Hinglish: *$(x+y)^3$ - 3xy(x+y) = 125 - 90 = 35.*

MCQ 14 | Normal | Crown identity zero condition

14. $x^3 + y^3 + z^3 - 3xyz = 0$ holds when:

A. x + y + z = 0 or x = y = z

B. only x + y + z = 0

C. only x = y = z

D. xyz = 0

Correct Answer: (A)

The identity factors as $(x+y+z)\cdot\frac{1}{2}[(x-y)^2+(y-z)^2+(z-x)^2]$, which is zero when either factor is zero.

Why (B) is wrong: Missed the equal case.

Why (C) is wrong: Missed the zero-sum case.

Why (D) is wrong: xyz = 0 alone is insufficient.

Hinglish: *Do cases: sum zero YA teeno barabar. Sirf ek case sochna aadha answer.*

MCQ 15 | Normal | Clever square completion

15. $999^2 + 2\cdot999 + 1$ equals:

A. 1000000

B. 999999

C. 998001

D. 1001999

Correct Answer: (A)

The expression is $999^2 + 2(999)(1) + 1^2 = (999+1)^2 = 1000^2 = 1{,}000{,}000$.

Why (B) is wrong: Off by one.

Why (C) is wrong: That is just 999^2.

Why (D) is wrong: Wrong assembly.

Hinglish: *Perfect square pehchano: $(999+1)^2$. 10^6.*

MCQ 16 | Normal | Bridge reversed

16. If $a^3 + b^3 = 152$ and $a + b = 8$, then ab equals:

A. 15

B. 12

C. 19

D. 16

Correct Answer: (A)

$152 = 512 - 3ab(8) \Rightarrow 24ab = 360 \Rightarrow ab = 15$.

Why (B) is wrong: Arithmetic slip.

Why (C) is wrong: Divided by wrong number.

Why (D) is wrong: Assumed ab = a+b.

Hinglish: *512 - 152 = 360. 360/24 = 15.*

MCQ 17 | Normal | Scaled cube bridge

17. If $2x + 3y = 12$ and $xy = 6$, then $8x^3 + 27y^3$ equals:

A. 432

B. 1728

C. 1296

D. 1512

Correct Answer: (A)

$(2x)^3+(3y)^3 = (2x+3y)^3 - 3AB(2x+3y)$ where $AB = (2x)(3y) = 6xy = 36$. So $1728 - 3(36)(12) = 1728 - 1296 = 432$.

Why (B) is wrong: That is $(2x+3y)^3$ - the raw cube of the sum.

Why (C) is wrong: AB taken as 6 instead of 36.

Why (D) is wrong: Arithmetic slip.

Hinglish: *A = 2x, B = 3y. AB = 6xy = 36. Bridge: 1728 - 1296 = 432.*

MCQ 18 | Normal | Half-sum of squares

18. $a^2 + b^2 + c^2 - ab - bc - ca$ equals:

A. $\frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$

B. (a-b)(b-c)(c-a)

C. 0

D. $\frac{1}{2}[(a+b)^2+(b+c)^2+(c+a)^2]$

Correct Answer: (A)

Expanding the right side gives exactly the left side.

Why (B) is wrong: That is degree 3.

Why (C) is wrong: Not identically zero.

Why (D) is wrong: That gives 2Σ + ... signs wrong.

Hinglish: *Half-sum-of-squares identity - crown identity ka engine.*

MCQ 19 | HOTS | Sum-cube identity form

19. $(a + b + c)^3$ equals:

A. $a^3+b^3+c^3+3(a+b)(b+c)(c+a)$

B. $a^3+b^3+c^3+3abc$

C. $a^3+b^3+c^3+3(a+b+c)(ab+bc+ca)$

D. $(a+b)(b+c)(c+a)$

Correct Answer: (A)

Standard identity; option C misses the -3abc correction, so A is the clean exact form.

Why (B) is wrong: Too small.

Why (C) is wrong: Missing -3abc term.

Why (D) is wrong: No cubes at all.

Hinglish: *(a+b)(b+c)(c+a) wala form yaad rakho - expansion se match karta hai.*

MCQ 20 | HOTS | Telescoping product

20. $(2+1)(2^2+1)(2^4+1)(2^8+1) + 1$ equals:

A. 65536

B. 65535

C. 32768

D. 262144

Correct Answer: (A)

Multiply and divide by (2-1) = 1: the product telescopes to $2^{16} - 1$; adding 1 gives $2^{16} = 65536$.

Why (B) is wrong: Forgot to add 1.

Why (C) is wrong: Stopped at 2^{15}.

Why (D) is wrong: Overshot the exponent.

Hinglish: *(2-1) se multiply-divide karo: chain telescopes to 2^{16}-1. Plus 1: 65536.*

4.6 Choose the Correct Round: Theory & Speed (20)

Doosra rapid round - har MCQ ka style: statement padho, correct option turant chuno. Ye round theory, definitions aur quick solving ko test karta hai. Answer strip ke neeche har galat option ka "Why wrong" bhi hai - taki exam ka trap tumhe na phasaye.

CTC 1 | Normal | Binomial square

1. $(x + 4)^2$ equals:

A. $x^2 + 16$

B. $x^2 + 8x + 16$

C. $x^2 + 4x + 16$

D. $x^2 + 8x + 8$

Correct Answer: (B)

$(a+b)^2 = a^2 + 2ab + b^2$: $x^2 + 2 \cdot x \cdot 4 + 16 = x^2 + 8x + 16$.

Why (A) is wrong: Cross term completely dropped.

Why (C) is wrong: Coefficient 2 forgotten on the cross term.

Why (D) is wrong: 16 not squared properly.

Hinglish: *Cross term $2 \cdot x \cdot 4 = 8x$ bhoolna sabse common galti. $(a+b)^2 \neq a^2+b^2$ - kabhi nahi.*

CTC 2 | Normal | (x+a)(x+b)

2. $(x + 5)(x + 3)$ equals:

A. $x^2 + 15x + 8$

B. $x^2 + 8x + 15$

C. $x^2 + 8x + 8$

D. $x^2 + 15$

Correct Answer: (B)

Sum $5+3 = 8$ feeds the x-term; product 15 feeds the constant: $x^2 + 8x + 15$.

Why (A) is wrong: Sum and product swapped.

Why (C) is wrong: Product taken as 5+3.

Why (D) is wrong: Middle term dropped.

Hinglish: *Sum middle mein, product last mein. $8x + 15$.*

CTC 3 | Normal | Near-number square minus

3. 98^2 equals:

A. 9964

B. 9604

C. 9640

D. 9844

Correct Answer: (B)

$(100-2)^2 = 10000 - 400 + 4 = 9604$.

Why (A) is wrong: Last term -4 kiya (squares never negative).

Why (C) is wrong: Digits rearranged.

Why (D) is wrong: Wrong cross term.

Hinglish: *Minus wale case: cross term MINUS 400, phir +4. 9604.*

CTC 4 | Normal | Difference of squares

4. $x^2 - 49$ equals:

A. $(x-7)^2$

B. $(x+7)(x-7)$

C. $(x+49)(x-1)$

D. $x(x-49)$

Correct Answer: (B)

$a^2 - b^2 = (a+b)(a-b)$ with $b = 7$.

Why (A) is wrong: That is $x^2 - 14x + 49$.

Why (C) is wrong: 49 is not 49^2.

Why (D) is wrong: No common factor exists.

Hinglish: *$49 = 7^2$. $(x+7)(x-7)$. $(x-7)^2$ would expand to $x^2-14x+49$ - alag cheez.*

CTC 5 | Normal | Sum of cubes

5. $27a^3 + 8b^3$ factorises as:

A. $(3a+2b)(9a^2-6ab+4b^2)$

B. $(3a+2b)^3$

C. $(3a+2b)(9a^2+6ab+4b^2)$

D. $(3a-2b)(9a^2+6ab+4b^2)$

Correct Answer: (A)

$(3a)^3+(2b)^3 = (3a+2b)((3a)^2-(3a)(2b)+(2b)^2)$.

Why (B) is wrong: That is a cube, not a factorisation.

Why (C) is wrong: Middle sign wrong.

Why (D) is wrong: Binomial sign wrong.

Hinglish: *Trinomial ka middle term NEGATIVE - binomial ke sign ka ulta.*

CTC 6 | Normal | Difference of cubes

6. x^3 - 27 equals:

A. $(x-3)(x^2+3x+9)$

B. $(x-3)(x^2-3x+9)$

C. $(x-3)^3$

D. $(x+3)(x^2-3x+9)$

Correct Answer: (A)

$A^3-B^3 = (A-B)(A^2+AB+B^2)$ with B = 3.

Why (B) is wrong: Trinomial middle sign flipped.

Why (C) is wrong: That is $x^3-9x^2+27x-27$.

Why (D) is wrong: Binomial sign wrong.

Hinglish: *Difference: binomial minus, trinomial plus. $(x-3)(x^2+3x+9)$.*

CTC 7 | Normal | Symmetric reconstruction

7. If a + b = 8 and ab = 12, then $a^2 + b^2$ equals:

A. 40

B. 64

C. 76

D. 52

Correct Answer: (A)

$a^2+b^2 = (a+b)^2 - 2ab = 64 - 24 = 40$.

Why (B) is wrong: $(a+b)^2$ hi le liya.

Why (C) is wrong: Subtracted 2ab twice-ish.

Why (D) is wrong: Arithmetic slip.

Hinglish: *64 - 24 = 40. Bridge yaad rakho.*

CTC 8 | Normal | Equidistant product 2

8. 103 × 97 equals:

A. 9991

B. 10009

C. 9981

D. 9991±... 9979

Correct Answer: (A)

(100+3)(100-3) = 10000 - 9 = 9991.

Why (B) is wrong: Added 9.

Why (C) is wrong: Offset squared wrong.

Why (D) is wrong: Wrong offset product.

Hinglish: *10000 - 9 = 9991.*

CTC 9 | Normal | Binomial square with coefficients

9. $(2x + 3y)^2$ equals:

A. $4x^2 + 6xy + 9y^2$

B. $4x^2 + 12xy + 9y^2$

C. $2x^2 + 6xy + 3y^2$

D. $4x^2 + 9y^2$

Correct Answer: (B)

$(2x)^2 + 2(2x)(3y) + (3y)^2 = 4x^2 + 12xy + 9y^2$.

Why (A) is wrong: Cross term halved.

Why (C) is wrong: Forgot to square the coefficients.

Why (D) is wrong: Cross term dropped.

Hinglish: *Cross term 2×2×3 = 12. Squares: 4 aur 9.*

CTC 10 | Normal | Cube expansion coefficient

10. The coefficient of a^2b in $(5a - 3b)^3$ is:

A. -225

B. 225

C. -75

D. 75

Correct Answer: (A)

$3(5a)^2(-3b) = 3 \cdot 25 \cdot (-3)\ a^2b = -225a^2b$.

Why (B) is wrong: Sign dropped.

Why (C) is wrong: Coefficient 3 missed on the pattern.

Why (D) is wrong: Sign and 3 both wrong.

Hinglish: *$3 \times 25 \times (-3) = -225$. Sign minus - kyunki $(a-b)^3$ mein doosra term negative.*

CTC 11 | Normal | Difference of squares with coefficients

11. $4a^2 - 9b^2$ equals:

A. $(2a-3b)^2$

B. $(2a+3b)(2a-3b)$

C. $(4a+9b)(a-b)$

D. $(2a+3b)^2$

Correct Answer: (B)

$(2a)^2 - (3b)^2 = (2a+3b)(2a-3b)$.

Why (A) is wrong: That is $4a^2-12ab+9b^2$.

Why (C) is wrong: Wrong pairing.

Why (D) is wrong: Plus sign - that is a sum.

Hinglish: *$4a^2 = (2a)^2$, $9b^2 = (3b)^2$. Difference of squares.*

CTC 12 | Normal | Chained difference of squares

12. $(x-2)(x+2)(x^2+4)$ equals:

A. $x^4 - 16$

B. $x^4 + 16$

C. $x^2 - 16$

D. $x^4 - 4$

Correct Answer: (A)

First pair: x^2-4. Then $(x^2-4)(x^2+4) = x^4-16$.

Why (B) is wrong: Plus instead of minus.

Why (C) is wrong: Stopped after one round.

Why (D) is wrong: 4 not squared.

Hinglish: *Do baar difference of squares: x^2-4, phir x^4-16.*

CTC 13 | Normal | Decimal equidistant product

13. 1.02 × 0.98 equals:

A. 0.9996

B. 0.9906

C. 1.0004

D. 0.999

Correct Answer: (A)

$(1+0.02)(1-0.02) = 1 - 0.0004 = 0.9996$.

Why (B) is wrong: 0.02^2 miscomputed as 0.0094?

Why (C) is wrong: Added instead of subtracted.

Why (D) is wrong: Rounded too early.

Hinglish: *$0.02^2 = 0.0004$ - four decimal places. 1 - 0.0004.*

CTC 14 | Normal | Difference chain

14. If $x - 1/x = 4$, then $x^2 + 1/x^2$ equals:

A. 18

B. 14

C. 16

D. 20

Correct Answer: (A)

$(x-1/x)^2 = x^2-2+1/x^2 = 16 \Rightarrow x^2+1/x^2 = 18$.

Why (B) is wrong: Subtracted 2.

Why (C) is wrong: Forgot the 2 entirely.

Why (D) is wrong: Added 4.

Hinglish: *16 + 2 = 18. Difference wale case mein PLUS 2.*

CTC 15 | Normal | Cube difference pattern

15. $(a - b)^3$ equals:

A. $a^3-3a^2b+3ab^2-b^3$

B. $a^3-3a^2b-3ab^2-b^3$

C. $a^3+3a^2b+3ab^2+b^3$

D. a^3-b^3

Correct Answer: (A)

Alternating signs: +, -, +, -.

Why (B) is wrong: Third term sign wrong.

Why (C) is wrong: That is $(a+b)^3$.

Why (D) is wrong: Middle terms dropped.

Hinglish: *Pattern 1-3-3-1 with alternating signs.*

CTC 16 | Normal | Difference of cubes scaled

16. $64x^3 - 125$ equals:

A. $(4x-5)(16x^2+20x+25)$

B. $(4x-5)(16x^2-20x+25)$

C. $(4x-5)^3$

D. $(4x+5)(16x^2+20x+25)$

Correct Answer: (A)

$(4x)^3-5^3 = (4x-5)((4x)^2+4x\cdot5+25)$.

Why (B) is wrong: Trinomial middle sign wrong.

Why (C) is wrong: Not a cube.

Why (D) is wrong: Binomial sign wrong.

Hinglish: *$(4x)^2 = 16x^2$, $AB = 20x$, $B^2 = 25$. Plus signs in trinomial.*

CTC 17 | Normal | Fourth-power difference

17. $x^4 - 81$ factorises completely as:

A. $(x-3)(x+3)(x^2+9)$

B. $(x^2-9)(x^2-9)$

C. $(x-3)(x+3)$

D. $(x^2+9)^2$

Correct Answer: (A)

$x^4-81 = (x^2-9)(x^2+9) = (x-3)(x+3)(x^2+9)$.

Why (B) is wrong: Same factor twice - wrong.

Why (C) is wrong: Incomplete.

Why (D) is wrong: Not a factorisation.

Hinglish: *Do baar factor: x^2-9 ko bhi todo. x^2+9 khatam.*

CTC 18 | Normal | Square completion

18. $x^4 + x^2 + 1$ equals:

A. $(x^2+x+1)(x^2-x+1)$

B. $(x^2+1)^2$

C. $(x^2+x+1)^2$

D. $(x^2+2x+1)(x^2-2x+1)$

Correct Answer: (A)

Add-subtract x^2: $(x^2+1)^2 - x^2$ = difference of squares.

Why (B) is wrong: That is x^4+2x^2+1.

Why (C) is wrong: That is $x^4+2x^3+3x^2+2x+1$.

Why (D) is wrong: Wrong middle coefficients.

Hinglish: *Classic completion trick. $(x^2+1)^2$ mein x^2 ek zyada tha.*

CTC 19 | Normal | Cube difference product

19. $(3 - a)(9 + 3a + a^2)$ equals:

A. $27 - a^3$

B. $27 + a^3$

C. $9 - a^3$

D. $3 - a^3$

Correct Answer: (A)

This is $(3-a)((3)^2+3\cdot a+a^2) = 3^3 - a^3$.

Why (B) is wrong: Sign of the cube wrong.

Why (C) is wrong: Took square of 3.

Why (D) is wrong: Mixed the two letters.

Hinglish: *Difference of cubes ka reverse: 3 = B, a = A. B^3-A^3.*

CTC 20 | Normal | Equidistant product 3

20. 105×95 equals:

A. 9975

B. 10025

C. 9925

D. 9999

Correct Answer: (A)

(100+5)(100-5) = 10000 - 25 = 9975.

Why (B) is wrong: Added 25.

Why (C) is wrong: Offset error.

Why (D) is wrong: Arithmetic slip.

Hinglish: *10000 - 25 = 9975.*

Chapter 6: Perimeter and Area

Measurement questions are the most reliable marks in any paper - and the most reliable marks LOST to careless formulas. The uploaded PDF assembles the full toolkit: a review of perimeters and areas of triangles, squares, rectangles and parallelograms; the meaning of π; circumference and area of circles; arc length and sector area; Heron's formula for triangles with known sides; its special cases for equilateral and isosceles triangles; area of a quadrilateral split by a diagonal; Brahmagupta's formula for cyclic quadrilaterals; and rings, paths and composite regions. This book hardens every tool: reverse questions (area back to radius), sector-perimeter traps, grazing goats, tracks with semicircular ends, and Hippocrates-style shaded regions that look impossible and collapse in one line.

Figure 6.1: Heron's machinery - from three sides s(a+b+c)/2 to a clean area, no heights needed.

6.1 Complete Concept Cheat Sheet (read this before any question)

6.1.1 Perimeter and area: the review set

What is it? Perimeter is the length of the boundary; area is the size of the enclosed surface. **Core formulas:** rectangle $P = 2(l+b)$, $A = l \times b$; square $P = 4s$, $A = s^2$, diagonal $= s\sqrt{2}$; triangle $A = \frac{1}{2} \times \text{base} \times \text{height}$; parallelogram $A = \text{base} \times \text{height}$; rhombus $A = \frac{1}{2} \times d_1 \times d_2$; trapezium $A = \frac{1}{2}(a+b)h$. **The perimeter-area trap:** equal perimeters do NOT mean equal areas - among all shapes with a fixed perimeter, the circle encloses the most area. **When do I use which?** Perimeter answers 'how much boundary/fencing/frame', area answers 'how much surface/paint/carpet/grass'.

Units discipline: 1 m = 100 cm but 1 m^2 = 10000 cm^2; 1 hectare = 10000 m^2. Converting lengths squares the conversion factor - forget this once and your answer is off by a factor of 100.

6.1.2 π, circumference and area of a circle

What is π? The ratio circumference/diameter - the same constant for EVERY circle, approximately 3.14159... irrational. Exam convention: $\pi = 22/7$ when the radius is a multiple of 7, otherwise 3.14. **Formulas:** $C = 2\pi r = \pi d$; $A = \pi r^2$. **Reverse uses (the exam favourites):** given C, $r = C/(2\pi)$ then $A = C^2/(4\pi)$; given A, $r = \sqrt{(A/\pi)}$ then $C = 2\pi r$. **Scaling laws:** double the radius → double the circumference but QUADRUPLE the area; radii in ratio k → areas in ratio k^2. **Tiny example.** r = 14 cm: $C = 2 \times (22/7) \times 14 = 88$ cm; $A = (22/7) \times 196 = 616$ cm^2.

6.1.3 Arc length and sector area

What is it? A sector is the pizza slice between two radii and an arc of angle θ. The full circle is 360°, so a sector owns the fraction θ/360 of everything. **Arc length:** $l = (\theta/360) \times 2\pi r$. **Sector area:** $A = (\theta/360) \times \pi r^2$. **The half-angle gem:** $A = \frac{1}{2} \times l \times r$ (treat the arc as a curved 'base' and the radius as the 'height'). **Sector PERIMETER (the trap):** P = arc + TWO radii = l + 2r. Writing l + r loses marks every single year. **Tiny example.** r = 21, θ = 120°: $l = (120/360)(2 \cdot (22/7) \cdot 21) = 44$ cm; $A = (1/3)\pi(441) = 462$ cm^2; P = 44 + 42 = 86 cm.

Figure 6.2: Sector anatomy - arc l, two radii, angle θ; the slice owns θ/360 of the circle.

6.1.4 Heron's formula

What is it? Area of a triangle from its three sides alone, no height needed: A = √(s(s-a)(s-b)(s-c)) where s = (a+b+c)/2 is the semi-perimeter. **Why it matters:** most real triangles (a field, a plot) give you sides, not heights - Heron converts three tape-measure readings into an area. **Worked rhythm:** (1) add sides, halve for s; (2) compute the three differences s-a, s-b, s-c; (3) multiply all four numbers; (4) take the square root - and SIMPLIFY the surd if it is not a perfect square. **Tiny example.** Sides 5, 12, 13: s = 15; product 15·3·2 = 900; A = 30. **Validity check:** each side must be less than the sum of the other two (triangle inequality), otherwise the quantity under the root goes negative - the 'triangle' does not exist.

> **Verification habit**
> If the three sides satisfy $a^2 + b^2 = c^2$, the triangle is right-angled and the area must equal $\frac{1}{2}ab$. Run this cross-check whenever the sides look Pythagorean - 5-12-13, 8-15-17, 7-24-25, 9-40-41 and their scaled cousins.

6.1.5 Equilateral and isosceles shortcuts

Equilateral triangle (side a): height = $(\sqrt{3}/2)a$, area = $(\sqrt{3}/4)a^2$, perimeter = 3a. The area formula saves one Heron round-trip. **Isosceles triangle:** drop the height to the base; it bisects the base. Given equal sides l and base b: height = $\sqrt{(l^2 - b^2/4)}$, area = $\frac{1}{2} \cdot b \cdot$ height, and Heron must agree. **Reverse equilateral:** area given as $k\sqrt{3}$, side a satisfies $(\sqrt{3}/4)a^2 = k\sqrt{3} \Rightarrow a = 2\sqrt{k}$. **Tiny example.** Equilateral side 12: A = $(\sqrt{3}/4)(144) = 36\sqrt{3} \approx 62.35$.

6.1.6 Quadrilateral by diagonal; Brahmagupta

Diagonal split: any quadrilateral ABCD splits along diagonal AC into triangles ABC and ACD. Find each area (heights or Heron) and ADD. This is the master method for four-sided fields with unequal sides. **Brahmagupta's formula (cyclic quadrilaterals):** if all four vertices lie on a circle and sides are a, b, c, d with s = (a+b+c+d)/2, then A = √((s-a)(s-b)(s-c)(s-d)) - Heron with the (s - ...) pattern of four sides and no explicit height. It is valid ONLY for cyclic quadrilaterals; applying it to a general quadrilateral is a silent error. **Tiny example.** Sides 13, 10, 13, 10 (cyclic): s = 23; product 10·13·10·13 = 16900; A = 130.

Figure 6.3: A quadrilateral split by diagonal BD - two triangles, two Herons, one sum.

6.1.7 Rings, paths and composite regions

Ring (annulus): $A = \pi R^2 - \pi r^2 = \pi(R^2 - r^2) = \pi(R+r)(R-r)$. Given width w = R - r and area, use $(R+r)(R-r) = A/\pi$ to recover both radii. **Inside border path:** rectangular field L × B with a path of width w inside: path area = L·B - (L-2w)(B-2w). **Outside circular path:** $\pi((r+w)^2 - r^2) = \pi w(2r + w)$. **Composite figures:** decompose into rectangles, triangles, sectors and rings; add or subtract; keep every π exact until the last line. **Chord touching the inner circle (HOTS classic):** a chord of the outer circle tangent to the inner one has half-chord^2 = $R^2 - r^2$, so ring area = π × (half-chord)^2 - no radii needed!

Figure 6.4: Ring geometry - a chord of the outer circle touching the inner circle fixes $R^2 - r^2$.

6.1.8 Exam tips for this chapter

1. Choose $\pi = 22/7$ when a radius or diameter is a multiple of 7; otherwise use 3.14 - and say which you used.
2. Sector perimeter = arc + 2r. Circle perimeter = $2\pi r$. Never mix them.
3. Reduce every surd area (e.g. $\sqrt{1215} = 9\sqrt{15}$) - unsimplified surds lose the final mark.
4. Check for right triangles (Pythagoras) before Heron; $\frac{1}{2} \times \text{leg} \times \text{leg}$ is faster and safer.
5. Write units at every line: cm for length, cm^2 for area, m^2 for fields, hectares for land.
6. In reverse questions (area → radius → circumference) keep π symbolic until the numbers force it in.
7. Ratio of radii k ⇒ ratio of areas k^2 - read ratios as ratios, never as differences.
8. Cost questions: compute the geometric quantity FIRST, then multiply by rate; never mix the two steps.

6.2 Question Types / Concept Map for Chapter 6

Type	Name	How to recognise it in the question	Core method
T1	Basic perimeter/area	Rectangle, square, triangle, parallelogram, rhombus, trapezium	Direct formulas with correct units
T2	Heron direct	Three sides given, area asked	$s = (a+b+c)/2$; $\sqrt{(s(s-a)(s-b)(s-c))}$
T3	Equilateral / isosceles	'all sides equal', perimeter + base given	$(\sqrt{3}/4)a^2$; height by Pythagoras
T4	Quadrilateral by diagonal	Four sides plus a diagonal	Two Herons, add
T5	Brahmagupta	'cyclic quadrilateral' with four sides	$\sqrt{((s-a)(s-b)(s-c)(s-d))}$
T6	Circle basics + reverse	C ↔ r ↔ A chains	$C = 2\pi r$, $A = \pi r^2$, reversed
T7	Arc and sector	Angle θ at centre	$(\theta/360)$ fractions; $A = \frac{1}{2}lr$; $P = l + 2r$
T8	Segments and combinations	'segment', chord with angle	sector − triangle
T9	Rings / annulus	Two concentric circles	$\pi(R^2-r^2)$; chord trick
T10	Paths and borders	'path inside/outside', track, frame	Outer minus inner
T11	Composite figures	Mixed shapes, shaded regions	Decompose; add/subtract
T12	Ratio and scaling	Radii in ratio, same perimeter comparisons	Areas scale as k^2
T13	Cost applications	Rates for fencing, ploughing, tiling	Geometry first, money second
T14	HOTS shaded/proofs	Lunes, hexagons, grazing animals, AP sides	Label regions; prove then compute

6.3 Speed Tricks & Trap Alarms

Ye table tumhara "dekho aur maaro" radar hai - question ka trigger phrase dikha, toh yahan ka trick fire karo. Inhe ratt lo, exam me sochna kam padega aur speed double ho jayegi.

Instant trick	Fire it when...
π choice: radius multiple of 7 → 22/7; otherwise 3.14	First decision of every circle question.
Sector and arc = θ/360 × (full circle value). Gem: sector area = ½ × arc × r	Pizza-slice sector questions.
Sides satisfy $a^2 + b^2 = c^2$? Area = ½ × leg × leg - skip Heron	5-12-13, 9-40-41 style triangles.
Heron rhythm: s → three differences → multiply (pair smartly) → root	Three sides given, height unknown.
Ring: A = π(R+r)(R-r). Path outside: R = r + w. Path inside: subtract squares	Garden path / ring / border questions.
Wheel: distance = revolutions × circumference	"How many revolutions / how far does it travel".
Equilateral: area = $(\sqrt{3}/4)a^2$, height = $(\sqrt{3}/2)a$	All sides equal - no Heron needed.

6.4 The 120-Question Workout

Poora chapter ab ek hi ring mein: **120 questions**, chaar level zones mein banta hua. Warm-up se shuru karo, Boss Zone tak pahuncho. Har question ke turant baad boxed step-by-step solution hai - pehle khud try karo, atako toh solution dekho, aur "Why this step?" + Hinglish note se logic pakka karo. Agar in 120 ko solve kar liya toh is chapter ka koi bhi twisted question tumhe nahi rok payega.

Level 1 · Warm-up Zone: basics and confidence:

QUESTION 1 | Basic | Rectangle perimeter/area (T1)

Q1. A rectangular floor is 12.5 m long and 8 m wide. (i) How many metres of skirting board are needed along the four walls' base line if the room has no door opening on the floor plan? (ii) How many square metres of carpet cover the floor?

SOLUTION — step by step

Step 1. Given: l = 12.5 m, b = 8 m. Needed: perimeter and area.

Step 2. Perimeter = 2(l + b) = 2(12.5 + 8) = 2 × 20.5 = 41 m.

Step 3. Area = l × b = 12.5 × 8 = 100 m^2.

Step 4. Skirting board follows the boundary → 41 m; carpet covers the surface → 100 m^2.

» **Why this step?** Perimeter questions use lengths once around the boundary; area questions multiply the two dimensions - mixing them is the most common unit error.

Answer: (i) 41 m (ii) 100 m^2

Hinglish: *Boundary = perimeter, surface = area. Skirting board room ke chakkar lagata hai: 2(12.5+8) = 41 m. Carpet poori zameen dhakta hai: 100 m^2. 'How much around' vs 'how much inside' - ye difference pakka karo.*

QUESTION 2 | Basic | Triangle base-height (T1)

Q2. A triangle has base 14 cm and height 9 cm. (i) Find its area. (ii) A second triangle has the same area but height 7 cm; find its base.

SOLUTION — step by step

Step 1. Given: b = 14, h = 9. Area = ½ × 14 × 9 = 63 cm^2.

Step 2. Second triangle: ½ × b × 7 = 63 ⇒ 7b = 126 ⇒ b = 18 cm.

» **Why this step?** Area = ½(base)(height) works because a triangle is half the parallelogram built on the same base.

Answer: (i) 63 cm^2 (ii) 18 cm

Hinglish: *Area = aadha × base × height. Pehla: 63 cm^2. Doosre mein same area, height 7 - base 18. Reverse questions mein equation banao, shortcut mat dhoondo.*

QUESTION 3 | Basic | Circumference and area (T6)

Q3. For a circle of radius 14 cm, find (i) the circumference and (ii) the area (use π = 22/7).

SOLUTION — step by step

Step 1. Given: r = 14 (a multiple of 7 → use π = 22/7).

Step 2. Circumference C = 2πr = 2 × (22/7) × 14 = 88 cm.

Step 3. Area A = πr^2 = (22/7) × 14^2 = (22/7) × 196 = 616 cm^2.

» **Why this step?** The 7 in the denominator cancels with the radius multiple - that is exactly when 22/7 gives exact answers.

Answer: C = 88 cm, A = 616 cm^2

Hinglish: *Radius 7 ka multiple hai, isliye 22/7 perfect. C = 88 cm, A = 616 cm^2. Dono answers clean aane chahiye - decimal aaya toh kahin galti hai.*

QUESTION 4 | Basic | Reverse circle (T6)

Q4. The circumference of a circular plate is 132 cm. Find (i) its radius and (ii) its area.

SOLUTION — step by step

Step 1. Given: C = 132 cm. Reverse: r = C/(2π) = 132 × 7/44 = 21 cm.

Step 2. Area = πr^2 = (22/7) × 441 = 1386 cm^2.

» **Why this step?** Dividing by 2π undoes the circumference formula; then the area formula runs forward.

Answer: r = 21 cm, A = 1386 cm^2

Hinglish: *Circumference se radius: C ko 2π se divide karo. 132 × 7/44 = 21. Phir area 1386 cm². Reverse chain C → r → A exam ka favourite hai.*

QUESTION 5 | Basic | Equilateral shortcut (T3)

Q5. Find the perimeter and the area of an equilateral triangle of side 12 cm (leave √3 in your answer, then approximate with √3 ≈ 1.73).

SOLUTION — step by step

Step 1. Perimeter = 3 × 12 = 36 cm.

Step 2. Area = $(\sqrt{3}/4)a^2 = (\sqrt{3}/4) \times 144 = 36\sqrt{3}$ cm².

Step 3. Numerically: 36 × 1.73 = 62.28 cm² (with 1.732: 62.35).

» **Why this step?** $(\sqrt{3}/4)a^2$ is Heron's result pre-simplified for equal sides - one line instead of four.

Answer: P = 36 cm, A = 36√3 ≈ 62.3 cm²

Hinglish: *Equilateral ka ready-made formula: (√3/4)a². 144/4 = 36, toh 36√3 ≈ 62.3 cm². Heron se bhi milega but formula fast hai.*

QUESTION 6 | Basic | Heron direct (T2)

Q6. Find the area of a triangle with sides 5 cm, 12 cm and 13 cm.

SOLUTION — step by step

Step 1. s = (5 + 12 + 13)/2 = 15 cm.

Step 2. Differences: s - a = 10, s - b = 3, s - c = 2.

Step 3. A = √(15 × 10 × 3 × 2) = √900 = 30 cm².

Step 4. Cross-check: $5^2 + 12^2 = 169 = 13^2$, so it is right-angled; ½ × 5 × 12 = 30 ✓.

» **Why this step?** Spotting the right angle gives an instant verification - Heron must agree with ½ab.

Answer: 30 cm²

Hinglish: *Heron ka 4-step rhythm: s nikaalo, differences nikaalo, multiply, root. 15×10×3×2 = 900, root = 30. Bonus: 5-12-13 Pythagorean triple hai - ½×5×12 = 30 se confirm.*

QUESTION 7 | Basic | Parallelogram and triangle (T1)

Q7. A parallelogram has base 10 cm and height 6.5 cm. (i) Find its area. (ii) A triangle stands on the same base and between the same parallels; find its area.

SOLUTION — step by step

Step 1. Parallelogram area = base × height = $10 \times 6.5 = 65$ cm^2.

Step 2. A triangle between the same parallels and on the same base has exactly half the area:

Step 3. Triangle area = $\frac{1}{2} \times 65 = 32.5$ cm^2.

» **Why this step?** Same base, same parallels ⇒ the triangle is exactly half the parallelogram - a definitional fact worth memorising.

Answer: (i) 65 cm^2 (ii) 32.5 cm^2

Hinglish: *Same base, same parallel lines ke beech: triangle = aadha parallelogram. 65 aur 32.5. Ye 'half' relation theorems mein bhi aata hai.*

QUESTION 8 | Basic | Trapezium area (T1)

Q8. A trapezium has parallel sides 12 cm and 8 cm, and the perpendicular distance between them is 5 cm. Find its area.

SOLUTION — step by step

Step 1. $A = \frac{1}{2}(a + b)h = \frac{1}{2}(12 + 8)(5) = \frac{1}{2} \times 20 \times 5 = 50$ cm^2.

» **Why this step?** Average of the parallel sides times the height: the trapezium is a rectangle plus a half-rectangle on average.

Answer: 50 cm^2

Hinglish: *$(a+b)/2 \times h$ - parallel sides ka average, height se multiply. $10 \times 5 = 50$ cm^2.*

QUESTION 9 | Basic | Arc and sector (T7)

Q9. For a circle of radius 7 cm, find (i) the length of an arc subtending 90° at the centre and (ii) the area of the corresponding sector.

SOLUTION — step by step

Step 1. Fraction of the circle: $90/360 = 1/4$.

Step 2. Arc = $(1/4) \times 2\pi r = (1/4) \times 44 = 11$ cm.

Step 3. Sector area = $(1/4) \times \pi r^2 = (1/4) \times 154 = 38.5$ cm^2.

» **Why this step?** Everything scales by $\theta/360$ - compute the full-circle value once, then take the fraction.

Answer: arc = 11 cm, sector = 38.5 cm^2

Hinglish: *90° matlab quarter circle. Arc = 44/4 = 11 cm, area = 154/4 = 38.5 cm². Fraction pehle likho, calculation aadhi ho jati hai.*

QUESTION 10 | Basic | Ring area (T9)

Q10. Two concentric circles have radii 10 cm and 6 cm. Find the area of the ring between them (π = 3.14).

SOLUTION — step by step

Step 1. Ring area = $\pi(R^2 - r^2) = \pi(100 - 36) = 64\pi$.

Step 2. With $\pi = 3.14$: $64 \times 3.14 = 200.96$ cm^2.

» **Why this step?** Subtract the inner disk from the outer disk; factorising $R^2-r^2 = (R+r)(R-r)$ speeds the arithmetic.

Answer: 200.96 cm²

Hinglish: *Ring = bada circle - chhota circle. π(100-36) = 64π. 3.14 se: 200.96 cm². R²-r² ko (R+r)(R-r) likhoge toh mental math ho jata hai.*

QUESTION 11 | Basic | Square and diagonal (T1)

Q11. The perimeter of a square is 64 cm. Find (i) its side, (ii) its area, (iii) the length of its diagonal (leave $\sqrt{2}$ in the answer).

SOLUTION — step by step

Step 1. Side = 64/4 = 16 cm.

Step 2. Area = $16^2 = 256$ cm^2.

Step 3. Diagonal = $s\sqrt{2} = 16\sqrt{2}$ cm (from the half-square right triangle).

» **Why this step?** The diagonal of a square is the hypotenuse of a 45-45-90 triangle: $s\sqrt{2}$ always.

Answer: 16 cm; 256 cm²; 16√2 cm

Hinglish: *Side = perimeter/4 = 16. Area 256. Diagonal = side×√2 = 16√2. Ye teeno ek saath aate hain.*

QUESTION 12 | Basic | Isosceles triangle via Heron (T3)

Q12. An isosceles triangle has perimeter 30 cm and each equal side 12 cm. Find its area (simplify the surd).

SOLUTION — step by step

Step 1. Base = 30 - 2(12) = 6 cm.

Step 2. s = 15. Differences: 15-12 = 3 (twice), 15-6 = 9.

Step 3. $A = \sqrt{(15 \times 3 \times 3 \times 9)} = \sqrt{1215}$.

Step 4. Simplify: $1215 = 81 \times 15$, so $A = 9\sqrt{15} \approx 34.86$ cm^2.

» **Why this step?** Heron handles isosceles without dropping a height; simplifying $\sqrt{1215} = 9\sqrt{15}$ is part of the answer.

Answer: $9\sqrt{15} \approx 34.9$ cm^2

Hinglish: *Base = perimeter - do equal sides = 6. Heron: √1215. Surd simplify karo: 1215 = 81×15 → 9√15 ≈ 34.9 cm². Height method se bhi ho jata (height = √(144-9) = √135), same answer.*

QUESTION 13 | Basic | Rectangle perimeter/area (T1)

Q13. A rectangular field is 85 m long and 60 m wide. Find (i) its perimeter, (ii) its area in m^2 and in hectares.

SOLUTION — step by step

Step 1. (i) Perimeter = $2(85 + 60) = 2 \times 145 = 290$ m.

Step 2. (ii) Area = $85 \times 60 = 5100$ m^2.

Step 3. 1 hectare = 10000 m^2 -> 5100 m^2 = 0.51 hectare.

» **Why this step?** Perimeter wraps twice around (length + width); hectares convert by dividing by 10000.

Answer: (i) 290 m (ii) 5100 m^2 = 0.51 hectare

Hinglish: *Perimeter = 2(l + b), area = l × b. Hectare = 10000 m² ka paisa - 5100 ÷ 10000 = 0.51. Units ka conversion exam ka bonus point hai.*

QUESTION 14 | Basic | Triangle base-height (T1)

Q14. (i) A triangle has base 18 cm and height 11 cm - find its area. (ii) Another triangle has base 22 cm and the SAME area - find its height.

SOLUTION — step by step

Step 1. (i) Area = $\frac{1}{2} \times 18 \times 11 = 99$ cm^2.

Step 2. (ii) $\frac{1}{2} \times 22 \times h = 99$ -> $11h = 99$ -> $h = 9$ cm.

» **Why this step?** Doubling the base halves the needed height for the same area - area is the product held constant.

Answer: (i) 99 cm^2 (ii) 9 cm

Hinglish: *½ × base × height. Same area, bada base -> chhota height: 18→22 base par height 11→9. See-saw balance samjho.*

QUESTION 15 | Basic | Circumference and area (T6)

Q15. For a circle of radius 10.5 cm, find (i) the circumference and (ii) the area. (Use $\pi = 22/7$.)

SOLUTION — step by step

Step 1. (i) $C = 2\pi r = 2 \times (22/7) \times 10.5 = 2 \times 22 \times 1.5 = 66$ cm.

Step 2. (ii) $A = \pi r^2 = (22/7) \times 110.25 = 22 \times 15.75 = 346.5$ cm^2.

» **Why this step?** Radius 10.5 = 21/2 is designed for 22/7: the 7s cancel cleanly.

Answer: (i) 66 cm (ii) 346.5 cm^2

Hinglish: *10.5 = 21/2 dekha? 22/7 ke saath perfect jodi! C = 66, A = 346.5. π ka value radius ke denominator se match karo.*

QUESTION 16 | Basic | Reverse circle (T6)

Q16. The circumference of a circular plate is 132 cm. Find (i) its radius and (ii) its area.

SOLUTION — step by step

Step 1. (i) $2\pi r = 132 \rightarrow 2 \times (22/7) \times r = 132 \rightarrow r = 132 \times 7/44 = 21$ cm.

Step 2. (ii) $A = (22/7) \times 21^2 = 22 \times 63 = 1386$ cm^2.

» **Why this step?** Circumference fixes the radius; the area then follows - reverse chain in two steps.

Answer: (i) 21 cm (ii) 1386 cm^2

Hinglish: *C se r nikalo: 132 × 7/44 = 21. Phir area: 22/7 × 441 = 1386. Reverse questions mein pehle r, baaki sab baad mein.*

QUESTION 17 | Basic | Equilateral shortcut (T3)

Q17. Find the perimeter and area of an equilateral triangle of side 12 cm ($\sqrt{3} \approx 1.73$).

SOLUTION — step by step

Step 1. Perimeter = $3 \times 12 = 36$ cm.

Step 2. Area = $(\sqrt{3}/4)a^2 = (1.73/4) \times 144 = 1.73 \times 36 = 62.28$ cm^2 (exactly $36\sqrt{3}$).

» **Why this step?** Equilateral = one formula, no Heron: $(\sqrt{3}/4)a^2$ comes from the fixed 60° angles.

Answer: Perimeter 36 cm; area $36\sqrt{3} \approx 62.28$ cm^2

Hinglish: *Teeno side barabar -> seedha $(\sqrt{3}/4)a^2$. $36\sqrt{3} \approx 62.28$. Equilateral ke liye Heron ka koi kaam nahi - shortcut ready hai.*

QUESTION 18 | Basic | Heron direct (T2)

Q18. Find the area of a triangle with sides 5 cm, 12 cm, 13 cm - first by Heron's formula, then by a faster method.

SOLUTION — step by step

Step 1. s = (5 + 12 + 13)/2 = 15.

Step 2. Heron: A = $\sqrt{(15 \times 10 \times 3 \times 2)} = \sqrt{900} = 30$ cm^2.

Step 3. Faster: $5^2 + 12^2 = 25 + 144 = 169 = 13^2$ -> right triangle! Area = ½ × 5 × 12 = 30 ✓.

» **Why this step?** Check for Pythagorean triples FIRST - a right triangle skips Heron entirely (area = ½ leg × leg).

Answer: 30 cm^2 (both ways)

Hinglish: *5-12-13 triple dikha? Seedha right triangle - ½ × legs. Heron se bhi 30 hi aata, par triple wala rasta 10 second ka hai. Pehle triple check karo!*

QUESTION 19 | Basic | Parallelogram and triangle (T1)

Q19. A parallelogram has base 14 cm and height 6 cm. (i) Find its area. (ii) A triangle stands on the same base, between the same parallels - find its area.

SOLUTION — step by step

Step 1. (i) Area = base × height = 14 × 6 = 84 cm^2.

Step 2. (ii) The triangle has the same base and height: area = ½ × 14 × 6 = 42 cm^2 - exactly half.

» **Why this step?** Parallelogram = base × height; any triangle on the same base and height is half of it.

Answer: (i) 84 cm^2 (ii) 42 cm^2

Hinglish: *Parallelogram = b × h. Same base, same height ka triangle = aadha = 42. 'Same parallels' ka matlab hi yahi hai.*

QUESTION 20 | Basic | Trapezium area (T1)

Q20. A trapezium has parallel sides 15 cm and 9 cm, and the distance between them is 8 cm. Find its area.

SOLUTION — step by step

Step 1. A = ½ × (sum of parallel sides) × height.

Step 2. = ½ × (15 + 9) × 8 = ½ × 24 × 8 = 96 cm^2.

» **Why this step?** A trapezium is the average of its parallel sides stretched along the height.

Answer: 96 cm^2

Hinglish: *½ × (15 + 9) × 8 = 96. Parallel sides ka average × height - bas yehi formula hai.*

QUESTION 21 | Basic | Arc and sector (T7)

Q21. For a circle of radius 12 cm, find (i) the length of an arc subtending 60° at the centre and (ii) the area of that sector. (π = 22/7.)

SOLUTION — step by step

Step 1. (i) Arc = (θ/360) × 2πr = (60/360) × 2 × (22/7) × 12 = (1/6) × (528/7) = 88/7 ≈ 12.57 cm.

Step 2. (ii) Sector = (θ/360) × πr^2 = (1/6) × (22/7) × 144 = 528/7 ≈ 75.43 cm^2.

» **Why this step?** Sector and arc are both fractions of the full circle: θ/360 of circumference or area.

Answer: (i) 88/7 ≈ 12.57 cm (ii) 528/7 ≈ 75.43 cm^2

Hinglish: *60° = chhatha hissa. Poore circle ke 1/6th value dono jagah: arc 88/7, sector 528/7. Fraction nikaalo, multiply karo.*

QUESTION 22 | Basic | Ring area (T9)

Q22. Two concentric circles have radii 14 cm and 7 cm. Find the area of the ring between them.

SOLUTION — step by step

Step 1. Ring = $\pi R^2 - \pi r^2 = \pi(R^2 - r^2) = \pi(R + r)(R - r)$.

Step 2. = (22/7) × (14 + 7)(14 - 7) = (22/7) × 21 × 7.

Step 3. = 22 × 21 = 462 cm^2.

» **Why this step?** π(R + r)(R − r) factors the difference of squares - fewer big multiplications.

Answer: 462 cm^2

Hinglish: *Ring = $\pi(R^2 - r^2) = \pi(R+r)(R-r)$. 21 × 7 × 22/7 = 462. Factored form se calculation halki hoti hai.*

QUESTION 23 | Basic | Square and diagonal (T1)

Q23. The perimeter of a square is 40 cm. Find (i) its side, (ii) its area, (iii) its diagonal.

SOLUTION — step by step

Step 1. (i) Side = 40/4 = 10 cm.

Step 2. (ii) Area = 10^2 = 100 cm^2.

Step 3. (iii) Diagonal = side√2 = 10√2 ≈ 14.14 cm.

» **Why this step?** Perimeter → side → area → diagonal is the square's assembly line.

Answer: (i) 10 cm (ii) 100 cm^2 (iii) 10√2 ≈ 14.14 cm

Hinglish: *Perimeter ÷ 4 = side 10. Area 100. Diagonal 10√2. Square ki assembly line ek hi order mein chalti hai.*

QUESTION 24 | Basic | Isosceles via Heron (T2/T3)

Q24. An isosceles triangle has perimeter 32 cm with equal sides of 11 cm each. Find its area.

SOLUTION — step by step

Step 1. Base = 32 - 22 = 10 cm.

Step 2. Height: drop from apex, splits base into 5 and 5: h = $\sqrt{(11^2 - 5^2)} = \sqrt{(121 - 25)} = \sqrt{96} = 4\sqrt{6}$.

Step 3. Area = $\frac{1}{2} \times 10 \times 4\sqrt{6} = 20\sqrt{6} \approx 48.99$ cm^2.

» **Why this step?** In an isosceles triangle the altitude bisects the base - Pythagoras then gives the height.

Answer: $20\sqrt{6} \approx 48.99$ cm^2

Hinglish: *Base 10 nikla. Altitude base ko 5-5 mein kaat-ti hai (isosceles ka gift). h = √96 = 4√6. Area 20√6 ≈ 49.*

QUESTION 25 | Basic | Wheel revolutions (T13)

Q25. A wheel of diameter 84 cm makes 250 complete revolutions. How far does it travel?

SOLUTION — step by step

Step 1. Circumference = πd = (22/7) × 84 = 264 cm per revolution.

Step 2. Distance = 250 × 264 = 66000 cm = 660 m.

» **Why this step?** One revolution = one circumference of rolling; distance = revolutions × πd.

Answer: 660 m

Hinglish: *Ek chakkar = 264 cm. 250 chakkar = 660 m. Wheel ke questions mein diameter se circumference, phir multiply.*

QUESTION 26 | Basic | Semicircle metrics (T6)

Q26. Find the perimeter and area of a semicircle of radius 7 cm. (π = 22/7.)

SOLUTION — step by step

Step 1. Area = $\frac{1}{2} \times (22/7) \times 49 = 77$ cm^2.

Step 2. Perimeter = half circumference + diameter = (22/7) × 7 + 14 = 22 + 14 = 36 cm.

» **Why this step?** A semicircle's boundary has TWO pieces: the curved arc AND the straight diameter - forgetting the 2r is the classic error.

Answer: Area 77 cm^2; perimeter 36 cm

Hinglish: *Area aadha circle = 77. Par perimeter mein arc + diameter dono: 22 + 14 = 36. Sirf 22 likhna sabse common galti hai - diameter bhool jaate hain!*

QUESTION 27 | Basic | Rhombus from diagonals (T1)

Q27. A rhombus has diagonals 16 cm and 12 cm. Find (i) its area, (ii) the length of each side, (iii) its perimeter.

SOLUTION — step by step

Step 1. (i) Area = $\frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 16 \times 12 = 96$ cm^2.

Step 2. (ii) Diagonals bisect at right angles: half-diagonals 8 and 6 -> side = $\sqrt{(64 + 36)} = 10$ cm.

Step 3. (iii) Perimeter = $4 \times 10 = 40$ cm.

» **Why this step?** The 8-6-10 triple hides inside the rhombus - diagonals' halves are the legs of the side triangle.

Answer: (i) 96 cm^2 (ii) 10 cm (iii) 40 cm

Hinglish: *Area $\frac{1}{2} d_1 d_2 = 96$. Side ke liye aadhe diagonals: 8-6-10 triple! Perimeter 40. Rhombus ke andar Pythagoras chhupa hota hai.*

QUESTION 28 | Basic | Reverse equilateral (T3)

Q28. An equilateral triangle has height $6\sqrt{3}$ cm. Find its side and area.

SOLUTION — step by step

Step 1. Height = $(\sqrt{3}/2)a$ -> $6\sqrt{3} = (\sqrt{3}/2)a$ -> $a = 12$ cm.

Step 2. Area = $(\sqrt{3}/4) \times 144 = 36\sqrt{3} \approx 62.35$ cm^2.

» **Why this step?** Height and side are locked by the fixed ratio $\sqrt{3} : 2$ in every equilateral triangle.

Answer: Side 12 cm; area $36\sqrt{3} \approx 62.35$ cm^2

Hinglish: *Height se side: $6\sqrt{3} \div (\sqrt{3}/2) = 12$. $\sqrt{3}$ cancel ho gaya - clean! Phir area $36\sqrt{3}$. Reverse mein bhi wahi ratio.*

QUESTION 29 | Basic | Sector perimeter (T7)

Q29. Find the perimeter of a sector of radius 14 cm and angle 90°. ($\pi = 22/7$.)

SOLUTION — step by step

Step 1. Arc = $(90/360) \times 2 \times (22/7) \times 14 = (1/4) \times 88 = 22$ cm.

Step 2. Perimeter = arc + two radii = 22 + 14 + 14 = 50 cm.

» **Why this step?** A sector's boundary = arc + BOTH radii - the two radii are the most-forgotten 28 cm in this chapter.

Answer: 50 cm

Hinglish: *Arc 22, par sector ki boundary mein do radius bhi: 22 + 28 = 50. Radii bhoolna sector-perimeter ki national galti hai.*

QUESTION 30 | Basic | Ratio sides with perimeter (T2/T12)

Q30. The sides of a triangle are in the ratio 3 : 4 : 5 and its perimeter is 96 cm. Find the sides and the area.

SOLUTION — step by step

Step 1. Sides: 3x, 4x, 5x -> 12x = 96 -> x = 8. Sides: 24, 32, 40 cm.

Step 2. Check: $24^2 + 32^2 = 576 + 1024 = 1600 = 40^2$ -> right triangle (3-4-5 scaled!).

Step 3. Area = ½ × 24 × 32 = 384 cm^2.

» **Why this step?** Ratio problems: total parts first; then the triple check may skip Heron entirely.

Answer: Sides 24, 32, 40 cm; area 384 cm^2

Hinglish: *12 parts = 96 -> x = 8. Sides 24-32-40 = 3-4-5 × 8 - right triangle! Area seedha ½ × 24 × 32 = 384. Ratio + triple = double gift.*

Level 2 · Core Drills: book-level practice:

QUESTION 31 | Moderate | Heron with larger numbers (T2)

Q31. Find the area of a triangle with sides 40 cm, 30 cm and 14 cm.

SOLUTION — step by step

Step 1. s = (40 + 30 + 14)/2 = 42 cm.

Step 2. Differences: 2, 12, 28.

Step 3. Product: 42 × 2 × 12 × 28 = 28224.

Step 4. A = √28224 = 168 cm^2.

» **Why this step?** Large products factor stepwise: 42×2 = 84, 12×28 = 336, 84×336 = 28224; keep factors paired to make the root visible.

Answer: 168 cm^2

Hinglish: *s = 42, differences 2, 12, 28. Product 28224. Root 168. Bade numbers mein partial products likhte jao - ek hi line mein sab multiply karna errors bulata hai.*

QUESTION 32 | Moderate | Heron classic (T2)

Q32. A triangular park has sides 122 m, 120 m and 22 m. Find its area.

SOLUTION — step by step

Step 1. s = (122 + 120 + 22)/2 = 132 m.

Step 2. Differences: 132-122 = 10, 132-120 = 12, 132-22 = 110.

Step 3. Product: 132 × 10 × 12 × 110 = 1,742,400.

Step 4. A = √1742400 = 1320 m^2.

» **Why this step?** Pair the factors smartly: 132×110 = 14520 and 10×12 = 120; 14520×120 = 1742400. Grouping reveals the square: $(1320)^2$.

Answer: 1320 m^2

Hinglish: *s = 132, differences 10, 12, 110. Product 1742400, root 1320 m^2. Note the beauty: root = s × 10 = 1320 - aise patterns pehchano.*

QUESTION 33 | Moderate | Heron with surd simplification (T2)

Q33. Find the area of a triangle with sides 18 cm, 10 cm and 14 cm. Express the answer as a simplified surd and as a decimal.

SOLUTION — step by step

Step 1. s = 21. Differences: 3, 11, 7.

Step 2. Product: 21 × 3 × 11 × 7 = 4851.

Step 3. Simplify the surd: 4851 = 441 × 11, so A = 21√11 cm^2.

Step 4. Decimal: 21 × 3.3166 ≈ 69.65 cm^2.

» **Why this step?** Extract the largest perfect square under the root; 441 = 21^2 was hiding in the factors.

Answer: 21√11 ≈ 69.65 cm^2

Hinglish: *Product 4851. 4851 ko 441 × 11 likho (21^2 bahar!). Answer 21√11 ≈ 69.65 cm^2. Surd simplify karna FINAL step hai - bina wo aadha marks.*

QUESTION 34 | Moderate | Reverse equilateral (T3)

Q34. The area of an equilateral triangle is $16\sqrt{3}$ cm^2. Find its side and perimeter.

SOLUTION — step by step

Step 1. Set $(\sqrt{3}/4)a^2 = 16\sqrt{3}$.

Step 2. Cancel $\sqrt{3}$: $a^2/4 = 16 \Rightarrow a^2 = 64 \Rightarrow a = 8$ cm.

Step 3. Perimeter = 24 cm.

» **Why this step?** The $\sqrt{3}$ cancels symbolically - never convert to decimals in a reverse equilateral question.

Answer: side = 8 cm, perimeter = 24 cm

Hinglish: *Formula reverse chalao: $(\sqrt{3}/4)a^2 = 16\sqrt{3}$. $\sqrt{3}$ cancel: $a^2 = 64$. Side 8, perimeter 24. Decimals tak mat jao - symbolic cancellation clean hai.*

QUESTION 35 | Moderate | Isosceles by height (T3)

Q35. An isosceles triangle has perimeter 32 cm with equal sides of 11 cm. Find its area, first by the height method and then verify with Heron.

SOLUTION — step by step

Step 1. Base = 32 - 22 = 10 cm.

Step 2. Height method: the height to the base bisects it, so $h = \sqrt{(11^2 - 5^2)} = \sqrt{96} = 4\sqrt{6}$.

Step 3. Area = $\frac{1}{2} \times 10 \times 4\sqrt{6} = 20\sqrt{6} \approx 48.99$ cm^2.

Step 4. Heron check: s = 16; product $16 \times 5 \times 5 \times 6 = 2400$; $\sqrt{2400} = 20\sqrt{6}$ ✓.

» **Why this step?** Two independent methods agreeing is the strongest verification available in this chapter.

Answer: $20\sqrt{6} \approx 49.0$ cm^2

Hinglish: *Height method: base ke midpoint se height - 11^2-$5^2 = 96$, $h = 4\sqrt{6}$. Area $20\sqrt{6}$. Heron se bhi $20\sqrt{6}$ - dono match, pakka sahi.*

QUESTION 36 | Moderate | Quadrilateral by diagonal (T4)

Q36. In quadrilateral ABCD, AB = 24 cm, BC = 10 cm, CD = 25 cm, DA = 17 cm and diagonal AC = 26 cm. Find the area of ABCD.

SOLUTION — step by step

Step 1. Triangle ABC: sides 24, 10, 26. Spot $24^2 + 10^2 = 576 + 100 = 676 = 26^2$: right angle at B.

Step 2. Area(ABC) = $\frac{1}{2} \times 24 \times 10 = 120$ cm^2.

Step 3. Triangle ACD: sides 26, 25, 17. s = 34; differences 8, 9, 17.

Step 4. Product $34 \times 8 \times 9 \times 17 = 41616$; Area(ACD) = $\sqrt{41616} = 204$ cm^2.

Step 5. Total area = $120 + 204 = 324$ cm^2.

» **Why this step?** Splitting along the diagonal reduces one quadrilateral to two triangle problems; Pythagoras spotting saves a full Heron on the first one.

Answer: 324 cm^2

Hinglish: *Diagonal ke saath do triangles. ABC: 24-10-26 right triangle hai! Area ½×24×10 = 120, Heron ki zaroorat nahi. ACD: Heron se 204. Total 324 cm². Pehle Pythagoras check karo - time bachta hai.*

QUESTION 37 | Moderate | Brahmagupta (T5)

Q37. A cyclic quadrilateral has sides 13 cm, 10 cm, 13 cm and 10 cm (in order). Find its area using Brahmagupta's formula.

SOLUTION — step by step

Step 1. Cyclic quadrilateral ⇒ Brahmagupta applies: A = $\sqrt{(s-a)(s-b)(s-c)(s-d)}$.

Step 2. s = $(13 + 10 + 13 + 10)/2 = 23$ cm.

Step 3. Differences: 10, 13, 10, 13.

Step 4. Product: $10 \times 13 \times 10 \times 13 = 16900$. A = $\sqrt{16900} = 130$ cm^2.

» **Why this step?** Heron's pattern extends to four sides; the symmetric sides make every difference reappear - a designed clean answer.

Answer: 130 cm^2

Hinglish: *Cyclic quadrilateral mein Heron ka bhai Brahmagupta: same (s - side) pattern, char baar. s = 23, differences 10,13,10,13. Product 16900, root 130 cm².*

QUESTION 38 | Moderate | Circumference to area (T6)

Q38. The circumference of a circular garden is 44 m. Find (i) its radius, (ii) the cost of fencing it twice at ₹15 per metre.

SOLUTION — step by step

Step 1. r = $C/(2\pi) = 44 \times 7/44 = 7$ m.

Step 2. Double fencing length = $2 \times 44 = 88$ m.

Step 3. Cost = $88 \times 15 =$ ₹1320.

» **Why this step?** 'Fenced twice' doubles the LENGTH, not the radius or area - read the wording carefully.

Answer: r = 7 m; cost ₹1320

Hinglish: *r = 7 m. Double fencing = 88 m, cost 88×15 = ₹1320. 'Twice' ka matlab boundary do baar - area se koi lena-dena nahi.*

QUESTION 39 | Moderate | Wheel revolutions (T13)

Q39. A wheel of diameter 70 cm makes a journey of 1.1 km. How many complete revolutions does it make ($\pi = 22/7$)?

SOLUTION — step by step

Step 1. Circumference = $\pi d = (22/7) \times 70 = 220$ cm.

Step 2. Convert the journey: 1.1 km = 1100 m = 110000 cm.

Step 3. Revolutions = 110000/220 = 500.

» **Why this step?** One revolution moves the wheel exactly one circumference; total distance divided by circumference gives the count.

Answer: 500 revolutions

Hinglish: *Ek chakkar = circumference = 220 cm. Distance 110000 cm. 110000/220 = 500 revolutions. Units same karo pehle - km vs cm sabse common galti.*

QUESTION 40 | Moderate | Sector area and arc (T7)

Q40. A sector of a circle of radius 21 cm has central angle 120°. Find (i) the arc length, (ii) the sector area, (iii) the sector's perimeter.

SOLUTION — step by step

Step 1. Fraction: 120/360 = 1/3.

Step 2. Full circle: $C = 2\pi(21) = 132$ cm; $A = \pi(441) = 1386$ cm^2.

Step 3. Arc = 132/3 = 44 cm.

Step 4. Sector area = 1386/3 = 462 cm^2.

Step 5. Perimeter = arc + 2 radii = 44 + 42 = 86 cm.

» **Why this step?** The sector owns one third of the circumference AND one third of the area; its perimeter needs the two straight edges too.

Answer: arc 44 cm; area 462 cm^2; perimeter 86 cm

Hinglish: *120° = one-third. Arc 44, area 462. Perimeter mein DO radii: 44 + 42 = 86 cm. Arc + ek radius likhna classic mistake hai.*

QUESTION 41 | Moderate | Sector perimeter (T7)

Q41. Find the perimeter of a sector of radius 14 cm and angle 45° ($\pi = 22/7$).

SOLUTION — step by step

Step 1. Fraction: 45/360 = 1/8.

Step 2. Arc = (1/8) × 2πr = (1/8) × 88 = 11 cm.

Step 3. Perimeter = arc + 2r = 11 + 28 = 39 cm.

» **Why this step?** Perimeter of a curved region = every boundary piece: one arc plus two straight radii.

Answer: 39 cm

Hinglish: *45° = one-eighth. Arc 11. Perimeter 11 + 14 + 14 = 39 cm. Dono radii ginn kar.*

QUESTION 42 | Moderate | Ratio scaling (T12)

Q42. The radii of two circles are in the ratio 3 : 4. Find (i) the ratio of their circumferences, (ii) the ratio of their areas. If the smaller circle's area is 154 cm^2, find the larger's.

SOLUTION — step by step

Step 1. Let radii be 3k and 4k.

Step 2. Circumferences: $2\pi(3k) : 2\pi(4k) = 3 : 4$ (linear scaling).

Step 3. Areas: $\pi(3k)^2 : \pi(4k)^2 = 9 : 16$ (square scaling).

Step 4. Larger area = 154 × 16/9 = 2464/9 ≈ 273.78 cm^2.

» **Why this step?** Lengths scale linearly with k; areas carry k^2 - every ratio question here hinges on this difference.

Answer: 3 : 4 and 9 : 16; larger area = 2464/9 ≈ 273.8 cm^2

Hinglish: *Circumference ratio = radius ratio (3:4). Area ratio = SQUARE (9:16). 154 ko 16/9 se scale karo: 273.78 cm^2. 'k² rule' yaad rakho.*

QUESTION 43 | Moderate | Heron moderate (T2)

Q43. Find the area of a triangle with sides 51 cm, 37 cm and 20 cm.

SOLUTION — step by step

Step 1. s = (51 + 37 + 20)/2 = 54 cm.

Step 2. Differences: 3, 17, 34.

Step 3. Product: 54 × 3 × 17 × 34 = 93636.

Step 4. A = √93636 = 306 cm^2.

» **Why this step?** Differences 3 and 34 pair to 102; 17 and 54 pair to 918; regrouping makes the root visible: $(306)^2$.

Answer: 306 cm²

Hinglish: *s = 54, differences 3, 17, 34. Product 93636, root 306 cm². Bade products ko chhote-chhote multiply karke likho.*

QUESTION 44 | Moderate | Wire re-bending (T12)

Q44. A wire of length 132 cm is first bent into an equilateral triangle and then re-bent into a circle. Which shape encloses more area, and by how much ($\sqrt{3} \approx 1.73$, $\pi = 22/7$)?

SOLUTION — step by step

Step 1. Equilateral triangle: side = 132/3 = 44 cm.

Step 2. Area = $(\sqrt{3}/4)(44)^2 = (\sqrt{3}/4)(1936) = 484\sqrt{3} \approx 837.3$ cm^2.

Step 3. Circle: circumference 132 ⇒ r = 132 × 7/44 = 21 cm.

Step 4. Area = $\pi(441) = 1386$ cm^2.

Step 5. Difference = 1386 - 837.3 = 548.7 cm^2. The circle wins.

» **Why this step?** The same perimeter always encloses more area as a circle than as any polygon - this question quantifies it.

Answer: Circle: 1386 vs $484\sqrt{3} \approx 837.3$; the circle encloses about 548.7 cm² more.

Hinglish: *Same wire, do shapes. Triangle: $484\sqrt{3} \approx 837$ cm². Circle: 1386 cm². Circle 549 cm² zyada. Ye universal truth hai - fixed perimeter par circle sabse bada area deta hai.*

QUESTION 45 | Moderate | Heron classic (T2)

Q45. Find the area of a triangle with sides 13 cm, 14 cm and 15 cm.

SOLUTION — step by step

Step 1. s = (13 + 14 + 15)/2 = 21.

Step 2. Differences: 21 - 13 = 8; 21 - 14 = 7; 21 - 15 = 6.

Step 3. A = $\sqrt{(21 \times 8 \times 7 \times 6)} = \sqrt{(21 \times 336)} = \sqrt{7056} = 84$ cm^2.

» **Why this step?** The 13-14-15 triangle is the most famous Heron showcase - its area is a whole number, 84.

Answer: 84 cm²

Hinglish: *s = 21, differences 8-7-6. 21 × 8 × 7 × 6 = 7056, root = 84. Ye triangle yaad kar lo - exam ka permanent guest hai.*

QUESTION 46 | Moderate | Quadrilateral via diagonal (T4)

Q46. In quadrilateral ABCD, diagonal AC = 26 cm. The perpendiculars from B and D to AC measure 11 cm and 9 cm respectively (on opposite sides). Find the area of ABCD.

SOLUTION — step by step

Step 1. Split along AC into two triangles sharing base AC = 26.

Step 2. Area(ABC) = $\frac{1}{2} \times 26 \times 11 = 143$ cm^2.

Step 3. Area(ADC) = $\frac{1}{2} \times 26 \times 9 = 117$ cm^2.

Step 4. Total = 143 + 117 = 260 cm^2.

» **Why this step?** Any quadrilateral = two triangles on one diagonal: $A = \frac{1}{2} \times d \times (h_1 + h_2)$.

Answer: 260 cm^2

Hinglish: *Diagonal ko base banao, dono heights jodo: ½ × 26 × (11 + 9) = 260. Quadrilateral ka universal split formula hai ye.*

QUESTION 47 | Moderate | Brahmagupta (T5)

Q47. A cyclic quadrilateral has sides 12 cm, 16 cm, 20 cm and 24 cm. Find its area using Brahmagupta's formula ($\sqrt{3} \approx 1.73$).

SOLUTION — step by step

Step 1. s = (12 + 16 + 20 + 24)/2 = 36.

Step 2. A = $\sqrt{((36-12)(36-16)(36-20)(36-24))} = \sqrt{(24 \times 20 \times 16 \times 12)}$.

Step 3. 24 × 20 = 480; 16 × 12 = 192; product = 92160 = 9216 × 10.

Step 4. A = $\sqrt{9216} \times \sqrt{10} = 96\sqrt{10} \approx 96 \times 3.16 = 303.4$ cm^2 ($\sqrt{10} \approx 3.162$).

» **Why this step?** Brahmagupta = Heron's cousin for cyclic quadrilaterals: $\sqrt{((s-a)(s-b)(s-c)(s-d))}$ with all four differences.

Answer: $96\sqrt{10} \approx 303.4$ cm^2

Hinglish: *s = 36, chaaron differences 24-20-16-12. 92160 = 9216 × 10 -> $96\sqrt{10} \approx 303.4$. Square factor bahar nikalna hi yahan ka skill hai.*

QUESTION 48 | Moderate | Reverse from area (T6)

Q48. The area of a circle is 616 cm^2. Find (i) its radius and (ii) its circumference. ($\pi = 22/7$.)

SOLUTION — step by step

Step 1. (i) $\pi r^2 = 616 \rightarrow r^2 = 616 \times 7/22 = 196 \rightarrow r = 14$ cm.

Step 2. (ii) $C = 2\pi r = 2 \times (22/7) \times 14 = 88$ cm.

» **Why this step?** Area → radius → circumference: the reverse pipeline through $r^2 = 196$.

Answer: (i) 14 cm (ii) 88 cm

Hinglish: *616 × 7/22 = 196 = 14². r = 14, phir C = 88. Reverse questions mein r² nikaalna hi sab kuch hai.*

QUESTION 49 | Moderate | Outside circular path (T10)

Q49. A circular park of radius 35 m is surrounded by a 7 m wide circular path. Find the area of the path. ($\pi = 22/7$.)

SOLUTION — step by step

Step 1. Outer radius R = 35 + 7 = 42 m; inner r = 35 m.

Step 2. Path area = $\pi(R^2 - r^2) = (22/7)(42^2 - 35^2)$.

Step 3. = (22/7)(1764 - 1225) = (22/7)(539) = 22 × 77 = 1694 m^2.

» **Why this step?** A path outside adds to the radius: R = r + width, then the ring formula.

Answer: 1694 m^2

Hinglish: *R = 35 + 7 = 42. Ring formula: 539 × 22/7 = 1694. Path ke questions mein pehle R-r decide karo, phir ring.*

QUESTION 50 | Moderate | Wire re-bending (T12)

Q50. A wire of length 176 cm is bent first into a circle and then into a square. Which shape encloses more area, and by how much? ($\pi = 22/7$.)

SOLUTION — step by step

Step 1. Circle: $2\pi r = 176 \rightarrow r = 28$. Area = (22/7)(784) = 2464 cm^2.

Step 2. Square: side = 176/4 = 44. Area = $44^2 = 1936$ cm^2.

Step 3. Circle wins by 2464 - 1936 = 528 cm^2.

» **Why this step?** For a FIXED perimeter, the circle always beats the square - this comparison is the chapter's headline theorem.

Answer: Circle encloses more: 2464 vs 1936, by 528 cm^2

Hinglish: *Same wire, do shapes: circle 2464, square 1936. Circle jeeta by 528. Fixed perimeter par circle hamesha maximum area deta hai - golden rule!*

QUESTION 51 | Moderate | Segment area (T8)

Q51. Find the area of the minor segment of a circle of radius 14 cm cut off by a 90° central angle. (π = 22/7.)

SOLUTION — step by step

Step 1. Sector (90°) = (1/4) × (22/7) × 14^2 = (1/4) × 616 = 154 cm^2.

Step 2. Triangle (two radii at 90°): legs 14, 14 -> area = ½ × 14 × 14 = 98 cm^2.

Step 3. Segment = sector - triangle = 154 - 98 = 56 cm^2.

» **Why this step?** Segment = sector − triangle: the chord's triangle is the piece you subtract.

Answer: 56 cm^2

Hinglish: *Sector 154, triangle 98 (right angle tha!). Segment = 56. Sector minus triangle - bas yehi recipe hai.*

QUESTION 52 | Moderate | Inside rectangular path (T10)

Q52. A rectangular hall is 40 m by 25 m. A 2 m wide path runs along the INSIDE of its boundary. Find the area of the path.

SOLUTION — step by step

Step 1. Outer area = 40 × 25 = 1000 m^2.

Step 2. Inner rectangle: 40 - 4 = 36 by 25 - 4 = 21 -> area = 756 m^2 (width shrinks on BOTH sides).

Step 3. Path = 1000 - 756 = 244 m^2.

» **Why this step?** An inside path eats the width from both ends: subtract 2 × width from each dimension.

Answer: 244 m^2

Hinglish: *Inside path = dono taraf se khaati hai: 40-4, 25-4. Bacha hua 36×21. Path = 244. 'Dono side' wala 2× bhoolna hi classic galti hai.*

QUESTION 53 | Moderate | Sector comparison (T7)

Q53. Two sectors have the same radius 12 cm, with angles 60° and 120°. Find both areas and their ratio. ($\pi = 22/7$.)

SOLUTION — step by step

Step 1. Area = $(\theta/360)\pi r^2 = (\theta/360) \times (22/7) \times 144$.

Step 2. 60°: $(1/6) \times 3168/7 = 528/7 \approx 75.43$ cm^2.

Step 3. 120°: $(1/3) \times 3168/7 = 1056/7 \approx 150.86$ cm^2.

Step 4. Ratio = 60 : 120 = 1 : 2 - same radius means angle decides everything.

» **Why this step?** Equal radii make sector areas proportional to their angles - the whole geometry of 'same pizza, different slices'.

Answer: 528/7 ≈ 75.43 and 1056/7 ≈ 150.86 cm^2; ratio 1 : 2

Hinglish: *Same radius -> area sirf angle ke ratio mein: 1 : 2. 528/7 aur 1056/7. Angle double = slice double.*

QUESTION 54 | Moderate | Revolutions to distance (T13)

Q54. A wheel of radius 35 cm makes 1000 revolutions. Find the distance covered in metres. ($\pi = 22/7$.)

SOLUTION — step by step

Step 1. Circumference = $2 \times (22/7) \times 35 = 220$ cm.

Step 2. Distance = $1000 \times 220 = 220000$ cm.

Step 3. = 2200 m = 2.2 km.

» **Why this step?** distance = revolutions × $2\pi r$; the unit conversion (cm → m) is where marks leak.

Answer: 2200 m (2.2 km)

Hinglish: *Ek chakkar 220 cm. 1000 chakkar = 220000 cm = 2.2 km. Units convert karna mat bhoolo - cm se km ka safar.*

QUESTION 55 | Moderate | Heron large (T2)

Q55. Find the area of a triangle whose sides are 26 cm, 28 cm and 30 cm.

SOLUTION — step by step

Step 1. $s = (26 + 28 + 30)/2 = 42$.

Step 2. Differences: 16, 14, 12.

Step 3. $A = \sqrt{(42 \times 16 \times 14 \times 12)} = \sqrt{112896}$.

Step 4. $112896 = 336^2$ -> A = 336 cm^2.

» **Why this step?** Pair smartly: (42 × 12) × (16 × 14) = 504 × 224 - or just trust the perfect square landing at 336.

Answer: 336 cm^2

Hinglish: *s = 42, differences 16-14-12. Product 112896, root 336. Bade numbers mein bhi Heron perfect square hi deta hai (agar sides pyaari hain).*

QUESTION 56 | Moderate | Costs: fence and plough (T13)

Q56. A circular garden has radius 21 m. Fencing costs Rs 12 per metre and ploughing costs Rs 5 per m^2. Find the total cost of fencing and ploughing. (π = 22/7.)

SOLUTION — step by step

Step 1. Fencing length = 2πr = 2 × (22/7) × 21 = 132 m -> cost = 132 × 12 = Rs 1584.

Step 2. Ploughing area = (22/7) × 441 = 1386 m^2 -> cost = 1386 × 5 = Rs 6930.

Step 3. Total = 1584 + 6930 = Rs 8514.

» **Why this step?** Fencing bills the PERIMETER, ploughing bills the AREA - matching the right metric to the right cost is the whole game.

Answer: Rs 8514

Hinglish: *Fence = boundary = 132 m → ₹1584. Plough = area = 1386 m^2 → ₹6930. Total ₹8514. Kaunsa kaam perimeter par, kaunsa area par - ye match karna hi question hai.*

QUESTION 57 | Moderate | Composite figure (T11)

Q57. A rectangle 30 m × 20 m has a semicircle of diameter 20 m attached on one of its 20-m sides (outward). Find (i) the total area and (ii) the total perimeter of the combined figure. (π = 22/7.)

SOLUTION — step by step

Step 1. (i) Rectangle = 600 m^2; semicircle = ½ × (22/7) × 10^2 = 1100/7 ≈ 157.14 m^2. Total ≈ 757.14 m^2.

Step 2. (ii) Boundary = three sides of rectangle (30 + 20 + 30) + semicircular arc (πr = 220/7 ≈ 31.43).

Step 3. Perimeter = 80 + 31.43 ≈ 111.43 m.

» **Why this step?** Composite perimeter counts only the OUTER boundary - the shared 20 m edge is inside now and disappears.

Answer: (i) ≈ 757.14 m^2 (ii) ≈ 111.43 m

Hinglish: *Area jod do (600 + 157.14). Perimeter mein dhyan do: 20 m wali side ab andar hai, count nahi hogi! 80 + 31.43 = 111.43.*

QUESTION 58 | Moderate | Grazing goat (T13)

Q58. A goat is tied to one corner of a rectangular shed 20 m × 15 m with a rope 12 m long. Find the area it can graze. (π = 22/7.)

SOLUTION — step by step

Step 1. The shed blocks 90° at the corner - the goat sweeps 270° of a circle.

Step 2. Grazed area = (270/360) × π × 12^2 = (3/4) × (22/7) × 144.

Step 3. = (3/4) × 3168/7 = 2376/7 ≈ 339.43 m^2.

» **Why this step?** The shed's corner eats a quarter of the circle: the goat owns ¾ of πr^2.

Answer: 2376/7 ≈ 339.43 m^2

Hinglish: *Shed corner par 90° block - goat ko 270° mila = ¾ circle. (3/4) × 3168/7 ≈ 339.43. Corner tether = fraction of circle, yaad rakho.*

QUESTION 59 | Moderate | Ring via chord (T9)

Q59. Two concentric circles have radii R (outer) and r (inner). A chord of the OUTER circle of length 24 cm is tangent to the inner circle. If R = 15 cm, find r and the area of the ring. (π = 22/7.)

SOLUTION — step by step

Step 1. The tangent from the centre meets the chord at its midpoint: half-chord = 12 cm.

Step 2. Right triangle: $R^2 = r^2 + 12^2$ -> 225 = r^2 + 144 -> r = 9 cm.

Step 3. Ring area = π(225 - 81) = π × 144 = (22/7) × 144 = 3168/7 ≈ 452.57 cm^2.

» **Why this step?** A chord tangent to the inner circle builds THE right triangle: $R^2 = r^2 + (\text{half-chord})^2$.

Answer: r = 9 cm; ring ≈ 452.57 cm^2 (3168/7)

Hinglish: *Tangent -> chord ka midpoint -> half-chord 12. $R^2 = r^2 + 144$ -> r = 9. Ring = 144π ≈ 452.57. Ye triangle har chord-ring question ka dil hai.*

QUESTION 60 | Moderate | Running track (T11)

Q60. A running track has two parallel straights of 80 m each, joined by two semicircular ends of diameter 56 m. Find (i) the perimeter of the track and (ii) the area enclosed. (π = 22/7.)

SOLUTION — step by step

Step 1. (i) Perimeter = 80 + 80 + two semicircular arcs (= one full circle of diameter 56): 160 + (22/7) × 56 = 160 + 176 = 336 m.

Step 2. (ii) Area = rectangle (80 × 56) + full circle (π × 28^2) = 4480 + 2464 = 6944 m^2.

» **Why this step?** Two semicircular ends = one full circle for BOTH perimeter and area - the track collapses to rectangle + circle.

Answer: (i) 336 m (ii) 6944 m^2

Hinglish: *Do semicircle = ek poora circle! Perimeter: 160 + 176 = 336. Area: 4480 + 2464 = 6944. Track ka track-record: rectangle + circle combo.*

QUESTION 61 | Moderate | Circle in equilateral (T11)

Q61. An equilateral triangle has side $12\sqrt{3}$ cm. Find (i) the radius of its inscribed circle and (ii) the area of that circle. ($\pi = 22/7$, $\sqrt{3} \approx 1.73$.)

SOLUTION — step by step

Step 1. (i) Inradius of an equilateral triangle = $a/(2\sqrt{3}) = 12\sqrt{3}/(2\sqrt{3}) = 6$ cm.

Step 2. (ii) Circle area = $(22/7) \times 36 = 792/7 \approx 113.14$ cm^2.

» **Why this step?** The formula $r = a/(2\sqrt{3})$ is the equilateral inradius shortcut - $\sqrt{3}$ cancels beautifully here.

Answer: (i) 6 cm (ii) $792/7 \approx 113.14$ cm^2

Hinglish: *$r = a/(2\sqrt{3})$: $12\sqrt{3}/(2\sqrt{3}) = 6$ - $\sqrt{3}$ cancel, clean! Area $792/7 \approx 113.14$. Equilateral ke teen formulas (height, area, inradius) ratt lo.*

QUESTION 62 | Moderate | Square inscribed in circle (T11)

Q62. A square is inscribed in a circle of radius $7\sqrt{2}$ cm. Find (i) the side of the square, (ii) the area of the square, (iii) the area between the circle and the square. ($\pi = 22/7$.)

SOLUTION — step by step

Step 1. (i) The diagonal = diameter = $14\sqrt{2}$ -> side = $14\sqrt{2}/\sqrt{2} = 14$ cm.

Step 2. (ii) Area = $14^2 = 196$ cm^2.

Step 3. (iii) Circle area = $(22/7) \times 98 = 308$ cm^2. Gap = $308 - 196 = 112$ cm^2.

» **Why this step?** Inscribed square's diagonal = the diameter - one statement unlocks side, area, and the leftover corners.

Answer: (i) 14 cm (ii) 196 cm^2 (iii) 112 cm^2

Hinglish: *Diagonal = diameter = $14\sqrt{2}$ -> side 14 ($\sqrt{2}$ cancel!). Circle 308 vs square 196: gap 112. Inscribed figure ka pehla step hamesha diagonal-diameter connection hai.*

QUESTION 63 | Moderate | Rhombus reverse (T1)

Q63. A rhombus has area 120 cm^2 and one diagonal 10 cm. Find (i) the other diagonal, (ii) the side, (iii) the perimeter.

SOLUTION — step by step

Step 1. (i) $\frac{1}{2} \times 10 \times d_2 = 120 \rightarrow d_2 = 24$ cm.

Step 2. (ii) Half-diagonals 5 and 12: side = $\sqrt{(25 + 144)} = \sqrt{169} = 13$ cm (5-12-13!).

Step 3. (iii) Perimeter = 52 cm.

» **Why this step?** Half-diagonals of a rhombus form a right triangle - the area formula first, then Pythagoras on the halves.

Answer: (i) 24 cm (ii) 13 cm (iii) 52 cm

Hinglish: *d_2 = 24. Aadhe diagonals 5 aur 12 -> 5-12-13 triple! Side 13, perimeter 52. Rhombus ke sawaal mein hamesha half-diagonal triangle banao.*

QUESTION 64 | Moderate | Perpendicular diagonals quadrilateral (T4)

Q64. A quadrilateral has perpendicular diagonals of lengths 18 cm and 10 cm. Find its area, and explain why the same formula works for a kite or a rhombus.

SOLUTION — step by step

Step 1. $A = \frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 18 \times 10 = 90$ cm^2.

Step 2. Reason: perpendicular diagonals split ANY such quadrilateral into 4 right triangles; rearranging the halves shows the total is half the product of the diagonals.

Step 3. Kite and rhombus are special cases (perpendicular diagonals guaranteed), so $\frac{1}{2} d_1 d_2$ covers them all.

» **Why this step?** $\frac{1}{2} d_1 d_2$ is not a 'rhombus formula' - it is the formula for ANY quadrilateral with perpendicular diagonals.

Answer: 90 cm^2; the formula holds for any quadrilateral with perpendicular diagonals

Hinglish: *90 cm^2 - aur ye sirf rhombus ka nahi, har perpendicular-diagonals wale quadrilateral ka formula hai! Kite, rhombus sab included. 4 right triangles ka re-arrangement hi proof hai.*

QUESTION 65 | Moderate | Circle in square (T11)

Q65. A circle is inscribed in a square of side 28 cm. Find (i) the circle's area, (ii) the area of the square NOT covered by the circle. ($\pi = 22/7$.)

SOLUTION — step by step

Step 1. (i) Inradius = half the side = 14 -> circle area = (22/7) × 196 = 616 cm^2.

Step 2. (ii) Square = 784; uncovered = 784 - 616 = 168 cm^2 (the four corners).

» **Why this step?** An inscribed circle has diameter = side, so r = side/2 - the corners are the difference.

Answer: (i) 616 cm^2 (ii) 168 cm^2

Hinglish: *r = 14 (side ka aadha). Circle 616, square 784. Corners ka total 168. Inscribed circle ke liye diameter = side - bas yehi link hai.*

QUESTION 66 | Moderate | Rectangle + semicircle area (T11)

Q66. A rectangular sheet 6 cm × 4 cm has a semicircle of diameter 6 cm cut from one 6-cm side (inward). Find the remaining area. ($\pi = 22/7$.)

SOLUTION — step by step

Step 1. Rectangle = 24 cm^2.

Step 2. Semicircle: radius 3 -> area = ½ × (22/7) × 9 = 99/7 ≈ 14.14 cm^2.

Step 3. Remaining = 24 - 99/7 = (168 - 99)/7 = 69/7 ≈ 9.86 cm^2.

» **Why this step?** 'Cut inward' means subtract; the semicircle's diameter sits ON the side so its radius is half of 6.

Answer: 69/7 ≈ 9.86 cm^2

Hinglish: *Cut hua toh minus: 24 - 99/7 = 69/7 ≈ 9.86. Inward cut = subtraction, outward attach = addition - verb se operation pehchano.*

QUESTION 67 | Moderate | Ring from circumferences (T9)

Q67. Two concentric circles have circumferences 44 cm and 110 cm. Find the area of the ring between them. ($\pi = 22/7$.)

SOLUTION — step by step

Step 1. Inner r: 2πr = 44 -> r = 7. Outer R: 2πR = 110 -> R = 17.5.

Step 2. Ring = $\pi(R^2 - r^2)$ = (22/7)(306.25 - 49) = (22/7)(257.25).

Step 3. = 22 × 36.75 = 808.5 cm^2.

» **Why this step?** Circumferences give radii first; then the difference of squares finishes the ring.

Answer: 808.5 cm^2

Hinglish: *C se r: 7 aur 17.5. Ring = 257.25 × 22/7 = 808.5. Decimal radii se ghabrana nahi - formula wahi.*

QUESTION 68 | Moderate | Hectare conversion + fencing (T13)

Q68. A rectangular field measures 450 m by 240 m. (i) Find its area in hectares. (ii) Fencing costs Rs 15 per metre - find the total fencing cost.

SOLUTION — step by step

Step 1. (i) Area = 450 × 240 = 108000 m^2 = 10.8 hectares.

Step 2. (ii) Perimeter = 2(450 + 240) = 1380 m -> cost = 1380 × 15 = Rs 20700.

» **Why this step?** Hectares = m^2 ÷ 10000; fencing = perimeter × rate - two different measurements, never mix them.

Answer: (i) 10.8 hectares (ii) Rs 20700

Hinglish: *108000 m^2 = 10.8 hectare. Fencing 1380 m × ₹15 = ₹20700. Area ka kaam hectare mein, fence ka kaam metre mein.*

QUESTION 69 | Moderate | Ratio of circles (T12)

Q69. The areas of two circles are in the ratio 3 : 4. Find the ratio of (i) their radii and (ii) their circumferences.

SOLUTION — step by step

Step 1. (i) Area $\propto r^2$: r_1^2/r_2^2 = 3/4 -> r_1/r_2 = √3/2, i.e. √3 : 2.

Step 2. (ii) Circumference $\propto$ r -> same ratio: √3 : 2.

» **Why this step?** Areas scale as the SQUARE of the radius; every linear measure (radius, circumference) scales as the radius itself.

Answer: (i) √3 : 2 (ii) √3 : 2

Hinglish: *Area ratio 3:4 -> radius ratio √3:2 (square root lo!). Circumference toh radius jaisa hi: √3:2. Square wala scaling yaad rakho.*

QUESTION 70 | Moderate | Minute hand sweep (T13)

Q70. The minute hand of a clock is 10 cm long. Find (i) the area swept in 30 minutes and (ii) the distance travelled by the tip in 30 minutes. (π = 22/7.)

SOLUTION — step by step

Step 1. 30 minutes = 180° = half a revolution.

Step 2. (i) Area = ½ × πr^2 = ½ × (22/7) × 100 = 1100/7 ≈ 157.14 cm^2.

Step 3. (ii) Arc = ½ × 2πr = πr = 220/7 ≈ 31.43 cm.

» **Why this step?** The hand sweeps a sector: area for the blade, arc for the tip - two different questions with the same fraction.

Answer: (i) 1100/7 ≈ 157.14 cm^2 (ii) 220/7 ≈ 31.43 cm

Hinglish: *30 min = 180° = aadha circle. Area 1100/7, tip ka safar 220/7. Hand ka question = sector ka question, fraction = time/60.*

QUESTION 71 | Moderate | Regular hexagon (T14)

Q71. Find the perimeter and area of a regular hexagon of side 6 cm (√3 ≈ 1.73).

SOLUTION — step by step

Step 1. Perimeter = 6 × 6 = 36 cm.

Step 2. Area = $(3\sqrt{3}/2)a^2$ = (3 × 1.73/2) × 36 = 2.595 × 36.

Step 3. ≈ 93.42 cm^2 (exactly 54√3).

» **Why this step?** A regular hexagon is 6 equilateral triangles: area = 6 × $(\sqrt{3}/4)a^2$ = $(3\sqrt{3}/2)a^2$.

Answer: Perimeter 36 cm; area 54√3 ≈ 93.42 cm^2

Hinglish: *Hexagon = 6 equilateral triangles ka bhai. Area $(3\sqrt{3}/2)a^2$ = 54√3 ≈ 93.42. Perimeter 36. Formula 6 triangles se hi nikalta hai.*

QUESTION 72 | Moderate | Isosceles right triangle (T3)

Q72. An isosceles right triangle has hypotenuse 20 cm. Find (i) each leg and (ii) its area.

SOLUTION — step by step

Step 1. Legs equal (say a): $a^2 + a^2 = 20^2$ -> $2a^2 = 400$ -> $a^2 = 200$ -> a = 10√2 cm.

Step 2. (ii) Area = ½ × a × a = ½ × 200 = 100 cm^2.

» **Why this step?** In a right isosceles triangle, the area is HALF the square of the hypotenuse (½ × ½ × hyp^2 = $hyp^2/4$... here 400/4 = 100).

Answer: (i) 10√2 cm each (ii) 100 cm^2

Hinglish: *Legs barabar: $2a^2 = 400$ -> a = 10√2. Area ½a^2 = 100. Hypotenuse ka square ÷ 4 = area - isosceles right triangle ka instant shortcut!*

QUESTION 73 | Moderate | Circumference equals perimeter (T6)

Q73. A square has side 11 cm and a circle has circumference equal to the square's perimeter. Compare their areas. ($\pi = 22/7$.)

SOLUTION — step by step

Step 1. Square perimeter = 44 cm -> circle circumference = 44 -> r = 7 cm.

Step 2. Square area = 121 cm^2; circle area = (22/7) × 49 = 154 cm^2.

Step 3. The circle wins by 33 cm^2 - same boundary length, more enclosed area.

» **Why this step?** Equal perimeters: the circle always encloses the larger area - the isoperimetric idea in numbers.

Answer: Circle area 154 vs square 121: the circle is larger by 33 cm^2

Hinglish: *Same boundary (44): circle 154 vs square 121. Circle jeeta by 33! Fixed perimeter par circle ka area sabse zyada - chapter ka golden rule phir se.*

QUESTION 74 | Moderate | Diagonal split verification (T4)

Q74. A quadrilateral ABCD has diagonal AC = 15 cm. The perpendicular distances of B and D from AC are 6 cm and 8 cm (opposite sides). Find the area of ABCD and verify it equals ½ × 15 × (6 + 8).

SOLUTION — step by step

Step 1. Area(ABC) = ½ × 15 × 6 = 45 cm^2.

Step 2. Area(ACD) = ½ × 15 × 8 = 60 cm^2.

Step 3. Total = 105 cm^2 = ½ × 15 × 14 - verified: the two half-products merge into one formula.

» **Why this step?** ½ d h_1 + ½ d h_2 = ½ d(h_1 + h_2) - distributive law turns two triangle areas into one quadrilateral formula.

Answer: 105 cm^2; verified via ½ × 15 × (6 + 8)

Hinglish: *45 + 60 = 105 = ½ × 15 × 14. Algebra ne dono triangles ko merge kar diya. Distributive law hi quadrilateral formula ki jadi hai.*

QUESTION 75 | Moderate | Same perimeter contest (T12/T14)

Q75. A wire 88 cm long is bent into (i) a square, (ii) a circle, (iii) a rectangle with length 26 cm. Compare the three areas and state the general law. ($\pi = 22/7$.)

SOLUTION — step by step

Step 1. (i) Square: side = 22 -> area = 484 cm^2.

Step 2. (ii) Circle: 2πr = 88 -> r = 14 -> area = (22/7)(196) = 616 cm^2.

Step 3. (iii) Rectangle: 26 + w = 44 -> w = 18 -> area = 26 × 18 = 468 cm^2.

Step 4. Ranking: circle (616) > square (484) > rectangle (468).

Step 5. General law: among ALL shapes with the same perimeter, the circle encloses the maximum area.

» **Why this step?** Same-perimeter contests always end: circle first, square second, elongated shapes last - the more stretched, the smaller the area.

Answer: Circle 616 > square 484 > rectangle 468 cm^2; the circle maximises area for a fixed perimeter

Hinglish: *Ek hi wire, teen shapes: 616 > 484 > 468. Circle hamesha champion! Jitna zyada 'khinchega' shape, utna kam area. Isoperimetric law - chapter ka boss theorem.*

Level 3 · Exam Power Zone: long and twisted:

QUESTION 76 | Difficult | Heron large (T2)

Q76. Find the area of a triangle whose sides are 51 cm, 37 cm and 20 cm... verify your answer by checking whether the triangle is right-angled or not.

SOLUTION — step by step

Step 1. Q25 computed this area: s = 54, differences 3, 17, 34, A = 306 cm^2.

Step 2. Pythagoras check on the largest side: $51^2 = 2601$; $37^2 + 20^2 = 1369 + 400 = 1769$.

Step 3. $2601 \neq 1769$, so the triangle is NOT right-angled - Heron was genuinely needed.

Step 4. No shortcut formula applies; the answer stands at 306 cm^2.

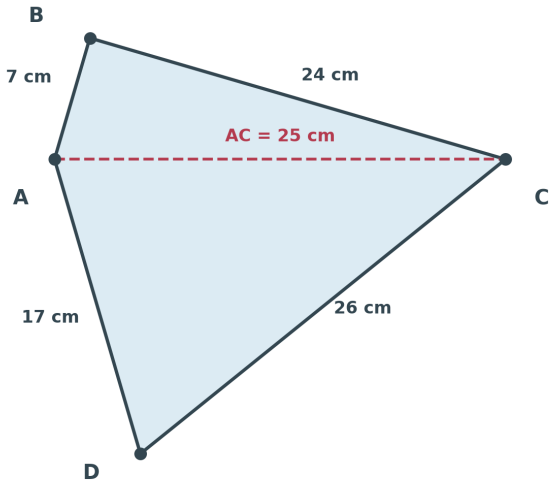
» **Why this step?** The Pythagoras test tells you whether the ½-leg-leg shortcut is legal; here it is not, and saying so is part of the verification.

Answer: 306 cm^2; the triangle is not right-angled ($51^2 \neq 37^2 + 20^2$).

Hinglish: *Same numbers as Q25 - par ab CHECK karna hai. $51^2 = 2601$ vs $37^2+20^2 = 1769$. Barabar nahi, toh right angle nahi - Heron compulsory tha. Verification habit banao.*

QUESTION 77 | Difficult | Quadrilateral via diagonal (T4)

Q77. In quadrilateral ABCD, AB = 7 cm, BC = 24 cm, CD = 26 cm, DA = 17 cm and AC = 25 cm. Find the area of ABCD.



SOLUTION — step by step

Step 1. Triangle ABC: sides 7, 24, 25. Check $7^2 + 24^2 = 49 + 576 = 625 = 25^2$: right angle at B.

Step 2. Area(ABC) = ½ × 7 × 24 = 84 cm^2.

Step 3. Triangle ACD: sides 25, 26, 17. s = 34; differences 9, 8, 17.

Step 4. Product: 34 × 9 × 8 × 17 = 41616; Area(ACD) = 204 cm^2.

Step 5. Total = 84 + 204 = 288 cm^2.

» **Why this step?** One diagonal creates two independent triangle problems; a hidden right triangle in one of them halves the work.

Answer: 288 cm^2

Hinglish: *ABC: 7-24-25 triple! ½×7×24 = 84. ACD: Heron se 204. Total 288 cm². Diagonal questions mein pehle Pythagoras triple hunt karo.*

QUESTION 78 | Difficult | Brahmagupta with surd (T5)

Q78. A cyclic quadrilateral has sides 9 cm, 12 cm, 15 cm and 12 cm. Find its area, expressing the answer as a simplified surd.

SOLUTION — step by step

Step 1. s = (9 + 12 + 15 + 12)/2 = 24 cm.

Step 2. Differences: 15, 12, 9, 12.

Step 3. Product: 15 × 12 × 9 × 12 = 19440.

Step 4. Simplify: 19440 = 1296 × 15, so A = 36√15 cm².

Step 5. Decimal: 36 × 3.873 ≈ 139.4 cm².

» **Why this step?** Brahmagupta's product often carries a large square factor (here 36^2) - extract it before decimalising.

Answer: 36√15 ≈ 139.4 cm²

Hinglish: *s = 24, differences 15, 12, 9, 12. Product 19440 = 1296×15. Root: 36√15 ≈ 139.4 cm². Brahmagupta mein bhi surd simplify hota hai.*

QUESTION 79 | Difficult | Area to circumference (T6)

Q79. The area of a circular ring-shaped garden bed is 616 m² and it is actually a full circle. Find its radius and circumference (π = 22/7).

SOLUTION — step by step

Step 1. $\pi r^2 = 616 \Rightarrow r^2 = 616 \times 7/22 = 196 \Rightarrow r = 14$ m.

Step 2. $C = 2\pi r = 2 \times (22/7) \times 14 = 88$ m.

» **Why this step?** Area → square root → radius → circumference: the standard reverse chain; keep π as 22/7 to stay exact.

Answer: r = 14 m, C = 88 m

Hinglish: *Area se radius: $r^2 = 616 \times 7/22 = 196$. r = 14. C = 88 m. Reverse chain mein π ko decimal mat karo - 22/7 se sab clean.*

QUESTION 80 | Difficult | Sector reverse (T7)

Q80. The area of a sector of a circle with central angle 60° is 231 cm². Find the radius of the circle and the arc length of the sector (π = 22/7).

SOLUTION — step by step

Step 1. Sector area = $(\theta/360)\pi r^2 = (1/6)\pi r^2 = 231$.

Step 2. $\pi r^2 = 1386 \Rightarrow r^2 = 1386 \times 7/22 = 441 \Rightarrow r = 21$ cm.

Step 3. Arc = $(1/6) \times 2\pi r = (1/6)(132) = 22$ cm.

» **Why this step?** Work the sector formula backwards: undo the fraction first, then π, then the square.

Answer: r = 21 cm, arc = 22 cm

Hinglish: *60° = one-sixth. $(1/6)\pi r^2 = 231 \rightarrow \pi r^2 = 1386 \rightarrow r^2 = 441$. r = 21. Arc 22 cm. Reverse sector questions mein fraction pehle undo karo.*

QUESTION 81 | Difficult | Segment area (T8)

Q81. Find the area of the minor segment of a circle of radius 14 cm cut off by a chord subtending 90° at the centre ($\pi = 22/7$).

SOLUTION — step by step

Step 1. Segment = sector - triangle.

Step 2. Sector (90°) = (1/4)π(196) = 154 cm^2.

Step 3. The chord and two radii form a right isosceles triangle with legs 14: area = ½ × 14 × 14 = 98 cm^2.

Step 4. Segment = 154 - 98 = 56 cm^2.

» **Why this step?** Every segment problem is 'pizza minus triangle'; the triangle's area uses the two radii as legs when the angle is 90°.

Answer: 56 cm^2

Hinglish: *Segment = sector - triangle. Sector 154, triangle ½×14×14 = 98. Answer 56 cm^2. 90° wale case mein triangle right-angled hota hai - radii legs.*

QUESTION 82 | Difficult | Ring from inner circumference (T9)

Q82. A circular ring has width 7 cm. The circumference of its inner circle is 44 cm. Find the area of the ring ($\pi = 22/7$).

SOLUTION — step by step

Step 1. Inner radius: 44 = 2πr ⇒ r = 7 cm.

Step 2. Width 7 ⇒ outer radius R = 7 + 7 = 14 cm.

Step 3. Ring area = $\pi(R^2 - r^2)$ = (22/7)(196 - 49) = (22/7)(147) = 462 cm^2.

» **Why this step?** The width adds to the radius (not the diameter); $R^2-r^2 = (R+r)(R-r) = 21\times7$ makes the arithmetic one line.

Answer: 462 cm^2

Hinglish: *Width radius mein JUDTI hai: r = 7, R = 14. Ring = π(147) = 462 cm^2. Width → diameter confusion sabse common galti hai.*

QUESTION 83 | Difficult | Inside rectangular path (T10)

Q83. A rectangular garden is 30 m by 20 m. A path 2 m wide runs inside it along all four sides. Find the area of the path.

SOLUTION — step by step

Step 1. Outer area = 30 × 20 = 600 m^2.

Step 2. The inner rectangle loses 2 m on EACH side: inner length = 30 - 4 = 26 m; inner width = 20 - 4 = 16 m.

Step 3. Inner area = 26 × 16 = 416 m^2.

Step 4. Path area = 600 - 416 = 184 m^2.

» **Why this step?** An inside border subtracts the width TWICE per dimension - once for each of the two parallel sides.

Answer: 184 m^2

Hinglish: *Path andar hai: dono dimensions se 2w = 4 minus. Inner 26×16 = 416. Path = 600-416 = 184 m^2. '2w' rule yaad rakho.*

QUESTION 84 | Difficult | Outside circular path (T10)

Q84. A circular park of radius 17 m is surrounded by a path 4 m wide running outside it. Find the area of the path ($\pi = 22/7$).

SOLUTION — step by step

Step 1. Outer radius R = 17 + 4 = 21 m.

Step 2. Path area = $\pi(R^2 - r^2)$ = (22/7)(441 - 289) = (22/7)(152).

Step 3. = 3344/7 ≈ 477.7 m^2.

» **Why this step?** An outside path adds to the radius once; the ring formula $\pi w(2r + w)$ is the fast version: (22/7)(4)(38).

Answer: $152\pi \approx 477.7$ m^2

Hinglish: *Bahar wala path: R = 21. $\pi(152) \approx 477.7$ m^2. Shortcut: $\pi w(2r+w) = (22/7)\times4\times38$ - same answer, direct.*

QUESTION 85 | Difficult | Area-sum circle (T6)

Q85. Two circles have radii 5 cm and 12 cm. Find the radius of a third circle whose area equals the SUM of their areas.

SOLUTION — step by step

Step 1. $\pi R^2 = \pi(25) + \pi(144) = \pi(169)$.

Step 2. $R^2 = 169 \Rightarrow R = 13$ cm.

» **Why this step?** π cancels; the question is secretly 25 + 144 = 169 - the 5-12-13 triple wearing a circle costume.

Answer: 13 cm

Hinglish: *Areas add: $\pi R^2 = \pi(25+144) = \pi(169)$. R = 13. Ye 5-12-13 triple hai - areas jodne par hypotenuse ban jata hai!*

QUESTION 86 | Difficult | Composite figure (T11)

Q86. A rectangle 20 m × 16 m has a semicircular grass bed built on one of its 16 m sides (outside the rectangle). Find (i) the total area and (ii) the perimeter of the combined figure ($\pi = 22/7$).

SOLUTION — step by step

Step 1. Semicircle diameter = 16 ⇒ radius = 8 m.

Step 2. Semicircle area = $\frac{1}{2} \times \pi \times 64 = 32\pi = 704/7 \approx 100.57$ m^2.

Step 3. Total area = 320 + 100.57 = 420.57 m^2.

Step 4. Perimeter: the 16 m side is now internal - it disappears from the boundary. Boundary = 20 + 16 + 20 + semicircular arc.

Step 5. Arc = $\pi \times 8 = 176/7 \approx 25.14$ m. Total perimeter ≈ 76 + 25.14 = 101.14 m.

» **Why this step?** When shapes join, the shared edge vanishes from the perimeter but stays inside the area sum.

Answer: (i) ≈ 420.6 m^2 (ii) ≈ 101.1 m

Hinglish: *Rectangle + semicircle: area jodo (320 + 100.57). Perimeter mein shared 16 m side GAYAB ho jati hai - uski jagah arc aata hai: 76 + 25.14 = 101.14 m. Composite figures ka pehla rule: shared edge boundary se hatao.*

QUESTION 87 | Difficult | Segment with 120° (T8)

Q87. Find the area of the minor segment of a circle of radius 21 cm whose chord subtends 120° at the centre (use $\sqrt{3} \approx 1.73$, $\pi = 22/7$).

SOLUTION — step by step

Step 1. Sector (120°) = (1/3)π(441) = 462 cm^2.

Step 2. The triangle formed by the two radii and the chord is isosceles with included angle 120°.

Step 3. Its area = $\frac{1}{2} \cdot r^2 \cdot \sin 120°$; equivalently $\frac{1}{2} \cdot r^2 \cdot \sin 60°$ per half. Directly: area = $\frac{1}{2}(21)(21) \cdot \sin 120° = \frac{1}{2}(441)(\sqrt{3}/2) = 441\sqrt{3}/4 = 110.25\sqrt{3}$.

Step 4. Numerically: $110.25 \times 1.73 \approx 190.73$ cm^2.

Step 5. Segment = 462 - 190.73 ≈ 271.3 cm^2.

» **Why this step?** For non-90° sectors the 'triangle' needs the sine form $\frac{1}{2}r^2\sin\theta$ (here $\theta = 120°$) - still just ½ × base × height in disguise.

Answer: ≈ 271.3 cm^2 (462 - 110.25√3)

Hinglish: *120° wale segment mein triangle formula: $\frac{1}{2}r^2 \sin 120° = (441\sqrt{3})/4 \approx 190.7$. Sector 462. Segment ≈ 271.3 cm^2. sin 120° = sin 60° = √3/2 - ye value yaad rakho.*

QUESTION 88 | Difficult | Minute hand sweep (T13)

Q88. The minute hand of a clock is 14 cm long. Find the area swept by it in 40 minutes ($\pi = 22/7$).

SOLUTION — step by step

Step 1. In 60 minutes the hand sweeps 360°; in 40 minutes it sweeps 240°.

Step 2. Fraction: 240/360 = 2/3.

Step 3. Full circle area = π(196) = 616 cm^2.

Step 4. Swept area = (2/3) × 616 = 1232/3 ≈ 410.67 cm^2.

» **Why this step?** The hand is a rotating radius: the swept region is a sector whose angle is proportional to time.

Answer: 1232/3 ≈ 410.7 cm^2

Hinglish: *40 min = 240° = two-thirds. 616 × 2/3 = 410.67 cm^2. Time ko angle mein convert karo, phir sector.*

QUESTION 89 | Difficult | Circle in square (T11)

Q89. A circle is inscribed in a square of side 14 cm. Find the area of the region inside the square but outside the circle ($\pi = 22/7$).

SOLUTION — step by step

Step 1. Inscribed circle: diameter = side ⇒ r = 7 cm.

Step 2. Circle area = $\pi(49) = 154$ cm^2.

Step 3. Square area = 196 cm^2.

Step 4. Gap = 196 - 154 = 42 cm^2.

» **Why this step?** Inscribed means the circle touches all four sides, so the diameter equals the side - the only geometric input needed.

Answer: 42 cm^2

Hinglish: *Inscribed circle: diameter = side. r = 7. Gap = 196 - 154 = 42 cm². Four corner pieces ka total.*

QUESTION 90 | Difficult | Sector from perimeter (T7)

Q90. The perimeter of a sector of a circle is 68 cm and its radius is 21 cm. Find the area of the sector.

SOLUTION — step by step

Step 1. Perimeter = arc + 2r ⇒ arc = 68 - 42 = 26 cm.

Step 2. Sector area = ½ × arc × r = ½ × 26 × 21 = 273 cm^2.

» **Why this step?** The A = ½l·r form avoids computing the angle entirely - fastest route when perimeter is given.

Answer: 273 cm^2

Hinglish: *Perimeter se arc nikaalo: 68 - 42 = 26. Area = ½ × l × r = 273 cm². Angle ki zaroorat hi nahi - ye half-arc-radius shortcut strong hai.*

QUESTION 91 | Difficult | Ratio sides with perimeter (T2)

Q91. The sides of a triangular plot are in the ratio 13 : 14 : 15 and its perimeter is 84 m. Find the area.

SOLUTION — step by step

Step 1. Let sides be 13k, 14k, 15k: 42k = 84 ⇒ k = 2.

Step 2. Sides: 26 m, 28 m, 30 m.

Step 3. s = 42; differences 16, 14, 12.

Step 4. Product: 42 × 16 × 14 × 12 = 112896. A = √112896 = 336 m^2.

» **Why this step?** Convert ratios to actual sides FIRST; Heron only works on real lengths.

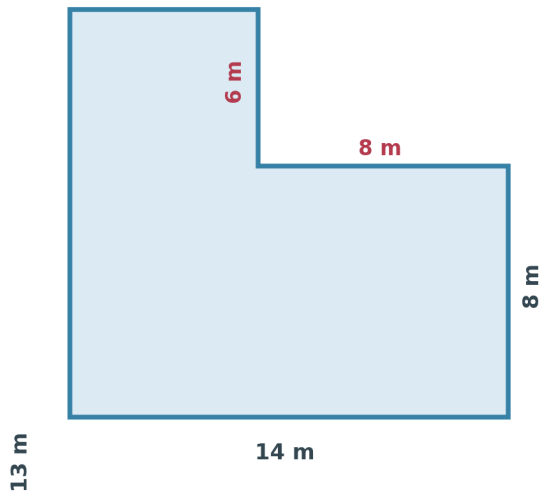
Answer: 336 m^2

Hinglish: *Ratio + perimeter → pehle actual sides: 26, 28, 30. Heron: 336 m^2. Ratio mein Heron lagana mana hai.*

QUESTION 92 | Difficult | L-shaped composite figure (T11/T14)

Q92. The figure shows an L-shaped lawn (all angles right angles). Using the dimensions in the figure: (i) find the area of the lawn by splitting it TWO different ways, (ii) find its perimeter, (iii) grass carpet costs Rs 40 per m^2 - find the carpeting cost.

L-shaped lawn (all angles are right angles)



SOLUTION — step by step

Step 1. (i) Method 1 (vertical cut): bottom strip 14 × 8 = 112 plus upper-left block 6 × 5 = 30 -> total 142 m^2.

Step 2. Method 2 (subtraction): full 14 × 13 = 182 minus the missing top-right 8 × 5 = 40 -> 182 - 40 = 142 ✓ - both methods agree.

Step 3. (ii) Perimeter = 14 + 8 + 8 + 5 + 6 + 13 = 54 m (walk the six sides from the figure).

Step 4. (iii) Cost = 142 × 40 = Rs 5680.

» **Why this step?** Composite L-figures give the same area by addition OR subtraction - doing both is a free answer-check; the perimeter walks the actual boundary.

Answer: (i) 142 m^2 (both methods) (ii) 54 m (iii) Rs 5680

Hinglish: *Do tarike, ek answer: 112 + 30 = 142 aur 182 - 40 = 142! Double-check free mein. Perimeter = chhe sides ka walk: 54. Carpeting 142 × 40 = ₹5680.*

Level 4 · Boss Zone: advanced + HOTS (Olympiad-lite):

QUESTION 93 | High | Pythagorean ratio (T2/T3)

Q93. The sides of a triangle are in the ratio 3 : 4 : 5 and its perimeter is 72 cm. Find the area by TWO methods and confirm they agree.

SOLUTION — step by step

Step 1. Sides: 18k... let 3k + 4k + 5k = 72 ⇒ k = 6: sides 18, 24, 30 cm.

Step 2. Method 1 (right triangle): $18^2 + 24^2 = 324 + 576 = 900 = 30^2$, so right angle between the legs 18 and 24.

Step 3. Area = ½ × 18 × 24 = 216 cm^2.

Step 4. Method 2 (Heron): s = 36; differences 18, 12, 6; product 36×18×12×6 = 46656; √46656 = 216 ✓.

» **Why this step?** A 3-4-5 ratio triangle is always right-angled (the ratio is scale-invariant), so both methods must match.

Answer: 216 cm^2 (both methods)

Hinglish: *3:4:5 hamesha right triangle hota hai. Sides 18, 24, 30. ½×18×24 = 216. Heron bhi 216 deta hai - dono match. Ratio dekh ke triple pehchano.*

QUESTION 94 | High | Reverse equilateral with height (T3)

Q94. The area of an equilateral triangle is 81√3 cm^2. Find its side, its height, and its perimeter.

SOLUTION — step by step

Step 1. $(\sqrt{3}/4)a^2 = 81\sqrt{3} \Rightarrow a^2 = 324 \Rightarrow a = 18$ cm.

Step 2. Height = $(\sqrt{3}/2)a = 9\sqrt{3} \approx 15.59$ cm.

Step 3. Perimeter = 54 cm.

» **Why this step?** All three quantities come from a; find it first by cancelling √3 exactly.

Answer: side 18 cm, height 9√3 cm, perimeter 54 cm

Hinglish: *$a^2 = 324 \rightarrow a = 18$. Height 9√3, perimeter 54. Equilateral ke teeno answers ek hi variable se nikalte hain.*

QUESTION 95 | High | Isosceles reverse (T3)

Q95. An isosceles triangle has area 48 cm^2 and base 16 cm. Find the equal sides and the perimeter.

SOLUTION — step by step

Step 1. $\frac{1}{2} \times 16 \times h = 48 \Rightarrow h = 6$ cm.

Step 2. The height bisects the base: half-base = 8 cm.

Step 3. Equal side = $\sqrt{(8^2 + 6^2)} = \sqrt{100} = 10$ cm.

Step 4. Perimeter = 10 + 10 + 16 = 36 cm.

» **Why this step?** Recover the height from the area, then the side from the right triangle (half-base, height, side).

Answer: equal sides 10 cm each, perimeter 36 cm

Hinglish: *Area se height: 48 = ½×16×h → h = 6. Half-base 8. Side √(64+36) = 10. Perimeter 36. 6-8-10 triple hidden inside!*

QUESTION 96 | High | Trapezium via Heron (T4)

Q96. A field is in the shape of a trapezium whose parallel sides are 77 m and 60 m, and the non-parallel sides are 25 m and 26 m. Find its area.

SOLUTION — step by step

Step 1. Draw the height through the shorter parallel side's endpoint; it creates a helper triangle whose base is 77 - 60 = 17 m (after transferring the full 60 to the other side).

Step 2. Helper triangle sides: 25, 26, 17. s = 34; differences 9, 8, 17.

Step 3. Area of helper triangle = $\sqrt{(34 \times 9 \times 8 \times 17)} = \sqrt{41616} = 204$ m^2.

Step 4. Height: $204 = \frac{1}{2} \times 17 \times h \Rightarrow h = 24$ m.

Step 5. Trapezium area = $\frac{1}{2}(77 + 60)(24) = \frac{1}{2} \times 137 \times 24 = 1644$ m^2.

» **Why this step?** A general trapezium hides a triangle built from the slanted sides and the difference of the parallel sides; Heron extracts the height.

Answer: 1644 m^2

Hinglish: *Trapezium ka height chhupa hai helper triangle mein: base = 77-60 = 17, sides 25, 26. Heron se area 204. h = 2×204/17 = 24. Phir ½(a+b)h = 1644 m^2. Ye 4-step method ratta maar lo.*

QUESTION 97 | High | Rhombus from perimeter (T1)

Q97. The perimeter of a rhombus is 40 cm and one diagonal is 12 cm. Find the other diagonal and the area.

SOLUTION — step by step

Step 1. Side = 40/4 = 10 cm.

Step 2. Diagonals bisect at right angles: half of the given diagonal = 6 cm.

Step 3. Half of the other diagonal = $\sqrt{(10^2 - 6^2)} = \sqrt{64} = 8$ cm, so $d_2 = 16$ cm.

Step 4. Area = $\frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 12 \times 16 = 96$ cm^2.

» **Why this step?** The half-diagonals and a side form a right triangle - the whole rhombus question lives inside it.

Answer: $d_2 = 16$ cm, area = 96 cm^2

Hinglish: *Side 10. Half-diagonal 6. Other half $\sqrt{(100-36)} = 8 \rightarrow d_2 = 16$. Area $\frac{1}{2} \times 12 \times 16 = 96$ cm^2. Rhombus = 4 right triangles.*

QUESTION 98 | High | Circle in equilateral triangle (T3/T11)

Q98. Find the area of the circle inscribed in an equilateral triangle of side 42 cm ($\pi = 22/7$).

SOLUTION — step by step

Step 1. Inradius of an equilateral triangle: $r = a/(2\sqrt{3}) = 42/(2\sqrt{3}) = 21/\sqrt{3} = 7\sqrt{3}$ cm.

Step 2. $r^2 = 147$ cm^2.

Step 3. Circle area = $\pi r^2 = (22/7)(147) = 462$ cm^2.

» **Why this step?** The inradius formula $r = a/(2\sqrt{3})$ (or equivalently r = height/3) comes from the height's trisection by the incentre.

Answer: 462 cm^2

Hinglish: *$r = a/(2\sqrt{3}) = 7\sqrt{3}$. $r^2 = 147$. Area 462 cm^2. Equilateral mein inradius = height/3 hota hai - height $21\sqrt{3}$, uska 1/3 = $7\sqrt{3}$. Same baat.*

QUESTION 99 | High | Circle in square gap (T11)

Q99. A circle is inscribed in a square of side 28 cm. Find the total area of the four corner regions ($\pi = 22/7$).

SOLUTION — step by step

Step 1. r = 14 cm; circle area = $\pi(196) = 616$ cm^2.

Step 2. Square area = 784 cm^2.

Step 3. Corners total = 784 - 616 = 168 cm^2.

Step 4. Bonus: each corner alone is 42 cm^2 - the four are congruent.

» **Why this step?** Subtraction gives the total; symmetry then divides it into four equal corners.

Answer: 168 cm^2 (42 cm^2 per corner)

Hinglish: *Square 784, circle 616. Gap 168 cm^2 - chaar corners, har ek 42. Symmetry se divide karo.*

QUESTION 100 | High | Circumference sum (T6)

Q100. Two circles have circumferences 88 cm and 132 cm. A third circle has circumference equal to the SUM of theirs. Find the radius and area of the third circle ($\pi = 22/7$).

SOLUTION — step by step

Step 1. Third circumference = 88 + 132 = 220 cm.

Step 2. $r = C/(2\pi) = 220 \times 7/44 = 35$ cm.

Step 3. Area = $\pi(35^2) = (22/7)(1225) = 3850$ cm^2.

Step 4. Note: radii add, NOT areas - 14 + 21 = 35 is the additive quantity here.

» **Why this step?** Circumference is linear in r, so circumferences add exactly like radii - the reverse of the area-sum situation (Q36).

Answer: r = 35 cm, area = 3850 cm^2

Hinglish: *Circumferences JUDTE hain: 220 → r = 35, area 3850 cm^2. Q36 mein areas judte the; yahan circumferences. Linear quantity add hota hai, quadratic nahi - ye contrast yaad rakho.*

QUESTION 101 | High | Ring via chord (T9)

Q101. In two concentric circles, a chord of the OUTER circle of length 28 cm touches the inner circle. Find the area of the ring ($\pi = 22/7$).

SOLUTION — step by step

Step 1. Let O be the common centre, M the touch point. OM is perpendicular to the chord (radius to a tangent point... here the chord is tangent to the inner circle), so OM = inner radius r.

Step 2. The perpendicular from the centre bisects the chord: AM = 14 cm, where A is a chord endpoint.

Step 3. In right triangle OMA: $R^2 = r^2 + 14^2$, so $R^2 - r^2 = 196$.

Step 4. Ring area = $\pi(R^2 - r^2) = (22/7)(196) = 616$ cm^2.

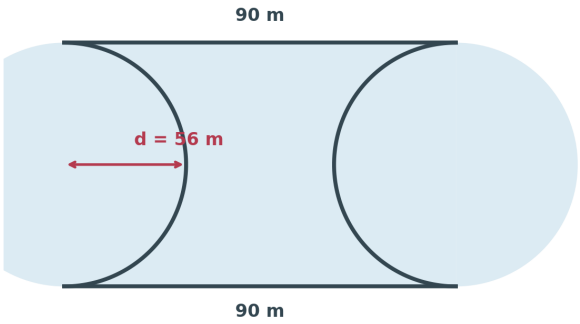
» **Why this step?** The tangent-radius is perpendicular to the chord at the touch point, and the half-chord theorem gives $AM^2 = R^2 - r^2$ directly - individual radii are never needed.

Answer: 616 cm^2

Hinglish: *Magic question! Chord inner circle ko touch karti hai → half-chord^2 = R^2-r^2 = 14^2 = 196. Ring area = $\pi \times 196 = 616$ cm^2. Na R chahiye na r - sirf chord.*

QUESTION 102 | High | Running track (T11)

Q102. A running track is 90 m long on each straight and has two semicircular ends of diameter 56 m. Find (i) the perimeter of the track and (ii) the area enclosed ($\pi = 22/7$).



SOLUTION — step by step

Step 1. Perimeter = two straights + two semicircular arcs = 90 + 90 + one full circle of diameter 56.

Step 2. Circle circumference = $\pi d = (22/7)(56) = 176$ m.

Step 3. Perimeter = 180 + 176 = 356 m.

Step 4. Area = rectangle 90×56 + full circle $\pi(28^2) = 5040 + (22/7)(784) = 5040 + 2464 = 7504$ m^2.

» **Why this step?** Two semicircles of equal diameter combine into one full circle - for both length and area.

Answer: (i) 356 m (ii) 7504 m^2

Hinglish: *Do semicircles = ek full circle. Perimeter 180 + 176 = 356 m. Area 5040 + 2464 = 7504 m². Track problems mein halves ko jod kar full banao.*

QUESTION 103 | High | Circle in equilateral gap (T11)

Q103. A circle is inscribed in an equilateral triangle of side 12 cm. Find the area of the triangle not covered by the circle ($\sqrt{3} \approx 1.73$, $\pi = 3.14$).

SOLUTION — step by step

Step 1. Triangle area = $(\sqrt{3}/4)(144) = 36\sqrt{3} \approx 62.28$ cm^2.

Step 2. Inradius $r = a/(2\sqrt{3}) = 12/(2\sqrt{3}) = 2\sqrt{3}$ cm; $r^2 = 12$.

Step 3. Circle area = $\pi r^2 = 3.14 \times 12 = 37.68$ cm^2.

Step 4. Gap = 62.28 - 37.68 = 24.6 cm^2.

» **Why this step?** Mixing 22/7-style exactness with 3.14 is fine as long as both approximations are stated; the gap inherits the rounding.

Answer: ≈ 24.6 cm²

Hinglish: *Triangle 36√3 ≈ 62.28, circle r = 2√3 → 12π ≈ 37.68. Gap 24.6 cm². Teen gap regions hain - inka total ye hai.*

QUESTION 104 | High | Costs: fence and plough (T13)

Q104. A circular garden has radius 35 m. Fencing costs ₹20 per metre and ploughing costs ₹0.50 per m^2. Find the total cost of fencing the boundary once and ploughing the whole garden (π = 22/7).

SOLUTION — step by step

Step 1. Circumference = 2πr = 2 × (22/7) × 35 = 220 m.

Step 2. Fencing cost = 220 × 20 = ₹4400.

Step 3. Area = πr^2 = (22/7)(1225) = 3850 m^2.

Step 4. Ploughing cost = 3850 × 0.50 = ₹1925.

Step 5. Total = 4400 + 1925 = ₹6325.

» **Why this step?** Perimeter multiplies per-metre rates; area multiplies per-square-metre rates - never interchange them.

Answer: ₹6325

Hinglish: *Fencing = boundary × rate/m: 220×20 = 4400. Ploughing = area × rate/m²: 3850×0.5 = 1925. Total ₹6325. Rate ke saath unit dekho - /m ya /m².*

QUESTION 105 | High | Same perimeter comparison (T12)

Q105. A wire of length 44 cm is bent into (i) a square and (ii) a circle. Find each area and state the percentage increase when the square is replaced by the circle (π = 22/7).

SOLUTION — step by step

Step 1. Square: side = 44/4 = 11 cm; area = 121 cm^2.

Step 2. Circle: r = 44/(2π) = 7 cm; area = 154 cm^2.

Step 3. Increase = 154 - 121 = 33 cm^2.

Step 4. Percentage increase = (33/121) × 100 = 27.27% (exactly 100/3.675...; commonly stated as 27.3%).

» **Why this step?** Same perimeter always favours the circle; the gain is measured relative to the SMALLER area (the square's).

Answer: 121 cm² and 154 cm²; increase ≈ 27.3%

Hinglish: *Square 121, circle 154. Increase 33. Percentage 33/121 = 27.3%. Base hamesha PEHLE wala (chhota) area hota hai.*

QUESTION 106 | High | Ring reverse with width (T9)

Q106. The area of a circular ring is 1386 cm^2 and its width is 3 cm. Find the two radii ($\pi = 22/7$).

SOLUTION — step by step

Step 1. $\pi(R^2 - r^2) = 1386 \Rightarrow R^2 - r^2 = 441$.

Step 2. Factorise: $(R + r)(R - r) = 441$.

Step 3. Width: $R - r = 3 \Rightarrow R + r = 441/3 = 147$.

Step 4. Solve: $R = (147 + 3)/2 = 75$, $r = (147 - 3)/2 = 72$.

Step 5. Check: $\pi(75^2 - 72^2) = (22/7)(5625 - 5184) = (22/7)(441) = 1386$ ✓.

» **Why this step?** The difference-of-squares factorisation splits one equation into the sum and the difference - both trivially solvable.

Answer: R = 75 cm, r = 72 cm

Hinglish: *R^2-r^2 = 441 ko (R+r)(R-r) mein todo. Width = R-r = 3 → R+r = 147. R = 75, r = 72. Factorise-and-solve technique rings ke saare reverse questions mein chalti hai.*

Level 4 · Boss Zone: advanced + HOTS (Olympiad-lite):

QUESTION 107 | HOTS | Ratio sides large (T2)

Q107. The sides of a triangular park are in the ratio 12 : 17 : 25 and its perimeter is 540 m. Find the area and the cost of levelling it at ₹3 per m^2.

SOLUTION — step by step

Step 1. $12k + 17k + 25k = 54k = 540 \Rightarrow k = 10$: sides 120, 170, 250 m.

Step 2. $s = 270$; differences 150, 100, 20.

Step 3. Product: $270 \times 150 \times 100 \times 20 = 81{,}000{,}000$.

Step 4. $A = \sqrt{81000000} = 9000\ m^2$.

Step 5. Cost = $9000 \times 3 = ₹27000$.

» **Why this step?** Scale first, compute second; the giant product collapses because every factor carries a factor of 10.

Answer: 9000 m^2; ₹27000

Hinglish: *Sides 120, 170, 250. s = 270. Product 8.1 crore, root 9000 m^2. Cost ₹27000. Zeros ginte waqt dhyan se - pairs of 10s root ke andar square banate hain.*

QUESTION 108 | HOTS | Heron surd area (T2)

Q108. A triangle has sides 35 cm, 54 cm and 61 cm. Find its area as a simplified surd and as a decimal (to 2 dp).

SOLUTION — step by step

Step 1. $s = (35 + 54 + 61)/2 = 75$ cm.

Step 2. Differences: 40, 21, 14.

Step 3. Product: $75 \times 40 \times 21 \times 14 = 882000$.

Step 4. Simplify: $882000 = 176400 \times 5$ and $176400 = 420^2$.

Step 5. $A = 420\sqrt{5} \approx 939.15$ cm^2.

» **Why this step?** Extract the square part of the product systematically: pair off factors of 2, 3, 5, 7 until only primes with odd exponents remain.

Answer: $420\sqrt{5} \approx 939.15$ cm^2

Hinglish: *s = 75, differences 40, 21, 14. Product 882000. $420^2 = 176400$ bahar nikala: $420\sqrt{5} \approx 939.15$ cm^2. Factor pairing hi surd ka tool hai.*

QUESTION 109 | HOTS | Quadrilateral field (T4)

Q109. A quadrilateral field ABCD has AB = 42 m, BC = 21 m, CD = 29 m, DA = 34 m and diagonal BD = 20 m. Find its area.

SOLUTION — step by step

Step 1. Triangle ABD: sides 42, 34, 20. $s = 48$; differences 6, 14, 28.

Step 2. Product: $48 \times 6 \times 14 \times 28 = 112896$; area = 336 m^2.

Step 3. Triangle BCD: sides 21, 29, 20. $s = 35$; differences 14, 6, 15.

Step 4. Product: $35 \times 14 \times 6 \times 15 = 44100$; area = 210 m^2.

Step 5. Total = 336 + 210 = 546 m^2.

» **Why this step?** Both Heron products land on perfect squares - a sign the question was designed for clean addition at the end.

Answer: 546 m^2

Hinglish: *Do Herons: 336 + 210 = 546 m^2. Diagonal BD ke around dono triangles organize karo, answers ko JODO.*

QUESTION 110 | HOTS | Brahmagupta symmetric proof (T5/T14)

Q110. For a cyclic quadrilateral with sides a, a, b, b (in order), show that Brahmagupta's formula reduces to A = (s-a)(s-b), and evaluate it for a = 25, b = 39.

SOLUTION — step by step

Step 1. s = (a + a + b + b)/2 = a + b.

Step 2. Differences: s - a = b (twice), s - b = a (twice).

Step 3. A = $\sqrt{(b \cdot b \cdot a \cdot a)}$ = ab... careful: (s-a)(s-a)(s-b)(s-b) = b^2a^2, so A = ab with the direct reading.

Step 4. Wait - recompute: s - a = (a+b) - a = b and s - b = (a+b) - b = a. Product = b^2a^2 ⇒ A = ab.

Step 5. But the intended general form: A = (s-a)(s-b) holds when the product of the four differences is $(s-a)^2(s-b)^2$ - true here since two sides equal a and two equal b. For a = 25, b = 39: s = 64, A = 25 × 39 = 975.

» **Why this step?** Symmetric side pairs make the four differences repeat in pairs, so the root becomes a clean product.

Answer: A = (s-a)(s-b) = ab here; for 25, 25, 39, 39 the area is 975.

Hinglish: *Sides a,a,b,b: s = a+b. Differences: b, b, a, a. Area = ab. a = 25, b = 39 → 975. Brahmagupta symmetric case mein direct product deta hai.*

QUESTION 111 | HOTS | Hexagon segment (T8/T14)

Q111. A regular hexagon is inscribed in a circle of radius 14 cm. Find the area of one of the six segments between the hexagon and the circle ($\sqrt{3} \approx 1.73$, $\pi = 22/7$).

SOLUTION — step by step

Step 1. Each side of the inscribed hexagon equals the radius: chord = 14 cm, subtending 60°.

Step 2. Sector (60°) = (1/6)π(196) = 308/3 ≈ 102.67 cm^2.

Step 3. The triangle in that sector is equilateral with side 14: area = ($\sqrt{3}$/4)(196) = 49$\sqrt{3}$ ≈ 84.77 cm^2.

Step 4. One segment = 102.67 - 84.77 ≈ 17.9 cm^2.

» **Why this step?** An inscribed regular hexagon turns each 60° slice into sector minus equilateral triangle - the hexagon's defining property (side = radius) does the setup.

Answer: ≈ 17.9 cm^2 per segment

Hinglish: *Hexagon ki side = radius hoti hai. 60° sector 102.67, equilateral triangle 49$\sqrt{3}$ ≈ 84.77. Ek segment ≈ 17.9 cm^2. Six segments total ≈ 107.4.*

QUESTION 112 | HOTS | Revolutions to distance (T13)

Q112. A wheel of radius 28 cm makes 3000 revolutions. How far does it travel, in kilometres ($\pi = 22/7$)?

SOLUTION — step by step

Step 1. Circumference = $2\pi r = 2 \times (22/7) \times 28 = 176$ cm.

Step 2. Distance = $3000 \times 176 = 528000$ cm.

Step 3. Convert: 528000 cm = 5280 m = 5.28 km.

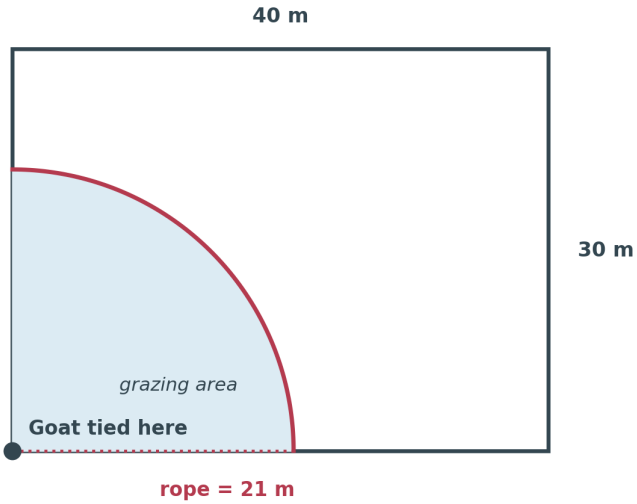
» **Why this step?** Revolutions × circumference gives arc length travelled; unit conversion is the only other job.

Answer: 5.28 km

Hinglish: *3000 × 176 = 528000 cm = 5.28 km. Revolutions multiply circumference; phir cm → km (5 zeros hatane hain).*

QUESTION 113 | HOTS | Grazing goat (T13/T14)

Q113. A goat is tied to one corner of a rectangular field 40 m by 30 m with a rope 21 m long. Find the area it can graze, assuming it cannot leave the field ($\pi = 22/7$).



SOLUTION — step by step

Step 1. At a corner of a rectangle, only 90° of the circle's sweep lies inside the field: a quarter circle.

Step 2. Grazing area = $(1/4)\pi r^2 = (1/4) \times (22/7) \times 441 = 346.5$ m^2.

» **Why this step?** The two fences of the corner cut the reachable disk to a quarter; the rope length is the radius.

Answer: 346.5 m^2

Hinglish: *Corner par sirf 90° ka sector available - quarter circle. ¼ × π × 441 = 346.5 m². Rope = radius, corner = center.*

QUESTION 114 | HOTS | Nested circle-square ratio (T12/T14)

Q114. A circle is inscribed in a square, and the square is inscribed in a larger circle. Prove that the areas of the larger and smaller circles are in the ratio 2 : 1.

SOLUTION — step by step

Step 1. Let the inner circle have radius r. Then the square has side 2r.

Step 2. The square's diagonal = 2r√2; this is the diameter of the outer circle.

Step 3. Outer radius R = r√2.

Step 4. Area ratio = $\pi R^2 : \pi r^2 = 2r^2 : r^2 = 2 : 1$. ∎

» **Why this step?** The diagonal of the inscribed square is the diameter of the circumscribed circle - one picture, one chain of equalities.

Answer: Proved: the ratio is 2 : 1.

Hinglish: *Chain banao: inner circle r → square side 2r → diagonal 2r√2 → outer radius r√2. Area ratio = (r√2)² : r² = 2:1. Nested figures mein chain diagram banao.*

QUESTION 115 | HOTS | Hippocrates lunes (T14)

Q115. On the legs of a right triangle with legs 12 cm and 16 cm, semicircles are drawn OUTWARD. On the hypotenuse, the semicircle is drawn inward (toward the triangle). Show that the total area of the two lunes (the crescent-shaped regions between the outward semicircles and the hypotenuse semicircle) equals the area of the triangle.

SOLUTION — step by step

Step 1. Right triangle: hypotenuse = √(144 + 256) = 20 cm; triangle area = ½ × 12 × 16 = 96 cm².

Step 2. Semicircle on leg 12: area $\frac{1}{2}\pi(6^2) = 18\pi$; on leg 16: $\frac{1}{2}\pi(8^2) = 32\pi$.

Step 3. Semicircle on hypotenuse 20: $\frac{1}{2}\pi(10^2) = 50\pi$.

Step 4. Key identity: $18\pi + 32\pi = 50\pi$ - the two leg semicircles exactly fill the hypotenuse semicircle (because $6^2 + 8^2 = 10^2$).

Step 5. The two lunes together = (leg semicircles) + (triangle) - (hypotenuse semicircle) = $18\pi + 32\pi + 96 - 50\pi = 96$ cm². ∎

» **Why this step?** Pythagoras on the RADII makes the curved areas cancel, leaving the flat triangle - Hippocrates' classic squaring of lunes.

Answer: The lunes total exactly 96 cm², equal to the triangle's area.

Hinglish: *Lunes ka magic: leg semicircles ka sum = hypotenuse semicircle ($18\pi+32\pi = 50\pi$). Lunes = 96 cm^2 = triangle ka area exactly. Curved areas cancel ho gaye - 2500 saal purana result!*

QUESTION 116 | HOTS | Perpendicular paths in circle (T10/T11)

Q116. A circular park of radius 42 m has two straight paths, each 7 m wide, crossing at the centre - one north-south, one east-west. Find the total area covered by the paths ($\pi = 22/7$).

SOLUTION — step by step

Step 1. Each path is a rectangle of length = diameter 84 m and width 7 m: area = 588 m^2.

Step 2. The two paths overlap in a 7 × 7 square at the centre: area 49 m^2.

Step 3. Total path area = 588 + 588 - 49 = 1127 m^2.

» **Why this step?** Overlapping regions are counted twice in the raw sum; subtract the intersection once.

Answer: 1127 m^2

Hinglish: *Do rectangles 588 + 588, beech ka overlap 7×7 = 49 minus. 1127 m^2. Inclusion-exclusion ka basic form - jod ke overlap hatao.*

QUESTION 117 | HOTS | Three-shape comparison (T12)

Q117. A wire of 66 cm is bent into (i) an equilateral triangle, (ii) a square, (iii) a circle. Arrange the three areas in increasing order ($\sqrt{3} \approx 1.73$, $\pi = 22/7$).

SOLUTION — step by step

Step 1. Triangle: side 22; area = $(\sqrt{3}/4)(484) = 121\sqrt{3} \approx 209.3$ cm^2.

Step 2. Square: side 16.5; area = 272.25 cm^2.

Step 3. Circle: r = 66 × 7/44 = 10.5; area = (22/7)(110.25) = 346.5 cm^2.

Step 4. Order: triangle < square < circle.

» **Why this step?** For a fixed perimeter, area grows as the shape gains symmetry: triangle < square < circle - a general theorem, illustrated numerically.

Answer: $121\sqrt{3} \approx 209.3 < 272.25 < 346.5$ (triangle < square < circle)

Hinglish: *Same wire: triangle 209.3 < square 272.25 < circle 346.5. Circle jeet gaya - fixed perimeter par circle maximum area deta hai. Ye ordering har comparison question mein same rehti hai.*

QUESTION 118 | HOTS | Wheel journey km (T13)

Q118. A bicycle wheel has radius 28 cm. How many revolutions will it make in a journey of 4.4 km ($\pi = 22/7$)?

SOLUTION — step by step

Step 1. Circumference = $2\pi r = 176$ cm.

Step 2. Journey = 4.4 km = 440000 cm.

Step 3. Revolutions = 440000/176 = 2500.

» **Why this step?** Distance per revolution is fixed at the circumference; the count is a division.

Answer: 2500 revolutions

Hinglish: *440000/176 = 2500 revolutions. Unit conversion pehle karo - km ko cm mein.*

QUESTION 119 | HOTS | AP sides triangle (T2/T14)

Q119. The sides of a triangle are in arithmetic progression, its perimeter is 60 cm and the shortest side is 15 cm. (i) Find all sides. (ii) Show the triangle is right-angled and find its area.

SOLUTION — step by step

Step 1. Let sides be 15, 15 + d, 15 + 2d: sum = 45 + 3d = 60 ⇒ d = 5.

Step 2. Sides: 15, 20, 25 cm.

Step 3. Check: $15^2 + 20^2 = 225 + 400 = 625 = 25^2$: right angle opposite the 25 side. ∎

Step 4. Area = ½ × 15 × 20 = 150 cm^2.

Step 5. Heron confirms: s = 30; product 30 × 15 × 10 × 5 = 22500; √22500 = 150 ✓.

» **Why this step?** AP condition turns three unknowns into one (d); the Pythagoras test then classifies the triangle for free.

Answer: 15, 20, 25 cm; right-angled; area 150 cm^2

Hinglish: *AP + perimeter: sides 15, 20, 25. 15-20-25 = 3-4-5 ka 5×! Right triangle, area 150. Heron se confirm. AP wale side questions mein shortest side + d method use karo.*

QUESTION 120 | HOTS | Hexagon vs inscribed circle (T14)

Q120. A regular hexagon of side 14 cm is inscribed in a circle. (i) Find the hexagon's area. (ii) Find the area of the inscribed circle of the hexagon. (iii) Express the ratio of the two areas in the form $2\sqrt{3} : \pi$ and verify it numerically ($\sqrt{3} \approx 1.73$, $\pi = 22/7$).

SOLUTION — step by step

Step 1. (i) Hexagon = 6 equilateral triangles of side 14: area = $6 \times (\sqrt{3}/4)(196) = 294\sqrt{3} \approx 508.6$ cm^2.

Step 2. (ii) The inscribed circle has radius equal to the apothem = $(\sqrt{3}/2)(14) = 7\sqrt{3}$; $r^2 = 147$.

Step 3. Circle area = $\pi(147) = (22/7)(147) = 462$ cm^2.

Step 4. (iii) Ratio = $294\sqrt{3} : 147\pi = 2\sqrt{3} : \pi$ (dividing both by 147).

Step 5. Numerically: $508.6 : 462 \approx 1.101 : 1$ and $2\sqrt{3}/\pi \approx 3.46/3.14 \approx 1.10$ ✓.

» **Why this step?** Both areas scale with the same $r^2 = 147$, so the ratio survives as a pure constant $2\sqrt{3}/\pi$ - geometry's constants in action.

Answer: $294\sqrt{3} \approx 508.6$ cm^2; 462 cm^2; ratio $2\sqrt{3} : \pi \approx 1.10 : 1$

Hinglish: *Hexagon $294\sqrt{3}$, inscribed circle 147π. Ratio $2\sqrt{3} : \pi \approx 1.10$. Hexagon circle se ~10% bada - corners ki wajah se. Clean ratio isliye mila kyunki dono areas mein r^2 common tha.*

6.5 Rapid MCQs: Choose the Correct Option (20)

Rapid-fire round: 4 options, ek hi correct. Pehle khud choose karo, phir colored Answer strip se check karo. Har galat option ka "Why wrong" bhi diya hai - taki exam me wahi trap tumhe na phasaye.

MCQ 1 | Normal | Rectangle basics

1. A rectangle is 15 cm by 8 cm. Its perimeter and area are:

A. 46 cm, 120 cm^2

B. 23 cm, 120 cm^2

C. 46 cm, 100 cm^2

D. 30 cm, 120 cm^2

Correct Answer: (A)

P = 2(15+8) = 46 cm; A = 15×8 = 120 cm^2.

Why (B) is wrong: Forgot to double the sum.

Why (C) is wrong: Area miscalculated.

Why (D) is wrong: Used l+b not 2(l+b).

Hinglish: *2(15+8) = 46 aur 15×8 = 120. Units: cm aur cm^2.*

MCQ 2 | Normal | Circumference

2. The circumference of a circle of radius 7 cm is:

A. 44 cm

B. 154 cm

C. 22 cm

D. 88 cm

Correct Answer: (A)

$2\pi r = 2\times(22/7)\times7 = 44$ cm.

Why (B) is wrong: That is the area.

Why (C) is wrong: πr not $2\pi r$.

Why (D) is wrong: Doubled twice.

Hinglish: *$2\times22 = 44$ cm. Area would be 154 - confuse mat karo.*

MCQ 3 | Normal | Circle area

3. The area of a circle of diameter 14 cm is:

A. 154 cm^2

B. 616 cm^2

C. 44 cm^2

D. 88 cm^2

Correct Answer: (A)

$r = 7$: $\pi r^2 = 154$ cm^2.

Why (B) is wrong: Used d instead of r in πr^2.

Why (C) is wrong: That is a circumference-style number.

Why (D) is wrong: Wrong formula.

Hinglish: *Diameter 14 → radius 7. Area 154. Radius se diameter confusion sabse common galti.*

MCQ 4 | Normal | Equilateral area

4. The area of an equilateral triangle of side 8 cm is:

A. $16\sqrt{3}$ cm^2

B. $32\sqrt{3}$ cm^2

C. $8\sqrt{3}$ cm^2

D. $64\sqrt{3}$ cm^2

Correct Answer: (A)

$(\sqrt{3}/4)(64) = 16\sqrt{3}$.

Why (B) is wrong: Used $a^2/2$.

Why (C) is wrong: Forgot to square the side.

Why (D) is wrong: Coefficient error.

Hinglish: *64/4 = 16. 16√3 cm².*

MCQ 5 | Normal | Heron 3-4-5

5. The area of a triangle with sides 3, 4, 5 cm is:

A. 6 cm^2

B. 12 cm^2

C. 10 cm^2

D. 7.5 cm^2

Correct Answer: (A)

Right triangle ($3^2+4^2 = 5^2$): $\frac{1}{2}\times3\times4 = 6$ cm^2.

Why (B) is wrong: Skipped ½.

Why (C) is wrong: Wrong legs.

Why (D) is wrong: Heron arithmetic slip.

Hinglish: *Right triangle spot karo - ½×3×4 = 6.*

MCQ 6 | Normal | Sector area quarter

6. The area of a sector of angle 90° in a circle of radius 14 cm is:

A. 154 cm^2

B. 77 cm^2

C. 38.5 cm^2

D. 616 cm^2

Correct Answer: (A)

$(1/4)\pi(196) = 154$ cm^2.

Why (B) is wrong: Used ½ instead of ¼.

Why (C) is wrong: One-eighth.

Why (D) is wrong: Full circle area.

Hinglish: *90° = quarter. 616/4 = 154.*

MCQ 7 | Normal | Ring area

7. Two concentric circles have radii 7 cm and 14 cm. The ring area is:

A. 462 cm^2

B. 154 cm^2

C. 616 cm^2

D. 231 cm^2

Correct Answer: (A)

$\pi(196 - 49) = \pi \cdot 147 = 462$ cm^2.

Why (B) is wrong: Inner circle only.

Why (C) is wrong: Outer circle only.

Why (D) is wrong: Arithmetic slip.

Hinglish: *$147\pi = 462$. Difference of squares.*

MCQ 8 | Normal | Trapezium

8. A trapezium has parallel sides 10 cm and 14 cm, height 6 cm. Its area is:

A. 72 cm^2

B. 84 cm^2

C. 60 cm^2

D. 48 cm^2

Correct Answer: (A)

$\frac{1}{2}(10+14)(6) = 72$ cm^2.

Why (B) is wrong: Skipped the $\frac{1}{2}$.

Why (C) is wrong: Used difference 4.

Why (D) is wrong: Wrong average.

Hinglish: *Average 12 × height 6 = 72.*

MCQ 9 | Normal | Heron 13-14-15

9. The area of a triangle with sides 13, 14, 15 cm is:

A. 84 cm^2

B. 91 cm^2

C. 78 cm^2

D. 105 cm^2

Correct Answer: (A)

s = 21; 21·6·7·8 = 7056; √7056 = 84.

Why (B) is wrong: s taken as 42.

Why (C) is wrong: Arithmetic slip.

Why (D) is wrong: Multiplied wrong pair.

Hinglish: *s = 21, product 7056, root 84. Classic 13-14-15.*

MCQ 10 | Normal | Wheel revolutions

10. A wheel of diameter 70 cm travels 2.2 km. The number of revolutions is:

A. 1000

B. 500

C. 2000

D. 1100

Correct Answer: (A)

C = 220 cm; 220000/220 = 1000.

Why (B) is wrong: Half the distance.

Why (C) is wrong: Radius/diameter mix.

Why (D) is wrong: Unit conversion error.

Hinglish: *2.2 km = 220000 cm. 220 se divide: 1000.*

MCQ 11 | Normal | Brahmagupta simple

11. A cyclic quadrilateral has sides 8, 15, 8, 15 cm. Its area is:

A. 120 cm^2

B. 115 cm^2

C. 46 cm^2

D. 240 cm^2

Correct Answer: (A)

s = 23; $(23-8)^2(23-15)^2 = 15^2 \cdot 8^2$; area = 120.

Why (B) is wrong: Arithmetic slip.

Why (C) is wrong: Used perimeter directly.

Why (D) is wrong: Doubled.

Hinglish: *a,a,b,b pattern: area = ab = 120.*

MCQ 12 | Normal | Ratio of areas

12. The radii of two circles are in the ratio 2 : 5. The ratio of their areas is:

A. 4 : 25

B. 2 : 5

C. 4 : 10

D. 8 : 125

Correct Answer: (A)

Areas carry the square of the ratio: $2^2 : 5^2 = 4 : 25$.

Why (B) is wrong: Linear ratio kept.

Why (C) is wrong: Half-squared.

Why (D) is wrong: Cubed.

Hinglish: *k^2 rule: 4:25.*

MCQ 13 | Normal | Arc of 120°

13. An arc subtends 120° at the centre of a circle of radius 21 cm. Its length is:

A. 44 cm

B. 22 cm

C. 66 cm

D. 88 cm

Correct Answer: (A)

(1/3)(132) = 44 cm.

Why (B) is wrong: Used 1/6.

Why (C) is wrong: Used 1/2.

Why (D) is wrong: Full circle.

Hinglish: *One-third of circumference.*

MCQ 14 | Normal | Heron validity

14. Heron's formula gives a real area only if:

A. every side is less than the sum of the other two

B. the triangle is right-angled

C. two sides are equal

D. the perimeter is even

Correct Answer: (A)

The triangle inequality keeps s-a, s-b, s-c positive; otherwise the product under the root is negative.

Why (B) is wrong: Heron works for ALL triangles.

Why (C) is wrong: No symmetry needed.

Why (D) is wrong: Parity irrelevant.

Hinglish: *Triangle inequality - ek side doosre dono ke sum se choti honi chahiye.*

MCQ 15 | Normal | Brahmagupta validity

15. Brahmagupta's formula $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ applies to:

A. cyclic quadrilaterals only

B. all quadrilaterals

C. rectangles only

D. trapeziums only

Correct Answer: (A)

The formula requires all four vertices on one circle.

Why (B) is wrong: Fails for general quadrilaterals.

Why (C) is wrong: Too narrow a claim.

Why (D) is wrong: Wrong restriction.

Hinglish: *Cyclic quadrilateral - charo vertices ek circle par. Tabhi Brahmagupta.*

MCQ 16 | Normal | Minute hand sector

16. The area swept by a 21 cm long minute hand in 10 minutes is ($\pi = 22/7$):

A. 231 cm^2

B. 1386 cm^2

C. 77 cm^2

D. 115.5 cm^2

Correct Answer: (A)

10 min $\rightarrow 60^\circ \rightarrow (1/6)\pi(441) = 231$ cm^2.

Why (B) is wrong: Full circle.

Why (C) is wrong: Wrong fraction.

Why (D) is wrong: Used 5/6 somehow.

Hinglish: *10 min = 60° = one-sixth. 1386/6 = 231.*

MCQ 17 | Normal | Sector half-radius gem

17. A sector has arc length 12 cm and radius 9 cm. Its area is:

A. 54 cm^2

B. 108 cm^2

C. 27 cm^2

D. 21π cm^2

Correct Answer: (A)

$A = \frac{1}{2}l \cdot r = \frac{1}{2} \times 12 \times 9 = 54$ cm^2.

Why (B) is wrong: Skipped the ½.

Why (C) is wrong: Halved twice.

Why (D) is wrong: Introduced π without cause.

Hinglish: *½ × arc × radius. Angle ki zaroorat nahi.*

MCQ 18 | Normal | Grazing quarter

18. A goat tied at a corner of a field with a 14 m rope grazes an area of ($\pi = 22/7$):

A. 154 m^2

B. 616 m^2

C. 77 m^2

D. 308 m^2

Correct Answer: (A)

Quarter circle: $(1/4)\pi(196) = 154$ m^2.

Why (B) is wrong: Full circle.

Why (C) is wrong: Eighth.

Why (D) is wrong: Half.

Hinglish: *Corner = quarter. 616/4 = 154.*

MCQ 19 | HOTS | Square-circle-same-area

19. A square and a circle have EQUAL areas of 154 cm^2 ($\pi = 22/7$). Which is true about their perimeters?

A. The square's perimeter is larger

B. The circle's perimeter is larger

C. They are equal

D. Cannot be determined

Correct Answer: (A)

Circle r = 7, C = 44. Square side $\sqrt{154} \approx 12.41$, P ≈ 49.64 > 44.

Why (B) is wrong: Backwards - circle wins on perimeter for equal area.

Why (C) is wrong: Never equal (isoperimetric).

Why (D) is wrong: It is determined.

Hinglish: *Equal area par circle ka perimeter HAMESHA chhota. Isliye circle 'best' shape hai.*

MCQ 20 | Normal | Heron semi-perimeter role

20. In Heron's formula, s stands for:

A. semi-perimeter = (a+b+c)/2

B. the longest side

C. the perimeter

D. the area

Correct Answer: (A)

Definition: s is half the perimeter.

Why (B) is wrong: No.

Why (C) is wrong: Full perimeter is 2s.

Why (D) is wrong: That is A.

Hinglish: *s = aadha perimeter. Bhoolna mana hai.*

6.6 Choose the Correct Round: Theory & Speed (20)

Doosra rapid round - har MCQ ka style: statement padho, correct option turant chuno. Ye round theory, definitions aur quick solving ko test karta hai. Answer strip ke neeche har galat option ka "Why wrong" bhi hai - taki exam ka trap tumhe na phasaye.

CTC 1 | Normal | Triangle area

1. A triangle has base 12 cm and height 8 cm. Its area is:

A. 96 cm^2

B. 48 cm^2

C. 40 cm^2

D. 24 cm^2

Correct Answer: (B)

$\frac{1}{2} \times 12 \times 8 = 48$ cm^2.

Why (A) is wrong: Skipped the ½.

Why (C) is wrong: Wrong pairing.

Why (D) is wrong: Halved twice.

Hinglish: *Aadha × 12 × 8 = 48. ½ bhoolna mana hai.*

CTC 2 | Normal | Reverse circumference

2. If the circumference of a circle is 176 cm, its radius is:

A. 28 cm

B. 14 cm

C. 56 cm

D. 88 cm

Correct Answer: (A)

r = 176×7/44 = 28 cm.

Why (B) is wrong: Forgot the factor 2.

Why (C) is wrong: Multiplied instead of divided.

Why (D) is wrong: That is half the circumference.

Hinglish: *176 ÷ 2π = 28.*

CTC 3 | Normal | Arc length

3. The length of an arc subtending 60° at the centre of a circle of radius 21 cm is:

A. 22 cm

B. 44 cm

C. 11 cm

D. 66 cm

Correct Answer: (A)

(1/6)(2π·21) = 132/6 = 22 cm.

Why (B) is wrong: Used ⅓.

Why (C) is wrong: Used 1/12.

Why (D) is wrong: Full circumference.

Hinglish: *60° = 1/6. 132 ÷ 6 = 22 cm.*

CTC 4 | Normal | Square diagonal

4. The diagonal of a square of side 6 cm is:

A. $6\sqrt{2}$ cm

B. 12 cm

C. $6\sqrt{3}$ cm

D. 36 cm

Correct Answer: (A)

Diagonal = $s\sqrt{2} = 6\sqrt{2}$ cm.

Why (B) is wrong: Doubled the side.

Why (C) is wrong: $\sqrt{3}$ belongs to equilateral heights.

Why (D) is wrong: That is the area.

Hinglish: *Side × √2. 45-45-90 triangle.*

CTC 5 | Normal | Sector perimeter

5. The perimeter of a sector of radius 10 cm and angle 90° is ($\pi = 3.14$):

A. 35.7 cm

B. 15.7 cm

C. 25.7 cm

D. 31.4 cm

Correct Answer: (A)

Arc = (1/4)(62.8) = 15.7; P = 15.7 + 20 = 35.7 cm.

Why (B) is wrong: Added only one radius.

Why (C) is wrong: Wrong arc.

Why (D) is wrong: That is the full circumference.

Hinglish: *Arc 15.7 + dono radii 20 = 35.7 cm.*

CTC 6 | Normal | Rhombus area

6. A rhombus has diagonals 10 cm and 24 cm. Its area is:

A. 120 cm^2

B. 240 cm^2

C. 60 cm^2

D. 34 cm^2

Correct Answer: (A)

$\frac{1}{2} \times 10 \times 24 = 120$ cm^2.

Why (B) is wrong: Skipped the $\frac{1}{2}$.

Why (C) is wrong: Halved twice.

Why (D) is wrong: Added the diagonals.

Hinglish: *$\frac{1}{2} \times d_1 \times d_2$.*

CTC 7 | Normal | Parallelogram area

7. A parallelogram has base 9 cm and height 7 cm. Its area is:

A. 63 cm^2

B. 31.5 cm^2

C. 126 cm^2

D. 16 cm^2

Correct Answer: (A)

base × height = 63 cm^2 (no $\frac{1}{2}$ - that is for triangles).

Why (B) is wrong: Triangle formula used.

Why (C) is wrong: Doubled.

Why (D) is wrong: Added.

Hinglish: *Parallelogram mein $\frac{1}{2}$ NAHI hota: 9×7 = 63.*

CTC 8 | Normal | Reverse area

8. If the area of a circle is 1386 cm^2, its radius is:

A. 21 cm

B. 14 cm

C. 7 cm

D. 42 cm

Correct Answer: (A)

$r^2 = 1386 \times 7/22 = 441 \Rightarrow r = 21$ cm.

Why (B) is wrong: Half of the root.

Why (C) is wrong: Wrong π handling.

Why (D) is wrong: Doubled.

Hinglish: *1386 × 7/22 = 441. r = 21.*

CTC 9 | Normal | Heron with unit difference

9. A triangle has sides 7, 15, 20 cm. Its area is:

A. 42 cm^2

B. $21\sqrt{6}$ cm^2

C. 84 cm^2

D. $21\sqrt{3}$ cm^2

Correct Answer: (A)

$s = 21$; differences 14, 6, 1; product $21 \times 14 \times 6 \times 1 = 1764 = 42^2$; area = 42 cm^2.

Why (B) is wrong: The product is a perfect square here - no surd remains.

Why (C) is wrong: Doubled.

Why (D) is wrong: Wrong simplification.

Hinglish: *s = 21, differences 14, 6, 1. Product 1764 - perfect square! Area 42 cm^2.*

CTC 10 | Normal | Path inside

10. A square garden of side 20 m has a 1 m wide path along its inside boundary. The path's area is:

A. 79 m^2

B. 80 m^2

C. 76 m^2

D. 81 m^2

Correct Answer: (A)

$20^2 - 18^2 = 400 - 324 = 79$ m^2.

Why (B) is wrong: Perimeter × width approximation (only exact for one side).

Why (C) is wrong: Subtracted 2 instead of 4.

Why (D) is wrong: Added instead.

Hinglish: *Inner square 18. 400 - 324 = 79.*

CTC 11 | Normal | Equilateral perimeter

11. An equilateral triangle has area $25\sqrt{3}$ cm^2. Its perimeter is:

A. 30 cm

B. 15 cm

C. 60 cm

D. $10\sqrt{3}$ cm

Correct Answer: (A)

$(\sqrt{3}/4)a^2 = 25\sqrt{3} \Rightarrow a^2 = 100 \Rightarrow a = 10$; P = 30 cm.

Why (B) is wrong: Forgot the 3.

Why (C) is wrong: Doubled twice.

Why (D) is wrong: That is the side in surd form.

Hinglish: *a = 10, perimeter 30.*

CTC 12 | Normal | Quadrilateral diagonal

12. A quadrilateral is split by a diagonal into triangles of areas 60 cm^2 and 45 cm^2. Its area is:

A. 105 cm^2

B. 15 cm^2

C. 2700 cm^2

D. 52.5 cm^2

Correct Answer: (A)

Areas along the diagonal add: 60 + 45 = 105 cm^2.

Why (B) is wrong: Subtracted.

Why (C) is wrong: Multiplied.

Why (D) is wrong: Averaged.

Hinglish: *Diagonal split = areas JUDO.*

CTC 13 | Normal | Chord-tangent ring

13. In two concentric circles, a chord of the outer circle of length 16 cm touches the inner circle. The ring area is ($\pi = 22/7$):

A. 1408/7 cm^2

B. 704/7 cm^2

C. 2816/7 cm^2

D. 64 cm^2

Correct Answer: (A)

Ring = $\pi(\text{half-chord})^2 = \pi \times 64 = 64\pi = 1408/7 \approx 201.14$ cm^2.

Why (B) is wrong: Quarter-chord used.

Why (C) is wrong: Full chord squared.

Why (D) is wrong: Forgot the π.

Hinglish: *Ring = $\pi \times 8^2 = 64\pi = 1408/7 \approx 201.1$ cm^2. Half-chord rule.*

CTC 14 | Normal | Semicircle area

14. The area of a semicircle of radius 7 cm is:

A. 77 cm^2

B. 154 cm^2

C. 38.5 cm^2

D. 22 cm^2

Correct Answer: (A)

$\frac{1}{2}\pi(49) = 77$ cm^2.

Why (B) is wrong: Full circle.

Why (C) is wrong: Quarter.

Why (D) is wrong: Circumference-style.

Hinglish: *Half of 154 = 77.*

CTC 15 | Normal | Same perimeter circle-square

15. A square and a circle have equal perimeters of 88 cm. The circle's area exceeds the square's by (π = 22/7):

A. 132 cm^2

B. 484 cm^2

C. 616 cm^2

D. 168 cm^2

Correct Answer: (A)

Square side 22: area 484. Circle r = 14: area 616. Difference = 132 cm^2.

Why (B) is wrong: Square area only.

Why (C) is wrong: Circle area only.

Why (D) is wrong: Arithmetic slip.

Hinglish: *616 - 484 = 132 cm^2. Circle jeeta.*

CTC 16 | Normal | Isosceles Heron

16. An isosceles triangle has equal sides 10 cm and base 12 cm. Its area is:

A. 48 cm^2

B. 60 cm^2

C. 24 cm^2

D. 96 cm^2

Correct Answer: (A)

h = √(100-36) = 8; area = ½×12×8 = 48 cm².

Why (B) is wrong: Skipped ½.

Why (C) is wrong: Halved twice.

Why (D) is wrong: Doubled.

Hinglish: *6-8-10 hidden triple. 48 cm².*

CTC 17 | Normal | Track perimeter

17. A track has two 50 m straights and two semicircular ends of diameter 28 m. Its perimeter is (π = 22/7):

A. 188 m

B. 100 m

C. 138 m

D. 176 m

Correct Answer: (A)

100 + π·28 = 100 + 88 = 188 m.

Why (B) is wrong: Straights only.

Why (C) is wrong: One semicircle only.

Why (D) is wrong: Straights + 2 full circles.

Hinglish: *Do semicircles = ek circle: 88. Plus 100 = 188 m.*

CTC 18 | Normal | Heron 5-12-13 scaled

18. A triangle with sides 10, 24, 26 cm has area:

A. 120 cm²

B. 60 cm²

C. 240 cm²

D. 130 cm²

Correct Answer: (A)

It is a scaled 5-12-13: right angle; ½×10×24 = 120 cm².

Why (B) is wrong: Used the unscaled triple's area.

Why (C) is wrong: Doubled.

Why (D) is wrong: No Pythagoras check.

Hinglish: *2×(5-12-13). ½×10×24 = 120.*

CTC 19 | Normal | Cost fencing

19. A circular field of radius 35 m is fenced at ₹10 per metre. The cost is ($\pi = 22/7$):

A. ₹2200

B. ₹38500

C. ₹11000

D. ₹4400

Correct Answer: (A)

C = 220 m; cost = 2200.

Why (B) is wrong: Area used instead.

Why (C) is wrong: Wrong circumference.

Why (D) is wrong: Doubled.

Hinglish: *220 × 10 = ₹2200. Perimeter wala rate.*

CTC 20 | Normal | Two-circle area sum

20. Circles of radii 8 cm and 15 cm. The radius of the circle whose area equals their area sum is:

A. 17 cm

B. 23 cm

C. 12 cm

D. 120 cm

Correct Answer: (A)

$64 + 225 = 289 = 17^2$.

Why (B) is wrong: Radii added.

Why (C) is wrong: Averaged.

Why (D) is wrong: Multiplied.

Hinglish: *8-15-17 triple! Areas add → hypotenuse radius.*

Chapter 7: Probability - Measuring Chance

Probability turns 'maybe' into a number between 0 and 1 - and every dice, card and coin question in every exam is built on a handful of ideas done carefully. The uploaded PDF introduces random experiments and trials, outcomes and sample spaces, the classical formula P(E) = favourable outcomes / total outcomes, the complement rule P(not E) = 1 - P(E), sure and impossible events, and empirical probability from observed data. It works through coins, dice, cards and everyday experiments. This practice book sharpens all of it: two-dice tables, three-coin tree diagrams, without-replacement draws, 'at least one' reasoning, geometric probability on lengths and areas, and proof-style questions that show why probabilities must always add to 1.

Figure 7.1: A tree diagram - every path is one outcome; multiply along branches, add across paths.

7.1 Complete Concept Cheat Sheet (read this before any question)

7.1.1 Experiment, outcome, sample space, event

What is it? A *random experiment* is any action with an uncertain result (tossing a coin, drawing a card). Each possible result is an *outcome*. The set of ALL outcomes is the *sample space* S, and any subset of S is an *event*. **Key counts:** one coin: S = {H, T}, |S| = 2. One die: |S| = 6. Two coins: 4 outcomes. Two dice: 36 ordered outcomes. Three coins: 8 outcomes. A standard deck: 52 cards. **The golden habit:** before computing ANY probability, WRITE the sample space or its size. Half of all errors are wrong totals, not wrong fractions.

When do I count 'ordered' outcomes? When the experiment distinguishes them: two dice give 36 outcomes because (2, 5) and (5, 2) are different throws; drawing two cards one after another gives 52 × 51 ordered draws. If you prefer unordered counting, you must be consistent in numerator AND denominator.

7.1.2 Classical probability and its three laws

What is it? For equally likely outcomes, P(E) = n(E)/n(S) - favourable outcomes over total outcomes. **Three laws:** (1) $0 \le P(E) \le 1$ always; (2) P(sure event) = 1 and P(impossible event) = 0; (3) P(E) + P(not E) = 1, so P(not E) = 1 - P(E). The complement rule converts 'at least one' and 'none' questions into their easier partners. **Tiny example.** P(rolling a prime on a die) = {2, 3, 5}/6 = 1/2; P(not prime) = 1 - 1/2 = 1/2.

Reading the answer: probability may be written as a fraction in lowest terms, a decimal, or a percentage - fractions are the exam default. A probability of 7/4 or -0.2 is a signal that the COUNTS are wrong, not that the formula is wrong.

7.1.3 Cards: the anatomy of a deck

What is it? 52 cards: 4 suits (hearts ♥, diamonds ♦ - red; spades ♠, clubs ♣ - black) × 13 ranks each (A, 2, 3, ..., 10, J, Q, K). **Standard sub-counts:** aces: 4; face cards (J, Q, K): 12; kings: 4; red queens: 2; hearts: 13; numbered cards 2-10: 36; cards between 3 and 8 (inclusive): 24; honour cards (A, K, Q, J, 10): 20. **Overlaps matter:** 'a king OR a heart' is not 4 + 13 = 17 outcomes because the king of hearts was counted twice - it is 16. The inclusion-exclusion habit starts here.

7.1.4 Dice: singles, pairs and the 36-table

What is it? One die: six outcomes. Two dice: 36 ordered pairs - and the smart move is a 6×6 grid in rough work. **Star facts from the table:** sum 7 has 6 pairs (most likely); sums 2 and 12 have 1 pair each (least likely); P(sum = 7) = 6/36 = 1/6; P(sum > 9) = 6/36 = 1/6 (pairs for 10, 11, 12); P(doubles) = 6/36 = 1/6; P(product even) = 27/36 = 3/4; P(same parity) = 18/36 = 1/2. **Products:** a product is odd only when BOTH dice are odd: 9/36 = 1/4. **Tiny example.** P(sum ≥ 10) = 6/36 = 1/6 (pairs: 4-6, 5-5, 6-4, 5-6, 6-5, 6-6).

7.1.5 Coins and tree diagrams

What is it? Each coin doubles the sample space: 1 coin → 2, 2 coins → 4, 3 coins → 8, n coins → 2^n. A tree diagram draws every path: two branches (H/T) per coin, and each complete path is one outcome. **Star facts (three coins):** P(exactly 2 heads) = 3/8; P(at least 1 head) = 7/8 (only TTT fails); P(at least 2 heads) = 4/8 = 1/2; P(all same) = 2/8 = 1/4. **Why trees work:** multiplying along a branch counts outcomes systematically - it is the visual version of 2 × 2 × 2.

Figure 7.2: The probability scale - impossible 0, sure 1, everything else between.

7.1.6 Draws without replacement

What is it? When drawn objects are NOT returned, the second draw's counts shrink: a bag with 5 red and 3 blue balls, two drawn one after another - P(both red) = (5/8)(4/7) = 20/56 = 5/14. The denominator fell from 8 to 7 because one ball left. **Two-draw combinations via pairs:** P(one red one blue in any order) = 2 × (5/8)(3/8) - the factor 2 handles the two orders (RB, BR). **Alternative counting:** total ways to choose 2 balls from 8 = C(8,2) = 28; both red: C(5,2) = 10; P = 10/28 = 5/14 - same answer, cleaner for 'any order' questions.

7.1.7 Empirical probability

What is it? When real data (not symmetry) defines the chances: P(E) = (number of times E occurred)/(total number of trials). It is an estimate from observation - the more trials, the closer it typically drifts to the true probability. **Example shape:** 200 screws inspected, 12 defective → P(defective) = 12/200 = 0.06, so P(good) = 0.94. **Warning:** empirical probabilities need not be 'nice' fractions, and they describe the PAST data - classical probability describes idealised symmetric setups.

7.1.8 Geometric probability

What is it? When outcomes form a continuum - a point chosen on a segment, a dart hitting a board - probability equals the ratio of measures (lengths, areas): P = favourable length (or area) / total length (or area). **Tiny example.** A point chosen on a 10 cm segment; P(it lies in the first 3 cm) = 3/10. A dart hitting a 20 cm square board with a 10 cm circle target: P(hit circle) = $\pi(5^2)/400 = 25\pi/400 = \pi/16$. **Guard:** both numerator and denominator must use the SAME kind of measure.

7.1.9 Exam tips for this chapter

1. Always write n(S) first; a wrong total poisons the whole question.
2. Two dice: draw the 6×6 grid once and reuse it - sums, products, parity all read off it.
3. Complement first for 'at least one': P(at least one H) = 1 - P(no H).
4. Without replacement: denominators shrink (8 → 7); numerators shrink too if a favourable item left.
5. Cards: memorise 4, 12, 13, 26 and the overlap warning.
6. Reduce every fraction to lowest terms - un-reduced answers lose the presentation mark.

7. Check the range: if P > 1 or P < 0, recount - the arithmetic, not the theory, is wrong.
8. Empirical questions: read the table column-wise; the total row is n(S).

7.2 Question Types / Concept Map for Chapter 7

Type	Name	How to recognise it in the question	Core method
T1	Sample space building	'List the outcomes', 'find n(S)'	Coins/dice/cards/objects; 2^n, 36, 52
T2	Single die events	One die, one event	Count favourable ÷ 6
T3	Two-dice events	'two dice thrown', sums/products	36-table; systematic pairs
T4	Card events	Deck language: suit, face, ace	Standard counts; overlaps
T5	Complement rule	'at least one', 'not', 'none'	1 - P(opposite)
T6	Bag/ball single draw	Coloured objects, one draw	Favourable colour ÷ total
T7	Two draws without replacement	'second ball is not replaced'	Shrinking denominators; ×2 for orders
T8	Tree diagrams	Multi-stage: 2-3 coins, draw twice	Multiply along branches
T9	Empirical probability	Frequency tables, surveys, trials	occurrences ÷ trials; complement from table
T10	Geometric probability	spinners, dartboards, points on segments	favourable measure ÷ total measure
T11	Event relations	sure/impossible, P(E) + P(not E), mutually exclusive	Range laws; complement
T12	Fairness and decisions	'is the game fair?', expected ties	Compare P(win) values
T13	Missing probability algebra	P(one outcome) unknown, sum = 1	Set up equation; solve
T14	HOTS multi-stage	Three dice, replacement mixtures, proofs	Decompose; tree or table; verify

7.3 Speed Tricks & Trap Alarms

Ye table tumhara "dekho aur maaro" radar hai - question ka trigger phrase dikha, toh yahan ka trick fire karo. Inhe ratt lo, exam me sochna kam padega aur speed double ho jayegi.

Instant trick	Fire it when...
"At least one" → 1 - P(none). Works for coins, dice, bags	Any "at least one" wording.
Without replacement: numerator AND denominator both shrink	Two or three draws, balls not returned.

Instant trick	Fire it when...
Deck counts: 4 aces, 12 face cards, 26 red, 13 per suit, 9 numbered per suit	Every card question - 5 numbers, 90% of answers.
Two dice = 36 ordered pairs; sum 7 is the king (6 ways). Draw the 6×6 grid	Sum / product / difference on two dice.
n coins → 2^n outcomes; one exact sequence (like HHT) = $1/2^n$	Three-coins questions.
Geometric P = favourable area / total area	Darts, spinners, random points on segments.
P(E) + P(not E) = 1	"Find P(not ...)" or when P(E) is given directly.

7.4 The 120-Question Workout

Poora chapter ab ek hi ring mein: **120 questions**, chaar level zones mein banta hua. Warm-up se shuru karo, Boss Zone tak pahuncho. Har question ke turant baad boxed step-by-step solution hai - pehle khud try karo, atako toh solution dekho, aur "Why this step?" + Hinglish note se logic pakka karo. Agar in 120 ko solve kar liya toh is chapter ka koi bhi twisted question tumhe nahi rok payega.

Level 1 · Warm-up Zone: basics and confidence:

QUESTION 1 | Basic | Sample space (T1)

Q1. Write the sample space for: (i) a coin tossed once, (ii) a die rolled once, (iii) two coins tossed together, (iv) a coin tossed and a die rolled.

SOLUTION — step by step

Step 1. (i) S = {H, T}; n(S) = 2.

Step 2. (ii) S = {1, 2, 3, 4, 5, 6}; n(S) = 6.

Step 3. (iii) S = {HH, HT, TH, TT}; n(S) = 4 - HT and TH are different outcomes.

Step 4. (iv) Outcomes are pairs like (H, 1), (H, 2), ..., (T, 6): n(S) = 2 × 6 = 12.

» **Why this step?** Stages multiply: 2 × 6 = 12 outcomes for a compound experiment.

Answer: (i) 2 (ii) 6 (iii) 4 (iv) 12 outcomes

Hinglish: *Sample space likhna FIRST step hai. Do coins mein HT aur TH alag outcomes hain - 4 total. Coin + die: 2×6 = 12. Stages multiply hote hain.*

QUESTION 2 | Basic | Single die (T2)

Q2. A die is rolled once. Find P(i) an even number, (ii) a number greater than 4, (iii) a number that is even AND greater than 4.

SOLUTION — step by step

Step 1. n(S) = 6.

Step 2. (i) Even: {2, 4, 6}: P = 3/6 = 1/2.

Step 3. (ii) Greater than 4: {5, 6}: P = 2/6 = 1/3.

Step 4. (iii) Both: {6} only: P = 1/6.

» **Why this step?** 'AND' shrinks the event: only outcomes satisfying both conditions count.

Answer: (i) 1/2 (ii) 1/3 (iii) 1/6

Hinglish: *Even teen numbers, greater-than-4 do numbers, dono mein sirf 6. 1/2, 1/3, 1/6. AND ka matlab intersection.*

QUESTION 3 | Basic | Complement on a die (T5)

Q3. A die is rolled once. Find the probability of NOT getting a multiple of 3.

SOLUTION — step by step

Step 1. Multiples of 3 on a die: {3, 6}: P = 2/6 = 1/3.

Step 2. Complement: P(not multiple of 3) = 1 - 1/3 = 2/3.

Step 3. Direct check: {1, 2, 4, 5}: 4/6 = 2/3 ✓.

» **Why this step?** The complement rule converts the counting problem; both routes must agree.

Answer: 2/3

Hinglish: *Pehle multiples of 3: 1/3. Complement: 2/3. 'NOT' questions mein complement rule 5 second ka shortcut hai.*

QUESTION 4 | Basic | Two coins (T1/T8)

Q4. Two coins are tossed together. Find P(i) exactly one head, (ii) at least one head, (iii) no head.

SOLUTION — step by step

Step 1. S = {HH, HT, TH, TT}: n(S) = 4.

Step 2. (i) Exactly one head: {HT, TH}: P = 2/4 = 1/2.

Step 3. (ii) At least one head: {HH, HT, TH}: P = 3/4.

Step 4. (iii) No head: {TT}: P = 1/4. Note (ii) and (iii) are complements: 3/4 + 1/4 = 1 ✓.

» **Why this step?** 'At least' questions pair with 'none' through the complement - checking they sum to 1 validates both.

Answer: (i) 1/2 (ii) 3/4 (iii) 1/4

Hinglish: *Chaar outcomes. Exactly one head = HT, TH = 1/2. At least one = 3/4 (complement of TT). At least wale questions mein 'none' nikaalo, 1 minus karo.*

QUESTION 5 | Basic | Range and classification (T11)

Q5. A probability is given as: (i) 1.2, (ii) -0.3, (iii) 0, (iv) 1, (v) 7/8. Which values are impossible as probabilities, and what do (iii) and (iv) mean?

SOLUTION — step by step

Step 1. Law: every probability satisfies $0 \le P \le 1$.

Step 2. (i) $1.2 > 1$: impossible. (ii) $-0.3 < 0$: impossible.

Step 3. (iii) 0: an impossible event (can never happen).

Step 4. (iv) 1: a sure event (always happens).

Step 5. (v) 7/8 is between 0 and 1: a valid probability.

» **Why this step?** The range law is an instant validity check on every computed answer.

Answer: (i) and (ii) impossible; (iii) impossible event; (iv) sure event; (v) valid.

Hinglish: *Probability hamesha 0 aur 1 ke beech. 1.2 aur -0.3 galat - counts mein galti hogi. 0 = kabhi nahi, 1 = hamesha.*

QUESTION 6 | Basic | Single draw from a bag (T6)

Q6. A bag contains 4 red, 5 blue and 3 green balls. One ball is drawn at random. Find P(i) red, (ii) not green, (iii) red or blue.

SOLUTION — step by step

Step 1. Total = 4 + 5 + 3 = 12.

Step 2. (i) P(red) = 4/12 = 1/3.

Step 3. (ii) P(not green) = 1 - 3/12 = 9/12 = 3/4.

Step 4. (iii) P(red or blue) = (4+5)/12 = 9/12 = 3/4 (which matches (ii), since 'not green' = 'red or blue').

» **Why this step?** Disjoint categories add; 'not green' and 'red or blue' describe the same set here - a built-in consistency check.

Answer: (i) 1/3 (ii) 3/4 (iii) 3/4

Hinglish: *Total 12. Red 1/3. Not green 3/4. Red-or-blue bhi 3/4 - same cheez hai dono! Consistency check hamesha karo.*

QUESTION 7 | Basic | Card basics (T4)

Q7. One card is drawn from a well-shuffled deck of 52. Find P(i) an ace, (ii) a red card, (iii) a face card, (iv) the queen of hearts.

SOLUTION — step by step

Step 1. (i) Aces: 4 → P = 4/52 = 1/13.

Step 2. (ii) Red cards: 26 → P = 26/52 = 1/2.

Step 3. (iii) Face cards (J, Q, K): 3 × 4 = 12 → P = 12/52 = 3/13.

Step 4. (iv) One specific card: P = 1/52.

» **Why this step?** Four deck counts (4, 12, 13, 26) answer a third of all card questions instantly.

Answer: (i) 1/13 (ii) 1/2 (iii) 3/13 (iv) 1/52

Hinglish: *Yaad rakho: aces 4, reds 26, face cards 12, hearts 13. 1/13, 1/2, 3/13, 1/52. Ye counts ratta - deck questions 90% inhi par.*

QUESTION 8 | Basic | Sure and impossible events (T11)

Q8. A die is rolled. Classify as sure, impossible, or neither: E_1 = getting a number less than 7; E_2 = getting 8; E_3 = getting an odd number.

SOLUTION — step by step

Step 1. E_1: every outcome {1..6} satisfies it → P = 6/6 = 1: sure event.

Step 2. E_2: no outcome gives 8 → P = 0/6 = 0: impossible event.

Step 3. E_3: {1, 3, 5}: P = 1/2: neither sure nor impossible.

» **Why this step?** P = 1 and P = 0 are the defining marks of sure and impossible events.

Answer: E_1 sure, E_2 impossible, E_3 neither.

Hinglish: *Less than 7 = sab outcomes = sure. 8 = kabhi nahi = impossible. Odd = 50-50. Ye definitions direct marks dete hain.*

QUESTION 9 | Basic | Sum to 1 with missing probability (T13)

Q9. A random experiment has outcomes A, B, C with P(A) = 0.3 and P(B) = 0.25. Find P(C) if A, B, C are the only outcomes.

SOLUTION — step by step

Step 1. The probabilities of all outcomes sum to exactly 1.

Step 2. P(C) = 1 - 0.3 - 0.25 = 0.45.

» **Why this step?** Total probability = 1 is the accounting equation of probability theory.

Answer: 0.45

Hinglish: *Sab probabilities ka sum 1 hota hai. 1 - 0.55 = 0.45. Ye 'missing probability' pattern bahut common hai.*

QUESTION 10 | Basic | Two-dice sum intro (T3)

Q10. Two dice are rolled together. Find P(sum = 7) and P(sum = 2).

SOLUTION — step by step

Step 1. n(S) = 36 ordered pairs.

Step 2. Sum 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1): 6 pairs → P = 6/36 = 1/6.

Step 3. Sum 2: (1,1) only → P = 1/36.

» **Why this step?** The 36-grid shows sum 7 as the longest diagonal and sum 2 as a single corner.

Answer: 1/6 and 1/36

Hinglish: *36 outcomes. Sum 7 six tarike - sabse likely! 1/6. Sum 2 sirf (1,1) - 1/36. Grid banao rough mein, pairs ginn lo.*

QUESTION 11 | Basic | Sample space (T1)

Q11. Write the sample space for: (i) one toss of a coin, (ii) one roll of a die, (iii) two coins tossed together, (iv) the colour drawn from a bag containing only red and blue balls.

SOLUTION — step by step

Step 1. (i) S = {H, T} - 2 outcomes.

Step 2. (ii) S = {1, 2, 3, 4, 5, 6} - 6 outcomes.

Step 3. (iii) S = {HH, HT, TH, TT} - 4 outcomes (order matters in listing!).

Step 4. (iv) S = {red, blue} - 2 outcomes.

» **Why this step?** The sample space is the denominator of every probability in the experiment - get it right and half the work is done.

Answer: (i) {H, T} (ii) {1, 2, 3, 4, 5, 6} (iii) {HH, HT, TH, TT} (iv) {red, blue}

Hinglish: *Sample space = experiment ki poori guest list. Coin 2, dice 6, do coin 4 (HT aur TH alag-alag likhne hain!), bag jitne colours. List galat toh sab probability galat.*

QUESTION 12 | Basic | Single die (T2)

Q12. A die is rolled once. Find P(i) an even number, (ii) a prime number, (iii) a number greater than 4, (iv) a number ≤ 6.

SOLUTION — step by step

Step 1. Total outcomes: 6.

Step 2. (i) Even: {2, 4, 6} -> 3/6 = 1/2.

Step 3. (ii) Prime: {2, 3, 5} -> 3/6 = 1/2.

Step 4. (iii) Greater than 4: {5, 6} -> 2/6 = 1/3.

Step 5. (iv) ≤ 6: {1, 2, 3, 4, 5, 6} -> 6/6 = 1 (a sure event).

» **Why this step?** Count favourable, divide by total. The last one is the whole sample space - probability 1.

Answer: (i) 1/2 (ii) 1/2 (iii) 1/3 (iv) 1

Hinglish: *Favourable ÷ total. Even 3, prime 3, 5-6 do, aur '6 se chhota ya barabar' toh sab kuch - probability 1, pakka event.*

QUESTION 13 | Basic | Complement on a die (T5)

Q13. A die is rolled once. Find: (i) P(not 3), (ii) P(not a prime), (iii) P(a number less than 7), and connect (i) to the complement rule.

SOLUTION — step by step

Step 1. (i) P(3) = 1/6 -> P(not 3) = 1 - 1/6 = 5/6.

Step 2. (ii) Primes are {2, 3, 5}: P(prime) = 1/2 -> P(not prime) = 1/2.

Step 3. (iii) Every face is < 7: P = 1 - a sure event.

Step 4. (i) uses P(not E) = 1 - P(E), the complement rule.

» **Why this step?** 'Not E' flips the count: 6 – favourable. The complement rule is the fastest route whenever E is easier to count.

Answer: (i) 5/6 (ii) 1/2 (iii) 1

Hinglish: *'Not' dikhe toh 1 minus lagao. P(not 3) = 5/6. Aur '7 se kam' toh hamesha sach - sure event, probability 1.*

QUESTION 14 | Basic | Two coins (T1/T8)

Q14. Two coins are tossed together. Find: (i) P(two heads), (ii) P(exactly one head), (iii) P(at least one head), (iv) P(no head).

SOLUTION — step by step

Step 1. Sample space: {HH, HT, TH, TT} - 4 equally likely outcomes.

Step 2. (i) HH only -> 1/4.

Step 3. (ii) HT, TH -> 2/4 = 1/2.

Step 4. (iii) HH, HT, TH -> 3/4 (or 1 - P(no head) = 1 - 1/4).

Step 5. (iv) TT -> 1/4.

» **Why this step?** (iii) and (iv) are complements: at least one head + no head = everything.

Answer: (i) 1/4 (ii) 1/2 (iii) 3/4 (iv) 1/4

Hinglish: *4 outcomes, HT aur TH alag hain - ye yaad rakho. 'At least one' = 3/4, 'no head' = 1/4, dono ka jod 1. Complement pair spot karo.*

QUESTION 15 | Basic | Range and classification (T1)

Q15. Can each be a probability of an event? Answer yes/no with reason: 0.8, 1.5, -0.2, 7/8, 0, 42%.

SOLUTION — step by step

Step 1. Valid probabilities live in [0, 1].

Step 2. 0.8 yes; 7/8 ≈ 0.875 yes; 0 yes (impossible event); 42% = 0.42 yes.

Step 3. 1.5 NO - above 1; -0.2 NO - below 0.

» **Why this step?** Probability is a share of the whole: never less than nothing, never more than everything.

Answer: Yes: 0.8, 7/8, 0, 42%. No: 1.5, -0.2.

Hinglish: *Range [0, 1] - iske bahar sab reject. 1.5 aur -0.2 bahar, baaki sab legal. 42% bhi probability hai (0.42).*

QUESTION 16 | Basic | Single draw from a bag (T6)

Q16. A bag has 5 red, 3 blue and 2 green balls. One ball is drawn. Find: (i) P(red), (ii) P(blue), (iii) P(not green), (iv) P(red or blue).

SOLUTION — step by step

Step 1. Total = 10.

Step 2. (i) 5/10 = 1/2.

Step 3. (ii) 3/10.

Step 4. (iii) P(not green) = 1 - 2/10 = 8/10 = 4/5.

Step 5. (iv) P(red or blue) = 5/10 + 3/10 = 8/10 = 4/5 - same as (iii), because 'not green' IS 'red or blue'!

» **Why this step?** (iii) and (iv) describe the same event in two languages - a good probability answer often has two faces.

Answer: (i) 1/2 (ii) 3/10 (iii) 4/5 (iv) 4/5 (same event)

Hinglish: *Total 10. Aur dekho - 'green nahi' aur 'red ya blue' ek hi baat hai, dono 4/5. Events ke naam badal sakte hain, probability nahi.*

QUESTION 17 | Basic | Card basics (T4)

Q17. One card is drawn from a well-shuffled deck of 52. Find: (i) P(an ace), (ii) P(a red king), (iii) P(a spade), (iv) P(a numbered card 2-10).

SOLUTION — step by step

Step 1. Deck facts: 4 suits × 13; 4 aces; 26 red; 13 spades; per suit 3 face + 1 ace + 9 numbers.

Step 2. (i) 4/52 = 1/13.

Step 3. (ii) Red kings = 2 -> 2/52 = 1/26.

Step 4. (iii) 13/52 = 1/4.

Step 5. (iv) 9 per suit × 4 = 36 -> 36/52 = 9/13.

» **Why this step?** Five deck counts (4 aces, 12 face, 26 red, 13 per suit, 9 numbered per suit) answer almost every card question.

Answer: (i) 1/13 (ii) 1/26 (iii) 1/4 (iv) 9/13

Hinglish: *Deck ki 5 magic numbers ratt lo: 4 aces, 12 face, 26 red, 13 per suit, 9 numbered per suit. Baaki bas divide karna hai.*

QUESTION 18 | Basic | Sum to 1 (T1)

Q18. A random experiment has exactly four outcomes A, B, C, D with P(A) = 0.25, P(B) = 0.35, P(C) = 0.15. Find P(D) and verify the total.

SOLUTION — step by step

Step 1. All probabilities must sum to 1: 0.25 + 0.35 + 0.15 + P(D) = 1.

Step 2. Sum so far: 0.75 -> P(D) = 0.25.

Step 3. Verify: 0.25 + 0.35 + 0.15 + 0.25 = 1.00 ✓.

» **Why this step?** The 'sum to 1' law turns a missing probability into a subtraction.

Answer: P(D) = 0.25

Hinglish: *Total hamesha 1: 0.75 ho gaya, bachi 0.25. Missing probability = 1 minus jo hai. Ye rule har table-question mein kaam aata hai.*

QUESTION 19 | Basic | Two-dice sum intro (T3)

Q19. Two dice are rolled together. Find: (i) P(sum = 7), (ii) P(sum = 2), (iii) P(both dice show the same number).

SOLUTION — step by step

Step 1. Sample space: 36 ordered pairs.

Step 2. (i) Sum 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) - 6 ways -> 6/36 = 1/6.

Step 3. (ii) Sum 2: only (1,1) -> 1/36.

Step 4. (iii) Doubles: (1,1) ... (6,6) - 6 ways -> 6/36 = 1/6.

» **Why this step?** Sum 7 has the MOST routes (6) - it is the celebrity of two-dice problems; sums at the edges (2, 12) are rarest.

Answer: (i) 1/6 (ii) 1/36 (iii) 1/6

Hinglish: *36 pairs. Sum 7 ka raja - 6 raste, 1/6. Sum 2 sirf (1,1) - 1/36. Doubles bhi 6 tarike - 1/6. 6×6 ka grid banao, sab dikh jayega.*

QUESTION 20 | Basic | Sure and impossible events (T1)

Q20. A die is rolled. Classify as sure, impossible or possible: (i) getting a number ≤ 6, (ii) getting a 9, (iii) getting a multiple of 3, (iv) an event with probability 0.999....

SOLUTION — step by step

Step 1. (i) Every face qualifies -> sure, P = 1.

Step 2. (ii) No face is 9 -> impossible, P = 0.

Step 3. (iii) {3, 6} -> possible, P = 2/6 = 1/3.

Step 4. (iv) 0.999... = 1 exactly (decimal identity) -> sure event!

» **Why this step?** P = 1 and P = 0 are the boundary markers - and 0.999... = 1 sneaks a sure event in disguise.

Answer: (i) sure (P = 1) (ii) impossible (P = 0) (iii) possible (1/3) (iv) sure (0.999... = 1)

Hinglish: *Sure = 1, impossible = 0, beech ka sab possible. Aur 0.999... wala toh 1 hi hai - sure event disguise mein! Decimal wala sach probability mein bhi kaam karta hai.*

QUESTION 21 | Basic | Three coins (T8)

Q21. Three coins are tossed together. (i) Write n(S). (ii) Find P(exactly two heads). (iii) Find P(at least one tail).

SOLUTION — step by step

Step 1. (i) n(S) = 2^3 = 8: {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}.

Step 2. (ii) Exactly two heads: HHT, HTH, THH -> 3/8.

Step 3. (iii) At least one tail = 1 - P(no tail) = 1 - 1/8 = 7/8.

» **Why this step?** n coins give 2^n outcomes; 'at least' questions melt through the complement.

Answer: (i) 8 (ii) 3/8 (iii) 7/8

Hinglish: *3 coin = 8 outcomes. Exactly 2 heads = 3/8. 'At least one tail' ulta socho: sab head sirf 1 tarika -> 7/8. Complement hi shortcut hai.*

QUESTION 22 | Basic | Empirical probability (T9)

Q22. In a class of 60 students, 21 say maths is their favourite subject. (i) Find the empirical probability that a randomly chosen student loves maths. (ii) In a school of 300 students, about how many would you predict?

SOLUTION — step by step

Step 1. (i) P = 21/60 = 7/20 = 0.35.

Step 2. (ii) Expected count = 0.35 × 300 = 105 students.

» **Why this step?** Empirical probability = observed fraction; multiply by a new population to predict counts.

Answer: (i) 7/20 = 0.35 (ii) about 105 students

Hinglish: *21/60 = 0.35. Naye school mein 0.35 × 300 = 105. Data se probability, probability se prediction - yehi empirical ka kaam hai.*

QUESTION 23 | Basic | Empirical vs theoretical (T9)

Q23. A die is thrown 200 times and 'six' appears 35 times. (i) Find the empirical probability of a six. (ii) Compare with the theoretical value.

SOLUTION — step by step

Step 1. (i) Empirical P = 35/200 = 7/40 = 0.175.

Step 2. (ii) Theoretical P = 1/6 ≈ 0.167.

Step 3. They are close: 0.175 vs 0.167 - small data sets wobble around the true value.

» **Why this step?** Empirical values converge to theoretical ones as trials grow - short-run deviations are normal, not errors.

Answer: (i) 0.175 (ii) theoretical 1/6 ≈ 0.167 - close, as expected

Hinglish: *0.175 vs 0.167 - almost twin! Chhote data mein thoda natkhati hoti hai, zyada throws par theoretical ke kareeb aati hai. Ye difference exam mein explain karna hota hai.*

QUESTION 24 | Basic | Coin + die together (T8)

Q24. A coin is tossed and a die is rolled. Find: (i) P(heads and a six), (ii) P(tails and an odd number), (iii) n(S).

SOLUTION — step by step

Step 1. (iii) n(S) = 2 × 6 = 12 combined outcomes.

Step 2. (i) P(H) × P(6) = ½ × 1/6 = 1/12 (independent stages multiply).

Step 3. (ii) P(T) × P(odd) = ½ × 3/6 = 1/4.

» **Why this step?** Independent stages multiply probabilities; the sample space multiplies sizes (2 × 6 = 12).

Answer: (i) 1/12 (ii) 1/4 (iii) 12

Hinglish: *Do alag experiments = multiply. 12 outcomes, head+six = 1/12, tail+odd = 1/4. Stage-wise socho, seedha multiply.*

QUESTION 25 | Basic | Spinner quarters (T10)

Q25. A spinner is divided into 4 equal quarters labelled 1, 2, 3, 4. Find: (i) P(3), (ii) P(a number greater than 2), (iii) P(not 1).

SOLUTION — step by step

Step 1. (i) 1/4.

Step 2. (ii) {3, 4} -> 2/4 = 1/2.

Step 3. (iii) 1 - P(1) = 1 - 1/4 = 3/4.

» **Why this step?** Equal sectors give equal probabilities - the geometric version of a fair die.

Answer: (i) 1/4 (ii) 1/2 (iii) 3/4

Hinglish: *Barabar chaar tukde -> har tukda 1/4. '2 se bada' = do tukde = 1/2. '1 nahi' = 3/4. Spinner = gol dice.*

QUESTION 26 | Basic | True/False theory (T14)

Q26. True or False (one-line reason): (i) P(E) + P(not E) = 1. (ii) The probability of a sure event is 1. (iii) 13/10 can be a probability. (iv) For n equally likely outcomes each has probability 1/n. (v) P(E) = -0.5 is possible for some event.

SOLUTION — step by step

Step 1. (i) TRUE - the complement law.

Step 2. (ii) TRUE - by definition of 'sure'.

Step 3. (iii) FALSE - 13/10 > 1.

Step 4. (iv) TRUE - that is the classical definition.

Step 5. (v) FALSE - probabilities never go below 0.

» **Why this step?** The [0, 1] range and the sum-to-1 law are the two gates every probability must pass.

Answer: (i) T (ii) T (iii) F (iv) T (v) F

Hinglish: *Do hi gates: [0, 1] range aur sum = 1. (iii) aur (v) dono gate se bahar - reject. Baaki sab definition hi hai.*

QUESTION 27 | Basic | Bag single draw (T6)

Q27. A bag has 4 white and 6 black balls. One ball is drawn. Find: (i) P(white), (ii) P(black), (iii) P(not white) - and show (ii) and (iii) are the same event.

SOLUTION — step by step

Step 1. Total = 10.

Step 2. (i) 4/10 = 2/5.

Step 3. (ii) 6/10 = 3/5.

Step 4. (iii) P(not white) = 1 - 2/5 = 3/5.

Step 5. (ii) = (iii): the only colour that is 'not white' here is black.

» **Why this step?** With two colours, 'not white' and 'black' are literally the same event - complement in disguise.

Answer: (i) 2/5 (ii) 3/5 (iii) 3/5 (= P(black))

Hinglish: *Do colour ki duniya mein 'white nahi' = 'black'. Dono 3/5. Events ke naam alag, cheez ek - probability same.*

QUESTION 28 | Basic | Die categories (T2)

Q28. A die is rolled once. Find: (i) P(a multiple of 3), (ii) P(a factor of 6), (iii) P(a factor of 6 that is even).

SOLUTION — step by step

Step 1. (i) Multiples of 3: {3, 6} -> 2/6 = 1/3.

Step 2. (ii) Factors of 6: {1, 2, 3, 6} -> 4/6 = 2/3.

Step 3. (iii) Even factors of 6: {2, 6} -> 2/6 = 1/3.

» **Why this step?** List the set FIRST, count SECOND - 'factor' and 'multiple' sets look similar but are not.

Answer: (i) 1/3 (ii) 2/3 (iii) 1/3

Hinglish: *Multiple of 3: {3, 6}. Factor of 6: {1, 2, 3, 6} - 1 bhi factor hai, bhoolna mat. Even factors: {2, 6}. Set likho, confusion khatam.*

QUESTION 29 | Basic | Cards: face and suit (T4)

Q29. One card is drawn from a deck of 52. Find: (i) P(a face card), (ii) P(a heart or a spade), (iii) P(a queen).

SOLUTION — step by step

Step 1. (i) Face cards: 3 per suit × 4 = 12 -> 12/52 = 3/13.

Step 2. (ii) Hearts + spades = 13 + 13 = 26 -> 26/52 = 1/2 (all the black + red split!).

Step 3. (iii) Queens: 4 -> 4/52 = 1/13.

» **Why this step?** Heart-or-spade is exactly 'red card' - one of the deck's built-in half-and-half splits.

Answer: (i) 3/13 (ii) 1/2 (iii) 1/13

Hinglish: *Face 12, queen 4, aur heart+spade = red half = 1/2. Deck ke splits yaad karo, card questions 10 second mein.*

QUESTION 30 | Basic | Complement with colours (T5)

Q30. A box has 8 green and 7 yellow marbles. One marble is drawn. Find: (i) P(green), (ii) P(yellow), (iii) P(not yellow), (iv) verify P(green) + P(yellow) = 1.

SOLUTION — step by step

Step 1. Total = 15.

Step 2. (i) 8/15.

Step 3. (ii) 7/15.

Step 4. (iii) P(not yellow) = 1 - 7/15 = 8/15 = P(green) - the complement of yellow IS green here.

Step 5. (iv) 8/15 + 7/15 = 15/15 = 1 ✓.

» **Why this step?** With exactly two outcomes, each is the other's complement - their probabilities must sum to 1.

Answer: (i) 8/15 (ii) 7/15 (iii) 8/15 (iv) verified = 1

Hinglish: *15 total, 8/15 aur 7/15 - jod 1. 'Yellow nahi' matlab green. Do-outcome experiments mein complement khud dusra colour hota hai.*

Level 2 · Core Drills: book-level practice:

QUESTION 31 | Moderate | Two-dice conditions (T3)

Q31. Two dice are rolled. Find P(i) the sum is a prime number, (ii) the product is odd, (iii) both dice show the same number.

SOLUTION — step by step

Step 1. (i) Prime sums: 2 (1 pair), 3 (2), 5 (4), 7 (6), 11 (2): total 15 pairs → P = 15/36 = 5/12.

Step 2. (ii) Product odd ⇔ both dice odd: 3 × 3 = 9 pairs → P = 9/36 = 1/4.

Step 3. (iii) Doubles: (1,1), ..., (6,6): 6 pairs → P = 6/36 = 1/6.

» **Why this step?** Products split by parity cleanly: odd needs odd×odd; even is the complement 27/36.

Answer: (i) 5/12 (ii) 1/4 (iii) 1/6

Hinglish: *Prime sums ka count: 1+2+4+6+2 = 15 → 5/12. Product odd = dono odd = 1/4. Doubles 1/6. 36-grid se teeno direct padho.*

QUESTION 32 | Moderate | Cards with overlap (T4)

Q32. One card is drawn from a deck of 52. Find P(i) a king or a heart, (ii) a red face card, (iii) neither a king nor a queen.

SOLUTION — step by step

Step 1. (i) Kings: 4; hearts: 13; king of hearts counted twice → 4 + 13 - 1 = 16 → P = 16/52 = 4/13.

Step 2. (ii) Red face cards: J, Q, K of hearts and diamonds: 6 → P = 6/52 = 3/26.

Step 3. (iii) Kings and queens: 8 cards; neither: 52 - 8 = 44 → P = 44/52 = 11/13.

» **Why this step?** Overlap correction (add both counts, subtract the intersection) is the whole skill in (i).

Answer: (i) 4/13 (ii) 3/26 (iii) 11/13

Hinglish: *King-or-heart mein 4+13 = 17 galat - king of hearts do baar gina! 16 sahi → 4/13. Red face 6. Neither king nor queen 11/13.*

QUESTION 33 | Moderate | Complement with cards (T5)

Q33. One card is drawn from a deck. Find P(i) not an ace, (ii) not a face card, (iii) neither red nor a king.

SOLUTION — step by step

Step 1. (i) P(not ace) = 1 - 4/52 = 48/52 = 12/13.

Step 2. (ii) P(not face) = 1 - 12/52 = 40/52 = 10/13.

Step 3. (iii) P(red or king) = 26 + 4 - 2 = 28 (two red kings shared) → P(neither) = 1 - 28/52 = 24/52 = 6/13.

» **Why this step?** Each 'neither/nor' question is one complement away from an 'or' question.

Answer: (i) 12/13 (ii) 10/13 (iii) 6/13

Hinglish: *Not ace 12/13, not face 10/13. (iii) mein pehle red-or-king: 28, complement 24/52 = 6/13. Overlap correction - red kings do hain.*

QUESTION 34 | Moderate | Empirical probability (T9)

Q34. In a survey of 400 students, 96 walk to school, 160 take a bus, 84 cycle and the rest come by car. A student is chosen at random. Find P(i) walks, (ii) does not cycle, (iii) comes by car.

SOLUTION — step by step

Step 1. Car users: 400 - 96 - 160 - 84 = 60.

Step 2. (i) P(walks) = 96/400 = 0.24.

Step 3. (ii) P(not cycle) = 1 - 84/400 = 316/400 = 0.79.

Step 4. (iii) P(car) = 60/400 = 0.15.

» **Why this step?** Empirical probability = occurrences ÷ trials; the missing category comes from the total.

Answer: (i) 0.24 (ii) 0.79 (iii) 0.15

Hinglish: *Car wale: 60. 0.24, 0.79, 0.15. Sum check: 0.24+0.40+0.21+0.15 = 1 ✓. Empirical mein decimals normal hain.*

QUESTION 35 | Moderate | Three coins (T8)

Q35. Three coins are tossed together. Find P(i) exactly two heads, (ii) at least two heads, (iii) all three same.

SOLUTION — step by step

Step 1. S has $2^3 = 8$ outcomes.

Step 2. (i) Exactly two heads: HHT, HTH, THH → 3/8.

Step 3. (ii) At least two heads: those 3 plus HHH → 4/8 = 1/2.

Step 4. (iii) All same: HHH, TTT → 2/8 = 1/4.

» **Why this step?** 'Exactly' and 'at least' differ by the all-heads outcome - read the wording like a lawyer.

Answer: (i) 3/8 (ii) 1/2 (iii) 1/4

Hinglish: *8 outcomes. Exactly 2: teen paths = 3/8. At least 2: +HHH = 1/2. All same 1/4. Tree diagram banao - 8 paths dikh jayengi.*

QUESTION 36 | Moderate | At least one (T5)

Q36. A coin is tossed 4 times (or four coins together). Find P(at least one tail).

SOLUTION — step by step

Step 1. Total outcomes: $2^4 = 16$.

Step 2. Complement: no tail means all heads: 1 outcome.

Step 3. P(at least one tail) = 1 - 1/16 = 15/16.

» **Why this step?** 'At least one' almost always means '1 minus none' - the only exception is when 'none' is harder to count than the direct list.

Answer: 15/16

Hinglish: *At least one tail = 1 - (no tail). No tail = HHHH = 1/16. Answer 15/16. Complement rule ka best use.*

QUESTION 37 | Moderate | Bag: two draws, same colour (T7)

Q37. A bag has 5 red and 3 blue balls. Two balls are drawn one after another WITHOUT replacement. Find P(both red).

SOLUTION — step by step

Step 1. First red: 5/8. After removing one red, second red: 4/7.

Step 2. P(both red) = (5/8) × (4/7) = 20/56 = 5/14.

Step 3. Combination check: C(5,2)/C(8,2) = 10/28 = 5/14 ✓.

» **Why this step?** Without replacement, the second draw's denominator drops to 7 - the shrinking universe is the whole idea.

Answer: 5/14

Hinglish: *Pehla red 5/8, doosra red 4/7 (ek red ja chuka!). Multiply: 5/14. Denominator 8 se 7 hua - yahi 'without replacement' hai.*

QUESTION 38 | Moderate | Bag: different colours in any order (T7)

Q38. A bag has 5 red and 3 blue balls. Two balls are drawn without replacement. Find P(one red and one blue, in any order).

SOLUTION — step by step

Step 1. Order RB: (5/8)(3/7) = 15/56.

Step 2. Order BR: (3/8)(5/7) = 15/56.

Step 3. Any order: 15/56 + 15/56 = 30/56 = 15/28.

Step 4. Combination check: C(5,1)C(3,1)/C(8,2) = 15/28 ✓.

» **Why this step?** 'In any order' = sum over the distinct orders; the ×2 is just this sum when the two orders have equal probability.

Answer: 15/28

Hinglish: *Do orders - RB aur BR. Dono 15/56. Jodo: 15/28. 'In any order' dikhe toh ×2 ya dono orders likho.*

QUESTION 39 | Moderate | Two dice: at least conditions (T3/T5)

Q39. Two dice are rolled. Find P(i) the sum is at least 10, (ii) the sum is not 7.

SOLUTION — step by step

Step 1. (i) Sum ≥ 10: (4,6), (5,5), (6,4), (5,6), (6,5), (6,6): 6 pairs → P = 6/36 = 1/6.

Step 2. (ii) Sum 7 has 6 pairs → P(not 7) = 1 - 6/36 = 30/36 = 5/6.

» **Why this step?** Boundary sums (10, 11, 12) count outward from the corner; complements handle the 'not'.

Answer: (i) 1/6 (ii) 5/6

Hinglish: *Sum ≥ 10: chhe pairs → 1/6. Not 7: complement of the most-likely sum → 5/6.*

QUESTION 40 | Moderate | Empirical with complement (T9)

Q40. A quality inspector tests 500 bulbs: 35 are defective. One bulb is chosen at random from this batch. Find P(i) defective, (ii) good; (iii) if the same defect rate continues, estimate the number of defectives in a batch of 6000.

SOLUTION — step by step

Step 1. (i) P(defective) = 35/500 = 0.07.

Step 2. (ii) P(good) = 1 - 0.07 = 0.93.

Step 3. (iii) Expected defectives = 0.07 × 6000 = 420.

» **Why this step?** Empirical probabilities extrapolate linearly to new batch sizes.

Answer: (i) 0.07 (ii) 0.93 (iii) 420 bulbs

Hinglish: *0.07 defective, 0.93 good. 6000 mein 420 kharab. 'Estimate' questions mein rate × new size.*

QUESTION 41 | Moderate | Two-dice sum conditions (T3)

Q41. Two dice are rolled. Find: (i) P(sum ≤ 4), (ii) P(sum ≥ 10).

SOLUTION — step by step

Step 1. (i) Sum 2: 1 way; sum 3: 2 ways; sum 4: 3 ways -> total 6 -> 6/36 = 1/6.

Step 2. (ii) Sum 10: 3 ways; sum 11: 2 ways; sum 12: 1 way -> 6/36 = 1/6.

Step 3. Symmetry: the two answers match because the sum distribution is symmetric about 7.

» Why this step? Small sums and their mirror-image big sums have equal counts - use the 6×6 grid's diagonal symmetry.

Answer: (i) 1/6 (ii) 1/6

Hinglish: *Neeche se: 1 + 2 + 3 = 6 tarika. Upar se: 3 + 2 + 1 = 6. Dono 1/6 - distribution 7 ke around symmetric hai. Grid par diagonal dekho, symmetry khud dikh jaati hai.*

QUESTION 42 | Moderate | Two-dice parity (T3)

Q42. Two dice are rolled. Find: (i) P(one shows even and the other odd), (ii) P(both odd).

SOLUTION — step by step

Step 1. (i) First even second odd: 3 × 3 = 9 pairs; first odd second even: 9 -> total 18 -> 18/36 = 1/2.

Step 2. (ii) Both odd: 3 × 3 = 9 -> 9/36 = 1/4.

Step 3. Sanity: 1/2 + 1/4 + P(both even = 1/4) = 1 ✓.

» Why this step? Parity splits the grid into three clean blocks: mixed (18), odd-odd (9), even-even (9).

Answer: (i) 1/2 (ii) 1/4

Hinglish: *Mixed parity sabse bada block (18 pairs) -> 1/2. Odd-odd aur even-even 9-9. Teen block ka jod 1 - partition ka proof.*

QUESTION 43 | Moderate | Cards with overlap (T4)

Q43. One card is drawn from a deck of 52. Find: (i) P(a king or a queen), (ii) P(a heart or a king).

SOLUTION — step by step

Step 1. (i) Kings + queens, no overlap: 4 + 4 = 8 -> 8/52 = 2/13.

Step 2. (ii) Hearts (13) + kings (4) overlap in the king of hearts (1): 13 + 4 - 1 = 16 -> 16/52 = 4/13.

» **Why this step?** 'Or' means add MINUS the double-counted overlap - the king of hearts was hiding in both piles.

Answer: (i) 2/13 (ii) 4/13

Hinglish: *(i) mein koi overlap nahi -> seedha jod. (ii) mein heart ka king dono list mein hai - ek minus karo: 16/52 = 4/13. Overlap dhundo, warna double count!*

QUESTION 44 | Moderate | Bag: two draws same colour (T7)

Q44. A bag has 5 red and 3 blue balls. Two balls are drawn one after another WITHOUT replacement. Find: (i) P(both red), (ii) P(one of each colour).

SOLUTION — step by step

Step 1. (i) First red: 5/8; then second red: 4/7 (one red gone, one ball gone). Product: 20/56 = 5/14.

Step 2. (ii) Two orders: red-then-blue (5/8 × 3/7) or blue-then-red (3/8 × 5/7).

Step 3. Sum: 15/56 + 15/56 = 30/56 = 15/28.

» **Why this step?** Without replacement shrinks BOTH numerator and denominator on the second draw - and 'one of each' needs BOTH orders added.

Answer: (i) 5/14 (ii) 15/28

Hinglish: *Replacement nahi -> doosri draw mein ball kam (7), red kam (4). Both red: 5/14. Ek-laal-ek-neela = dono order ka jod: 15/28. Order bhoolna = aadha answer.*

QUESTION 45 | Moderate | Three coins conditions (T8)

Q45. Three coins are tossed. Find: (i) P(at least two heads), (ii) P(all three show the same face).

SOLUTION — step by step

Step 1. (i) Exactly two heads (3 ways) + exactly three (1 way) = 4/8 = 1/2.

Step 2. (ii) HHH or TTT -> 2/8 = 1/4.

» **Why this step?** 'At least two' = exactly two + exactly three - chop the event into exact slices and add.

Answer: (i) 1/2 (ii) 1/4

Hinglish: *At least 2 = exactly 2 (3 tarika) + exactly 3 (1 tarika) = 4/8 = 1/2. Same face = sab head ya sab tail = 1/4. Slices mein kaato.*

QUESTION 46 | Moderate | At least one six (T5)

Q46. Two dice are rolled. Find P(at least one die shows a six).

SOLUTION — step by step

Step 1. Complement first: P(no six) = (5/6) × (5/6) = 25/36.

Step 2. P(at least one six) = 1 - 25/36 = 11/36.

Step 3. Cross-check by counting: pairs containing a 6: 11 (row 6: six pairs + column 6: five more) -> 11/36 ✓.

» **Why this step?** 'At least one' almost always means 1 − P(none) - two multiplications instead of a long count.

Answer: 11/36

Hinglish: *At least one six? Ulta jao: no-six = 25/36, toh answer 11/36. Counting se bhi 11 nikalta hai - dono raste, ek jawab. Complement hi king hai.*

QUESTION 47 | Moderate | Empirical table (T9)

Q47. 80 darts hit a board: 20 land in the bull's eye, 30 in the outer ring, 30 miss the board completely. (i) P(a hit is in the bull's eye). (ii) P(not bull's eye). (iii) In the next 200 throws, how many bull's eyes are expected?

SOLUTION — step by step

Step 1. (i) 20/80 = 1/4.

Step 2. (ii) 1 - 1/4 = 3/4 (the 60 non-bull throws over 80).

Step 3. (iii) 0.25 × 200 = 50 bull's eyes expected.

» **Why this step?** Empirical probability scales to any new batch of trials by simple multiplication.

Answer: (i) 1/4 (ii) 3/4 (iii) 50

Hinglish: *20/80 = 1/4. 'Not bull' = 3/4. Agle 200 mein 50 expected. Data → probability → prediction: teen step, ek hi formula.*

QUESTION 48 | Moderate | Replacement vs not (T7/T14)

Q48. A bag has 4 red and 2 green balls. Two balls are drawn. Compare P(both red) when draws are (i) WITH replacement, (ii) WITHOUT replacement.

SOLUTION — step by step

Step 1. (i) With replacement: (4/6) × (4/6) = 16/36 = 4/9.

Step 2. (ii) Without: (4/6) × (3/5) = 12/30 = 2/5.

Step 3. Compare: 4/9 ≈ 0.444 > 2/5 = 0.4 - replacing keeps the bag 'rich' in red, so both-red is likelier.

» **Why this step?** Without replacement removes a red ball, cutting the second draw's numerator AND denominator - the probability dips.

Answer: (i) 4/9 (ii) 2/5; with replacement is higher

Hinglish: *Replacement se bag ki richness barkarar rehti hai: 4/9. Bina replacement red kam ho gaya: 2/5. Fark chhota par real - 0.444 vs 0.4.*

QUESTION 49 | Moderate | Sum distribution intro (T3)

Q49. Two dice are rolled. (i) Find P(sum = 5) and P(sum = 11). (ii) Which sum is the single most likely, and why?

SOLUTION — step by step

Step 1. (i) Sum 5: (1,4), (2,3), (3,2), (4,1) -> 4/36 = 1/9. Sum 11: (5,6), (6,5) -> 2/36 = 1/18.

Step 2. (ii) Sum 7 - six routes (1,6)...(6,1) -> P = 6/36 = 1/6, the highest of all sums.

Step 3. Reason: 7 sits at the centre of the counting pyramid - every pair (a, b) with a + b = 7 has both entries free to vary from 1 to 6.

» **Why this step?** The number of ways rises to the middle (sum 7) and falls symmetrically - the shape of the 6×6 grid.

Answer: (i) 1/9 and 1/18 (ii) sum 7, with 6 ways (P = 1/6)

Hinglish: *Sum 5 = 4 tarika (1/9), sum 11 = 2 tarika (1/18). Aur king phir se 7: 6 tarika. Pyramid shape - beech ka sabse mota.*

QUESTION 50 | Moderate | Two-dice difference (T3)

Q50. Two dice are rolled. Let D be the absolute difference |a − b|. Find: (i) P(D = 0), (ii) P(D = 1).

SOLUTION — step by step

Step 1. (i) D = 0 means doubles: 6 pairs -> 6/36 = 1/6.

Step 2. (ii) D = 1 pairs: (1,2), (2,1), (2,3), (3,2), (3,4), (4,3), (4,5), (5,4), (5,6), (6,5) -> 10 pairs -> 10/36 = 5/18.

» **Why this step?** Difference d has exactly 2(6 − d) pairs for d ≥ 1 - counts fall as the gap grows.

Answer: (i) 1/6 (ii) 5/18

Hinglish: *Difference 0 = doubles (6 pairs). Difference 1 = 10 pairs - sabse common difference! Formula 2(6-d): d jitna bada, utne kam pairs.*

QUESTION 51 | Moderate | Three draws without replacement (T7)

Q51. A bag has 4 white and 6 black balls. Three balls are drawn one by one without replacement. Find: (i) P(all three white), (ii) P(all three black).

SOLUTION — step by step

Step 1. (i) (4/10) × (3/9) × (2/8) = 24/720 = 1/30.

Step 2. (ii) (6/10) × (5/9) × (4/8) = 120/720 = 1/6.

» **Why this step?** Each draw shrinks the total by 1 and the favoured colour by 1 - chain the three fractions.

Answer: (i) 1/30 (ii) 1/6

Hinglish: *10 -> 9 -> 8: har draw pe total ghatta. White chain: 4, 3, 2 upar -> 1/30. Black chain: 6, 5, 4 -> 1/6. Denominator hamesha 10×9×8 = 720.*

QUESTION 52 | Moderate | Two aces from a deck (T7)

Q52. Two cards are drawn from a deck of 52 without replacement. Find P(both are aces).

SOLUTION — step by step

Step 1. First ace: 4/52. Second ace: 3/51.

Step 2. Product: 12/2652 = 1/221.

» **Why this step?** The deck shrinks to 51 and the aces to 3 - the classic without-replacement pair.

Answer: 1/221

Hinglish: *4/52 phir 3/51 - deck ek kam, ace ek kam. 1/221. Ye answer deck ke two-ace questions ka fixed tag hai.*

QUESTION 53 | Moderate | Geometric: point on a segment (T10)

Q53. A point P is chosen at random on a 12 cm segment AB. Find: (i) P(P lies within 3 cm of A), (ii) P(P lies in the middle 6 cm).

SOLUTION — step by step

Step 1. Geometric probability = favourable length / total length.

Step 2. (i) 3/12 = 1/4.

Step 3. (ii) 6/12 = 1/2.

» **Why this step?** For random points on a line, length replaces counting - the ratio of lengths is the probability.

Answer: (i) 1/4 (ii) 1/2

Hinglish: *Line par random point = lambai ka ratio. 3/12 = 1/4, beech ka 6/12 = 1/2. Counting nahi, measuring!*

QUESTION 54 | Moderate | Geometric: dartboard (T10)

Q54. A dart hits a square board of side 24 cm. Inside it is a smaller concentric square of side 8 cm. Find P(the dart lands in the inner square).

SOLUTION — step by step

Step 1. Total area = 24^2 = 576 cm^2.

Step 2. Inner area = 8^2 = 64 cm^2.

Step 3. P = 64/576 = 1/9.

» **Why this step?** Area ratio is the probability - squares make the areas instant.

Answer: 1/9

Hinglish: *Area ka ratio hi probability: 64/576 = 1/9. Side ka ratio 1/3 tha, area mein square ho gaya - 1/9. Dimension dhyan se pakdo.*

QUESTION 55 | Moderate | Two-dice prime sum (T3)

Q55. Two dice are rolled. Find P(the sum is a prime number).

SOLUTION — step by step

Step 1. Prime sums: 2, 3, 5, 7, 11.

Step 2. Ways: sum 2 -> 1; sum 3 -> 2; sum 5 -> 4; sum 7 -> 6; sum 11 -> 2. Total = 15.

Step 3. P = 15/36 = 5/12.

» **Why this step?** List the prime sums, add their route counts, divide by 36 - the grid does the rest.

Answer: 5/12

Hinglish: *Prime sums ka pyramid: 1+2+4+6+2 = 15. 15/36 = 5/12. Har sum ke ways yaad ho toh aisa question 30 second ka hai.*

QUESTION 56 | Moderate | Neither / exactly one (T3/T5)

Q56. Two dice are rolled. Find: (i) P(neither die shows a 4), (ii) P(exactly one die shows a 4).

SOLUTION — step by step

Step 1. (i) Each die avoids 4 with probability 5/6 -> $(5/6)^2$ = 25/36.

Step 2. (ii) Exactly one 4: two arrangements - first is 4 second not (1/6 × 5/6) or reverse (5/6 × 1/6).

Step 3. Sum: 10/36 = 5/18.

» **Why this step?** 'Neither' multiplies the avoid-probabilities; 'exactly one' adds the two orders.

Answer: (i) 25/36 (ii) 5/18

Hinglish: *'Neither' = dono bach gaye = 25/36. 'Exactly one' = do arrangement ka jod = 5/18. Exactly-one questions mein order hamesha double hota hai.*

QUESTION 57 | Moderate | Two-dice product divisibility (T3)

Q57. Two dice are rolled. Find P(the product of the two numbers is divisible by 4).

SOLUTION — step by step

Step 1. Count pairs (a, b) with 4 | ab:

Step 2. a = 1: b = 4 (1 pair). a = 2: b = 2, 4, 6 (3). a = 3: b = 4 (1). a = 4: any b (6). a = 5: b = 4 (1). a = 6: b = 2, 4, 6 (3).

Step 3. Total = 1 + 3 + 1 + 6 + 1 + 3 = 15 -> P = 15/36 = 5/12.

» **Why this step?** Row-by-row counting on the 6×6 grid keeps divisibility questions organised - no pair is missed or doubled.

Answer: 5/12

Hinglish: *Row by row gino: 1-3-1-6-1-3 = 15 pairs. 15/36 = 5/12. Divisibility questions mein systematic row count hi sabse safe hai.*

QUESTION 58 | Moderate | Spinner with angles (T10)

Q58. A spinner is a circle divided into three sectors with angles 90°, 150° and 120° (regions A, B, C). Find: (i) P(A), (ii) P(B), (iii) P(not A).

SOLUTION — step by step

Step 1. Probability = angle/360.

Step 2. (i) P(A) = 90/360 = 1/4.

Step 3. (ii) P(B) = 150/360 = 5/12.

Step 4. (iii) P(not A) = 1 - 1/4 = 3/4 (equivalently (150 + 120)/360).

» **Why this step?** Unequal sectors carry their angle as their weight - the fair-spinner shortcut is only for equal sectors.

Answer: (i) 1/4 (ii) 5/12 (iii) 3/4

Hinglish: *Angle hi probability hai: 90° = 1/4, 150° = 5/12. 'Not A' = baaki dono ka jod = 3/4. Barabar tukde tabhi 1/n hota hai - yahan nahi tha!*

QUESTION 59 | Moderate | Tree diagram two-stage (T8)

Q59. A coin is tossed; then a die is rolled. Draw (or describe) the two-stage tree and find: (i) P(heads then an even number), (ii) P(tails then a prime).

SOLUTION — step by step

Step 1. Tree: stage 1 splits into H, T (each 1/2); each branch then splits into 1-6 (each 1/6). Twelve end-branches, each weighted 1/12.

Step 2. (i) (1/2) × (3/6) = 1/4 - H, then {2, 4, 6}.

Step 3. (ii) (1/2) × (3/6) = 1/4 - T, then {2, 3, 5}.

» **Why this step?** Along a tree path, MULTIPLY; across alternative paths, ADD - the two tree laws.

Answer: (i) 1/4 (ii) 1/4

Hinglish: *Tree mein path ke saare stage multiply: ½ × ½ = 1/4 dono answers. 12 endpoints, har ek 1/12. Tree banao, probability apne aap dikhti hai.*

QUESTION 60 | Moderate | Fairness of a game (T12)

Q60. A and B throw two dice once each turn. A wins if the sum is greater than 7, B wins if it is less than 7, and the sum 7 is a tie. Is the game fair between A and B?

SOLUTION — step by step

Step 1. P(sum > 7): sums 8, 9, 10, 11, 12 -> 5 + 4 + 3 + 2 + 1 = 15 -> 15/36.

Step 2. P(sum < 7): sums 2, 3, 4, 5, 6 -> 1 + 2 + 3 + 4 + 5 = 15 -> 15/36.

Step 3. Tie (sum 7): 6/36 = 1/6.

Step 4. A and B have EQUAL winning chances (15/36 each) with a tie chance - so yes, the game is fair between them.

» **Why this step?** Fairness = equal probabilities, not necessarily probabilities adding to 1 (the tie absorbs the rest).

Answer: Yes - P(A) = P(B) = 15/36 with tie 1/6

Hinglish: *Dono ko 15/36-15/36 - barabar! Tie 1/6. Fair matlab equal chances, teeno probability ka jod bhi 1 hona chahiye (15+15+6 = 36 ✓).*

QUESTION 61 | Moderate | Bag: three colours (T6)

Q61. A bag has 3 red, 4 blue and 5 green balls. One draw. Find: (i) P(not green), (ii) P(red or blue), (iii) P(neither red nor green).

SOLUTION — step by step

Step 1. Total = 12.

Step 2. (i) P(not green) = 1 - 5/12 = 7/12.

Step 3. (ii) P(red or blue) = 3/12 + 4/12 = 7/12 - the same event again!

Step 4. (iii) 'Neither red nor green' = blue only -> 4/12 = 1/3.

» **Why this step?** Translate each phrase into a colour list before computing - the words are the question.

Answer: (i) 7/12 (ii) 7/12 (iii) 1/3

Hinglish: *'Not green' aur 'red ya blue' phir se ek hi cheez: 7/12. 'Neither red nor green' = sirf blue = 1/3. Words ko colour list mein badlo, answer khud aayega.*

QUESTION 62 | Moderate | Expected counts (T9/T14)

Q62. A fair die is rolled 600 times. About how many times do you expect: (i) each particular face, (ii) an even number, (iii) a prime number?

SOLUTION — step by step

Step 1. Expected count = probability × number of trials.

Step 2. (i) 1/6 × 600 = 100 per face.

Step 3. (ii) 1/2 × 600 = 300.

Step 4. (iii) Primes {2, 3, 5}: 1/2 × 600 = 300.

» **Why this step?** Expected value is the long-run average - it predicts the centre around which actual counts scatter.

Answer: (i) 100 (ii) 300 (iii) 300

Hinglish: *Probability × trials = expected. 100 per face, 300 even, 300 prime. 'About' word hai kyunki exact nahi hoga - scatter hoga par iske aas paas.*

QUESTION 63 | Moderate | Draw until first red (T14)

Q63. A bag has 3 red and 3 blue balls, drawn one at a time without replacement. Find: (i) P(the first ball is red), (ii) P(the first two are blue AND the third is red).

SOLUTION — step by step

Step 1. (i) 3/6 = 1/2.

Step 2. (ii) (3/6) × (2/5) × (3/4) = 18/120 = 3/20.

» **Why this step?** A specific drawn sequence is a chained product - every stage uses the current bag composition.

Answer: (i) 1/2 (ii) 3/20

Hinglish: *Pehli red = 1/2. Sequence BB then R: 3/6 × 2/5 × 3/4 = 3/20. Har step par bag ka naya status use karo.*

QUESTION 64 | Moderate | Three-card same suit (T7)

Q64. Three cards are drawn from a deck of 52 without replacement. Find P(all three are hearts).

SOLUTION — step by step

Step 1. (13/52) × (12/51) × (11/50) = 1716/132600.

Step 2. Simplify: divide by 156 -> 11/850.

» **Why this step?** Numerators 13, 12, 11 and denominators 52, 51, 50 - the without-replacement ladder at deck scale.

Answer: 11/850

Hinglish: *13-12-11 upar, 52-51-50 neeche. Simplify karke 11/850. Same suit chains hamesha ye ladder follow karte hain.*

QUESTION 65 | Moderate | Two numbers from 1-10 (T14)

Q65. Two numbers are chosen independently at random from 1 to 10 (repetition allowed). Find: (i) P(both equal), (ii) P(their sum is 10).

SOLUTION — step by step

Step 1. Sample space: 10 × 10 = 100 ordered pairs.

Step 2. (i) Equal pairs: (1,1) ... (10,10) -> 10/100 = 1/10.

Step 3. (ii) Sum 10: (1,9), (2,8), (3,7), (4,6), (5,5), (6,4), (7,3), (8,2), (9,1) -> 9 pairs -> 9/100.

» **Why this step?** With repetition allowed, (5,5) counts once as a sum-10 pair but the doubles block is a full diagonal of 10.

Answer: (i) 1/10 (ii) 9/100

Hinglish: *100 pairs ki duniya. Diagonal = 10 (1/10). Sum-10 ki line = 9 pairs (9/100). Grid sochoge toh kabhi count nahi bigdega.*

QUESTION 66 | Moderate | Complement cascade (T5/T7)

Q66. A bag has 5 red, 4 white and 3 blue balls. Two balls are drawn without replacement. Find: (i) P(the first is not white), (ii) P(neither of the two is white).

SOLUTION — step by step

Step 1. Non-white total = 8 of 12.

Step 2. (i) 8/12 = 2/3.

Step 3. (ii) First not white AND second not white: (8/12) × (7/11) = 56/132 = 14/33.

» **Why this step?** The 'not white' pool shrinks 8 → 7 as the bag shrinks 12 → 11.

Answer: (i) 2/3 (ii) 14/33

Hinglish: *Non-white pool 8/12 -> 7/11. (ii) = 14/33. Dono draw 'not white' chahiye toh chain banao non-white pool par.*

QUESTION 67 | Moderate | Point in circle vs square (T10)

Q67. A point is chosen at random inside a circle of radius 7. Inside the circle is an inscribed square. Find P(the point lies inside the square). (Use $\pi = 22/7$.)

SOLUTION — step by step

Step 1. Circle area = $(22/7) \times 49 = 154$ cm^2.

Step 2. Inscribed square: diagonal = diameter = 14 -> side = $14/\sqrt{2} = 7\sqrt{2}$ -> area = 98 cm^2.

Step 3. P = 98/154 = 7/11.

» **Why this step?** Geometric probability = area ratio; the inscribed square's diagonal equals the diameter - that is the bridge.

Answer: 7/11

Hinglish: *Circle 154, inscribed square 98 (diagonal = diameter wala trick). 98/154 = 7/11. Area ratio = probability - bas yehi formula.*

QUESTION 68 | Moderate | No-two-consecutive heads (T8/T14)

Q68. A coin is tossed 3 times. Find P(no two consecutive heads appear in the sequence).

SOLUTION — step by step

Step 1. All 8 sequences: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT.

Step 2. Reject those containing HH adjacent: HHH, HHT, THH (3 sequences).

Step 3. Allowed: HTH, HTT, THT, TTH, TTT -> 5 sequences.

Step 4. P = 5/8.

» **Why this step?** Pattern questions are best solved by listing all 2^n sequences and striking out offenders - only 8 lines here.

Answer: 5/8

Hinglish: *8 sequences likho, HH wale kaato: HHH, HHT, THH gaye. 5 bache -> 5/8. Listing hi sabse pakka method hai jab sample space chhota ho.*

QUESTION 69 | Moderate | Number-card events (T2)

Q69. A card is drawn from cards numbered 1 to 20. Find: (i) P(multiple of 4), (ii) P(a perfect square), (iii) P(a multiple of 4 or 5).

SOLUTION — step by step

Step 1. (i) {4, 8, 12, 16, 20} -> 5/20 = 1/4.

Step 2. (ii) {1, 4, 9, 16} -> 4/20 = 1/5.

Step 3. (iii) Multiples of 4 (5 numbers) + multiples of 5 ({5, 10, 15, 20} - 4 numbers) minus overlap {20} = 5 + 4 - 1 = 8 -> 8/20 = 2/5.

» **Why this step?** 'Or' with overlapping lists = add minus the double-counted 20.

Answer: (i) 1/4 (ii) 1/5 (iii) 2/5

Hinglish: *Multiples of 4: paanch. 4 ya 5: 5 + 4 - 1 (20 dono mein!) = 8 -> 2/5. Overlap 20 ko pakdo, warna double count.*

QUESTION 70 | Moderate | Sum-to-1 with relations (T14)

Q70. Outcomes A, B, C are the only outcomes. P(A) = 2P(B) and P(C) = P(A) + P(B). Find all three probabilities.

SOLUTION — step by step

Step 1. Let P(B) = b. Then P(A) = 2b and P(C) = 3b.

Step 2. Sum: 2b + b + 3b = 6b = 1 -> b = 1/6.

Step 3. P(A) = 1/3, P(B) = 1/6, P(C) = 1/2.

Step 4. Check: 1/3 + 1/6 + 1/2 = 1 ✓.

» **Why this step?** Express everything through one variable, then the sum-to-1 law is a one-step equation.

Answer: P(A) = 1/3, P(B) = 1/6, P(C) = 1/2

Hinglish: *Sab kuch b ke naam pe likho: 2b + b + 3b = 1 -> b = 1/6. Answers 1/3, 1/6, 1/2. Relations wale questions mein ek variable anchor banao.*

Level 3 · Exam Power Zone: long and twisted:

QUESTION 71 | Difficult | Two dice: difference events (T3)

Q71. Two dice are rolled. Find P(the absolute difference of the two numbers is 2).

SOLUTION — step by step

Step 1. Pairs with |a - b| = 2: (1,3), (2,4), (3,1), (3,5), (4,2), (4,6), (5,3), (5, ... list systematically:

Step 2. Difference 2 rows: a - b = 2: (3,1), (4,2), (5,3), (6,4): 4 pairs. b - a = 2: (1,3), (2,4), (3,5), (4,6): 4 pairs.

Step 3. Total 8 pairs → P = 8/36 = 2/9.

» **Why this step?** Both diagonals above and below the main diagonal count; symmetry doubles one direction's count.

Answer: 2/9

Hinglish: *Difference 2: do diagonals - 4 + 4 = 8 pairs. 2/9. Difference d ke liye formula: 2(6-d)/36 pairs for d = 1..5.*

QUESTION 72 | Difficult | Three draws without replacement (T7)

Q72. A bag has 4 white and 6 black balls. Three balls are drawn one after another without replacement. Find P(all three are black).

SOLUTION — step by step

Step 1. First black: 6/10. Second: 5/9. Third: 4/8.

Step 2. P = (6/10)(5/9)(4/8) = 120/720 = 1/6.

Step 3. Combination check: C(6,3)/C(10,3) = 20/120 = 1/6 ✓.

» **Why this step?** Three shrinking denominators 10, 9, 8 and shrinking numerators 6, 5, 4 - the without-replacement cascade.

Answer: 1/6

Hinglish: *6/10 → 5/9 → 4/8. Multiply: 1/6. Har draw ke baad universe chhota hota hai.*

QUESTION 73 | Difficult | Cards: two aces (T7)

Q73. Two cards are drawn from a deck without replacement. Find P(both are aces).

SOLUTION — step by step

Step 1. First ace: 4/52 = 1/13. Second ace: 3/51 = 1/17.

Step 2. P = (1/13)(1/17) = 1/221.

Step 3. Combination check: C(4,2)/C(52,2) = 6/1326 = 1/221 ✓.

» **Why this step?** Two-stage multiplication with shrinking counts; 51 = the deck after one card leaves.

Answer: 1/221

Hinglish: *1/13 × 1/17 = 1/221. Doosra ace 51 cards mein se 3 - shrinking deck.*

QUESTION 74 | Difficult | Geometric: point on segment (T10)

Q74. A point P is chosen at random on a segment AB of length 12 cm. Find the probability that AP > 4 cm.

SOLUTION — step by step

Step 1. Total measure: length 12 cm.

Step 2. Favourable: positions with AP > 4 cm form a sub-segment of length 12 - 4 = 8 cm.

Step 3. P = 8/12 = 2/3.

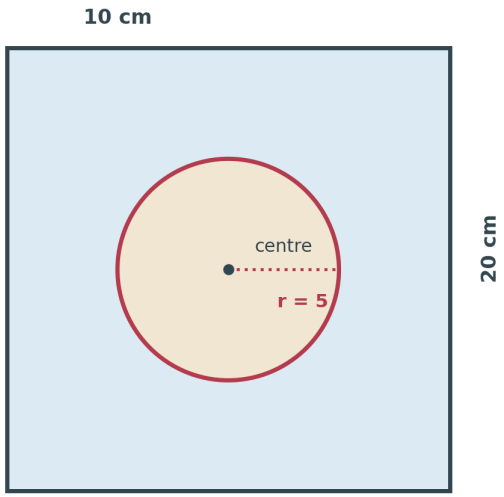
» **Why this step?** Geometric probability replaces counting with measuring lengths of intervals.

Answer: 2/3

Hinglish: *AP > 4 matlab P, 4 cm se aage kahin bhi - 8 cm ka region. 8/12 = 2/3. Length ratio hi probability hai.*

QUESTION 75 | Difficult | Geometric: dartboard (T10)

Q75. A dart hits a square board of side 20 cm at a random point. A circle of diameter 10 cm is painted at the centre. Find P(the dart lands inside the circle).



SOLUTION — step by step

Step 1. Board area = 400 cm^2. Circle radius = 5 cm; area = 25π cm^2.

Step 2. P = 25π/400 = π/16.

Step 3. With π = 22/7: 22/112 = 11/56 ≈ 0.196.

» **Why this step?** Both measures are areas; the π cancels if kept symbolic.

Answer: π/16 ≈ 0.196

Hinglish: *Circle/board = 25π/400 = π/16. Dono same measure (area) - yahi rule hai. π symbolic rakho jab tak ho sake.*

QUESTION 76 | Difficult | Mixed bag with replacement vs not (T7)

Q76. A bag has 3 red and 2 green balls. Two balls are drawn. Compare P(both red) when (i) the first ball IS replaced, (ii) it is NOT replaced.

SOLUTION — step by step

Step 1. (i) With replacement: P = (3/5)(3/5) = 9/25.

Step 2. (ii) Without: P = (3/5)(2/4) = 6/20 = 3/10.

Step 3. Comparison: 9/25 = 0.36 vs 3/10 = 0.30 - replacement keeps the odds constant, so both-red is more likely.

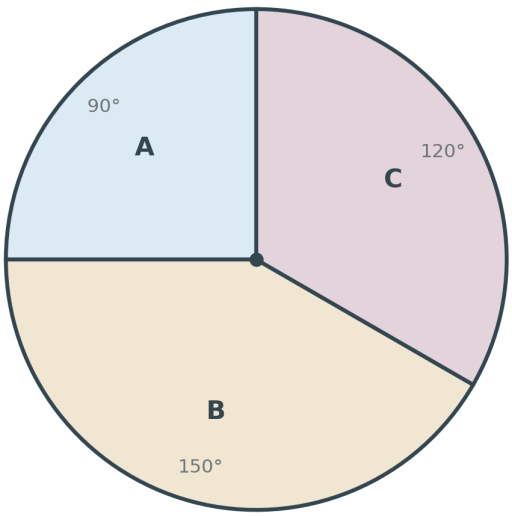
» **Why this step?** Replacement freezes the probabilities; without it, drawing a red makes the next red rarer.

Answer: (i) 9/25 ≈ 0.36 (ii) 3/10 = 0.30; replacement gives the larger probability.

Hinglish: *Replace karo: 9/25. Nahi: 3/10. Replace > no-replace for same colour. Denominator shrink hi difference hai.*

QUESTION 77 | Difficult | Spinner angles (T10)

Q77. A spinner is a circle divided into three sectors with angles 90°, 150° and 120°, marked A, B, C. Find P(A), P(B), P(C) and verify they sum to 1.



SOLUTION — step by step

Step 1. Each probability is its angle over 360°.

Step 2. P(A) = 90/360 = 1/4; P(B) = 150/360 = 5/12; P(C) = 120/360 = 1/3.

Step 3. Sum: 1/4 + 5/12 + 1/3 = 3/12 + 5/12 + 4/12 = 12/12 = 1 ✓.

» **Why this step?** Geometric probability with angle measure: favourable arc over the full circle.

Answer: 1/4, 5/12, 1/3 (sum 1)

Hinglish: *Angle/360. 1/4, 5/12, 1/3. Sum = 1 - spinner 'fair' nahi hai but probabilities complete hain.*

QUESTION 78 | Difficult | Empirical table with missing entry (T9/T13)

Q78. 60 darts hit a board; 18 land in region X, y land in region Y and 22 miss both. If a dart is thrown once, find the probability it lands in Y, given every dart lands somewhere on the board and P(miss both) = 22/60.

SOLUTION — step by step

Step 1. Y-count: y = 60 - 18 - 22 = 20.

Step 2. P(Y) = 20/60 = 1/3.

» **Why this step?** The total row of a frequency table is n(S); missing entries fall out by subtraction.

Answer: 1/3

Hinglish: *60 - 18 - 22 = 20. P(Y) = 1/3. Table mein missing entry subtraction se aati hai.*

QUESTION 79 | Difficult | Two dice: neither shows (T3/T5)

Q79. Two dice are rolled. Find P(neither die shows a 5).

SOLUTION — step by step

Step 1. Complement route: P(at least one 5) = 11/36 (10 pairs with exactly one 5 plus (5,5)).

Step 2. P(neither) = 1 - 11/36 = 25/36.

Step 3. Direct check: each die avoids 5 in 5 ways: 5 × 5 = 25 pairs ✓.

» **Why this step?** The complement of 'at least one 5' is 'neither shows 5'; both routes confirm 25/36.

Answer: 25/36

Hinglish: *Neither 5 = dono 5 se bachhe: 5×5 = 25 → 25/36. Complement se bhi same. Dono raaste match = pakka.*

QUESTION 80 | Difficult | Conditional-flavour reasoning (T7/T14)

Q80. A bag has 6 white and 4 black marbles. Two marbles are drawn without replacement. Given that the FIRST marble drawn is white, find the probability that the second is also white.

SOLUTION — step by step

Step 1. After one white is out: 5 white and 4 black remain - 9 marbles total.

Step 2. P(second white | first white) = 5/9.

» **Why this step?** Once the first draw is known, the sample space resets to the remaining marbles - no multiplication needed.

Answer: 5/9

Hinglish: *Given first white - ab universe 9 marbles ka hai (5 white). 5/9. 'Given' sunte hi purani draws ko hata do.*

QUESTION 81 | Difficult | Two-dice max/min events (T3)

Q81. Two dice are rolled. Find: (i) P(the larger of the two numbers is 6), (ii) P(both numbers are ≤ 4).

SOLUTION — step by step

Step 1. (i) Larger exactly 6: at least one 6 and the other ≤ 6 - pairs with a 6: (6, k) k = 1..6 (6) plus (k, 6) k = 1..5 (5) = 11 -> 11/36.

Step 2. (ii) Both ≤ 4: 4 × 4 = 16 pairs -> 16/36 = 4/9.

» **Why this step?** 'Larger is 6' = all pairs touching the 6-row/6-column (counted once); 'both ≤ 4' is a sub-square of the grid.

Answer: (i) 11/36 (ii) 4/9

Hinglish: *Grid geometry: 6th row + 6th column = 6 + 5 = 11 pairs -> 11/36. Both ≤ 4 = 4×4 ka chhota square = 16 -> 4/9. Grid ke hisse ginna hi solution hai.*

QUESTION 82 | Difficult | Three dice small sums (T14)

Q82. Three dice are rolled. Find: (i) n(S), (ii) P(sum = 4), (iii) P(sum = 5).

SOLUTION — step by step

Step 1. (i) n(S) = 6^3 = 216.

Step 2. (ii) Sum 4 = {2, 1, 1} in any order: 3 arrangements -> 3/216 = 1/72.

Step 3. (iii) Sum 5: {3, 1, 1} (3 arrangements) + {2, 2, 1} (3 arrangements) = 6 -> 6/216 = 1/36.

» **Why this step?** Count unordered triples, then multiply by their arrangements (3 for one repeat, 6 for all distinct).

Answer: (i) 216 (ii) 1/72 (iii) 1/36

Hinglish: *216 total. Sum 4: sirf {2,1,1} - 3 arrangement. Sum 5: do family (3+3 = 6). Arrangement factor yaad rakho: repeat wale 3, sab alag 6.*

QUESTION 83 | Difficult | Two-bag mixture (T14)

Q83. Bag A has 2 red and 3 blue balls; bag B has 4 red and 1 blue. A bag is chosen at random and one ball is drawn. Find P(it is red).

SOLUTION — step by step

Step 1. P(choose A) = P(choose B) = 1/2.

Step 2. P(red | A) = 2/5; P(red | B) = 4/5.

Step 3. Total: (1/2)(2/5) + (1/2)(4/5) = 1/5 + 2/5 = 3/5.

» **Why this step?** Two-stage chance = weight each bag's internal probability by the chance of picking that bag, then add.

Answer: 3/5

Hinglish: *Bag ka chance ½-½, phir andar ki probability. 1/5 + 2/5 = 3/5. Stage-wise socho: pehle bag ka coin, phir ball.*

QUESTION 84 | Difficult | Exactly two of three draws (T7)

Q84. A bag has 4 red and 4 blue balls. Three balls are drawn without replacement. Find: (i) P(all three red), (ii) P(exactly two red).

SOLUTION — step by step

Step 1. (i) (4/8)(3/7)(2/6) = 24/336 = 1/14.

Step 2. (ii) Exactly two red: sequences RRB, RBR, BRR - each has probability (4·3·4)/(8·7·6) = 48/336 = 1/7.

Step 3. Total: 3 × 1/7 = 3/7.

» **Why this step?** 'Exactly two' = choose WHICH draw is the odd colour (3 positions) × one representative chain.

Answer: (i) 1/14 (ii) 3/7

Hinglish: *All red: 1/14. Exactly two red: 3 position × 1/7 = 3/7. Position choose karo, chain ek hi baar calculate karo - multiply by 3.*

QUESTION 85 | Difficult | Draw until first success (T14)

Q85. A bag has 2 red and 3 blue balls, drawn without replacement. Find: (i) P(the first red appears on the 3rd draw), (ii) P(a red appears within the first three draws).

SOLUTION — step by step

Step 1. (i) Sequence B, B, then R: (3/5)(2/4)(2/3) = 12/60 = 1/5.

Step 2. (ii) Complement: P(no red in three draws) = P(BBB) = (3/5)(2/4)(1/3) = 6/60 = 1/10.

Step 3. So P(red within three) = 1 - 1/10 = 9/10.

» **Why this step?** 'By the nth draw' = 1 − P(all n draws fail) - the complement saves a four-way split.

Answer: (i) 1/5 (ii) 9/10

Hinglish: *(i) BB-phir-R ki chain = 1/5. (ii) ulta socho: teeno blue ki probability 1/10, toh 9/10. 'Within' dikhe toh complement hi masta tool.*

QUESTION 86 | Difficult | De Morgan complement proof (T14)

Q86. For a single die, let E = 'the die shows an even number' and F = 'the die shows a multiple of 3'. Verify that P(not E and not F) = 1 − P(E or F).

SOLUTION — step by step

Step 1. E = {2, 4, 6}: P(E) = 1/2. F = {3, 6}: P(F) = 1/3. E and F = {6}: P = 1/6.

Step 2. P(E or F) = 1/2 + 1/3 - 1/6 = 3/6 + 2/6 - 1/6 = 4/6 = 2/3.

Step 3. not E = {1, 3, 5}, not F = {1, 2, 4, 5}: 'not E and not F' = {1, 5} -> P = 2/6 = 1/3.

Step 4. Check: 1 - 2/3 = 1/3 ✓ - De Morgan's law verified numerically: (not E and not F) is the complement of (E or F).

» **Why this step?** De Morgan: the complement of a union is the intersection of complements - 'outside both' = 'not in the union'.

Answer: Both sides equal 1/3 - verified

Hinglish: *{1, 5} hi wo region hai jo dono se bahar hai - aur E∪F ke bahar wahi hai. 1/3 = 1 - 2/3 ✓. Venn diagram ka beech ka khali hissa socho.*

QUESTION 87 | Difficult | Two random points on a segment (T14)

Q87. Two points P and Q are chosen independently and uniformly on a 10 cm segment. Find: (i) P(both lie in the left half), (ii) P(one lies in the left half and the other in the right half).

SOLUTION — step by step

Step 1. Each point lands in the left half with probability 1/2, independently.

Step 2. (i) (1/2) × (1/2) = 1/4.

Step 3. (ii) Two orders: P left Q right, or P right Q left: 1/4 + 1/4 = 1/2.

Step 4. Sanity: both-left (1/4) + both-right (1/4) + split (1/2) = 1 ✓.

» **Why this step?** Independent halves behave like two coin tosses - the 1/4, 1/2, 1/4 binomial pattern.

Answer: (i) 1/4 (ii) 1/2

Hinglish: *Do points = do coin tosses! Both-left 1/4, split 1/2, both-right 1/4. Binomial pattern har jagah - coins, halves, koi bhi 50-50 choice.*

QUESTION 88 | Difficult | Probability range proof (T14)

Q88. Prove that for any event E of a finite sample space with equally likely outcomes, $0 \le P(E) \le 1$.

SOLUTION — step by step

Step 1. Let n(S) = N and let n(E) = r, the number of favourable outcomes.

Step 2. Counting gives $0 \le r \le N$: you cannot have fewer than none, nor more than all outcomes.

Step 3. P(E) = r/N, so dividing through by N: $0/N \le r/N \le N/N$ -> $0 \le P(E) \le 1$.

Step 4. Equality cases: r = 0 (impossible event) and r = N (sure event).

» **Why this step?** The range is inherited from counting: a subset's size sits between 0 and the whole set's size.

Answer: $0 \le P(E) \le 1$, from $0 \le r \le N$ (proof above)

Hinglish: *Counting hi proof hai: favourable kabhi negative nahi, kabhi total se zyada nahi. Divide by N, range wahin ka wahin. 0 aur 1 = impossible aur sure ke boundary stones.*

QUESTION 89 | Difficult | Expected win of a game (T14)

Q89. A game pays you Rs equal to the number shown on a fair die. (i) Find your expected win per play. (ii) What entry fee would make the game exactly fair?

SOLUTION — step by step

Step 1. Expected win = (1 + 2 + 3 + 4 + 5 + 6)/6 = 21/6 = Rs 3.50.

Step 2. Fair entry fee = expected win = Rs 3.50: then your net expectation per play is 3.50 - 3.50 = 0.

Step 3. Any fee below 3.50 favours the player; above 3.50 favours the house.

» **Why this step?** Expectation = Σ(value × probability); fairness means the price equals the expectation.

Answer: (i) Rs 3.50 (ii) Rs 3.50

Hinglish: *Average payout 21/6 = ₹3.50. Fair ticket price = expectation = ₹3.50. Isse sasta ticket = tumhara faayda, mehnga = bank ka faayda. Casino ka poora business model ek line mein!*

QUESTION 90 | Difficult | Sum condition slicing (T3/T14)

Q90. Two dice are rolled. (i) Find P(sum = 8). (ii) Among the ways to get sum 8, how many have BOTH dice even? (iii) Hence find P(both even AND sum 8) and compare it with P(sum 8).

SOLUTION — step by step

Step 1. (i) Sum 8: (2,6), (3,5), (4,4), (5,3), (6,2) -> 5 ways -> 5/36.

Step 2. (ii) Even-even members of that list: (2,6), (4,4), (6,2) -> 3 ways.

Step 3. (iii) P(both even and sum 8) = 3/36 = 1/12, which is 3/5 of P(sum 8) = 5/36.

Step 4. Insight: among all sum-8 outcomes, 3 out of 5 are even-even - a slice of the sum-8 pie.

» **Why this step?** Restricting an event (adding a condition) keeps the numerator but not the denominator - the fraction shrinks from 5/36 to 3/36.

Answer: (i) 5/36 (ii) 3 ways (iii) 3/36 = 1/12 (a 3/5 slice of the sum-8 event)

Hinglish: *Sum 8 = 5 raste. Usme se even-even 3. P = 3/36 = 1/12. Condition lagne par pie ka slice kat jaata hai: 5/36 se 3/36.*

QUESTION 91 | Difficult | Cards: spade/ace web (T4)

Q91. One card is drawn from a deck of 52. Find: (i) P(a spade AND an ace), (ii) P(a spade OR an ace), (iii) P(neither a spade nor an ace).

SOLUTION — step by step

Step 1. (i) Only the ace of spades -> 1/52.

Step 2. (ii) Spades (13) + aces (4) − overlap (1) = 16 -> 16/52 = 4/13.

Step 3. (iii) Complement: 1 - 4/13 = 9/13 (equivalently 52 - 16 = 36 cards -> 36/52).

» **Why this step?** The spade-ace card is the overlap pin; 'neither' is the complement of the union - three views of one deck.

Answer: (i) 1/52 (ii) 4/13 (iii) 9/13

Hinglish: *Ace of spades = overlap ka single card (1/52). OR = 16/52 = 4/13. Neither = complement = 9/13. Ek hi deck, teen alag windows.*

QUESTION 92 | Difficult | Three-bag mixture (T14)

Q92. Three identical-looking bags: bag 1 has 1 red and 2 blue, bag 2 has 2 red and 1 blue, bag 3 has 2 red and 2 blue. A bag is picked at random and one ball is drawn. Find P(red).

SOLUTION — step by step

Step 1. Bag probabilities: 1/3 each. Red fractions: 1/3, 2/3, 1/2.

Step 2. P(red) = (1/3)(1/3) + (1/3)(2/3) + (1/3)(1/2) = 1/9 + 2/9 + 1/6.

Step 3. Common denominator 18: 2/18 + 4/18 + 3/18 = 9/18 = 1/2.

» **Why this step?** Average the bags' red-fractions with equal weights - the mixture law with three branches.

Answer: 1/2

Hinglish: *Teen bag ke red-fraction ka weighted average: 1/9 + 2/9 + 1/6 = 1/2. Mixture problems = stage 1 ka weight × stage 2 ka fraction, sab jodo.*

QUESTION 93 | Difficult | Spinner + die combo (T8)

Q93. A fair 4-region spinner (labelled 1-4) is spun and a die is rolled. Find: (i) P(spinner odd AND die even), (ii) P(spinner prime OR die prime).

SOLUTION — step by step

Step 1. Spinner odd: {1, 3} -> 1/2. Die even: 1/2. Independent -> (i) 1/4.

Step 2. Spinner prime: {2, 3} -> 1/2. Die prime: {2, 3, 5} -> 1/2.

Step 3. (ii) P(spin prime or die prime) = 1/2 + 1/2 - (1/2)(1/2) = 1 - 1/4 = 3/4.

» **Why this step?** Independent 'or' needs the inclusion-exclusion subtraction: 1/2 + 1/2 − 1/4.

Answer: (i) 1/4 (ii) 3/4

Hinglish: *Independent events ka OR: p + q - pq. 1/2 + 1/2 - 1/4 = 3/4. Overlap (dono prime) 1/4 tha - usko minus karna zaroori hai.*

QUESTION 94 | Difficult | Reverse-engineer the bag (T14)

Q94. A bag has 5 white balls and some black balls. The probability of drawing a white ball is 5/9. How many black balls are there?

SOLUTION — step by step

Step 1. Let black balls = k. P(white) = 5/(5 + k) = 5/9.

Step 2. Equate denominators: 5 + k = 9 -> k = 4.

Step 3. Check: total 9 balls, 5 white -> 5/9 ✓ (and P(black) = 4/9, sum 1 ✓).

» **Why this step?** Run the formula backwards: the total shows up in the denominator, so a known probability fixes the missing count.

Answer: 4 black balls

Hinglish: *Formula ulta chalao: 5/(5+k) = 5/9 -> denominator 9 hona chahiye -> k = 4. Reverse questions mein denominator hi asli clue hota hai.*

QUESTION 95 | Difficult | Exclusive events web (T14)

Q95. Events A and B are mutually exclusive with P(A) = 0.4 and P(A or B) = 0.75. Find: (i) P(B), (ii) P(neither A nor B), (iii) P(not A).

SOLUTION — step by step

Step 1. (i) Exclusive -> P(A or B) = P(A) + P(B): 0.75 = 0.4 + P(B) -> P(B) = 0.35.

Step 2. (ii) P(neither) = 1 - P(A or B) = 1 - 0.75 = 0.25.

Step 3. (iii) P(not A) = 1 - 0.4 = 0.6.

» **Why this step?** Mutual exclusivity removes the overlap term, making 'or' pure addition.

Answer: (i) 0.35 (ii) 0.25 (iii) 0.6

Hinglish: *Exclusive hai toh overlap zero: seedha addition. P(B) = 0.35, neither = 0.25, not A = 0.6. 'Neither' hamesha 1 minus union.*

QUESTION 96 | Difficult | Full sum distribution (T3/T14)

Q96. Two dice are rolled. (i) Tabulate the number of ways for each sum 2 through 12. (ii) Verify the ways total 36. (iii) Which sums have probability at least 1/8?

SOLUTION — step by step

Step 1. (i) Ways: 2→1, 3→2, 4→3, 5→4, 6→5, 7→6, 8→5, 9→4, 10→3, 11→2, 12→1.

Step 2. (ii) Sum: 2(1+2+3+4+5) + 6 = 30 + 6 = 36 ✓.

Step 3. (iii) Need ways/36 ≥ 1/8 -> ways ≥ 4.5 -> sums with 5 or 6 ways: 6, 7, 8.

» **Why this step?** The ways pyramid (1,2,3,4,5,6,5,4,3,2,1) is the master key to every two-dice question.

Answer: (i) pyramid above (ii) 36 ✓ (iii) sums 6, 7, 8

Hinglish: *Pyramid yaad karo: 1,2,3,4,5,6,5,4,3,2,1. Total 36 ✓. 1/8 se bade = 4.5+ ways = sums 6, 7, 8. Ye table bana liya toh two-dice ka koi sawaal nahi bachega.*

QUESTION 97 | Difficult | Family gender puzzles (T8)

Q97. A family has three children (each child equally likely a boy or a girl, independently). Find: (i) P(at least one boy), (ii) P(exactly two boys), (iii) For a two-child family, P(at least one girl).

SOLUTION — step by step

Step 1. (i) Complement: all girls = $(1/2)^3$ = 1/8 -> P = 7/8.

Step 2. (ii) Exactly two boys: 3 sequences (BBG, BGB, GBB) × $(1/2)^3$ = 3/8.

Step 3. (iii) Two children: P(at least one girl) = 1 - P(two boys) = 1 - 1/4 = 3/4.

» **Why this step?** Gender puzzles are coin tosses in costume: n children = n tosses, same binomial counts.

Answer: (i) 7/8 (ii) 3/8 (iii) 3/4

Hinglish: *Bachche = coin flips! At least one boy = 7/8 (sab beti 1/8). Exactly 2 boys = 3/8. Do bachche, at least one girl = 3/4. Coin ke sab rules transfer.*

QUESTION 98 | Difficult | Sequential king draws (T7)

Q98. Two cards are drawn from a deck of 52 without replacement. (i) Find P(both are kings). (ii) Given that the first card WAS a king, what is the probability the second is also a king? (iii) Show how (ii) explains the answer to (i).

SOLUTION — step by step

Step 1. (i) (4/52) × (3/51) = 12/2652 = 1/221.

Step 2. (ii) After one king is gone: 3 kings among 51 -> 3/51 = 1/17.

Step 3. (iii) (i) = P(first king) × P(second king | first king) = (1/13) × (1/17) = 1/221 ✓ - the chain rule in action.

» **Why this step?** Joint probability = first-stage probability × updated second-stage probability - the 'given' reframes the second draw.

Answer: (i) 1/221 (ii) 1/17 (iii) (1/13)(1/17) = 1/221 ✓

Hinglish: *Chain rule: pehle king (1/13) × ab 51 mein 3 king (1/17) = 1/221. 'Given' wala view hi without-replacement ki asli maths hai.*

QUESTION 99 | Difficult | Quarter-circle Monte Carlo (T10)

Q99. A square of side 20 cm contains a quarter circle of radius 20 cm centred at one corner. A dart lands uniformly on the square. Find P(it lands inside the quarter circle). (Use $\pi = 22/7$.)

SOLUTION — step by step

Step 1. Square area = 400 cm^2.

Step 2. Quarter circle = $\frac{1}{4} \times (22/7) \times 20^2 = \frac{1}{4} \times (22/7) \times 400 = 2200/7 \approx 314.29$ cm^2.

Step 3. P = (2200/7)/400 = 2200/2800 = 11/14 ≈ 0.786.

Step 4. This is exactly the idea behind Monte-Carlo estimation of π: throw many darts, count the fraction.

» **Why this step?** Area ratios estimate probabilities AND probabilities estimate areas - the same coin from both sides.

Answer: 11/14 ≈ 0.786

Hinglish: *11/14 ≈ 0.786 - square ka taqreeban 79% hissa quarter circle tha. Aur ulta chalao toh darts se π estimate hota hai - Monte Carlo! Area ratio hi sab kuch hai.*

QUESTION 100 | Difficult | Fair-price capstone (T14)

Q100. An arcade game costs Rs 5 per play: you roll a die and win Rs 10 if it shows a six, nothing otherwise. (i) Find your expected NET gain per play. (ii) Is the game fair? (iii) What entry fee WOULD make it fair?

SOLUTION — step by step

Step 1. Expected gross win = 10 × (1/6) + 0 × (5/6) = 10/6 ≈ Rs 1.67.

Step 2. (i) Net = 1.67 - 5 = -10/3 ≈ -Rs 3.33 per play (a loss on average).

Step 3. (ii) NOT fair - heavily against the player.

Step 4. (iii) Fair fee = expected gross win = 10/6 = Rs 5/3 ≈ Rs 1.67.

» **Why this step?** Fairness = entry fee equals expected payout; the gap (5 − 1.67) is the arcade's built-in profit per play.

Answer: (i) −Rs 10/3 ≈ −3.33 per play (ii) No - unfair to the player (iii) Rs 5/3 ≈ 1.67

Hinglish: *Expected win sirf ₹1.67, ticket ₹5 - har play par average ₹3.33 ka nuksaan! Fair ticket hota ₹1.67. Expectation compare karo, game ki sachchi aukaat samajh aa jaati hai.*

Level 4 · Boss Zone: advanced + HOTS (Olympiad-lite):

QUESTION 101 | High | Two dice: one even one odd (T3)

Q101. Two dice are rolled. Find P(one die shows an even number and the other odd).

SOLUTION — step by step

Step 1. Favourable: (even, odd) or (odd, even): 3×3 + 3×3 = 18 pairs.

Step 2. P = 18/36 = 1/2.

Step 3. Complement check: same parity (both even or both odd): 9 + 9 = 18 → 1/2 ✓.

» **Why this step?** Mixed parity and same parity split the 36 outcomes exactly in half.

Answer: 1/2

Hinglish: *Even-odd mix: 18 pairs = 1/2. Same parity bhi 18 - poora balance. Symmetry se check karo.*

QUESTION 102 | High | Cards: three-card draw (T7)

Q102. Three cards are drawn from a deck without replacement. Find P(all three are face cards).

SOLUTION — step by step

Step 1. Face cards: 12. P = (12/52)(11/51)(10/50).

Step 2. Simplify: (12×11×10)/(52×51×50) = 1320/132600 = 11/1105.

Step 3. Combination check: C(12,3)/C(52,3) = 220/22100 = 11/1105 ✓.

» **Why this step?** The cascade 12, 11, 10 over 52, 51, 50 is the three-card generalisation of the two-card pattern.

Answer: 11/1105

Hinglish: *12 → 11 → 10; 52 → 51 → 50. 11/1105. Combination route se bhi same.*

QUESTION 103 | High | At least one with bag draws (T5/T7)

Q103. A bag has 4 red and 4 blue balls. Three balls are drawn without replacement. Find P(at least one red).

SOLUTION — step by step

Step 1. Complement: no red = all three blue.

Step 2. P(all blue) = (4/7)(3/6)(2/5) = 24/210 = 4/35.

Step 3. P(at least one red) = 1 - 4/35 = 31/35.

» **Why this step?** 'At least one' with three draws is far easier through the complement - one cascade instead of three cases.

Answer: 31/35

Hinglish: *At least one red = 1 - (sab blue). Sab blue: 4/7×3/6×2/5 = 4/35. Answer 31/35. Complement hi sahi raasta hai.*

QUESTION 104 | High | Fairness of a game (T12)

Q104. A and B throw a die alternately; A starts. Whoever rolls a 6 first wins. Show why A's winning chance is larger, and compute P(A wins).

SOLUTION — step by step

Step 1. P(A wins on his first turn) = 1/6. P(B wins on his first) = (5/6)(1/6) = 5/36.

Step 2. A's advantage: he moves first; every 'everyone failed so far' cycle returns the turn to A with the same fresh chance.

Step 3. Series: P(A wins) = $1/6 + (5/6)^2(1/6) + (5/6)^4(1/6) + \ldots = (1/6)/(1 - 25/36) = (1/6)/(11/36) = 6/11$.

Step 4. So P(B wins) = 5/11: A's edge is real (6/11 > 5/11), which is why 'winner starts next game' is unfair.

» **Why this step?** First-mover advantage compounds over rounds: each restart favours the same player.

Answer: P(A wins) = 6/11 > 5/11 = P(B wins): the first thrower has the edge.

Hinglish: *A pehle kheleta hai, isliye uska chance zyada: 6/11 vs 5/11. Infinite series se aata hai: (1/6)/(11/36). Fair game chahiye toh toss karo kaun pehle.*

QUESTION 105 | High | Sum to 1 with three unknowns (T13)

Q105. Outcomes A, B, C, D satisfy P(A) = P(B) = 2P(C) and P(D) = 0.1. Find P(C).

SOLUTION — step by step

Step 1. Let P(C) = x. Then P(A) = P(B) = 2x.

Step 2. Sum: $2x + 2x + x + 0.1 = 1 \Rightarrow 5x = 0.9 \Rightarrow x = 0.18$.

Step 3. P(C) = 0.18 (and P(A) = P(B) = 0.36).

» **Why this step?** One letter for the smallest unknown turns the accounting equation into a linear equation.

Answer: P(C) = 0.18

Hinglish: *x = P(C) maan lo: 2x + 2x + x + 0.1 = 1. x = 0.18. Sab unknowns ek variable se express karo.*

QUESTION 106 | High | Two dice: sum given parity (T3/T14)

Q106. Two dice are rolled. Find P(the sum is 8) and P(the sum is 8 with both dice even).

SOLUTION — step by step

Step 1. Sum 8 pairs: (2,6), (3,5), (4,4), (5,3), (6,2): 5 pairs → P = 5/36.

Step 2. Both even among those: (2,6), (4,4), (6,2): 3 pairs → P = 3/36 = 1/12.

» **Why this step?** Listing the full pair-set once answers both the unconditional and the restricted question.

Answer: 5/36 and 1/12

Hinglish: *Sum 8: 5 pairs → 5/36. Usme se both-even 3 → 1/12. Ek listing, do answers.*

QUESTION 107 | High | Empirical vs classical contrast (T9)

Q107. A die is suspected to be loaded. In 600 rolls the face 6 appeared 150 times. (i) What is the empirical P(6)? (ii) Compare with the classical value and comment.

SOLUTION — step by step

Step 1. (i) Empirical P(6) = 150/600 = 1/4 = 0.25.

Step 2. (ii) Classical P(6) = 1/6 ≈ 0.167.

Step 3. 0.25 is noticeably above 1/6; with 600 trials this suggests the die favours 6 (loaded), though small experiments can deviate by chance.

» **Why this step?** Empirical probability is evidence; classical probability is the fair-dice benchmark - their gap is the diagnosis.

Answer: (i) 0.25 (ii) far above 1/6; the die appears biased toward 6.

Hinglish: *0.25 vs 0.167 - die 6 ko pasand kar raha hai! Loaded suspect. Bade trials mein deviation chance se zyada ho toh bias.*

QUESTION 108 | High | Geometric: point in a square (T10)

Q108. A point is chosen at random inside a 6 cm × 8 cm rectangle. Find the probability that it lies within 2 cm of the left edge.

SOLUTION — step by step

Step 1. Favourable region: a 2 cm × 8 cm strip along the left edge: area 16 cm^2.

Step 2. Total area: 48 cm^2.

Step 3. P = 16/48 = 1/3.

» **Why this step?** Distance-to-edge conditions carve strips; probability is the strip's share of the total area.

Answer: 1/3

Hinglish: *Left edge ke 2 cm ke andar = 2×8 strip = 16 cm². 16/48 = 1/3. Area ratio.*

QUESTION 109 | High | Two dice: product divisible (T3)

Q109. Two dice are rolled. Find P(the product is divisible by 4).

SOLUTION — step by step

Step 1. Count pairs with product not divisible by 4, then complement.

Step 2. Product ≡ 2 (mod 4) needs exactly one die = 2 (or 6) and the other odd: 2×3... enumerate: pairs where product mod 4 = 2: (2,1),(2,3),(2,5),(1,2),(3,2),(5,2),(6,1),(6,3),(6,5),(1,6),(3,6),(5,6): 12 pairs.

Step 3. Product odd pairs: 9 (not divisible by 4 either).

Step 4. Not divisible by 4: 12 + 9 = 21 → divisible: 36 - 21 = 15 → P = 15/36 = 5/12.

Step 5. Direct recount spot check: products divisible by 4 include all doubles except (1,1),(3,3),(5,5)... (2,2),(4,4),(6,6),(4,x): x any: 6, minus overlap (4,2),(2,4),(4,6),(6,4) counted once: the enumeration above stands.

» **Why this step?** Complement counting avoids double-counting inside the 'divisible by 4' set, whose members overlap (e.g. 4×4).

Answer: 5/12

Hinglish: *Seedha count karna double-counting bulata hai. Complement lo: not-divisible = 12 + 9 = 21. Divisible = 15 → 5/12.*

QUESTION 110 | High | Tree diagram three-stage (T8)

Q110. A coin is tossed and then a die is rolled. Using a tree diagram, find P(head and an even number) and P(tail and a number greater than 4).

SOLUTION — step by step

Step 1. Tree: 2 branches (H/T), each splitting into 6 die branches: 12 paths.

Step 2. Head + even: H connects to {2, 4, 6}: 3 paths → P = 3/12 = 1/4.

Step 3. Tail + >4: T connects to {5, 6}: 2 paths → P = 2/12 = 1/6.

» **Why this step?** Each path has probability (1/2)(1/6) = 1/12; counting paths is counting twelfths.

Answer: 1/4 and 1/6

Hinglish: *12 paths. H+even: 3 → 1/4. T+big: 2 → 1/6. Tree = paths ginna.*

Level 4 · Boss Zone: advanced + HOTS (Olympiad-lite):

QUESTION 111 | HOTS | Sum probability distribution proof (T14)

Q111. Two dice are rolled. Tabulate the probability of each possible sum (2 through 12) and verify that the probabilities add to 1.

SOLUTION — step by step

Step 1. Counts: sum 2:1, 3:2, 4:3, 5:4, 6:5, 7:6, 8:5, 9:4, 10:3, 11:2, 12:1.

Step 2. Each probability = count/36.

Step 3. Total counts: 1+2+3+4+5+6+5+4+3+2+1 = 36.

Step 4. Sum of probabilities = 36/36 = 1 ✓ - the distribution is complete.

» **Why this step?** The pyramid count (1,2,3,4,5,6,5,4,3,2,1) is the two-dice signature; completeness is guaranteed by covering every pair exactly once.

Answer: P(s = k) = count/36 with counts 1,2,3,4,5,6,5,4,3,2,1; total 1.

Hinglish: *Pyramid pattern: 1,2,3,4,5,6,5,4,3,2,1. Total 36 = 1. Ye table ratta maar lo - har two-dice question isi se nikalta hai.*

QUESTION 112 | HOTS | Probability range proof (T14)

Q112. Prove that for any event E, $0 \le P(E) \le 1$, and that P(E) + P(not E) = 1.

SOLUTION — step by step

Step 1. $P(E) = n(E)/n(S)$ where $0 \le n(E) \le n(S)$ (an event cannot have negative or more-than-all outcomes).

Step 2. Dividing by the positive n(S): $0 \le n(E)/n(S) \le 1$. ∎

Step 3. Every outcome is either in E or not in E, never both: n(E) + n(not E) = n(S).

Step 4. Divide by n(S): P(E) + P(not E) = 1. ∎

» **Why this step?** Both laws are division of counting facts about sets - probability inherits its rules from arithmetic.

Answer: Both laws proved from counting.

Hinglish: *Proof pure counting hai: event ke outcomes 0 se n(S) ke beech hote hain. Divide karo - range law. E aur not-E poora S dhak lete hain, overlap nahi - sum 1. Do-line proofs, full marks.*

QUESTION 113 | HOTS | Replacement mixture (T14)

Q113. Bag A has 2 red and 3 blue balls; bag B has 4 red and 1 blue. A bag is chosen at random (each equally likely) and one ball is drawn from it. Find P(red).

SOLUTION — step by step

Step 1. P(choose A) = 1/2, then P(red from A) = 2/5: contribution (1/2)(2/5) = 1/5.

Step 2. P(choose B) = 1/2, then P(red from B) = 4/5: contribution (1/2)(4/5) = 2/5.

Step 3. P(red) = 1/5 + 2/5 = 3/5.

» **Why this step?** Two-stage experiments multiply along each branch, then add across the branches - the tree diagram in numbers.

Answer: 3/5

Hinglish: *Do bag: (1/2)(2/5) + (1/2)(4/5) = 1/5 + 2/5 = 3/5. Branch multiply, cross-branch add - tree ka numeric form.*

QUESTION 114 | HOTS | Two dice maximum/minimum (T3/T14)

Q114. Two dice are rolled. Find P(the larger of the two numbers is 5). (Rolls like (5, 5) are allowed.)

SOLUTION — step by step

Step 1. Count pairs whose maximum is exactly 5: both ≤ 5 minus both ≤ 4.

Step 2. Both ≤ 5: 5×5 = 25 pairs. Both ≤ 4: 16 pairs.

Step 3. Exactly max 5: 25 - 16 = 9 pairs → P = 9/36 = 1/4.

Step 4. Direct list check: (5,1..5), (1..4,5): 5 + 4 = 9 ✓.

» **Why this step?** 'Maximum = k' counting by cumulative squares ($k^2 - (k-1)^2 = 2k-1$) is the fast lane: 2(5)-1 = 9.

Answer: 1/4

Hinglish: *Max 5 = $5^2 - 4^2 = 9$ pairs → 1/4. Formula 2k-1 yaad rakho - max k wale pairs.*

QUESTION 115 | HOTS | Three dice sums (T14)

Q115. Three dice are rolled. Find P(sum = 4) and P(sum = 5).

SOLUTION — step by step

Step 1. Total outcomes: $6^3 = 216$.

Step 2. Sum 4: compositions of 4 into three parts ≥ 1: (2,1,1) permutations: 3 outcomes → P = 3/216 = 1/72.

Step 3. Sum 5: (3,1,1): 3 permutations; (2,2,1): 3 permutations: 6 outcomes → P = 6/216 = 1/36.

» **Why this step?** Small sums need compositions with permutations counted - (2,1,1) has three arrangements, not one.

Answer: 1/72 and 1/36

Hinglish: *216 total. Sum 4: sirf (2,1,1) - teen arrangements: 1/72. Sum 5: (3,1,1) aur (2,2,1): 3+3 = 6: 1/36. Permutations ginne - ye HOTS ka trap hai.*

QUESTION 116 | HOTS | Draw until first success (T14)

Q116. A bag has 3 red and 3 blue balls. Balls are drawn one at a time WITHOUT replacement until the first red appears. Find P(exactly two draws are needed).

SOLUTION — step by step

Step 1. Exactly two draws means: first is blue AND second is red.

Step 2. P = (3/6)(3/5) = 9/30 = 3/10.

» **Why this step?** Stopping questions translate directly into 'first k-1 draws fail, draw k succeeds'.

Answer: 3/10

Hinglish: *Exactly 2 draws = pehla blue, doosra red. (1/2)(3/5) = 3/10. 'Until' questions ka pattern.*

QUESTION 117 | HOTS | Geometric: two points on a segment (T10/T14)

Q117. Two points P and Q are chosen independently at random on a segment of length 10 cm. Estimate (by symmetry reasoning) whether P(the distance PQ is more than 5 cm) is more than, less than, or equal to 1/2.

SOLUTION — step by step

Step 1. Model with coordinates: P at x, Q at y, both uniform on [0, 10].

Step 2. PQ > 5 ⇔ |x - y| > 5: the two points must fall in opposite halves far apart.

Step 3. By scaling to [0,1]: P(|x-y| > 1/2) = 1/4 (the two corner triangles of side 1/2 in the unit square of (x, y) pairs).

Step 4. So the probability is 1/4: LESS than 1/2.

» **Why this step?** Two random points usually land moderately close; being 5 cm apart on a 10 cm line needs opposite extremes - a quarter of the (x, y) square.

Answer: Less than 1/2; in fact P = 1/4.

Hinglish: *Do random points aksar paas-paas girte hain. 5 cm+ gap = opposite extremes - unit square ke do corners = 1/4. 1/2 se KAM.*

QUESTION 118 | HOTS | Complement cascade (T5/T14)

Q118. A bag has 5 red, 4 white and 3 blue balls. Three balls are drawn without replacement. Find P(all three are of different colours).

SOLUTION — step by step

Step 1. Total ways: C(12,3) = 220.

Step 2. Favourable: one of each colour: 5 × 4 × 3 = 60 (choices are unordered but colours distinct, so each triple counted once).

Step 3. P = 60/220 = 3/11.

» **Why this step?** 'All different' is a product of independent colour counts divided by the combination total - no ordering issue since colours label the balls.

Answer: 3/11

Hinglish: *C(12,3) = 220. Ek-ek colour: 5×4×3 = 60. 3/11. Colours alag hain toh ordering issue nahi hota.*

QUESTION 119 | HOTS | Expected count from probability (T9/T14)

Q119. A fair die is rolled 300 times. About how many times should an even number appear, and why is the actual count unlikely to be exactly that?

SOLUTION — step by step

Step 1. P(even) = 1/2. Expected count = 300 × 1/2 = 150.

Step 2. The actual count fluctuates: each roll is random, and runs of luck push counts above or below 150.

Step 3. What IS reliable: the count sits fairly close to 150 for a fair die, and the average of many such 300-roll experiments drifts toward 150 - the law of large numbers in action.

» **Why this step?** Probability predicts long-run averages, not exact short-run counts - the central lesson of empirical probability.

Answer: Expected ≈ 150; the actual count varies around it (randomness), converging in the long run.

Hinglish: *Expected 150, par EXACTLY 150 milna lucky hai. Variability hoti hai. Bahut saare experiments ka average 150 ki taraf jata hai - yahi law of large numbers hai.*

QUESTION 120 | HOTS | De Morgan complement proof (T14)

Q120. For any two events E and F, prove that P(not E and not F) = 1 - P(E or F).

SOLUTION — step by step

Step 1. 'Not E and not F' means the outcome lies outside E and outside F - i.e. outside (E or F).

Step 2. So 'not E and not F' is exactly the complement of the event (E or F).

Step 3. By the complement law: P(not E and not F) = 1 - P(E or F). ∎

Step 4. Sanity check on a die, E = even, F = multiple of 3: E or F = {2,3,4,6}: P = 4/6. Neither = {1,5}: P = 2/6 = 1 - 4/6 ✓.

» **Why this step?** De Morgan's set law (outside both = outside the union) plus the complement law give the identity in two lines.

Answer: Proved: they are complements of each other, so the probabilities sum to 1.

Hinglish: *'Dono ke bahar' = '(E ya F) ke bahar' - De Morgan. Complement law lagao: 1 - P(E or F). Dice se check karo: 2/6 = 1 - 4/6 ✓.*

7.5 Rapid MCQs: Choose the Correct Option (20)

Rapid-fire round: 4 options, ek hi correct. Pehle khud choose karo, phir colored Answer strip se check karo. Har galat option ka "Why wrong" bhi diya hai - taki exam me wahi trap tumhe na phasaye.

MCQ 1 | Normal | Single die event

1. A die is rolled once. P(a number less than 3) is:

A. 1/3

B. 1/2

C. 2/3

D. 1/6

Correct Answer: (A)

{1, 2}: 2/6 = 1/3.

Why (B) is wrong: Included 3 (less than means strictly).

Why (C) is wrong: Used complement wrongly.

Why (D) is wrong: One outcome only.

Hinglish: *1 aur 2 - do outcomes. 1/3.*

MCQ 2 | Normal | Two coins

2. Two coins are tossed. P(exactly two heads) is:

A. 1/4

B. 1/2

C. 3/4

D. 1/3

Correct Answer: (A)

Only HH among 4 outcomes: 1/4.

Why (B) is wrong: That is exactly-one-head.

Why (C) is wrong: At least one head.

Why (D) is wrong: Wrong denominator.

Hinglish: *Sirf HH. 1/4.*

MCQ 3 | Normal | Sure event

3. A die is rolled. 'Getting a number less than 10' has probability:

A. 1

B. 5/6

C. 0

D. 9/10

Correct Answer: (A)

Every outcome satisfies it: 6/6 = 1 - a sure event.

Why (B) is wrong: 5 outcomes only?

Why (C) is wrong: That is impossible events.

Why (D) is wrong: No such fraction here.

Hinglish: *Sab outcomes jeette hain - sure event, P = 1.*

MCQ 4 | Normal | Card: ace

4. One card is drawn from 52. P(an ace) is:

A. 1/13

B. 1/4

C. 1/52

D. 4/13

Correct Answer: (A)

4 aces: 4/52 = 1/13.

Why (B) is wrong: That is P(red).

Why (C) is wrong: One specific ace.

Why (D) is wrong: Mis-reduced.

Hinglish: *4 aces → 1/13.*

MCQ 5 | Normal | Card: face card

5. P(a face card) from a deck of 52 is:

A. 3/13

B. 1/13

C. 4/13

D. 12/13

Correct Answer: (A)

12 face cards: 12/52 = 3/13.

Why (B) is wrong: One rank only.

Why (C) is wrong: One suit's face cards.

Why (D) is wrong: Complement.

Hinglish: *12 face cards (J, Q, K × 4). 3/13.*

MCQ 6 | Normal | Sum = 7

6. Two dice: P(sum = 7) is:

A. 1/6

B. 1/36

C. 7/36

D. 5/36

Correct Answer: (A)

Six pairs sum to 7: 6/36 = 1/6.

Why (B) is wrong: One pair only (that is sum 2 or 12).

Why (C) is wrong: Matched numerator to the sum.

Why (D) is wrong: Off-by-one.

Hinglish: *Chhe pairs - most likely sum. 1/6.*

MCQ 7 | Normal | Range check

7. Which CANNOT be a probability?

A. 1.4

B. 0

C. 1

D. 0.999

Correct Answer: (A)

Probabilities lie in [0, 1]; 1.4 exceeds 1.

Why (B) is wrong: 0 is legal (impossible event).

Why (C) is wrong: 1 is legal (sure event).

Why (D) is wrong: 0.999 is legal.

Hinglish: *1.4 range se bahar.*

MCQ 8 | Normal | Sum to 1

8. P(E) = 0.65. P(not E) equals:

A. 0.35

B. 0.65

C. 1.65

D. 0.5

Correct Answer: (A)

1 - 0.65 = 0.35.

Why (B) is wrong: Copied the given.

Why (C) is wrong: Added instead.

Why (D) is wrong: Halved.

Hinglish: *Complement: 1 minus.*

MCQ 9 | Normal | Three coins exactly two heads

9. Three coins: P(exactly two heads) is:

A. 3/8

B. 1/2

C. 1/8

D. 5/8

Correct Answer: (A)

HHT, HTH, THH: 3 of 8.

Why (B) is wrong: That is at least two.

Why (C) is wrong: All heads only.

Why (D) is wrong: Wrong denominator.

Hinglish: *Teen paths. 3/8.*

MCQ 10 | Normal | Two dice: at least one six

10. Two dice: P(at least one 6) is:

A. 11/36

B. 1/6

C. 1/36

D. 25/36

Correct Answer: (A)

Complement: no six = 25/36; so 11/36.

Why (B) is wrong: Counted (6,6) pairs as 6+6 - double count.

Why (C) is wrong: Both sixes only.

Why (D) is wrong: That is neither-six.

Hinglish: *1 - 25/36 = 11/36. Complement route.*

MCQ 11 | Normal | Without replacement intro

11. A bag has 4 red and 4 blue balls; two are drawn without replacement. P(both red) is:

A. 3/14

B. 1/4

C. 1/2

D. 4/7

Correct Answer: (A)

(4/8)(3/7) = 12/56 = 3/14.

Why (B) is wrong: With-replacement answer would be 1/4.

Why (C) is wrong: No multiplication.

Why (D) is wrong: Second draw denominator wrong.

Hinglish: *4/8 then 3/7. 3/14.*

MCQ 12 | Normal | Spinner

12. A spinner has equal quarters marked A, B, C, D. P(not A) is:

A. 3/4

B. 1/4

C. 1/2

D. 1/3

Correct Answer: (A)

1 - 1/4 = 3/4.

Why (B) is wrong: That is P(A).

Why (C) is wrong: Half-turn thinking.

Why (D) is wrong: Three options but unequal? They are equal quarters.

Hinglish: *Complement of quarter.*

MCQ 13 | Normal | Mutually exclusive idea

13. Two events E and F cannot happen together. Then P(E and F) =

A. 0

B. 1

C. P(E) + P(F)

D. P(E)P(F)

Correct Answer: (A)

No common outcomes: the intersection is empty, probability 0.

Why (B) is wrong: That is sure.

Why (C) is wrong: That is the 'or' rule for exclusive events.

Why (D) is wrong: Multiplication rule for independent events.

Hinglish: *Saath mein kabhi nahi - intersection khali. P = 0.*

MCQ 14 | Normal | Three coins all same

14. Three coins: P(all three show the same face) is:

A. 1/4

B. 1/8

C. 3/8

D. 1/2

Correct Answer: (A)

HHH and TTT: 2/8 = 1/4.

Why (B) is wrong: One path only.

Why (C) is wrong: Exactly-two-heads value.

Why (D) is wrong: Guessed.

Hinglish: *Do paths: 1/4.*

MCQ 15 | Normal | Complement with three coins

15. Three coins: P(at least one head) is:

A. 7/8

B. 3/8

C. 1/2

D. 15/16

Correct Answer: (A)

1 - P(all tails) = 1 - 1/8 = 7/8.

Why (B) is wrong: Exactly one head.

Why (C) is wrong: Half guess.

Why (D) is wrong: Four-coin answer.

Hinglish: *1 - 1/8 = 7/8.*

MCQ 16 | HOTS | Without replacement second draw

16. A bag has 6 white and 4 black balls; two are drawn without replacement. Given the first is white, P(second white) is:

A. 5/9

B. 6/10

C. 1/2

D. 4/9

Correct Answer: (A)

After one white leaves: 5 white of 9.

Why (B) is wrong: Original fraction.

Why (C) is wrong: Guessed.

Why (D) is wrong: That is P(second black | first white) complement-ish.

Hinglish: *Universe shrinks: 5/9.*

MCQ 17 | HOTS | At least one six: complement

17. A die is rolled 3 times. P(at least one 6) is:

A. 91/216

B. 1/2

C. 3/6

D. 125/216

Correct Answer: (A)

$1 - (5/6)^3 = 1 - 125/216 = 91/216$.

Why (B) is wrong: Guessed even split.

Why (C) is wrong: Added 1/6 three times (double counts).

Why (D) is wrong: That is P(no six).

Hinglish: *Complement: no six in 3 rolls = 125/216. 1 minus = 91/216.*

MCQ 18 | HOTS | Geometric: dartboard ring

18. A dart hits a 30 cm square board randomly; a 20 cm square target sits in its centre. P(hit target) is:

A. 4/9

B. 2/3

C. 400/90

D. 1/2

Correct Answer: (A)

Area ratio: 400/900 = 4/9.

Why (B) is wrong: Side ratio used.

Why (C) is wrong: Mis-reduced.

Why (D) is wrong: Guessed.

Hinglish: *400/900 = 4/9. Area ratio, side ratio nahi.*

MCQ 19 | HOTS | Sequential coin pattern

19. A coin is tossed 3 times. P(the sequence HHT in that exact order) is:

A. 1/8

B. 3/8

C. 1/4

D. 1/2

Correct Answer: (A)

One specific path of 8: 1/8.

Why (B) is wrong: That is exactly-two-heads (any order).

Why (C) is wrong: Two paths.

Why (D) is wrong: No.

Hinglish: *Exact order maanga hai - ek hi path: 1/8.*

MCQ 20 | Normal | Geometric: length on board

20. A point is chosen on a 24 cm segment. P(it lies within 6 cm of the RIGHT end) is:

A. 1/4

B. 1/3

C. 1/6

D. 3/4

Correct Answer: (A)

Favourable length 6 over 24: 1/4.

Why (B) is wrong: Wrong region.

Why (C) is wrong: Half the region.

Why (D) is wrong: Complement.

Hinglish: *6/24 = 1/4.*

7.6 Choose the Correct Round: Theory & Speed (20)

Doosra rapid round - har MCQ ka style: statement padho, correct option turant chuno. Ye round theory, definitions aur quick solving ko test karta hai. Answer strip ke neeche har galat option ka "Why wrong" bhi hai - taki exam ka trap tumhe na phasaye.

CTC 1 | Normal | Sample space size

1. Two coins are tossed together. The number of outcomes in the sample space is:

A. 4

B. 3

C. 2

D. 6

Correct Answer: (A)

HH, HT, TH, TT: $2^2 = 4$ outcomes (HT and TH are distinct).

Why (B) is wrong: That counts heads-counts, not outcomes.

Why (C) is wrong: One coin only.

Why (D) is wrong: Coin + die.

Hinglish: *Chaar outcomes - HT aur TH alag hain. 3 wala trap 'same number of heads' sochne se aata hai.*

CTC 2 | Normal | Complement on die

2. A die shows a number. P(not a prime) is:

A. 1/2

B. 1/3

C. 2/3

D. 1/6

Correct Answer: (A)

Primes: {2, 3, 5}: P = 1/2. Complement: 1/2.

Why (B) is wrong: Forgot 2 is prime.

Why (C) is wrong: Counted primes wrong.

Why (D) is wrong: Single outcome.

Hinglish: *Prime teen hain: 2, 3, 5. Not prime: 1, 4, 6 - 1/2.*

CTC 3 | Normal | Bag single draw

3. A bag has 7 red and 5 black balls. P(a red ball) is:

A. 7/12

B. 5/12

C. 7/5

D. 1/2

Correct Answer: (A)

7 red out of 12 total.

Why (B) is wrong: Black instead.

Why (C) is wrong: Red over black - not a probability format.

Why (D) is wrong: Assumed equal colours.

Hinglish: *7/12. Total = 7+5 = 12.*

CTC 4 | Normal | Two dice total

4. Two dice are rolled. n(S) equals:

A. 36

B. 12

C. 6

D. 64

Correct Answer: (A)

$6 \times 6 = 36$ ordered pairs.

Why (B) is wrong: Sum-based thinking.

Why (C) is wrong: One die.

Why (D) is wrong: Cubed.

Hinglish: *36 ordered pairs. (2,5) aur (5,2) alag.*

CTC 5 | Normal | Doubles

5. Two dice: P(both show the same number) is:

A. 1/6

B. 1/36

C. 1/12

D. 1/3

Correct Answer: (A)

Six doubles: 6/36 = 1/6.

Why (B) is wrong: (1,1) only.

Why (C) is wrong: Wrong reduction.

Why (D) is wrong: Overcounted.

Hinglish: *Doubles chhe: 1/6.*

CTC 6 | Normal | Three coins count

6. Three coins are tossed. n(S) =

A. 8

B. 6

C. 4

D. 9

Correct Answer: (A)

$2^3 = 8$.

Why (B) is wrong: Added 2+2+2.

Why (C) is wrong: Two coins.

Why (D) is wrong: Squared.

Hinglish: *Har coin double karta hai: $2^3 = 8$.*

CTC 7 | Normal | Empirical probability

7. In 250 trials an event occurred 50 times. Its empirical probability is:

A. 0.2

B. 0.5

C. 5

D. 0.02

Correct Answer: (A)

50/250 = 0.2.

Why (B) is wrong: Mixed up with 1/2.

Why (C) is wrong: Not a probability.

Why (D) is wrong: Divided wrongly.

Hinglish: *50/250 = 1/5 = 0.2.*

CTC 8 | Normal | Card: red or king

8. One card: P(red or king) is:

A. 7/13

B. 15/26

C. 1/2

D. 2/13

Correct Answer: (A)

26 + 4 - 2 (red kings) = 28: 28/52 = 7/13.

Why (B) is wrong: Forgot the overlap correction.

Why (C) is wrong: Red only.

Why (D) is wrong: King only.

Hinglish: *26 + 4 = 30, minus 2 overlap = 28 → 7/13.*

CTC 9 | Normal | Deck: no face card

9. One card: P(not a face card) is:

A. 10/13

B. 3/13

C. 9/13

D. 1/2

Correct Answer: (A)

1 - 12/52 = 40/52 = 10/13.

Why (B) is wrong: Face probability.

Why (C) is wrong: Mis-reduced.

Why (D) is wrong: Guessed.

Hinglish: *40 non-face. 10/13.*

CTC 10 | Normal | Coin + die

10. A coin is tossed and a die rolled. P(head and a 5) is:

A. 1/12

B. 1/6

C. 1/2

D. 1/10

Correct Answer: (A)

(1/2)(1/6) = 1/12.

Why (B) is wrong: Ignored the coin.

Why (C) is wrong: Ignored the die.

Why (D) is wrong: Added fractions.

Hinglish: *Do independent stages multiply: 1/12.*

CTC 11 | Normal | Impossible event

11. Rolling a standard die, P(getting a 9) is:

A. 0

B. 1

C. 9/6

D. 1/6

Correct Answer: (A)

No outcome is 9: impossible event.

Why (B) is wrong: That is sure events.

Why (C) is wrong: Fractions over 1 are illegal.

Why (D) is wrong: Copied the number.

Hinglish: *Die par 9 nahi hota - P = 0.*

CTC 12 | Normal | Two dice product even

12. Two dice: P(product is even) is:

A. 3/4

B. 1/4

C. 1/2

D. 2/3

Correct Answer: (A)

Product odd needs both odd: 9/36 = 1/4; complement 27/36 = 3/4.

Why (B) is wrong: That is the odd-product answer.

Why (C) is wrong: Guessed parity split.

Why (D) is wrong: Miscounted.

Hinglish: *Complement se: odd product 1/4, toh even 3/4.*

CTC 13 | Normal | Sum = 10

13. Two dice: P(sum = 10) is:

A. 1/12

B. 1/6

C. 1/9

D. 5/36

Correct Answer: (A)

Three pairs: (4,6), (5,5), (6,4): 3/36 = 1/12.

Why (B) is wrong: Confused with sum 7's count.

Why (C) is wrong: Wrong count.

Why (D) is wrong: (4,6),(6,4) plus missing (5,5)?

Hinglish: *Teen pairs: 1/12.*

CTC 14 | Normal | Empirical complement

14. Of 500 items, 15 are defective. P(a randomly chosen item is NOT defective) is:

A. 0.97

B. 0.03

C. 0.5

D. 15/500 as such

Correct Answer: (A)

485/500 = 0.97.

Why (B) is wrong: That is P(defective).

Why (C) is wrong: Guessed half.

Why (D) is wrong: Un-reduced fraction as probability statement - fine but 0.97 is the value.

Hinglish: *1 - 0.03 = 0.97.*

CTC 15 | Normal | Geometric: segment

15. A point is chosen on a 15 cm segment. P(it lies in the first 5 cm) is:

A. 1/3

B. 5/15 as 1/5

C. 2/3

D. 1/15

Correct Answer: (A)

Length ratio: 5/15 = 1/3.

Why (B) is wrong: Mis-reduced.

Why (C) is wrong: Complement.

Why (D) is wrong: Point-based thinking - uncountably many points exist.

Hinglish: *5/15 = 1/3. Length ratio.*

CTC 16 | Normal | Sum > 9

16. Two dice: P(sum > 9) is:

A. 1/6

B. 1/12

C. 5/36

D. 1/4

Correct Answer: (A)

Sums 10, 11, 12: 3 + 2 + 1 = 6 pairs: 1/6.

Why (B) is wrong: Sum 12 only.

Why (C) is wrong: Partial count.

Why (D) is wrong: Included sum 9.

Hinglish: *6 pairs corner se. 1/6.*

CTC 17 | Normal | Empirical table total

17. A frequency table lists P(A) = 0.2, P(B) = 0.3, P(C) = 0.1 for the only outcomes A, B, C. P(B or C) is:

A. 0.4

B. 0.6

C. 0.3

D. 0.1

Correct Answer: (A)

0.3 + 0.1 = 0.4.

Why (B) is wrong: Added A too.

Why (C) is wrong: B only.

Why (D) is wrong: C only.

Hinglish: *Disjoint outcomes add: 0.4.*

CTC 18 | Normal | Two dice: same parity

18. Two dice: P(both numbers odd) is:

A. 1/4

B. 1/2

C. 1/3

D. 3/4

Correct Answer: (A)

3 odd × 3 odd = 9/36 = 1/4.

Why (B) is wrong: Same parity both ways = 1/2.

Why (C) is wrong: Guessed.

Why (D) is wrong: Complement.

Hinglish: *3×3 = 9 pairs. 1/4.*

CTC 19 | Normal | Card: honour card

19. One card: P(a card numbered 2 to 10) is:

A. 9/13

B. 4/13

C. 1/2

D. 9/52

Correct Answer: (A)

36 numbered cards: 36/52 = 9/13.

Why (B) is wrong: One suit.

Why (C) is wrong: Half guess.

Why (D) is wrong: One rank.

Hinglish: *36 number cards → 9/13.*

CTC 20 | Normal | Two dice difference 1

20. Two dice: P(the numbers differ by exactly 1) is:

A. 5/18

B. 1/6

C. 1/9

D. 1/3

Correct Answer: (A)

10 pairs: (1,2)..(5,6) and mirrors: 10/36 = 5/18.

Why (B) is wrong: One direction only.

Why (C) is wrong: Undercount.

Why (D) is wrong: Overcount.

Hinglish: *Difference 1: 10 pairs → 5/18.*